Worked Solutions to Casella & Berger, Statistical Inference (2e)

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Chapter 1

Exercise 1.1

Central idea. Match the sample space to the type of measurement: finite sequences for repeated trials, integers for counts, intervals for continuous measurements.

Solution. (a) $S = \{(x_1, x_2, x_3, x_4) : x_i \in \{H, T\}\}$, $2^4 = 16$ outcomes. (b) $S = \{0, 1, 2, \dots\}$. (c) $S = \{t : t \geq 0\}$ (hours). (d) $S = \{w : w > 0\}$ (grams, say; a vector in $(0, \infty)^n$ if n rats are weighed). (e) $S = \{p : 0 \leq p \leq 1\}$.

Exercise 1.2

Central idea. Element-chasing: x belongs to the left side iff it belongs to the right side; A^c is "not in A".

Solution. (a) $x \in A \setminus B \iff x \in A$ and $x \notin B$. Also $x \in A \setminus (A \cap B) \iff x \in A$ and $x \notin A \cap B \iff x \in A$ and ($x \notin A$ or $x \notin B$) $\iff x \in A$ and $x \notin B$. And $x \in A \cap B^c \iff x \in A$ and $x \notin B$. All three coincide. (b) $B = B \cap S = B \cap (A \cup A^c) = (B \cap A) \cup (B \cap A^c)$ by distributivity; the union is disjoint. (c) Same as (a) with roles swapped. (d) By (c), $B \cap A^c = B \setminus A$, and $A \cup B = A \cup (B \setminus A)$ since every element of B is either in A or not.

Exercise 1.3

Central idea. Membership in a union (intersection) is the logical "or" ("and") of membership statements, which is commutative and associative; complements flip "or" to "and".

Solution. (a) $x \in A \cup B \iff (x \in A$ or $x \in B) \iff (x \in B$ or $x \in A) \iff x \in B \cup A$. Same with "and" for $\cap$. (b) $x \in A \cup (B \cup C) \iff (x \in A$ or $(x \in B$ or $x \in C)) \iff ((x \in A$ or $x \in B)$ or $x \in C)$ $\iff x \in (A \cup B) \cup C$, using associativity of "or". Same with "and". (c) $x \in (A \cup B)^c \iff$ not ($x \in A$ or $x \in B) \iff (x \notin A$ and $x \notin B) \iff x \in A^c \cap B^c$. The second law is dual.

Exercise 1.4

Central idea. Translate the words into set operations, then use additivity on disjoint pieces (inclusion–exclusion).

Solution. (a) $A \cup B$: $P(A) + P(B) - P(A \cap B)$. (b) Symmetric difference $A \,\Delta\, B = (A \setminus B) \cup (B \setminus A)$, disjoint: $[P(A) - P(A \cap B)] + [P(B) - P(A \cap B)] = P(A) + P(B) - 2P(A \cap B)$. (c) Same as (a): $P(A) + P(B) - P(A \cap B)$. (d) Complement of $A \cap B$: $1 - P(A \cap B)$.

Exercise 1.5

Central idea. Chain the conditional probabilities along the nested sequence $C \supset B \supset (A \cap B)$: $P(A \cap B \cap C) = P(C)\,P(B \mid C)\,P(A \mid B \cap C)$.

Solution. (a) $A \cap B \cap C$ is the event that a U.S. birth results in identical twin females (all three conditions hold at once). (b) $P(C) = 1/90$. Given twins, $P(B \mid C) = 1/3$. Given identical twins, $P(A \mid B \cap C) = 1/2$ (same sex, equally likely). Hence

$$P(A \cap B \cap C) = \frac{1}{90} \cdot \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{540}.$$

Exercise 1.6

Central idea. If the three probabilities are equal they must each be $1/3$; this forces $u+w = 1$ and $uw = 1/3$, whose discriminant is negative.

Solution. By independence, $p_0 = (1-u)(1-w)$, $p_1 = u(1-w)+w(1-u)$, $p_2 = uw$. If $p_0 = p_1 = p_2 = p$, then $3p = 1$ so $p = 1/3$. From $p_2 = 1/3$, $uw = 1/3$; from $p_0 = 1/3$, $1-u-w+uw = 1/3$, so $u+w = 2/3+uw = 1$. Thus u, w are roots of $t^2 - t + 1/3 = 0$, with discriminant $1 - 4/3 = -1/3 < 0$. No real (hence no legitimate) u, w exist.

Exercise 1.7

Central idea. Geometric probability: scale every area by the wall area A; conditioning on hitting the board cancels the common factor $\pi r^2/A$.

Solution. Let the board have radius r (area πr^2). Hitting any region has probability proportional to its area, and the wall is hit with probability 1, so $P(\text{hit board}) = \pi r^2/A$. (a) For $i = 1, \dots, 5$, with $p_i = \frac{(6-i)^2-(5-i)^2}{25}$ from Example 1.2.7,

$$P(\text{scoring } i) = P(\text{board hit})\,p_i = \frac{\pi r^2}{A} \cdot \frac{(6-i)^2 - (5-i)^2}{25},$$

and $P(\text{scoring } 0) = 1 - \pi r^2/A$ (dart hits the wall but misses the board). These are nonnegative and sum to $\frac{\pi r^2}{A}\sum_{i=1}^{5} p_i + 1 - \frac{\pi r^2}{A} = 1$. (b) $P(\text{scoring } i \mid \text{board hit}) = \frac{(\pi r^2/A)\,p_i}{\pi r^2/A} = p_i$, exactly Example 1.2.7.

Exercise 1.8

Central idea. Ring areas are differences of squares; higher scores live in smaller (inner) rings.

Solution. The board has radius r with rings spaced $r/5$; i points are scored in the ring between radii $(5-i)r/5$ and $(6-i)r/5$. (a)

$$P(\text{scoring } i) = \frac{\pi\left(\frac{(6-i)r}{5}\right)^2 - \pi\left(\frac{(5-i)r}{5}\right)^2}{\pi r^2} = \frac{(6-i)^2 - (5-i)^2}{25} = \frac{11-2i}{25}, \quad i = 1, \dots, 5,$$

independent of r. (b) $\frac{11-2i}{25}$ is strictly decreasing in i (values $9/25, 7/25, 5/25, 3/25, 1/25$): the bullseye is smallest. (c) Each value is ≥ 0 and $\sum_{i=1}^{5} \frac{11-2i}{25} = \frac{9+7+5+3+1}{25} = 1$; with the sample space $\{1, \dots, 5\}$ this satisfies the three axioms (Theorem 1.2.6).

Exercise 1.9

Central idea. "Not in at least one" is "in none"; "not in all" is "missing from at least one" — element-chasing with quantifiers.

Solution. (a) $x \in (\cup_\alpha A_\alpha)^c \iff x \notin \cup_\alpha A_\alpha \iff$ there is no α with $x \in A_\alpha \iff$ for every α, $x \notin A_\alpha \iff$ for every α, $x \in A_\alpha^c \iff x \in \cap_\alpha A_\alpha^c$. (b) $x \in (\cap_\alpha A_\alpha)^c \iff$ not (for every α, $x \in A_\alpha$) $\iff$ there exists α with $x \notin A_\alpha \iff x \in \cup_\alpha A_\alpha^c$.

Exercise 1.10

Central idea. Induction on n, using the two-set laws as the base case.

Solution. Claim: $(\cup_{i=1}^n A_i)^c = \cap_{i=1}^n A_i^c$ and $(\cap_{i=1}^n A_i)^c = \cup_{i=1}^n A_i^c$. For $n = 2$ this is Problem 1.3(c). Assume true for n; then $(\cup_{i=1}^{n+1} A_i)^c = \big((\cup_{i=1}^n A_i) \cup A_{n+1}\big)^c = (\cup_{i=1}^n A_i)^c \cap A_{n+1}^c = (\cap_{i=1}^n A_i^c) \cap A_{n+1}^c = \cap_{i=1}^{n+1} A_i^c$, using the $n = 2$ law and the induction hypothesis. The second law is dual.

Exercise 1.11

Central idea. Check the three defining properties directly; for (c), a set in the intersection inherits each property from both algebras.

Solution. (a) $S \in \mathcal{B}$; complements: $\emptyset^c = S$, $S^c = \emptyset$; countable unions of $\emptyset$s and Ss are $\emptyset$ or S. (b) The power set: S is a subset of itself; complements and arbitrary unions of subsets are subsets. (c) Let $\mathcal{B}_1, \mathcal{B}_2$ be sigma algebras and $\mathcal{B} = \mathcal{B}_1 \cap \mathcal{B}_2$. Then $S \in \mathcal{B}$; if $A \in \mathcal{B}$ then $A^c \in \mathcal{B}_1$ and $A^c \in \mathcal{B}_2$, so $A^c \in \mathcal{B}$; if $A_i \in \mathcal{B}$ for all i, then $\cup_i A_i$ is in each $\mathcal{B}_j$, hence in $\mathcal{B}$.

Exercise 1.12

Central idea. (a) Pad a finite disjoint collection with empty sets. (b) Write the infinite union as a finite union plus a "tail", and let continuity kill the tail since the tails decrease to $\emptyset$.

Solution. (a) If $A_1, \ldots, A_n$ are disjoint, set $A_{n+1} = A_{n+2} = \cdots = \emptyset$; countable additivity gives $P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i) + \sum_{i>n} P(\emptyset) = \sum_{i=1}^n P(A_i)$. (b) Let $\{A_i\}$ be disjoint and put $S_n = \cup_{i=1}^n A_i$, $S = \cup_{i=1}^\infty A_i$, $T_n = S \setminus S_n = \cup_{i=n+1}^\infty A_i$. Then $T_n \downarrow \emptyset$, and by finite additivity $P(S) = P(S_n) + P(T_n) = \sum_{i=1}^n P(A_i) + P(T_n)$. By continuity, $P(T_n) \to 0$, so $P(S) = \sum_{i=1}^\infty P(A_i)$.

Exercise 1.13

Central idea. Disjointness would force $P(A \cup B) = P(A) + P(B) > 1$, contradicting the axioms.

Solution. $P(B) = 1 - 1/4 = 3/4$. If A and B were disjoint, $P(A \cup B) = P(A) + P(B) = 1/3 + 3/4 = 13/12 > 1$, impossible since probabilities are at most 1. Hence A and B cannot be disjoint.

Exercise 1.14

Central idea. Each element independently is either in or out of a subset — n binary choices.

Solution. A subset is determined by, for each of the n elements, the choice "include" or "exclude": 2 choices per element, $2 \times \cdots \times 2 = 2^n$ by the multiplication principle. (Equivalently, $\sum_{k=0}^n \binom{n}{k} = (1+1)^n = 2^n$.)

Exercise 1.15

Central idea. The Fundamental Theorem of Counting: group the first k tasks into one super-task, then apply the $k = 2$ case.

Solution. Theorem 1.2.14: if a job consists of k tasks with n_i ways to do task i, there are $n_1 \cdots n_k$ ways total. The text proves $k = 2$. Assume the result for k tasks. For $k+1$ tasks, view the first k tasks as a single composite task, doable in $n_1 \cdots n_k$ ways by the induction hypothesis; pairing each with the n_{k+1} ways for task $k+1$ gives $(n_1 \cdots n_k) \cdot n_{k+1}$ by the $k = 2$ case.

Exercise 1.16

Central idea. Multiplication principle for a fixed pattern, addition principle across mutually exclusive patterns.

Solution. Each name position has 26 choices. (a) Surname $\times$ 2 given names: $26 \cdot 26 \cdot 26 = 26^3$. (b) Either the (a) pattern or surname $\times$ 1 given name: $26^3 + 26^2$. (c) Add the three-given-name pattern: $26^4 + 26^3 + 26^2$.

Exercise 1.17

Central idea. Unordered sampling with repetition — stars and bars, or "doubles plus mixed pairs".

Solution. A piece is an unordered pair $\{i, j\}$ with $1 \le i \le j \le n$: $\binom{n+2-1}{2} = \binom{n+1}{2} = n(n+1)/2$. Equivalently, n doubles plus $\binom{n}{2}$ mixed pairs.

Exercise 1.18

Central idea. Exactly one empty cell forces exactly one doubled cell; count: choose the empty cell, choose the doubled cell, choose its 2 balls, permute the rest.

Solution. Total outcomes: n^n. For exactly one empty cell, some other cell holds exactly 2 balls and the rest hold 1 each. Choose the empty cell (n ways), the doubled cell ($n-1$ ways), the 2 balls in it ($\binom{n}{2}$ ways), and arrange the remaining $n-2$ balls in the remaining $n-2$ cells ($(n-2)!$ ways). Favorable: $n(n-1)\binom{n}{2}(n-2)! = \binom{n}{2}n!$. Hence $P = \binom{n}{2}n!/n^n$.

Exercise 1.19

Central idea. An rth partial is a multiset of r choices from n variables — stars and bars.

Solution. (a) A fourth partial is determined by how many times each of the 3 variables is differentiated, $k_1+k_2+k_3 = 4$: $\binom{3+4-1}{4} = \binom{6}{4} = 15$. (b) An rth partial corresponds to $(k_1, \dots, k_n)$ with $k_i \ge 0$ and $\sum k_i = r$ (order irrelevant by continuity); the number of such multisets is $\binom{n+r-1}{r}$.

Exercise 1.20

Central idea. Inclusion–exclusion over the 7 "day i gets no calls" events (onto-function counting).

Solution. Let $A_i = \{\text{day } i \text{ gets no calls}\}$. Then $P(\cap_i A_i^c) = \sum_{k=0}^{7}(-1)^k\binom{7}{k}\left(\frac{7-k}{7}\right)^{12} \approx 0.2285$: choose the k empty days, and all 12 calls land in the remaining $7-k$ days.

Exercise 1.21

Central idea. No matching pair means the $2r$ shoes come from $2r$ distinct pairs — choose the pairs, then choose left/right from each.

Solution. Total: $\binom{2n}{2r}$. Favorable: choose which $2r$ pairs contribute ($\binom{n}{2r}$), then for each pair choose one of its 2 shoes (2^{2r}). Hence $P = \binom{n}{2r}2^{2r}/\binom{2n}{2r}$.

Exercise 1.22

Central idea. Hypergeometric counting: (a) 15 days from each month; (b) all 30 from the 336 non-September days.

Solution. (a) "Evenly distributed" means 15 days from each of the 12 months. If month m has d_m days, favorable $= \prod_{m=1}^{12}\binom{d_m}{15}$; total $= \binom{366}{180}$. The ratio is $\approx 0.167 \times 10^{-8}$. (b) September has 30 days, leaving 336; $P = \binom{336}{30}/\binom{366}{30}$.

Exercise 1.23

Central idea. Sum over k of $P(\text{both get } k)$ and recognize Vandermonde's identity $\sum_k \binom{n}{k}^2 = \binom{2n}{n}$.

Solution. $P(\text{both score } k \text{ heads}) = \left[\binom{n}{k}(1/2)^n\right]^2$. Summing over k,

$$P(\text{tie}) = \frac{1}{4^n}\sum_{k=0}^{n}\binom{n}{k}^2 = \frac{1}{4^n}\binom{2n}{n},$$

since $\sum_k \binom{n}{k}\binom{n}{n-k} = \binom{2n}{n}$ (choose n from $2n$ by splitting into two groups of n).

Exercise 1.24

Central idea. A wins iff the first head lands on an odd toss — a geometric series over odd terms; the first-move advantage is strict for every p.

Solution. Let $E_i = \{\text{first head on toss } i\}$, $P(E_i) = p(1-p)^{i-1}$. A wins on odd tosses:

$$P(\text{A wins}) = \sum_{k=0}^{\infty} p(1-p)^{2k} = \frac{p}{1-(1-p)^2} = \frac{1}{2-p}.$$

(a) $p = 1/2$: $\frac{1/2}{1-1/4} = 2/3$. (b) As above. (c) $\frac{1}{2-p} > \frac{1}{2} \iff 2 > 2-p \iff p > 0$.

Exercise 1.25

Central idea. Condition on the reduced sample space $\{BB, BG, GB\}$ — the conditioning event, not the family, is what matters.

Solution. A priori $\{BB, BG, GB, GG\}$ are equally likely. Given "at least one boy", the sample space shrinks to $\{BB, BG, GB\}$, exactly one of which is BB. Hence $P(\text{both boys} \mid \text{at least one boy}) = 1/3$.

Exercise 1.26

Central idea. "More than five casts" means the first five casts all fail — a geometric tail.

Solution. $P(\text{more than 5 casts}) = P(\text{no 6 in first 5 casts}) = (5/6)^5 \approx 0.402$.

Exercise 1.27

Central idea. Differentiate the binomial theorem, then substitute $x = 1$ or $x = -1$. The scanned source has k, not $1/k$, in (c); the earlier erratum claim was a transcription error.

Solution. (a) $(1-1)^n = \sum_{k=0}^{n}(-1)^k\binom{n}{k} = 0$. (b) From $n(1+x)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}x^{k-1}$, put $x = 1$ to obtain $n2^{n-1}$. (c) From $nx(1+x)^{n-1} = \sum_{k=1}^{n} k\binom{n}{k}x^k$, put $x = -1$ and negate: $\sum_{k=1}^{n}(-1)^{k+1}k\binom{n}{k} = 0$ for $n \geq 2$. The unrelated identity with $1/k$ equals H_n, but it is not the statement in the inspected textbook.

Exercise 1.28

Central idea. Squeeze $\log n!$ between integrals; the gap between $\log n!$ and the averaged integral is a convergent series (the per-term error is $O(1/k^2)$), so the ratio tends to a constant.

Solution. Summing the hint's inequalities gives $L_n < \log n! < U_n$ with $L_n = \int_0^n \log x\,dx = n\log n - n$ and $U_n = \int_1^{n+1} \log x\,dx$. Write $\log k$ as the average $\frac{1}{2}(\int_{k-1}^{k} + \int_{k}^{k+1})\log x\,dx$ plus an error ε_k. By Taylor expansion of $\log x$ about $x = k$, $\varepsilon_k = O(1/k^2)$, hence $\sum_{k=1}^{\infty}\varepsilon_k$ converges. Therefore $\log n! - \frac{1}{2}(L_n + U_n)$ converges to a finite limit c; since $\frac{1}{2}(L_n + U_n) = (n+\frac{1}{2})\log n - n + \frac{1}{2} + o(1)$, exponentiating gives $n!/(n^{n+1/2}e^{-n}) \to e^{c+1/2}$, a constant.

Exercise 1.29

Central idea. An unordered sample has multinomial multiplicity. Summing multiplicities counts ordered samples; summing ones counts unordered samples.

Solution. (a) The six permutations of $(4,4,12,12)$ are $(4,4,12,12)$, $(4,12,4,12)$, $(4,12,12,4)$, $(12,4,4,12)$, $(12,4,12,4)$, and $(12,12,4,4)$. The twelve permutations of $(2,9,9,12)$ are

$$(2,9,9,12),(2,9,12,9),(2,12,9,9),$$
$$(9,2,9,12),(9,2,12,9),(9,9,2,12),(9,9,12,2),$$
$$(9,12,2,9),(9,12,9,2),$$
$$(12,2,9,9),(12,9,2,9),(12,9,9,2).$$

(b) There are $6!/(1!2!1!2!) = 180$ favorable ordered samples out of 6^6, giving $5/1296$. (c) Choose positions for each repeated value: successive binomial coefficients multiply to $k!/(k_1!\cdots k_m!)$. (d) As printed, the identity is false: the weighted sum equals m^k by the multinomial theorem, not $\binom{k+m-1}{k}$ (e.g. $m = k = 2$ gives $1+2+1 = 4 \neq 3$). Stars and bars counts the *summands*: $\sum_{k_1,\ldots,k_m \geq 0} 1\{k_1+\cdots+k_m = k\} = \binom{k+m-1}{k}$, by placing $m-1$ separators among k identical objects.

Exercise 1.30

Central idea. Enumerate all $6^6 = 46656$ ordered samples (the natural reading: sample size = collection size), tally the average of each, and weight by multiplicity — the sampling distribution of the mean.

Solution. There are $6^6 = 46656$ equally likely ordered samples of size 6. Computing $\bar{x}$ for each gives 85 distinct averages from 1 to 20, roughly bell-shaped around the population mean $26/3 \approx 8.67$. The most likely average is $26/3$ itself (2520 samples, probability 0.0540); e.g., $P(\bar{X} \leq 4) \approx 0.0347$ and $P(\bar{X} \geq 14) \approx 0.0284$. A histogram plots these 85 average values on the horizontal axis against their probabilities (count/46656) as bar heights.

Exercise 1.31

Central idea. Multinomial probabilities are maximized by the most "spread out" allocation; Stirling turns $n!/n^n$ into closed form; missing x_i in all n draws is $((n-1)/n)^n$.

Solution. (a) An unordered outcome with composition $(k_1,\ldots,k_n)$, $\sum k_i = n$, has probability $\frac{n!}{k_1!\cdots k_n!}/n^n$. Since $k_i! \geq 1$ with equality iff all $k_i = 1$, the maximum is attained uniquely (as a multiset) at $(1,\ldots,1)$ — the sample $\{x_1,\ldots,x_n\}$ with average $(\sum x_i)/n$ — with probability $n!/n^n$. (b) By Stirling, $n! \approx \sqrt{2\pi n}\,(n/e)^n$, so $n!/n^n \approx \sqrt{2\pi n}\,e^{-n}$. (c) Each draw misses x_i with probability $(n-1)/n$, independently: $((n-1)/n)^n = (1-1/n)^n \to e^{-1}$.

Exercise 1.32

Central idea. Being hired at trial i implies being the best among the first i; by symmetry, $P(\text{best of } i \text{ is best of } N) = i/N$.

Solution. Let $R_i = \{\text{candidate } i \text{ is the best among the first } i\}$; hiring at trial i implies R_i. $P(R_i) = 1/i$ (all i positions equally likely to hold the max of the first i). Also $P(R_i \cap \{\text{candidate } i \text{ is best overall}\}) = P(\text{candidate } i \text{ is best overall}) = 1/N$ (a global best is automatically a record). Hence

$$P(\text{best hired} \mid \text{hired at } i) = \frac{1/N}{1/i} = \frac{i}{N}.$$

Exercise 1.33

Central idea. Bayes' theorem with equal priors — the likelihood ratio $0.05/0.0025 = 20$ dominates.

Solution. Let $M = \text{male}$, $C = \text{color-blind}$. $P(C \mid M) = 0.05$, $P(C \mid M^c) = 0.0025$, $P(M) = 1/2$. Then

$$P(M \mid C) = \frac{0.05 \cdot \frac{1}{2}}{0.05 \cdot \frac{1}{2} + 0.0025 \cdot \frac{1}{2}} = \frac{0.05}{0.0525} = \frac{20}{21} \approx 0.952.$$

Exercise 1.34

Central idea. Two-stage experiment: law of total probability for (a), then Bayes' theorem to invert the conditioning for (b).

Solution. Let $L_i = \{\text{litter } i \text{ chosen}\}$, $P(L_1) = P(L_2) = 1/2$; $P(\text{brown} \mid L_1) = 2/3$, $P(\text{brown} \mid L_2) = 3/5$. (a) $P(\text{brown}) = \frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{3}{5} = \frac{1}{3} + \frac{3}{10} = \frac{19}{30}$. (b) $P(L_1 \mid \text{brown}) = \frac{P(\text{brown}|L_1)P(L_1)}{P(\text{brown})} = \frac{1/3}{19/30} = \frac{10}{19}$.

Exercise 1.35

Central idea. $P(A \mid B) = P(A \cap B)/P(B)$ inherits nonnegativity, normalization, and additivity from P, since intersecting with B preserves disjointness.

Solution. Nonnegativity: $P(A \mid B) \geq 0$ as a ratio of nonnegative numbers. Normalization: $P(S \mid B) = P(S \cap B)/P(B) = P(B)/P(B) = 1$. Countable additivity: if A_i are disjoint, so are $A_i \cap B$, hence $P(\cup_i A_i \mid B) = P(\cup_i (A_i \cap B))/P(B) = \sum_i P(A_i \cap B)/P(B) = \sum_i P(A_i \mid B)$.

Exercise 1.36

Central idea. Binomial model; use complements ($1 - P(0) - P(1)$) and the definition of conditional probability.

Solution. Let $X \sim \text{Binomial}(10, 1/5)$. $P(X = 0) = (4/5)^{10}$, $P(X = 1) = 10(1/5)(4/5)^9$. Then

$$P(X \geq 2) = 1 - (4/5)^{10} - 10(1/5)(4/5)^9 \approx 0.624,$$

and since $\{X \geq 2\} \subset \{X \geq 1\}$,

$$P(X \geq 2 \mid X \geq 1) = \frac{P(X \geq 2)}{P(X \geq 1)} = \frac{P(X \geq 2)}{1 - (4/5)^{10}} \approx \frac{0.624}{0.893} \approx 0.699.$$

Exercise 1.37

Central idea. Bayes' theorem with the warden's tie-breaking rule made explicit; swapping fates transfers C's posterior survival probability to A.

Solution. $P(\text{A pardoned}) = P(\text{B}) = P(\text{C}) = 1/3$. Given $W = \{\text{warden says B dies}\}$: $P(W \mid A) = \gamma$, $P(W \mid B) = 0$ (warden must name C), $P(W \mid C) = 1$ (warden must name B). (a) $P(W) = \gamma/3 + 0 + 1/3 = (1+\gamma)/3$, so

$$P(A \mid W) = \frac{\gamma/3}{(1+\gamma)/3} = \frac{\gamma}{1+\gamma}.$$

$\frac{\gamma}{1+\gamma} < \frac{1}{3} \iff 3\gamma < 1 + \gamma \iff \gamma < 1/2$; equal iff $\gamma = 1/2$; greater iff $\gamma > 1/2$. (b) With $\gamma = 1/2$, $P(W) = 1/2$. Swapping fates with C, A survives iff C was pardoned: $P(C \mid W) = \frac{P(W|C)P(C)}{P(W)} = \frac{1/3}{1/2} = 2/3$. (Indeed $P(B \mid W) = 0$, so $P(C \mid W) = 1 - P(A \mid W) = 2/3$.)

Exercise 1.38

Central idea. All are one-line consequences of $P(X \mid Y) = P(X \cap Y)/P(Y)$ plus basic set algebra.

Solution. (a) $P(A \cap B) = P(A) - P(A \cap B^c) \geq P(A) - P(B^c) = P(A)$, so $P(A \cap B) = P(A)$ and $P(A \mid B) = P(A)/1$. (b) $A \subset B$ gives $A \cap B = A$: $P(B \mid A) = P(A)/P(A) = 1$ and $P(A \mid B) = P(A)/P(B)$. (c) Disjointness gives $A \cap (A \cup B) = A$ and $P(A \cup B) = P(A) + P(B)$. (d) Chain: $P(A \cap B \cap C) = \frac{P(A\cap B\cap C)}{P(B\cap C)} \cdot \frac{P(B\cap C)}{P(C)} \cdot P(C)$.

Exercise 1.39

Central idea. Mutual exclusivity forces $P(A \cap B) = 0$ while independence (with positive probabilities) forces $P(A \cap B) > 0$ — direct contradiction.

Solution. (a) Mutually exclusive $\Rightarrow P(A \cap B) = 0$, but independence would give $P(A \cap B) = P(A)P(B) > 0$. Contradiction. (b) Contrapositive of (a): independence gives $P(A \cap B) = P(A)P(B) > 0 \neq 0$, so they cannot be mutually exclusive.

Exercise 1.40

Central idea. Theorem 1.3.9: independence is preserved under complementation; write the target intersection as a difference (or use DeMorgan) and apply independence of A and B.

Solution. Theorem 1.3.9 states that if A and B are independent, so are the pairs in (a) A, B^c (proved in text), (b) A^c, B, (c) A^c, B^c. (b) $P(A^c \cap B) = P(B) - P(A \cap B) = P(B) - P(A)P(B) = P(B)(1 - P(A)) = P(B)P(A^c)$. (c) $P(A^c \cap B^c) = 1 - P(A \cup B) = 1 - P(A) - P(B) + P(A \cap B) = (1 - P(A))(1 - P(B)) = P(A^c)P(B^c)$.

Exercise 1.41

Central idea. Bayes' theorem on the channel; for (b), posteriors are proportional to prior $\times$ likelihood, computed per two-signal message.

Solution. Priors: $P(\text{dot sent}) = 3/7$, $P(\text{dash sent}) = 4/7$. Channel: $P(\text{dash rec} \mid \text{dot sent}) = 1/4$, $P(\text{dot rec} \mid \text{dot sent}) = 3/4$, $P(\text{dot rec} \mid \text{dash sent}) = 1/3$, $P(\text{dash rec} \mid \text{dash sent}) = 2/3$. (a) $P(\text{dash rec}) = \frac{1}{4} \cdot \frac{3}{7} + \frac{2}{3} \cdot \frac{4}{7} = \frac{41}{84}$. Hence

$$P(\text{dash sent} \mid \text{dash rec}) = \frac{\frac{2}{3} \cdot \frac{4}{7}}{41/84} = \frac{32}{41} \approx 0.78.$$

(b) For each sent message m, posterior $\propto P(m)\, P(\text{dot-dot rec} \mid m)$:

$$\begin{aligned} \text{dot-dot:} &\quad (3/7)^2(3/4)^2 = 81/784, \\ \text{dot-dash:} &\quad (3/7)(4/7)(3/4)(1/3) = 48/784, \\ \text{dash-dot:} &\quad (4/7)(3/7)(1/3)(3/4) = 48/784, \\ \text{dash-dash:} &\quad (4/7)^2(1/3)^2 = 16/441. \end{aligned}$$

Normalizing by $1849/7056$: $P(\text{dot-dot} \mid \text{rec}) = 729/1849$, $P(\text{dot-dash} \mid \text{rec}) = P(\text{dash-dot} \mid \text{rec}) = 432/1849$, $P(\text{dash-dash} \mid \text{rec}) = 256/1849$.

Exercise 1.42

Central idea. Use indicators to count membership pointwise, then integrate. This works for continuous as well as discrete sample spaces.

Solution. Let $N(\omega) = \sum_{i=1}^{n} 1_{A_i}(\omega)$ and $E_k = \{N = k\}$. (a) The $E_1, \ldots, E_n$ partition the union, so $P(\bigcup_i A_i) = \sum_{k=1}^{n} P(E_k)$. (b) On E_1, exactly one indicator is one: $1_{E_1} \sum_i 1_{A_i} = 1_{E_1}$. The notation about the probability of a fixed point should be read as a counting explanation, not a new probability operation. (c) On E_k, exactly $\binom{k}{j}$ of the j-fold intersection indicators are one. Consequently $P_j = E\binom{N}{j} = \sum_{k=j}^{n} \binom{k}{j} P(E_k)$. (d) $\sum_{j=1}^{k} (-1)^{j+1} \binom{k}{j} = 1 - (1-1)^k = 1$ for $k \geq 1$. (e) Substitute (c), interchange the finite sums, and use (d): $\sum_{j=1}^{n} (-1)^{j+1} P_j = \sum_{k=1}^{n} P(E_k) = P(\bigcup_i A_i)$. The reference in (e) needs (d) as well.

Exercise 1.43

Central idea. Truncating the alternating inclusion–exclusion sum gives alternating upper/lower bounds (Bonferroni); the remainder has sign $(-1)^m$ by the binomial tail identity. *Caveat on (b): as stated it is not correct in general — see below.*

Solution. (a) With $S_m = P_1 - P_2 + \cdots \pm P_m$, Boole's inequality is $P(\cup_{i=1}^n A_i) \le P_1 = S_1$ (Theorem 1.2.11), and $P(\cup_{i=1}^n A_i) \ge P_1 - P_2 = S_2$; both are the $m = 1, 2$ cases of the alternating bounds below. The book's Bonferroni inequality (1.2.10) is the intersection form: applying Boole to complements, $P(\cap_{i=1}^n A_i) = 1 - P(\cup_{i=1}^n A_i^c) \ge 1 - \sum_{i=1}^n P(A_i^c) = \sum_{i=1}^n P(A_i) - (n-1)$. For the bounds: from 1.42, writing $P_j = \sum_{k \ge j} \binom{k}{j} P(E_k)$,

$$P(\cup_{i=1}^n A_i) - S_m = (-1)^m \sum_{k>m} P(E_k) \binom{k-1}{m},$$

since $1 - \sum_{j=1}^m (-1)^{j+1} \binom{k}{j} = (-1)^m \binom{k-1}{m}$ (for $k > m$; it is 0 for $k \le m$). The sum is nonnegative, so the error has sign $(-1)^m$: odd partial sums are upper bounds, even ones lower bounds. (b) *As stated this is not true in general — neither direction holds.* Counterexample to the exercise's direction: disjoint A_i with $P(A_i) > 0$ gives $P_1 > 0 = P_2$. Counterexample to the reverse direction: $A_1 = \cdots = A_n = S$ with $n \ge 4$ gives $P_2 = \binom{n}{2} > n = P_1$. What *is* true and relevant: the *normalized* averages $P_k / \binom{n}{k}$ are decreasing in k (each k-fold intersection contains the $(k+1)$-fold ones: $\binom{l}{k} P_l \le \binom{n-k}{l-k} P_k$ for $k < l$). The alternating bounds above are valid regardless; they "improve" in the typical (rare-event) applications, but not universally — see (c). (c) If $A_i = A$ for all i with $P(A) = p$, then $P_k = \binom{n}{k} p$ and $S_m = p \sum_{j=1}^m (-1)^{j+1} \binom{n}{j}$. These oscillate with *growing* amplitude (e.g., $n = 10$: $S_1 = 10p$, $S_2 = -35p$, $S_3 = 85p$, …), since the binomial coefficients peak mid-sum. The bounds remain valid ($S_{\text{odd}} \ge p \ge S_{\text{even}}$) but are useless until nearly all n terms are included — the extreme illustration of Schwager's caution about highly correlated events.

Exercise 1.44

Central idea. Binomial tail: $X \sim \text{Binomial}(20, 1/4)$, sum the upper tail.

Solution.

$$P(X \ge 10) = \sum_{k=10}^{20} \binom{20}{k} \left(\tfrac{1}{4}\right)^k \left(\tfrac{3}{4}\right)^{20-k} \approx 0.0139.$$

Exercise 1.45

Central idea. The induced law $P_X(B) = P(X^{-1}(B))$ inherits everything from P because preimages preserve set operations (unions, disjointness).

Solution. Recall (1.4.1): $P_X(X = x_i) = P(\{s_j \in S : X(s_j) = x_i\})$, i.e., $P_X(B) = P(X^{-1}(B))$. Nonnegativity is inherited. $P_X(\mathbb{R}) = P(X^{-1}(\mathbb{R})) = P(S) = 1$. If B_i are disjoint, so are $X^{-1}(B_i)$ (a point can't map into two disjoint sets), and $X^{-1}(\cup_i B_i) = \cup_i X^{-1}(B_i)$; hence $P_X(\cup_i B_i) = P(\cup_i X^{-1}(B_i)) = \sum_i P(X^{-1}(B_i)) = \sum_i P_X(B_i)$.

Exercise 1.46

Central idea. $X_3 \in \{0, 1, 2\}$; count directly: choose the triple-occupied cells, choose their balls (multinomial), and place the rest avoiding new triples.

Solution. Total outcomes: $7^7 = 823543$. - $X_3 = 2$: choose 2 cells $\binom{7}{2}$; assign balls $3+3+1$: $\binom{7}{3}\binom{4}{3}\binom{1}{1} = 140$; the single ball goes to one of the other 5 cells. Count $= 21 \cdot 140 \cdot 5 = 14700$; $P = 14700/7^7 \approx 0.0178$. - $X_3 = 1$: choose the cell (7), its 3 balls ($\binom{7}{3}$); distribute the remaining 4 balls into the other 6 cells with no cell getting exactly 3: $6^4 - 6\binom{4}{3} \cdot 5 = 1296 - 120 = 1176$. Count $= 7 \cdot 35 \cdot 1176 = 288120$; $P = 288120/7^7 \approx 0.3499$. - $X_3 = 0$: the rest, $P = 1 - 288120/7^7 - 14700/7^7 = 520723/7^7 \approx 0.6323$. (Three triple-cells would need 9 balls, so $x \ge 3$ is impossible.)

Exercise 1.47

Central idea. Each is nondecreasing (positive derivative), right-continuous, with limits 0 and 1 — the cdf checklist.

Solution. In each case F is continuous (hence right-continuous) with $F'(x) \geq 0$. (a) $\tan^{-1}$ increases from $-\pi/2$ to $\pi/2$: $F : 0 \to 1$ (Cauchy). (b) e^{-x} decreases $\infty \to 0$: $F : 0 \to 1$ (logistic). (c) e^{-x} decreases, so $-e^{-x}$ increases and $e^{-e^{-x}}$ increases from 0 to 1 (Gumbel). (d) On $(0, \infty)$ (with $F = 0$ for $x \leq 0$): increasing from 0 to 1 (exponential). (e) From (1.5.6), $F_Y(y) = \frac{1-\varepsilon}{1+e^{-y}}$ for $y < 0$ and $\varepsilon + \frac{1-\varepsilon}{1+e^{-y}}$ for $y \geq 0$: the logistic pieces increase, there is an upward jump of height $\varepsilon > 0$ at $y = 0$ (right-continuous there), $\lim_{y\to-\infty} F_Y = 0$, $\lim_{y\to\infty} F_Y = \varepsilon + (1 - \varepsilon) = 1$.

Exercise 1.48

Central idea. Theorem 1.5.3 characterizes cdfs (nondecreasing, right-continuous, limits 0/1); each property follows from the corresponding set relation plus continuity of probability.

Solution. Let $F(x) = P(X \leq x)$. Nondecreasing: $x_1 < x_2 \Rightarrow \{X \leq x_1\} \subset \{X \leq x_2\}$. Right continuity: if $x_n \downarrow x$ then $\{X \leq x_n\} \downarrow \{X \leq x\}$, so $F(x_n) \to F(x)$ by continuity of P. Limits: $\{X \leq -n\} \downarrow \emptyset$ gives $F(x) \to 0$ as $x \to -\infty$; $\{X \leq n\} \uparrow S$ gives $F(x) \to 1$ as $x \to \infty$.

Exercise 1.49

Central idea. Tail probabilities are one minus the cdf, so the cdf inequality reverses.

Solution. $P(X > t) = 1 - F_X(t) \geq 1 - F_Y(t) = P(Y > t)$ for every t, with strict inequality at any t where $F_X(t) < F_Y(t)$ (which exists by hypothesis).

Exercise 1.50

Central idea. Induction on n, or the multiply-by-$(1 - t)$ telescoping trick.

Solution. (1.5.4): $\sum_{k=1}^{n} t^{k-1} = \frac{1-t^n}{1-t}$ $(t \neq 1)$. By induction: true for $n = 1$; assuming it for n, $\sum_{k=1}^{n+1} t^{k-1} = \frac{1-t^n}{1-t} + t^n = \frac{1-t^{n+1}}{1-t}$. (Equivalently, $(1 - t)\sum_{k=1}^{n} t^{k-1} = 1 - t^n$ telescopes.)

Exercise 1.51

Central idea. Sampling without replacement from a finite population: hypergeometric; the cdf is a step function jumping by the pmf values.

Solution. $X \sim$ Hypergeometric$(N = 30, K = 5, n = 4)$:

$$P(X = x) = \frac{\binom{5}{x}\binom{25}{4-x}}{\binom{30}{4}}, \quad x = 0, 1, 2, 3, 4,$$

i.e. ($\binom{30}{4} = 27405$):

x	0	1	2	3	4
$P(X = x)$	0.4616	0.4196	0.1095	0.0091	0.0002
$F(x)$	0.4616	0.8812	0.9907	0.9998	1.0000

The cdf plot is a right-continuous step function: 0 for $x < 0$, then flat segments at the tabulated $F(x)$ values with jumps of height $P(X = x)$ at $x = 0, 1, 2, 3, 4$, reaching 1 at $x = 4$.

Exercise 1.52

Central idea. g is the conditional density of X given $X \geq x_0$ — just renormalize the tail to integrate to 1.

Solution. $g(x) \geq 0$ since $f \geq 0$ and $1 - F(x_0) > 0$. Moreover,

$$\int_{-\infty}^{\infty} g(x)\,dx = \frac{1}{1 - F(x_0)} \int_{x_0}^{\infty} f(x)\,dx = \frac{1 - F(x_0)}{1 - F(x_0)} = 1.$$

Exercise 1.53

Central idea. Differentiate for the pdf; for (c), push the transformation through the cdf: $F_Z(z) = F_Y(1 + z/10)$.

Solution. (a) With $F_Y(y) = 0$ for $y < 1$: F_Y is continuous, $F_Y'(y) = 2/y^3 > 0$ on $[1, \infty)$ so nondecreasing, $\lim_{y\downarrow 1} F_Y(y) = 0$, $\lim_{y\to\infty} F_Y(y) = 1$. (b) $f_Y(y) = F_Y'(y) = 2/y^3$, $y \geq 1$ (0 otherwise). (c) For $z \geq 0$,

$$F_Z(z) = P(10(Y-1) \leq z) = P\left(Y \leq 1 + \frac{z}{10}\right) = 1 - \frac{1}{(1+z/10)^2} = 1 - \frac{100}{(10+z)^2},$$

and $F_Z(z) = 0$ for $z < 0$.

Exercise 1.54

Central idea. Choose c as the reciprocal of the unnormalized integral.

Solution. (a) $\int_0^{\pi/2} \sin x\, dx = 1$, so $c = 1$. (b) $\int_{-\infty}^{\infty} e^{-|x|}\, dx = 2\int_0^{\infty} e^{-x}\, dx = 2$, so $c = 1/2$ (Laplace/double-exponential).

Exercise 1.55

Central idea. V is mixed: an atom at 5 (when $T < 3$) plus a continuous tail (when $T \geq 3$); build the cdf piecewise from T's exponential law.

Solution. $T \sim \text{Exp}(\text{mean } 1.5)$, $F_T(t) = 1 - e^{-t/1.5}$. $P(V = 5) = P(T < 3) = 1 - e^{-2}$. For $v < 5$: $F_V(v) = 0$. At $v = 5$ the cdf jumps to $1 - e^{-2}$. For $5 \leq v < 6$ there is no mass ($V = 2T \geq 6$ when $T \geq 3$), so $F_V(v) = 1 - e^{-2}$. For $v \geq 6$: $\{V \leq v\} = \{T \leq v/2\}$, so $F_V(v) = 1 - e^{-(v/2)/1.5} = 1 - e^{-v/3}$. In summary,

$$F_V(v) = \begin{cases} 0 & v < 5, \\ 1 - e^{-2} & 5 \leq v < 6, \\ 1 - e^{-v/3} & v \geq 6. \end{cases}$$

Chapter 2

Exercise 2.1

Central idea. Each g is strictly monotone on the support, so $f_Y(y) = f_X(g^{-1}(y))\,|dg^{-1}/dy|$; the integral checks are beta integrals in disguise.

Solution. (a) $x = y^{1/3}$, $dx/dy = \frac{1}{3}y^{-2/3}$, $0 < y < 1$:

$$f_Y(y) = 42y^{5/3}(1 - y^{1/3}) \cdot \tfrac{1}{3}y^{-2/3} = 14y(1 - y^{1/3}), \quad 0 < y < 1.$$

$\int_0^1 14y(1 - y^{1/3})\, dy = 14[\frac{1}{2} - \frac{3}{7}] = 1$. (b) $x = (y-3)/4$, $dx/dy = 1/4$, $y > 3$:

$$f_Y(y) = 7e^{-7(y-3)/4} \cdot \tfrac{1}{4} = \tfrac{7}{4}e^{-7(y-3)/4}, \quad y > 3.$$

With $u = 7(y-3)/4$, $\int_3^{\infty} f_Y\, dy = \int_0^{\infty} e^{-u}\, du = 1$. (c) $x = \sqrt{y}$, $dx/dy = 1/(2\sqrt{y})$, $0 < y < 1$:

$$f_Y(y) = 30y(1 - \sqrt{y})^2 \cdot \tfrac{1}{2\sqrt{y}} = 15\sqrt{y}\,(1 - \sqrt{y})^2, \quad 0 < y < 1.$$

With $t = \sqrt{y}$, $\int_0^1 f_Y\, dy = 30\int_0^1 t^2(1-t)^2\, dt = 30B(3,3) = 30 \cdot \frac{2!\,2!}{5!} = 1$.

Exercise 2.2

Central idea. Same monotone change-of-variables formula; track how the support transforms ($x \in (0,1)$ sends $-\log x$ to $(0,\infty)$, while e^x of a positive variable lands in $(1,\infty)$).

Solution. (a) $x = \sqrt{y}$, $0 < y < 1$: $f_Y(y) = 1 \cdot \frac{1}{2\sqrt{y}} = \frac{1}{2\sqrt{y}}$. (b) $x = e^{-y}$, $|dx/dy| = e^{-y}$, $y > 0$:

$$f_Y(y) = \frac{(n+m+1)!}{n!m!} e^{-ny}(1-e^{-y})^m e^{-y} = \frac{(n+m+1)!}{n!m!} e^{-(n+1)y}(1-e^{-y})^m, \quad y > 0.$$

(c) $x = \log y$, $dx/dy = 1/y$, $y > 1$:

$$f_Y(y) = \frac{1}{\sigma^2}(\log y) e^{-(\log y/\sigma)^2/2} \cdot \frac{1}{y}, \quad y > 1.$$

Exercise 2.3

Central idea. $g(x) = x/(x+1)$ is strictly increasing on $[0,\infty)$, so Y is just a relabeling of the mass points: $f_Y(y) = f_X(g^{-1}(y))$ on the image values.

Solution. $y = x/(x+1) \iff x = y/(1-y)$, giving the values $y \in \{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots\} = \{\frac{k}{k+1} : k = 0, 1, 2, \dots\}$. Hence

$$f_Y(y) = \frac{1}{3}\left(\frac{2}{3}\right)^{y/(1-y)}, \qquad y = \frac{k}{k+1} \ (k = 0, 1, 2, \dots),$$

and $f_Y(y) = 0$ otherwise. This sums to 1 since it is f_X composed with the inverse bijection.

Exercise 2.4

Central idea. This is the Laplace (double-exponential) law; symmetry about 0 halves every computation.

Solution. (a) $f \geq 0$, and $\int_{-\infty}^{0} \frac{1}{2}\lambda e^{\lambda x} dx = \frac{1}{2} = \int_0^{\infty} \frac{1}{2}\lambda e^{-\lambda x} dx$, so the total integral is 1. (b) For $t \leq 0$: $P(X < t) = \int_{-\infty}^{t} \frac{1}{2}\lambda e^{\lambda x} dx = \frac{1}{2} e^{\lambda t}$. For $t > 0$: $P(X < t) = \frac{1}{2} + \int_0^t \frac{1}{2}\lambda e^{-\lambda x} dx = 1 - \frac{1}{2} e^{-\lambda t}$. (c) For $t \leq 0$: $P(|X| < t) = 0$. For $t > 0$, by symmetry, $P(|X| < t) = 2\int_0^t \frac{1}{2}\lambda e^{-\lambda x} dx = 1 - e^{-\lambda t}$.

Exercise 2.5

Central idea. $\sin^2$ has four monotone branches on $(0, 2\pi)$, each contributing the identical Jacobian $1/(2\sqrt{y(1-y)})$; the cdf route differentiates $F_Y(y) = 2\arcsin(\sqrt{y})/\pi$.

Solution. In Example 2.1.2, $X \sim \text{uniform}(0, 2\pi)$ and $Y = \sin^2 X$. Partition $(0, 2\pi)$ (up to the null set of endpoints) into $A_1 = (0, \pi/2)$, $A_2 = (\pi/2, \pi)$, $A_3 = (\pi, 3\pi/2)$, $A_4 = (3\pi/2, 2\pi)$, on each of which $\sin^2$ is monotone with range $\mathcal{Y} = (0, 1)$. The inverses are $\arcsin\sqrt{y}$, $\pi - \arcsin\sqrt{y}$, $\pi + \arcsin\sqrt{y}$, $2\pi - \arcsin\sqrt{y}$, each with $|dx/dy| = 1/(2\sqrt{y(1-y)})$. By Theorem 2.1.8,

$$f_Y(y) = \sum_{i=1}^{4} \frac{1}{2\pi} \cdot \frac{1}{2\sqrt{y(1-y)}} = \frac{1}{\pi\sqrt{y(1-y)}}, \qquad 0 < y < 1,$$

the arcsine law. From (2.1.6), with $x_1 = \arcsin\sqrt{y}$ and $x_2 = \pi - \arcsin\sqrt{y}$ the two roots of $\sin^2 x = y$ in $(0, \pi)$,

$$F_Y(y) = 2\frac{x_1}{2\pi} + 2\frac{\pi - x_2}{2\pi} = \frac{2\arcsin\sqrt{y}}{\pi}, \quad 0 < y < 1,$$

(0 for $y \leq 0$, 1 for $y \geq 1$), and differentiating gives $f_Y(y) = \frac{2}{\pi} \cdot \frac{1}{\sqrt{1-y}} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\pi\sqrt{y(1-y)}}$, the same.

Exercise 2.6

Central idea. Fold the two sides of a non-monotone (or piecewise) transformation together; in (c) the two pieces have different inverses but the same range $(0,1)$.

Solution. (a) For $y>0$, $F_Y(y)=P(|X|\le y^{1/3})=P(-y^{1/3}\le X\le y^{1/3})$, so

$$f_Y(y)=2\cdot\tfrac{1}{2}e^{-y^{1/3}}\cdot\tfrac{1}{3}y^{-2/3}=\tfrac{1}{3}y^{-2/3}e^{-y^{1/3}},\quad y>0.$$

With $t=y^{1/3}$ ($dy=3t^2dt$), $\int_0^\infty f_Y\,dy=\int_0^\infty e^{-t}dt=1$. (b) For $0<y<1$, $F_Y(y)=P(X\le-\sqrt{1-y})+P(X\ge\sqrt{1-y})$, so with $t=\sqrt{1-y}$,

$$f_Y(y)=\frac{f_X(-t)+f_X(t)}{2t}=\frac{\frac{3}{8}(1-t)^2+\frac{3}{8}(1+t)^2}{2t}=\frac{3(2-y)}{8\sqrt{1-y}},\quad 0<y<1.$$

With $u=1-y$, $\int_0^1 f_Y\,dy=\frac{3}{8}\int_0^1\frac{1+u}{\sqrt{u}}\,du=\frac{3}{8}(2+\frac{2}{3})=1$. (c) On $(-1,0]$, $g_1(x)=1-x^2$ increases from 0 to 1 with inverse $-\sqrt{1-y}$; on $(0,1)$, $g_2(x)=1-x$ decreases from 1 to 0 with inverse $1-y$. For $0<y<1$,

$$f_Y(y)=f_X(-\sqrt{1-y})\frac{1}{2\sqrt{1-y}}+f_X(1-y)\cdot 1=\frac{3}{16}\frac{(1-\sqrt{1-y})^2}{\sqrt{1-y}}+\frac{3}{8}(2-y)^2.$$

With $t=\sqrt{1-y}$ ($dy=-2t\,dt$), the first term integrates to $\frac{3}{8}\int_0^1(1-t)^2dt=\frac{1}{8}$ and the second to $\frac{3}{8}\int_0^1(2-y)^2dy=\frac{7}{8}$, totaling 1.

Exercise 2.7

Central idea. x^2 is monotone on $(-1,0)$ and $(0,2)$ but the two images $(0,1)$ and $(0,4)$ differ, violating condition (iii); letting the pieces carry their own ranges repairs the formula.

Solution. (a) $F_X(x)=\frac{1}{9}(x+1)^2$ on $[-1,2]$. For $0<y<1$, $F_Y(y)=P(-\sqrt{y}\le X\le\sqrt{y})=F_X(\sqrt{y})-F_X(-\sqrt{y})=\frac{4\sqrt{y}}{9}$; for $1\le y<4$, $F_Y(y)=P(X\le\sqrt{y})=F_X(\sqrt{y})=\frac{1}{9}(\sqrt{y}+1)^2$. Differentiating,

$$f_Y(y)=\begin{cases}\dfrac{2}{9\sqrt{y}}, & 0<y<1,\\[2mm] \dfrac{1}{9}\left(1+\dfrac{1}{\sqrt{y}}\right), & 1<y<4,\end{cases}$$

(0 otherwise). Theorem 2.1.8 is not directly applicable because on $(-1,0)$ the image of x^2 is $(0,1)$ while on $(0,2)$ it is $(0,4)$ — not the same $\mathcal{Y}$ for each piece. (b) If the A_i partition a superset of $\mathcal{X}$, the proof of Theorem 2.1.8 goes through unchanged: $P(X\in A_i\setminus\mathcal{X})=0$, and differentiating $F_Y(y)=\sum_i P(g_i(X)\le y,\,X\in A_i)$ gives $f_Y(y)=\sum_i f_X(g_i^{-1}(y))\,|dg_i^{-1}/dy|$, each summand restricted to its own range $\mathcal{Y}_i$. With $A_1=(-2,0)$, $A_2=(0,2)$: $g_1^{-1}(y)=-\sqrt{y}$, $g_2^{-1}(y)=\sqrt{y}$, each with derivative magnitude $1/(2\sqrt{y})$. Since $f_X(-\sqrt{y})=\frac{2}{9}(1-\sqrt{y})$ is nonzero only for $0<y<1$ while $f_X(\sqrt{y})=\frac{2}{9}(1+\sqrt{y})$ is nonzero for $0<y<4$: for $0<y<1$, $f_Y(y)=\frac{1}{9\sqrt{y}}[(1-\sqrt{y})+(1+\sqrt{y})]=\frac{2}{9\sqrt{y}}$; for $1<y<4$, $f_Y(y)=\frac{1}{9\sqrt{y}}(1+\sqrt{y})$ — agreeing with (a).

Exercise 2.8

Central idea. Check monotonicity, right-continuity, and the 0/1 limits; the generalized inverse $F_X^{-1}(y)=\inf\{x:F_X(x)\ge y\}$ inverts flat stretches to their left endpoint and jumps to the jump location.

Solution. Each function is nondecreasing, right-continuous, with limits 0 at $-\infty$ and 1 at $+\infty$, hence a cdf. (a) Continuous and strictly increasing on $[0,\infty)$: $F_X^{-1}(y)=-\log(1-y)$, $0<y<1$. (b) Flat at $1/2$ on $[0,1)$: $F_X^{-1}(y)=\log(2y)$ for $0<y\le\frac{1}{2}$ (solving $e^x/2=y$), and $F_X^{-1}(y)=1-\log(2(1-y))$ for $\frac{1}{2}<y<1$ (solving $1-e^{1-x}/2=y$); at $y=\frac{1}{2}$, $\inf\{x:F_X(x)\ge\frac{1}{2}\}=0$. (c) Jump of size $1/2$ at 0 ($P(X=0)=1/2$), yet right-continuous. $F_X^{-1}(y)=\log(4y)$ for $0<y\le\frac{1}{4}$; $=0$ for $\frac{1}{4}<y\le\frac{3}{4}$ (every y in the jump gap inverts to the jump point 0); $=-\log(4(1-y))$ for $\frac{3}{4}<y<1$.

Exercise 2.9

Central idea. The probability integral transform: the cdf itself is the monotone map that uniformizes a continuous variable.

Solution. $F_X(x) = \int_1^x \frac{t-1}{2}\,dt = \frac{(x-1)^2}{4}$, $1 < x < 3$, is continuous and strictly increasing. Take $u(x) = F_X(x) = \frac{(x-1)^2}{4}$; then $Y = u(X) \sim \text{uniform}(0,1)$ by Theorem 2.1.10.

Exercise 2.10

Central idea. Couple the discrete variable to a uniform through its generalized inverse; then $F_X(X) \geq U$ pointwise.

Solution. Let $U \sim U(0,1)$ and $Q(u) = \inf\{x : F_X(x) \geq u\}$. Then $Q(U)$ has the law of X, and right-continuity gives $F_X(Q(u)) \geq u$ for $0 < u < 1$. Hence

$$P(F_X(X) \leq y) = P(F_X(Q(U)) \leq y) \leq P(U \leq y) = y.$$

For strictness, take an atom x_0 of mass $p > 0$ and set $a = F_X(x_0-)$, $b = F_X(x_0) = a + p$. For $a < y < b$, $F_X(X) \leq y$ implies $X < x_0$, so $P(F_X(X) \leq y) \leq a < y$. Taking complements proves (a). The earlier supremum proof was invalid: for a Bernoulli variable with $P(X = 0) = 1/2$ and $y = 3/4$, $\sup\{x : F_X(x) \leq y\} = 1$ but $F_X(1) = 1 > y$.

Exercise 2.11

Central idea. Integration by parts kills the boundary term for EX^2; $|X|$ folds the normal in half (half-normal).

Solution. (a) Directly,

$$EX^2 = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \left[-x \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 0 + 1 = 1.$$

From Example 2.1.7, $f_Y(y) = \frac{1}{2\sqrt{y}}(f_X(\sqrt{y}) + f_X(-\sqrt{y})) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}$, $y > 0$, so $EY = \frac{1}{\sqrt{2\pi}} \int_0^{\infty} \sqrt{y}\, e^{-y/2} dy$; with $t = y/2$ this equals $\frac{2}{\sqrt{\pi}}\Gamma(\frac{3}{2}) = 1$. (b) For $y > 0$, $F_Y(y) = P(-y \leq X \leq y) = 2\Phi(y) - 1$, so $f_Y(y) = 2\phi(y) = \sqrt{2/\pi}\, e^{-y^2/2}$, $y > 0$. Then $E|X| = \sqrt{2/\pi} \int_0^{\infty} y e^{-y^2/2} dy = \sqrt{2/\pi}\big[-e^{-y^2/2}\big]_0^{\infty} = \sqrt{2/\pi}$, and $\text{Var}|X| = EX^2 - (E|X|)^2 = 1 - 2/\pi$.

Exercise 2.12

Central idea. The figure gives height $Y = d\tan X$, a strictly increasing transformation; its tail is too heavy for a finite mean.

Solution. The triangle has adjacent side d and angle X at the origin, so $Y = d\tan X$, strictly increasing on $(0, \pi/2)$ with range $(0, \infty)$. With $x = \arctan(y/d)$, $dx/dy = d/(d^2 + y^2)$, and $f_X = 2/\pi$,

$$f_Y(y) = \frac{2}{\pi} \cdot \frac{d}{d^2 + y^2}, \qquad y > 0.$$

$EY = \int_0^{\infty} y \frac{2d}{\pi(d^2+y^2)} dy = +\infty$ (the integrand is $\sim \frac{2d}{\pi y}$, logarithmically divergent), so the mean does not exist as a finite number.

Exercise 2.13

Central idea. $\{X = k\}$ is the first k flips agreeing and flip $k+1$ breaking the run — a two-sided geometric law.

Solution. $P(X=k)=p^k(1-p)+(1-p)^k p$, $k=1,2,\ldots$ (a run of k heads broken by a tail, or k tails broken by a head). This sums to $(1-p)\frac{p}{1-p}+p\frac{1-p}{p}=1$. Using $\sum_{k\geq 1} kr^k = r/(1-r)^2$,

$$EX=(1-p)\frac{p}{(1-p)^2}+p\frac{1-p}{p^2}=\frac{p}{1-p}+\frac{1-p}{p}.$$

Exercise 2.14

Central idea. Write the survival function as an integral (sum) of the density (mass), then swap the order: the mean is the total mass "above" each level.

Solution. (a) $1-F_X(x)=\int_x^\infty f_X(t)\,dt$, so by Tonelli on $\{0\leq x\leq t<\infty\}$,

$$\int_0^\infty [1-F_X(x)]dx=\int_0^\infty\int_x^\infty f_X(t)\,dt\,dx=\int_0^\infty\int_0^t dx\,f_X(t)\,dt=\int_0^\infty t f_X(t)\,dt=EX.$$

(b) $1-F_X(k)=P(X>k)=\sum_{j>k} f_X(j)$, so

$$\sum_{k\geq 0}(1-F_X(k))=\sum_{k\geq 0}\sum_{j\geq k+1} f_X(j)=\sum_{j\geq 1} j\,f_X(j)=EX,$$

each $j\geq 1$ being counted exactly j times (for $k=0,\ldots,j-1$). Part (b) is the discrete analogue of (a): the integral of the survival function becomes a sum of survival probabilities, with $F_X(k)=P(X\leq k)$ in the role of $F_X(x)$.

Exercise 2.15

Central idea. Pointwise, one of X,Y is the max and the other the min, so their sum splits; linearity of expectation does the rest — no independence needed.

Solution. For every outcome, $\{X\vee Y,\ X\wedge Y\}=\{X,Y\}$ as multisets, so $X+Y=(X\vee Y)+(X\wedge Y)$ pointwise. Taking expectations (assumed to exist) and rearranging gives $E(X\vee Y)=EX+EY-E(X\wedge Y)$.

Exercise 2.16

Central idea. The survival function is a mixture of two exponential tails; integrate termwise.

Solution. By Exercise 2.14(a),

$$ET=\int_0^\infty P(T>t)\,dt=a\int_0^\infty e^{-\lambda t}dt+(1-a)\int_0^\infty e^{-\mu t}dt=\frac{a}{\lambda}+\frac{1-a}{\mu}.$$

Exercise 2.17

Central idea. For a continuous law the median solves $F(m)=1/2$; symmetry pins it down without computation in (b).

Solution. (a) $F(x)=x^3$ on $(0,1)$; $m^3=1/2$ gives $m=2^{-1/3}$. (b) The Cauchy density is symmetric about 0, so $\int_{-\infty}^0 f=\int_0^\infty f=1/2$ and $m=0$.

Exercise 2.18

Central idea. Differentiate $g(a)=E|X-a|$ under the integral: $g'(a)=2F_X(a)-1$, whose zero is the median, and g is convex.

Solution. Assume $E|X|<\infty$ and put $g(a)=\int_{-\infty}^a (a-x)f_X(x)dx+\int_a^\infty (x-a)f_X(x)dx$. By Leibniz' rule (the boundary terms vanish by continuity of X),

$$g'(a)=\int_{-\infty}^a f_X(x)dx-\int_a^\infty f_X(x)dx=2F_X(a)-1.$$

Hence $g'(a) = 0 \iff F_X(a) = 1/2$, i.e. at the median m; $g''(a) = 2f_X(a) \geq 0$, so this critical point is the global minimum.

Exercise 2.19

Central idea. $E(X-a)^2 = EX^2 - 2aEX + a^2$ is a quadratic in a; differentiating under the expectation is justified by a finite second moment.

Solution. Assume $EX^2 < \infty$ (hence $E|X| < \infty$). Then $h(a) = E(X-a)^2 = \int (x-a)^2\, dF_X(x)$ may be differentiated under the integral by dominated convergence, since $\frac{d}{da}(x-a)^2 = -2(x-a)$ is dominated near any a_0 by the integrable function $2|x| + 2(|a_0|+1)$:

$$h'(a) = E[-2(X-a)] = -2EX + 2a,$$

so $h'(a) = 0 \iff a = EX$. Moreover $h''(a) = 2 > 0$, so $a = EX$ is the unique global minimizer. Needed: X has finite second moment (so EX exists and h is finite); no continuity of F_X and no density f_X are required.

Exercise 2.20

Central idea. The number of trials to the first success is geometric with $p = 1/2$.

Solution. Let X be the number of children up to and including the first daughter; $P(X = k) = (1/2)^k$, $k \geq 1$, the geometric law with success probability $1/2$. Hence $EX = \sum_{k\geq 1} k(1/2)^k = \frac{1/2}{(1-1/2)^2} = 2$.

Exercise 2.21

Central idea. Substitute $y = g(x)$ in the integral defining $Eg(X)$; monotonicity makes the change of variables legal and produces exactly the pdf of Y.

Solution. For g increasing, $Eg(X) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$; with $y = g(x)$ ($dy = g'(x)dx$) this becomes $\int_{-\infty}^{\infty} y\, f_X(g^{-1}(y))(g^{-1})'(y) dy = \int y f_Y(y) dy = EY$ by the change-of-variables formula for f_Y (Theorem 2.1.5). If g is decreasing the limits swap and $|(g^{-1})'|$ absorbs the sign, giving the same conclusion.

Exercise 2.22

Central idea. The substitution $u = x^2/\beta^2$ turns every moment into a gamma integral.

Solution. (a) $f \geq 0$, and with $u = x^2/\beta^2$ ($x^2 dx = \frac{\beta^3}{2} u^{1/2} du$),

$$\int_0^{\infty} x^2 e^{-x^2/\beta^2} dx = \frac{\beta^3}{2}\Gamma(\tfrac{3}{2}) = \frac{\beta^3\sqrt{\pi}}{4},$$

so $\int_0^{\infty} f\, dx = 1$. (b) $EX = \frac{4}{\beta^3\sqrt{\pi}} \int_0^{\infty} x^3 e^{-x^2/\beta^2} dx = \frac{4}{\beta^3\sqrt{\pi}} \cdot \frac{\beta^4}{2}\Gamma(2) = \frac{2\beta}{\sqrt{\pi}}$. $EX^2 = \frac{4}{\beta^3\sqrt{\pi}} \cdot \frac{\beta^5}{2}\Gamma(\frac{5}{2}) = \frac{2\beta^2}{\sqrt{\pi}} \cdot \frac{3\sqrt{\pi}}{4} = \frac{3\beta^2}{2}$. Hence $\mathrm{Var}X = \frac{3\beta^2}{2} - \frac{4\beta^2}{\pi} = \beta^2\left(\frac{3}{2} - \frac{4}{\pi}\right)$.

Exercise 2.23

Central idea. Fold at 0 as in Example 2.1.7; the odd part of f_X cancels in the fold, and moments of Y are even moments of X.

Solution. (a) For $0 < y < 1$,

$$f_Y(y) = \frac{1}{2\sqrt{y}}[f_X(\sqrt{y}) + f_X(-\sqrt{y})] = \frac{1}{2\sqrt{y}} \cdot \tfrac{1}{2}[(1+\sqrt{y}) + (1-\sqrt{y})] = \frac{1}{2\sqrt{y}}.$$

(b) $EY = EX^2 = \frac{1}{2}\int_{-1}^{1} x^2(1+x)dx = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}$ (the x^3 term integrates to 0). $EY^2 = EX^4 = \frac{1}{2}\int_{-1}^{1} x^4 dx = \frac{1}{5}$. Thus $\mathrm{Var}Y = \frac{1}{5} - \frac{1}{9} = \frac{4}{45}$.

Exercise 2.24

Central idea. Direct moment integrals/sums; (c) is symmetric about 1, so center there.

Solution. (a) $EX = \int_0^1 ax^a\,dx = \frac{a}{a+1}$; $EX^2 = \frac{a}{a+2}$; $\mathrm{Var}X = \frac{a}{a+2} - \frac{a^2}{(a+1)^2} = \frac{a}{(a+1)^2(a+2)}$. (b) $EX = \frac{1}{n}\sum_{k=1}^n k = \frac{n+1}{2}$; $EX^2 = \frac{1}{n}\sum_{k=1}^n k^2 = \frac{(n+1)(2n+1)}{6}$; $\mathrm{Var}X = \frac{(n+1)(2n+1)}{6} - \frac{(n+1)^2}{4} = \frac{n^2-1}{12}$. (c) With $u = x-1$, f is symmetric about 1, so $EX = 1$. $EX^2 = \frac{3}{2}\int_{-1}^1 (u+1)^2u^2du = \frac{3}{2}(\frac{2}{5}+\frac{2}{3}) = \frac{8}{5}$; $\mathrm{Var}X = \frac{8}{5} - 1 = \frac{3}{5}$.

Exercise 2.25

Central idea. Evenness of the density is exactly invariance under $x \mapsto -x$, which passes to the mgf by the same change of variables.

Solution. (a) For continuous X, $f_{-X}(y) = f_X(-y) = f_X(y)$; identical pdfs give identical distributions. (b) $M_X(-t) = \int e^{-tx}f_X(x)dx$; with $u = -x$, $= \int e^{tu}f_X(-u)du = \int e^{tu}f_X(u)du = M_X(t)$.

Exercise 2.26

Central idea. Shifting by a reduces everything to the even case; the exponential's one-sided support cannot be mirrored.

Solution. (a) $n(a,1)$; uniform$(a-c, a+c)$ $(c>0)$; $\frac{1}{2}e^{-|x-a|}$ (Laplace centered at a). (b) $\int_{-\infty}^a f(x)dx = \int_0^\infty f(a-u)du = \int_0^\infty f(a+u)du = \int_a^\infty f(x)dx = \frac{1}{2}$, so $P(X \le a), P(X \ge a) \ge \frac{1}{2}$: a is a median. (c) $EX - a = \int_{-\infty}^\infty (x-a)f(x)dx = \int_0^\infty u\,[f(a+u) - f(a-u)]\,du = 0$ (the split is valid since $E|X| < \infty$). (d) If symmetric about a, the support must mirror about a. For $a \ge 0$: $f(a+(a+1)) = e^{-(2a+1)} > 0$ but $f(a-(a+1)) = f(-1) = 0$. For $a < 0$: $f(a+(1-a)) = e^{-1} > 0$ but $f(a-(1-a)) = f(2a-1) = 0$ since $2a-1 < 0$. No a works. (e) Median $m = \log 2 \approx 0.69 < 1 =$ mean.

Exercise 2.27

Central idea. Symmetry mirrors the two monotone sides onto each other; a flat density has every point as a mode.

Solution. (a) $n(0,1)$, with unique mode 0. (b) uniform$(0,1)$: every $a \in [0,1]$ satisfies the mode inequalities (with equality), so the mode is not unique. (c) Let f be symmetric about s (Exercise 2.26) and unimodal with mode m; assume $m \ge s$ (the case $m < s$ is analogous). Since m is a mode, $f(m) = \max f$, and by symmetry $f(2s-m) = f(m)$, so $2s-m$ is also a mode; note $2s - m \le s \le m$. Because f is nondecreasing on $(-\infty, m]$, every $x \in [2s-m, m]$ satisfies $f(x) \ge f(2s-m) = \max f$, i.e. f is constantly the global maximum on $[2s-m, m] \ni s$. Now check the mode inequalities at s: $f(s) = \max f$ dominates every value. For $s \ge x \ge y$: if $x \ge 2s-m$ then $f(x) = \max f \ge f(y)$; if $x < 2s-m \le m$ then $y \le x$ both lie on $(-\infty, m]$, where f is nondecreasing, so $f(y) \le f(x)$. For $s \le x \le y$: if $x \ge m$ use that f is nonincreasing on $[m, \infty)$; if $x < m$ then $x \in (2s-m, m)$ (since $x \ge s > 2s-m$) and $f(x) = \max f \ge f(y)$. Thus s satisfies the mode inequalities: the point of symmetry is a mode. (d) For $0 \le x \le y$: $f(0) = 1 \ge e^{-x} \ge e^{-y}$; for $x \le 0$, $f(x) = 0 \le f(0)$. Hence unimodal with (unique) mode $a = 0$.

Exercise 2.28

Central idea. Odd central moments vanish under symmetry; α_4 is scale-free, so the normal's 3 is the benchmark.

Solution. (a) $\mu_3 = E[(X-a)^3] = \int_{-\infty}^\infty u^3 f(a+u)\,du = 0$, since u^3 is odd and $f(a+u)$ is even in u (assuming μ_3 exists). Hence $\alpha_3 = 0$. (b) For Exp(1): $EX = 1$, $\mu_2 = 1$, $\mu_3 = E[(X-1)^3] = E[X^3] - 3E[X^2] + 3EX - 1 = 6 - 6 + 3 - 1 = 2$. So $\alpha_3 = 2 > 0$, right-skewed. (c) Normal: $\mu_4 = 3$, $\mu_2 = 1$, so $\alpha_4 = 3$ (the benchmark). Uniform$(-1,1)$: $\mu_2 = 1/3$, $\mu_4 = \frac{1}{2}\int_{-1}^1 x^4dx = \frac{1}{5}$, so $\alpha_4 = \frac{1/5}{1/9} = \frac{9}{5} = 1.8 < 3$ (flatter top, lighter tails than normal). Laplace: $\mu_2 = 2$, $\mu_4 = 24$, so $\alpha_4 = 24/4 = 6 > 3$ (sharper peak, heavier tails than normal).

Exercise 2.29

Central idea. Factorial moments absorb the combinatorial coefficients: $x(x-1)\binom{n}{x} = n(n-1)\binom{n-2}{x-2}$. *Note on (c): as printed it is incorrect — with the leading factor $a(y+a)$ the "probabilities" do not sum to 1 (e.g. for $n=2, a=b=1$ they sum to $14/3$). This is a typesetting erratum: the intended factor is $\frac{a}{y+a}$, which sums to 1 and equals $\binom{n}{y}\frac{B(y+a,n-y+b)}{B(a,b)}$.*

Solution. (a) Binomial(n,p): $E[X(X-1)] = \sum_x x(x-1)\binom{n}{x}p^x(1-p)^{n-x} = n(n-1)p^2\sum_y \binom{n-2}{y}p^y(1-p)^{n-2-y} = n(n-1)p^2$. Poisson$(\lambda)$: $E[X(X-1)] = \sum_{x\geq 2}\frac{e^{-\lambda}\lambda^x}{(x-2)!} = \lambda^2$. (b) $\mathrm{Var}X = E[X(X-1)]+EX-(EX)^2$: binomial gives $n(n-1)p^2+np-n^2p^2 = np(1-p)$; Poisson gives $\lambda^2+\lambda-\lambda^2 = \lambda$. (c, corrected) In beta form the pmf is $\binom{n}{y}\frac{B(y+a,n-y+b)}{B(a,b)} = \binom{n}{y}\frac{(a)_y(b)_{n-y}}{(a+b)_n}$, with $(a)_y$ the rising factorial. Then, with $(Y)_r = Y(Y-1)\cdots(Y-r+1)$,

$$E[(Y)_r] = \sum_y \frac{y!}{(y-r)!}\binom{n}{y}\frac{(a)_y(b)_{n-y}}{(a+b)_n} = \frac{n!}{(n-r)!}\frac{(a)_r}{(a+b)_n}\sum_z \binom{n-r}{z}(a+r)_z(b)_{n-r-z},$$

using $(a)_y = (a)_r(a+r)_{y-r}$ and $z = y-r$. By Chu–Vandermonde the sum is $(a+b+r)_{n-r}$, and $\frac{(a+b+r)_{n-r}}{(a+b)_n} = \frac{1}{(a+b)_r}$, so $E[(Y)_r] = \frac{n!}{(n-r)!}\frac{(a)_r}{(a+b)_r}$. Hence $EY = n\frac{a}{a+b}$ and $E[Y(Y-1)] = n(n-1)\frac{a(a+1)}{(a+b)(a+b+1)}$, giving

$$\mathrm{Var}Y = n(n-1)\frac{a(a+1)}{(a+b)(a+b+1)} + n\frac{a}{a+b} - n^2\frac{a^2}{(a+b)^2} = \frac{nab\,(n+a+b)}{(a+b)^2(a+b+1)},$$

the usual beta-binomial variance (overdispersed relative to binomial$(n, \frac{a}{a+b})$ by the factor $\frac{n+a+b}{a+b+1}$).

Exercise 2.30

Central idea. Integrate (or sum) e^{tx} against the density; (c) is a shifted Laplace, (d) a negative-binomial series.

Solution. (a) $M(t) = \frac{1}{c}\int_0^c e^{tx}dx = \frac{e^{ct}-1}{ct}$ for $t\neq 0$ ($M(0)=1$). (b) $M(t) = \frac{2}{c^2}\int_0^c xe^{tx}dx = \frac{2}{c^2}\cdot\frac{e^{ct}(ct-1)+1}{t^2} = \frac{2(e^{ct}(ct-1)+1)}{c^2t^2}$. (c) Shifting by α: $M(t) = e^{\alpha t}\int_{-\infty}^{\infty}e^{tx}\frac{1}{2\beta}e^{-|x|/\beta}dx = e^{\alpha t}\cdot\frac{1}{1-\beta^2t^2}$, $|t|<1/\beta$. (d) $M(t) = p^r\sum_{x\geq 0}\binom{r+x-1}{x}[(1-p)e^t]^x = p^r(1-(1-p)e^t)^{-r} = \left(\frac{p}{1-(1-p)e^t}\right)^r$, for $(1-p)e^t<1$ (i.e. $t<-\log(1-p)$).

Exercise 2.31

Central idea. Every mgf satisfies $M_X(0) = E[e^0] = 1$; this candidate gives 0.

Solution. No. For any random variable, $M_X(0) = E[1] = 1$, but $t/(1-t)|_{t=0} = 0 \neq 1$. Hence no distribution has this mgf.

Exercise 2.32

Central idea. Logarithmic differentiation turns the mgf's derivatives into the mean and the variance directly.

Solution. Assume M_X exists in a neighborhood of 0. $S'(t) = M_X'(t)/M_X(t)$, so $S'(0) = M_X'(0)/M_X(0) = EX/1 = EX$. $S''(t) = \frac{M_X''(t)M_X(t)-(M_X'(t))^2}{M_X(t)^2}$, so $S''(0) = EX^2-(EX)^2 = \mathrm{Var}X$.

Exercise 2.33

Central idea. Sum/integrate e^{tx} against the law (recognizing the exponential series in (a), a geometric series in (b), and completing the square in (c)), then read off $M'(0)$, $M''(0)$.

Solution. (a) $M(t) = \sum_{x\geq 0}e^{tx}\frac{e^{-\lambda}\lambda^x}{x!} = e^{-\lambda}\sum\frac{(\lambda e^t)^x}{x!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}$. $M'(t) = \lambda e^tM(t)$, $M'(0)=\lambda$; $M''(t) = (\lambda e^t+\lambda^2e^{2t})M(t)$, $M''(0) = \lambda+\lambda^2$; $\mathrm{Var}X = \lambda$. (b) With $q = 1-p$: $M(t) = p\sum_{x\geq 0}(qe^t)^x = \frac{p}{1-qe^t}$

for $qe^t < 1$. $M'(t) = \frac{pqe^t}{(1-qe^t)^2}$, $M'(0) = \frac{q}{p}$; $M''(t) = \frac{pqe^t}{(1-qe^t)^2} + \frac{2pq^2e^{2t}}{(1-qe^t)^3}$, $M''(0) = \frac{q}{p} + \frac{2q^2}{p^2}$; $\mathrm{Var}X = \frac{q}{p} + \frac{q^2}{p^2} = \frac{q}{p^2} = \frac{1-p}{p^2}$. (c) $tx - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{(x-(\mu+\sigma^2 t))^2}{2\sigma^2} + \mu t + \frac{\sigma^2 t^2}{2}$, so $\int \frac{1}{\sqrt{2\pi}\sigma} e^{tx} e^{-(x-\mu)^2/(2\sigma^2)} dx = e^{\mu t + \sigma^2 t^2/2}$. $M'(0) = \mu$, $M''(0) = \sigma^2 + \mu^2$, $\mathrm{Var}X = \sigma^2$.

Exercise 2.34

Central idea. Symmetry kills the odd moments on both sides; the even moments are matched by solving $EY^2 = 1$, $EY^4 = 3$ with a symmetric three-point law.

Solution. $EX^r = 0$ for odd r by symmetry; $EX^2 = 1$, $EX^4 = 3$. For Y: $EY^r = \frac{1}{6}(\sqrt{3})^r + \frac{1}{6}(-\sqrt{3})^r + \frac{2}{3}\cdot 0^r$, which is 0 for odd r; $EY^2 = \frac{1}{6}\cdot 3 + \frac{1}{6}\cdot 3 = 1$; $EY^4 = \frac{1}{6}\cdot 9 + \frac{1}{6}\cdot 9 = 3$. Hence $EX^r = EY^r$ for $r = 1, \dots, 5$.

Exercise 2.35

Central idea. f_1 is the law of e^Z, $Z \sim n(0,1)$, so moments are normal mgf values; the substitution $y = \log x - r$ turns the perturbation integral into an odd function over $\mathbb{R}$.

Solution. (a) $f_1(x) = \frac{1}{\sqrt{2\pi}x} e^{-(\log x)^2/2}$ $(0 \le x < \infty)$ is the pdf of $X_1 = e^Z$ with $Z \sim n(0,1)$. Hence $EX_1^r = E[e^{rZ}] = M_Z(r) = e^{r^2/2} < \infty$ for all $r \ge 0$. (b) With $y = \log x - r$ ($x = e^{y+r}$, $dx/x = dy$):

$$x^r f_1(x)\,dx = \frac{1}{\sqrt{2\pi}} e^{r(y+r)} e^{-(y+r)^2/2} dy = \frac{e^{r^2/2}}{\sqrt{2\pi}} e^{-y^2/2} dy,$$

and, r being an integer, $\sin(2\pi \log x) = \sin(2\pi(y+r)) = \sin(2\pi y)$. Thus

$$\int_0^\infty x^r f_1(x) \sin(2\pi \log x)\,dx = e^{r^2/2} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{-y^2/2} \sin(2\pi y)\,dy = 0,$$

the integrand being odd. Since $f_2(x) = f_1(x)[1 + \sin(2\pi \log x)]$, $EX_2^r = EX_1^r + 0 = EX_1^r$ for all r.

Exercise 2.36

Central idea. An mgf must converge in a two-sided neighborhood of 0; for $t > 0$ the factor e^{tx} eventually dominates the Gaussian-in-$\log x$ decay.

Solution. With $x = e^y$ ($dx/x = dy$), $M_X(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty e^{te^y} e^{-y^2/2} dy$. For fixed $t > 0$, $te^y - y^2/2 \to +\infty$ as $y \to \infty$ (the exponential dominates the quadratic), so the integrand is eventually ≥ 1 and the integral diverges. Hence $E[e^{tX}] = \infty$ for every $t > 0$; since an mgf must be finite on some interval $(-h, h)$ with $h > 0$, the lognormal has no mgf — despite having all moments finite.

Exercise 2.37

Central idea. The oscillatory perturbation $\frac{1}{2}\sin(2\pi x)$ is large on the density scale but its mgf contribution picks up the factor $e^{-2\pi^2} \approx 2.7 \times 10^{-9}$ — invisible numerically, present analytically.

Solution. In Miscellanea 2.6.3, f_1 is the $n(0,1)$ pdf and $f_2(x) = f_1(x)[1 + \frac{1}{2}\sin(2\pi x)]$, with $K_1(t) = t^2/2$ and $K_2(t) = K_1(t) + \log\left(1 + \frac{1}{2}e^{-2\pi^2}\sin(2\pi t)\right)$. (a) The densities differ visibly: f_2 oscillates around f_1 with relative amplitude up to 50% and period 1:

(b) The cgfs are indistinguishable to the eye — $\max_t |K_2(t) - K_1(t)| \le \log\left(1 + \frac{1}{2}e^{-2\pi^2}\right) \approx 1.34 \times 10^{-9}$:

(c) $M_1(t) = e^{t^2/2}$. Since $\int e^{tx}\phi(x)\sin(2\pi x)dx = \mathrm{Im}\, e^{(t+2\pi i)^2/2} = e^{t^2/2} e^{-2\pi^2}\sin(2\pi t)$,

$$M_2(t) = e^{t^2/2}\left(1 + \tfrac{1}{2}e^{-2\pi^2}\sin(2\pi t)\right).$$

Analytically different, numerically identical (relative difference $\le 1.34 \times 10^{-9}$).

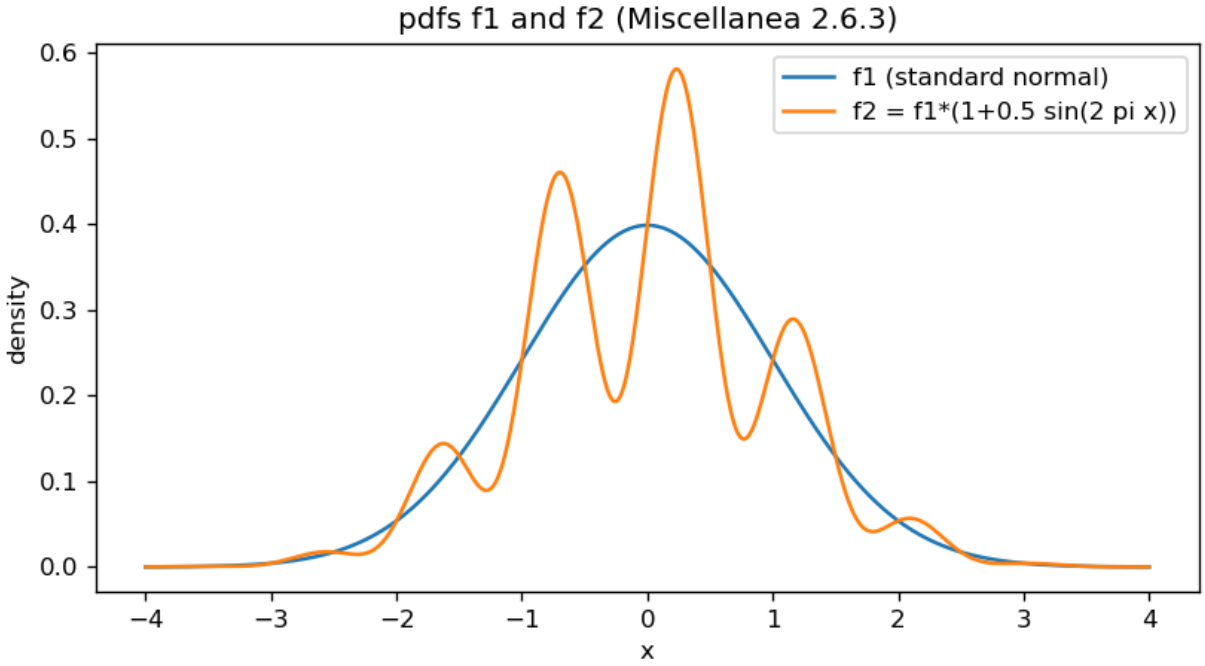


Figure 1: pdfs f1 and f2

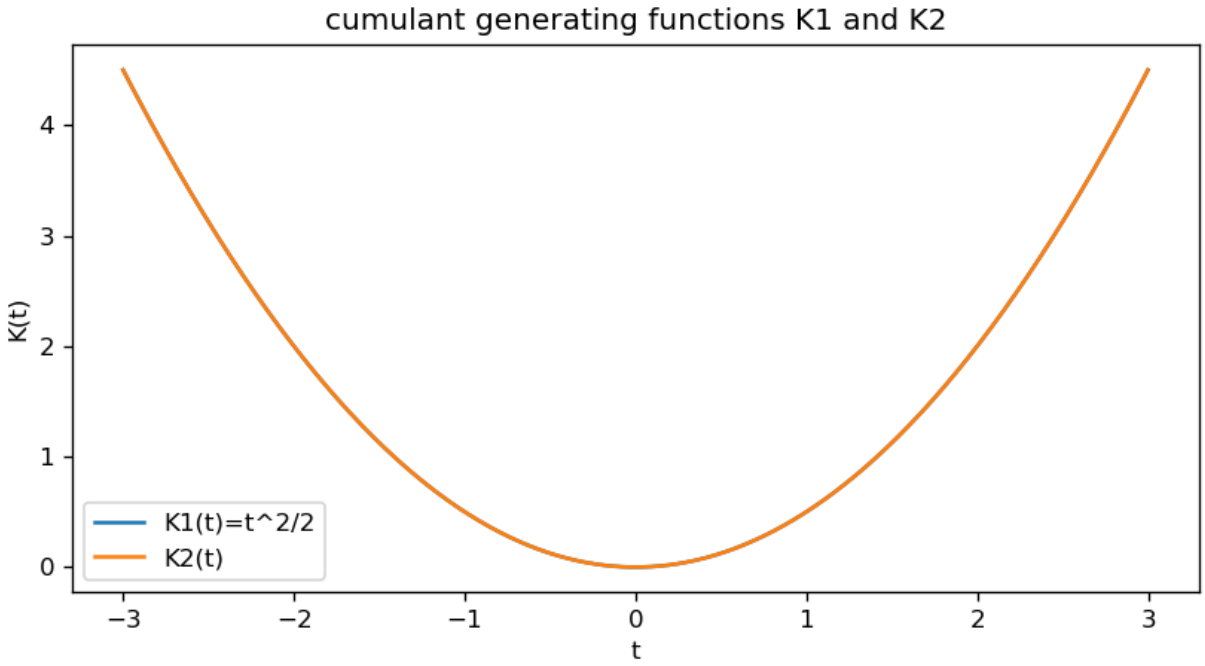


Figure 2: cgfs K1 and K2

(d) Same construction as Example 2.3.10: $f_2 = f_1 \cdot (1+\text{oscillation})$ with the oscillation tuned so a transform cannot see it. In 2.3.10 the sine is in $\log x$, making all polynomial moments exactly equal; here the sine is in x, making the mgfs exponentially close but not equal. Both show that moments/mgfs can fail — exactly or numerically — to distinguish visibly different densities.

Exercise 2.38

Central idea. Scale the negative binomial so its mean stays fixed as $p \to 0$; the mgf limit is read off termwise.

Solution. (a) From Exercise 2.30(d), $M_X(t) = \left(\frac{p}{1-(1-p)e^t}\right)^r$ for $t < -\log(1-p)$. (b) $M_Y(t) = M_X(2pt) = \left(\frac{p}{1-(1-p)e^{2pt}}\right)^r = \left(\frac{1}{\frac{1-e^{2pt}}{p}+e^{2pt}}\right)^r$. As $p \downarrow 0$, $\frac{1-e^{2pt}}{p} \to -2t$ and $e^{2pt} \to 1$, so the bracket tends to $\frac{1}{1-2t}$ for $|t| < 1/2$ (which also keeps $M_X(2pt)$ defined for small p). Hence $\lim_{p\to 0} M_Y(t) = \left(\frac{1}{1-2t}\right)^r$, the mgf of χ^2_{2r}.

Exercise 2.39

Central idea. (a) and (c) are the Fundamental Theorem of Calculus; (b) and (d) differentiate under the integral, justified by an integrable dominating derivative.

Solution. (a) $e^{-\lambda x}$, by FTC (integrand continuous). (b) For $\lambda \geq \delta > 0$, $\left|\frac{\partial}{\partial\lambda} e^{-\lambda t}\right| = te^{-\lambda t} \leq te^{-\delta t}$, integrable on $[0,\infty)$; hence $\frac{d}{d\lambda}\int_0^\infty e^{-\lambda t}dt = \int_0^\infty (-t)e^{-\lambda t}dt = -\frac{1}{\lambda^2}$ (also $\frac{d}{d\lambda}\frac{1}{\lambda}$). (c) $\int_t^1 x^{-2}dx = \frac{1}{t} - 1$ $(t > 0)$, so the derivative is $-\frac{1}{t^2}$ (FTC directly). (d) For $t < 1$, substitute $u = x - t$: $\int_1^\infty \frac{dx}{(x-t)^2} = \int_{1-t}^\infty u^{-2}du = \frac{1}{1-t}$, so the derivative is $\frac{1}{(1-t)^2}$. Justification: for $t \leq t_0 < 1$, $\left|\frac{\partial}{\partial t}(x-t)^{-2}\right| = \frac{2}{(x-t)^3} \leq \frac{2}{(x-t_0)^3}$, integrable on $[1,\infty)$; the integral diverges for $t \geq 1$.

Exercise 2.40

Central idea. Both sides vanish at $p = 1$ (for $x < n$) and have the same derivative: differentiating the binomial cdf telescopes to a single term matching the integral's derivative. *Note: as stated the identity needs $x < n$ — for $x = n$ the left side is 1 while the improper integral on the right diverges for $p < 1$.*

Solution. Let $L(p) = \sum_{k=0}^x \binom{n}{k} p^k(1-p)^{n-k}$ and $R(p) = (n-x)\binom{n}{x}\int_0^{1-p} t^{n-x-1}(1-t)^x dt$, for $0 \leq x < n$. Differentiating under the integral,

$$R'(p) = -(n-x)\binom{n}{x}(1-p)^{n-x-1}p^x.$$

For L, using $k\binom{n}{k} = n\binom{n-1}{k-1}$ and $(n-k)\binom{n}{k} = n\binom{n-1}{k}$,

$$\begin{aligned} L'(p) &= \sum_{k=0}^{x}\binom{n}{k}\left[kp^{k-1}(1-p)^{n-k} - (n-k)p^k(1-p)^{n-k-1}\right] \\ &= n\sum_{j=0}^{x-1}\binom{n-1}{j}p^j(1-p)^{n-1-j} - n\sum_{j=0}^{x}\binom{n-1}{j}p^j(1-p)^{n-1-j} \\ &= -n\binom{n-1}{x}p^x(1-p)^{n-1-x} = -(n-x)\binom{n}{x}p^x(1-p)^{n-x-1} = R'(p), \end{aligned}$$

the two sums telescoping (with $j = k-1$ in the first). Since $L(1) = P(\text{Bin}(n,1) \leq x) = 0 = R(1)$ for $x < n$, $L \equiv R$. (For $x = n$ and $p < 1$, the integral has a t^{-1} singularity at zero, so the displayed right side is an undefined $0 \cdot \infty$, not zero. The identity requires integer $0 \leq x \leq n-1$.)

Chapter 3

Exercise 3.1

Central idea. Shift to $\{0, 1, \dots, m\}$ with $m = N_1 - N_0$; the mean is the midpoint by symmetry, and the variance follows from $\sum_{k=0}^{m} k^2 = m(m+1)(2m+1)/6$.

Solution. $P(X = x) = 1/(m+1)$ on $m+1$ points, where $m = N_1 - N_0$. By symmetry,

$$EX = \frac{1}{m+1}\sum_{k=0}^{m}(N_0 + k) = N_0 + \frac{m}{2} = \frac{N_0 + N_1}{2}.$$

$\mathrm{Var}X = \mathrm{Var}(X - N_0)$, and $E[(X - N_0)^2] = \frac{1}{m+1} \cdot \frac{m(m+1)(2m+1)}{6} = \frac{m(2m+1)}{6}$, so

$$\mathrm{Var}\, X = \frac{m(2m+1)}{6} - \frac{m^2}{4} = \frac{m(m+2)}{12} = \frac{(N_1 - N_0 + 1)^2 - 1}{12}.$$

Exercise 3.2

Central idea. The worst case is the smallest unacceptable defect count ($D = 6$): the acceptance probability decreases as D grows, so bound it at $D = 6$ with the hypergeometric pmf.

Solution. Given D defectives, the number of defectives in the sample is hypergeometric$(100, D, K)$, and $P(\text{accept} \mid D)$ decreases in D (adding a defective to the lot can only increase the sample's defective count). Hence it suffices to require $P(\text{accept} \mid D = 6) < 0.10$. (a) $P(\text{accept} \mid D = 6) = \binom{94}{K}\big/\binom{100}{K}$. This equals 0.1006 at $K = 31$ and 0.0918 at $K = 32$, so $K = 32$. (b) $P(\text{accept} \mid D = 6) = \left[\binom{94}{K} + 6\binom{94}{K-1}\right]\big/\binom{100}{K}$. This equals 0.1022 at $K = 50$ and 0.0933 at $K = 51$, so $K = 51$.

Exercise 3.3

Central idea. "Wait exactly four seconds" means seconds 5, 6, 7 are the first three consecutive car-free seconds; this pins down disjoint blocks of seconds, and independence does the rest.

Solution. Let $p = P(\text{car in a given second})$. Waiting exactly 4 seconds means the checks at times $0, 1, 2, 3$ all fail (a car somewhere in the next three seconds) but the check at time 4 succeeds. Given no car in seconds $\{5, 6, 7\}$: the time-3 check fails iff there is a car in second 4 (which then also blocks the time-1 and time-2 checks), and the time-0 check fails iff there is a car in $\{1, 2, 3\}$. These events involve the disjoint second-sets $\{4\}$, $\{1, 2, 3\}$, $\{5, 6, 7\}$, hence are independent, and

$$P(\text{wait exactly 4 s}) = p \cdot (1 - (1-p)^3) \cdot (1-p)^3.$$

Exercise 3.4

Central idea. With replacement the trials are iid Bernoulli$(1/n)$ (geometric waiting time); without replacement the correct key is equally likely to sit in any position.

Solution. (a) Each trial succeeds independently with probability $1/n$, so the number of trials $N \sim$ geometric$(1/n)$ and $EN = n$. (b) Without replacement, the correct key is equally likely to be the 1st, 2nd, ..., nth key tried, i.e. uniform on $\{1, \dots, n\}$, so $EN = (n+1)/2$.

Exercise 3.5

Central idea. Under "equally effective", $X \sim$ Binomial$(100, 0.8)$; the p-value is the upper tail $P(X \geq 85)$.

Solution. Under the null, $P(X \geq 85) = \sum_{k=85}^{100}\binom{100}{k}(0.8)^k(0.2)^{100-k} \approx 0.1285$ (normal approximation: $\mu = 80$, $\sigma = 4$, $P \approx 1 - \Phi(1.125) \approx 0.130$). Since $0.1285 > 0.05$, 85 successes is well within the range of chance under the null — there is no convincing evidence that the new drug is superior.

Exercise 3.6

Central idea. Survivors $\approx$ Binomial$(2000, 0.01)$; with n large and p small, Poisson(20) is the practical approximation — and 100 is ≈ 18 standard deviations above the mean.

Solution. (a) $X \sim$ Binomial$(2000, 0.01)$ (each insect survives independently with probability 0.01), well approximated by Poisson$(\lambda = 20)$. (b) $P(X < 100) = \sum_{k=0}^{99} \binom{2000}{k}(0.01)^k(0.99)^{2000-k}$. (c) Poisson approximation: $\sum_{k=0}^{99} e^{-20}20^k/k! \approx 1$ to all practical precision (99 is about 17.7 standard deviations above the mean 20). So it is essentially certain that fewer than 100 survive — as expected, since $EX = 20$.

Exercise 3.7

Central idea. $P(X \geq 2) = 1 - e^{-\lambda}(1 + \lambda)$ is increasing in λ; invert it at 0.99.

Solution. $1-e^{-\lambda}(1+\lambda) > 0.99 \iff e^{-\lambda}(1+\lambda) < 0.01$. The left side decreases in λ; solving $e^{-\lambda}(1+\lambda) = 0.01$ gives $\lambda \approx 6.64$ (at $\lambda = 6.6$ the left side is 0.01034; at $\lambda = 6.64$ it is 0.00999). The smallest mean ensuring the requirement is $\lambda \approx 6.64$.

Exercise 3.8

Central idea. $X \sim$ Binomial$(1000, 1/2)$ counts theater-A choosers; "turn away" is $\{X > N\}$; invert the normal tail at 0.01.

Solution. (a) $P(X > N) = \sum_{k=N+1}^{1000} \binom{1000}{k}2^{-1000} < 0.01$; the required N is the smallest integer satisfying this. (b) With $\mu = 500$, $\sigma = \sqrt{250} \approx 15.81$: $P(X > N) \approx 1 - \Phi((N + 0.5 - 500)/15.81) < 0.01$ requires $(N-499.5)/15.81 \geq 2.326$, i.e. $N \geq 536.3$, so $N = 537$ (check: $1-\Phi(37.5/15.81) = 1-\Phi(2.372) \approx 0.0089 < 0.01$).

Exercise 3.9

Central idea. One school is a Binomial$(60, 1/90)$ tail event — rare but not absurd; multiply the opportunities (schools $\times$ counties $\times$ states $\times$ years) and the rarity evaporates: the law of truly large numbers.

Solution. (a) Treat each of the 60 entering children as an (approximately) independent twin-birth trial with $p = 1/90$; five sets of twins ≈ 5 or more successes. $P(X \geq 5) = \sum_{k=5}^{60} \binom{60}{k}(1/90)^k(89/90)^{60-k} \approx 5.6 \times 10^{-4} \approx 1/1800$ (Poisson$(2/3)$ approximation: 6.3×10^{-4}). For one pre-specified school this is rare — arguably newsworthy locally — but not a "statistical impossibility." (b) With $62 \times 5 = 310$ schools, $P(\text{some school has } \geq 5) = 1 - (1-p)^{310} \approx 0.16$. About a 1-in-6 chance somewhere in the state — hardly impossible. (c) With $310 \times 50 \times 10 = 155{,}000$ school-years, $1-(1-p)^{155000} \approx 1-e^{-86} \approx 1$: virtually certain to occur somewhere. So no — it is exactly the law of truly large numbers: with enough opportunities, even 1-in-1800 events are expected to happen.

Exercise 3.10

Central idea. Sequential sampling without replacement: the joint probability is the ratio of favorable ordered selections to all ordered selections (a hypergeometric ratio); the "innocence probability" is then maximized numerically over $N + M = 496$.

Solution. (a) Choose the 4 tested packets from the N cocaine packets and the 2 sold packets from the M non-cocaine packets, out of all ways to draw 4 then 2 from $N + M = 496$ packets: $\binom{N}{4}\binom{M}{2} \Big/ \left[\binom{496}{4}\binom{492}{2}\right]$. (b) With $M = 496 - N$ the denominator is constant, so maximize $g(N) = \binom{N}{4}\binom{496-N}{2}$. Numerically g peaks at $N = 331$ ($M = 165$), giving $0.02208 \approx 0.022$ (neighbors: $N = 330$ gives 0.022081, $N = 332$ gives 0.022080 — 331 is the unique maximizer).

Exercise 3.11

Central idea. Write the hypergeometric pmf as $\binom{K}{x}$ times a ratio of falling factorials; each fixed-length product tends to 1 while the powers tend to $p^x(1-p)^{K-x}$; the Poisson step is the ordinary binomial-to-Poisson limit (or Lemma 2.3.14 applied directly).

Solution. Write $(a)_b = a(a-1)\cdots(a-b+1)$. Then

$$P(X = x \mid N, M, K) = \binom{K}{x} \frac{(M)_x \, (N-M)_{K-x}}{(N)_K},$$

since $\binom{M}{x}\binom{N-M}{K-x}\big/\binom{N}{K} = \binom{K}{x}(M)_x(N-M)_{K-x}/(N)_K$. (a) $\frac{(M)_x(N-M)_{K-x}}{(N)_K} = \frac{M^x(N-M)^{K-x}}{N^K} \cdot \frac{\prod_{i<x}(1-i/M)\prod_{j<K-x}(1-j/(N-M))}{\prod_{l<K}(1-l/N)} \to p^x(1-p)^{K-x}$, because $M/N \to p$ and each product has a fixed number of factors, each tending to 1. (b) By (a), for $K \to \infty$ the hypergeometric tends to Binomial$(K, M/N)$; with $M/N \to 0$ and $K(M/N) \to \lambda$, the Poisson limit theorem gives $\binom{K}{x}(M/N)^x(1-M/N)^{K-x} \to e^{-\lambda}\lambda^x/x!$. (c) Directly: for fixed x, $\binom{K}{x} = \frac{K^x}{x!}(1+o(1))$ and $\frac{(M)_x}{(N)_x} = \frac{M^x}{N^x}(1+o(1))$, while, with $(N)_K = (N)_x(N-x)_{K-x}$,

$$\frac{(N-M)_{K-x}}{(N-x)_{K-x}} = \prod_{j=0}^{K-x-1}\left(1 - \frac{M-x}{N-x-j}\right) \to e^{-\lambda}$$

by Lemma 2.3.14 (the $K-x$ terms each tend to 0 and sum to $\frac{(K-x)(M-x)}{N-x} \to \lambda$). Multiplying gives $\frac{K^x}{x!}\frac{M^x}{N^x}e^{-\lambda} \to \frac{\lambda^x}{x!}e^{-\lambda}$.

Exercise 3.12

Central idea. $\{X \le r-1\}$ = "fewer than r successes in n trials" = "the rth success comes after trial n" = $\{Y > n-r\}$, where Y counts failures before the rth success.

Solution. $F_X(r-1) = P(X \le r-1)$, where X is the number of successes in n trials. $X \le r-1$ iff the rth success occurs after the nth trial, iff the number Y of failures before the rth success exceeds $n-r$. Hence $P(X \le r-1) = P(Y > n-r) = 1 - F_Y(n-r)$.

Exercise 3.13

Central idea. Truncation just renormalizes by $P(X>0)$; moments of X_T are moments of X divided by $P(X>0)$.

Solution. (a) $P(X_T = x) = \dfrac{e^{-\lambda}\lambda^x/x!}{1-e^{-\lambda}}$, $x = 1, 2, \ldots$. $EX_T = \dfrac{\lambda}{1-e^{-\lambda}}$; $E[X_T^2] = \dfrac{\lambda+\lambda^2}{1-e^{-\lambda}}$, so $\operatorname{Var} X_T = \dfrac{\lambda+\lambda^2}{1-e^{-\lambda}} - \dfrac{\lambda^2}{(1-e^{-\lambda})^2} = \dfrac{\lambda}{1-e^{-\lambda}}\left(1+\lambda - \dfrac{\lambda}{1-e^{-\lambda}}\right)$. (b) $P(X_T = x) = \dfrac{\binom{r+x-1}{x}p^r(1-p)^x}{1-p^r}$, $x = 1, 2, \ldots$. $EX_T = \dfrac{r(1-p)/p}{1-p^r}$; with $E[X^2] = \dfrac{r(1-p)}{p^2} + \dfrac{r^2(1-p)^2}{p^2}$,

$$\operatorname{Var} X_T = \frac{r(1-p)/p^2 + r^2(1-p)^2/p^2}{1-p^r} - \frac{r^2(1-p)^2/p^2}{(1-p^r)^2}.$$

Exercise 3.14

Central idea. $\sum_{x\ge1}(1-p)^x/x = -\log p$ (Mercator series) makes the terms sum to 1; moments come from geometric-series sums $\sum xq^x = q/(1-q)^2$.

Solution. (a) Terms are positive since $\log p < 0$; $\sum_{x=1}^{\infty}\frac{(1-p)^x}{x} = -\log(1-(1-p)) = -\log p$, so $\sum_{x\ge1}P(X=x) = \frac{-\log p}{-\log p} = 1$. (b) $EX = \frac{-1}{\log p}\sum_{x\ge1}(1-p)^x = \frac{-(1-p)}{p\log p}$; $E[X^2] = \frac{-1}{\log p}\sum_{x\ge1}x(1-p)^x = \frac{-(1-p)}{p^2\log p}$. Hence

$$\operatorname{Var} X = \frac{-(1-p)}{p^2\log p} - \frac{(1-p)^2}{p^2(\log p)^2} = \frac{-(1-p)}{p^2\log p}\left(1 + \frac{1-p}{\log p}\right) > 0,$$

positive because $\log p < p - 1$ for $0 < p < 1$.

Exercise 3.15

Central idea. Write the NB mgf as $\left[1 - \frac{r(1-p)}{rp}(e^t - 1)\right]^{-r}$ and use $(1 + a_r/r)^{-r} \to e^{-a}$.

Solution. $M_Y(t) = \left(\frac{p}{1-(1-p)e^t}\right)^r = \left[1 - \frac{1-p}{p}(e^t - 1)\right]^{-r}$ for $t < -\log(1-p)$. Since $r(1-p) \to \lambda$ and $p \to 1$, $\frac{r(1-p)}{p}(e^t - 1) \to \lambda(e^t - 1)$, so

$$M_Y(t) = \left[1 - \frac{\lambda(e^t - 1) + o(1)}{r}\right]^{-r} \longrightarrow \exp\left(\lambda(e^t - 1)\right),$$

the Poisson(λ) mgf, for each fixed t.

Exercise 3.16

Central idea. (a) is one integration by parts; (b) is the Gaussian integral in disguise ($u = \sqrt{x}$).

Solution. (a) $\Gamma(\alpha + 1) = \int_0^\infty x^\alpha e^{-x} dx = \left[-x^\alpha e^{-x}\right]_0^\infty + \alpha \int_0^\infty x^{\alpha-1} e^{-x} dx = \alpha\Gamma(\alpha)$. (b) $\Gamma(\frac{1}{2}) = \int_0^\infty x^{-1/2} e^{-x} dx$; with $u = \sqrt{x}$ ($dx = 2u\, du$), $= 2\int_0^\infty e^{-u^2} du = \int_{-\infty}^\infty e^{-u^2} du = \sqrt{\pi}$.

Exercise 3.17

Central idea. x^v times the gamma kernel is another gamma kernel; the normalizing constants give the ratio of gamma functions.

Solution. $EX^v = \frac{1}{\Gamma(\alpha)\beta^\alpha} \int_0^\infty x^{\alpha+v-1} e^{-x/\beta} dx = \frac{\Gamma(\alpha+v)\beta^{\alpha+v}}{\Gamma(\alpha)\beta^\alpha} = \frac{\beta^v \Gamma(v+\alpha)}{\Gamma(\alpha)}$, valid for $v > -\alpha$ (in particular for $v > 0$).

Exercise 3.18

Central idea. $M_{pY}(t) = M_Y(pt)$; as $p \to 0$, $e^{pt} = 1 + pt + o(p)$ collapses the NB mgf to $(1-t)^{-r}$.

Solution. $M_{pY}(t) = E[e^{tpY}] = M_Y(pt) = \left(\frac{p}{1-(1-p)e^{pt}}\right)^r$. Now $1 - (1-p)e^{pt} = 1 - (1-p)(1 + pt + o(p)) = p(1-t) + o(p)$, so $\frac{p}{p(1-t)+o(p)} \to \frac{1}{1-t}$ (for $t < 1$), and $M_{pY}(t) \to (1-t)^{-r}$, the mgf of gamma$(r, 1)$.

Exercise 3.19

Central idea. Integrate by parts to peel off one term of the Poisson sum at a time; probabilistically it is Poisson-process duality: the αth arrival comes after x iff fewer than α arrivals occurred by time x.

Solution. Let $I_\alpha(x) = \int_x^\infty \frac{z^{\alpha-1}e^{-z}}{\Gamma(\alpha)} dz$. Parts ($u = z^{\alpha-1}/\Gamma(\alpha)$, $dv = e^{-z}dz$): $I_\alpha(x) = \frac{x^{\alpha-1}e^{-x}}{(\alpha-1)!} + I_{\alpha-1}(x)$; $I_1(x) = e^{-x}$. Induction gives the sum. Probabilistic: if $Z \sim$ gamma$(\alpha, 1)$ is the waiting time for the αth event in a unit-rate Poisson process and $Y \sim$ Poisson(x) counts events by time x, both sides equal $P(Z > x) = P(Y \le \alpha - 1)$.

Exercise 3.20

Central idea. Folded normal $= |Z|$ for $Z \sim n(0, 1)$; squaring it gives $Z^2 \sim \chi^2_1 =$ gamma$(1/2, 2)$.

Solution. (a) $X \stackrel{d}{=} |Z|$: $EX = E|Z| = 2\int_0^\infty \frac{z}{\sqrt{2\pi}} e^{-z^2/2} dz = \sqrt{2/\pi}$. $E[X^2] = E[Z^2] = 1$, so $\mathrm{Var}X = 1 - 2/\pi$. (b) Take $g(x) = x^2$: $Y = X^2 \stackrel{d}{=} Z^2 \sim \chi^2_1$, whose pdf $\frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2} = \frac{1}{\Gamma(1/2)2^{1/2}} y^{-1/2} e^{-y/2}$ is gamma$(1/2, 2)$. So $\alpha = 1/2$, $\beta = 2$.

Exercise 3.21

Central idea. The mgf integrand is $e^{tx}/(1+x^2)$; for $t \neq 0$ the exponential beats the polynomial denominator on one tail — no surprise, since the Cauchy has no mean.

Solution. $M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{dx}{\pi(1+x^2)}$. For $t > 0$, $e^{tx}/(1+x^2) \to \infty$ as $x \to \infty$, so the integral diverges to $+\infty$; for $t < 0$ it diverges as $x \to -\infty$. At $t = 0$, $M_X(0) = 1$. So the integral is finite only at $t = 0$ — expected, since the Cauchy distribution has no mean, hence no mgf on any interval about 0.

Exercise 3.22

Central idea. Factorial moments $E[X(X-1)]$ (discrete) and gamma/beta integrals (continuous) give EX^2; subtract $(EX)^2$.

Solution. (a) $E[X(X-1)] = \sum_{x \geq 2} \frac{e^{-\lambda}\lambda^x}{(x-2)!} = \lambda^2$; $EX^2 = \lambda^2 + \lambda$; $\mathrm{Var}X = \lambda$. (b) $x(x-1)\binom{r+x-1}{x} = r(r+1)\binom{r+x-1}{x-2}$, so $E[X(X-1)] = r(r+1)(1-p)^2 p^{-2} \sum_{y \geq 0} \binom{r+y+1}{y} p^{r+2}(1-p)^y = \frac{r(r+1)(1-p)^2}{p^2}$ (the sum is a negative binomial$(r+2, p)$ pmf). Then $\mathrm{Var}X = \frac{r(r+1)(1-p)^2}{p^2} + \frac{r(1-p)}{p} - \frac{r^2(1-p)^2}{p^2} = \frac{r(1-p)}{p^2}$. (c) $EX = \alpha\beta$, $EX^2 = \beta^2(\alpha+1)\alpha$ by (3.17); $\mathrm{Var}X = \alpha\beta^2$. (d) $EX = \frac{B(\alpha+1,\beta)}{B(\alpha,\beta)} = \frac{\alpha}{\alpha+\beta}$; $EX^2 = \frac{(\alpha+1)\alpha}{(\alpha+\beta+1)(\alpha+\beta)}$; $\mathrm{Var}X = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$. (e) $EX = \mu$ by symmetry; $\mathrm{Var}X = E[(X-\mu)^2] = \int_0^{\infty} \frac{u^2}{\sigma} e^{-u/\sigma} du = 2\sigma^2$.

Exercise 3.23

Central idea. Power integrals $\int_{\alpha}^{\infty} x^{-k} dx$; moments exist only when the exponent beats 1.

Solution. (a) $\int_{\alpha}^{\infty} \frac{\beta\alpha^\beta}{x^{\beta+1}} dx = \beta\alpha^\beta \left[\frac{x^{-\beta}}{-\beta}\right]_{\alpha}^{\infty} = 1$. (b) $EX = \beta\alpha^\beta \int_{\alpha}^{\infty} x^{-\beta} dx = \frac{\beta\alpha}{\beta-1}$ $(\beta > 1)$; $EX^2 = \frac{\beta\alpha^2}{\beta-2}$ $(\beta > 2)$; $\mathrm{Var}X = \frac{\beta\alpha^2}{\beta-2} - \frac{\beta^2\alpha^2}{(\beta-1)^2} = \frac{\beta\alpha^2}{(\beta-1)^2(\beta-2)}$ $(\beta > 2)$. (c) For $\beta \leq 2$, $EX^2 = \int_{\alpha}^{\infty} \beta\alpha^\beta x^{1-\beta} dx = \infty$ (since $1 - \beta \geq -1$), so the variance does not exist (it is infinite).

Exercise 3.24

Central idea. Each is a smooth monotone transform of a gamma/exponential variable: push the pdf through the change of variables, then read off mean/variance from gamma integrals.

Solution. (a) $x = y^\gamma$: $f_Y(y) = \frac{1}{\beta} e^{-y^\gamma/\beta} \gamma y^{\gamma-1}$, $y > 0$ (integrates to 1 via $u = y^\gamma/\beta$). $EY = \beta^{1/\gamma}\Gamma(1+1/\gamma)$, $\mathrm{Var}Y = \beta^{2/\gamma}[\Gamma(1+2/\gamma) - \Gamma(1+1/\gamma)^2]$. (b) $x = \beta y^2/2$: $f_Y(y) = \frac{1}{\beta} e^{-y^2/2} \beta y = y e^{-y^2/2}$, $y > 0$ (Rayleigh; note β cancels). $EY = \sqrt{\pi/2}$, $EY^2 = 2$, $\mathrm{Var}Y = 2 - \pi/2$. (c) $x = 1/y$: $f_Y(y) = \frac{1}{\Gamma(a) b^a y^{a+1}} e^{-1/(by)}$, $y > 0$ (integrates to 1 via $u = 1/(by)$). $EY = \frac{1}{b(a-1)}$ $(a > 1)$, $\mathrm{Var}Y = \frac{1}{b^2(a-1)^2(a-2)}$ $(a > 2)$. (d) $x = \beta y^2$: $f_Y(y) = \frac{4}{\sqrt{\pi}} y^2 e^{-y^2}$, $y > 0$ (Maxwell; β cancels). $EY = 2/\sqrt{\pi}$, $EY^2 = 3/2$, $\mathrm{Var}Y = 3/2 - 4/\pi$. (e) $x = e^{(\alpha-y)/\gamma}$: $f_Y(y) = \frac{1}{\gamma} \exp\!\left(-\frac{y-\alpha}{\gamma} - e^{-(y-\alpha)/\gamma}\right)$, $y \in \mathbb{R}$ (Gumbel). $EY = \alpha + \gamma\eta$ ($\eta \approx 0.5772$ Euler–Mascheroni, since $E \log X = -\eta$ for Exp(1)), $\mathrm{Var}Y = \pi^2\gamma^2/6$.

Exercise 3.25

Central idea. The conditional probability is a difference quotient of F_T; the limit is $f_T(t)/(1 - F_T(t))$, which is $-\frac{d}{dt}\log(1 - F_T(t))$ by the chain rule.

Solution. $P(t \leq T < t + \delta \mid T \geq t) = \frac{F_T(t+\delta) - F_T(t)}{1 - F_T(t)}$. Dividing by δ and letting $\delta \downarrow 0$ gives $h_T(t) = \frac{f_T(t)}{1 - F_T(t)} = -\frac{d}{dt}\log(1 - F_T(t))$.

Exercise 3.26

Central idea. $h_T = f_T/(1 - F_T)$; for each family compute $1 - F_T$ and simplify.

Solution. (a) $f = \frac{1}{\beta}e^{-t/\beta}$, $1-F = e^{-t/\beta}$: $h_T = 1/\beta$ (memoryless $\iff$ constant hazard). (b) $f = \frac{\gamma}{\beta}t^{\gamma-1}e^{-t^\gamma/\beta}$, $1-F = e^{-t^\gamma/\beta}$: $h_T = (\gamma/\beta)t^{\gamma-1}$. (c) $f_T = F_T' = \frac{e^{-(t-\mu)/\beta}}{\beta(1+e^{-(t-\mu)/\beta})^2}$, $1-F_T = \frac{e^{-(t-\mu)/\beta}}{1+e^{-(t-\mu)/\beta}}$: $h_T = \frac{f_T}{1-F_T} = \frac{1}{\beta(1+e^{-(t-\mu)/\beta})} = \frac{1}{\beta}F_T(t)$.

Exercise 3.27

Central idea. Log-concavity (or the sign of $(\log f)'$) locates the mode; the beta with $\alpha, \beta < 1$ is U-shaped — the counterexample.

Solution. (Unimodal = nondecreasing then nonincreasing.) (a) Uniform: constant on (a,b), 0 outside — nondecreasing up to any $x^* \in [a,b]$, nonincreasing after: unimodal (flat top). (b) Gamma: $\frac{d}{dx}\log f = \frac{\alpha-1}{x} - \frac{1}{\beta}$; for $\alpha \geq 1$ increases to the mode $(\alpha-1)\beta$ then decreases; for $\alpha < 1$ strictly decreasing on $(0,\infty)$ with $\lim_{x\to 0^+} f(x) = \infty$ (no finite mode): all unimodal. (c) Normal: strictly increasing to μ, strictly decreasing after: unimodal. (d) No: for $\alpha < 1, \beta < 1$ the beta pdf is U-shaped with modes at both endpoints (e.g. $\alpha = \beta = 1/2$), hence not unimodal. (For $\alpha, \beta \geq 1$ it is unimodal with mode $(\alpha-1)/(\alpha+\beta-2)$, flat when $\alpha = \beta = 1$.)

Exercise 3.28

Central idea. Each pdf already separates into $h(x)\cdot c(\theta)\cdot \exp(w(\theta)\cdot t(x))$; read off the pieces.

Solution. (a) σ known: $f = \underbrace{\frac{e^{-x^2/(2\sigma^2)}}{\sqrt{2\pi}\sigma}}_{h(x)} \underbrace{e^{-\mu^2/(2\sigma^2)}}_{c(\mu)} \exp(\underbrace{\mu/\sigma^2}_{w(\mu)} \underbrace{x}_{t(x)})$. μ known: $f = \underbrace{1}_{h(x)} \underbrace{\frac{1}{\sqrt{2\pi}\sigma}}_{c(\sigma)} \exp(\underbrace{-1/(2\sigma^2)}_{w(\sigma)} \underbrace{(x-\mu)^2}_{t(x)})$.
(b) $f = \underbrace{x^{-1}}_{h(x)} \underbrace{\frac{1}{\Gamma(\alpha)\beta^\alpha}}_{c(\alpha,\beta)} \exp(\alpha\log x - \frac{1}{\beta}x)$; with one parameter fixed, absorb its term into h or c. (c) $f = \underbrace{\frac{1}{x(1-x)}}_{h(x)} \underbrace{\frac{1}{B(\alpha,\beta)}}_{c(\alpha,\beta)} \exp(\alpha\log x + \beta\log(1-x))$; similarly with one parameter fixed. (d) $f = \underbrace{\frac{1}{x!}}_{h(x)} \underbrace{e^{-\lambda}}_{c(\lambda)} \exp(\underbrace{\log\lambda}_{w(\lambda)} \underbrace{x}_{t(x)})$.
(e) $f = \underbrace{\binom{r+x-1}{x}}_{h(x)} \underbrace{p^r}_{c(p)} \exp(\underbrace{\log(1-p)}_{w(p)} \underbrace{x}_{t(x)})$.

Exercise 3.29

Central idea. The natural parameter is $\eta = w(\theta)$; its space is the image $w(\Theta)$ (which here coincides with the set where $\int h\, e^{\eta\cdot t} < \infty$).

Solution. (a) Normal: μ unknown: $\eta = \mu/\sigma^2 \in \mathbb{R}$; σ unknown: $\eta = -1/(2\sigma^2) \in (-\infty, 0)$. (b) Gamma: both unknown: $(\eta_1, \eta_2) = (\alpha, -1/\beta) \in (0,\infty)\times(-\infty,0)$; α known: $\eta = -1/\beta \in (-\infty, 0)$; β known: $\eta = \alpha \in (0,\infty)$. (c) Beta: both unknown: $(\alpha,\beta) \in (0,\infty)^2$; one known: the other ranges over $(0,\infty)$. (d) Poisson: $\eta = \log\lambda \in \mathbb{R}$. (e) Negative binomial (r known): $\eta = \log(1-p) \in (-\infty, 0)$.

Exercise 3.30

Central idea. (a) is one line from (3.4.4)/(3.4.5) with $w(p) = \log\frac{p}{1-p}$; (b) gets gamma moments from the identities and transfers them to beta via $U/(U+V)$.

Solution. (a) $f(x|p) = \binom{n}{x}(1-p)^n \exp(x\log\frac{p}{1-p})$: $w_1'(p) = \frac{1}{p(1-p)}$, $-c'/c = \frac{n}{1-p}$. (3.4.4) gives $EX = np$. (3.4.5): $w_1''(p)EX + (w_1'(p))^2 \mathrm{Var}\, X = -(\log c)''(p) = \frac{n}{(1-p)^2}$; with $w_1''(p) = -\frac{1}{p^2} + \frac{1}{(1-p)^2}$ this yields $\mathrm{Var}X = np(1-p)$. (b) The identities give log-moments of beta directly; for EX and $\mathrm{Var}X$ use the gamma bridge. The identities applied to gamma$(a,1)$ give $EU = a$, $\mathrm{Var}U = a$ (cf. 3.32(b)). With independent $U \sim$ gamma$(a,1)$, $V \sim$ gamma$(b,1)$: $X \overset{d}{=} U/(U+V) \sim$ beta(a,b), independent of $S = U+V \sim$ gamma$(a+b,1)$ (Jacobian factorization). Then $a = EU = E[XS] = EX\cdot ES = EX(a+b)$, so $EX = \frac{a}{a+b}$; and $E[U^2] = E[X^2]E[S^2]$ gives $E[X^2] = \frac{a(a+1)}{(a+b)(a+b+1)}$, so $\mathrm{Var}X = \frac{ab}{(a+b)^2(a+b+1)}$.

Exercise 3.31

Central idea. Differentiate $\int f(x|\theta)\,dx = 1$ under the integral sign: the first derivative gives (3.4.4); the second gives (3.4.5) after using (3.4.4) to cancel the squared terms.

Solution. (a) $0 = \frac{d}{d\theta}\int h(x)c(\theta)e^{\sum w_i(\theta)t_i(x)}dx = \int f(x|\theta)\left[\frac{c'(\theta)}{c(\theta)} + \sum_i w_i'(\theta)t_i(x)\right]dx$ (regularity allows differentiating under the integral), i.e. $\sum_i w_i'(\theta)E_\theta t_i(X) = -\frac{d}{d\theta}\log c(\theta)$, which is (3.4.4). (b) Differentiating again:

$$0 = E_\theta\left[\left(\frac{c'}{c} + \sum_i w_i' t_i\right)^2\right] + \frac{d^2}{d\theta^2}\log c(\theta) + \sum_i w_i''(\theta)E_\theta t_i(X),$$

using $\frac{d^2}{d\theta^2}\log c = \frac{c''}{c} - (\frac{c'}{c})^2$. Expanding the square and substituting $\sum_i w_i' E_\theta t_i = -c'/c$ from (a), the $(c'/c)^2$ terms cancel, leaving $\sum_i w_i'' E_\theta t_i + \sum_{i,j} w_i' w_j' \mathrm{Cov}_\theta(t_i, t_j) = -\frac{d^2}{d\theta^2}\log c(\theta)$, which is (3.4.5).

Exercise 3.32

Central idea. In the (3.4.7) parametrization $w_i(\eta) = \eta_i$, so $\partial w_i/\partial\eta_j = \delta_{ij}$ kills every term except the jth; for gamma read $c^*(\eta)$ off the pdf.

Solution. (a) With $w_i(\eta) = \eta_i$: $\frac{\partial w_i}{\partial \eta_j} = \delta_{ij}$ and $\frac{\partial^2 w_i}{\partial \eta_j^2} = 0$. Then (3.4.4) becomes $E_\eta t_j(X) = -\frac{\partial}{\partial\eta_j}\log c^*(\eta)$ and (3.4.5) becomes $\mathrm{Var}_\eta(t_j(X)) = -\frac{\partial^2}{\partial\eta_j^2}\log c^*(\eta)$. (b) $f(x) = \frac{1}{x}\cdot\frac{1}{\Gamma(\eta_1)(-\eta_2)^{\eta_1}}\exp(\eta_1\log x + \eta_2 x)$ with $\eta_1 = a$, $\eta_2 = -1/b$, $t_2(x) = x$. $\log c^*(\eta) = -\log\Gamma(\eta_1) + \eta_1\log(-\eta_2)$: $-\frac{\partial}{\partial\eta_2}\log c^* = -\frac{\eta_1}{\eta_2} = ab$, so $EX = ab$; $-\frac{\partial^2}{\partial\eta_j^2}\log c^* = \frac{\eta_1}{\eta_2^2} = ab^2$, so $\mathrm{Var}X = ab^2$.

Exercise 3.33

Central idea. Write each family in natural form $\eta(\theta)$; a one-dimensional θ tracing a genuine curve in a $k \geq 2$-dimensional η-space is a curved exponential family.

Solution. (a) $f = \frac{1}{\sqrt{2\pi\theta}}\exp(-\frac{x^2}{2\theta} + x - \frac{\theta}{2})$: exponential family with natural parameters $\eta_1 = -\frac{1}{2\theta}$ (statistic x^2), $\eta_2 = 1$ (statistic x). Curve: the horizontal ray $\{(\eta_1, 1) : \eta_1 < 0\}$ — degenerate (one natural coordinate constant) but 1-d in a 2-d space. (b) $f = \frac{1}{\sqrt{2\pi a}\,\theta}\exp(-\frac{x^2}{2a\theta^2} + \frac{x}{a\theta} - \frac{1}{2a})$: $\eta_1 = \frac{1}{a\theta}$, $\eta_2 = -\frac{1}{2a\theta^2}$. Eliminating θ: $\eta_2 = -\frac{a}{2}\eta_1^2$ with $\eta_1 > 0$, the right half of a downward parabola with vertex at the origin. (c) $f = \frac{\alpha^\alpha}{\Gamma(\alpha)}\frac{1}{x}\exp(\alpha\log x - \alpha x)$: $\eta_1 = \alpha$, $\eta_2 = -\alpha$: the ray $\eta_2 = -\eta_1$, $\eta_1 > 0$ (the mean-1 curved family). (d) $-(x-\theta)^4 = -x^4 + 4\theta x^3 - 6\theta^2 x^2 + 4\theta^3 x - \theta^4$, so $f = Ce^{-\theta^4}e^{-x^4}\exp(4\theta x^3 - 6\theta^2 x^2 + 4\theta^3 x)$ is an exponential family with $\eta(\theta) = (4\theta, -6\theta^2, 4\theta^3)$, $\theta \in \mathbb{R}$: the twisted cubic $\eta_2 = -\frac{3}{8}\eta_1^2$, $\eta_3 = \frac{1}{16}\eta_1^3$ in $\mathbb{R}^3$ — curved ($d = 1 < k = 3$). (Sketches: (a) horizontal ray at $\eta_2 = 1$ in the left half-plane; (b) right half of $\eta_2 = -\frac{a}{2}\eta_1^2$; (c) ray from the origin into the fourth quadrant along $\eta_1 + \eta_2 = 0$; (d) a twisted-cubic space curve through the origin, odd in θ.)

Exercise 3.34

Central idea. For $\mathrm{n}(\mu, \sigma^2)$ the natural parameters are $(\mu/\sigma^2, -1/(2\sigma^2))$; each approximation constrains (μ, σ^2) to a one-dimensional curve, hence a curve in η-space.

Solution. (a) $\eta_1 = \frac{\lambda}{\lambda/n} = n$ (constant!), $\eta_2 = -\frac{n}{2\lambda}$: the vertical ray $\{(\eta_1, \eta_2) : \eta_1 = n,\ \eta_2 < 0\}$. (b) $\eta_1 = \frac{n}{1-p}$, $\eta_2 = -\frac{n}{2p(1-p)}$: eliminating p, $\eta_2 = -\frac{\eta_1^2}{2(\eta_1 - n)}$ for $\eta_1 > n$ — a branch in the lower half-plane running from $(n^+, -\infty)$ (as $p \to 0^+$) toward the asymptote $\eta_2 = -(\eta_1 + n)/2$ (as $p \to 1^-$). (c) $\eta_1 = \frac{r(1-p)/p}{r(1-p)/(np^2)} = np$ and $\eta_2 = -\frac{np^2}{2r(1-p)}$, but p is the varying parameter, so η_1 is not constant. Eliminating $p = \eta_1/n$: $\eta_2 = -\frac{\eta_1^2}{2r(n-\eta_1)}$ for $0 < \eta_1 < n$ — a curve in the lower half-plane from $(0^+, 0^-)$ (as $p \to 0^+$) descending toward the asymptote $\eta_1 = n$ (as $p \to 1^-$).

Exercise 3.35

Central idea. Imposing $EX = \mu$ (or $\mu = \sigma^2$) cuts the full parameter space down to a curve; in natural parameters these constraints are straight rays.

Solution. (a) $\mu = \sigma^2 = e^\theta$: $\eta_1 = \mu/\sigma^2 = 1$, $\eta_2 = -1/(2e^\theta)$: the vertical ray at $\eta_1 = 1$, $\eta_2 < 0$. Comparison with 3.34(a): both are vertical rays in the natural parameter space, but 3.34(a)'s sits at $\eta_1 = n$ while this one sits at $\eta_1 = 1$ — the 3.34(a) family is $\mathrm{n}(\lambda, \lambda/n)$ (variance = mean/n, appropriate for a sample mean), whereas here variance = mean (appropriate for a single Poisson count). Same shape, different location; the n records the averaging. (b) $\alpha\beta = \mu$: with natural parameters $\eta_1 = \alpha$, $\eta_2 = -1/\beta$, the constraint is $\eta_2 = -\eta_1/\mu$: the straight ray $\{(\eta_1, -\eta_1/\mu) : \eta_1 > 0\}$ from the origin into the fourth quadrant, of slope $-1/\mu$. (c) The constraints $\alpha_i\beta_i = \mu$, $i = 1, \ldots, n$, cut out the n-dimensional manifold $\{(\alpha_i, \beta_i)_{i=1}^n : \alpha_i > 0,\ \beta_i > 0,\ \alpha_i\beta_i = \mu\}$ — a product of n hyperbolas — embedded in the $2n$-dimensional full gamma parameter space.

Exercise 3.36

Central idea. Location–scale: μ slides the graph, σ stretches it horizontally by σ and flattens it vertically by $1/\sigma$; the shape — a symmetric bump peaking at $\pm\sqrt{3}/2$ with height $1701/1024 \approx 1.66$ — is unchanged.

Solution. f is a symmetric bump on $(-1, 1)$ with $f'(x) = \frac{63}{2}x^5(3 - 4x^2)$, hence peaks at $x = \pm\sqrt{3}/2$ of height $f(\pm\sqrt{3}/2) = \frac{63}{4} \cdot \frac{27}{256} = \frac{1701}{1024} \approx 1.661$, and $f(0) = f(\pm 1) = 0$. (a) is f itself: support $(-1, 1)$, peaks at ± 0.866 of height 1.661. (b) shifts it rigidly: support $(2, 4)$, peaks at 3 ± 0.866, same height 1.661. (c) support $(1, 5)$, peaks at 3 ± 1.732, height halved to ≈ 0.831. On the same axes: (a) a tall narrow bump at the origin; (b) the identical bump centered at 3; (c) a twice-as-wide, half-as-tall bump centered at 3.

Exercise 3.37

Central idea. Substitute $z = (x - \mu)/\sigma$; symmetry of f about 0 makes the lower half-mass exactly $1/2$.

Solution. $P(X \le \mu) = \int_{-\infty}^{\mu} \frac{1}{\sigma} f(\frac{x-\mu}{\sigma})dx = \int_{-\infty}^{0} f(z)dz$. By symmetry $f(z) = f(-z)$, so $\int_{-\infty}^{0} f = \int_0^{\infty} f = 1/2$; likewise $P(X \ge \mu) = 1/2$. Hence μ is a median.

Exercise 3.38

Central idea. Standardize: $\{X > x_\alpha\} = \{Z > z_\alpha\}$ for $Z = (X - \mu)/\sigma$, which has pdf f.

Solution. $P(X > x_\alpha) = P\left(\frac{X-\mu}{\sigma} > \frac{x_\alpha - \mu}{\sigma}\right) = P(Z > z_\alpha) = \int_{z_\alpha}^{\infty} f(z)dz = \alpha$, since $Z = (X - \mu)/\sigma$ has pdf f and $(x_\alpha - \mu)/\sigma = z_\alpha$.

Exercise 3.39

Central idea. For $\mu = 0, \sigma = 1$ the tail integrals are arctans; Exercise 3.38 lifts the result to general (μ, σ).

Solution. For $\mu = 0, \sigma = 1$: $P(X \ge 0) = \int_0^{\infty} \frac{dz}{\pi(1+z^2)} = \frac{1}{\pi} \cdot \frac{\pi}{2} = \frac{1}{2}$, and by symmetry $P(X \le 0) = \frac{1}{2}$, proving (a). $P(X \ge 1) = \int_1^{\infty} \frac{dz}{\pi(1+z^2)} = \frac{1}{\pi}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \frac{1}{4}$; by symmetry $P(X \le -1) = \frac{1}{4}$, proving (b). For general μ, σ, put $z_{1/2} = 0$ and $z_{1/4} = 1$ (upper-tail notation); then $x_{1/2} = \mu$, $x_{1/4} = \mu + \sigma$ (and by symmetry $x_{3/4} = \mu - \sigma$), so Exercise 3.38 gives $P(X \ge \mu) = P(X \le \mu) = 1/2$ and $P(X \ge \mu + \sigma) = P(X \le \mu - \sigma) = 1/4$.

Exercise 3.40

Central idea. Standardize: $Z = (X - \mu)/\sigma$ has pdf $f^*(z) = \sigma f(\sigma z + \mu)$ with mean 0 and variance 1; it generates the location–scale family.

Solution. Let $Z = (X - \mu)/\sigma$; its pdf is $f^*(z) = \sigma f(\sigma z + \mu)$. Then $\int f^* = 1$, $EZ = \int z\sigma f(\sigma z + \mu)dz = \frac{1}{\sigma}E[X - \mu] = 0$, and $\mathrm{Var}Z = \frac{1}{\sigma^2}\mathrm{Var}\,X = 1$. The location–scale family $\frac{1}{\tau}f^*(\frac{x-\nu}{\tau})$ based on f^* has standard pdf f^* with mean 0 and variance 1.

Exercise 3.41

Central idea. "Stochastically increasing" = the cdf decreases pointwise in the parameter; for $\mathrm{n}(\mu, \sigma^2)$ and the gamma scale this is a direct monotonicity check.

Solution. $\theta_1 > \theta_2 \Rightarrow F(x|\theta_1) \le F(x|\theta_2)$ for all x is the requirement (stochastically greater means smaller cdf). (a) $F(x|\mu) = \Phi((x-\mu)/\sigma)$, which decreases in μ for fixed x, σ. (b) $F(x|\beta) = P(X \le x) = P(X/\beta \le x/\beta) = G(x/\beta)$, where G is the gamma$(\alpha, 1)$ cdf; for $x \ge 0$, x/β decreases in β and G is nondecreasing, so F decreases in β (for $x < 0$, $F \equiv 0$).

Exercise 3.42

Central idea. Location shifts and scale stretches move every quantile in the same direction as the parameter.

Solution. (a) $F(x|\theta) = F_0(x-\theta)$: $\theta_1 > \theta_2 \Rightarrow x - \theta_1 < x - \theta_2 \Rightarrow F(x|\theta_1) \le F(x|\theta_2)$. (b) On $[0, \infty)$, $F(x|\theta) = F_0(x/\theta)$: for $x \ge 0$, $\theta_1 > \theta_2 \Rightarrow x/\theta_1 < x/\theta_2 \Rightarrow F(x|\theta_1) \le F(x|\theta_2)$ (at $x = 0$ both equal $F_0(0)$).

Exercise 3.43

Central idea. Inversion $Y = 1/X$ and the reciprocal parametrization $\theta \mapsto 1/\theta$ both reverse order, turning "increasing" into "decreasing".

Solution. (a) For $y > 0$, $F_Y(y|\theta) = P(X \ge 1/y) = 1 - F_X((1/y) - |\theta)$. Positive support does not imply continuity; the left limit makes the argument valid for atoms as well. For $y \le 0$, $F_Y(y|\theta) = 0$. If $F_X(\cdot|\theta)$ is stochastically increasing, $\theta_1 > \theta_2 \Rightarrow F_X((1/y) - |\theta_1) \le F_X((1/y) - |\theta_2) \Rightarrow F_Y(y|\theta_1) \ge F_Y(y|\theta_2)$: stochastically decreasing. (b) $\theta_1 > \theta_2 > 0 \Rightarrow 1/\theta_1 < 1/\theta_2 \Rightarrow F_X(x|1/\theta_1) \ge F_X(x|1/\theta_2)$: $F_X(x|1/\theta)$ is stochastically decreasing in θ.

Exercise 3.44

Central idea. Both are Markov's inequality: $P(|X| \ge b) = P(X^2 \ge b^2) \le EX^2/b^2$ and $P(|X| \ge b) \le E|X|/b$; plug in $EX = 1$, $EX^2 = 2$ for Exp(1).

Solution. $P(|X| \ge b) = P(X^2 \ge b^2) \le EX^2/b^2$ and $P(|X| \ge b) \le E|X|/b$, both by Markov's inequality. For $f(x) = e^{-x}$ $(x > 0)$: $E|X| = 1$, $EX^2 = 2$, so the bounds are $2/b^2$ and $1/b$. At $b = 3$: $2/9 \approx 0.222 < 1/3 \approx 0.333$ — the EX^2/b^2 bound is better. At $b = \sqrt{2}$: $2/2 = 1 > 1/\sqrt{2} \approx 0.707$ — the $E|X|/b$ bound is better.

Exercise 3.45

Central idea. Chernoff's trick: $P(X \ge a) = P(e^{tX} \ge e^{ta}) \le e^{-ta}E[e^{tX}]$ for $t > 0$ (Markov on e^{tX}); any h dominating $\mathbf{1}_{[0,\infty)}$ works the same way.

Solution. (a) For $0 < t < h$: $\{X \ge a\} = \{e^{tX} \ge e^{ta}\}$, so $P(X \ge a) \le E[e^{tX}]/e^{ta} = e^{-at}M_X(t)$ by Markov's inequality. (b) For $-h < t < 0$: $\{X \le a\} = \{e^{tX} \ge e^{ta}\}$ (multiplying by negative t flips $\le$), so $P(X \le a) \le e^{-at}M_X(t)$. (c) $P(X \ge 0) \le E[e^{tX}]$, $t \ge 0$. In general, $P(X \ge 0) \le Eh(t, X)$ whenever $h(t, x) \ge \mathbf{1}_{[0,\infty)}(x)$ for all x (i.e. h is nonnegative everywhere and at least 1 on $[0, \infty)$): then $P(X \ge 0) = E[\mathbf{1}_{\{X \ge 0\}}] \le E[h(t, X)\mathbf{1}_{\{X \ge 0\}}] \le Eh(t, X)$.

Exercise 3.46

Central idea. Compute the exact two-sided tail from the cdf; Chebychev's $1/k^2$ is universal but loose — the exact probabilities are much smaller and hit 0.

Solution. Uniform$(0, 1)$: $\mu_X = 1/2$, $\sigma_X = 1/\sqrt{12}$, so $P(|X - \mu_X| \ge k\sigma_X) = P(|X - 1/2| \ge k/\sqrt{12}) = \max(0, 1 - k/\sqrt{3})$ (0 for $k \ge \sqrt{3}$). Exponential(λ) (mean λ): $\mu_X = \sigma_X = \lambda$, so $P(|X - \lambda| \ge k\lambda) = P(X \ge (1+k)\lambda) + P(X \le (1-k)\lambda) = e^{-(1+k)} + \max(0, 1 - e^{-(1-k)})$ (the second term is 0 for $k \ge 1$). Versus

Chebychev's $1/k^2$: e.g. at $k=2$: uniform 0, exponential $e^{-3}\approx 0.050$, bound 0.25; at $k=1$: uniform $1-1/\sqrt{3}\approx 0.423$, exponential $e^{-2}\approx 0.135$, bound 1. The bound is valid but very conservative.

Exercise 3.47

Central idea. Write the desired lower bound as $h(t)=(1+t^2)P(Z>t)-t\phi(t)\geq 0$; h is decreasing (by the upper bound of Example 3.6.3) and $h(t)\to 0$.

Solution. It suffices to show $P(Z>t)\geq \frac{t}{1+t^2}\phi(t)$, then double for $|Z|$. Put $h(t)=(1+t^2)P(Z>t)-t\phi(t)$ for $t\geq 0$. Then $h'(t)=2tP(Z>t)-(1+t^2)\phi(t)-\phi(t)+t^2\phi(t)=2(tP(Z>t)-\phi(t))\leq 0$ by Example 3.6.3 ($P(Z>t)\leq \phi(t)/t$). Also $h(t)\to 0$: $0\leq (1+t^2)P(Z>t)\leq (1+t^2)\phi(t)/t\to 0$ and $t\phi(t)\to 0$. A decreasing function with limit 0 is nonnegative: $h(t)\geq 0$, i.e. $P(Z>t)\geq \frac{t\phi(t)}{1+t^2}$. Doubling gives $P(|Z|\geq t)\geq \sqrt{\frac{2}{\pi}}\frac{t}{1+t^2}e^{-t^2/2}$.

Exercise 3.48

Central idea. Each pmf's consecutive ratio $p(x+1)/p(x)$ is a rational function; that ratio *is* the recursion.

Solution. Binomial: $\frac{p(x+1)}{p(x)}=\frac{\binom{n}{x+1}}{\binom{n}{x}}\frac{p}{1-p}$, i.e. $p(x+1)=\frac{n-x}{x+1}\cdot\frac{p}{1-p}\,p(x)$. Negative binomial: $\frac{p(y+1)}{p(y)}=\frac{\binom{r+y}{y+1}}{\binom{r+y-1}{y}}(1-p)$, i.e. $p(y+1)=\frac{r+y}{y+1}(1-p)\,p(y)$. Hypergeometric: $\frac{p(x+1)}{p(x)}=\frac{M-x}{x+1}\cdot\frac{K-x}{N-M-K+x+1}$, i.e. $p(x+1)=\frac{(M-x)(K-x)}{(x+1)(N-M-K+x+1)}\,p(x)$.

Exercise 3.49

Central idea. Stein's lemma is integration by parts against the pdf: arrange the integrand as g times the derivative of (pdf $\times$ something), so the boundary term dies and g' appears.

Solution. (a) $(x-\alpha\beta)x^{\alpha-1}e^{-x/\beta}=-\beta\frac{d}{dx}[x^\alpha e^{-x/\beta}]$. Hence $E[g(X)(X-\alpha\beta)]=\frac{-\beta}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty g(x)\frac{d}{dx}[x^\alpha e^{-x/\beta}]dx$; integrating by parts gives $\frac{\beta}{\Gamma(\alpha)\beta^\alpha}\int_0^\infty g'(x)x^\alpha e^{-x/\beta}dx=\beta E[Xg'(X)]$ (boundary terms vanish under the hypotheses on g). (b) $[\beta x-(\alpha-1)(1-x)]x^{\alpha-1}(1-x)^{\beta-1}=-x\frac{d}{dx}[x^{\alpha-1}(1-x)^\beta]$. Hence $E\left[g(X)\left(\beta-(\alpha-1)\frac{1-X}{X}\right)\right]=\frac{-1}{B(\alpha,\beta)}\int_0^1 g(x)\frac{d}{dx}[x^{\alpha-1}(1-x)^\beta]dx$; integrating by parts gives $\frac{1}{B(\alpha,\beta)}\int_0^1 g'(x)(1-x)x^{\alpha-1}(1-x)^{\beta-1}dx=E[(1-X)g'(X)]$.

Exercise 3.50

Central idea. It is the discrete Stein identity $E[(r+X)(1-p)g(X+1)]=E[Xg(X)]$: shift the summation index and use $(r+y-1)\binom{r+y-2}{y-1}=y\binom{r+y-1}{y}$ to absorb the weight into the pmf.

Solution. Theorem 3.6.8(b): for $X\sim$ negative binomial(r,p) and bounded g,

$$E\big[(r+X)(1-p)\,g(X+1)\big]=E\big[X\,g(X)\big].$$

Indeed,

$$E[(r+X)(1-p)g(X+1)]=\sum_{x=0}^{\infty}(r+x)(1-p)g(x+1)\binom{r+x-1}{x}p^r(1-p)^x.$$

Put $y=x+1$: $=\sum_{y=1}^{\infty}(r+y-1)(1-p)g(y)\binom{r+y-2}{y-1}p^r(1-p)^{y-1}$. Since $(r+y-1)\binom{r+y-2}{y-1}=\frac{(r+y-1)!}{(y-1)!\,(r-1)!}=y\binom{r+y-1}{y}$, this equals $\sum_{y=1}^{\infty}y\,g(y)\binom{r+y-1}{y}p^r(1-p)^y=\sum_{y\geq 1}y\,g(y)p(y)=E[Xg(X)]$ (the $y=0$ term is 0).

Chapter 4

Exercise 4.1

Central idea. For a uniform distribution, probabilities are area ratios; use symmetry (the square is centrally symmetric) and note that the complement of (c) is just two corner points.

Solution. (a) The region $\{X^2 + Y^2 < 1\}$ is the unit disk, area π, inside the square of area 4: $P = \pi/4$. (b) The line $Y = 2X$ passes through the center of the square, so it bisects it: $P = 1/2$. (c) On $[-1, 1]^2$, $|X+Y| \le 2$ always, with equality only at $(1, 1)$ and $(-1, -1)$ — a set of area zero. Hence $P(|X+Y| < 2) = 1$.

Exercise 4.2

Central idea. Expectation is a sum/integral, so linearity and order-preservation pass straight through; (c) and (d) reduce to (a) and (b).

Solution. (a) $\mathrm{E}(ag_1+bg_2+c) = \sum\sum(ag_1+bg_2+c)f = a\sum\sum g_1 f + b\sum\sum g_2 f + c\sum\sum f$ (integrals in the continuous case), giving the result since $\sum\sum f = 1$. (b) Every term $g_1(x, y)f(x, y) \ge 0$, so the sum/integral is ≥ 0. (c) Apply (b) to $g_1 - g_2 \ge 0$ and use (a): $\mathrm{E}(g_1) - \mathrm{E}(g_2) = \mathrm{E}(g_1 - g_2) \ge 0$. (d) From $g_1 \ge a$ and (c), $\mathrm{E}(g_1) \ge \mathrm{E}(a) = a$; from $g_1 \le b$, $\mathrm{E}(g_1) \le b$.

Exercise 4.3

Central idea. Count sample points: with two fair dice the 36 outcomes are equally likely, and X depends only on the first die while Y depends only on the second.

Solution. In Example 4.1.5, S is the 36-point toss-two-dice space, $X = 0$ if the first die shows at most 2 ($X = 1$ otherwise), and $Y = 0$ if the second die is odd ($Y = 1$ if even). By Definition 4.1.1, $f(x, y) = P(X = x, Y = y)$ is the fraction of sample points mapping to (x, y). $P(X = 0) = 12/36 = 1/3$, $P(X = 1) = 2/3$, $P(Y = 0) = P(Y = 1) = 1/2$, and the dice are independent, so $f(0, 0) = f(0, 1) = (1/3)(1/2) = 1/6$ and $f(1, 0) = f(1, 1) = (2/3)(1/2) = 1/3$, with $f(x, y) = 0$ elsewhere — exactly the pmf stated in the example.

Exercise 4.4

Central idea. Normalize by integrating over the rectangle; marginalize by integrating out y; for (d) use the cdf method on the monotone decreasing transform, differentiating via the chain rule.

Solution. (a) $1 = \int_0^2\int_0^1 C(x+2y)\,dy\,dx = C\int_0^2(x+1)\,dx = 4C$, so $C = 1/4$. (b) $f_X(x) = \int_0^1 \frac{x+2y}{4}\,dy = \frac{x+1}{4}$, $0 < x < 2$. (c) For $0 < x < 2$, $0 < y < 1$: $F(x, y) = \int_0^x\int_0^y \frac{s+2t}{4}\,dt\,ds = \frac{x^2y}{8} + \frac{xy^2}{4}$. Piecing together: $F(x, y) = 0$ if $x \le 0$ or $y \le 0$; $F(x, y) = \frac{x^2y}{8} + \frac{xy^2}{4}$ on $(0, 2) \times (0, 1)$; $F(x, y) = \frac{x^2}{8} + \frac{x}{4}$ for $0 < x < 2$, $y \ge 1$; $F(x, y) = \frac{y+y^2}{2}$ for $x \ge 2$, $0 < y < 1$; $F(x, y) = 1$ for $x \ge 2$, $y \ge 1$. (d) $X \in (0, 2)$ gives $Z \in (1, 9)$, and $z = 9/(x+1)^2$ is strictly decreasing in x. For $1 < z < 9$: $F_Z(z) = P(X \ge 3/\sqrt{z} - 1) = 1 - F_X(3/\sqrt{z} - 1)$, where $F_X(x) = (x^2 + 2x)/8$ on $(0, 2)$. Differentiating, $f_Z(z) = f_X(3/\sqrt{z} - 1) \cdot \frac{3}{2z^{3/2}} = \frac{3}{4\sqrt{z}} \cdot \frac{3}{2z^{3/2}} = \frac{9}{8z^2}$, $1 < z < 9$ (which integrates to 1).

Exercise 4.5

Central idea. Integrate the joint density over the described region, choosing the order that makes the inner limits simple.

Solution. (a) $\int_0^1\int_{\sqrt{y}}^1 (x+y)\,dx\,dy = \int_0^1 \left(\frac{1}{2} + \frac{y}{2} - y^{3/2}\right) dy = \frac{1}{2} + \frac{1}{4} - \frac{2}{5} = \frac{7}{20}$. (b) For $0 < x < 1$, y runs from x^2 to x: $\int_0^1\int_{x^2}^x 2x\,dy\,dx = \int_0^1 2x(x - x^2)\,dx = 2\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{1}{6}$.

Exercise 4.6

Central idea. With X, Y iid uniform$(0,1)$, A's wait is $W = \max(X - Y, 0)$: a point mass at 0 plus a continuous piece on $(0,1)$.

Solution. Let X, Y be the arrival times (hours after 1 PM), iid uniform$(0,1)$, and $W = \max(X - Y, 0)$. $P(W = 0) = P(X \le Y) = 1/2$. For $0 < w \le 1$, $F_W(w) = P(X - Y \le w) = 1 - (1-w)^2/2$. Hence W is mixed: $P(W = 0) = 1/2$, and on $(0,1)$ it has pdf $f_W(w) = F_W'(w) = 1 - w$. (In particular $\mathrm{E}W = \int_0^1 w(1-w)\,dw = 1/6$ hour $= 10$ minutes.)

Exercise 4.7

Central idea. (X, Y) is uniform on a 30×10 rectangle; the favorable region $\{X + Y < 60\}$ is a trapezoid whose area is a ratio away.

Solution. $X \sim$ uniform$(0, 30)$ (minutes after 8 AM), $Y \sim$ uniform$(40, 50)$, independent, so (X, Y) is uniform on a rectangle of area 300. She arrives before 9 iff $X + Y < 60$. For $0 \le x \le 10$ all $y \in [40, 50]$ qualify (area 100); for $10 \le x \le 20$, $y < 60 - x$ gives area $\int_{10}^{20}(20 - x)\,dx = 50$; for $x \ge 20$ none qualify. Favorable area $= 150$, so $P = 150/300 = 1/2$.

Exercise 4.8

Central idea. With a continuous prior density, the branch $X = 2M$ contributes a Jacobian factor $1/2$. Discrete prior masses use a different formula.

Solution. (a) Conditional on $M = m$, $X = m$ or $2m$, each with probability $1/2$. For a continuous prior density π of the smaller amount, the two contributions to the density of X at x are $\pi(x)/2$ and $\pi(x/2)/4$. Thus

$$P(M = x \mid X = x) = \frac{2\pi(x)}{2\pi(x) + \pi(x/2)}, \qquad P(M = x/2 \mid X = x) = \frac{\pi(x/2)}{2\pi(x) + \pi(x/2)}.$$

These conditional probabilities describe the two possible branches of M given $X = x$; they are not unconditional point probabilities. (b) Therefore

$$E(Y \mid X = x) = \frac{4x\pi(x) + (x/2)\pi(x/2)}{2\pi(x) + \pi(x/2)} > x \iff \pi(x/2) < 4\pi(x).$$

For the rate-λ density $\pi(m) = \lambda e^{-\lambda m}$, switch iff $x < 2\log 4/\lambda$. The statement's factor 2 and threshold $2\log 2/\lambda$ omit the Jacobian for a continuous prior. If π instead denotes *discrete masses*, those factor-2 formulas are appropriate wherever both amounts are in the discrete support. State which model is used. (c) With a fixed smaller amount m, the other envelope is $2m$ when yours is m, and m when yours is $2m$. Both envelopes have expectation $3m/2$ before observation, so switching unconditionally has no advantage.

Exercise 4.9

Central idea. A rectangle probability is an alternating sum of four cdf values; the product form factors that sum.

Solution. $P(a \le X \le b,\ c \le Y \le d) = F_{X,Y}(b,d) - F_{X,Y}(a,d) - F_{X,Y}(b,c) + F_{X,Y}(a,c)$. Substituting $F_{X,Y}(x,y) = F_X(x)F_Y(y)$ gives $F_X(b)F_Y(d) - F_X(a)F_Y(d) - F_X(b)F_Y(c) + F_X(a)F_Y(c) = (F_X(b) - F_X(a))(F_Y(d) - F_Y(c)) = P(a \le X \le b)P(c \le Y \le d)$.

Exercise 4.10

Central idea. Marginals come from summing rows/columns; dependence is one violated product $f(x,y) \ne f_X(x)f_Y(y)$, and independence with fixed marginals forces the product table.

Solution. Marginals: $f_X(1) = 1/4$, $f_X(2) = 1/2$, $f_X(3) = 1/4$; $f_Y(2) = f_Y(3) = f_Y(4) = 1/3$. (a) $f(1,3) = 1/6 \neq 1/12 = f_X(1)f_Y(3)$, so X and Y are dependent. (b) With U, V independent and the same marginals, $f_{U,V}(u,v) = f_X(u)f_Y(v)$:

	U 1	U 2	U 3
$V = 2$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
$V = 3$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$
$V = 4$	$\frac{1}{12}$	$\frac{1}{6}$	$\frac{1}{12}$

Exercise 4.11

Central idea. One joint probability versus the product of marginals settles it: $P(U = 1, V = 2) = P(HH) = 1/4 \neq (1/2)(1/4)$.

Solution. No. $P(U = 1) = 1/2$ and $P(V = 2) = P(HH) = 1/4$, but $P(U = 1, V = 2) = P(\text{first two tosses are } HH) = 1/4 \neq 1/8 = P(U = 1)P(V = 2)$. (Indeed $V > U$ always.)

Exercise 4.12

Central idea. Two iid uniform break points give three piece lengths; a triangle is possible iff every piece is shorter than half the stick, and each ordering of the breaks contributes the same triangular region.

Solution. Let the break points be X, Y iid uniform$(0,1)$. By symmetry take $X < Y$ (probability $1/2$ each way); the pieces are X, $Y - X$, $1 - Y$. They form a triangle iff each is $< 1/2$, i.e. $X < 1/2$, $Y - X < 1/2$, $Y > 1/2$: a triangle in $\{0 < x < y < 1\}$ with vertices $(0, 1/2), (1/2, 1/2), (1/2, 1)$, area $1/8$. Doubling for $Y < X$ gives $1/4$.

Exercise 4.13

Central idea. Add and subtract $\mathrm{E}(Y|X)$ inside the square; the cross term vanishes by iterated expectation, leaving a sum of a g-free term and a nonnegative term minimized at $g = \mathrm{E}(Y|X)$.

Solution. (a) $\mathrm{E}(Y - g(X))^2 = \mathrm{E}[(Y - \mathrm{E}(Y|X)) + (\mathrm{E}(Y|X) - g(X))]^2 = \mathrm{E}(Y - \mathrm{E}(Y|X))^2 + \mathrm{E}(\mathrm{E}(Y|X) - g(X))^2$, because the cross term is $2\,\mathrm{E}[(\mathrm{E}(Y|X) - g(X)) \cdot \mathrm{E}[Y - \mathrm{E}(Y|X)|X]] = 0$. The second term is ≥ 0 and equals 0 for $g(x) = \mathrm{E}(Y|X = x)$, which attains the minimum. (b) Equation (2.2.4), $\min_a \mathrm{E}(Y - a)^2 = \mathrm{Var}\, Y$, is the case where X is degenerate: g ranges over constants a and $\mathrm{E}(Y|X) = \mathrm{E}Y$.

Exercise 4.14

Central idea. $X^2 + Y^2 \sim \chi^2_2$ is exponential with mean 2; X^2 is the square of a standard normal, i.e. χ^2_1 by definition.

Solution. (a) $X^2 + Y^2 \sim \chi^2_2$, which is exponential with mean 2: $P(X^2 + Y^2 < 1) = 1 - e^{-1/2}$. (b) By definition χ^2_1 is the distribution of the square of an $n(0,1)$ variable, so $X^2 \sim \chi^2_1$. Then $P(X^2 < 1) = P(|X| < 1) = 2\Phi(1) - 1 \approx 0.6826$.

Exercise 4.15

Central idea. Condition on the sum: the ratio of the joint pmf to the Poisson$(\theta + \lambda)$ pmf of the sum collapses to a binomial pmf.

Solution. For $0 \leq k \leq n$,

$$P(X = k|X + Y = n) = \frac{e^{-\theta}\theta^k/k! \cdot e^{-\lambda}\lambda^{n-k}/(n-k)!}{e^{-(\theta+\lambda)}(\theta+\lambda)^n/n!} = \binom{n}{k}\left(\frac{\theta}{\theta+\lambda}\right)^k\left(\frac{\lambda}{\theta+\lambda}\right)^{n-k},$$

so $X|X + Y = n \sim \text{binomial}(n, \theta/(\theta + \lambda))$. By symmetry, $Y|X + Y = n \sim \text{binomial}(n, \lambda/(\theta + \lambda))$.

Exercise 4.16

Central idea. The joint pmf of (U, V) factors because the geometric tail is memoryless: $P(U = u, V = v) = p^2(1-p)^{2u+|v|-2}$ splits into a function of u times a function of v.

Solution. The book's geometric is $P(X = x) = p(1-p)^{x-1}$, $x = 1, 2, \ldots$ (p. 80), so $X+Y \geq 2$ and the "$Z = 0$ if $X+Y = 0$" clause in (b) is vacuous. (a) For $u \geq 1$, $v \in \mathbb{Z}$: if $v > 0$, $\{U = u, V = v\} = \{X = u+v, Y = u\}$; if $v < 0$, $\{U = u, V = v\} = \{X = u, Y = u - v\}$; if $v = 0$, $\{U = u, V = v\} = \{X = Y = u\}$. In all cases $P(U = u, V = v) = p^2(1-p)^{2u+|v|-2}$. Summing out: $P(U = u) = p^2(1-p)^{2u-2}\left[1 + 2\sum_{k\geq 1}(1-p)^k\right] = p(2-p)(1-p)^{2u-2}$ $(u \geq 1)$, and $P(V = v) = p^2(1-p)^{|v|}\sum_{u\geq 1}(1-p)^{2u-2} = p(1-p)^{|v|}/(2-p)$ $(v \in \mathbb{Z})$. Their product is $p^2(1-p)^{2u+|v|-2} = P(U = u, V = v)$, so U, V are independent. (b) Since $Y \geq 1$, $0 < Z = X/(X+Y) < 1$ always. For coprime $0 < s < t$, $\{Z = s/t\} = \bigcup_{k\geq 1}\{X = ks, Y = k(t-s)\}$, so $P(Z = s/t) = \sum_{k\geq 1} p^2(1-p)^{kt-2} = \frac{p^2(1-p)^{t-2}}{1-(1-p)^t}$. (c) With $W = X + Y$, $P(X = x, W = w) = P(X = x, Y = w - x) = p^2(1-p)^{w-2}$ for integers $1 \leq x \leq w-1$ (and 0 otherwise).

Exercise 4.17

Central idea. Discretizing an exponential at the integers gives a (shifted) geometric law; conditioning on $\{Y \geq 5\} = \{X \geq 4\}$ and subtracting 4 is exactly the memoryless property.

Solution. (a) $P(Y = i+1) = P(i \leq X < i+1) = e^{-i} - e^{-(i+1)} = (1 - e^{-1})(e^{-1})^i$, $i \geq 0$: $Y - 1$ is geometric$(1 - e^{-1})$ (failures), i.e. Y is the trials-until-first-success version with success probability $1 - e^{-1}$. (b) $\{Y \geq 5\} = \{X \geq 4\}$, and by memorylessness $X - 4 | X \geq 4 \sim$ exponential(1). So $X - 4 | Y \geq 5 \sim$ exponential(1).

Exercise 4.18

Central idea. Go to polar coordinates: the angular integral contributes $\pi/2$, the r in the area element cancels the $\sqrt{x^2+y^2}$ in the denominator, leaving $\int_0^\infty g = 1$.

Solution. $f \geq 0$ is clear. With $x = r\cos\theta$, $y = r\sin\theta$ $(r > 0,\ 0 < \theta < \pi/2)$, $dx\,dy = r\,dr\,d\theta$:

$$\iint_{x,y>0} f = \int_0^{\pi/2}\int_0^\infty \frac{2g(r)}{\pi r}\, r\,dr\,d\theta = \frac{2}{\pi}\cdot\frac{\pi}{2}\int_0^\infty g(r)\,dr = 1.$$

Exercise 4.19

Central idea. $(X_1 - X_2)/\sqrt{2} \sim n(0,1)$, so its square is χ^2_1; the gamma ratio is the classic beta transformation.

Solution. (a) $(X_1 - X_2)/\sqrt{2} \sim n(0,1)$, hence $(X_1 - X_2)^2/2 \sim \chi^2_1$ with pdf $f(z) = (2\pi z)^{-1/2}e^{-z/2}$, $z > 0$. (b) The transformation $(z_1, z_2) = (x_1 + x_2, x_1/(x_1 + x_2))$ has Jacobian z_1 and yields $f(z_1, z_2) = \text{gamma}(z_1; \alpha_1 + \alpha_2, 1) \times \text{beta}(z_2; \alpha_1, \alpha_2)$. Thus $X_1/(X_1 + X_2) \sim \text{beta}(\alpha_1, \alpha_2)$ and $X_2/(X_1 + X_2) \sim \text{beta}(\alpha_2, \alpha_1)$.

Exercise 4.20

Central idea. This is polar coordinates in disguise ($Y_1 = R^2$, $Y_2 = \cos\Theta$); the transformation is two-to-one (sign of X_2), and the joint density factors into a radial piece times an angular piece.

Solution. (a) Write $X_1 = Y_2\sqrt{Y_1}$, $X_2 = \pm\sqrt{Y_1(1 - Y_2^2)}$; each branch has Jacobian $|J| = 1/(2\sqrt{1 - Y_2^2})$. Summing both branches,

$$f_{Y_1,Y_2}(y_1, y_2) = \frac{1}{2\pi\sigma^2\sqrt{1 - y_2^2}} e^{-y_1/(2\sigma^2)}, \quad y_1 > 0,\ |y_2| < 1.$$

(b) This factors as $\left[\frac{1}{2\sigma^2}e^{-y_1/(2\sigma^2)}\right]\left[\frac{1}{\pi\sqrt{1-y_2^2}}\right]$: $Y_1 \sim$ exponential$(2\sigma^2)$ (i.e. gamma$(1, 2\sigma^2)$ with scale $2\sigma^2$) independent of Y_2, which has the arcsine distribution on $(-1, 1)$. Geometrically, $Y_1 = R^2$ is the squared radius and $Y_2 = \cos\Theta$ encodes the angle; for iid normals the radius is independent of the direction.

Exercise 4.21

Central idea. The χ^2_2 law for R^2 makes R Rayleigh, and the polar Jacobian r converts $re^{-r^2/2}/(2\pi)$ into the iid $n(0, 1)$ density — this is the Box–Muller construction.

Solution. $R^2 \sim \chi^2_2$ is exponential with mean 2, so $f_R(r) = re^{-r^2/2}$, $r > 0$, and $f_{R,\Theta}(r, \theta) = \frac{1}{2\pi}re^{-r^2/2}$. The polar transformation has Jacobian r, so $f_{X,Y}(x, y) = \frac{1}{2\pi}e^{-(x^2+y^2)/2}$: X and Y are independent $n(0, 1)$.

Exercise 4.22

Central idea. A separable linear rescaling has constant Jacobian $1/(ac)$.

Solution. The transformation $(x, y) = ((u-b)/a, (v-d)/c)$ is one-to-one with Jacobian $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \frac{1}{ac}$ (using $a, c > 0$). By the transformation formula, $f_{U,V}(u, v) = f((u-b)/a, (v-d)/c) \cdot \frac{1}{ac}$.

Exercise 4.23

Central idea. In Example 4.3.3, $X \sim$ beta(α, β), $Y \sim$ beta$(\alpha+\beta, \gamma)$ are independent; both transformations lead (after a change of variable) to the same V-integral, which is a beta integral giving $U \sim$ beta$(\alpha, \beta+\gamma)$.

Solution. (a) $x = u/v$, $y = v$, $|J| = 1/v$ on $0 < u < v < 1$. $f_{U,V}(u, v) = f_X(u/v)f_Y(v)/v$; integrating v from u to 1:

$$f_U(u) = \frac{\Gamma(\alpha+\beta+\gamma)}{\Gamma(\alpha)\Gamma(\beta)\Gamma(\gamma)}u^{\alpha-1}\int_u^1 (v-u)^{\beta-1}(1-v)^{\gamma-1}\,dv.$$

With $w = (v-u)/(1-u)$ the integral is $(1-u)^{\beta+\gamma-1}B(\beta, \gamma)$, so $f_U(u) = \frac{\Gamma(\alpha+\beta+\gamma)}{\Gamma(\alpha)\Gamma(\beta+\gamma)}u^{\alpha-1}(1-u)^{\beta+\gamma-1}$: $U \sim$ beta$(\alpha, \beta+\gamma)$. (b) $x = \sqrt{uv}$, $y = \sqrt{u/v}$, $|J| = 1/(2v)$ on $0 < u < 1$, $u < v < 1/u$. Substituting $t = \sqrt{u/v}$ (so $dt/t = -dv/(2v)$) turns $\int_u^{1/u} f_X(\sqrt{uv})f_Y(\sqrt{u/v})\frac{dv}{2v}$ into $\int_u^1 f_X(u/t)f_Y(t)\frac{dt}{t}$ — exactly the integral in (a). Hence $U \sim$ beta$(\alpha, \beta+\gamma)$ again.

Exercise 4.24

Central idea. The transformation $(x, y) \mapsto (x+y, x/(x+y))$ has Jacobian z_1, and the gamma kernels regroup into a gamma$(r+s)$ piece times a beta(r, s) piece.

Solution. With $x = z_1z_2$, $y = z_1(1-z_2)$, $|J| = z_1$:

$$f(z_1, z_2) = \frac{1}{\Gamma(r)\Gamma(s)}(z_1z_2)^{r-1}e^{-z_1z_2}(z_1(1-z_2))^{s-1}e^{-z_1(1-z_2)}z_1 = \frac{z_1^{r+s-1}e^{-z_1}}{\Gamma(r+s)} \cdot \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)}z_2^{r-1}(1-z_2)^{s-1},$$

$z_1 > 0$, $0 < z_2 < 1$. Hence $Z_1 \sim$ gamma$(r+s, 1)$, $Z_2 \sim$ beta(r, s), independent.

Exercise 4.25

Central idea. Both are unit-Jacobian shifts: $Y = X + Z$ gives $z = y - x$, and $Y = X^2 + Z$ gives $z = y - x^2$; substitute into the joint pdf of (X, Z) and read off the support.

Solution. In Example 4.5.8, $X \sim$ uniform$(0, 1)$, $Z \sim$ uniform$(0, 1/10)$ independent, so $f_{X,Z}(x, z) = 10$ on $(0, 1) \times (0, 1/10)$. With $Y = X + Z$, the transformation $(x, z) \mapsto (x, y)$ has $|J| = 1$ ($z = y - x$), giving $f_{X,Y}(x, y) = 10$ for $0 < x < 1$, $x < y < x + 1/10$. In Example 4.5.9, $X \sim$ uniform$(-1, 1)$, $Z \sim$ uniform$(0, 1/10)$ independent, $f_{X,Z}(x, z) = 5$ on $(-1, 1) \times (0, 1/10)$. With $Y = X^2 + Z$ ($z = y - x^2$, $|J| = 1$), $f_{X,Y}(x, y) = 5$ for $-1 < x < 1$, $x^2 < y < x^2 + 1/10$.

Exercise 4.26

Central idea. Competing exponentials (scale convention, as in the text): the minimum is exponential with scale $\lambda\mu/(\lambda+\mu)$, the winner is X with probability $\mu/(\lambda+\mu)$, and — by memorylessness — the minimum's distribution doesn't depend on who won.

Solution. (a) With $X \sim$ exponential(λ), $Y \sim$ exponential(μ) in the text's scale convention ($f_X(x) = \frac{1}{\lambda}e^{-x/\lambda}$), for $z > 0$: $P(Z > z, W = 1) = P(X > z, X < Y) = \int_z^\infty \frac{1}{\lambda}e^{-x/\lambda}e^{-x/\mu}\,dx = \frac{\mu}{\lambda+\mu}e^{-z(\lambda+\mu)/(\lambda\mu)}$. Hence $P(Z \le z, W = 1) = \frac{\mu}{\lambda+\mu}(1-e^{-z(\lambda+\mu)/(\lambda\mu)})$ and similarly $P(Z \le z, W = 0) = \frac{\lambda}{\lambda+\mu}(1-e^{-z(\lambda+\mu)/(\lambda\mu)})$. The joint density is $f(z,1) = \frac{1}{\lambda}e^{-z(\lambda+\mu)/(\lambda\mu)}$, $f(z,0) = \frac{1}{\mu}e^{-z(\lambda+\mu)/(\lambda\mu)}$, $z > 0$. (b) $P(W = 1) = \lambda/(\lambda+\mu)$, $P(W = 0) = \mu/(\lambda+\mu)$, and $P(Z \le z|W = i) = 1 - e^{-(\lambda+\mu)z} = P(Z \le z)$ for $i = 0, 1$. Thus $Z \sim$ exponential($\lambda+\mu$) independent of W.

Exercise 4.27

Central idea. Linear combinations of independent normals are jointly (bivariate) normal, and for jointly normal variables zero covariance means independence; here $\mathrm{Cov}(U,V) = \mathrm{Var}X - \mathrm{Var}Y = 0$.

Solution. (U,V) is a linear transformation of independent normals, hence bivariate normal. $\mathrm{Cov}(U,V) = \mathrm{Var}X - \mathrm{Var}Y = \sigma^2 - \sigma^2 = 0$, so U, V are independent. $U \sim n(\mu+\gamma, 2\sigma^2)$ and $V \sim n(\mu-\gamma, 2\sigma^2)$.

Exercise 4.28

Central idea. Y/X is standard Cauchy, and $X/(X+Y) = 1/(1+Y/X)$ inherits a Cauchy law; for (b), conditioning on Y and using symmetry gives the same Cauchy density — no surprise, because a symmetric Y satisfies $Y \overset{d}{=} S|Y|$.

Solution. (a) $T = Y/X \sim$ Cauchy$(0,1)$. With $W = X/(X+Y) = 1/(1+T)$, $f_W(w) = \frac{1}{\pi(1+((1-w)/w)^2)} \cdot \frac{1}{w^2} = \frac{1/2}{\pi((w-1/2)^2+1/4)}$: $W \sim$ Cauchy$(1/2, 1/2)$. (b) For $t > 0$, $P(X/|Y| \le t) = \mathrm{E}[\Phi(t|Y|)]$, so $f(t) = \mathrm{E}[\phi(t|Y|)|Y|] = \frac{1}{\pi(1+t^2)}$; by symmetry this holds for all t: $X/|Y| \sim$ Cauchy$(0,1)$, the same as X/Y. (c) Not surprising. General theorem: if X, Y are independent and symmetric about 0 ($X \overset{d}{=} -X$, $Y \overset{d}{=} -Y$), then $X/|Y| \overset{d}{=} X/Y$, since $Y \overset{d}{=} S|Y|$ for an independent Rademacher S, and $X/(S|Y|) = (XS)/|Y| \overset{d}{=} X/|Y|$.

Exercise 4.29

Central idea. Uniform angle suffices for the ratio; equality of the radial projections also requires independence of radius and angle.

Solution. (a) $X/Y = \cot\theta$ almost surely, and its cdf is $1/2 + \arctan(t)/\pi$, so the ratio is standard Cauchy. No independence assumption on R is needed here because $R > 0$ cancels. (b) Add the missing assumption $R \perp \theta$, as in the authors' seventh-printing errata. Then $2XY/\sqrt{X^2+Y^2} = R\sin 2\theta$. Since $2\theta \bmod 2\pi$ is uniform and independent of R, this has the same law as $R\cos\theta = X$. Independence is necessary for the argument: take $R = 1 + a\cos\theta$, $0 < a < 1$. Then $E(R\cos\theta) = a/2$, but $E(R\sin 2\theta) = 0$, disproving the unqualified claim. For iid $N(0,\sigma^2)$ variables the radius and angle are independent, and the transformed variable is $N(0,\sigma^2)$.

Exercise 4.30

Central idea. Tower laws: $\mathrm{E}Y = \mathrm{E}[\mathrm{E}(Y|X)]$, $\mathrm{Var}\,Y = \mathrm{E}[\mathrm{Var}(Y|X)] + \mathrm{Var}(\mathrm{E}(Y|X))$; and $Y/X|X = x \sim n(1,1)$ is free of x, which characterizes independence.

Solution. (a) $\mathrm{E}Y = \mathrm{E}[X] = 1/2$. $\mathrm{Var}\,Y = \mathrm{E}[X^2] + \mathrm{Var}(X) = 1/3 + 1/12 = 5/12$. $\mathrm{Cov}(X,Y) = \mathrm{E}[X \cdot \mathrm{E}(Y|X)] - \mathrm{E}X\,\mathrm{E}Y = \mathrm{E}[X^2] - 1/4 = 1/3 - 1/4 = 1/12$. (b) Given $X = x > 0$, $Y/X = (1/x)Y \sim n(1,1)$, independent of x. Hence $P(Y/X \le t|X = x) = \Phi(t-1)$ for all x, so $Y/X \sim n(1,1)$ independent of X.

Exercise 4.31

Central idea. Tower laws give the moments; the joint is binomial×uniform; integrating out x is a beta integral that collapses to the discrete uniform $1/(n+1)$.

Solution. (a) $\mathrm{E}Y = \mathrm{E}[nX] = n/2$. $\mathrm{Var}\,Y = \mathrm{E}[nX(1-X)] + \mathrm{Var}(nX) = n(1/2-1/3)+n^2/12 = n(n+2)/12$. (b) $f_{X,Y}(x,y) = \binom{n}{y}x^y(1-x)^{n-y}\cdot 1$, $0<x<1$, $y=0,\dots,n$. (c) $P(Y=y) = \binom{n}{y}\int_0^1 x^y(1-x)^{n-y}\,dx = \binom{n}{y}B(y+1,n-y+1) = \frac{1}{n+1}$: Y is discrete uniform on $\{0,\dots,n\}$.

Exercise 4.32

Central idea. (a) The Poisson–gamma mixture integral is a gamma integral yielding a negative binomial pmf. (b) *As stated this is incorrect* — even the means differ ($p\alpha\beta$ versus $\alpha\beta$); the three-stage marginal is negative binomial with a different success probability, coinciding with (a) only if $p=1$.

Solution. (a) (Here gamma(α,β) uses the scale parameterization, mean $\alpha\beta$.)

$$P(Y=y) = \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!}\,\frac{\lambda^{\alpha-1}e^{-\lambda/\beta}}{\Gamma(\alpha)\beta^\alpha}\,d\lambda = \frac{\Gamma(y+\alpha)}{y!\,\Gamma(\alpha)}\left(\frac{1}{\beta+1}\right)^\alpha\left(\frac{\beta}{\beta+1}\right)^y,$$

i.e. negative binomial (failures before the αth success) with success probability $1/(\beta+1)$ when α is an integer. $\mathrm{E}Y = \mathrm{E}\Lambda = \alpha\beta$; $\mathrm{Var}\,Y = \mathrm{E}\Lambda + \mathrm{Var}\Lambda = \alpha\beta+\alpha\beta^2$. (b) *Erratum — the claim is false as stated.* By binomial thinning, $Y|\Lambda \sim \text{Poisson}(p\Lambda)$, so $\mathrm{E}Y = p\,\mathrm{E}\Lambda = p\alpha\beta \neq \alpha\beta$ for $p\neq 1$; the marginals cannot be the same. What is true: the three-stage marginal is again negative binomial,

$$P(Y=y) = \frac{\Gamma(y+\alpha)}{y!\,\Gamma(\alpha)}\left(\frac{1}{1+\beta p}\right)^\alpha\left(\frac{\beta p}{1+\beta p}\right)^y,$$

i.e. NB$(\alpha, 1/(1+\beta p))$ in the failures parameterization — equal to (a)'s NB$(\alpha,1/(1+\beta))$ only when $p=1$.

Exercise 4.33

Central idea. The mgf of a randomly stopped sum is $M_H(t) = \exp(\lambda(M_X(t)-1))$; the logarithmic mgf $M_X(t) = \log(1-(1-p)e^t)/\log p$ turns this into $[p/(1-(1-p)e^t)]^r$ using $e^{-\lambda} = p^r$.

Solution. $M_{X_i}(t) = \frac{-1}{\log p}\sum_{t\geq 1}\frac{((1-p)e^t)^t}{t} = \frac{\log(1-(1-p)e^t)}{\log p}$. By Theorems 4.4.3/4.6.7, $M_H(t) = \mathrm{E}[(M_X(t))^N] = \exp(\lambda(M_X(t)-1)) = e^{-\lambda}(1-(1-p)e^t)^{\lambda/\log p}$. With $r = -\lambda/\log p > 0$ and $e^{-\lambda} = p^r$ (since $p^r = e^{r\log p} = e^{-\lambda}$), $M_H(t) = \left(\frac{p}{1-(1-p)e^t}\right)^r$, the mgf of the negative binomial(r,p) (failures before r successes). Hence $H \sim \mathrm{NB}(r,p)$.

Exercise 4.34

Central idea. Both are beta integrals against the beta(α,β) prior: $\int_0^1 p^{A-1}(1-p)^{B-1}dp = B(A,B)$; moments come from $\mathrm{E}[X] = \mathrm{E}[\mathrm{E}(X|P)]$ and the variance decomposition.

Solution. (a) In Example 4.4.6, $X|P \sim \text{binomial}(n,P)$, $P\sim\text{beta}(\alpha,\beta)$:

$$P(X=x) = \binom{n}{x}\frac{1}{B(\alpha,\beta)}\int_0^1 p^{x+\alpha-1}(1-p)^{n-x+\beta-1}\,dp = \binom{n}{x}\frac{B(x+\alpha,n-x+\beta)}{B(\alpha,\beta)},$$

which is the stated formula. (b) With $X|P\sim\mathrm{NB}(r,P)$ in the text's failures form (3.2.10), $P(X=x|P=p) = \binom{r+x-1}{x}p^r(1-p)^x$ $(x\geq 0)$:

$$P(X=x) = \binom{r+x-1}{x}\frac{B(r+\alpha,\,x+\beta)}{B(\alpha,\beta)} = \binom{r+x-1}{x}\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\frac{\Gamma(r+\alpha)\Gamma(x+\beta)}{\Gamma(\alpha+\beta+r+x)}.$$

$\mathrm{E}[X|P] = r(1-P)/P$, $\mathrm{Var}(X|P) = r(1-P)/P^2$. With $m_k = \mathrm{E}[1/P^k]$ ($m_1 = \frac{\alpha+\beta-1}{\alpha-1}$, $m_2 = \frac{(\alpha+\beta-1)(\alpha+\beta-2)}{(\alpha-1)(\alpha-2)}$): $\mathrm{E}[X] = r(m_1 - 1)$ ($\alpha > 1$), and

$$\mathrm{Var}\, X = \mathrm{E}\left[\frac{r(1-P)}{P^2}\right] + \mathrm{Var}\left(\frac{r(1-P)}{P}\right) = r(m_2 - m_1) + r^2(m_2 - m_1^2) \quad (\alpha > 2).$$

Exercise 4.35

Central idea. Both are the variance decomposition $\mathrm{Var} = \mathrm{E}[\mathrm{Var}(\cdot|\cdot)] + \mathrm{Var}(\mathrm{E}[\cdot|\cdot])$, regrouped to isolate the non-hierarchical variance plus the hierarchy's inflation.

Solution. (a) $\mathrm{Var}\, X = \mathrm{E}[nP(1-P)] + \mathrm{Var}(nP) = n(\mathrm{E}P - \mathrm{E}P^2) + n^2 \mathrm{Var}\, P$. Since $\mathrm{E}P^2 = (\mathrm{E}P)^2 + \mathrm{Var}\, P$, this is $n\mathrm{E}P(1-\mathrm{E}P) - n\mathrm{Var}\, P + n^2\mathrm{Var}\, P = n\mathrm{E}P(1-\mathrm{E}P) + n(n-1)\mathrm{Var}\, P$. (b) $\mathrm{Var}\, Y = \mathrm{E}[\Lambda] + \mathrm{Var}(\Lambda) = \alpha\beta + \alpha\beta^2$. With $\mu = \alpha\beta$, $\alpha\beta^2 = \mu^2/\alpha$, so $\mathrm{Var}\, Y = \mu + \mu^2/\alpha$. The "extra-Poisson" variation is $\mathrm{Var}\, \Lambda = \mu^2/\alpha$ — the variance injected by the random Λ.

Exercise 4.36

Central idea. Each trial gets its own independent P_i, so the X_i are iid Bernoulli($\alpha/(\alpha+\beta)$) — the hierarchy collapses to a plain binomial; (c) is the same computation with n_i trials per group.

Solution. The pairs (X_i, P_i) are independent across i, so the X_i are iid with $P(X_i = 1) = \mathrm{E}[P_i] = \alpha/(\alpha+\beta)$. (a) $\mathrm{E}Y = n\alpha/(\alpha+\beta)$. (b) $\mathrm{Var}\, X_i = \mathrm{E}[P_i(1-P_i)] + \mathrm{Var}(P_i) = \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)} + \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{\alpha\beta}{(\alpha+\beta)^2}$, so $\mathrm{Var}\, Y = n\alpha\beta/(\alpha+\beta)^2$. In fact $Y \sim$ binomial$(n, \alpha/(\alpha+\beta))$ exactly. (c) $\mathrm{E}Y = \sum_i n_i \mathrm{E}[P_i] = \frac{\alpha}{\alpha+\beta}\sum_i n_i$. The X_i are independent across i, so $\mathrm{Var}\, Y = \sum_i \mathrm{Var}\, X_i$ with $\mathrm{Var}\, X_i = \mathrm{E}[n_i P_i(1-P_i)] + \mathrm{Var}(n_i P_i) = n_i \frac{\alpha\beta}{(\alpha+\beta)(\alpha+\beta+1)} + n_i^2 \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = n_i \frac{\alpha\beta(\alpha+\beta+n_i)}{(\alpha+\beta)^2(\alpha+\beta+1)}$.

Exercise 4.37

Central idea. (a) *As stated for general n this is incorrect*: with $C = (1+P)/2$ the mixing variable, beta-binomial(n, α, β) would require $\mathrm{E}[C^r] = (\alpha)^{\overline{r}}/(\alpha+\beta)^{\overline{r}}$ for $r = 1, \ldots, n$, but $C \geq 1/2$ a.s. while a beta(α, β) charges $(0, 1/2)$ — and concretely, for $n = 3$, $P \sim$ uniform$(0,1)$, the first two moments force $(\alpha, \beta) = (6, 2)$ yet $\mathrm{E}[C^3] = 15/32 \neq 7/15$. (The claim is true for $n \leq 2$, where only two moments must match.) (b) The mean and variance are still elementary via iterated expectation.

Audit addition. The authors' first-printing correction asks for a *mixture* of beta-binomials. Introduce discrimination indicators $D_i \mid P \overset{iid}{\sim} \mathrm{Bernoulli}(P)$ and independent guesses $G_i \sim \mathrm{Bernoulli}(1/2)$, and put $X_i = D_i + (1-D_i)G_i$. Conditional on $J = \sum G_i = j$, the remaining $n-j$ trials mix over $P \sim \mathrm{Beta}(a, b)$, so

$$\sum X_i \mid J = j \overset{d}{=} j + \mathrm{BetaBinomial}(n-j, a, b), \qquad J \sim \mathrm{Binomial}(n, 1/2).$$

This gives the intended mixture of shifted beta-binomial laws and an explicit marginal. The counterexample below correctly disproves a single beta-binomial law. For a fixed finite n, differing mixing supports alone do not prove that; the moment counterexample does.

Solution. (a) *Erratum.* Let $C = (1+P)/2$, so $X_i|C$ are iid Bernoulli(C) and $\mathrm{E}[(\Sigma X_i)_{\underline{r}}] = (n)_{\underline{r}}\mathrm{E}[C^r]$. If ΣX_i were beta-binomial(n, α, β), we would need $\mathrm{E}[C^r] = (\alpha)^{\overline{r}}/(\alpha+\beta)^{\overline{r}}$ for $r = 1, \ldots, n$. Take $a = b = 1$ ($P \sim$ uniform$(0,1)$) and $n = 3$: $\mathrm{E}[C] = 3/4$, $\mathrm{E}[C^2] = 7/12$ force $\alpha/(\alpha+\beta) = 3/4$ and $\alpha(\alpha+1)/((\alpha+\beta)(\alpha+\beta+1)) = 7/12$, i.e. $(\alpha, \beta) = (6, 2)$ — but then beta$(6, 2)$ predicts $\mathrm{E}[C^3] = 6 \cdot 7 \cdot 8/(8 \cdot 9 \cdot 10) = 7/15$, whereas $\mathrm{E}[C^3] = \int_0^1 ((1+p)/2)^3 dp = 15/32$. Contradiction, so no beta-binomial fits. (For $n \leq 2$ the claim holds, since only the first two moments must match a beta.) (b) With $\mu = \mathrm{E}[C] = (2a+b)/(2(a+b))$: $\mathrm{E}[\Sigma X_i] = n\mu = \frac{n(2a+b)}{2(a+b)}$. $\mathrm{Var}(\Sigma X_i) = n\mu(1-\mu) + n(n-1)\mathrm{Var}(C)$ with $\mathrm{Var}(C) = \mathrm{Var}(P)/4 = \frac{ab}{4(a+b)^2(a+b+1)}$.

Exercise 4.38

Central idea. (a) The substitution $u = x/v - x/\lambda$ turns the v-integral into $\Gamma(1-r)$ times the gamma kernel. (b) $v = \lambda w$ gives a beta$(r, 1-r)$ integral. (c) A gamma(r,λ) density with $r > 1$ increases near 0 ($\frac{d}{dx}\log f = (r-1)/x - 1/\lambda > 0$), but any exponential mixture has $\frac{d}{dx}\log h(x) = h'(x)/h(x) < 0$ since $h'(x) = -\int v^{-2}e^{-x/v}q(v)\,dv < 0$.

Solution. (a) For $0 < r < 1$ (the case $r = 1$ is the degenerate point mass at $v = \lambda$), substitute $u = x/v - x/\lambda$ ($dv = -v^2 du/x$): the integrand becomes $\frac{1}{\Gamma(r)\Gamma(1-r)}x^{r-1}\lambda^{-r}e^{-x/\lambda}u^{-r}e^{-u}$, and $\int_0^\infty u^{-r}e^{-u}du = \Gamma(1-r)$, leaving $\frac{x^{r-1}e^{-x/\lambda}}{\Gamma(r)\lambda^r} = f(x)$. (b) With $v = \lambda w$: $\int_0^\lambda p_\lambda(v)dv = \frac{1}{\Gamma(r)\Gamma(1-r)}\int_0^1 w^{r-1}(1-w)^{-r}dw = \frac{B(r,1-r)}{\Gamma(r)\Gamma(1-r)} = 1$ (convergent for $r < 1$). (c) If $f(x) = h(x) := \int_0^\infty \frac{e^{-x/v}}{v}q_\lambda(v)dv$, then $h'(x) = -\int_0^\infty \frac{e^{-x/v}}{v^2}q_\lambda(v)dv < 0$, so $\frac{d}{dx}\log h(x) < 0$ for all $x > 0$. But for gamma(r,λ) with $r > 1$, $\frac{d}{dx}\log f(x) = \frac{r-1}{x} - \frac{1}{\lambda} > 0$ for $0 < x < \lambda(r-1)$ — contradiction.

Exercise 4.39

Central idea. Dividing the multinomial joint pmf by the binomial(m, p_j) marginal of X_j leaves a binomial pmf in x_i; the covariance follows from $\mathrm{E}[X_iX_j] = \mathrm{E}[X_j\mathrm{E}(X_i|X_j)]$.

Solution. $P(X_i = x_i, X_j = x_j) = \frac{m!}{x_i!x_j!(m-x_i-x_j)!}p_i^{x_i}p_j^{x_j}(1-p_i-p_j)^{m-x_i-x_j}$. Dividing by $P(X_j = x_j) = \binom{m}{x_j}p_j^{x_j}(1-p_j)^{m-x_j}$ gives $\binom{m-x_j}{x_i}\left(\frac{p_i}{1-p_j}\right)^{x_i}\left(\frac{1-p_i-p_j}{1-p_j}\right)^{m-x_j-x_i}$: binomial$(m - x_j, p_i/(1-p_j))$. $X_j \sim$ binomial(m, p_j) by the usual marginal argument. $\mathrm{E}[X_iX_j] = \mathrm{E}[X_j(m-X_j)p_i/(1-p_j)] = \frac{p_i}{1-p_j}\left(m^2p_j - mp_j(1-p_j) - m^2p_j^2\right) = m(m-1)p_ip_j$, so $\mathrm{Cov}(X_i, X_j) = m(m-1)p_ip_j - m^2p_ip_j = -mp_ip_j$.

Exercise 4.40

Central idea. Integrate y out with $y = (1-x)t$ to get beta integrals twice (giving C and the beta$(a, b+c)$ marginal); the conditional is read off as a scaled beta; $\mathrm{E}[XY] = \mathrm{E}[X(1-X)]\cdot b/(b+c)$.

Solution. The support is $x > 0$, $y > 0$, $x + y < 1$. (a) $\int_0^{1-x} y^{b-1}(1-x-y)^{c-1}dy = (1-x)^{b+c-1}B(b,c)$ (via $y = (1-x)t$), then $\int_0^1 x^{a-1}(1-x)^{b+c-1}dx = B(a, b+c)$; the product is $\frac{\Gamma(a)\Gamma(b)\Gamma(c)}{\Gamma(a+b+c)}$, so C is its reciprocal. (b) $f_X(x) = Cx^{a-1}(1-x)^{b+c-1}B(b,c) = \frac{\Gamma(a+b+c)}{\Gamma(a)\Gamma(b+c)}x^{a-1}(1-x)^{b+c-1}$: $X \sim$ beta$(a, b+c)$; similarly $Y \sim$ beta$(b, a+c)$. (c) $f_{Y|X}(y|x) = \frac{y^{b-1}(1-x-y)^{c-1}}{(1-x)^{b+c-1}B(b,c)}$, $0 < y < 1-x$; with $t = y/(1-x)$ this is the beta(b,c) density. So $Y/(1-x)|X = x \sim$ beta(b,c). (d) $\mathrm{E}[XY] = \mathrm{E}[X\cdot\mathrm{E}(Y|X)] = \mathrm{E}[X(1-X)]\frac{b}{b+c}$. With $X \sim$ beta$(a, b+c)$, $\mathrm{E}[X(1-X)] = \frac{a(b+c)}{(a+b+c)(a+b+c+1)}$, giving $\mathrm{E}[XY] = \frac{ab}{(a+b+c)(a+b+c+1)}$. Hence $\mathrm{Cov}(X,Y) = \frac{ab}{(a+b+c)(a+b+c+1)} - \frac{a}{a+b+c}\frac{b}{a+b+c} = -\frac{ab}{(a+b+c)^2(a+b+c+1)}$.

Exercise 4.41

Central idea. A constant has zero deviation from its mean, so the covariance integrand is identically zero.

Solution. For any X and constant c: $\mathrm{Cov}(X, c) = \mathrm{E}[(X - \mathrm{E}X)(c-c)] = 0$.

Exercise 4.42

Central idea. Independence gives $\mathrm{Cov}(XY, Y) = \mu_X\sigma_Y^2$ and $\mathrm{Var}(XY) = \mathrm{E}[X^2]\mathrm{E}[Y^2] - (\mathrm{E}X\,\mathrm{E}Y)^2$; assemble the correlation.

Solution. $\mathrm{Cov}(XY, Y) = \mathrm{E}[XY^2] - \mathrm{E}[XY]\mathrm{E}[Y] = \mathrm{E}X\,\mathrm{E}[Y^2] - (\mathrm{E}X)(\mathrm{E}Y)^2 = \mu_X\sigma_Y^2$. $\mathrm{Var}(XY) = (\sigma_X^2 + \mu_X^2)(\sigma_Y^2 + \mu_Y^2) - \mu_X^2\mu_Y^2 = \sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2$. Hence

$$\mathrm{Corr}(XY, Y) = \frac{\mu_X\sigma_Y}{\sqrt{\sigma_X^2\sigma_Y^2 + \sigma_X^2\mu_Y^2 + \sigma_Y^2\mu_X^2}}.$$

Exercise 4.43

Central idea. Bilinearity of covariance kills every cross term between distinct (uncorrelated) variables, leaving only the shared variable's variance.

Solution. $\operatorname{Cov}(X_1 + X_2, X_2 + X_3) = \operatorname{Cov}(X_2, X_2) = \sigma^2$ (all other pairs are uncorrelated). $\operatorname{Cov}(X_1 + X_2, X_1 - X_2) = \operatorname{Var} X_1 - \operatorname{Var} X_2 = \sigma^2 - \sigma^2 = 0$.

Exercise 4.44

Central idea. Expand the squared sum of centered variables; diagonal terms give variances, off-diagonal terms pair up into covariances.

Solution. With $\mu_i = \mathrm{E}X_i$,

$$\operatorname{Var}\left(\sum_i X_i\right) = \mathrm{E}\left[\left(\sum_i (X_i - \mu_i)\right)^2\right] = \sum_i \mathrm{E}[(X_i - \mu_i)^2] + 2\sum_{i<j} \mathrm{E}[(X_i - \mu_i)(X_j - \mu_j)],$$

which is $\sum_i \operatorname{Var} X_i + 2\sum_{i<j} \operatorname{Cov}(X_i, X_j)$.

Exercise 4.45

Central idea. All three follow from the bivariate normal mgf $M(s,t) = \exp(\mu_X s + \mu_Y t + \frac{1}{2}(\sigma_X^2 s^2 + 2\rho\sigma_X\sigma_Y st + \sigma_Y^2 t^2))$: set $t = 0$ for (a), divide joint by marginal (completing the square) for (b), and evaluate at (at, bt) for (c).

Solution. (a) $M_X(s) = M(s,0) = \exp(\mu_X s + \sigma_X^2 s^2/2)$: $X \sim n(\mu_X, \sigma_X^2)$; similarly for Y. (b) $f_{Y|X}(y|x) = f(x,y)/f_X(x)$; completing the square in y in the joint exponent gives the $n(\mu_Y + \rho(\sigma_Y/\sigma_X)(x-\mu_X), \sigma_Y^2(1-\rho^2))$ density. (c) $M_{aX+bY}(t) = M(at, bt) = \exp((a\mu_X + b\mu_Y)t + \frac{1}{2}(a^2\sigma_X^2 + 2ab\rho\sigma_X\sigma_Y + b^2\sigma_Y^2)t^2)$, the mgf of the stated normal.

Exercise 4.46

Central idea. A linear image of independent $n(0,1)$s is bivariate normal with moments read off the coefficients; the coefficient matrix is a square root of the covariance matrix, and square roots are unique only up to an orthogonal factor — infinitely many.

Solution. (a) Direct from $\mathrm{E}Z_i = 0$, $\operatorname{Var} Z_i = 1$, independence: e.g. $\operatorname{Var} X = a_X^2 + b_X^2$, $\operatorname{Cov}(X,Y) = a_X a_Y + b_X b_Y$. (b) $\operatorname{Var} X = \frac{1+\rho}{2}\sigma_X^2 + \frac{1-\rho}{2}\sigma_X^2 = \sigma_X^2$; similarly $\operatorname{Var} Y = \sigma_Y^2$; $\operatorname{Cov}(X,Y) = \frac{1+\rho}{2}\sigma_X\sigma_Y - \frac{1-\rho}{2}\sigma_X\sigma_Y = \rho\sigma_X\sigma_Y$, so $\rho_{XY} = \rho$; means are c_X, c_Y. (c) $M_{X,Y}(s,t) = e^{c_X s + c_Y t}\mathrm{E}[e^{(a_X s + a_Y t)Z_1 + (b_X s + b_Y t)Z_2}] = \exp(\mu_X s + \mu_Y t + \frac{1}{2}(\sigma_X^2 s^2 + 2\rho\sigma_X\sigma_Y st + \sigma_Y^2 t^2))$, the bivariate normal mgf. (d) Another solution: the Cholesky choice $a_X = \sigma_X$, $b_X = 0$, $a_Y = \rho\sigma_Y$, $b_Y = \sigma_Y\sqrt{1-\rho^2}$ (check: $a_X^2 + b_X^2 = \sigma_X^2$, $a_Y^2 + b_Y^2 = \sigma_Y^2$, $a_X a_Y + b_X b_Y = \rho\sigma_X\sigma_Y$). In general the coefficient matrix is $\Sigma^{1/2}Q$ for any 2×2 orthogonal Q — infinitely (a circle's worth of) solutions.

Exercise 4.47

Central idea. $Z = |X|\operatorname{sign}(Y) \overset{d}{=} S|X| \overset{d}{=} X$ since X is symmetric; but $ZY = |X||Y| > 0$ a.s., whereas a nondegenerate bivariate normal charges all four quadrants.

Solution. (a) $Z = X \cdot \operatorname{sign}(XY) = |X| \cdot \operatorname{sign}(Y)$. $\operatorname{sign}(Y) = \pm 1$ with probability $1/2$ each, independent of $|X|$, so $Z \overset{d}{=} S|X| \overset{d}{=} X \sim n(0,1)$. (b) $ZY = |X||Y| > 0$ a.s.: Z and Y always have the same sign. If (Z,Y) were bivariate normal and nondegenerate, each open quadrant would have positive probability; it is degenerate only if $|\mathrm{Corr}| = 1$, but $\operatorname{Corr}(Z,Y) = \mathrm{E}[ZY] = (\mathrm{E}|X|)^2 = 2/\pi \notin \{-1, 1\}$. Hence (Z,Y) is not bivariate normal.

Exercise 4.48

Central idea. In x, the exponent is quadratic — complete the square to read off the normal conditional; for (b) solve $\nabla \log f = 0$ to find three critical points, two of which are local maxima at $(4 \pm \sqrt{15},\, 4 \mp \sqrt{15})$.

Solution. (a) As a function of x, the exponent is $-\frac{1}{2}[(Ay^2+1)x^2 - 2(By+C)x + (y^2 - 2Dy)]$; completing the square gives $\exp(-\frac{Ay^2+1}{2}(x - \frac{By+C}{Ay^2+1})^2)$: $X|Y=y \sim n\left(\frac{By+C}{Ay^2+1}, \frac{1}{Ay^2+1}\right)$. Similarly $Y|X=x \sim n\left(\frac{Bx+D}{Ax^2+1}, \frac{1}{Ax^2+1}\right)$. (b) With $A=1, B=0, C=D=8$: $\nabla \log f = 0$ gives $x(y^2+1)=8$, $y(x^2+1)=8$. Solutions: $x=y \approx 1.83$ (unique positive root of $x^3+x=8$), and $xy=1$, $x+1/x=8$, i.e. $(4+\sqrt{15}, 4-\sqrt{15})$ and $(4-\sqrt{15}, 4+\sqrt{15})$. The Hessian of $\log f$ is $\begin{pmatrix} -(y^2+1) & -2xy \\ -2xy & -(x^2+1) \end{pmatrix}$: at $(4+\sqrt{15}, 4-\sqrt{15})$ it is $\begin{pmatrix} -1.02 & -2 \\ -2 & -63 \end{pmatrix}$ (determinant > 0, trace < 0: local max), likewise by symmetry at $(4-\sqrt{15}, 4+\sqrt{15})$, while $(1.83, 1.83)$ is a saddle (determinant < 0). Two local maxima: the joint distribution is bimodal.

Exercise 4.49

Central idea. This is a two-component mixture of independent pairs; the joint-minus-product-of-marginals factors as $a(1-a)(f_1-f_2)(g_1-g_2)$, and the covariance is the mixture covariance $a(1-a)(\mu_1-\mu_2)(\xi_1-\xi_2)$ — so equal component means ($\mu_1=\mu_2$) with $f_1 \neq f_2$ gives dependence with zero correlation.

Solution. (a) Integrating out y (resp. x) kills the g (resp. f) factors since they integrate to 1: $f_X(x) = af_1(x) + (1-a)f_2(x)$, $f_Y(y) = ag_1(y) + (1-a)g_2(y)$. (b) $f_{X,Y} - f_X f_Y = af_1g_1 + (1-a)f_2g_2 - [af_1 + (1-a)f_2][ag_1 + (1-a)g_2] = a(1-a)(f_1-f_2)(g_1-g_2)$. With $0 < a < 1$, independence holds iff $(f_1-f_2)(g_1-g_2) = 0$ a.e. (c) $\mathrm{E}[XY] = a\mu_1\xi_1 + (1-a)\mu_2\xi_2$, $\mathrm{E}X = a\mu_1 + (1-a)\mu_2$, $\mathrm{E}Y = a\xi_1 + (1-a)\xi_2$; subtracting gives $\mathrm{Cov}(X,Y) = a(1-a)(\mu_1-\mu_2)(\xi_1-\xi_2)$. For dependent but uncorrelated variables: take $\mu_1=\mu_2$ (so Cov $= 0$) but $f_1 \neq f_2$ and $g_1 \neq g_2$ with $\xi_1 \neq \xi_2$ (so $(f_1-f_2)(g_1-g_2) \neq 0$, dependent). (d) Independent: $f_1 = f_2$ (e.g. both binomial$(5, \frac{1}{2})$). Correlated: $f_1 = $ binomial$(5, \frac{1}{4})$, $f_2 = $ binomial$(5, \frac{3}{4})$, $g_1 = $ binomial$(5, \frac{1}{4})$, $g_2 = $ binomial$(5, \frac{3}{4})$ (means differ on both sides). Uncorrelated but dependent: $f_1 = $ binomial$(2, \frac{1}{2})$, $f_2 = $ binomial$(4, \frac{1}{4})$ (both mean 1, different pmfs), $g_1 = $ binomial$(4, \frac{1}{4})$, $g_2 = $ binomial$(4, \frac{3}{4})$ (means $1 \neq 3$).

Exercise 4.50

Central idea. Condition on X: $\mathrm{E}[XY] = \mathrm{E}[X \cdot \mathrm{E}(Y|X)] = \rho\mathrm{E}[X^2]$; and $\mathrm{E}[X^2Y^2] = \mathrm{E}[X^2(\rho^2X^2 + 1 - \rho^2)]$ using $Y|X=x \sim n(\rho x, 1-\rho^2)$.

Solution. Here X, Y are standard normal marginally with $Y|X=x \sim n(\rho x, 1-\rho^2)$. $\mathrm{E}[XY] = \mathrm{E}[X \cdot \rho X] = \rho$, so $\mathrm{Corr}(X,Y) = \rho$. $\mathrm{E}[X^2Y^2] = \mathrm{E}[X^2\mathrm{E}(Y^2|X)] = \mathrm{E}[X^2(\rho^2X^2 + 1 - \rho^2)] = 3\rho^2 + (1-\rho^2) = 1 + 2\rho^2$ (using $\mathrm{E}[X^4] = 3$). With $\mathrm{E}[X^2] = \mathrm{E}[Y^2] = 1$, $\mathrm{Var}(X^2) = 2$: $\mathrm{Corr}(X^2, Y^2) = \frac{1+2\rho^2-1}{2} = \rho^2$.

Exercise 4.51

Central idea. Condition on the denominator: $P(X/Y \le t) = \int_0^1 P(X \le ty)\, dy$; for products use the cdf $P(XY \le t) = t - t\log t$ on $(0,1)$ and iterate.

Solution. (a) $P(X/Y \le t) = \int_0^1 \min(ty, 1)\, dy$: for $0 < t \le 1$ this is $t/2$; for $t > 1$ it is $\int_0^{1/t} ty\, dy + (1 - 1/t) = 1 - \frac{1}{2t}$. $P(XY \le t) = \int_0^1 \min(t/y, 1)\, dy$: for $0 < t < 1$, $t + \int_t^1 \frac{t}{y} dy = t - t\log t$ (and 1 for $t \ge 1$). (b) $P(XY/Z \le t) = \mathrm{E}[P(XY \le tZ)]$. For $0 < t < 1$: $\int_0^1 tz(1 - \log t - \log z)\, dz = \frac{t}{2}(1 - \log t) + \frac{t}{4} = \frac{t(3-2\log t)}{4}$. For $t \ge 1$: $\int_0^{1/t} tz(1 - \log(tz))\, dz + (1 - 1/t) = \frac{3}{4t} + 1 - \frac{1}{t} = 1 - \frac{1}{4t}$.

Exercise 4.52

Central idea. The coordinate differences are iid $n(0,2)$, so the distance is $\sqrt{2}$ times a Rayleigh(1) variable.

Solution. With $(X_1, Y_1), (X_2, Y_2)$ iid, $X_1 - X_2$ and $Y_1 - Y_2$ are iid $n(0,2)$. The distance $D = \sqrt{(X_1-X_2)^2 + (Y_1-Y_2)^2} = \sqrt{2}\sqrt{Z_1^2 + Z_2^2}$ with Z_i iid $n(0,1)$; $\sqrt{Z_1^2+Z_2^2}$ is Rayleigh(1), so $f_D(d) =$

$\frac{d}{2}e^{-d^2/4}$, $d>0$ (Rayleigh with scale $\sqrt{2}$; mean $\sqrt{\pi}$).

Exercise 4.53

Central idea. Real roots $\iff B^2 \geq 4AC$; condition on A, C to get $\mathrm{E}[(1-2\sqrt{AC})_+]$, then $-\log A - \log C \sim$ gamma$(2,1)$ reduces it to a one-dimensional integral.

Solution. Real roots iff the discriminant $B^2 - 4AC \geq 0$. Given A, C, $P(B \geq 2\sqrt{AC}|A,C) = (1-2\sqrt{AC})_+$. Put $S = -\log A - \log C \sim$ gamma$(2,1)$ (rate 1): $\sqrt{AC} = e^{-S/2}$, and

$$P = \int_{\log 4}^{\infty} (1-2e^{-s/2}) s e^{-s}\, ds = \frac{\log 4 + 1}{4} - \left(\frac{\log 4}{6} + \frac{1}{9}\right) = \frac{\log 4}{12} + \frac{5}{36} = \frac{5 + \log 64}{36} \approx 0.254.$$

Exercise 4.54

Central idea. $-\log \prod X_i = \sum(-\log X_i) \sim$ gamma$(n,1)$; the cdf is the Poisson–gamma duality sum, whose derivative telescopes to $(-\log t)^{n-1}/(n-1)!$.

Solution. Let $P = \prod_{i=1}^n X_i$ and $G = -\log P = \sum_{i=1}^n (-\log X_i) \sim$ gamma$(n,1)$ (rate 1). For $0<t<1$, $F_P(t) = P(G \geq -\log t) = \sum_{k=0}^{n-1} \frac{t(-\log t)^k}{k!}$. Differentiating (telescoping sum): $f_P(t) = \frac{(-\log t)^{n-1}}{(n-1)!}$, $0<t<1$.

Exercise 4.55

Central idea. The cdf of a maximum is the product of the cdfs.

Solution. With $T = \max(X_1, X_2, X_3)$ and exponential(λ) in the text's scale convention, $F_T(t) = P(X_1 \leq t)^3 = (1-e^{-t/\lambda})^3$, $t>0$, with pdf $f_T(t) = \frac{3}{\lambda} e^{-t/\lambda}(1-e^{-t/\lambda})^2$.

Exercise 4.56

Central idea. A pool is positive iff at least one member is positive; each pool costs 1 test plus k more with probability $1-(1-p)^k$; as $p \to 0$ the per-person cost under (ii) tends to $1/k < 1$.

Solution. (a) $1-(1-p)^k$ (one minus the probability all k are negative). (b) Per pool, the number of tests is 1 with probability $(1-p)^k$ and $k+1$ otherwise: expected $1 + k(1-(1-p)^k)$. With $m = N/k$ pools, $\mathrm{E}X = m[1 + k(1-(1-p)^k)] = N\left[\frac{1}{k} + 1 - (1-p)^k\right]$. (c) Plan (ii). As $p \to 0$, $\mathrm{E}X/N = 1/k + 1 - (1-p)^k \to 1/k < 1 = \mathrm{E}X_{(i)}/N$: pooling uses about $1/k$ as many tests when positives are rare.

Exercise 4.57

Central idea. $A_r = \exp(\log A_r)$ transfers monotonicity; differentiating $\log A_r = \frac{1}{r}\log(\frac{1}{n}\sum x_i^r)$ and recognizing the weights $a_i = x_i^r / \sum x_j^r$ turns the derivative's sign into "entropy $\leq \log n$" — Jensen on the concave log.

Solution. (a) $A_1 = \frac{1}{n}\sum x_i$ (arithmetic); $A_{-1} = (\frac{1}{n}\sum x_i^{-1})^{-1} = n/\sum(1/x_i)$ (harmonic); $\log A_r = \frac{\log(\frac{1}{n}\sum x_i^r)}{r} \to \frac{d}{dr}\log(\frac{1}{n}\sum x_i^r)\big|_{r=0} = \frac{1}{n}\sum \log x_i$ by L'Hôpital, so $A_0 = (\prod x_i)^{1/n}$ (geometric; take $x_i > 0$). (b)(i) $A_r = e^{\log A_r}$ and e^t is increasing, so monotonicity of $\log A_r$ implies that of A_r. (ii) $\frac{d}{dr}\log A_r = \frac{d}{dr}\left[\frac{1}{r}(\log\sum x_i^r - \log n)\right] = -\frac{1}{r^2}\log\left(\frac{1}{n}\sum x_i^r\right) + \frac{1}{r}\frac{\sum x_i^r \log x_i}{\sum x_i^r}$, as claimed. (iii) With $a_i = x_i^r/\sum_j x_j^r$: $\log n - \sum a_i \log(1/a_i) = \log n + \sum a_i \log a_i = \log n + \sum a_i(r\log x_i - \log\sum x_j^r) = \frac{r\sum x_i^r \log x_i}{\sum x_i^r} - \log(\frac{1}{n}\sum x_i^r)$, the braced quantity. By Jensen, $\sum a_i \log(1/a_i) = \mathrm{E}\log(1/a) \leq \log(\mathrm{E}[1/a]) = \log(\sum a_i/a_i) = \log n$, so the braces are ≥ 0, $\frac{d}{dr}\log A_r \geq 0$, and $A_{-1} \leq A_0 \leq A_1$ is the harmonic–geometric–arithmetic inequality.

Exercise 4.58

Central idea. $\mathrm{E}[XY] = \mathrm{E}[X\,\mathrm{E}(Y|X)]$ by iterated expectation (given X, X is constant) — this makes the residual $Y - \mathrm{E}(Y|X)$ orthogonal to X and gives the variance decomposition.

Solution. (a) $\mathrm{E}[XY] = \mathrm{E}[\mathrm{E}(XY|X)] = \mathrm{E}[X\,\mathrm{E}(Y|X)]$, and $\mathrm{E}[\mathrm{E}(Y|X)] = \mathrm{E}Y$, so $\mathrm{E}[XY] - \mathrm{E}X\,\mathrm{E}Y = \mathrm{E}[X\,\mathrm{E}(Y|X)] - \mathrm{E}X\,\mathrm{E}[\mathrm{E}(Y|X)]$. (b) $\mathrm{Cov}(X, Y - \mathrm{E}(Y|X)) = \mathrm{Cov}(X,Y) - \mathrm{Cov}(X, \mathrm{E}(Y|X)) = 0$ by (a). (c) $\mathrm{E}[Y - \mathrm{E}(Y|X)] = 0$, so $\mathrm{Var}(Y - \mathrm{E}(Y|X)) = \mathrm{E}[(Y - \mathrm{E}(Y|X))^2] = \mathrm{E}[\mathrm{E}[(Y - \mathrm{E}(Y|X))^2|X]] = \mathrm{E}[\mathrm{Var}(Y|X)]$.

Exercise 4.59

Central idea. Expand $\mathrm{E}[\mathrm{Cov}(X,Y|Z)] = \mathrm{E}[XY] - \mathrm{E}[\mathrm{E}(X|Z)\mathrm{E}(Y|Z)]$ by iterated expectation; the subtracted term is exactly what $\mathrm{Cov}(\mathrm{E}(X|Z), \mathrm{E}(Y|Z))$ adds back, leaving $\mathrm{E}[XY] - \mathrm{E}X\,\mathrm{E}Y$.

Solution. $\mathrm{E}[\mathrm{Cov}(X,Y|Z)] = \mathrm{E}[\mathrm{E}(XY|Z) - \mathrm{E}(X|Z)\mathrm{E}(Y|Z)] = \mathrm{E}[XY] - \mathrm{E}[\mathrm{E}(X|Z)\mathrm{E}(Y|Z)]$. And $\mathrm{Cov}(\mathrm{E}(X|Z), \mathrm{E}(Y|Z)) = \mathrm{E}[\mathrm{E}(X|Z)\mathrm{E}(Y|Z)] - \mathrm{E}[\mathrm{E}(X|Z)]\mathrm{E}[\mathrm{E}(Y|Z)] = \mathrm{E}[\mathrm{E}(X|Z)\mathrm{E}(Y|Z)] - \mathrm{E}X\,\mathrm{E}Y$. Adding gives $\mathrm{E}[XY] - \mathrm{E}X\,\mathrm{E}Y = \mathrm{Cov}(X,Y)$.

Exercise 4.60

Central idea. This is the Borel paradox: conditioning on the probability-zero set $\{Y = X\}$ is ambiguous — approaching it via the difference $Y - X$, via the ratio Y/X, or via the indicator $I(Y = X)$ gives different (or undefined) answers.

Solution. Let X, Y be iid $n(0,1)$. (1) $Z_1 = Y - X = 0$: (X, Z_1) is bivariate normal with $\mathrm{Cov}(X, Z_1) = -1$, $\mathrm{Var}\, Z_1 = 2$, so $X|Z_1 = 0 \sim n(0, 1/2)$; on $\{Z_1 = 0\}$, $Y = X$, hence $Y|Z_1 = 0 \sim n(0, 1/2)$. (2) $Z_2 = Y/X = 1$: with $y = xz$, $|J| = |x|$, $f_{X,Z_2}(x,1) = |x|\frac{1}{2\pi}e^{-x^2}$ and $f_{Z_2}(1) = 1/(2\pi)$, so $f_{X|Z_2=1}(x) = |x|e^{-x^2}$; on $\{Z_2 = 1\}$ (minus a null set) $Y = X$, so $Y|Z_2 = 1$ has pdf $f(y) = |y|e^{-y^2}$, $y \in \mathbb{R}$ — different from (1). (3) $Z_3 = I(Y = X) = 1$: $P(Z_3 = 1) = P(Y = X) = 0$, and the indicator specifies no direction of approach, so there is no canonical conditional distribution — conditioning here is undefined (any version is valid). The three "correct" interpretations disagree: the Borel paradox.

Exercise 4.61

Central idea. Under $(x_1, z) = (x_1, (x_2 - 1)/x_1)$ the Jacobian $|J| = x_1$ weights the conditional density toward large x_1, giving $X_1|Z = 0 \sim \text{gamma}(2,1)$ versus $X_1|X_2 = 1 \sim \text{exponential}(1)$; the wedge B_1 widens with x_1 (informative) while the strip B_2 does not.

Solution. (a) With $x_2 = 1 + zx_1$, $|J| = x_1$: $f_{X_1,Z}(x_1, z) = x_1 e^{-x_1} e^{-(1+zx_1)}$ $(1 + zx_1 > 0)$. At $z = 0$: $f_{X_1|Z=0}(x_1) \propto x_1 e^{-x_1}$, i.e. $X_1|Z = 0 \sim \text{gamma}(2,1)$ (mean 2). But $X_1|X_2 = 1 \sim \text{exponential}(1)$ (mean 1) by independence — different distributions from "the same" information. (b) B_1 is a wedge $1 - \epsilon x_1 < x_2 < 1 + \epsilon x_1$ around the ray $x_2 = 1$, widening as x_1 grows; B_2 is the horizontal strip $1 - \epsilon < x_2 < 1 + \epsilon$. $P(B_1|X_1 = x_1) = P(1 - \epsilon x_1 < X_2 < 1 + \epsilon x_1) \approx 2\epsilon x_1 e^{-1}$ grows with x_1 (informative: observing B_1 favors large X_1), while $P(B_2|X_1 = x_1) = P(1 - \epsilon < X_2 < 1 + \epsilon)$ is constant in x_1 (uninformative, by independence). (c) $P(X_1 \le x|B_1) = \frac{\int_0^x (e^{-(1-\epsilon t)} - e^{-(1+\epsilon t)})e^{-t}dt}{\int_0^\infty (e^{-(1-\epsilon t)} - e^{-(1+\epsilon t)})e^{-t}dt} \to \frac{\int_0^x te^{-t}dt}{\int_0^\infty te^{-t}dt} = \int_0^x te^{-t}dt$ as $\epsilon \to 0$ (using $e^{-(1-\epsilon t)} - e^{-(1+\epsilon t)} \sim 2\epsilon t e^{-1}$): the gamma$(2,1)$ cdf, agreeing with (a). $P(X_1 \le x|B_2) = P(X_1 \le x)$ by independence $\to 1 - e^{-x}$, the exponential(1) cdf, agreeing with $X_1|X_2 = 1$.

Exercise 4.62

Central idea. The tangent line lies strictly below g except at the point of tangency, so $\mathrm{E}[g(X) - (a + bX)] > 0$ whenever X takes other values with positive probability.

Solution. Since $a + bx$ is tangent at $\mathrm{E}X$, $a + b\mathrm{E}X = g(\mathrm{E}X)$. Then $\mathrm{E}g(X) - g(\mathrm{E}X) = \mathrm{E}[g(X) - a - bX]$. The integrand is ≥ 0 everywhere and > 0 when $X \ne \mathrm{E}X$; if $P(X \ne \mathrm{E}X) > 0$ its expectation is > 0, giving $\mathrm{E}g(X) > g(\mathrm{E}X)$. Equality holds only if $P(X = \mathrm{E}X) = 1$.

Exercise 4.63

Central idea. Jensen on the convex exponential: $\mathrm{E}X = \mathrm{E}[e^Z] \ge e^{\mathrm{E}Z} = 1$, strict unless Z (hence X) is degenerate.

Solution. $EX = E[e^Z] \geq e^{EZ} = e^0 = 1$ by Jensen's inequality (e^t convex). So $EX \geq 1$, with equality iff $P(Z = 0) = 1$, i.e. $X = 1$ a.s.; otherwise $EX > 1$.

Exercise 4.64

Central idea. Square both sides for the pointwise inequality, then take expectations and use monotonicity of E.

Solution. (a) $|a+b|^2 = a^2+2ab+b^2 \leq a^2+2|ab|+b^2 = (|a|+|b|)^2$; taking square roots gives $|a+b| \leq |a|+|b|$. (b) Pointwise $|X+Y| \leq |X|+|Y|$; taking expectations and using Exercise 4.2(c): $E|X+Y| \leq E[|X|+|Y|] = E|X| + E|Y|$.

Exercise 4.65

Central idea. The text proves $|\text{Corr}| \leq 1$ from $0 \leq \text{Var}(X/\sigma_X \pm Y/\sigma_Y)$; the same idea with arbitrary coefficients — $0 \leq \text{Var}(tX + Y)$ as a quadratic in t — gives $|\text{Cov}(X, Y)| \leq \sigma_X\sigma_Y$ via the nonpositive discriminant.

Solution. For any real t, $0 \leq \text{Var}(tX+Y) = t^2\text{Var}\,X+2t\,\text{Cov}(X, Y)+\text{Var}\,Y$. A quadratic $at^2+bt+c \geq 0$ for all t has discriminant $b^2-4ac \leq 0$, so $4[\text{Cov}(X, Y)]^2-4\text{Var}\,X\,\text{Var}\,Y \leq 0$, i.e. $|\text{Cov}(X, Y)| \leq \sqrt{\text{Var}\,X\,\text{Var}\,Y} = \sigma_X\sigma_Y$.

Chapter 5

Exercise 5.1

Central idea. Work with the complement: $P(\text{at least one}) = 1 - P(\text{none}) = 1 - 0.99^n$.

Solution. With replacement, the number of color-blind people in the sample is Binomial$(n, 0.01)$. We need $1-0.99^n \geq 0.95$, i.e., $0.99^n \leq 0.05$, so $n \geq \log(0.05)/\log(0.99) \approx 298.07$. Hence $n = 299$ (the smallest integer doing the job).

Exercise 5.2

Central idea. For $N = n$ we need X_1 to be the largest of $X_1, \ldots, X_n$ but not of $X_1, \ldots, X_{n+1}$ — i.e., X_1 ranks exactly second among the first $n + 1$ observations, whose ranks are all equally likely.

Solution. (a) Let $N = \min\{n \geq 1 : X_{n+1} > X_1\}$ (with $N = 1$ if $X_2 > X_1$). Then

$$\{N = n\} = \{X_2 < X_1, \ldots, X_n < X_1, X_{n+1} > X_1\},$$

i.e., among $X_1, \ldots, X_{n+1}$, the largest is X_{n+1} and the second largest is X_1. Since the X_i are iid continuous, all $(n+1)!$ orderings of $X_1, \ldots, X_{n+1}$ are equally likely, and $(n-1)!$ of them have X_{n+1} largest and X_1 second largest. Hence

$$P(N = n) = \frac{(n-1)!}{(n+1)!} = \frac{1}{n(n+1)}, \qquad n = 1, 2, \ldots$$

(This sums to 1 since $\sum_{n\geq 1}[1/n - 1/(n+1)] = 1$, so X_1 is exceeded eventually with probability 1.) (b) $E[N] = \sum_{n=1}^{\infty} n \cdot \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty$, the harmonic series. The waiting time is almost surely finite but has infinite mean — a few enormous droughts dominate the average.

Exercise 5.3

Central idea. Each Y_i is Bernoulli; iid X_i give iid Y_i, so the sum is binomial — but the success probability is $P(X_1 > \mu)$, which is *not* 1/2 in general (mean $\neq$ median).

Solution. The Y_i are iid (functions of iid X_i) with $P(Y_i = 1) = P(X_i > \mu) =: p$ and $P(Y_i = 0) = 1 - p$. Therefore

$$\sum_{i=1}^{n} Y_i \sim \text{Binomial}(n, p), \qquad p = P(X_1 > \mu) = 1 - F_X(\mu).$$

In particular p need not equal $1/2$: $EX_1 = \mu$ says nothing about the median. (If the distribution is symmetric about μ, then $p = 1/2$ by continuity.)

Exercise 5.4

Central idea. Integrating out the random P gives a joint pmf depending on $(x_1, \dots, x_k)$ only through the sum t — hence permutation-invariant — but the X_i remain positively correlated through the shared P.

Solution. (a) Conditioning on $P = p$ and integrating over the uniform prior (density 1 on $(0, 1)$):

$$P(X_1 = x_1, \dots, X_k = x_k) = \int_0^1 \prod_{i=1}^{k} p^{x_i}(1-p)^{1-x_i}\, dp = \int_0^1 p^t (1-p)^{k-t}\, dp,$$

with $t = \sum x_i$. This is the Beta integral $B(t+1, k-t+1) = \frac{\Gamma(t+1)\Gamma(k-t+1)}{\Gamma(k+2)} = \frac{t!(k-t)!}{(k+1)!}$. The right side depends on $(x_1, \dots, x_k)$ only through t, hence is unchanged by permuting the coordinates; the same holds for any k-subset. So the Xs are exchangeable. (b) Each marginal is $P(X_i = 1) = \int_0^1 p\, dp = 1/2$, so $\prod_{i=1}^{n} P(X_i = x_i) = 2^{-n}$. But the joint is $t!(n-t)!/(n+1)!$ with $t = \sum x_i$; e.g., for $n = 2$, $x_1 = x_2 = 1$: joint $= 1/3 \neq 1/4$. Hence not independent (they are positively correlated: a large P makes all X_i simultaneously likely to be 1).

Exercise 5.5

Central idea. This is just the change-of-variables formula for the linear map $Y = n\bar{X}$; no mgf is needed.

Solution. Let $Y = \sum_{i=1}^{n} X_i = n\bar{X}$, so $\bar{X} = Y/n$. By the univariate transformation formula (Theorem 2.1.5), with $y = nx$,

$$f_{\bar{X}}(x) = f_Y(nx) \left| \frac{d}{dx}(nx) \right| = n\, f_{X_1 + \cdots + X_n}(nx).$$

Equivalently via cdfs: $P(\bar{X} \le x) = P(Y \le nx)$, and differentiating gives the same result. No mgf is used.

Exercise 5.6

Central idea. Write $F_Z(z)$ as an iterated integral over the joint density $f_X(x)f_Y(y)$, then differentiate under the integral sign (cf. the convolution formula (5.2.3)).

Solution. (a) $F_Z(z) = P(X - Y \le z) = \int_{-\infty}^{\infty} P(X \le z+y) f_Y(y)\, dy = \int_{-\infty}^{\infty} F_X(z+y) f_Y(y)\, dy$. Differentiating,

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(z+y) f_Y(y)\, dy = \int_{-\infty}^{\infty} f_X(x) f_Y(x-z)\, dx$$

(substituting $x = z + y$). (b) For z fixed, $P(XY \le z) = \int_{-\infty}^{\infty} P(X \le z/y) f_Y(y)\, dy$ requires care with the sign of y; differentiating the joint integral $\iint_{xy \le z} f_X(x) f_Y(y)\, dx\, dy$ with $u = xy$ gives

$$f_Z(z) = \int_{-\infty}^{\infty} \frac{1}{|y|} f_X(z/y) f_Y(y)\, dy = \int_{-\infty}^{\infty} \frac{1}{|x|} f_X(x) f_Y(z/x)\, dx.$$

(c) Similarly, with $u = x/y$,

$$f_Z(z) = \int_{-\infty}^{\infty} |y|\, f_X(zy) f_Y(y)\, dy.$$

(These are the standard difference/product/ratio convolution formulas; (5.2.3) is the special case $Z = X + Y$.)

Exercise 5.7

Central idea. Clear denominators to get a cubic polynomial identity in w; the four coefficients give four linear equations for A, B, C, D. The decomposition is engineered so the divergent log terms cancel ($A\sigma^2 = C\tau^2$), leaving arctan terms that assemble into the Cauchy$(0, \sigma+\tau)$ density.

Solution. (a) Multiplying through by $(w^2+\sigma^2)((z-w)^2+\tau^2)$ gives the polynomial identity

$$\sigma^2\tau^2 = \sigma^2(Aw+B)((z-w)^2+\tau^2) - \tau^2(Cw+D)(w^2+\sigma^2).$$

Equating coefficients of w^3, w^2, w^1, w^0:

$$\begin{aligned} \sigma^2 A - \tau^2 C &= 0, \\ \sigma^2(B-2Az) - \tau^2 D &= 0, \\ A(z^2+\tau^2-\sigma^2) - 2Bz &= 0, \\ B(z^2+\tau^2) - \tau^2 D &= \tau^2. \end{aligned}$$

With $u = z^2+\tau^2-\sigma^2$ and $\Delta = u^2+4\sigma^2z^2 = (z^2+(\sigma+\tau)^2)(z^2+(\sigma-\tau)^2)$, the solution is

$$A = \frac{2\tau^2 z}{\Delta}, \quad B = \frac{\tau^2 u}{\Delta}, \quad C = \frac{2\sigma^2 z}{\Delta}, \quad D = \frac{\sigma^2(u-4z^2)}{\Delta} = \frac{\sigma^2(\tau^2-\sigma^2-3z^2)}{\Delta}.$$

(b) The convolution is $f_W(z) = \frac{1}{\pi^2\sigma\tau}\int_{-\infty}^{\infty} \frac{dw}{[1+(w/\sigma)^2][1+((z-w)/\tau)^2]}$. Termwise antiderivatives:

$$\int \frac{Aw\,dw}{1+(w/\sigma)^2} = \tfrac{A\sigma^2}{2}\log\!\big(1+\tfrac{w^2}{\sigma^2}\big), \quad \int \frac{B\,dw}{1+(w/\sigma)^2} = B\sigma\arctan\!\big(\tfrac{w}{\sigma}\big),$$

$$\int \frac{Cw\,dw}{1+((z-w)/\tau)^2} = -C\tau z\arctan\!\big(\tfrac{z-w}{\tau}\big)+\tfrac{C\tau^2}{2}\log\!\Big(1+\tfrac{(z-w)^2}{\tau^2}\Big), \quad \int \frac{D\,dw}{1+((z-w)/\tau)^2} = -D\tau\arctan\!\big(\tfrac{z-w}{\tau}\big).$$

The two log terms are each logarithmically divergent, but their difference converges because $A\sigma^2 = C\tau^2$ (from $\sigma^2 A = \tau^2 C$), so the $\log|w|$ pieces cancel. Evaluating from $-R$ to R and letting $R\to\infty$ (using $\arctan(t)\to\pm\pi/2$), the integral equals $\pi(B\sigma - C\tau z - D\tau)$. Substituting the constants,

$$B\sigma - C\tau z - D\tau = \frac{\sigma\tau(\sigma+\tau)}{z^2+(\sigma+\tau)^2},$$

hence $f_W(z) = \frac{1}{\pi^2\sigma\tau}\cdot\pi(B\sigma - C\tau z - D\tau) = \frac{\sigma+\tau}{\pi\left(z^2+(\sigma+\tau)^2\right)}$, the Cauchy$(0,\sigma+\tau)$ pdf — verifying (5.2.5): the sum of independent Cauchy$(0,\sigma)$ and Cauchy$(0,\tau)$ is Cauchy$(0,\sigma+\tau)$.

Exercise 5.8

Central idea. (a) is algebra: double-centering the pairwise differences. For (b)–(c), center ($Y_i = X_i - \theta_1$) and expand moments of $\sum Y_i^2 - n\bar{Y}^2$, using independence to kill every term with an unmatched Y_i.

Solution. (a) $\sum_{i,j}(X_i-X_j)^2 = \sum_{i,j}\big[(X_i-\bar{X})-(X_j-\bar{X})\big]^2 = 2n\sum_i(X_i-\bar{X})^2 - 2\big(\sum_i(X_i-\bar{X})\big)^2 = 2n(n-1)S^2$, since $\sum_i(X_i-\bar{X}) = 0$. Dividing gives the result. (b) Put $Y_i = X_i - \theta_1$ (so $EY_i = 0$); then $S^2 = \frac{1}{n-1}(A-B)$ with $A = \sum Y_i^2$, $B = n\bar{Y}^2 = (\sum Y_i)^2/n$. Now $E[A^2] = n\theta_4 + n(n-1)\theta_2^2$, $E[AB] = \theta_4 + (n-1)\theta_2^2$ (the cross terms $E[(\sum_i Y_i^2)Y_jY_k]$, $j\neq k$, vanish since $EY_i = 0$), and $E[B^2] = \frac{1}{n^2}E[(\sum Y_i)^4] = \frac{1}{n^2}(n\theta_4 + 3n(n-1)\theta_2^2) = \frac{\theta_4}{n} + \frac{3(n-1)}{n}\theta_2^2$. Hence

$$E[(A-B)^2] = \frac{(n-1)^2}{n}\theta_4 + \frac{(n-1)(n^2-2n+3)}{n}\theta_2^2,$$

and subtracting $(E[A-B])^2 = (n-1)^2\theta_2^2$ gives $\mathrm{Var}(A-B) = \frac{n-1}{n}[(n-1)\theta_4 - (n-3)\theta_2^2]$. Dividing by $(n-1)^2$:

$$\mathrm{Var}\,S^2 = \frac{1}{n}\left(\theta_4 - \frac{n-3}{n-1}\,\theta_2^2\right).$$

(c) $\mathrm{Cov}(\bar{X}, S^2) = E[\bar{Y}S^2]$ (since $E\bar{Y} = 0$). Now $E[\bar{Y}\sum_j Y_j^2] = \frac{1}{n}E[(\sum_i Y_i)(\sum_j Y_j^2)] = \frac{1}{n}\cdot n\theta_3 = \theta_3$ (only $i = j$ terms survive), and $E[n\bar{Y}^3] = n\cdot\theta_3/n^2 = \theta_3/n$ (since $E[(\sum Y_i)^3] = n\theta_3$). Thus

$$\mathrm{Cov}(\bar{X}, S^2) = \frac{1}{n-1}\left(\theta_3 - \frac{\theta_3}{n}\right) = \frac{\theta_3}{n}.$$

So $\mathrm{Cov}(\bar{X}, S^2) = 0$ iff $\theta_3 = E(X_1 - \theta_1)^3 = 0$, e.g., whenever the population is symmetric about its mean (symmetry is sufficient, not necessary).

Exercise 5.9

Central idea. The difference of the two sides is a sum of squares, hence ≥ 0 (Cauchy–Schwarz); equality forces every 2×2 minor $a_ib_j - a_jb_i$ to vanish, i.e., all centered points proportional — collinearity.

Solution. Both sides expand to $\sum_{i,j}(a_i^2b_j^2 - a_ib_ia_jb_j)$: indeed,

$$\sum_{i,j}(a_i^2b_j^2 - a_ib_ia_jb_j) = \tfrac{1}{2}\sum_{i,j}(a_i^2b_j^2 + a_j^2b_i^2 - 2a_ib_ia_jb_j) = \tfrac{1}{2}\sum_{i,j}(a_ib_j - a_jb_i)^2 = \sum_{i<j}(a_ib_j - a_jb_i)^2,$$

since diagonal terms vanish and (i,j), (j,i) pairs coincide. (For $n = 2$ this is $(a_1^2+a_2^2)(b_1^2+b_2^2) - (a_1b_1+a_2b_2)^2 = (a_1b_2 - a_2b_1)^2$, and the general case follows by induction on n.) For the sample correlation, apply the identity to the centered values $a_i = x_i - \bar{x}$, $b_i = y_i - \bar{y}$: with $r = \sum a_ib_i/\sqrt{\sum a_i^2\sum b_i^2}$,

$$1 - r^2 = \frac{\sum_{i<j}(a_ib_j - a_jb_i)^2}{(\sum a_i^2)\,(\sum b_i^2)} \geq 0.$$

Thus $|r| = 1$ iff $a_ib_j - a_jb_i = 0$ for all $i < j$, i.e., (a_i, b_i) is proportional to (a_j, b_j) for all i, j — all points (x_i, y_i) lie on the line $y - \bar{y} = c(x - \bar{x})$ for some c. (The hint's "$= 1$" is $|r| = 1$ up to sign; $r = \pm 1$ both give straight lines.)

Exercise 5.10

Central idea. For a normal population the central moments are $\theta_2 = \sigma^2$, $\theta_3 = 0$, $\theta_4 = 3\sigma^4$; both routes must give $2\sigma^4/(n-1)$.

Solution. (a) $\theta_1 = \mu$, $\theta_2 = \sigma^2$, $\theta_3 = 0$ (symmetry), $\theta_4 = 3\sigma^4$ (standard normal fourth moment). (b) From 5.8(b): $\mathrm{Var}\,S^2 = \frac{1}{n}\left(3\sigma^4 - \frac{n-3}{n-1}\sigma^4\right) = \frac{\sigma^4}{n}\cdot\frac{3(n-1)-(n-3)}{n-1} = \frac{\sigma^4}{n}\cdot\frac{2n}{n-1} = \frac{2\sigma^4}{n-1}$. (c) $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ has variance $2(n-1)$, so $\mathrm{Var}\,S^2 = \frac{\sigma^4}{(n-1)^2}\cdot 2(n-1) = \frac{2\sigma^4}{n-1}$, agreeing with (b). (Also 5.8(c) gives $\mathrm{Cov}(\bar{X}, S^2) = \theta_3/n = 0$ here — the familiar normal independence in covariance form.)

Exercise 5.11

Central idea. Jensen's inequality for the strictly concave square root: $E\sqrt{S^2} \leq \sqrt{ES^2}$, with strict inequality unless S^2 is a.s. constant — and a non-degenerate sample forces S^2 to take both the value 0 and positive values with positive probability.

Solution. By Jensen, $ES = E\sqrt{S^2} \leq \sqrt{E[S^2]} = \sqrt{\sigma^2} = \sigma$, since $\sqrt{\cdot}$ is concave. For strict concavity, equality holds iff S^2 is a.s. constant. Suppose $\sigma^2 > 0$ and, to get a contradiction, $S^2 = c$ a.s. Then $c = ES^2 = \sigma^2 > 0$. But $E[S^2 \mid X_1] = \sigma^2$ a.s. is a quadratic equation in X_1 (expanding $S^2 = \frac{1}{n-1}(\sum X_i^2 - n\bar{X}^2)$ and conditioning, the X_1^2 coefficient is $1/n \neq 0$), so X_1 takes at most two values a.s. With X_1 non-degenerate two-valued, $S^2 = 0$ on the all-equal event (positive probability) and $S^2 > 0$ on $\{X_1 \neq X_2\}$ (positive probability) — not a.s. constant, a contradiction. Hence S^2 is non-degenerate and $ES < \sigma$: the sample standard deviation is always biased downward.

Exercise 5.12

Central idea. Y_1 is the absolute value of a $n(0, 1/n)$ variable; Y_2 averages iid $|X_i|$. The triangle inequality $|\sum X_i| \le \sum |X_i|$ gives $Y_1 \le Y_2$ pointwise.

Solution. $\bar{X} \sim n(0, 1/n)$, so $Y_1 = |\bar{X}| \overset{d}{=} |Z|/\sqrt{n}$ with $Z \sim n(0, 1)$. Since $E|Z| = \sqrt{2/\pi}$,

$$EY_1 = \frac{1}{\sqrt{n}}\sqrt{\frac{2}{\pi}} = \sqrt{\frac{2}{\pi n}}.$$

By linearity, $EY_2 = E|X_1| = \sqrt{2/\pi}$. By the triangle inequality, $Y_1 = \frac{1}{n}|\sum X_i| \le \frac{1}{n}\sum |X_i| = Y_2$ a.s., hence

$$EY_1 = \sqrt{\frac{2}{\pi n}} \le \sqrt{\frac{2}{\pi}} = EY_2,$$

strict for $n > 1$. (Averaging before taking absolute values cancels signs and shrinks the mean.)

Exercise 5.13

Central idea. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and $E\sqrt{\chi^2_{n-1}} = \sqrt{2}\,\Gamma(n/2)/\Gamma((n-1)/2)$; choose c to cancel this factor.

Solution. Take $g(S^2) = cS$. With $W = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$,

$$E[S] = \frac{\sigma}{\sqrt{n-1}}E[\sqrt{W}] = \frac{\sigma}{\sqrt{n-1}}\int_0^\infty \sqrt{w}\,\frac{w^{(n-1)/2-1}e^{-w/2}}{2^{(n-1)/2}\Gamma((n-1)/2)}\,dw = \sigma\sqrt{\frac{2}{n-1}}\,\frac{\Gamma(n/2)}{\Gamma((n-1)/2)}.$$

Thus $g(S^2) = cS$ with

$$c = \sqrt{\frac{n-1}{2}}\,\frac{\Gamma((n-1)/2)}{\Gamma(n/2)}$$

satisfies $Eg(S^2) = \sigma$: an unbiased estimator of σ (necessarily a slight inflation of S, since $ES < \sigma$ by 5.11).

Exercise 5.14

Central idea. Standardizing ($X_j = \sigma_j Z_j + \mu_j$) turns each linear form into an affine function of a linear form in iid $n(0, 1)$ variables; covariances are shift-invariant and scale predictably, and independence is preserved under affine maps.

Solution. (a) $\sum_j a_{ij}X_j = \sum_j (a_{ij}\sigma_j)Z_j + \sum_j a_{ij}\mu_j$: a linear form in the Z's plus a constant (similarly for the b's). Constants don't affect covariance or independence, and rescaling coefficients $a_{ij} \to a_{ij}\sigma_j$ is absorbed into the "arbitrary constants" of the special case. Hence zero covariance of the X-forms gives zero covariance of the corresponding Z-forms, which are then independent by the special case, and adding back constants preserves independence. (b) By bilinearity and independence of the X_j,

$$\operatorname{Cov}\left(\sum_j a_{ij}X_j, \sum_r b_{rj}X_r\right) = \sum_{j,r} a_{ij}b_{rj}\operatorname{Cov}(X_j, X_r) = \sum_{j=1}^n a_{ij}b_{ij}\,\sigma_j^2,$$

since $\operatorname{Cov}(X_j, X_r) = 0$ for $j \ne r$ and $= \sigma_j^2$ for $j = r$.

Exercise 5.15

Central idea. Write the new sum of squares about $\bar{X}_{n+1}$ by recentering the old sum of squares (about $\bar{X}_n$) — the cross term vanishes because deviations from $\bar{X}_n$ sum to zero.

Solution. (a) Immediate: $\bar{X}_{n+1} = \frac{1}{n+1}\sum_{i=1}^{n+1} X_i = \frac{n\bar{X}_n + X_{n+1}}{n+1}$. (b) $nS^2_{n+1} = \sum_{i=1}^{n+1}(X_i - \bar{X}_{n+1})^2$. For $i \le n$,

$$(X_i - \bar{X}_{n+1})^2 = (X_i - \bar{X}_n)^2 + 2(X_i - \bar{X}_n)(\bar{X}_n - \bar{X}_{n+1}) + (\bar{X}_n - \bar{X}_{n+1})^2,$$

and summing over $i \le n$ the middle term drops ($\sum_{i\le n}(X_i - \bar{X}_n) = 0$), giving $(n-1)S_n^2 + n(\bar{X}_n - \bar{X}_{n+1})^2$. Adding $i = n+1$ and using $\bar{X}_{n+1} - \bar{X}_n = \frac{X_{n+1} - \bar{X}_n}{n+1}$ (from (a)):

$$nS_{n+1}^2 = (n-1)S_n^2 + \left[n \cdot \frac{1}{(n+1)^2} + \frac{n^2}{(n+1)^2}\right](X_{n+1} - \bar{X}_n)^2 = (n-1)S_n^2 + \frac{n}{n+1}(X_{n+1} - \bar{X}_n)^2.$$

(One-pass updating of mean and variance without storing all data.)

Exercise 5.16

Central idea. Standardize first: $Z_i = (X_i - i)/i \sim$ iid $n(0,1)$; then use the definitions $\chi_p^2 = \sum Z_i^2$, $t_p = Z/\sqrt{\chi_p^2/p}$, $F_{p,q} = (\chi_p^2/p)/(\chi_q^2/q)$.

Solution. Put $Z_i = (X_i - i)/i$; the Z_i are iid $n(0,1)$. (a) $Z_1^2 + Z_2^2 + Z_3^2 = \sum_{i=1}^3 \left(\frac{X_i - i}{i}\right)^2 \sim \chi_3^2$. (b) $\dfrac{Z_1}{\sqrt{(Z_2^2 + Z_3^2)/2}} \sim t_2$ (numerator $n(0,1)$ independent of denominator $\sqrt{\chi_2^2/2}$). (c) $\dfrac{Z_1^2/1}{(Z_2^2 + Z_3^2)/2} = \dfrac{2(X_1 - 1)^2}{\left(\frac{X_2 - 2}{2}\right)^2 + \left(\frac{X_3 - 3}{3}\right)^2} \sim F_{1,2}$.

Exercise 5.17

Central idea. Write $X = (U/p)/(V/q)$ with $U \sim \chi_p^2 \perp V \sim \chi_q^2$; moments come from $E[V^{-k}]$ of a chi-square, and (d) is the classic $U/(U+V) \sim$ Beta fact in disguise.

Solution. Write $X = \frac{U/p}{V/q}$, $U \sim \chi_p^2 \perp V \sim \chi_q^2$. (a) Transforming $(U, V) \to (X, V)$ ($u = pxv/q$, Jacobian pv/q) and integrating out v:

$$f_X(x) = \frac{\Gamma((p+q)/2)}{\Gamma(p/2)\Gamma(q/2)} \left(\frac{p}{q}\right)^{p/2} x^{p/2-1} \left(1 + \frac{p}{q}x\right)^{-(p+q)/2}, \quad x > 0.$$

(b) $E[X] = E[U/p]\,E[q/V] = 1 \cdot q \cdot E[V^{-1}]$. For $V \sim \chi_q^2$, $E[V^{-k}] = 1/[(q-2)(q-4)\cdots(q-2k)]$ ($q > 2k$), so $E[X] = \frac{q}{q-2}$ ($q > 2$). Similarly $E[X^2] = E[(U/p)^2]E[(q/V)^2] = \frac{p+2}{p} \cdot \frac{q^2}{(q-2)(q-4)}$, giving

$$\operatorname{Var} X = \frac{2q^2(p+q-2)}{p(q-2)^2(q-4)}, \qquad q > 4.$$

(c) $1/X = (V/q)/(U/p)$ is exactly the F construction with p, q swapped: $1/X \sim F_{q,p}$. (d) $\frac{(p/q)X}{1+(p/q)X} = \frac{U/V}{1+U/V} = \frac{U}{U+V} \sim \text{Beta}(p/2, q/2)$, since for independent χ^2 variables the ratio $U/(U+V)$ is Beta (Example 4.5.4/Theorem 4.4.7).

Exercise 5.18

Central idea. $X = Z/\sqrt{V/p}$ with $Z \sim n(0,1) \perp V \sim \chi_p^2$; as $p \to \infty$, $V/p \to 1$ in probability, so the t collapses to the normal and $qF_{q,p} = U/(V/p) \to \chi_q^2$.

Solution. (a) $E[X] = E[Z]\,E[(V/p)^{-1/2}] = 0$ for $p > 1$ (for $p = 1$, Cauchy, no mean). $E[X^2] = E[Z^2]E[p/V] = p/(p-2)$ for $p > 2$, so $\operatorname{Var} X = p/(p-2)$ ($p > 2$). (b) $X^2 = \dfrac{Z^2/1}{V/p} \sim F_{1,p}$ by definition of F ($Z^2 \sim \chi_1^2 \perp V$). (c) $f(x|p) = \frac{\Gamma((p+1)/2)}{\sqrt{p\pi}\,\Gamma(p/2)} \left(1 + \frac{x^2}{p}\right)^{-(p+1)/2}$. By Stirling, $\frac{\Gamma(p/2+1/2)}{\Gamma(p/2)} \sim (p/2)^{1/2}$, so the constant $\to \frac{1}{\sqrt{\pi}} \cdot \frac{(p/2)^{1/2}}{\sqrt{p}} = \frac{1}{\sqrt{2\pi}}$; and $(1 + x^2/p)^{-(p+1)/2} \to e^{-x^2/2}$. Hence $f(x|p) \to \phi(x)$ pointwise. (d) $X^2 \sim F_{1,p} = \frac{\chi_1^2/1}{\chi_p^2/p}$; since $\chi_p^2/p \to 1$ in probability (WLLN), Slutsky gives $X^2 \to_d \chi_1^2/1 = \chi_1^2$. (e) $qF_{q,p} = \frac{\chi_q^2}{\chi_p^2/p} \to_d \chi_q^2$: conjecture $qF_{q,p} \to_d \chi_q^2$ as $p \to \infty$ (the F test statistic becomes the chi-square statistic in large samples).

Exercise 5.19

Central idea. Couple the variables on one probability space so the inequality holds pointwise: $\chi^2_p \overset{d}{=} \chi^2_q + \chi^2_{p-q} \geq \chi^2_q$ a.s.; the same coupling with a shared denominator lifts stochastic monotonicity to $kF_{k,v}$ and then to the scaled cutoffs.

Solution. (a) By gamma additivity, $\chi^2_p \overset{d}{=} \chi^2_q + \chi^2_{p-q}$ with independent summands, and $\chi^2_{p-q} > 0$ a.s. Hence, on a common space, $\chi^2_p > \chi^2_q$ a.s., so $P(\chi^2_p > a) \geq P(\chi^2_q > a)$ for all a, with strict inequality e.g. for a between typical values (indeed $P(\chi^2_p > a) - P(\chi^2_q > a) = P(\chi^2_q \leq a < \chi^2_q + \chi^2_{p-q}) > 0$ for every $a > 0$, since all densities are positive on $(0, \infty)$). (b) $kF_{k,v} \overset{d}{=} \chi^2_k/(\chi^2_v/v)$. Couple nested numerators $\chi^2_1 \subset \chi^2_2 \subset \cdots$ (partial sums of iid χ^2_1) with a single independent denominator $W \sim \chi^2_v$: then $kF_{k,v} \overset{d}{=} U_k/(W/v)$ is a.s. increasing in k, hence stochastically increasing. (c) Let $c_k = kF_{\alpha,k,v}$, so $P(kF_{k,v} > c_k) = \alpha$. If $c_k \leq c_{k-1}$, then $\alpha = P(kF_{k,v} > c_k) \geq P(kF_{k,v} > c_{k-1}) > P((k-1)F_{k-1,v} > c_{k-1}) = \alpha$, where the strict inequality is the strict stochastic dominance from (b) at the fixed point $c_{k-1} > 0$ (same coupling argument as in (a): the increment is a.s. positive with density support $(0, \infty)$). Contradiction; thus $c_k > c_{k-1}$.

Exercise 5.20

Central idea. Differentiating a mixture cdf brings down the inner density; the resulting $\int_0^\infty y^{a-1}e^{-by}\,dy$ is a gamma integral that collapses to exactly the t (resp. F) pdf.

Solution. (a) The integrand is $\frac{1}{\sqrt{2\pi v}}\frac{1}{\Gamma(v/2)2^{v/2}}\,x^{(v+1)/2-1}\exp\!\big[-\frac{x}{2}(1+t^2/v)\big]$. Integrating over $x > 0$ (gamma integral):

$$\int_0^\infty = \frac{1}{\sqrt{2\pi v}\,\Gamma(v/2)2^{v/2}} \cdot \Gamma\big(\tfrac{v+1}{2}\big)\left(\frac{2}{1+t^2/v}\right)^{(v+1)/2} = \frac{\Gamma((v+1)/2)}{\sqrt{\pi v}\,\Gamma(v/2)}\left(1+\frac{t^2}{v}\right)^{-(v+1)/2},$$

the t_v pdf. (b) Differentiating in t: $v\,f_{F_{1,v}}(vt) = \int_0^\infty y\,f_{\chi^2_1}(ty)\,f_v(y)\,dy$. The integrand is $\frac{t^{-1/2}}{\sqrt{2\pi}\,\Gamma(v/2)2^{v/2}}\,y^{(v+1)/2-1}e^{-y(t+1)/2}$; integrating gives $v\,f_{F_{1,v}}(vt) = \frac{\Gamma((v+1)/2)}{\sqrt{\pi}\,\Gamma(v/2)}\,t^{-1/2}(1+t)^{-(v+1)/2}$. With $s = vt$,

$$f_{F_{1,v}}(s) = \frac{\Gamma((v+1)/2)}{\sqrt{\pi}\,\Gamma(v/2)}\,v^{-1/2}s^{-1/2}\left(1+\frac{s}{v}\right)^{-(v+1)/2},$$

the $F_{1,v}$ pdf. (c) The mixture identity holds because $F_{m,v} = \frac{\chi^2_m/m}{\chi^2_v/v}$, so $\{F_{m,v} \leq \frac{v}{m}t\} = \{\chi^2_m \leq t\chi^2_v\}$ and conditioning on $\chi^2_v = y$ gives the integral. Differentiating,

$$\tfrac{v}{m}f_{F_{m,v}}(\tfrac{v}{m}t) = \int_0^\infty y\,f_{\chi^2_m}(ty)\,f_v(y)\,dy = \frac{t^{m/2-1}\Gamma((m+v)/2)}{\Gamma(m/2)\Gamma(v/2)}(1+t)^{-(m+v)/2},$$

again a gamma integral; with $s = \frac{v}{m}t$ this is the $F_{m,v}$ pdf $\frac{\Gamma((m+v)/2)}{\Gamma(m/2)\Gamma(v/2)}\,(\frac{m}{v})^{m/2}\,s^{m/2-1}\,(1+\frac{m}{v}s)^{-(m+v)/2}$.

Exercise 5.21

Central idea. The max exceeds the median unless *all* observations fall below it; each observation is below the median with probability $1/2$, independently.

Solution. Let m be the population median, $P(X_i \leq m) = 1/2$. Then $P(\max(X_1, X_2) > m) = 1 - P(X_1 \leq m, X_2 \leq m) = 1 - (1/2)^2 = 3/4$. In general, $P(\max_{1\leq i\leq n} X_i > m) = 1 - 2^{-n}$.

Exercise 5.22

Central idea. Differentiate $P(Z^2 \leq t) = P(-\sqrt{t} \leq Z \leq \sqrt{t})$ using $f_Z(z) = 2(1-\Phi(z))\phi(z)$; the Φ terms telescope to give the χ^2_1 density.

Solution. $F_Z(z) = 1-(1-\Phi(z))^2$, so $f_Z(z) = 2(1-\Phi(z))\phi(z)$. For $t>0$,

$$P(Z^2 \le t) = \int_{-\sqrt{t}}^{\sqrt{t}} 2(1-\Phi(z))\phi(z)\,dz,$$

and differentiating (Leibniz rule; the boundary derivatives are $\pm 1/(2\sqrt{t})$):

$$f_{Z^2}(t) = \frac{f_Z(\sqrt{t})+f_Z(-\sqrt{t})}{2\sqrt{t}} = \frac{\phi(\sqrt{t})}{\sqrt{t}}\big[(1-\Phi(\sqrt{t}))+(1-\Phi(-\sqrt{t}))\big] = \frac{\phi(\sqrt{t})}{\sqrt{t}} = \frac{t^{-1/2}e^{-t/2}}{\sqrt{2\pi}} = \frac{t^{1/2-1}e^{-t/2}}{\Gamma(1/2)2^{1/2}},$$

the χ^2_1 pdf, since $\Phi(\sqrt{t})+\Phi(-\sqrt{t}) = 1$.

Exercise 5.23

Central idea. Condition on $X=x$: $Z \mid X=x$ has pdf $x(1-z)^{x-1}$; mixing over x with weights $c/x!$ sums the exponential series in closed form.

Solution. Given $X=x$, Z is the minimum of x iid uniforms: $f_{Z|X}(z \mid x) = x(1-z)^{x-1}$, $0<z<1$. Hence

$$f_Z(z) = \sum_{x=1}^{\infty} x(1-z)^{x-1}\cdot\frac{c}{x!} = c\sum_{x=1}^{\infty}\frac{(1-z)^{x-1}}{(x-1)!} = c\,e^{1-z} = \frac{e^{1-z}}{e-1}, \qquad 0<z<1,$$

a truncated-exponential-type density (it integrates to 1 over $(0,1)$).

Exercise 5.24

Central idea. Transform the joint pdf of $(X_{(1)}, X_{(n)})$ via $(W,V) = (U/V, V)$; the Jacobian factor v combines with $(v-u)^{n-2}$ to split into a function of w times a function of v.

Solution. $f_{X_{(1)},X_{(n)}}(u,v) = n(n-1)(v-u)^{n-2}/\theta^n$, $0<u<v<\theta$. With $w=u/v$, $v=v$ (Jacobian $|J|=v$),

$$f_{W,V}(w,v) = n(n-1)\frac{(v-wv)^{n-2}}{\theta^n}\cdot v = \underbrace{(n-1)(1-w)^{n-2}\mathbf{1}_{(0,1)}(w)}_{\text{Beta}(1,n-1)\text{ pdf}}\cdot\underbrace{\frac{nv^{n-1}}{\theta^n}\mathbf{1}_{(0,\theta)}(v)}_{\text{pdf of } X_{(n)}},$$

which factors. Hence $X_{(1)}/X_{(n)} \sim \text{Beta}(1,n-1)$ independent of $X_{(n)}$.

Exercise 5.25

Central idea. The successive-ratio transformation $W_i = U_i/U_{i+1}$ ($W_n = U_n$) has triangular Jacobian $\prod w_j^{j-1}$, which turns the order-statistic joint pdf into a product of Beta$(ja,1)$-type factors.

Solution. $f_{X_{(1)},\dots,X_{(n)}}(u_1,\dots,u_n) = n!\,(\frac{a}{\theta^a})^n\prod_{i=1}^n u_i^{a-1}$, $0<u_1<\cdots<u_n<\theta$. Set $w_i = u_i/u_{i+1}$ ($i<n$), $w_n = u_n$; then $u_i = w_i\cdots w_n$ and $|J| = \prod_{j=1}^n w_j^{j-1}$ (upper triangular). Thus

$$f_W(w) = n!\left(\frac{a}{\theta^a}\right)^n\prod_{j=1}^n w_j^{j(a-1)}\cdot\prod_{j=1}^n w_j^{j-1} = n!\left(\frac{a}{\theta^a}\right)^n\prod_{j=1}^n w_j^{ja-1}.$$

This factors over j: for $j<n$, $W_j = X_{(j)}/X_{(j+1)}$ has pdf $ja\cdot w^{ja-1}$ on $(0,1)$ (i.e., Beta$(ja,1)$), and $W_n = X_{(n)}$ has pdf $\frac{na}{\theta^{an}}w_n^{na-1}$ on $(0,\theta)$ — indeed $P(X_{(n)}\le t) = (t^a/\theta^a)^n$. The n variables are mutually independent. (For $a=1$ this recovers 5.24, with $X_{(1)}/X_{(n)} = \prod_{j=1}^{n-1} W_j \perp X_{(n)}$.)

Exercise 5.26

Central idea. Order-statistic events are counting events: $\{X_{(i)} \le u\} = \{U \ge i\}$. The mixed partial of the multinomial sum telescopes — each interior term's u-derivative cancels a neighbor's — leaving only the (i, j)-cell term.

Solution. (a) Each X_ℓ falls independently into $(-\infty, u]$, $(u, v]$, (v, ∞) with probabilities $F_X(u)$, $F_X(v) - F_X(u)$, $1 - F_X(v)$; counting the n iid trials gives the multinomial pmf. (b) For $u < v$: $\{X_{(i)} \le u\} = \{U \ge i\}$ and $\{X_{(j)} \le v\} = \{U + V \ge j\}$, so $F_{X_{(i)},X_{(j)}}(u, v) = P(U \ge i, U + V \ge j)$. Splitting on whether $U \ge j$ or $i \le U \le j - 1$ (in which case $U + V \ge j$ needs $V \ge j - U$) and inserting the multinomial pmf from (a) gives the displayed sum. (c) Differentiating termwise: $\frac{\partial^2}{\partial v\, \partial u}$ of the (k, m) summand produces terms proportional to $f_X(u) f_X(v)$ times $F_X(u)^{k-1}(F_X(v) - F_X(u))^{m-1}(1 - F_X(v))^{n-k-m}$ with multinomial-type coefficients. Using $k \cdot \frac{n!}{k!m!(n-k-m)!} = n \cdot \frac{(n-1)!}{(k-1)!m!(n-k-m)!}$ and the analogous m-identity (the "(5.4.5) relationships"), each differentiated term splits into pieces that cancel the corresponding pieces from the $(k \pm 1, m \pm 1)$ neighbors — a telescoping sum. All interior terms cancel; only the boundary term $(k, m) = (i, j - i)$ survives:

$$f_{X_{(i)},X_{(j)}}(u, v) = \frac{n!}{(i-1)!\,(j-i-1)!\,(n-j)!}\,[F_X(u)]^{i-1}[F_X(v) - F_X(u)]^{j-i-1}[1 - F_X(v)]^{n-j} f_X(u) f_X(v),$$

for $u < v$ (and 0 otherwise), the usual joint pdf of two order statistics. The $P(U \ge j)$ term contributes nothing to the mixed partial.

Exercise 5.27

Central idea. (a) The joint-to-marginal ratio collapses to the pdf of the ith order statistic of a *truncated* sample: given $X_{(j)} = y$, the $j - 1$ smaller observations are iid from $F_X(\cdot)/F_X(y)$. (b) For a uniform sample, range $R = X_{(n)} - X_{(1)}$ and midrange $V = (X_{(1)} + X_{(n)})/2$ have constant joint density, so $V \mid R = r$ is uniform.

Solution. (a) For $i < j$ and $x < y$, dividing the joint pdf (5.26) by the marginal pdf of $X_{(j)}$,

$$f_{X_{(i)}|X_{(j)}}(x \mid y) = \frac{(j-1)!}{(i-1)!\,(j-i-1)!}\left[\frac{F_X(x)}{F_X(y)}\right]^{i-1}\left[\frac{F_X(y) - F_X(x)}{F_X(y)}\right]^{j-i-1}\frac{f_X(x)}{F_X(y)},$$

0 for $x \ge y$. This is exactly the pdf of the ith order statistic from a sample of size $j - 1$ drawn from the truncated distribution with cdf $F_X(\cdot)/F_X(y)$ on $(-\infty, y)$. (b) (Example 5.4.7: for a uniform$(0, 1)$ sample, $R = X_{(n)} - X_{(1)}$ is the range and $V = \frac{1}{2}(X_{(1)} + X_{(n)})$ the midrange.) From $f_{X_{(1)},X_{(n)}}(u, v) = n(n-1)(v-u)^{n-2}$, the transform $(r, m) = (v - u, (u + v)/2)$ (Jacobian 1) gives $f_{R,V}(r, m) = n(n-1)r^{n-2}$ on $\frac{r}{2} < m < 1 - \frac{r}{2}$, $0 < r < 1$. Hence $f_R(r) = n(n-1)r^{n-2}(1 - r)$ and

$$f_{V|R}(m \mid r) = \frac{1}{1 - r}, \qquad \frac{r}{2} < m < 1 - \frac{r}{2}\,:$$

given the range, the midrange is uniform over its feasible interval.

Exercise 5.28

Central idea. Multinomial counting over the gaps between successive order statistics (marginal); conditioning on one group fixes cut points, splitting the remaining sample into independent truncated subsamples in each gap (conditional).

Solution. (a) With $i_0 = 0$, $i_{l+1} = n + 1$, $x_0 = -\infty$, $x_{l+1} = +\infty$: the marginal cdf is the multinomial sum

$$P(X_{(i_1)} \le x_1, \ldots, X_{(i_l)} \le x_l) = \sum_{\text{valid } (c_0,\ldots,c_l)} \frac{n!}{\prod c_k!} \prod_{k=0}^{l} \left[F_X(x_{k+1}) - F_X(x_k)\right]^{c_k},$$

over count vectors with $\sum_{t\le k} c_t \ge i_{k+1}-1$-type constraints; differentiating gives the joint pdf for $x_1 < \cdots < x_l$:

$$f(x_1,\dots,x_l) = \frac{n!}{\prod_{k=0}^{l}(i_{k+1}-i_k-1)!}\,[F_X(x_1)]^{i_1-1}\prod_{k=2}^{l}[F_X(x_k)-F_X(x_{k-1})]^{i_k-i_{k-1}-1}[1-F_X(x_l)]^{n-i_l}\prod_{k=1}^{l} f_X(x_k).$$

(b) Given $X_{(j_1)} = y_1,\dots,X_{(j_m)} = y_m$, the y's partition the line into gaps; the $n-m$ unobserved X's fall independently into these gaps, and within each gap they are iid from F_X truncated to that gap. Each $X_{(i_t)}$ lying in a gap is then an order statistic of that gap's subsample from the truncated distribution. E.g., if all j's precede all i's, given $X_{(j_m)} = y$ the vector $(X_{(i_1)},\dots,X_{(i_l)})$ has the joint order-statistic distribution (as in (a)) of a sample of size $n-j_m$ from cdf $[F_X(x)-F_X(y)]/[1-F_X(y)]$, $x>y$, with indices $i_t - j_m$. The conditional cdf is the corresponding multinomial sum with these truncated cell probabilities.

Exercise 5.29

Central idea. $P(\sum X_i > 100.4) = P(\bar X > 1.004)$; by the CLT, $\bar X \approx n(1, 0.05^2/100)$, reducing the problem to a standard normal tail.

Solution. Assume the 100 booklet weights are iid with mean 1 and sd 0.05. Then $P(\text{total} > 100.4) = P(\bar X > 1.004)$. By the Central Limit Theorem, $\bar X \dot\sim n(1, 0.05^2/100) = n(1, 0.005^2)$, so

$$P(\bar X > 1.004) \approx P\left(Z > \frac{1.004-1}{0.005}\right) = P(Z > 0.8) \approx 0.2119.$$

The approximation needs $n = 100$ large enough for the CLT and iid weights; no distributional assumption on individual weights is needed. (Exact value would require the weight distribution; with only mean/variance, CLT is the tool.)

Exercise 5.30

Central idea. $\bar X_1 - \bar X_2 \dot\sim n(0, 2\sigma^2/n)$ by the CLT; invert $P(|Z| < \sqrt{n/50}) = 0.99$.

Solution. $E[\bar X_1 - \bar X_2] = 0$, $\mathrm{Var}(\bar X_1 - \bar X_2) = 2\sigma^2/n$, and by the CLT $\frac{\bar X_1 - \bar X_2}{\sigma\sqrt{2/n}} \dot\sim n(0,1)$. Hence

$$P\left(|\bar X_1 - \bar X_2| < \frac{\sigma}{5}\right) \approx P\left(|Z| < \frac{\sigma/5}{\sigma\sqrt{2/n}}\right) = P\left(|Z| < \sqrt{\frac{n}{50}}\right) \overset{!}{=} 0.99.$$

$P(|Z| < z) = 0.99$ gives $z = 2.5758$, so $n = 50(2.5758)^2 \approx 331.7$; take $n = 332$. (Uses CLT for the difference of means and independence of the samples.)

Exercise 5.31

Central idea. Chebyshev: $P(|\bar X - \mu| \ge k) \le \frac{9/100}{k^2}$ (distribution-free but loose); CLT: $|\bar X - \mu|/0.3 \dot\sim |Z|$ (tight but approximate and asymptotic).

Solution. $\mathrm{Var}(\bar X) = 9/100 = 0.09$. Chebyshev: $P(|\bar X - \mu| < k) \ge 1 - 0.09/k^2 = 0.90$ gives $k = \sqrt{0.9} \approx 0.949$: $\bar X - \mu \in (-0.949,\ 0.949)$ with probability at least .90, for *any* population with variance 9 — but the bound is very conservative. CLT: $P(|\bar X - \mu| < k) \approx P(|Z| < k/0.3) = 0.90$ gives $k/0.3 = 1.6449$, i.e., $k \approx 0.493$: $\bar X - \mu \in (-0.493,\ 0.493)$ approximately. The CLT interval is about half as wide but is only approximate (relying on $n = 100$ being large enough and the unknown shape of the population); Chebyshev is rigorous and universal but pays for it in width.

Exercise 5.32

Central idea. Continuous mapping: $\sqrt{\cdot}$ is continuous on $(0,\infty)$ (and at 0 from the right), $x \mapsto a/x$ is continuous at $a \ne 0$; then $S_n^2 \to_p \sigma^2$ (WLLN) gives $\sigma/S_n \to_p 1$.

Solution. (a) $Y_i = \sqrt{X_i} \to_p \sqrt{a}$: $\sqrt{\cdot}$ is continuous at every $a \geq 0$ (from the right at 0), and $X_i > 0$ a.s., so this is the continuous mapping theorem (Theorem 5.5.4). For $Y_i' = a/X_i$: if $a \neq 0$, $x \mapsto a/x$ is continuous at a, so $Y_i' \to_p a/a = 1$; if $a = 0$, then $Y_i' \equiv 0 \to_p 0$. Either way Y_i' converges in probability. (b) By the WLLN applied to $(X_i - \mu)^2$, $S_n^2 \to_p \sigma^2$ (Example 5.5.18; assume $\sigma > 0$). By (a) with the square root, $S_n = \sqrt{S_n^2} \to_p \sigma$, and applying the reciprocal form again, $\sigma/S_n \to_p \sigma/\sigma = 1$.

Exercise 5.33

Central idea. $\{X_n + Y_n \leq c\} \subseteq \{X_n \leq c - M\} \cup \{Y_n \leq M\}$: make $Y_n \leq M$ negligible by hypothesis, then send $M \to \infty$ through continuity points of F_X to kill $P(X_n \leq c - M)$.

Solution. For any fixed M,

$$P(X_n + Y_n \leq c) \leq P(X_n \leq c - M) + P(Y_n \leq M).$$

Let $M_k \to \infty$ with each $c - M_k$ a continuity point of F_X (possible since discontinuities are countable). Then $\limsup_n P(X_n + Y_n \leq c) \leq F_X(c - M_k) + 0$ for each k; letting $k \to \infty$, $F_X(c - M_k) \to 0$. Hence $P(X_n + Y_n \leq c) \to 0$, i.e., $P(X_n + Y_n > c) \to 1$. (Intuition: $Y_n \to_p \infty$ drags the sum to $+\infty$ regardless of X_n's limit.)

Exercise 5.34

Central idea. Linearity of expectation and $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$; the $\sqrt{n}/\sigma$ scaling is exactly what standardizes these.

Solution. $E[\bar{X}_n] = \mu$, so $E[\sqrt{n}(\bar{X}_n - \mu)/\sigma] = 0$. And $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, so

$$\mathrm{Var}\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} = \frac{n}{\sigma^2} \cdot \frac{\sigma^2}{n} = 1.$$

The CLT's normalization therefore matches the first two moments of its $n(0,1)$ limit exactly, for every n.

Exercise 5.35

Central idea. This is the CLT for exponential(1) (mean = sd = 1); $n\bar{X}_n = \sum X_i \sim \mathrm{Gamma}(n, 1)$, so differentiating the CLT approximation in x exposes $\Gamma(n)$.

Solution. (a) $EX_i = \mathrm{Var}(X_i) = 1$, so by the CLT $\frac{\bar{X}_n - 1}{1/\sqrt{n}} \to_d n(0,1)$. (b) Let $S_n = n\bar{X}_n \sim \mathrm{Gamma}(n,1)$ with pdf $s^{n-1}e^{-s}/\Gamma(n)$. Then $P\left(\frac{\bar{X}_n - 1}{1/\sqrt{n}} \leq x\right) = P(S_n \leq n + x\sqrt{n})$; differentiating in x,

$$\frac{d}{dx}P(S_n \leq n + x\sqrt{n}) = \sqrt{n}\, f_{S_n}(n + x\sqrt{n}) = \frac{\sqrt{n}}{\Gamma(n)}(x\sqrt{n} + n)^{n-1}e^{-(x\sqrt{n}+n)} \approx \frac{d}{dx}\Phi(x) = \frac{e^{-x^2/2}}{\sqrt{2\pi}}.$$

At $x = 0$: $\frac{\sqrt{n}}{\Gamma(n)}n^{n-1}e^{-n} \approx \frac{1}{\sqrt{2\pi}}$, i.e., $\Gamma(n) \approx \sqrt{2\pi}\, n^{n-1/2}e^{-n}$, and $n! = n\Gamma(n) \approx \sqrt{2\pi}\, n^{n+1/2}e^{-n}$ — Stirling's formula.

Exercise 5.36

Central idea. Tower laws: $E[Y \mid N] = 2N$, $\mathrm{Var}(Y \mid N) = 4N$; the mgf $M_Y(t) = \exp(\theta \cdot 2t/(1-2t))$ then yields the CLT by the usual mgf expansion.

Solution. (a) $EY = E[E[Y \mid N]] = E[2N] = 2\theta$. $\mathrm{Var}\, Y = E[\mathrm{Var}(Y \mid N)] + \mathrm{Var}(E[Y \mid N]) = E[4N] + \mathrm{Var}(2N) = 4\theta + 4\theta = 8\theta$. (b) $M_Y(t) = E[E[e^{tY} \mid N]] = E[(1-2t)^{-N}] = E[e^{-N\log(1-2t)}] = \exp\left(\theta(e^{-\log(1-2t)} - 1)\right) = \exp\left(\theta \cdot \frac{2t}{1-2t}\right)$, $t < 1/2$. With $s = t/\sqrt{8\theta}$,

$$M_{(Y-2\theta)/\sqrt{8\theta}}(t) = \exp\left(\theta \cdot \frac{2s}{1-2s} - 2\theta s\right) = \exp\left(\frac{4\theta s^2}{1-2s}\right) = \exp\left(\frac{t^2/2}{1 - 2t/\sqrt{8\theta}}\right) \to e^{t^2/2},$$

the $n(0,1)$ mgf. Hence $(Y - EY)/\sqrt{\mathrm{Var}\, Y} \to_d n(0,1)$.

Exercise 5.37

Central idea. The observations in Example 5.5.16 are negative binomial, counting failures before r successes. Thus $V_n \sim \text{NB}(nr, p) = \text{NB}(20, .7)$, not binomial. Approximate its cdf by a normal cdf and difference consecutive values, treating the lower support boundary explicitly.

Solution. The exact pmf is $P(V_n = v) = \binom{v+19}{19}(.7)^{20}(.3)^v$. The approximating normal has

$$\mu = \frac{20(.3)}{.7} = 8.5714285714, \qquad \sigma = \sqrt{\frac{20(.3)}{.7^2}} = 3.4992710611.$$

For $v \geq 1$, the uncorrected approximation is $\Phi((v-\mu)/\sigma) - \Phi((v-1-\mu)/\sigma)$; the continuity-corrected version is $\Phi((v+1/2-\mu)/\sigma) - \Phi((v-1/2-\mu)/\sigma)$. At $v = 0$, use $P(V_n = 0) = P(V_n \leq 0)$: the corresponding cdf approximations are $\Phi(-\mu/\sigma)$ and $\Phi((1/2-\mu)/\sigma)$.

v	(a) Exact	(b) Normal, no CC	(c) Normal, with CC
0	0.0008	0.0072	0.0105
1	0.0048	0.0081	0.0111
2	0.0151	0.0150	0.0197
3	0.0332	0.0255	0.0323
4	0.0572	0.0400	0.0487
5	0.0824	0.0580	0.0677
6	0.1030	0.0775	0.0869
7	0.1148	0.0955	0.1028
8	0.1162	0.1085	0.1121
9	0.1085	0.1136	0.1128
10	0.0944	0.1097	0.1046

Another convention approximates the zero cell by a finite normal interval too: $(-1, 0]$ without correction and $(-1/2, 1/2]$ with correction. Those give 0.0040 and 0.0058, respectively, and explain the zero row in the newly supplied draft. The table above instead uses the support-aware cdf convention. Neither normal approximation is exact at this boundary. The continuity correction improves many interior entries (at $v = 10$, 0.1097 becomes 0.1046, compared with exact 0.0944), but does not improve every entry (for example $v = 2$).

Exercise 5.38

Central idea. Chernoff bounds give $C\rho^n$ for fixed thresholds. At threshold zero the prefactor is 1; for arbitrary thresholds it can be absorbed into c^n for all sufficiently large n, but not necessarily for every n.

Solution. (a) For $t > 0$, Markov gives $P(S_n > a) \leq e^{-ta} M_X(t)^n$. For $t < 0$, the map $s \mapsto e^{ts}$ is decreasing, so $P(S_n \leq a) \leq e^{-ta} M_X(t)^n$.

(b) If $EX < 0$, then $M_X(t) = 1 + tEX + o(t) < 1$ for some small positive t. Put $\rho = M_X(t) \in (0, 1)$ and $C = e^{-ta}$. Then $P(S_n > a) \leq C\rho^n$. For any $\rho < c < 1$, $C\rho^n \leq c^n$ once n is sufficiently large. If $a \geq 0$, $C \leq 1$ and $c = \rho$ works for every n. The unrestricted all-n statement in (b) needs qualification: taking $X_i \equiv -1$ and $a = -2$ gives $P(S_1 > a) = 1$, incompatible with $c < 1$. No adjustment of finitely many n can repair a probability equal to 1. Similarly, if $EX > 0$, a small negative t gives $P(S_n \leq a) \leq C\rho^n$, and an eventual bound c^n; for $a \leq 0$ the prefactor is at most 1 and the bound holds for every n.

(c) Apply (b) at threshold zero to $Y_i = X_i - \mu - \varepsilon$, whose mean is $-\varepsilon$. This gives $P(\bar{X}_n - \mu > \varepsilon) \leq c_+^n$ for every n.

(d) Apply the same argument to $-X_i + \mu - \varepsilon$ to get $P(\bar{X}_n - \mu < -\varepsilon) \leq c_-^n$. With $c = \max(c_+, c_-) < 1$, the union bound gives $P(|\bar{X}_n - \mu| > \varepsilon) \leq 2c^n$. Summability and Borel–Cantelli, followed by a countable sequence of positive ε tending to zero, give the SLLN under the stated mgf assumption.

Exercise 5.39

Central idea. (a) Continuous mapping for constant limits: $\{|h(X_n)-h(x)|>\varepsilon\}\subseteq\{|X_n-x|\geq\delta\}$. (b) The "sliding hump" $Y_n\to_p 0$ but not a.s.; freezing the hump at the left edge gives $Y_{m,1}=\mathbf{1}_{[0,2^{-m})}\to 0$ pointwise.

Solution. (a) (Theorem 5.5.4: if $X_n\to_p x$, constant, and h is continuous at x, then $h(X_n)\to_p h(x)$.) Given $\varepsilon>0$, choose $\delta>0$ with $|x_n-x|<\delta\implies|h(x_n)-h(x)|<\varepsilon$. Then $P(|h(X_n)-h(x)|>\varepsilon)\leq P(|X_n-x|\geq\delta)\to 0$. (b) (Example 5.5.8 is the "typewriter"/sliding-hump sequence: $Y_{m,k}=\mathbf{1}_{[(k-1)/2^m,\,k/2^m)}$ on $(0,1)$, $k=1,\ldots,2^m$, $m=1,2,\ldots$, so $Y_n\to_p 0$ but $Y_n(\omega)=1$ infinitely often for every ω.) Take the subsequence $Y_{m,1}=\mathbf{1}_{[0,1/2^m)}$: for each fixed $\omega>0$, $Y_{m,1}(\omega)=0$ for all large m, so $Y_{m,1}\to 0$ pointwise (hence a.s.).

Exercise 5.40

Central idea. Convergence in probability implies convergence in distribution: sandwich $\{X_n\leq t\}$ between $\{X\leq t\pm\varepsilon\}$ up to the vanishing event $\{|X_n-X|\geq\varepsilon\}$, then let $\varepsilon\to 0$ through continuity points.

Solution. (a) $\{X\leq t-\varepsilon\}\subseteq\{X_n\leq t\}\cup\{|X_n-X|\geq\varepsilon\}$: if $X\leq t-\varepsilon$ but $|X_n-X|<\varepsilon$, then $X_n<X+\varepsilon\leq t$. Taking probabilities and rearranging, $P(X_n\leq t)\geq P(X\leq t-\varepsilon)-P(|X_n-X|\geq\varepsilon)$. (b) Similarly $\{X_n\leq t\}\subseteq\{X\leq t+\varepsilon\}\cup\{|X_n-X|\geq\varepsilon\}$, so $P(X_n\leq t)\leq P(X\leq t+\varepsilon)+P(|X_n-X|\geq\varepsilon)$. (c) With $X_n\to_p X$, $P(|X_n-X|\geq\varepsilon)\to 0$; hence $\liminf P(X_n\leq t)\geq P(X\leq t-\varepsilon)$ and $\limsup P(X_n\leq t)\leq P(X\leq t+\varepsilon)$ for every $\varepsilon>0$. For continuous X every t is a continuity point of F_X, so letting $\varepsilon\to 0$ pinches: $P(X_n\leq t)\to P(X\leq t)$.

Exercise 5.41

Central idea. Convergence in probability to μ is convergence in distribution to the degenerate law at μ; both directions are inclusions of the relevant events.

Solution. (a) ($\Rightarrow$) Let $x>\mu$, $\varepsilon=x-\mu>0$: $\{|X_n-\mu|\leq\varepsilon\}=\{\mu-\varepsilon\leq X_n\leq\mu+\varepsilon\}\subseteq\{X_n\leq x\}$, so $P(X_n\leq x)\geq P(|X_n-\mu|\leq\varepsilon)\to 1$. Let $x<\mu$, $\varepsilon=\mu-x>0$: $\{X_n\leq x\}\subseteq\{|X_n-\mu|\geq\varepsilon\}$, so $P(X_n\leq x)\leq P(|X_n-\mu|\geq\varepsilon)\to 0$. (b) ($\Leftarrow$) $\{|X_n-\mu|>\varepsilon\}=\{X_n<\mu-\varepsilon\}\cup\{X_n>\mu+\varepsilon\}\subseteq\{X_n\leq\mu-\varepsilon\}\cup\{X_n>\mu+\varepsilon\}$. Hence $P(|X_n-\mu|>\varepsilon)\leq P(X_n\leq\mu-\varepsilon)+1-P(X_n\leq\mu+\varepsilon)\to 0+1-1=0$, using the assumed cdf convergence at $\mu-\varepsilon<\mu$ and $\mu+\varepsilon\geq\mu$.

Exercise 5.42

Central idea. $P(X_{(n)}\leq x)=F(x)^n$; choose the scaling so that $n(1-F(\text{threshold}))$ tends to a constant, and $(1-c/n)^n\to e^{-c}$ produces the extreme-value limit.

Solution. (a) For Beta$(1,\beta)$, $F(x)=1-(1-x)^\beta$ on $(0,1)$. For $t>0$,

$$P(n^v(1-X_{(n)})\leq t)=1-P\big(X_{(n)}<1-\tfrac{t}{n^v}\big)=1-\Big[1-\big(\tfrac{t}{n^v}\big)^\beta\Big]^n=1-\left[1-\frac{t^\beta}{n^{v\beta}}\right]^n.$$

With $v=1/\beta$, this is $1-(1-t^\beta/n)^n\to 1-e^{-t^\beta}$, a Weibull cdf. So $v=1/\beta$ works. (b) For exponential(1), $P(X_{(n)}-a_n\leq t)=(1-e^{-(t+a_n)})^n$. With $a_n=\log n$,

$$\left(1-\frac{e^{-t}}{n}\right)^n\to e^{-e^{-t}},$$

the Gumbel cdf. So $a_n=\log n$ works (the classic exponential-maximum normalization).

Exercise 5.43

Central idea. (a) $\sqrt{n}$-consistency implies consistency: $P(|Y_n-\mu|>\varepsilon)=P(|\sqrt{n}(Y_n-\mu)|>\varepsilon\sqrt{n})\to 0$ by tightness. (b) The delta method: factor $\sqrt{n}(g(Y_n)-g(\theta))=\frac{g(Y_n)-g(\theta)}{Y_n-\theta}\cdot\sqrt{n}(Y_n-\theta)$ and apply Slutsky to the (constant) $\times$ (asymptotically normal) product.

Solution. (a) $P(|Y_n-\mu|>\varepsilon)=P(|\sqrt{n}(Y_n-\mu)|>\varepsilon\sqrt{n})$. Given $M>0$, for large n, $\varepsilon\sqrt{n}>M$, so $\limsup_n P(|Y_n-\mu|>\varepsilon)\le P(|Z|>M)$ with $Z\sim n(0,\sigma^2)$; letting $M\to\infty$ gives 0. Hence $Y_n\to_p\mu$. (b) (Theorem 5.5.24 is the delta method.) Write, for $Y_n\neq\theta$,

$$\sqrt{n}(g(Y_n)-g(\theta))=\frac{g(Y_n)-g(\theta)}{Y_n-\theta}\cdot\sqrt{n}(Y_n-\theta)$$

(defining the ratio as $g'(\theta)$ when $Y_n=\theta$). By (a), $Y_n\to_p\theta$, and differentiability of g at θ gives $\frac{g(Y_n)-g(\theta)}{Y_n-\theta}\to_p g'(\theta)$ (given $\eta>0$, the ratio is within η of $g'(\theta)$ whenever $0<|Y_n-\theta|<\delta$, an event of probability $\to 1$). Slutsky's theorem (5.5.17) then yields $\sqrt{n}(g(Y_n)-g(\theta))\to_d g'(\theta)\cdot n(0,\sigma^2)=n(0,\sigma^2[g'(\theta)]^2)$.

Exercise 5.44

Central idea. (a) is the CLT; (b) is the delta method with $g(t)=t(1-t)$, $g'(p)=1-2p$; (c) is the second-order delta method since $g'(1/2)=0$: $Y_n(1-Y_n)-1/4=-(Y_n-1/2)^2$.

Solution. (a) Immediate from the CLT: $E[Y_n]=p$, $\mathrm{Var}(Y_n)=p(1-p)/n$. (b) With $g(t)=t(1-t)$, $g'(t)=1-2t\neq 0$ at $t=p\neq 1/2$; the delta method (5.43) gives $\sqrt{n}(g(Y_n)-g(p))\to_d n(0,[g'(p)]^2p(1-p))=n(0,(1-2p)^2p(1-p))$. (c) For $p=1/2$, $g'(1/2)=0$ kills the first-order term. Note $Y_n(1-Y_n)-\frac14=-(Y_n-\frac12)^2$, so

$$n\left[Y_n(1-Y_n)-\tfrac14\right]=-\left(\sqrt{n}(Y_n-\tfrac12)\right)^2\to_d -Z^2,$$

where $Z\sim n(0,1/4)$ by (a); $-Z^2\overset{d}{=}-\frac14\chi_1^2$. Equivalently $4n[\frac14-Y_n(1-Y_n)]=(2\sqrt{n}(Y_n-\frac12))^2\to_d\chi_1^2$.

Exercise 5.45

Central idea. Each component survives 150h with probability $p=P(X>150)$ (Erlang/Poisson sum for integer a, numerical otherwise); the number surviving is Binomial(c,p), and we need its upper tail at $0.75c$.

Solution. (C&B's gamma(a,b) uses b as the *scale*: $E[X]=ab$.) Let $p=P(X>150)$; the number of survivors is Binomial(c,p). (a) $a=4$ integer: $p=P(\mathrm{Gamma}(4,5)>150)=\sum_{j=0}^{3}e^{-30}\frac{30^j}{j!}=4981e^{-30}\approx 4.66\times10^{-10}$. Then $P(\mathrm{Bin}(300,p)\ge 225)\approx 0$ (about 10^{-2100} — numerically zero). (b) $a=20$ integer: $p=\sum_{j=0}^{19}e^{-30}\frac{30^j}{j!}=P(\mathrm{Poisson}(30)\le 19)\approx 0.0219$. Then $P(\mathrm{Bin}(100,p)\ge 75)\approx 4.4\times10^{-102}\approx 0$. (c) $a=20.7$: no closed form; numerical integration gives $p=P(\mathrm{Gamma}(20.7,5)>150)\approx 0.0307$. Then $P(\mathrm{Bin}(100,p)\ge 75)\approx 4.0\times10^{-91}\approx 0$. In all three cases the required probability is essentially zero: with these gamma lifetimes, getting 75% of components past 150 hours is virtually impossible (the mean lifetimes are 20, 100, 103.5 hours respectively, far below 150).

Exercise 5.46

Central idea. $P(\mathrm{Bin}(c,p)\ge 0.75c)\approx 1-\Phi\left(\frac{0.75c-cp}{\sqrt{cp(1-p)}}\right)$; with $0.75c$ dozens of standard deviations above the mean, this is $1-\Phi(\text{huge})=0$, agreeing with 5.45.

Solution. With p from 5.45: (a) $1-\Phi((225-300p)/\sqrt{300p(1-p)})\approx 1-\Phi(4.9\times10^4)=0$; (b) $1-\Phi((75-100p)/\sqrt{100p(1-p)})\approx 1-\Phi(50.6)=0$; (c) $1-\Phi((75-100p)/\sqrt{100p(1-p)})\approx 1-\Phi(42.4)=0$. The normal approximation agrees with the exact ≈ 0 in all three cases — here the event is so far in the tail that both methods just say "essentially impossible."

Exercise 5.47

Central idea. (5.6.5) gives the standard uniform-based generators: inverse-cdf for the exponential, $-2\log U\sim\chi_2^2$, and sums thereof; each is verified by cdf differentiation or mgf.

Solution. (5.6.5) collects the basic constructions from uniforms $U,U_1,U_2,\ldots\sim$ iid uniform$(0,1)$: - $X=-\log U\sim$ exponential(1): $P(-\log U\le x)=P(U\ge e^{-x})=1-e^{-x}$, $x>0$. - $Y=-2\log U\sim\chi_2^2$: $P(-2\log U\le y)=P(U\ge e^{-y/2})=1-e^{-y/2}$, the exponential$(1/2)=\chi_2^2$ cdf. - $\sum_{i=1}^{n}(-\log U_i)\sim$

Gamma$(n,1)$: sum of n iid exponential(1) variables (mgf $(1-t)^{-n}$). - $-2\sum_{i=1}^{k}\log U_i \sim \chi^2_{2k}$: sum of k iid χ^2_2 variables (mgf $(1-2t)^{-k}$). - $-\beta\sum_{j=1}^{a}\log U_j \sim$ gamma(a,β) (scale β): $-\log U_j \sim$ exponential(1), so $-\beta\log U_j \sim$ exponential(β), and the sum of a iid exponential(β) variables is gamma(a,β). - $\frac{\sum_{j=1}^{a}\log U_j}{\sum_{j=1}^{a+b}\log U_j} \sim$ beta(a,b): writing the sums as $-G_a$ and $-(G_a+G_b)$ with $G_a \sim$ gamma$(a,1) \perp G_b \sim$ gamma$(b,1)$, the ratio is $G_a/(G_a+G_b) \sim$ beta(a,b). These are the building blocks for generating gamma, chi-square, and beta variables (used in 5.48).

Exercise 5.48

Central idea. $F_{m,n} = (\chi^2_m/m)/(\chi^2_n/n)$, and for even df a chi-square is a sum of $-2\log U_i$ terms by (5.6.5).

Solution. Generate $U_1,\dots,U_{m/2},V_1,\dots,V_{n/2}$ iid uniform$(0,1)$. Put

$$A = -2\sum_{i=1}^{m/2}\log U_i \sim \chi^2_m, \qquad B = -2\sum_{i=1}^{n/2}\log V_i \sim \chi^2_n,$$

independent by (5.6.5). Then

$$F = \frac{A/m}{B/n} \sim F_{m,n}.$$

(Even degrees of freedom are exactly what make the $-2\log U$ construction give the right df.)

Exercise 5.49

Central idea. Inverse-cdf method throughout: $-\log U$ inverts the exponential cdf, $\log\frac{u}{1-u}$ inverts the logistic cdf $e^x/(1+e^x)$, and location-scale does the rest.

Solution. (a) $P(-\log U \le x) = P(U \ge e^{-x}) = 1-e^{-x}$ $(x \ge 0)$; $1-U \overset{d}{=} U$, so $-\log(1-U)$ is exponential(1) too. (b) $P(X \le x) = P\left(\frac{U}{1-U} \le e^x\right) = P\left(U \le \frac{e^x}{1+e^x}\right) = \frac{e^x}{1+e^x} = \frac{1}{1+e^{-x}}$, the logistic$(0,1)$ cdf. (c) If $Z \sim$ logistic$(0,1)$, then $X = \mu+\beta Z$ has cdf $1/(1+e^{-(x-\mu)/\beta})$, i.e., logistic(μ,β). So generate $U \sim$ uniform$(0,1)$ and return $X = \mu+\beta\log\frac{U}{1-U}$.

Exercise 5.50

Central idea. In polar coordinates $(R,\Theta) = (\sqrt{-2\log U_2}, 2\pi U_1)$, R is Rayleigh and Θ uniform$(0,2\pi)$, independent; the polar-to-Cartesian Jacobian r converts their joint density into the product of two standard normals.

Solution. Put $R = \sqrt{-2\log U_2}$, $\Theta = 2\pi U_1$. Then $R^2 = -2\log U_2 \sim \chi^2_2$ (by 5.47), i.e., R has Rayleigh density $f_R(r) = re^{-r^2/2}$, $r>0$, independent of $\Theta \sim$ uniform$(0,2\pi)$. The map $(r,\theta) \mapsto (x_1,x_2) = (r\cos\theta, r\sin\theta)$ has Jacobian r, so

$$f_{X_1,X_2}(x_1,x_2) = f_{R,\Theta}(r,\theta)\cdot\frac{1}{r} = \frac{1}{2\pi}e^{-r^2/2} = \frac{1}{2\pi}e^{-(x_1^2+x_2^2)/2} = \phi(x_1)\phi(x_2),$$

with $r^2 = x_1^2+x_2^2$. Hence X_1, X_2 are independent $n(0,1)$.

Exercise 5.51

Central idea. $E[U_i] = 1/2$, $\mathrm{Var}(U_i) = 1/12$: the sum has exactly mean 0 and variance 1, and 12 summands is enough for the CLT to bite — but the support stays $[-6,6]$ and the kurtosis is $2.9 \neq 3$.

Solution. (a) $E[X] = 12\cdot\frac{1}{2} - 6 = 0$, $\mathrm{Var}(X) = 12\cdot\frac{1}{12} = 1$; by the CLT, a sum of 12 iid variables is approximately normal, so $X \overset{\cdot}{\sim} n(0,1)$. (b) X is bounded: $|X| \le 6$ always, whereas $P(|Z|>6) > 0$ for

$Z \sim n(0,1)$ — the tails are completely wrong (also the sum is exactly, not asymptotically, mean-0 variance-1, which is deceptive). (c) By symmetry $E[X] = E[X^3] = 0$ and $\mathrm{Var}(X) = 1$, matching $n(0,1)$. With $Y_i = U_i - 1/2$ ($E[Y_i^2] = 1/12$, $E[Y_i^4] = 1/80$),

$$E[X^4] = E\Big[\Big(\sum Y_i\Big)^4\Big] = \sum_i E[Y_i^4] + 3\sum_{i\neq j} E[Y_i^2]E[Y_j^2] = 12\cdot\frac{1}{80} + 3\cdot 12\cdot 11\cdot\frac{1}{144} = 0.15 + 2.75 = \frac{29}{10} = 2.9,$$

versus 3 for $n(0,1)$: slightly platykurtic (lighter tails), as expected from a bounded sum.

Exercise 5.52

Central idea. Binomial = sum of Bernoulli trials; hypergeometric = sequential draws without replacement using conditional success probabilities; negative binomial = Bernoulli trials until the 5th success (or a sum of 5 geometrics).

Solution. (a) For $i = 1, \ldots, 8$: generate $U_i \sim \mathrm{uniform}(0,1)$, set $B_i = \mathbf{1}_{\{U_i < 2/3\}}$; return $Y = \sum_{i=1}^{8} B_i$. (b) Sequential sampling without replacement: set $r = 8$ (successes remaining), $t = 10$ (draws remaining), $Y = 0$; for $j = 1, \ldots, 4$: generate $U \sim \mathrm{uniform}(0,1)$; if $U < r/t$ then $Y \leftarrow Y + 1$, $r \leftarrow r - 1$; $t \leftarrow t - 1$. Return Y. (c) Repeat Bernoulli(1/3) trials (success if $U < 1/3$) until 5 successes occur; return the number of failures. Equivalently, return $\sum_{i=1}^{5} \lfloor \frac{\log U_i}{\log(2/3)} \rfloor$ (sum of 5 iid geometric(1/3) "failures-before-success" variables, generated by inversion).

Exercise 5.53

Central idea. A simulation sanity check: with 1000 draws the sample moments should land within a standard error or two of the theoretical mean/variance, and the histogram should track the pmf.

Solution. Using the algorithms of 5.52 (seed 20261004, 1000 draws each):

Distribution	Sample mean (theoretical)	Sample variance (theoretical)
Binomial(8, $\frac{2}{3}$)	5.374 (5.333)	1.814 (1.778)
Hypergeometric(10, 8, 4)	3.176 (3.200)	0.415 (0.427)
Neg. binomial(5, $\frac{1}{3}$)	9.572 (10.000)	26.203 (30.000)

Theoretical values: binomial mean $8\cdot\frac{2}{3} = \frac{16}{3}$, var $8\cdot\frac{2}{3}\cdot\frac{1}{3} = \frac{16}{9}$; hypergeometric mean $4\cdot\frac{8}{10} = 3.2$, var $4\cdot\frac{8}{10}\cdot\frac{2}{10}\cdot\frac{10-4}{10-1} = 0.4267$; negative binomial (failures before 5th success) mean $5\cdot\frac{2/3}{1/3} = 10$, var $5\cdot\frac{2/3}{(1/3)^2} = 30$. All sample moments agree with theory within simulation error (e.g., se of the binomial mean $\approx \sqrt{1.78/1000} \approx 0.042$), and the histograms (not shown) track the pmfs — the generators are working.

Exercise 5.54

Central idea. Under Poisson, $E[S^2] = \bar{x} \approx 142$, but the observed $S^2 \approx 6371$ is astronomically larger — the simulated Poisson S^2 distribution never gets near it. A gamma–Poisson (negative binomial) mixture adds the extra variance; method-of-moments gives $r \approx 3.2$, and the reported simulation tails support several possible shapes, including $r = 5$ under a conventional two-sided tail check.

Solution. Here $n = 13$, $\bar{x} = 141.54$, $S^2_{\mathrm{obs}} = 6370.6$. (a) Simulate 20,000 samples of size 13 from Poisson(141.54) and compute S^2 each time: *none* reached 6370.6 (estimated $P(S^2 \geq S^2_{\mathrm{obs}}) \approx 0$). The Poisson assumption is untenable — the data are wildly overdispersed ($S^2/\bar{x} \approx 45$). (b) With $\mu = \bar{x}$ and $\mathrm{Var} = \mu + \mu^2/r$, matching S^2_{obs} gives $\hat{r} = \bar{x}^2/(S^2_{\mathrm{obs}} - \bar{x}) \approx 3.22$. Simulating S^2 under NB(r,p) with mean $\bar{x}$ (via the gamma–Poisson mixture): $P(S^2 \geq S^2_{\mathrm{obs}}) \approx 0.90, 0.70, 0.41, 0.13, 0.002, 0.000$ for $r = 1, 2, 3.2, 5, 10, 20$. A consistency cutoff requires specifying a test. For a two-sided equal-tail check at level 0.05, all displayed shapes $r = 1, 2, 3.2, 5$ have tail probabilities between 0.025 and 0.975 and are not excluded; $r = 10$ and 20

are excluded by the displayed lower tail values. The earlier $r \lesssim 4$ cutoff was not justified by the simulation results. (Smaller r = more clumping = more overdispersion.)

Exercise 5.55

Central idea. The inversion method (5.6.7) locates Y by sequential search through the cumulative probabilities; finding $Y = i$ costs exactly $i + 1$ comparisons, so the count is $Y + 1$ a.s.

Solution. In (5.6.7), $U \sim \text{uniform}(0, 1)$ is compared successively against p_0, $p_0 + p_1$, ... until the interval containing U is found. If the outcome is $Y = i$ (i.e., $y_i = i$), exactly $i + 1$ comparisons were performed. Hence the number of comparisons C satisfies $C = Y + 1$ a.s., and $E[C] = E[Y + 1]$ (equivalently, by Exercise 2.14's tail-sum formula, $E[Y + 1] = \sum_{k\geq 0} P(Y \geq k)$).

Exercise 5.56

Central idea. As printed this is incorrect — the stated f_Y is not a pdf and $\tan^{-1}$ is not its integral (details below); with the missing $1/\pi$ restored, (a) is a one-line integration and (b) is the inverse-cdf method.

Solution. Erratum (genuine). As stated, (a) is false, for two independent reasons. First, $f_Y(y) = 1/(1 + y^2)$ is not a pdf: $\int_{-\infty}^{\infty} \frac{dy}{1+y^2} = \pi \neq 1$. Second, even formally, $\int_{-\infty}^{y} \frac{dt}{1+t^2} = \tan^{-1}(y) + \frac{\pi}{2} \neq \tan^{-1}(y)$; moreover $\tan^{-1}$ alone cannot be a cdf (its range is $(-\pi/2, \pi/2)$, and $F_Y(-\infty) = -\pi/2 \neq 0$). The intended Cauchy$(0, 1)$ pdf is $f_Y(y) = \frac{1}{\pi(1+y^2)}$, for which

$$F_Y(y) = \int_{-\infty}^{y} \frac{dt}{\pi(1 + t^2)} = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(y), \qquad -\infty < y < \infty,$$

which is (a) corrected. For (b): with $U \sim \text{uniform}(0, 1)$, the inverse-cdf method gives $X = F^{-1}(U)$. Inverting $u = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(\frac{x-a}{b})$ yields

$$X = a + b \tan(\pi(U - \tfrac{1}{2})) \sim \text{Cauchy}(a, b),$$

since $P(X \leq x) = P(U \leq \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(\frac{x-a}{b})) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(\frac{x-a}{b})$, the Cauchy$(a, b)$ cdf.

Exercise 5.57

Central idea. The shared X_3 is the only source of dependence: it contributes $\text{Var}(X_3) = \lambda_3$ to the covariance, and $\{Z_1 = Z_2 = 1\} = \{X_1 = X_2 = X_3 = 0\}$. The correlation bound follows from $e^{\lambda_3} - 1 \leq e^{\lambda_i+\lambda_3} - 1 = (1 - p_i)/p_i$.

Solution. (a) $\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 + X_3, X_2 + X_3) = \text{Var}(X_3) = \lambda_3$ (all cross terms vanish by independence). (b) $Y_i \sim \text{Poisson}(\lambda_i + \lambda_3)$, so $Z_i = \mathbf{1}_{\{Y_i=0\}} \sim \text{Bernoulli}(p_i)$ with $p_i = e^{-(\lambda_i+\lambda_3)}$. Now $P(Z_1 = Z_2 = 1) = P(Y_1 = Y_2 = 0) = P(X_1 = X_2 = X_3 = 0) = e^{-(\lambda_1+\lambda_2+\lambda_3)} = p_1 p_2 e^{\lambda_3}$. Hence

$$\text{Corr}(Z_1, Z_2) = \frac{p_1 p_2 e^{\lambda_3} - p_1 p_2}{\sqrt{p_1(1 - p_1)p_2(1 - p_2)}} = \frac{p_1 p_2 (e^{\lambda_3} - 1)}{\sqrt{p_1(1 - p_1)}\sqrt{p_2(1 - p_2)}} \geq 0.$$

(c) $\text{Corr}(Z_1, Z_2) = \sqrt{\frac{p_1 p_2}{(1-p_1)(1-p_2)}}\,(e^{\lambda_3} - 1)$. Since $\lambda_1 \geq 0$, $e^{\lambda_3} - 1 \leq e^{\lambda_1+\lambda_3} - 1 = \frac{1-p_1}{p_1}$, giving Corr $\leq \sqrt{\frac{p_1 p_2}{(1-p_1)(1-p_2)}} \cdot \frac{1-p_1}{p_1} = \sqrt{\frac{p_2(1-p_1)}{p_1(1-p_2)}}$; symmetrically using $\lambda_2 \geq 0$ gives the other bound. So the correlation lies in $[0, \min\{\cdot\}]$ — always nonnegative and bounded away from 1 unless the p_i match.

Exercise 5.58

Central idea. (a) is induction via convolution; $\{N = n\} = \{S_{n-1} \leq 1\} \setminus \{S_n \leq 1\}$ since partial sums increase; then $E[N] = \sum_{n\geq 0} P(N > n) = \sum 1/n! = e$.

Solution. (a) For $0 \leq t \leq 1$: $P(S_1 \leq t) = t$, and $P(S_k \leq t) = \int_0^t P(S_{k-1} \leq t - u)\, du = \int_0^t \frac{(t-u)^{k-1}}{(k-1)!}\, du = \frac{t^k}{k!}$ by induction. (b) $\{N = n\} = \{S_{n-1} \leq 1 < S_n\} = \{S_{n-1} \leq 1\} \setminus \{S_n \leq 1\}$ (since $S_n \geq S_{n-1}$, $\{S_n \leq 1\} \subseteq$

$\{S_{n-1} \le 1\}$), so $P(N=n) = P(S_{n-1} \le 1) - P(S_n \le 1) = \frac{1}{(n-1)!} - \frac{1}{n!}$ (with $P(N=1) = 0$ since $S_1 = U_1 \le 1$ a.s.). For a positive integer variable, $E[N] = \sum_{n\ge 0} P(N > n) = \sum_{n\ge 0} P(S_n \le 1) = \sum_{n=0}^{\infty} \frac{1}{n!} = e$. (c) Simulate: repeatedly generate uniforms until their sum exceeds 1, record N; the average of many N's estimates $e \approx 2.71828$. (d) $\mathrm{Var}(N) = E[N^2] - e^2$ with $E[N^2] = \sum_{n\ge 0}(2n+1)P(N > n) = \sum_{n\ge 0} \frac{2n+1}{n!} = 3e$, so $\mathrm{Var}(N) = e(3-e) \approx 0.766$. For the mean of n replicates, $1.96\sqrt{\mathrm{Var}(N)/n} \le 0.0005$ (four digits = ± 0.0005) gives $n \ge \left(\frac{1.96\sqrt{0.766}}{0.0005}\right)^2 \approx 1.18 \times 10^7$ — about 12 million replications.

Exercise 5.59

Central idea. Accept-reject with a uniform$(0,1)$ candidate: $P(X \le x, \text{accept}) = \int_0^x f(t)/c\, dt$ and $P(\text{accept}) = 1/c$; the ratio is $\int_0^x f$.

Solution. (Example 5.6.7: generate $X \sim$ uniform$(0,1)$, $Y \sim$ uniform$(0,c)$ with $c = \max_{[0,1]} f$, f the beta(a,b) pdf; keep X if $Y < f(X)$.) For $0 \le x \le 1$,

$$P(X \le x,\ Y < f(X)) = \int_0^x P(Y < f(t))\, dt = \int_0^x \frac{f(t)}{c}\, dt,$$

and $P(\text{accept}) = P(Y < f(X)) = \int_0^1 f(t)/c\, dt = 1/c$. Hence

$$P(X \le x \mid \text{accept}) = \frac{\int_0^x f(t)/c\, dt}{1/c} = \int_0^x f(t)\, dt,$$

so accepted draws have pdf f: beta(a,b). (Requires $a, b \ge 1$ so f is bounded on $[0,1]$.)

Exercise 5.60

Central idea. (X,Y) is uniform over the rectangle $[a,b] \times [0,c]$; accepting iff the point falls under the curve $y = f(x)$ makes the accepted X's density proportional to f.

Solution. (a) For $a \le w \le b$: $P(W \le w) = P(X \le w,\ Y < f(X)) = \int_a^w P(Y < f(t)) \cdot \frac{dt}{b-a} = \int_a^w \frac{f(t)}{c(b-a)}\, dt$. (b) $P(W \le b) = \int_a^b f/[c(b-a)] = 1/[c(b-a)]$ (since f is a pdf). Conditioning on acceptance $\{W \le b\}$,

$$P(W \le w \mid W \le b) = \frac{\int_a^w f(t)\, dt/[c(b-a)]}{1/[c(b-a)]} = \int_a^w f(t)\, dt,$$

so accepted W's have pdf f. Geometrically: accept the uniform rectangle point iff it lies under $y = f(x)$; the x-marginal of the sub-curve region has density $\propto f(x)$. Repeat (generate new (X,Y)) until $W \le b$ — this is the accept-reject method, with acceptance probability $1/[c(b-a)]$.

Exercise 5.61

Central idea. The proposal must dominate both endpoints. For gamma targets, increase the proposal scale as well as choosing a shape no larger than the target shape.

Solution. (a) Assume $a, b > 1$ so $k = \lfloor a \rfloor$, $l = \lfloor b \rfloor$ are positive. The density ratio is $B(k,l)y^{a-k}(1-y)^{b-l}/B(a,b)$, which is bounded. If either shape is below one, the proposed zero-shape beta distribution does not exist; that is another missing restriction. (b) With the *same* scale b, the ratio for $V \sim \mathrm{Gamma}(k,b)$ is proportional to y^{a-k} and is unbounded. The authors' seventh-printing errata specify $V \sim \mathrm{Gamma}(\lfloor a \rfloor, b+1)$. More generally, for $V \sim \mathrm{Gamma}(k,c)$ with $0 < k \le a$ and $c > b$,

$$\frac{f_Y(y)}{f_V(y)} = \frac{\Gamma(k)c^k}{\Gamma(a)b^a} y^{a-k} \exp\{-y(1/b - 1/c)\}.$$

For $k < a$ it peaks at $y_* = (a-k)/(1/b - 1/c)$ and has finite maximum

$$M(k,c) = \frac{\Gamma(k)c^k}{\Gamma(a)b^a}\left[\frac{a-k}{e(1/b-1/c)}\right]^{a-k}.$$

Thus the earlier claim that no single integer-shape gamma proposal works was false; it holds only when the scale is fixed at b. (c) Shape $\lfloor a \rfloor + 1 > a$ makes the ratio diverge at zero for gamma proposals of any scale and for the indicated beta proposal. (d) For beta integer shapes, minimize

$$M(k,l) = \frac{B(k,l)}{B(a,b)}\left(\frac{a-k}{a+b-k-l}\right)^{a-k}\left(\frac{b-l}{a+b-k-l}\right)^{b-l}$$

over $1 \le k \le \lfloor a \rfloor$, $1 \le l \le \lfloor b \rfloor$. Floors are not always optimal: at $(a,b) = (4.2, 2.9)$, $M(4,2) = 1.81954$ but $M(3,2) = 1.21757$. For a gamma proposal of fixed shape $k < a$, minimize over its scale: differentiating $\log M$ gives $c_* = ab/k$. The optimized envelope is

$$M_*(k) = \frac{\Gamma(k)a^a e^{-(a-k)}}{\Gamma(a)k^k}.$$

For integer k, $\log M_*(k+1) - \log M_*(k) = 1 - (k+1)\log(1+1/k) < 0$. Consequently the optimal positive integer shape is $k = \lfloor a \rfloor$ with scale $c = ab/\lfloor a \rfloor$, assuming $a > 1$. If the proposal scale is fixed specifically at $b+1$, minimize $M(k, b+1)$ over the finite admissible integer grid instead. If $0 < a < 1$, no positive integer shape can dominate the singularity at zero. In all admissible cases $E(N) = M$.

Exercise 5.62

Central idea. $M = \sup_x \phi(x)/g(x)$: for Cauchy the ratio $\propto (1+x^2)e^{-x^2/2}$ peaks at $x = \pm 1$; for Laplace $\propto e^{|x|-x^2/2}$ also peaks at $|x| = 1$ but lower.

Solution. With $f = \phi$: (a) $g(x) = \frac{1}{\pi(1+x^2)}$: $\frac{f(x)}{g(x)} = \sqrt{\frac{\pi}{2}}(1+x^2)e^{-x^2/2}$, maximized at $x = \pm 1$ (derivative $\propto x(1-x^2)e^{-x^2/2}$). $M = \sqrt{\frac{\pi}{2}} \cdot 2e^{-1/2} = \sqrt{\frac{2\pi}{e}} \approx 1.520$. (b) $g(x) = \frac{1}{2}e^{-|x|}$: $\frac{f(x)}{g(x)} = \sqrt{\frac{2}{\pi}}e^{|x|-x^2/2}$, maximized at $|x| = 1$. $M = \sqrt{\frac{2}{\pi}}e^{1/2} = \sqrt{\frac{2e}{\pi}} \approx 1.315$. (c) The double-exponential candidate has the smaller M ($1.315 < 1.520$), i.e., acceptance probability $1/M \approx 0.76$ vs 0.66 — about 15% fewer wasted draws. Recommend the double-exponential (Laplace) candidate: its tails match the normal's better than Cauchy's heavy tails do.

Exercise 5.63

Central idea. The signed exponential(λ) is Laplace$(0, 1/\lambda)$; $M(\lambda) = \sup_{v>0} \frac{\sqrt{2/\pi}\, e^{-v^2/2}}{\lambda e^{-\lambda v}} \propto e^{\lambda^2/2}/\lambda$, minimized at $\lambda = 1$.

Solution. By symmetry it suffices to dominate $|Y|$: target $f(v) = \sqrt{2/\pi}\, e^{-v^2/2}$ ($v > 0$), candidate $g(v) = \lambda e^{-\lambda v}$. $M(\lambda) = \sup_{v>0} \frac{f(v)}{g(v)} = \frac{1}{\lambda}\sqrt{\frac{2}{\pi}} \sup_{v>0} e^{\lambda v - v^2/2} = \sqrt{\frac{2}{\pi}}\, \frac{e^{\lambda^2/2}}{\lambda}$ (max at $v = \lambda$). Minimizing: $\frac{d}{d\lambda}\frac{e^{\lambda^2/2}}{\lambda} = \frac{e^{\lambda^2/2}(\lambda^2-1)}{\lambda^2} = 0$ gives $\lambda = 1$. The optimal algorithm uses exponential(1) (i.e., the Laplace candidate of 5.62(b)), with $M = \sqrt{2e/\pi} \approx 1.315$.

Exercise 5.64

Central idea. The weight f/g converts a g-expectation into an f-expectation (change of measure); self-normalizing divides by weights summing to $\to 1$, and for constant h the weights cancel exactly, giving zero variance.

Solution. (a) $E_g\left[\frac{f(Y)}{g(Y)}h(Y)\right] = \int \frac{f(y)}{g(y)}h(y)g(y)\,dy = \int f(y)h(y)\,dy = E_f[h(X)]$; average over m iid copies. (b) By the WLLN the average $\to_p E_g[(f(Y)/g(Y))h(Y)] = E_f[h(X)]$ (integrability: $E_g|(f/g)h| = E_f|h| \le$

$\sqrt{E_f[h^2]} < \infty$). (c) Write the estimator as $\frac{\frac{1}{m}\sum w_i h(Y_i)}{\frac{1}{m}\sum w_j}$ with $w_i = f(Y_i)/g(Y_i)$. Numerator $\to_p E_f[h(X)]$ by (b); denominator $\to_p E_g[w] = \int f = 1$; Slutsky gives $\to_p E_f[h(X)]$. If $h \equiv c$, the estimator equals $c\,\frac{\sum w_i}{\sum w_j} = c$ exactly — zero variance — whereas (a)'s estimator is $c \cdot \frac{1}{m}\sum w_i$ with positive variance. Hence superior.

Exercise 5.65

Central idea. Conditionally on the Y's, $P(X^* \le x) = \frac{\frac{1}{m}\sum w_i \mathbf{1}_{\{Y_i \le x\}}}{\frac{1}{m}\sum w_j}$; numerator $\to_p \int_{-\infty}^{x} f = F(x)$, denominator $\to_p 1$, so $X^* \to_d f$ — and r such resamples are asymptotically iid from f.

Solution. Conditionally on $Y_1, \dots, Y_m$,

$$P(X^* \le x \mid Y_1, \dots, Y_m) = \sum_{i=1}^{m} q_i \mathbf{1}_{\{Y_i \le x\}} = \frac{\frac{1}{m}\sum_{i=1}^{m} w_i \mathbf{1}_{\{Y_i \le x\}}}{\frac{1}{m}\sum_{j=1}^{m} w_j}, \quad w_i = \frac{f(Y_i)}{g(Y_i)}.$$

By the WLLN, the numerator $\to_p E_g[w(Y)\mathbf{1}_{\{Y \le x\}}] = \int_{-\infty}^{x} f(t)\,dt = F(x)$ and the denominator $\to_p E_g[w(Y)] = 1$. Hence $P(X^* \le x \mid Y\text{'s}) \to_p F(x)$, i.e., $X^* \to_d f$. Drawing $X_1^*, \dots, X_r^*$ iid from this conditional discrete law, they are conditionally independent given the Y's with marginals $\to_d f$ — approximately an iid sample from f for large m.

Exercise 5.66

Central idea. Medians are equivariant under increasing affine maps, so standardize to $n(0,1)$; the asymptotic variance $1/(4f(0)^2) = \pi/2$ can be checked directly against simulation.

Solution. (a) $x \mapsto (x-\mu)/\sigma$ is strictly increasing, hence preserves order: the median (any usual definition, e.g., $X_{((n+1)/2)}$ for odd n) transforms by the same map. So it suffices to study $n(0,1)$. (b) Simulating 60,000 samples of size 15 from $n(0,1)$ and recording the sample median (8th order statistic): the medians are centered near 0 with sample sd ≈ 0.318; the histogram is bell-shaped, slightly heavier-tailed than normal. (c) The asymptotic law gives $\mathrm{sd}(m) \approx \sqrt{\frac{1}{4f(0)^2 n}} = \sqrt{\frac{\pi/2}{15}} \approx 0.324$ (since $f(0) = 1/\sqrt{2\pi}$). The simulated 0.318 is remarkably close — yes, $n = 15$ is already large enough for the asymptotics to be quite accurate here (the normal population is the friendliest case for the median's CLT).

Exercise 5.67

Central idea. Random-walk Metropolis needs only the *unnormalized* target density: propose $X^* = X_t + \varepsilon$ (ε symmetric, e.g., normal), accept with probability $\min(1, \pi(X^*)/\pi(X_t))$; normalizing constants cancel.

Solution. In each case run a symmetric random-walk Metropolis chain: at state x, propose $x^* = x + z$, $z \sim n(0, \tau^2)$ (tune τ for $\approx$ 25–40% acceptance); accept (move to x^*) with probability $\alpha = \min\!\left(1, \frac{\pi(x^*)}{\pi(x)}\right)$, else stay at x. Only ratios of the target π are needed: (a) $\pi(x) \propto \left(1 + \frac{1}{v}\left(\frac{x-\mu}{\sigma}\right)^2\right)^{-(v+1)/2}$ (Student's t, location μ, scale σ); $\alpha = \min\!\left(1, \left[\frac{1+(x^*-\mu)^2/(v\sigma^2)}{1+(x-\mu)^2/(v\sigma^2)}\right]^{-(v+1)/2}\right)$. (b) $\pi(x) \propto \frac{1}{x}\exp\!\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right)$, $x > 0$; reject proposals $x^* \le 0$ (or run the chain on $\log X$ with $\pi_{\log}(y) \propto \exp(-(y-\mu)^2/(2\sigma^2))$ and exponentiate — cleaner, no boundary). (c) $\pi(x) \propto x^{\alpha-1}\exp(-(x/\beta)^\alpha)$, $x > 0$; again reject $x^* \le 0$ or work on the log scale. Discard an initial burn-in (e.g., first few hundred iterates), then take 100 (thinned, if autocorrelation is high) draws as the approximate sample. The chains are valid because the random-walk proposal is symmetric, so detailed balance holds with the stated α (cf. 5.69).

Exercise 5.68

Central idea. (a) Random-walk Metropolis with $n(0,1)$ steps targeting t_v; (b) accept-reject with Cauchy candidate ($M_v = \sup t_v/\mathrm{Cauchy} < \infty$ since the t_v density has tails $|x|^{-(v+1)}$, no heavier than Cauchy when $v \ge 1$); (c) $T = Z/\sqrt{V/v}$ is exact. Simulation confirms: direct is exact, iid, and fastest.

Solution. (a) Target $\pi(x) \propto (1 + x^2/v)^{-(v+1)/2}$. Start $X_0 \sim n(0,1)$; iterate: propose $X^* = X_t + Z$, $Z \sim n(0,1)$; accept with prob $\min(1, \pi(X^*)/\pi(X_t))$. After burn-in the chain is approximately t_v. (b) This envelope argument needs $v \geq 1$; for $0 < v < 1$ the density ratio diverges at infinity. Candidate $g(x) = \frac{1}{\pi(1+x^2)}$ (Cauchy). $M_v = \sup_x f_v(x)/g(x) < \infty$: e.g., $M_2 \approx 1.209$ (ratio $\propto (1 + x^2)(1 + x^2/2)^{-3/2}$, max at $|x| = 1$). Generate $C \sim$ Cauchy, $U \sim$ uniform$(0,1)$; accept C if $U < f_v(C)/(M_v g(C))$; accepted draws are exactly t_v. (c) Generate $Z \sim n(0,1)$ (Box–Muller) and $V \sim \chi^2_v$ (sum of v squared normals, or $-2\sum \log U_i$ if v even) independently; $T = Z/\sqrt{V/v} \sim t_v$ exactly. (d) Samples of 100 (seed 11):

v	Direct (c)	Accept/Reject (b)	Metropolis (a)
2	mean 0.33, var 6.35 (th: var $= \infty$)	mean 0.38, var 5.98, $M = 1.21$, 1.17 draws/accept	mean -0.02, var 5.11, acc. 0.76
10	mean 0.23, var 1.61 (th 1.25)	mean -0.09, var 1.64, $M = 1.45$, 1.68 draws/accept	mean 0.06, var 1.25, acc. 0.72
25	mean 0.11, var 1.10 (th 1.09)	mean -0.08, var 1.33, $M = 1.49$, 1.38 draws/accept	mean 0.16, var 1.20, acc. 0.71

All three produce reasonable t_v samples, but the **direct transformation (c)** is preferred: it is exact (iid draws, no burn-in, no autocorrelation) and fastest (100 draws in ~1ms vs. ~700ms for accept/reject with its Cauchy draws and M evaluation, ~5ms for Metropolis plus burn-in). Accept/reject is exact but wasteful; Metropolis is only approximate and its draws are dependent — its advantage (no supremum needed) doesn't buy anything here since the supremum is easy.

Exercise 5.69

Central idea. Detailed balance: $f(y)q(z \mid y)\alpha(y,z) = f(z)q(y \mid z)\alpha(z,y)$ (both equal $\min(f(y)q(z \mid y), f(z)q(y \mid z))$); integrating out the current state leaves f invariant.

Solution. Let the proposal density be $q(z \mid y)$ and $\alpha(y,z) = \min\!\big(1, \frac{f_Y(z)q(y|z)}{f_Y(y)q(z|y)}\big)$. The chain's kernel is $K(y, dz) = q(z \mid y)\alpha(y,z)\,dz + r(y)\delta_y(dz)$ with $r(y) = 1 - \int q(z \mid y)\alpha(y,z)\,dz$. If $Z_i \sim f_Y$,

$$P(Z_{i+1} \in A) = \iint_A f_Y(y)q(z \mid y)\alpha(y,z)\,dz\,dy + \int_A f_Y(y)r(y)\,dy.$$

Detailed balance: $f_Y(y)q(z \mid y)\alpha(y,z) = \min\big(f_Y(y)q(z \mid y),\, f_Y(z)q(y \mid z)\big) = f_Y(z)q(y \mid z)\alpha(z,y)$, symmetric in y, z. Hence the first term equals $\int_A f_Y(z)\,[\int q(y \mid z)\alpha(z,y)\,dy]\,dz = \int_A f_Y(z)(1 - r(z))\,dz$. Adding the rejection term $\int_A f_Y(z)r(z)\,dz$ gives $\int_A f_Y(z)\,dz = P(Z_i \in A)$. So $Z_{i+1} \sim f_Y$: the target is a stationary (stable) distribution of the Metropolis chain.

Chapter 6

Exercise 6.1

Central idea. The normal density is symmetric, so the likelihood depends on x only through $x^2 = |x|^2$ — a direct factorization.

Solution. The pdf is $f(x \mid \sigma) = (2\pi\sigma^2)^{-1/2}\exp(-x^2/(2\sigma^2)) = g(|x| \mid \sigma) \cdot 1$ with $g(t \mid \sigma) = (2\pi\sigma^2)^{-1/2}\exp(-t^2/(2\sigma^2))$. By the Factorization Theorem, $|X|$ is sufficient for σ (equivalently, $P(X = x \mid |X| = t)$ is $\pm t$ with probability $1/2$ each, free of σ). Yes.

Exercise 6.2

Central idea. The n different support constraints $X_i \geq i\theta$ collapse into the single constraint $\theta \leq \min_i(X_i/i)$; everything else in the likelihood is θ-free.

Solution. The joint density is

$$\prod_{i=1}^{n} e^{i\theta - x_i} I(x_i \geq i\theta) = \exp\left(\theta \sum_{i=1}^{n} i - \sum_{i=1}^{n} x_i\right) I\left(\theta \leq \min_i \frac{x_i}{i}\right) = g(T(\mathbf{x}) \mid \theta)\, h(\mathbf{x}),$$

with $T(\mathbf{x}) = \min_i(x_i/i)$, $g(t \mid \theta) = e^{\theta n(n+1)/2} I(t \geq \theta)$, $h(\mathbf{x}) = e^{-\sum x_i}$. By the Factorization Theorem, T is sufficient.

Exercise 6.3

Central idea. The support constraint involves only the minimum, and the exponent involves only the sum — the two pieces separate cleanly.

Solution. The joint pdf is

$$\sigma^{-n} \exp\left(-\frac{1}{\sigma}\sum_{i=1}^{n}(x_i - \mu)\right) I(x_{(1)} > \mu) = \sigma^{-n} \exp\left(-\frac{n(\bar{x} - \mu)}{\sigma}\right) I(x_{(1)} \geq \mu),$$

which is $g(\bar{x}, x_{(1)} \mid \mu, \sigma) \cdot 1$. Hence $(\bar{X}, X_{(1)})$ — equivalently $(\sum X_i, \min_i X_i)$ — is a two-dimensional sufficient statistic for (μ, σ).

Exercise 6.4

Central idea. Multiply the n exponential-family densities: the θ-dependent part collects into sums of the $t_i(X_j)$, and the rest factors off as $h(\mathbf{x})$.

Solution. The joint pdf/pmf is

$$\begin{aligned}\prod_{j=1}^{n} f(x_j \mid \theta) &= \prod_{j=1}^{n} h(x_j)\, c(\theta) \exp\left(\sum_{i=1}^{k} w_i(\theta) t_i(x_j)\right) \\ &= \underbrace{[c(\theta)]^n \exp\left(\sum_{i=1}^{k} w_i(\theta) \sum_{j=1}^{n} t_i(x_j)\right)}_{g(T(\mathbf{x})|\theta)} \underbrace{\prod_{j=1}^{n} h(x_j)}_{h(\mathbf{x})}.\end{aligned}$$

This is the factorization $f(\mathbf{x} \mid \theta) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x})$, so $T(\mathbf{X})$ is sufficient by the Factorization Theorem.

Exercise 6.5

Central idea. Each observation constrains $\theta > |x_i/i - 1|$; the n constraints combine into one involving only the max and min of X_i/i.

Solution. The joint pdf is $(\prod_{i=1}^{n} \frac{1}{2i})\theta^{-n} \prod_{i=1}^{n} I(-i(\theta-1) < x_i < i(\theta+1))$. Now

$$-i(\theta-1) < x_i < i(\theta+1) \iff \theta > 1 - \frac{x_i}{i} \text{ and } \theta > \frac{x_i}{i} - 1 \iff \theta > \left|\frac{x_i}{i} - 1\right|,$$

so the indicator is $I(\theta > \max_i |x_i/i - 1|)$. Since $\max_i |x_i/i - 1| = \max(\max_i(x_i/i) - 1,\ 1 - \min_i(x_i/i))$, with $T_1 = \max_i(X_i/i)$ and $T_2 = \min_i(X_i/i)$ the joint pdf equals $g(T_1, T_2 \mid \theta) \cdot \text{const}$. Thus $(T_1, T_2) = (\max_i(X_i/i),\ \min_i(X_i/i))$ is a two-dimensional sufficient statistic for θ.

Exercise 6.6

Central idea. The gamma density is a two-parameter exponential family with natural statistics x and $\log x$.

Solution. With $f(x \mid \alpha, \beta) = x^{\alpha-1}e^{-x/\beta}/(\Gamma(\alpha)\beta^\alpha)$,

$$\prod_{i=1}^n f(x_i \mid \alpha, \beta) = \frac{\left(\prod_{i=1}^n x_i\right)^{\alpha-1} e^{-\sum x_i/\beta}}{[\Gamma(\alpha)]^n \beta^{n\alpha}} = g(\textstyle\prod X_i, \sum X_i \mid \alpha, \beta) \cdot 1.$$

Hence $(\prod_{i=1}^n X_i,\ \sum_{i=1}^n X_i)$ — equivalently $(\sum_{i=1}^n \log X_i,\ \sum_{i=1}^n X_i)$ — is a two-dimensional sufficient statistic for (α, β) (this is Theorem 6.2.10 with $t_1(x) = \log x$, $t_2(x) = x$).

Exercise 6.7

Central idea. The likelihood is constant on a rectangle-shaped support; the support's four edges are exactly the four extreme order statistics.

Solution. The joint pdf is

$$\frac{1}{[(\theta_3 - \theta_1)(\theta_4 - \theta_2)]^n}\, I(\theta_1 < x_{(1)},\ x_{(n)} < \theta_3)\, I(\theta_2 < y_{(1)},\ y_{(n)} < \theta_4),$$

where $x_{(1)} = \min_i x_i$, etc. This is $g(\min_i X_i,\ \min_i Y_i,\ \max_i X_i,\ \max_i Y_i \mid \theta_1, \dots, \theta_4)\cdot 1$, so $(\min_i X_i,\ \min_i Y_i,\ \max_i X_i,\ \max_i Y_i)$ is a four-dimensional sufficient statistic.

Exercise 6.8

Central idea. The likelihood $\prod f(x_i - \theta)$ is symmetric in the sample, hence a function of the order statistics; since nothing is assumed about f, the permutation partition is the finest θ-free partition available.

Solution. Sufficiency: $\prod_{i=1}^n f(x_i - \theta) = \prod_{i=1}^n f(x_{(i)} - \theta) = g(T(\mathbf{x}) \mid \theta) \cdot 1$, so the order statistics are sufficient by factorization. For maximality, note no structure of f is assumed. If S were sufficient and strictly coarser than the order statistics, there would exist $\mathbf{x}, \mathbf{y}$ with $S(\mathbf{x}) = S(\mathbf{y})$ but different multisets of values, forcing $f(\mathbf{x} \mid \theta)/f(\mathbf{y} \mid \theta)$ to be θ-free for every location family. But for the double-exponential location family the ratio is θ-free only when the multisets agree (Exercise 6.9(e)), a contradiction. Hence, with f arbitrary, no further reduction is possible — the order statistics are minimal sufficient. (For a specific f, e.g. normal, further reduction is of course possible.)

Exercise 6.9

Central idea. Apply the ratio criterion (Theorem 6.2.13): the likelihood ratio is θ-free exactly on the level sets of the minimal sufficient statistic.

Solution. (a) $\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \exp\left(\theta(\sum x_i - \sum y_i) - \frac{1}{2}(\sum x_i^2 - \sum y_i^2)\right)$, constant in θ iff $\sum x_i = \sum y_i$. Minimal sufficient: $\sum X_i$ (or $\bar{X}$). (b) $\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = e^{-\sum(x_i - y_i)} \frac{I(\theta < x_{(1)})}{I(\theta < y_{(1)})}$, constant in θ iff $x_{(1)} = y_{(1)}$ (otherwise the ratio jumps between 0 and a nonzero value as θ crosses the minima). Minimal sufficient: $X_{(1)}$. (c) $\log \frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = -\sum(x_i - y_i) + 2\sum_i \left[\log(1 + e^{\theta - y_i}) - \log(1 + e^{\theta - x_i})\right]$. If the ratio is θ-free, differentiating and expanding $\sigma(\theta - c) = 1 - e^{-(\theta-c)} + O(e^{-2(\theta-c)})$ as $\theta \to \infty$ gives $\sum e^{kx_i} = \sum e^{ky_i}$ for all $k \geq 1$, whence $\{e^{x_i}\}$ and $\{e^{y_i}\}$ have the same power sums and are the same multiset. Conversely equal multisets give ratio 1. Minimal sufficient: the order statistics $(X_{(1)}, \dots, X_{(n)})$. (d) $\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \prod_{i=1}^n \frac{1+(y_i-\theta)^2}{1+(x_i-\theta)^2}$. If constant ($= c$) for all θ, then $\prod_i((\theta - y_i)^2 + 1) = c\prod_i((\theta - x_i)^2 + 1)$ as polynomials; leading coefficients force $c = 1$, and unique factorization over $\mathbb{C}$ (roots $y_i \pm i$, $x_i \pm i$) forces $\{x_i\} = \{y_i\}$ as multisets. Minimal sufficient: the order statistics. (e) $\log \frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \sum|y_i - \theta| - \sum|x_i - \theta|$, piecewise linear in θ with slope jumping by 2 at each data point. If constant for all θ, the slope $-\#\{x_i > \theta\} + \#\{x_i < \theta\} + \#\{y_i > \theta\} - \#\{y_i < \theta\} \equiv 0$, so the empirical distributions agree at all continuity points: same multiset. Minimal sufficient: the order statistics.

Exercise 6.10

Central idea. The range $X_{(n)} - X_{(1)}$ has a θ-free mean, so centering it gives a nonzero unbiased estimator of zero.

Solution. By Example 6.2.15, $T = (X_{(1)}, X_{(n)})$ is minimal sufficient. For X_i iid uniform$(\theta, \theta+1)$, $\mathrm{E}_\theta X_{(1)} = \theta + \frac{1}{n+1}$ and $\mathrm{E}_\theta X_{(n)} = \theta + \frac{n}{n+1}$, so $\mathrm{E}_\theta[X_{(n)} - X_{(1)}] = \frac{n-1}{n+1}$ for all θ. Hence $g(X_{(1)}, X_{(n)}) = X_{(n)} - X_{(1)} - \frac{n-1}{n+1}$ is not identically zero but satisfies $\mathrm{E}_\theta g = 0$ for all θ: T is not complete. (For $n = 1$ the witness is vacuous since $X_{(1)} = X_{(n)}$ a.s.; the argument needs $n \geq 2$.)

Exercise 6.11

Central idea. All five are location families, so writing $X_i = Z_i + \theta$ makes the θ cancel in every difference; independence then hinges on whether the minimal sufficient statistic is complete (Basu) or already contains the Y_i.

Solution. (a) Each pdf in 6.9 has the form $f_0(x - \theta)$. Write $X_i = Z_i + \theta$ with Z_i iid from f_0 (free of θ). Then $Y_i = X_{(n)} - X_{(i)} = (Z_{(n)} + \theta) - (Z_{(i)} + \theta) = Z_{(n)} - Z_{(i)}$, whose distribution does not involve θ. Hence $(Y_1, \ldots, Y_{n-1})$ is ancillary. General theorem (as in Example 6.2.18): in a location family, any statistic invariant under simultaneous translation $x_i \mapsto x_i + c$ (in particular any function of the differences $X_i - X_j$) is ancillary. (b) (a) Normal: minimal sufficient $\bar{X}$ is complete sufficient, (Y_i) ancillary $\Rightarrow$ independent by Basu's Theorem. (b) Location exponential: minimal sufficient $X_{(1)}$ is complete sufficient (Exercise 6.30(a)), (Y_i) ancillary $\Rightarrow$ independent by Basu. (c) Logistic, (d) Cauchy, (e) double exponential: the minimal sufficient statistic is the full set of order statistics, of which $(Y_1, \ldots, Y_{n-1})$ is a (nontrivial) function — so they are not independent (indeed $Y_{n-1} = X_{(n)} - X_{(n-1)}$ is determined by the minimal sufficient statistic).

Exercise 6.12

Central idea. Condition on N: the joint pmf factors into a θ-free part times $\theta^x(1-\theta)^{n-x}$, and the likelihood ratio pins down both coordinates.

Solution. (a) $P_\theta(X = x, N = n) = p_n \binom{n}{x} \theta^x (1-\theta)^{n-x}$. For $(x, n), (y, m)$,

$$\frac{P_\theta(x, n)}{P_\theta(y, m)} = \frac{p_n \binom{n}{x}}{p_m \binom{m}{y}} \, \theta^{x-y} (1-\theta)^{(n-x)-(m-y)},$$

constant in $\theta \in (0, 1)$ iff $x = y$ and $n - x = m - y$, i.e. $(x, n) = (y, m)$. By Theorem 6.2.13, (X, N) is minimal sufficient. N has distribution $P(N = n) = p_n$, free of θ: ancillary. (b) $\mathrm{E}_\theta[X/N] = \mathrm{E}[\mathrm{E}[X/N \mid N]] = \mathrm{E}[(N\theta)/N] = \theta$. By the conditional variance formula, $\mathrm{Var}(X/N) = \mathrm{E}[\mathrm{Var}(X/N \mid N)] + \mathrm{Var}(\mathrm{E}[X/N \mid N]) = \mathrm{E}[(1/N^2)N\theta(1-\theta)] + \mathrm{Var}(\theta) = \theta(1-\theta)\mathrm{E}(1/N)$.

Exercise 6.13

Central idea. $X^\alpha \sim$ exponential(1) regardless of α; taking logs turns the power $1/\alpha$ into a factor that cancels in the ratio.

Solution. For $X \sim f(\cdot \mid \alpha)$, $P(X^\alpha > t) = P(X > t^{1/\alpha}) = \int_{t^{1/\alpha}}^\infty \alpha x^{\alpha-1} e^{-x^\alpha} \, dx = e^{-t}$, so $Y_i = X_i^\alpha$ are iid exponential(1), free of α. Then $\log X_i = \frac{1}{\alpha} \log Y_i$, and

$$\frac{\log X_1}{\log X_2} = \frac{\log Y_1}{\log Y_2},$$

whose distribution does not depend on α. Hence it is ancillary.

Exercise 6.14

Central idea. A location shift moves the median and the mean by the same amount, so their difference is shift-invariant.

Solution. Write $X_i = Z_i + \theta$ with Z_i iid from the base distribution (free of θ). The sample median satisfies $M = M_Z + \theta$ and $\bar{X} = \bar{Z} + \theta$, so $M - \bar{X} = M_Z - \bar{Z}$, free of θ. Hence $M - \bar{X}$ is ancillary (a special case of the general location-family result in Exercise 6.11(a)).

Exercise 6.15

Central idea. The mean–variance pair $(\theta, a\theta^2)$ traces a one-dimensional parabola in $\mathfrak{R}^2$; sufficiency comes from the exponential-family form, while incompleteness comes from an unbiased estimator of zero built from $\mathrm{E}\bar{X}^2$ and $\mathrm{E}S^2$.

Solution. (a) The parameter space $\{(\theta, a\theta^2) : \theta > 0\}$ is the parabola $\sigma^2 = a\mu^2$ ($\mu > 0$), a one-dimensional curve in $\mathfrak{R}^2$; it contains no nonempty open subset of $\mathfrak{R}^2$. (b) The joint pdf is $(2\pi a\theta^2)^{-n/2} \exp\big(-\sum(x_i - \theta)^2/(2a\theta^2)\big) = (2\pi a\theta^2)^{-n/2} \exp\big(-\sum x_i^2/(2a\theta^2) + \sum x_i/(a\theta) - n/(2a)\big)$, i.e. an exponential family in $(\sum X_i^2, \sum X_i)$, equivalently in $(\bar{X}, S^2)$ since $\sum X_i^2 = (n-1)S^2 + n\bar{X}^2$. By Theorem 6.2.10, $T = (\bar{X}, S^2)$ is sufficient. It is not complete: $\mathrm{E}_\theta S^2 = a\theta^2$ and $\mathrm{E}_\theta \bar{X}^2 = \theta^2 + a\theta^2/n = \theta^2(n+a)/n$, so $g(\bar{X}, S^2) = a\bar{X}^2 - \frac{n+a}{n} S^2$ is nonzero with $\mathrm{E}_\theta g = a\theta^2 \frac{n+a}{n} - \frac{n+a}{n} a\theta^2 = 0$ for all $\theta > 0$.

Exercise 6.16

Central idea. The multinomial is an exponential family, but the four cell probabilities trace a one-dimensional curve (not an open set) in the natural parameter space; the ratio criterion then merges the two symmetric cells.

Solution. (a) The pmf is $\frac{n!}{x_1!x_2!x_3!x_4!} \exp\big(x_1 \log(\frac{1}{2} + \frac{\theta}{4}) + (x_2 + x_3)\log(\frac{1-\theta}{4}) + x_4 \log(\frac{\theta}{4})\big)$, an exponential family with natural statistics $(X_1, X_2 + X_3, X_4)$. The natural parameters $\eta_1 = \log(\frac{1}{2} + \frac{\theta}{4})$, $\eta_2 = \log(\frac{1-\theta}{4})$, $\eta_3 = \log(\frac{\theta}{4})$ trace a one-dimensional curve in $\mathfrak{R}^3$ as $\theta \in (0,1)$ varies (e.g. $\eta_2 - \eta_3 = \log((1-\theta)/\theta)$ ranges over $\mathfrak{R}$ while nonlinearly tied to η_1), containing no open set: curved. (b) By Theorem 6.2.10 (or factorization), (X_1, X_2, X_3, X_4) — and hence $(X_1, X_2 + X_3, X_4)$ — is sufficient. (c) By Theorem 6.2.13, with $d_1 = x_1 - y_1$, $d_{23} = (x_2 + x_3) - (y_2 + y_3)$, $d_4 = x_4 - y_4$,

$$\log \frac{p(\mathbf{x} \mid \theta)}{p(\mathbf{y} \mid \theta)} = \text{const} + d_1 \log(2 + \theta) + d_{23} \log(1 - \theta) + d_4 \log \theta,$$

constant in $\theta \in (0,1)$ iff $d_1 = d_{23} = d_4 = 0$ (differentiate: $d_1/(2+\theta) - d_{23}/(1-\theta) + d_4/\theta \equiv 0$ forces all d's to be 0). Thus $(X_1, X_2 + X_3, X_4)$ is minimal sufficient; since $X_1 + X_2 + X_3 + X_4 = n$, this is equivalent to the two-dimensional (X_1, X_4).

Exercise 6.17

Central idea. The joint pmf is $\theta^n(1-\theta)^{\sum x_i - n}$ — an exponential family in $\sum X_i$, whose distribution is negative binomial with an open natural parameter space.

Solution. $\prod_{i=1}^n \theta(1-\theta)^{x_i - 1} = \theta^n(1-\theta)^{\sum x_i - n} = g(\sum x_i \mid \theta) \cdot 1$, so $\sum X_i$ is sufficient by factorization. $S = \sum X_i$ is the trial of the nth success: $P_\theta(S = s) = \binom{s-1}{n-1}\theta^n(1-\theta)^{s-n}$, $s = n, n+1, \ldots$ (negative binomial). The family is complete: if $\mathrm{E}_\theta[g(S)] = \theta^n \sum_{k \geq 0} g(k+n)\binom{k+n-1}{n-1}(1-\theta)^k = 0$ for all $\theta \in (0,1)$, the power series in $(1-\theta)$ is identically zero, so $g \equiv 0$. (Equivalently, it is a full exponential family with natural parameter $\log(1-\theta) \in (-\infty, 0)$, open, so Theorem 6.2.25 applies.)

Exercise 6.18

Central idea. $\sum X_i \sim \text{Poisson}(n\lambda)$; a zero expectation for all $\lambda > 0$ is a power series in λ identically zero, forcing every coefficient to vanish.

Solution. $T = \sum X_i \sim \text{Poisson}(n\lambda)$. If $\mathrm{E}_\lambda[g(T)] = \sum_{k=0}^\infty g(k) e^{-n\lambda}(n\lambda)^k/k! = 0$ for all $\lambda > 0$, then $\sum_{k=0}^\infty \frac{g(k) n^k}{k!} \lambda^k = 0$ for all $\lambda > 0$. A power series vanishing on $(0, \infty)$ has all coefficients zero, so $g(k) = 0$ for every k: T is complete.

Exercise 6.19

Central idea. $\mathrm{E}_p[g(X)] = 0$ is a polynomial in p; it vanishes on an interval iff every coefficient vanishes — count the free coefficients against the three values of g.

Solution. Distribution 1: $\mathrm{E}_p[g(X)] = g(0)p + 3g(1)p + g(2)(1-4p) = g(2) + p\,[g(0) + 3g(1) - 4g(2)] \equiv 0$ on $(0, \frac{1}{4})$ gives $g(2) = 0$ and $g(0) + 3g(1) = 0$, with nonzero solutions (e.g. $g(0) = 3$, $g(1) = -1$, $g(2) = 0$). Not complete. Distribution 2: $\mathrm{E}_p[g(X)] = g(2) + p\,[g(0) - g(2)] + p^2\,[g(1) - g(2)] \equiv 0$ on $(0, \frac{1}{2})$ forces $g(2) = 0$, $g(0) = g(2)$, $g(1) = g(2)$, i.e. $g \equiv 0$. Complete.

Exercise 6.20

Central idea. (a) is a nonregular family handled by differentiating the zero-expectation integral; (b), (c), (e) are full exponential families with open natural parameter spaces; (d) is a one-parameter exponential family in disguise ($t(x) = e^{-x}$).

Solution. (a) Joint $= 2^n(\prod x_i)\theta^{-2n} I(x_{(n)} < \theta)$: $X_{(n)}$ sufficient. Its pdf is $2nt^{2n-1}/\theta^{2n}$ on $(0, \theta)$. If $\mathrm{E}_\theta[g(X_{(n)})] = \theta^{-2n}\int_0^\theta g(t)2nt^{2n-1}\,dt = 0$ for all $\theta > 0$, then $H(\theta) = \int_0^\theta g(t)t^{2n-1}\,dt \equiv 0$, so $H'(\theta) = g(\theta)\theta^{2n-1} = 0$ a.e.: $g = 0$ a.e. $X_{(n)}$ is complete sufficient. (b) Joint $= \theta^n \exp(-(1+\theta)\sum \log(1+x_i))$: $T = \sum \log(1+X_i)$ sufficient. Since $\log f = -\theta \log(1+x) + \log\theta - \log(1+x)$, this is a full exponential family with natural parameter $-\theta \in (-\infty, 0)$, open: complete by Theorem 6.2.25. (c) Joint $= (\log\theta)^n \theta^{\sum x_i}/(\theta-1)^n$: $\sum X_i$ sufficient; $\log f = (\log\theta)x + \log(\log\theta/(\theta-1))$, natural parameter $\log\theta \in (0, \infty)$, open: complete by Theorem 6.2.25. (d) Joint $= \exp(n\theta - \sum x_i - e^\theta \sum e^{-x_i})$: $T = \sum e^{-X_i}$ sufficient, and $\log f = -e^\theta e^{-x} + \theta - x$ is a one-parameter exponential family in $t(x) = e^{-x}$ with natural parameter $-e^\theta \in (-\infty, 0)$, open: complete by Theorem 6.2.25. (Indeed the ratio criterion gives minimal sufficiency of $\sum e^{-X_i}$ directly.) (e) Each $X_i \sim$ binomial$(2, \theta)$, so $T = \sum X_i \sim$ binomial$(2n, \theta)$ is sufficient. If $\mathrm{E}_\theta[g(T)] = \sum_{k=0}^{2n} g(k)\binom{2n}{k}\theta^k(1-\theta)^{2n-k} \equiv 0$ on $[0, 1]$, dividing by $(1-\theta)^{2n}$ gives a degree-$2n$ polynomial in $\theta/(1-\theta)$ vanishing on $[0, \infty)$, so all $g(k)\binom{2n}{k} = 0$: $g \equiv 0$. $\sum X_i$ is complete sufficient.

Exercise 6.21

Central idea. $|X| \sim$ Bernoulli(θ) carries all the θ-information (the sign is a fair coin flip); the exponential representation breaks at the endpoints $\theta = 0, 1$.

Solution. Here $P_\theta(X = 0) = 1 - \theta$, $P_\theta(X = \pm 1) = \theta/2$. (a) X is sufficient (the whole sample always is). $\mathrm{E}_\theta[g(X)] = g(0) + \theta\big[\frac{g(-1)+g(1)}{2} - g(0)\big] \equiv 0$ on $[0, 1]$ gives $g(0) = 0$, $g(-1) + g(1) = 0$, with nonzero solutions (e.g. $g(-1) = 1$, $g(1) = -1$, $g(0) = 0$). Not complete. (b) $|X|$ takes values $0, 1$ with $P_\theta(|X| = 1) = \theta$: given $|X|$, X is 0 or ± 1 with probability $1/2$ each, free of θ, so $|X|$ is sufficient. $\mathrm{E}_\theta[g(|X|)] = g(0) + \theta[g(1) - g(0)] \equiv 0$ forces $g(0) = g(1) = 0$: complete. Yes. (c) No. On $0 < \theta < 1$ one can write $f(x \mid \theta) = (1-\theta)\exp\left(|x| \log\frac{\theta}{2(1-\theta)}\right)$, but no representation $h(x)c(\theta)e^{w(\theta)|x|}$ works on all of $[0, 1]$: at $\theta = 0$, $f(0 \mid 0) = 1$ forces $c(0), h(0) > 0$, while $f(1 \mid 0) = 0$ then forces $h(1) = 0$, contradicting $f(1 \mid \theta) = \theta/2 > 0$ for $\theta \in (0, 1)$.

Exercise 6.22

Central idea. The likelihood depends on the data only through $\sum \log X_i$; $\sum X_i$ does not determine $\sum \log X_i$, and $\sum \log X_i$ is the natural statistic of a full exponential family.

Solution. (a) No. The joint pdf is $\theta^n \exp\left((\theta - 1)\sum \log x_i\right)$, so by Theorem 6.2.13 the ratio is θ-free iff $\sum \log x_i = \sum \log y_i$. But $\sum x_i = \sum y_i$ does not imply $\sum \log x_i = \sum \log y_i$ (e.g. $n = 2$, $\mathbf{x} = (0.5, 0.5)$, $\mathbf{y} = (0.25, 0.75)$), so $\sum X_i$ is not sufficient. (b) $T = \sum_{i=1}^n \log X_i$ (equivalently $\prod X_i$) is sufficient by factorization, and $\log f(x \mid \theta) = (\theta - 1)\log x + \log\theta$ is a one-parameter exponential family with natural parameter $\theta - 1 \in (-1, \infty)$, an open set: complete by Theorem 6.2.25.

Exercise 6.23

Central idea. The support $(\theta, 2\theta)$ brackets θ between $X_{(n)}/2$ and $X_{(1)}$; the ratio criterion shows both extremes are needed, and a linear combination has constant (zero) mean.

Solution. The joint pdf is $\theta^{-n} I(x_{(n)}/2 < \theta < x_{(1)})$. For sample points $\mathbf{x}, \mathbf{y}$ the ratio $f(\mathbf{x} \mid \theta)/f(\mathbf{y} \mid \theta)$ equals 1 on $\theta > \max(x_{(n)}, y_{(n)})/2$ intersected with the respective upper bounds; it is θ-free (where defined) iff $x_{(1)} = y_{(1)}$ and $x_{(n)} = y_{(n)}$. By Theorem 6.2.13, $(X_{(1)}, X_{(n)})$ is minimal sufficient. It is not complete: $\mathrm{E}_\theta X_{(1)} = \theta\frac{n+2}{n+1}$ and $\mathrm{E}_\theta X_{(n)} = \theta\frac{2n+1}{n+1}$, so $g = (2n+1)X_{(1)} - (n+2)X_{(n)}$ is nonzero with $\mathrm{E}_\theta g \equiv 0$. (For $n = 1$, $X_{(1)} = X_{(n)}$ a.s. and the witness vanishes; the argument needs $n \geq 2$.)

Exercise 6.24

Central idea. With only two parameter values, $\mathrm{E}_\lambda[g(X)] = 0$ is just two linear constraints on infinitely many values $g(x)$ — plenty of room for a nonzero solution.

Solution. P_0 is degenerate at 0, so $\mathrm{E}_0[g(X)] = g(0) = 0$. Also $\mathrm{E}_1[g(X)] = e^{-1}\sum_{x\geq 1} g(x)/x! = 0$ is one linear constraint on $\{g(x)\}_{x\geq 1}$, satisfied by e.g. $g(1) = 1$, $g(2) = -2$, $g(x) = 0$ otherwise (since $1/1! - 2/2! = 0$). This nonzero g has $\mathrm{E}_\lambda g(X) = 0$ for $\lambda = 0, 1$: $\mathcal{P}$ is not complete. (With $\lambda > 0$ continuous, Exercise 6.18 shows the Poisson family is complete — the range matters.)

Exercise 6.25

Central idea. Write the log-likelihood ratio as a function of μ (resp. (μ, σ^2)) and read off which sums must match for it to be parameter-free; in $n(\mu, \mu)$ the $\sum x_i$ terms cancel, leaving only $\sum x_i^2$.

Solution. Put $A = \sum x_i^2 - \sum y_i^2$, $B = \sum x_i - \sum y_i$. (a)–(b) For $n(\mu, \mu)$: $\log\frac{f(\mathbf{x}|\mu)}{f(\mathbf{y}|\mu)} = -\frac{A}{2\mu} + B$, constant in $\mu > 0$ iff $A = 0$ (B is already μ-free). By Theorem 6.2.13, $\sum X_i^2$ is minimal sufficient. $(\sum X_i, \sum X_i^2)$ is sufficient by Theorem 6.2.10 but not minimal, since $\sum X_i^2$ is sufficient and $(\sum X_i, \sum X_i^2)$ is not a function of $\sum X_i^2$ alone. (c) For $n(\mu, \mu^2)$ ($\mu \neq 0$): $\log\frac{f(\mathbf{x}|\mu)}{f(\mathbf{y}|\mu)} = -\frac{A}{2\mu^2} + \frac{B}{\mu}$, constant in μ iff $A = B = 0$ (multiply by $2\mu^2$: $-A + 2B\mu$ constant forces $A = B = 0$). Minimal sufficient: $(\sum X_i, \sum X_i^2)$. (d) For $n(\mu, \sigma^2)$: $\log\frac{f(\mathbf{x}|\mu,\sigma^2)}{f(\mathbf{y}|\mu,\sigma^2)} = -\frac{A}{2\sigma^2} + \frac{\mu B}{\sigma^2}$, constant on the open set $\mathfrak{R} \times (0, \infty)$ iff $A = B = 0$ (vary μ at fixed σ^2 to get $B = 0$, then $A = 0$). Minimal sufficient: $(\sum X_i, \sum X_i^2)$.

Exercise 6.26

Central idea. Theorem 6.6.5 says the likelihood-ratio vector for a finite subfamily with common support is minimal sufficient for that subfamily; if it is 1–1 with the candidate statistic and the candidate is sufficient for the whole family, minimality lifts. (For (c) the common-support hypothesis fails, so the ratio criterion Theorem 6.2.13 — the engine behind 6.6.5, cf. Exercise 6.28 — is used directly; for (d)–(e) ratios at enough generic parameter values determine the order statistics, so 6.6.5 applies directly.)

Solution. (a) Take f_0, f_1 to be the joint $n(0, 1)$ and $n(1, 1)$ densities (common support $\mathfrak{R}^n$). $\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = \exp(n(\bar{x} - \frac{1}{2}))$, 1–1 in $\bar{x}$. $\bar{X}$ is sufficient for $\{n(\theta, 1)\}$, so by Theorem 6.6.5(b) the ratio — hence $\bar{X}$ — is minimal sufficient. (b) Take f_0, f_1 to be the joint gamma$(\alpha, 1)$ and gamma$(\alpha, 2)$ densities (common support $(0, \infty)^n$). $\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} = 2^{-n\alpha} e^{\sum x_i/2}$, 1–1 in $\sum x_i$. $\sum X_i$ is sufficient (Theorem 6.2.10), so by Theorem 6.6.5(b) it is minimal sufficient. (c) The uniform$(0, \theta)$ family has nested, not common, supports, so Theorem 6.6.5 does not apply directly; use Theorem 6.2.13 instead. Here $\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \frac{I(x_{(n)}<\theta)}{I(y_{(n)}<\theta)}$: on $\theta > \max(x_{(n)}, y_{(n)})$ both densities equal θ^{-n} and the ratio is 1, but as a function of θ the ratio is constant iff $x_{(n)} = y_{(n)}$ (if $x_{(n)} < y_{(n)}$, the ratio is 0 for $\theta \in (x_{(n)}, y_{(n)}]$ but 1 beyond — not constant). Hence $\max X_i = X_{(n)}$ is minimal sufficient. (d) For Cauchy$(\theta, 1)$, $\frac{f(\mathbf{x}|\theta)}{f(\mathbf{y}|\theta)} = \prod_i \frac{1+(y_i-\theta)^2}{1+(x_i-\theta)^2}$; constancy in $\theta \in \mathfrak{R}$ forces $\prod_i((\theta - y_i)^2 + 1) \equiv \prod_i((\theta - x_i)^2 + 1)$ as polynomials, and unique factorization (roots $x_i \pm i$) gives identical multisets (Exercise 6.9(d)). Evaluating the ratio at $2n$ generic θ values determines this degree-$2n$ polynomial, hence the multiset: the finite ratio vector is 1–1 with the order statistics, which are sufficient (joint density symmetric), so by Theorem 6.6.5

they are minimal sufficient. (e) For logistic(μ,β) the same argument as Exercise 6.9(c) (with $\theta=\mu$) shows ratios at n generic μ values determine the multiset; the order statistics are sufficient and hence minimal sufficient by Theorem 6.6.5.

Exercise 6.27

Central idea. Expanding $(x-\mu)^2/(\mu^2 x) = x/\mu^2 - 2/\mu + 1/x$ reveals a two-parameter exponential family in $(\sum X_i, \sum 1/X_i)$ with an open natural parameter space; for $n=2$ the $(S,U)\mapsto(\bar X, W)$ Jacobian computation factors the joint density.

Solution. (a) Since $\frac{(x-\mu)^2}{\mu^2 x} = \frac{x}{\mu^2} - \frac{2}{\mu} + \frac{1}{x}$,

$$\prod_{i=1}^n f(x_i \mid \mu,\lambda) = \left(\frac{\lambda}{2\pi}\right)^{n/2}\Big(\prod x_i^{-3/2}\Big)\exp\!\left(-\frac{\lambda}{2\mu^2}\sum x_i + \frac{n\lambda}{\mu} - \frac{\lambda}{2}\sum\frac{1}{x_i}\right),$$

a two-parameter exponential family in $(\sum X_i, \sum \frac{1}{X_i})$ with natural parameters $\eta_1 = -\lambda/(2\mu^2)$, $\eta_2 = -\lambda/2$ ranging over the open set $(-\infty,0)^2$. $(\bar X, T)$ is 1–1 with $(\sum X_i, \sum 1/X_i)$ (via $\sum 1/X_i = n/T + 1/\bar X$), hence sufficient by Theorem 6.2.10 and complete by Theorem 6.2.25. (b) With the corrected T^*, put $S = X_1 + X_2 = 2\bar X$ and $W = \lambda(\frac{1}{X_1} + \frac{1}{X_2} - \frac{4}{S}) = 2\lambda/T^*$. Writing $X_1 = SU$, $X_2 = S(1-U)$ (Jacobian S) and changing $U \to W$ (two symmetric branches), the joint density becomes

$$h(s,w) = \underbrace{\sqrt{\frac{2\lambda}{\pi}}\, s^{-3/2}\exp\!\left(-\frac{\lambda s}{2\mu^2} + \frac{2\lambda}{\mu} - \frac{2\lambda}{s}\right)}_{\text{density of } S=2\bar X \text{ (so } \bar X = S/2 \sim \mathrm{IG}(\mu,2\lambda))} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} w^{-1/2} e^{-w/2}}_{\chi^2_1 \text{ density}},$$

using $\frac{1}{U(1-U)} = \frac{sw}{\lambda} + 4$ and $|dU/dW| = \frac{s}{\lambda}\big/\big[(\frac{sw}{\lambda}+4)^{3/2}\sqrt{\frac{sw}{\lambda}}\big]$. Thus $\bar X \sim \mathrm{IG}(\mu, 2\lambda)$, $W = n\lambda/T^* \sim \chi^2_1 = \chi^2_{n-1}$, and they are independent.

Exercise 6.28

Central idea. For a finite family the parameter is just the index, so the ratio criterion (6.2.13) applies verbatim; minimality then lifts to any larger family for which the ratio statistic remains sufficient.

Solution. (a) Let the parameter be $i \in \{0,\dots,k\}$ with density f_i. For sample points $\mathbf{x},\mathbf{y}$,

$$\frac{f_i(\mathbf{x})}{f_i(\mathbf{y})} = \frac{f_i(\mathbf{x})/f_0(\mathbf{x})}{f_i(\mathbf{y})/f_0(\mathbf{y})}\cdot\frac{f_0(\mathbf{x})}{f_0(\mathbf{y})},$$

so $\frac{f_i(\mathbf{x})}{f_i(\mathbf{y})}$ is constant in i (including $i=0$) iff $\frac{f_i(\mathbf{x})/f_0(\mathbf{x})}{f_i(\mathbf{y})/f_0(\mathbf{y})} = 1$ for all $i=1,\dots,k$, i.e. iff $T(\mathbf{x}) = T(\mathbf{y})$ where $T(\mathbf{x}) = (\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})},\dots,\frac{f_k(\mathbf{x})}{f_0(\mathbf{x})})$ (common support keeps all ratios defined and positive). By Theorem 6.2.13, $T(\mathbf{X})$ is minimal sufficient for $\{f_0,\dots,f_k\}$. (b) Let T be the ratio statistic of (a), now assumed sufficient for $\mathcal{F} \supset \{f_0,\dots,f_k\}$. If U is any sufficient statistic for $\mathcal{F}$, then U is sufficient for $\{f_0,\dots,f_k\}$, so by minimality in the finite family T is a function of U. Since U was arbitrary, T is minimal sufficient for $\mathcal{F}$.

Exercise 6.29

Central idea. The order statistics are sufficient for every f by symmetry; they are minimal sufficient for the logistic location subfamily (6.26(e)), and any statistic sufficient for all densities is sufficient for that subfamily.

Solution. Let $\mathcal{F}$ be (essentially) all densities. The joint density $\prod_{i=1}^n f(x_i) = \prod_{i=1}^n f(x_{(i)})$ is symmetric, hence a function of the order statistics $S = (X_{(1)},\dots,X_{(n)})$: S is sufficient for $\mathcal{F}$. Let U be any statistic sufficient for $\mathcal{F}$. Then U is sufficient for the logistic location subfamily $\{f(\cdot \mid \mu)\} \subset \mathcal{F}$, for which S is minimal sufficient (Exercise 6.26(e), via Theorem 6.6.5). Hence S is a function of U. Since U was arbitrary, S is minimal sufficient for $\mathcal{F}$: with f completely unknown, the order statistics achieve the maximal data reduction.

Exercise 6.30

Central idea. The joint density splits into $e^{n\mu}I(\mu < x_{(1)})$ times a μ-free factor; completeness follows by differentiating the zero-expectation integral, and S^2 is location-invariant hence ancillary.

Solution. (a) $\prod_{i=1}^n e^{-(x_i-\mu)}I(x_i > \mu) = e^{n\mu}I(\mu < x_{(1)})\, e^{-\sum x_i}$: $X_{(1)}$ sufficient by factorization. Moreover $X_{(1)} = \mu + Z$ with $Z \sim$ exponential(n) ($P(Z > t) = e^{-nt}$). If $\mathrm{E}_\mu[g(X_{(1)})] = 0$ for all μ, then $\int_\mu^\infty g(u)ne^{-n(u-\mu)}\,du = 0$, i.e. $H(\mu) = \int_\mu^\infty g(u)e^{-nu}\,du \equiv 0$; differentiating, $H'(\mu) = -g(\mu)e^{-n\mu} = 0$ a.e., so $g = 0$ a.e.: $X_{(1)}$ is complete. (b) $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ depends on the X_i only through $X_i - \bar{X} = Z_i - \bar{Z}$, free of μ: ancillary. By Basu's Theorem (complete sufficient $X_{(1)}$, ancillary S^2), they are independent.

Exercise 6.31

Central idea. Basu's Theorem turns "complete sufficient + ancillary" into independence for free: (a) gives $\bar{X} \perp S^2$; (b) kills the covariance term in the swindle; (c) makes the ratio $X_{(i)}/\sum X_i$ ancillary, hence independent of the complete sufficient sum.

Solution. (a)(i) $\bar{X} \sim n(\mu, \sigma^2/n)$ is sufficient (factorization) and complete (full exponential family, open parameter space) for μ. S^2 depends only on $X_i - \bar{X}$, free of μ: ancillary. By Basu, $\bar{X} \perp S^2$. (ii) The joint law is $\bar{X} \sim n(\mu, \sigma^2/n) \perp (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ for every $\sigma^2 > 0$; "knowing" σ^2 changes nothing about this factorization, so independence holds whether or not σ^2 is known. (b)(i) With σ^2 known, $\bar{X}$ is complete sufficient for μ and $M - \bar{X}$ is ancillary for μ (location family), so $\mathrm{Cov}(M - \bar{X}, \bar{X}) = 0$ by Basu. Hence $\mathrm{Var}(M) = \mathrm{Var}((M - \bar{X}) + \bar{X}) = \mathrm{Var}(M - \bar{X}) + \mathrm{Var}(\bar{X})$. (ii) Estimating $\mathrm{Var}(M)$ by the sample variance of N simulated medians has variance $\approx 2[\mathrm{Var}(M)]^2/(N-1)$ (the χ^2-type variance of a sample variance); estimating $\mathrm{Var}(M-\bar{X})$ analogously has variance $\approx 2[\mathrm{Var}(M-\bar{X})]^2/(N-1)$. Since $\mathrm{Var}(M-\bar{X}) = \mathrm{Var}(M) - \sigma^2/n < \mathrm{Var}(M)$, the swindle estimator $\widehat{\mathrm{Var}}(M-\bar{X}) + \sigma^2/n$ has strictly smaller Monte Carlo error. (c)(i) $\mathrm{E}[X^k] = \mathrm{E}[(X/Y)^k Y^k] = \mathrm{E}[(X/Y)^k]\,\mathrm{E}[Y^k]$ by independence; dividing gives $\mathrm{E}[(X/Y)^k] = \mathrm{E}[X^k]/\mathrm{E}[Y^k]$ (moments assumed finite, $Y \neq 0$). (ii) Put $Y = \sum X_i$ (complete sufficient for β, α known) and $W_i = X_{(i)}/Y$. The vector $(X_1/Y, \ldots, X_n/Y)$ is scale-invariant, hence ancillary for β; W_i, a function of it, is ancillary too, so $W_i \perp Y$ by Basu. Then $\mathrm{E}[X_{(i)} \mid Y] = \mathrm{E}[W_i Y \mid Y] = Y\,\mathrm{E}[W_i] = Y\,\mathrm{E}[X_{(i)}/Y]$, and by (i) with $X = X_{(i)}$, $\mathrm{E}[X_{(i)}/Y] = \mathrm{E}[X_{(i)}]/\mathrm{E}[Y]$. Hence $\mathrm{E}[X_{(i)} \mid \sum X_i] = \sum X_i \cdot \mathrm{E}[X_{(i)}]/\mathrm{E}[\sum X_i]$.

Exercise 6.32

Central idea. Embed two experiments with proportional likelihoods in a mixture experiment, collapse the two data points with a sufficient statistic, and let the Sufficiency and Conditionality Principles force the evidences equal.

Solution. Let E_1, E_2 have data $\mathbf{x}_1^*, \mathbf{x}_2^*$ with $L_1(\theta \mid \mathbf{x}_1^*) \propto L_2(\theta \mid \mathbf{x}_2^*)$. Form the mixture experiment E^* (perform E_j with probability $1/2$) and the statistic T collapsing $(1, \mathbf{x}_1^*)$ and $(2, \mathbf{x}_2^*)$ to one value (identity elsewhere); T is sufficient in E^* (Exercise 6.33(a)). By the Formal Sufficiency Principle, $\mathrm{Ev}(E^*, (1, \mathbf{x}_1^*)) = \mathrm{Ev}(E^*, (2, \mathbf{x}_2^*))$. By the Conditionality Principle, $\mathrm{Ev}(E^*, (j, \mathbf{x}_j^*)) = \mathrm{Ev}(E_j, \mathbf{x}_j^*)$. Hence $\mathrm{Ev}(E_1, \mathbf{x}_1^*) = \mathrm{Ev}(E_2, \mathbf{x}_2^*)$: the evidence depends on $(E, \mathbf{x})$ only through the likelihood function up to proportionality.

Exercise 6.33

Central idea. The constant C is the likelihood proportionality factor; it absorbs the difference between the two collapsed points so the factorization goes through, and each principle then contributes one equality.

Solution. (a) In E^*, $f^*((j, \mathbf{x}_j) \mid \theta) = \frac{1}{2} f_j(\mathbf{x}_j \mid \theta)$. By hypothesis $f_1(\mathbf{x}_1^* \mid \theta) = c\, f_2(\mathbf{x}_2^* \mid \theta)$ for a θ-free c; take $C = f_2(\mathbf{x}_2^* \mid \theta)/f_1(\mathbf{x}_1^* \mid \theta) = 1/c$, free of θ. Recall $T(1, \mathbf{x}_1^*) = T(2, \mathbf{x}_2^*) = (1, \mathbf{x}_1^*)$. If $(j, \mathbf{x}_j) = (2, \mathbf{x}_2^*)$: $g(T \mid \theta)h = g((1, \mathbf{x}_1^*) \mid \theta)\, C = \frac{1}{2} f_1(\mathbf{x}_1^* \mid \theta)\, C = \frac{1}{2} f_2(\mathbf{x}_2^* \mid \theta) = f^*((2, \mathbf{x}_2^*) \mid \theta)$. If $(j, \mathbf{x}_j) = (1, \mathbf{x}_1^*)$: $g(T \mid \theta)h = \frac{1}{2} f_1(\mathbf{x}_1^* \mid \theta) \cdot 1 = f^*$. Otherwise $T(j, \mathbf{x}_j) = (j, \mathbf{x}_j)$, $h = 1$, and $g(T \mid \theta)h = f^*$. Thus $f^* = g(T \mid \theta)h$ with h θ-free: T is sufficient by factorization. (b) $T(1, \mathbf{x}_1^*) = T(2, \mathbf{x}_2^*)$, so the Formal Sufficiency Principle gives $\mathrm{Ev}(E^*, (1, \mathbf{x}_1^*)) = \mathrm{Ev}(E^*, (2, \mathbf{x}_2^*))$, i.e. (6.3.4). The Conditionality Principle ("only the performed experiment

matters") gives $\mathrm{Ev}(E^*, (j, \mathbf{x}_j^*)) = \mathrm{Ev}(E_j, \mathbf{x}_j^*)$, i.e. (6.3.5). Together: $\mathrm{Ev}(E_1, \mathbf{x}_1^*) = \mathrm{Ev}(E_2, \mathbf{x}_2^*)$ whenever the likelihoods are proportional — the Formal Likelihood Principle. (c) Apply the Formal Likelihood Principle to E^* vs E_j at $(j, \mathbf{x}_j)$: the E^*-likelihood $\frac{1}{2} f_j(\mathbf{x}_j \mid \theta)$ is proportional (in θ) to the E_j-likelihood $f_j(\mathbf{x}_j \mid \theta)$, so $\mathrm{Ev}(E^*, (j, \mathbf{x}_j)) = \mathrm{Ev}(E_j, \mathbf{x}_j)$: the Conditionality Principle. If T is sufficient and $T(\mathbf{x}) = T(\mathbf{y})$, factorization gives $L(\theta \mid \mathbf{x}) = g(T(\mathbf{x}) \mid \theta) h(\mathbf{x}) \propto g(T(\mathbf{y}) \mid \theta) h(\mathbf{y}) = L(\theta \mid \mathbf{y})$; the Formal Likelihood Principle then yields $\mathrm{Ev}(E, \mathbf{x}) = \mathrm{Ev}(E, \mathbf{y})$: the Formal Sufficiency Principle.

Exercise 6.34

Central idea. The random-n factor p_n is θ-free, so both likelihoods are proportional to $\theta^x (1-\theta)^{n-x}$.

Solution. In Exercise 6.12, $L(\theta \mid (n, x)) = p_n \binom{n}{x} \theta^x (1-\theta)^{n-x} \propto \theta^x (1-\theta)^{n-x}$ as a function of θ. In a fixed-n binomial(n, θ) experiment, $L(\theta \mid x) = \binom{n}{x} \theta^x (1-\theta)^{n-x} \propto \theta^x (1-\theta)^{n-x}$. The two likelihoods are proportional, so by the Formal Likelihood Principle the evidence about θ is identical: the randomness of N is irrelevant to inference about θ.

Exercise 6.35

Central idea. Stopping after the first failure gives four possible histories; each has the same θ-dependence as a fixed-n binomial outcome with the same success count.

Solution. The four sample points (stopping at the first failure) are (0), $(1,0)$, $(1,1,0)$, $(1,1,1)$, with likelihoods $1-p, p(1-p), p^2(1-p), p^3$. These are proportional (in p) to fixed-sample-size binomial likelihoods: binomial$(1, p)$ with 0 successes ($\propto 1-p$), binomial$(2, p)$ with 1 success ($\propto p(1-p)$), binomial$(3, p)$ with 2 successes ($\propto p^2(1-p)$), binomial$(3, p)$ with 3 successes ($\propto p^3$). By the Formal Likelihood Principle, each yields the same evidence about p as its fixed-n counterpart: the data-dependent stopping rule does not affect inference about p.

Exercise 6.36

Central idea. Minimal sufficiency makes T_2 a function of T_1, so Rao–Blackwellizing twice collapses to once — and the conditional variance decomposition shows the second Rao–Blackwellization can only remove variance.

Solution. (a) Since T_2 is minimal sufficient and T_1 is sufficient, $T_2 = r(T_1)$ for some r, so $\sigma(T_2) \subset \sigma(T_1)$. By the tower property, $E(U_1 \mid T_2) = E(E(U \mid T_1) \mid T_2) = E(U \mid T_2) = U_2$. (b) By Theorem 4.4.7, $\mathrm{Var}(U_1) = \mathrm{Var}(E(U_1 \mid T_2)) + \mathrm{E}[\mathrm{Var}(U_1 \mid T_2)] = \mathrm{Var}(U_2) + \mathrm{E}[\mathrm{Var}(U_1 \mid T_2)] \geq \mathrm{Var}(U_2)$, since the second term is nonnegative. Thus Rao–Blackwellizing with the minimal sufficient statistic gives the smallest variance.

Exercise 6.37

Central idea. $X \sim \mathrm{Exp}(\theta) \perp Y \sim \mathrm{Exp}(1/\theta)$; (T, U) is 1–1 with $(\sum X_i, \sum Y_i)$ (preserving information and sufficiency), while $T = \theta\sqrt{A/B}$ is a scale statistic whose information is computed via the Beta(n, n) law of $A/(A+B)$.

Solution. (a) f factors as $\theta e^{-\theta x} \cdot \frac{1}{\theta} e^{-y/\theta}$: $X \sim$ exponential$(\theta) \perp Y \sim$ exponential$(1/\theta)$. $\frac{\partial^2}{\partial \theta^2} \log f = -\frac{2y}{\theta^3}$, so $I_1(\theta) = \frac{2\mathrm{E}_\theta Y}{\theta^3} = \frac{2\theta}{\theta^3} = \frac{2}{\theta^2}$; for n iid pairs, $I(\theta) = 2n/\theta^2$. (b) Note $(\sum X_i, \sum Y_i) \leftrightarrow (T, U)$ is 1–1 on $(0, \infty)^2$: $\sum X_i = U/T$, $\sum Y_i = UT$. (ii) $(\sum X_i, \sum Y_i)$ is sufficient (two-parameter exponential family), and 1–1 transformations preserve Fisher information, so $I_{(T,U)}(\theta) = I(\theta) = 2n/\theta^2$. (i) $T^2 = \frac{\sum Y_i}{\sum X_i} = \theta^2 \frac{A}{B}$ with $A = \frac{1}{\theta} \sum Y_i$, $B = \theta \sum X_i$ iid Gamma$(n, 1)$. From $f_{T,U}(t, u) = \frac{2}{\Gamma(n)^2} u^{2n-1} t^{-1} e^{-u(\theta/t + t/\theta)}$ (Jacobian $2u/t$), $f_T(t) \propto \theta^{2n} t^{2n-1} / (\theta^2 + t^2)^{2n}$, giving score $\frac{2n(T^2 - \theta^2)}{\theta(T^2 + \theta^2)}$. Hence $I_T(\theta) = \frac{4n^2}{\theta^2} \mathrm{E}[\frac{(V-1)^2}{(V+1)^2}]$ with $V = T^2/\theta^2 = A/B$. Since $\frac{V-1}{V+1} = \frac{A-B}{A+B} = 2P - 1$ with $P = \frac{A}{A+B} \sim$ Beta(n, n), $\mathrm{E}[\frac{(V-1)^2}{(V+1)^2}] = 4\mathrm{Var}(P) = \frac{1}{2n+1}$. Thus $I_T(\theta) = \frac{4n^2}{(2n+1)\theta^2} = \frac{2n}{2n+1} I(\theta)$. (iii) (T, U) is a 1–1 function of the sufficient $(\sum X_i, \sum Y_i)$, hence sufficient. Not complete: $\mathrm{E}_\theta[U^2] = \mathrm{E}_\theta[\sum X_i] \, \mathrm{E}_\theta[\sum Y_i] = \frac{n}{\theta} \cdot n\theta = n^2$ for all θ, so $U^2 - n^2$ is a nonzero unbiased estimator of 0.

Exercise 6.38

Central idea. The inverse of any group element, composed with the element via (ii), must be the identity function — and (ii) puts that composition back in G.

Solution. Definition 6.4.2: G is a group of transformations of $\mathcal{X}$ if (i) every $g \in G$ has an inverse $g' \in G$ with $g'(g(x)) = x$ for all x, and (ii) for $g, g' \in G$ there is $g'' \in G$ with $g'(g(x)) = g''(x)$ for all x; (iii) asserts the identity $e(x) = x$ belongs to G. Let G be nonempty and satisfy (i)–(ii). Pick $g \in G$ and let g' be its inverse from (i): $g'(g(x)) = x$ for all x. Apply (ii) to this pair (g, g'): there is $g'' \in G$ with $g'(g(x)) = g''(x)$ for all x. Hence $g''(x) = x$ for all x, i.e. $g'' = e$, so $e \in G$: (iii) holds.

Exercise 6.39

Central idea. The same physical measurement is $y = \frac{9}{5}x + 32$; converting the Fahrenheit estimate back to Celsius reproduces $T(x)$ — but the two rules are different functions, so formal invariance fails.

Solution. (a) The same data in Fahrenheit is $y = \frac{9}{5}x + 32$. Then

$$\tfrac{5}{9}(T^*(y) - 32) = \tfrac{5}{9}(\tfrac{1}{2}y + \tfrac{1}{2}(212) - 32) = \tfrac{5}{9}(\tfrac{1}{2}y + 74).$$

Substituting $y = \frac{9}{5}x + 32$: $\frac{1}{2}(\frac{9}{5}x + 32) + 74 = \frac{9}{10}x + 90$, and $\frac{5}{9}(\frac{9}{10}x + 90) = \frac{1}{2}x + 50 = T(x)$. The converted estimate agrees with $T(x)$: measurement equivariance holds. (b) $T(x) = .5x + 50$ while $T^*(x) = .5x + 106$: $T(x) \neq T^*(x)$. The estimation rule changes with the scale, violating formal invariance, so the estimators are not equivariant in the sense of the Equivariance Principle.

Exercise 6.40

Central idea. Both statistics scale by a and ignore b, so their ratio kills both the location and the scale parameters.

Solution. (a) Write $X_i = \sigma Z_i + \mu$ with Z_i iid from the base distribution. Then $T_i(\mathbf{X}) = T_i(\sigma\mathbf{Z} + \mu) = \sigma T_i(\mathbf{Z})$ by the hypothesis, so $\frac{T_1(\mathbf{X})}{T_2(\mathbf{X})} = \frac{T_1(\mathbf{Z})}{T_2(\mathbf{Z})}$, free of (μ, σ): ancillary (where $T_2 \neq 0$). (b) $R(a\mathbf{x} + b) = (ax_{(n)} + b) - (ax_{(1)} + b) = aR(\mathbf{x})$, and $S(a\mathbf{x} + b) = aS(\mathbf{x})$ for $a > 0$ (translation cancels in deviations, scaling factors out). Hence R/S is ancillary by (a).

Exercise 6.41

Central idea. Measurement equivariance matches estimates across scales for the same physical data; formal invariance forces one rule for all scales, yielding the location-equivariance equation $T(\mathbf{x} + a) = T(\mathbf{x}) + a$.

Solution. (a) $\mathbf{y} = g_a(\mathbf{x}) = \mathbf{x} + a$ represents the same physical data in the shifted scale. Estimating $\mu + a$ from $\mathbf{y}$ by $T^*(\mathbf{y})$ and converting back gives the μ-estimate $T^*(\mathbf{y}) - a$; measurement equivariance demands this equal the direct estimate $T(\mathbf{x})$. (b) Formal invariance (the Equivariance Principle) requires the same estimation rule in every scale: $T^* = T$ as functions, i.e. $T(\mathbf{x}) = T^*(\mathbf{x})$ for all $\mathbf{x}$. With (a), $T(\mathbf{x}) = T(\mathbf{x} + a\mathbf{1}) - a$, i.e. $T(x_1 + a, \ldots, x_n + a) = T(x_1, \ldots, x_n) + a$. (c) $W(\mathbf{x} + a\mathbf{1}) = \bar{x} + a = W(\mathbf{x}) + a$: equivariant. With $X_i = Z_i + \theta$, $\mathrm{E}_\theta W = \mathrm{E}_\theta \bar{X} = \theta + \mathrm{E}_0 Z_1 = \theta$.

Exercise 6.42

Central idea. Location shifts add a to $\bar{x} + k$ (matching $\theta \mapsto \theta + a$), but a scale change multiplies the stray constant k while the induced parameter map sends $\theta \mapsto c\theta + a$ — the k breaks equivariance unless it vanishes.

Solution. (a) $\mathcal{G}_2$ induces $\bar{\theta} = \theta + a$ on the parameter. $W(g_a(\mathbf{x})) = \bar{x} + a + k = W(\mathbf{x}) + a$: equivariant. $\mathcal{G}_1$ induces $\bar{\theta} = c\theta + a$: $W(g_{a,c}(\mathbf{x})) = c\bar{x} + a + k$, whereas equivariance requires $cW(\mathbf{x}) + a = c\bar{x} + ck + a$; these agree iff $ck = k$, i.e. $c = 1$, and with $k \neq 0$ they differ for $c \neq 1$. Not equivariant under $\mathcal{G}_1$. (b) For $\mathcal{G}_1$-equivariant W ($W(c\mathbf{x} + a\mathbf{1}) = cW(\mathbf{x}) + a$): with $X_i = \sigma Z_i + \theta$, $\mathrm{E}_{\theta,\sigma}[W] = \theta + \sigma\,\mathrm{E}_{0,1}[W(\mathbf{Z})]$, unbiased for

all (θ, σ) iff $\mathrm{E}_{0,1}[W(\mathbf{Z})] = 0$. For $\mathcal{G}_2$-equivariant W ($W(\mathbf{x} + a\mathbf{1}) = W(\mathbf{x}) + a$): $\mathrm{E}_{\theta,\sigma}[W] = \theta + \mathrm{E}_{0,\sigma}[W(\sigma\mathbf{Z})]$, unbiased for all (θ, σ) iff $\mathrm{E}_{0,\sigma}[W] = 0$ for every $\sigma > 0$ (location 0).

Exercise 6.43

Central idea. S^2 is translation-invariant and scales by c^2, matching $\sigma^2 \mapsto \sigma^2$ under $\mathcal{G}_2$ and $\sigma^2 \mapsto c^2\sigma^2$ under $\mathcal{G}_1, \mathcal{G}_3$; the ratio $\bar{X}/S$ is scale-invariant but not translation-invariant, so a nonconstant ϕ survives only $\mathcal{G}_3$.

Solution. The induced maps on σ^2 are: $\mathcal{G}_1$: $\sigma^2 \mapsto c^2\sigma^2$; $\mathcal{G}_2$: $\sigma^2 \mapsto \sigma^2$; $\mathcal{G}_3$: $\sigma^2 \mapsto c^2\sigma^2$. (a) $S^2(\mathbf{x} + a\mathbf{1}) = S^2(\mathbf{x})$ and $S^2(c\mathbf{x} + a\mathbf{1}) = c^2 S^2(\mathbf{x})$. Hence $kS^2(g_a(\mathbf{x})) = kS^2(\mathbf{x})$ (invariant under $\mathcal{G}_2$, equivariance with the identity parameter map), and $kS^2(g_{a,c}(\mathbf{x})) = c^2 kS^2(\mathbf{x})$, $kS^2(g_c(\mathbf{x})) = c^2 kS^2(\mathbf{x})$ (equivariant under $\mathcal{G}_1, \mathcal{G}_3$). (b) Under $\mathcal{G}_3$: $\bar{X}(c\mathbf{x})/S(c\mathbf{x}) = \bar{X}/S$ and $S^2(c\mathbf{x}) = c^2S^2$, so $W(c\mathbf{x}) = \phi(\bar{X}/S)c^2S^2 = c^2W(\mathbf{x})$: equivariant. Under $\mathcal{G}_2$ (identity on σ^2), equivariance needs $W(\mathbf{x} + a\mathbf{1}) = W(\mathbf{x})$, i.e. $\phi((\bar{X} + a)/S) = \phi(\bar{X}/S)$ for all a, forcing ϕ constant on the range of $(\bar{X} + a)/S$. Under $\mathcal{G}_1$, equivariance needs $W(c\mathbf{x} + a\mathbf{1}) = c^2W(\mathbf{x})$, i.e. $\phi((c\bar{X} + a)/(cS)) = \phi(\bar{X}/S)$ for all a and $c > 0$, again forcing ϕ constant. If $\phi \equiv k$, $W = kS^2$ is equivariant under all three groups by (a).

Chapter 7

Exercise 7.1

Central idea. With a finite parameter space, the MLE is found by direct comparison: for the observed x, pick the θ maximizing that row of the table. **Solution.** For each observed x, $\hat{\theta}(x) = \arg\max_{\theta \in \{1,2,3\}} f(x \mid \theta)$: - $x = 0$: values $(1/3, 1/4, 0)$, max at $\theta = 1$. - $x = 1$: values $(1/3, 1/4, 0)$, max at $\theta = 1$. - $x = 2$: values $(0, 1/4, 1/4)$, max (tied) at $\theta = 2$ or 3. - $x = 3$: values $(1/6, 1/4, 1/2)$, max at $\theta = 3$. - $x = 4$: values $(1/6, 0, 1/4)$, max at $\theta = 3$.

Hence $\hat{\theta}(0) = \hat{\theta}(1) = 1$, $\hat{\theta}(3) = \hat{\theta}(4) = 3$, and $\hat{\theta}(2)$ is not unique (2 or 3 both maximize the likelihood).

Exercise 7.2

Central idea. Profile the likelihood: substitute the closed-form $\hat{\beta}(\alpha)$ into $\log L$ and maximize the resulting univariate function of α (whose first-order condition is $\log\alpha - \psi(\alpha) = \log(\bar{x}/\tilde{x})$). **Solution.** The gamma pdf is $f(x \mid \alpha, \beta) = \Gamma(\alpha)^{-1}\beta^{-\alpha}x^{\alpha-1}e^{-x/\beta}$, $x > 0$. (a) $\log L = -n\log\Gamma(\alpha) - n\alpha\log\beta + (\alpha - 1)\sum \log x_i - \sum x_i/\beta$. Then $\partial \log L/\partial\beta = -n\alpha/\beta + \sum x_i/\beta^2 = 0$ gives $\hat{\beta} = \bar{x}/\alpha$ (a maximum since $\partial^2 \log L/\partial\beta^2 < 0$ at $\hat{\beta}$). (b) Substituting $\hat{\beta}(\alpha) = \bar{x}/\alpha$ gives the profile log-likelihood

$$l_p(\alpha) = -n\log\Gamma(\alpha) - n\alpha\log(\bar{x}/\alpha) + (\alpha - 1)\sum \log x_i - n\alpha,$$

maximized numerically. Equivalently $\hat{\alpha}$ solves $\log\alpha - \psi(\alpha) = \log(\bar{x}/\tilde{x})$, where $\tilde{x}$ is the geometric mean and $\psi = \Gamma'/\Gamma$. For the 14 egg-length observations of Exercise 7.10(c) ($\bar{x} = 23.114$, $\sum \log x_i = 43.953$), this gives $\hat{\alpha} \approx 514.3$ and $\hat{\beta} = \bar{x}/\hat{\alpha} \approx 0.0449$.

Exercise 7.3

Central idea. The logarithm is strictly increasing, so it preserves order — and hence maximizers. **Solution.** Since $L(\theta \mid \mathbf{x}) \geq 0$ and log is strictly increasing on $(0, \infty)$, for any θ_1, θ_2 with positive likelihood, $L(\theta_1 \mid \mathbf{x}) \leq L(\theta_2 \mid \mathbf{x})$ iff $\log L(\theta_1 \mid \mathbf{x}) \leq \log L(\theta_2 \mid \mathbf{x})$. Thus both functions attain their maximum at the same $\hat{\theta}$ (points with $L = 0$ cannot be maximizers unless $L \equiv 0$).

Exercise 7.4

Central idea. The unrestricted maximizer is $\bar{x}$; if it violates the constraint $\theta \geq 0$, the constrained maximum occurs at the boundary because the likelihood is monotone on either side of $\bar{x}$. **Solution.** In Example 7.2.8

the log-likelihood (normal with known variance, or any location family with log-concave likelihood) is strictly increasing for $\theta < \bar{x}$ and strictly decreasing for $\theta > \bar{x}$, with unique unconstrained maximizer $\bar{x}$. If $\bar{x} \geq 0$, the constraint $\theta \geq 0$ is inactive and $\hat{\theta} = \bar{x}$. If $\bar{x} < 0$, then on the feasible set $\theta \geq 0 > \bar{x}$ the log-likelihood is strictly decreasing, so the maximum is at the boundary $\hat{\theta} = 0$. Hence $\hat{\theta} = \max(\bar{x}, 0)$, as asserted.

Exercise 7.5

Central idea. The likelihood-ratio inequalities pin k between consecutive values of $1/z$; solving $(1-p)^n = \prod_{i=1}^n (1 - x_i z)$ for $\hat{z}$ and flooring $1/\hat{z}$ gives the integer MLE. **Solution.** (a) In Example 7.2.9 the likelihood ratios give $L(k)/L(k-1) \geq 1$ iff $k \leq 1/z$ (for the z solving the likelihood equation), and the function $\prod_i (1 - x_i z)$ is strictly decreasing in z, so there is a unique $\hat{z}$ with $(1-p)^n = \prod_i (1 - x_i \hat{z})$. The integer k maximizing $L(k)$ is then the largest integer with $k \leq 1/\hat{z}$, i.e. $\hat{k} = \lfloor 1/\hat{z} \rfloor$. (b) Solve $(1/2)^4 = (1 - 0 \cdot z)(1 - 20z)(1 - z)(1 - 19z)$, i.e. $(1 - 20z)(1 - z)(1 - 19z) = 1/16$. Numerically $\hat{z} \approx 0.0382$, so $1/\hat{z} \approx 26.20$ and $\hat{k} = \lfloor 26.20 \rfloor = 26$.

Exercise 7.6

Central idea. This is a "Pareto-type" family with threshold θ: the likelihood forces $\theta \leq X_{(1)}$, and $E_\theta X_1 = \infty$, so moments fail while the MLE is trivial. **Solution.** (a) $L(\theta \mid \mathbf{x}) = \theta^n (\prod x_i)^{-2} \mathbf{1}_{[\theta \leq x_{(1)}]}$, which factors through $X_{(1)}$; hence $X_{(1)}$ is sufficient (indeed minimal sufficient). (b) L is strictly increasing in θ on $0 < \theta \leq x_{(1)}$, so $\hat{\theta} = X_{(1)}$. (c) $E_\theta X_1 = \int_\theta^\infty \theta x^{-1}\, dx = \infty$; the first population moment does not exist, so there is no method-of-moments estimator based on $\bar{X}$ (equating $\bar{x} = E_\theta X_1$ is impossible).

Exercise 7.7

Central idea. With only two parameter values, compare the two likelihood values directly — the MLE is the one giving the larger likelihood. **Solution.** For $\mathbf{x} \in (0,1)^n$, $L(0 \mid \mathbf{x}) = 1$ and $L(1 \mid \mathbf{x}) = \prod_{i=1}^n (2\sqrt{x_i})^{-1} = 2^{-n} (\prod x_i)^{-1/2}$. Hence $L(1 \mid \mathbf{x}) > L(0 \mid \mathbf{x})$ iff $(\prod x_i)^{-1/2} > 2^n$ iff $\prod_{i=1}^n x_i < 4^{-n}$. So $\hat{\theta} = 1$ if $\prod X_i < 4^{-n}$ and $\hat{\theta} = 0$ otherwise (either value may be taken on the null set where equality holds).

Exercise 7.8

Central idea. $EX^2 = \sigma^2$ gives unbiasedness immediately; the MLE follows from maximizing in σ; moments can match $E|X|$ instead of EX^2. **Solution.** (a) $E_\sigma[X^2] = \mathrm{Var}(X) = \sigma^2$, so $\delta(X) = X^2$ is unbiased for σ^2. (b) $L(\sigma \mid x) = (2\pi\sigma^2)^{-1/2} \exp(-x^2/(2\sigma^2))$; $\partial \log L / \partial \sigma = -\sigma^{-1} + x^2 \sigma^{-3} = 0$ gives $\hat{\sigma} = |x|$, a maximum. So the MLE of σ is $|X|$. (c) The usual MOM equation $E_\sigma X = \bar{x}$ is useless ($E_\sigma X \equiv 0$). Instead match a nonzero moment, e.g. $E_\sigma |X| = \sigma\sqrt{2/\pi}$ (half-normal mean), giving the MOM estimator $\tilde{\sigma} = \sqrt{\pi/2}\, |X|$; equivalently match $E_\sigma X^2 = \sigma^2$ to get $\tilde{\sigma} = |X|$.

Exercise 7.9

Central idea. MOM gives $2\bar{X}$ while the MLE is $X_{(n)}$; both are (essentially) unbiased, but the MLE has much smaller variance — it is sufficient, while $2\bar{X}$ is not. **Solution.** MOM: $E_\theta X_1 = \theta/2$, so $\tilde{\theta} = 2\bar{X}$, with $E\tilde{\theta} = \theta$ and $\mathrm{Var}(\tilde{\theta}) = 4 \cdot \theta^2/(12n) = \theta^2/(3n)$. MLE: $L(\theta \mid \mathbf{x}) = \theta^{-n} \mathbf{1}_{[\theta \geq x_{(n)}]}$, decreasing in θ, so $\hat{\theta} = X_{(n)}$, with $E\hat{\theta} = n\theta/(n+1)$ and $\mathrm{Var}(\hat{\theta}) = n\theta^2/[(n+1)^2(n+2)]$. The MLE is preferred: it is sufficient and $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + (E\hat{\theta} - \theta)^2 = 2\theta^2/[(n+1)(n+2)] \leq \theta^2/(3n) = \mathrm{MSE}(\tilde{\theta})$, with equality at $n = 1, 2$ and strict inequality for $n \geq 3$ (and $\hat{\theta}$ is asymptotically unbiased).

Exercise 7.10

Central idea. The density $f(x \mid \alpha, \beta) = \alpha x^{\alpha-1}/\beta^\alpha \mathbf{1}_{[0 \leq x \leq \beta]}$ is a threshold family: β is estimated by the largest observation and α from the log-spacings below it. **Solution.** The pdf is

$f(x \mid \alpha, \beta) = \alpha x^{\alpha-1}\beta^{-\alpha}\mathbf{1}_{[0\le x\le\beta]}$. (a) $L(\alpha,\beta \mid \mathbf{x}) = \alpha^n(\prod x_i)^{\alpha-1}\beta^{-n\alpha}\mathbf{1}_{[\beta\ge x_{(n)}]}$ factors through $(\prod X_i, X_{(n)})$, a two-dimensional sufficient statistic. (b) For fixed α, L is decreasing in β on $\beta \ge x_{(n)}$, so $\hat\beta = X_{(n)}$. Then $\log L = n\log\alpha + (\alpha-1)\sum\log x_i - n\alpha\log\hat\beta$ gives $\hat\alpha = n/(n\log\hat\beta - \sum\log x_i) = -n/\sum\log(x_i/\hat\beta)$. (c) Here $x_{(n)} = 25.0$, so $\hat\beta = 25.0$, and $\hat\alpha = 14/(14\log 25.0 - 43.953) \approx 12.59$.

Exercise 7.11

Central idea. $-\log X_i \sim$ exponential(θ), so the MLE is $n/\sum(-\log X_i)$ — an inverse-gamma-scaled statistic whose variance is explicitly $O(1/n)$. **Solution.** (a) $\log L = n\log\theta + (\theta-1)\sum\log x_i$; $\partial/\partial\theta$: $n/\theta + \sum\log x_i = 0$, so $\hat\theta = -n/\sum_{i=1}^n \log X_i$ (a maximum). Since $-\log X_i \sim$ exponential(θ) (rate θ), $S := \sum(-\log X_i) \sim$ gamma(n,θ), and $\hat\theta = n/S$ has $E\hat\theta = n\theta/(n-1)$, $\mathrm{Var}(\hat\theta) = n^2\theta^2/[(n-1)^2(n-2)] \to 0$ as $n\to\infty$. (b) $E_\theta X_1 = \theta/(\theta+1)$; solving $\bar x = \tilde\theta/(\tilde\theta+1)$ gives $\tilde\theta = \bar X/(1-\bar X)$.

Exercise 7.12

Central idea. The parameter space is truncated at $1/2$: MOM ignores the boundary ($\bar X$), while the MLE truncates ($\min(\bar X, 1/2)$) and therefore dominates in MSE. **Solution.** (a) $E_\theta X_1 = \theta$, so the MOM estimator is $\tilde\theta = \bar X$. The unrestricted maximizer of the Bernoulli likelihood is $\bar X$; constrained to $[0,1/2]$, the MLE is $\hat\theta = \min(\bar X, 1/2)$ (if $\bar X > 1/2$ the likelihood is increasing on $[0,1/2]$). (b) $\mathrm{MSE}(\tilde\theta) = \theta(1-\theta)/n$. Writing $D = (\bar X - 1/2)_+ \ge 0$ so $\hat\theta = \bar X - D$,

$$\mathrm{MSE}(\hat\theta) = \mathrm{MSE}(\tilde\theta) - 2E[(\bar X-\theta)D] + E[D^2] \le \mathrm{MSE}(\tilde\theta),$$

since on $\{D>0\}$ (i.e. $\bar X > 1/2$), $2(\bar X-\theta) - D = \bar X - 2\theta + 1/2 \ge 1 - 2\theta \ge 0$ (as $\theta \le 1/2$), so $2E[(\bar X-\theta)D] - E[D^2] = E[(2(\bar X-\theta)-D)D] \ge 0$. Strict unless $P_\theta(\bar X > 1/2) = 0$. For $\theta < 1/2$ the correction vanishes as $n\to\infty$ ($P_\theta(\bar X > 1/2) \to 0$ exponentially), so both MSEs are $\theta(1-\theta)/n + o(1/n)$. (c) The MLE is preferred: it has uniformly smaller MSE and always lies in the parameter space, whereas $\tilde\theta$ can exceed $1/2$.

Exercise 7.13

Central idea. Maximizing the likelihood is minimizing $\sum|x_i-\theta|$ — the least-absolute-deviations problem, whose solution is any median. **Solution.** $\log L(\theta \mid \mathbf{x}) = -n\log 2 - \sum_{i=1}^n |x_i-\theta|$; maximizing L is minimizing $g(\theta) = \sum|x_i-\theta|$. Now g is convex and piecewise linear with slope $g'(\theta) = \#\{x_i < \theta\} - \#\{x_i > \theta\}$ (where defined), which changes sign at any median. Hence: if $n = 2m+1$ is odd, the unique minimizer is $\hat\theta = X_{(m+1)}$ (the sample median); if $n = 2m$ is even, g is flat on $[X_{(m)}, X_{(m+1)}]$, so any $\hat\theta \in [X_{(m)}, X_{(m+1)}]$ (e.g. the midpoint) is an MLE.

Exercise 7.14

Central idea. Reparameterize by the rates $a = 1/\lambda$, $b = 1/\mu$: then $Z \sim$ exponential$(a+b)$ independently of $W \sim$ Bernoulli$(a/(a+b))$, and the log-likelihood separates. **Solution.** From Exercise 4.26, Z and W are independent with $Z \sim$ exponential of mean $(1/\lambda + 1/\mu)^{-1}$ and $P(W=1) = \mu/(\lambda+\mu)$. With $a = 1/\lambda$, $b = 1/\mu$: $Z \sim$ exponential$(a+b)$ (rate) and $W \sim$ Bernoulli$(a/(a+b))$, independently. The log-likelihood is

$$\log L = n\log(a+b) - (a+b)\sum z_i + \Big(\sum w_i\Big)\log\frac{a}{a+b} + \Big(n - \sum w_i\Big)\log\frac{b}{a+b}$$

$$= -(a+b)\sum z_i + \Big(\sum w_i\Big)\log a + \Big(n - \sum w_i\Big)\log b.$$

Thus $\hat a = (\sum w_i)/(\sum z_i) = \bar w/\bar z$ and $\hat b = (n - \sum w_i)/(\sum z_i) = (1-\bar w)/\bar z$, i.e. $\hat\lambda = \bar Z/\bar W$ and $\hat\mu = \bar Z/(1-\bar W)$ (for $0 < \bar w < 1$; at the boundaries the corresponding MLE is $+\infty$).

Exercise 7.15

Central idea. Profile λ out of the log-likelihood; the profile is minimized by $\bar{X}$ via $\sum(x_i-\mu)^2/(\mu^2 x_i) = \sum x_i/\mu^2 - 2n/\mu + \sum x_i^{-1}$. For (b), the algebraic identity $\sum(x_i-\mu)^2/x_i = n(\hat{\mu}_n-\mu)^2/\hat{\mu}_n + \mu^2 n/\hat{\lambda}_n$ makes the joint density factor after a triangular change of variables. **Solution.** (a) $\log L = \frac{n}{2}\log\lambda - \frac{3}{2}\sum\log x_i - \frac{\lambda}{2\mu^2}\sum\frac{(x_i-\mu)^2}{x_i} + c$. From $\partial\log L/\partial\lambda = 0$, $\hat{\lambda}(\mu) = n\mu^2/\sum(x_i-\mu)^2/x_i$. The profile log-likelihood is maximized by minimizing $S(\mu)/\mu^2 = \sum(x_i-\mu)^2/(\mu^2 x_i) = \mu^{-2}\sum x_i - 2n\mu^{-1} + \sum x_i^{-1}$, whose derivative $-2\sum x_i/\mu^3 + 2n/\mu^2$ vanishes at $\hat{\mu}_n = \bar{X}$ (a minimum). Since $\sum(x_i-\bar{x})^2/x_i = \bar{x}^2\sum(1/x_i - 1/\bar{x})$, $\hat{\lambda}_n = n\bar{x}^2/\sum(x_i-\bar{x})^2/x_i = n/[\sum x_i^{-1} - 1/\bar{x}]$. (b)(i) Put $m = (x_1+x_2)/2 = \hat{\mu}_2$ and $u = x_1^{-1} + x_2^{-1} - 2/m = 2/\hat{\lambda}_2$. The key identity is

$$\frac{(x_1-\mu)^2}{x_1} + \frac{(x_2-\mu)^2}{x_2} = \frac{2(m-\mu)^2}{m} + \mu^2 u,$$

(since $(x-\mu)^2/x = x - 2\mu + \mu^2/x$). Transform $(x_1,x_2)\to(m, p = x_1x_2)$ (Jacobian factor $2/\sqrt{m^2-p}$ for the two orderings) and then $(m,p)\to(m, l=\hat{\lambda}_2)$ via $p = m^2 l/(m+l)$ (Jacobian $m^3/(m+l)^2$). Using $x_1x_2 = m^2 l/(m+l)$ and $u = 2/l$, the joint density becomes

$$f(m,l) = \frac{\lambda}{\pi} m^{-3/2} l^{-3/2} \exp\left\{-\frac{\lambda(m-\mu)^2}{\mu^2 m} - \frac{\lambda}{l}\right\},$$

which factors as $\mathrm{IG}(\mu, 2\lambda)$ in m times the density of l under $2\lambda/l \sim \chi^2_1$ (check: $(\lambda/(\pi m^3))^{1/2} e^{-\lambda(m-\mu)^2/(\mu^2 m)} \times (2\pi)^{-1/2}(2\lambda)^{1/2} l^{-3/2} e^{-\lambda/l}$). Hence $\hat{\mu}_2 \sim \mathrm{IG}(\mu, 2\lambda) \perp\!\!\!\perp 2\lambda/\hat{\lambda}_2 \sim \chi^2_1$. (ii) Write $m_k = \hat{\mu}_k$, $v_k = k/\hat{\lambda}_k = \sum_{i\le k}(x_i^{-1} - m_k^{-1})$. The recursive updates are $m_{k+1} = (km_k + x)/(k+1)$ and $(k+1)/l_{k+1} = k/l_k + k/m_k + 1/x - (k+1)/m_{k+1}$. By the induction hypothesis and independence of x, $f(x, m_k, l_k) = \mathrm{IG}(x \mid \mu, \lambda)\,\mathrm{IG}(m_k \mid \mu, k\lambda)\, f_{\chi^2_{k-1}}(k\lambda/l_k)\,(k\lambda/l_k^2)$. The exponent decomposes as

$$-\frac{k\lambda(m_k-\mu)^2}{2\mu^2 m_k} - \frac{\lambda(x-\mu)^2}{2\mu^2 x} - \frac{k\lambda}{2l_k} = -\frac{(k+1)\lambda(m_{k+1}-\mu)^2}{2\mu^2 m_{k+1}} - \frac{(k+1)\lambda}{2l_{k+1}} - [\text{terms free of } (\mu,\lambda)],$$

using $\sum_{i\le k+1}(x_i-\mu)^2/x_i = (k+1)(m_{k+1}-\mu)^2/m_{k+1} + \mu^2(k+1)/l_{k+1}$. The transformation $(m_k, l_k)\to(m_{k+1}, l_{k+1})$ (at fixed x) is triangular with Jacobian combining with the $m^{-3/2}l^{-3/2}$-type prefactors to produce exactly the $\mathrm{IG}(\mu,(k+1)\lambda)$ and χ^2_k normalizing constants. Thus $f(x, m_{k+1}, l_{k+1})$ factors as $[\mathrm{IG}(\mu,(k+1)\lambda)$ in $m_{k+1}] \times [\chi^2_k$ in $(k+1)\lambda/l_{k+1}] \times [$free of $(\mu,\lambda)]$, proving $\hat{\mu}_{k+1} \sim \mathrm{IG}(\mu,(k+1)\lambda) \perp\!\!\!\perp (k+1)\lambda/\hat{\lambda}_{k+1} \sim \chi^2_k$.

Exercise 7.16

Central idea. Least squares on the transformed scale is minimized when $h(a)$ is the ordinary mean of the $h(x_i)$; for an exponential family the likelihood equation is $\sum h(x_i) = nH'(\theta)$, i.e. the same generalized mean. **Solution.** (a) $\frac{d}{da}\sum[h(x_i)-h(a)]^2 = -2h'(a)\sum[h(x_i)-h(a)] = 0$ gives (for $h' \neq 0$) $h(a) = \frac{1}{n}\sum h(x_i)$, i.e. $a = h^{-1}(\frac{1}{n}\sum h(x_i))$ (a minimum by convexity in $h(a)$). (b) $h(x) = x$: $a = \bar{x}$. $h(x) = \log x$: $a = \exp(\frac{1}{n}\sum\log x_i) = (\prod x_i)^{1/n}$, the geometric mean. $h(x) = 1/x$: $a = [\frac{1}{n}\sum x_i^{-1}]^{-1}$, the harmonic mean. (c) The Box–Cox transform is (a monotone function of) x^λ; (a) gives $a = [\frac{1}{n}\sum x_i^\lambda]^{1/\lambda}$, the generalized mean with $h(x) = x^\lambda$. (d) $\log X_i \sim \mathrm{n}(\mu,\sigma^2)$, so $\hat{\mu} = \frac{1}{n}\sum\log X_i$; by equivariance the MLE of e^μ (the median) is $\exp(\hat{\mu}) = (\prod X_i)^{1/n}$, the geometric mean. (e)(i) $\log L = \theta\sum h(x_i) - nH(\theta) + \sum\log g(x_i)$; FOC: $\sum h(x_i) - nH'(\theta) = 0$. Since $H' = h$, $h(\hat{\theta}) = \frac{1}{n}\sum h(x_i)$, so $\hat{\theta} = h^{-1}(\frac{1}{n}\sum h(x_i))$ (h increasing makes this the maximizer). (ii) Normal $\mathrm{n}(\theta,1)$: $f = \exp\{\theta x - \theta^2/2\}g(x)$, $h(x) = x = H'(x)$; $\hat{\theta} = \bar{x}$ (arithmetic mean). Inverted gamma: $f = \exp\{-\theta(1/x) + \log\theta\}x^{-2}$, i.e. with sign change $h(x) = -1/x$, $H(\theta) = -\log\theta$, $H'(\theta) = -1/\theta = h(\theta)$; $\hat{\theta} = h^{-1}(-\frac{1}{n}\sum x_i^{-1}) = n/\sum x_i^{-1}$ (harmonic mean).

Exercise 7.17

Central idea. $\{Z=0\} = \{X_2 = 1\}$ is the same observation, but the two likelihoods differ by a Jacobian factor — the same null-event equality, but different observed statistics and limiting neighborhoods. **Solution.**

(a) $L(\theta \mid x_2) = \theta^{-1}e^{-x_2/\theta}$ is maximized at $\hat{\theta} = x_2$. (b) With $x_2 = 1 + zx_1$, $|\partial(x_1, x_2)/\partial(x_1, z)| = x_1$, so $f_{X_1,Z}(x_1, z) = \theta^{-2}x_1 \exp\{-(x_1(1 + z) + 1)/\theta\}$ on $x_1 > 0$, $1 + zx_1 > 0$. For $z \geq 0$, $L(\theta \mid z) = \int_0^\infty \theta^{-2}x_1 e^{-(x_1(1+z)+1)/\theta} dx_1 = e^{-1/\theta}/(1 + z)^2$. For $z < 0$ let $L = -1/z$ and $a = 1 + z$. The support requires $0 < x_1 < L$, and the remaining likelihood is

$$f_Z(z \mid \theta) = \begin{cases} \dfrac{e^{-1/\theta}}{a^2}\left[1 - \left(1 + \dfrac{aL}{\theta}\right)e^{-aL/\theta}\right], & z < 0,\ z \neq -1, \\ \dfrac{e^{-1/\theta}}{2\theta^2}, & z = -1. \end{cases}$$

This follows by integrating $x_1 e^{-ax_1/\theta}$ on the finite interval and completes the negative-support case, including the continuous limit at $z = -1$. (c) At $X_2 = 1$ (i.e. $Z = 0$): (a) gives $L \propto \theta^{-1}e^{-1/\theta}$, $\hat{\theta} = 1$; (b) gives $L \propto e^{-1/\theta}$, which is strictly increasing in θ — no finite MLE (supremum at $\theta = \infty$). The equality of these probability-zero events does not make the observed statistics informationally equivalent; their likelihoods and limiting neighborhoods differ, illustrating the Borel phenomenon. (d) The posteriors are $\pi(\theta \mid x_2 = 1) \propto \pi(\theta)\theta^{-1}e^{-1/\theta}$ versus $\pi(\theta \mid z = 0) \propto \pi(\theta)e^{-1/\theta}$ — different posteriors from conditioning on different observed statistics at corresponding null-event values.

Exercise 7.18

Central idea. Factor the bivariate normal into marginal×conditional: the marginal pieces give $(\bar{x}, \frac{1}{n}\sum(x_i - \bar{x})^2)$ and $(\bar{y}, \frac{1}{n}\sum(y_i - \bar{y})^2)$, and the remaining profile likelihood in ρ is maximized at the sample correlation. **Solution.** (a) Equating population and sample moments through second order gives directly the stated estimators (using $E(X - \mu_X)(Y - \mu_Y) = \rho\sigma_X\sigma_Y$). (b) From $f(x, y) = f(y \mid x)f(x)$ with $X \sim \mathrm{n}(\mu_X, \sigma_X^2)$, the factor $f(x \mid \mu_X, \sigma_X^2)$ alone determines $\hat{\mu}_X = \bar{x}$, $\hat{\sigma}_X^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$; symmetrically $\hat{\mu}_Y = \bar{y}$, $\hat{\sigma}_Y^2 = \frac{1}{n}\sum(y_i - \bar{y})^2$. Conditionally, $Y \mid x \sim \mathrm{n}(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2))$; substituting the four MLEs, the profile log-likelihood in ρ is $-\frac{n}{2}\log(1 - \rho^2) - \frac{1}{2(1-\rho^2)}\sum\left(\frac{y_i - \bar{y}}{\hat{\sigma}_Y} - \rho\frac{x_i - \bar{x}}{\hat{\sigma}_X}\right)^2$. Differentiating (or writing it as a regression through the standardized variables) gives $\hat{\rho} = \sum(x_i - \bar{x})(y_i - \bar{y})/\sqrt{\sum(x_i - \bar{x})^2\sum(y_i - \bar{y})^2}$, the sample correlation — identical to the MOM estimators.

Exercise 7.19

Central idea. Through-the-origin regression: the likelihood depends on (β, σ^2) only through $\sum(Y_i - \beta x_i)^2$, minimized at $\sum x_i Y_i / \sum x_i^2$. **Solution.** (a) $L \propto (\sigma^2)^{-n/2}\exp\{-\sum(Y_i - \beta x_i)^2/(2\sigma^2)\}$ and $\sum(Y_i - \beta x_i)^2 = \sum Y_i^2 - 2\beta\sum x_i Y_i + \beta^2\sum x_i^2$; hence $(\sum x_i Y_i, \sum Y_i^2)$ is sufficient for (β, σ^2). (b) Minimizing $\sum(Y_i - \beta x_i)^2$ gives $\hat{\beta} = \sum x_i Y_i / \sum x_i^2$. Since $E[\sum x_i Y_i] = \beta\sum x_i^2$, $E\hat{\beta} = \beta$: unbiased. (c) $\hat{\beta}$ is a linear combination of independent normals, so $\hat{\beta} \sim \mathrm{n}(\beta, \sigma^2/\sum x_i^2)$.

Exercise 7.20

Central idea. Cauchy–Schwarz gives $(\sum x_i)^2 \leq n\sum x_i^2$, so the "ratio of sums" estimator has variance at least as large as the MLE's. **Solution.** (a) $E[\sum Y_i/\sum x_i] = \beta\sum x_i/\sum x_i = \beta$. (b) $\mathrm{Var}(\sum Y_i/\sum x_i) = n\sigma^2/(\sum x_i)^2$, versus $\mathrm{Var}(\hat{\beta}_{\mathrm{MLE}}) = \sigma^2/\sum x_i^2$. By Cauchy–Schwarz, $(\sum x_i)^2 \leq n\sum x_i^2$, so $n\sigma^2/(\sum x_i)^2 \geq \sigma^2/\sum x_i^2$, with equality iff all x_i are equal. The MLE has uniformly smaller variance.

Exercise 7.21

Central idea. Averaging the individual ratios Y_i/x_i is unbiased but inefficient: Cauchy–Schwarz again shows its variance dominates the MLE's. **Solution.** (a) $E[\frac{1}{n}\sum\frac{Y_i}{x_i}] = \frac{1}{n}\sum(\beta + \frac{E\varepsilon_i}{x_i}) = \beta$ (for $x_i \neq 0$). (b) $\mathrm{Var}(\frac{1}{n}\sum Y_i/x_i) = \frac{\sigma^2}{n^2}\sum x_i^{-2}$. By Cauchy–Schwarz, $(\sum x_i^{-2})(\sum x_i^2) \geq n^2$, so $\frac{\sigma^2}{n^2}\sum x_i^{-2} \geq \sigma^2/\sum x_i^2 = \mathrm{Var}(\hat{\beta}_{\mathrm{MLE}})$. Hence the MLE of Exercise 7.19 dominates both this estimator and that of Exercise 7.20 (the latter two have no uniform ordering between themselves).

Exercise 7.22

Central idea. $\bar{X} \mid \theta \sim \mathrm{n}(\theta, \sigma^2/n)$ and $\theta \sim \mathrm{n}(\mu, \tau^2)$: multiply, complete the square in θ — the normal–normal conjugate update. **Solution.** (a) $f(\bar{x}, \theta) = f(\bar{x} \mid \theta)\pi(\theta) = \mathrm{n}(\bar{x} \mid \theta, \sigma^2/n) \cdot \mathrm{n}(\theta \mid \mu, \tau^2)$. (b) $\bar{X} = \theta + (\bar{X} - \theta)$ with $\theta \sim \mathrm{n}(\mu, \tau^2)$ independent of $\bar{X} - \theta \sim \mathrm{n}(0, \sigma^2/n)$; hence m is $\mathrm{n}(\mu, \sigma^2/n + \tau^2)$. (c) $\pi(\theta \mid \bar{x}) \propto \exp\{-\frac{n}{2\sigma^2}(\bar{x} - \theta)^2 - \frac{1}{2\tau^2}(\theta - \mu)^2\}$; completing the square in θ gives a normal with precision $n/\sigma^2 + 1/\tau^2$ and mean $\frac{\tau^2}{\tau^2 + \sigma^2/n}\bar{x} + \frac{\sigma^2/n}{\tau^2 + \sigma^2/n}\mu$, i.e. (7.2.10).

Exercise 7.23

Central idea. The χ^2_{n-1} likelihood kernel $(\sigma^2)^{-(n-1)/2}\exp(-(n-1)S^2/(2\sigma^2))$ multiplies the IG prior kernel — conjugacy just adds exponents. **Solution.** $\pi(\sigma^2 \mid \mathbf{x}) \propto (\sigma^2)^{-(n-1)/2} e^{-(n-1)S^2/(2\sigma^2)} \cdot (\sigma^2)^{-(\alpha+1)} e^{-1/(\beta\sigma^2)} = (\sigma^2)^{-(\alpha+(n-1)/2+1)} \exp\{-[\frac{(n-1)S^2}{2} + \frac{1}{\beta}]/\sigma^2\}$, i.e. $\mathrm{IG}(\alpha + \frac{n-1}{2}, [\frac{(n-1)S^2}{2} + \frac{1}{\beta}]^{-1})$ in the stated parameterization (shape $\alpha' = \alpha + (n-1)/2$, "scale" $1/\beta' = \frac{(n-1)S^2}{2} + \frac{1}{\beta}$). Since $E[\sigma^2] = \beta'^{-1}/(\alpha' - 1)$ for this parameterization, the Bayes estimator (posterior mean) is $\dfrac{\frac{(n-1)S^2}{2} + \frac{1}{\beta}}{\alpha + \frac{n-1}{2} - 1}$.

Exercise 7.24

Central idea. Poisson–gamma conjugacy: add the total count to the shape and the sample size to the rate. **Solution.** (a) $\pi(\lambda \mid \mathbf{x}) \propto \lambda^{\sum x_i} e^{-n\lambda} \cdot \lambda^{\alpha-1} e^{-\lambda/\beta} = \lambda^{\alpha + n\bar{x} - 1} \exp\{-\lambda(n + 1/\beta)\}$, i.e. $\mathrm{gamma}(\alpha + n\bar{x}, \beta/(n\beta+1))$ (shape–scale). (b) Posterior mean $\dfrac{\alpha\beta + n\beta\bar{x}}{n\beta + 1}$ (a weighted average of the prior mean $\alpha\beta$ and $\bar{x}$) and posterior variance $(\alpha + n\bar{x})(\frac{\beta}{n\beta+1})^2$.

Exercise 7.25

Central idea. The pairs (X_i, θ_i) are iid across i; marginalizing an iid sequence coordinate-wise preserves iidness, and the normal case is a convolution. **Solution.** (a) $X_i = \theta_i + (X_i - \theta_i)$ with $\theta_i \sim \mathrm{n}(\mu, \tau^2)$ independent of $X_i - \theta_i \sim \mathrm{n}(0, \sigma^2)$, so $X_i \sim \mathrm{n}(\mu, \sigma^2 + \tau^2)$ marginally; since the pairs (X_i, θ_i) are independent across i with a common joint law, the marginals X_i are iid. (b) The vectors (X_i, θ_i) are iid across i (independent pairs, each with joint density $f(x \mid \theta)\pi(\theta \mid \tau)$); hence their X_i-marginals, $m(x \mid \tau) = \int f(x \mid \theta)\pi(\theta \mid \tau)\, d\theta$, are iid.

Exercise 7.26

Central idea. Reduce by sufficiency to $\bar{X}$, then complete the square separately on $\theta > 0$ and $\theta < 0$ — the posterior mean is a Φ/ϕ-weighted average of two shifted normals. **Solution.** By sufficiency, $\pi(\theta \mid \mathbf{x}) \propto \exp\{-\frac{n}{2\sigma^2}(\theta - \bar{x})^2 - \frac{|\theta|}{a}\}$. Hence, with $\sigma_n^2 = \sigma^2/n$,

$$E[\theta \mid \mathbf{x}] = \frac{\int_{-\infty}^{\infty} \theta \exp\{-\frac{(\theta-\bar{x})^2}{2\sigma_n^2} - \frac{|\theta|}{a}\}\, d\theta}{\int_{-\infty}^{\infty} \exp\{-\frac{(\theta-\bar{x})^2}{2\sigma_n^2} - \frac{|\theta|}{a}\}\, d\theta}.$$

Completing the square on each half-line with $\mu_\pm = \bar{x} \mp \sigma_n^2/a$ gives, with ϕ, Φ the standard normal pdf/cdf,

$$E[\theta \mid \mathbf{x}] = \frac{e^{-(\bar{x}^2 - \mu_+^2)/(2\sigma_n^2)}[\mu_+\Phi(\mu_+/\sigma_n) + \sigma_n\phi(\mu_+/\sigma_n)] + e^{-(\bar{x}^2 - \mu_-^2)/(2\sigma_n^2)}[\mu_-\Phi(-\mu_-/\sigma_n) - \sigma_n\phi(\mu_-/\sigma_n)]}{e^{-(\bar{x}^2 - \mu_+^2)/(2\sigma_n^2)}\Phi(\mu_+/\sigma_n) + e^{-(\bar{x}^2 - \mu_-^2)/(2\sigma_n^2)}\Phi(-\mu_-/\sigma_n)}.$$

Exercise 7.27

Central idea. The complete-data log-likelihood separates in (β, τ); with x_1 missing the β score involves only $i \geq 2$, giving the closed form directly. **Solution.** (a) From (7.2.11), $\partial l/\partial\beta = -\sum \tau_i + \sum y_i/\beta = 0$ and $\partial l/\partial\tau_j = -(\beta + 1) + (x_j + y_j)/\tau_j = 0$; solving jointly gives $\hat{\beta} = \sum y_i / \sum x_i$, $\hat{\tau}_j = (x_j + y_j)/(\hat{\beta} + 1)$,

i.e. (7.2.12). (b) At a fixed point of (7.2.23), $\hat\beta = \sum y_i/(\hat\tau_1 + \sum_{i\ge 2} x_i)$ and $\hat\tau_j = (x_j + y_j)/(\hat\beta + 1)$ $(j \ge 2)$, $\hat\tau_1 = (\hat\beta\hat\tau_1 + y_1)/(\hat\beta + 1)$; these are exactly the incomplete-data likelihood equations (7.2.16). (c) With x_1 missing, the incomplete-data log-likelihood drops the x_1 terms: $\partial l/\partial\tau_1 = -\beta + y_1/\tau_1 = 0$ gives $\hat\tau_1 = y_1/\hat\beta$; $\partial l/\partial\tau_j = -(\beta{+}1){+}(x_j{+}y_j)/\tau_j = 0$ gives $\hat\tau_j = (x_j{+}y_j)/(\hat\beta{+}1)$ $(j \ge 2)$; and $\partial l/\partial\beta = -\sum_{i\ge2}\tau_i{+}\sum_{i\ge2} y_i/\beta = 0$ with $\sum_{i\ge2}\hat\tau_i = (\sum_{i\ge2}(x_i{+}y_i))/(\hat\beta{+}1)$ yields $\hat\beta = \sum_{i=2}^n y_i/\sum_{i=2}^n x_i$. This direct solution of (7.2.16) is unique, so the EM limit from part (b) — which satisfies (7.2.16) — coincides with these MLEs.

Exercise 7.28

Central idea. This is the Poisson model of Example 7.2.17 in action: the complete-data MLEs are ratios of sums, and each missing-data pattern deletes one term from one score equation. **Solution.** Here $n = 18$, $\sum x_i = 55{,}860$, $\sum y_i = 47$. (a) Full data: $\hat\beta = \sum y_i/\sum x_i = 47/55{,}860 \approx 8.414\times10^{-4}$, $\hat\tau_j = (x_j{+}y_j)/(\hat\beta{+}1)$. With x_1 missing (Exercise 7.27(c)): $\hat\beta = \sum_{i=2}^{18} y_i/\sum_{i=2}^{18} x_i = 44/52{,}320 \approx 8.410\times10^{-4}$, $\hat\tau_1 = y_1/\hat\beta \approx 3567.3$, $\hat\tau_j = (x_j + y_j)/(\hat\beta + 1)$ for $j \ge 2$. The estimates are nearly identical to the full-data fit. (b) With y_1 missing, the E-step imputes $E[Y_1 \mid \cdot] = \beta^{(r)}\tau_1^{(r)}$ and the M-step gives $\beta^{(r+1)} = (E[Y_1] + \sum_{i\ge2} y_i)/\sum x_i$, $\tau_j^{(r+1)} = (x_j + \tilde y_j)/(\beta^{(r+1)} + 1)$. Iterating to convergence: $\hat\beta = 44/52{,}320 \approx 8.410\times10^{-4}$ (same $\hat\beta$ as in (a)), $\hat\tau_1 = x_1 = 3{,}540$ exactly, and $\hat\tau_j = (x_j{+}y_j)/(\hat\beta{+}1)$ for $j \ge 2$ — again essentially the full-data fit, as expected since only one count of 47 is missing.

Exercise 7.29

Central idea. Independence of the Poisson and multinomial pieces gives the joint pmf; the τ score blends the X and Y counts, and the EM plugs $E[X_1] = m\tau_1^{(r)}$ into the complete-data MLE formulas. **Solution.** (a) By independence, the joint pmf is the product of the n Poisson pmfs and the multinomial pmf $m!\prod_i \tau_i^{x_i}/x_i!$. (As typeset, the $m!$ appears inside the $\prod_{i=1}^n$; the intended reading — consistent with "$(X_1,\dots,X_n) \sim$ multinomial$(m;\tau)$" — has a single factor $m!$ outside the product, since $\prod_i[m!\tau_i^{x_i}/x_i!]$ would carry $(m!)^n$ and would not be a valid pmf. The MLEs are unaffected, as m is known.) (b) $\log L = \sum_i[-m\beta\tau_i + y_i\log(m\beta\tau_i)] + \sum_i x_i\log\tau_i + C$ with $\sum\tau_i = 1$. With Lagrange multiplier (or $\phi_i = \beta\tau_i$), the joint MLEs are $\hat\beta = \sum y_i/m$ and $\hat\tau_j = (x_j + y_j)/(m + \sum y_i)$. (c) E-step: $E[X_1 \mid \mathbf{y}, \mathbf{x}_{-1}, \theta^{(r)}] = m\hat\tau_1^{(r)}$. The M-step plugs the expected sufficient statistics ($E[X_1] = m\hat\tau_1^{(r)}$, $E[\sum X_i] = m\hat\tau_1^{(r)} + \sum_{i\ge2} x_i$) into the complete-data MLEs from (b), giving the stated sequence (for $j = 1$, x_1 is replaced by $m\hat\tau_1^{(r)}$). (d) Full data ($m = 55{,}860$, $\sum y_i = 47$): $\hat\beta = 47/55{,}860 \approx 8.414\times10^{-4}$, $\hat\tau_j = (x_j + y_j)/55{,}907$ (e.g. $\hat\tau_1 = 3543/55{,}907 \approx 0.06337$). With x_1 missing, iterating the EM sequence converges to $\hat\beta \approx 8.414\times10^{-4}$, $\hat\tau_1 \approx 0.06337$ — essentially the full-data MLEs, since the EM reconstructs x_1.

Exercise 7.30

Central idea. The component labels are the missing data: the E-step computes posterior membership probabilities (Bayes' rule), and the M-step averages them. **Solution.** (a) By the hierarchical specification, $f(x_i, z_i \mid p) = [pf(x_i)]^{z_i}[(1{-}p)g(x_i)]^{1-z_i}$ for $z_i \in \{0,1\}$; independence across i gives the product. (b) $P(Z_i = 1 \mid x_i, p) = \frac{pf(x_i)}{pf(x_i)+(1-p)g(x_i)}$ by Bayes' rule (the denominator is the mixture marginal), so $Z_i \mid x_i, p \sim$ Bernoulli with that success probability. (c) The complete-data log-likelihood is $\sum_i[z_i\log(pf(x_i)) + (1-z_i)\log((1-p)g(x_i))]$. Its expectation under $Z_i \mid x_i, \hat p^{(r)}$ replaces z_i by $\hat z_i^{(r)} = \hat p^{(r)}f(x_i)/[\hat p^{(r)}f(x_i) + (1-\hat p^{(r)})g(x_i)]$. Maximizing $\sum_i[\hat z_i^{(r)}\log p + (1-\hat z_i^{(r)})\log(1-p)]$ in p gives $\hat p^{(r+1)} = \frac{1}{n}\sum_i \hat z_i^{(r)}$, as claimed.

Exercise 7.31

Central idea. Split the incomplete-data log-likelihood into the EM objective Q minus a Kullback–Leibler-type term; the M-step raises Q while Jensen shows the KL term cannot rise — so the likelihood never decreases. **Solution.** (a) From (7.2.19), $L(\theta \mid \mathbf{y}, \mathbf{x}) = L(\theta \mid \mathbf{y})\, k(\mathbf{x} \mid \theta, \mathbf{y})$. Taking logs and then $E[\cdot \mid$

$\widehat{\theta}^{(r)}, \mathbf{y}]$ gives the decomposition (the k-term is the conditional entropy term). The M-step chooses $\widehat{\theta}^{(r+1)}$ to maximize $Q(\theta \mid \widehat{\theta}^{(r)}, \mathbf{y}) = E[\log L(\theta \mid \mathbf{y}, \mathbf{X}) \mid \widehat{\theta}^{(r)}, \mathbf{y}]$, hence the inequality; it is an equality iff $\widehat{\theta}^{(r)}$ already maximizes Q (e.g. at a fixed point/convergence). (b) The difference of the two k-expectations is $E\big[\log \frac{k(\mathbf{X}|\widehat{\theta}^{(r+1)}, \mathbf{y})}{k(\mathbf{X}|\widehat{\theta}^{(r)}, \mathbf{y})} \mid \widehat{\theta}^{(r)}, \mathbf{y}\big] \le \log E\big[\frac{k(\mathbf{X}|\widehat{\theta}^{(r+1)}, \mathbf{y})}{k(\mathbf{X}|\widehat{\theta}^{(r)}, \mathbf{y})} \mid \widehat{\theta}^{(r)}, \mathbf{y}\big] = \log 1 = 0$ by Jensen (concavity of log), using the hint with $f = k(\cdot \mid \widehat{\theta}^{(r+1)}, \mathbf{y})$ and $g = k(\cdot \mid \widehat{\theta}^{(r)}, \mathbf{y})$. Thus the k-term does not increase, while Q does not decrease, so $\log L(\widehat{\theta}^{(r+1)} \mid \mathbf{y}) \ge \log L(\widehat{\theta}^{(r)} \mid \mathbf{y})$: the EM sequence is monotone.

Exercise 7.32

Central idea. This is sampling–importance–resampling: the self-normalized weights converge to the posterior, and conditioning the resample on the prior draws can only reduce variance. **Solution.** (a) $P(\theta^* \le t \mid \theta_{1:m}) = \sum_{i:\theta_i \le t} q_i = \frac{\frac{1}{m}\sum_i \mathbf{1}_{\{\theta_i \le t\}} L(\theta_i|\mathbf{x})}{\frac{1}{m}\sum_i L(\theta_i|\mathbf{x})} \to \frac{\int_{-\infty}^{t} L(\theta|\mathbf{x})\pi(\theta)\,d\theta}{\int L(\theta|\mathbf{x})\pi(\theta)\,d\theta} = \int_{-\infty}^{t} \pi(\theta \mid \mathbf{x})\,d\theta$ a.s. as $m \to \infty$, by the SLLN (numerator and denominator). (b) Conditionally on $\theta_{1:m}$, the θ_j^* are iid from the discrete law $\{q_i\}$; as $m, r \to \infty$, $\frac{1}{r}\sum_j h(\theta_j^*) \to E[h(\theta) \mid \mathbf{x}]$ by the SLLN and part (a). (c) Given $\theta_{1:m}$, θ_j^* equals θ_i with probability q_i, so $E[h(\theta_j^*) \mid \theta_{1:m}] = \sum_i q_i h(\theta_i)$, the self-normalized importance-sampling estimator. It has the same unconditional mean as $\frac{1}{r}\sum_j h(\theta_j^*)$ (tower property), and $\mathrm{Var}(\frac{1}{r}\sum_j h(\theta_j^*)) = \mathrm{Var}(E[\cdot \mid \theta_{1:m}]) + E[\mathrm{Var}(\cdot \mid \theta_{1:m})] \ge \mathrm{Var}(\sum_i q_i h(\theta_i))$.

Exercise 7.33

Central idea. With $\alpha = \beta = c$, the MSE numerator is $c^2 + (n - 4c^2)p(1-p)$; choosing $4c^2 = n$ kills the p-dependence. **Solution.** From Example 7.3.5, with $Y \sim \text{binomial}(n, p)$ and $\widehat{p}_B = (Y + \alpha)/(n + \alpha + \beta)$,

$$\mathrm{MSE}(\widehat{p}_B) = \frac{np(1-p) + (\alpha - (\alpha+\beta)p)^2}{(n+\alpha+\beta)^2}.$$

With $\alpha = \beta = c$: numerator $= np(1-p) + c^2(1-2p)^2 = c^2 + (n-4c^2)p(1-p)$. Taking $c = \sqrt{n/4} = \sqrt{n}/2$ (so $4c^2 = n$) makes the numerator identically $n/4$, hence $\mathrm{MSE}(\widehat{p}_B) = \frac{n/4}{(n+\sqrt{n})^2}$, constant in p.

Exercise 7.34

Central idea. The flip group forces $\delta(x) + \delta(n-x) = 1$; among $\widehat{p}_B = (x+\alpha)/(n+\alpha+\beta)$ this singles out $\alpha = \beta$, and the minimax choice is the constant-risk prior of Exercise 7.33. **Solution.** (a) Reading the problem as a single observation $X \sim \text{binomial}(n, p)$ — the reading parts (b)–(c) require, since the Bayes class there is Example 7.2.14's, indexed by n — Example 6.4.1's group (the flip $x \leftrightarrow n - x$, $p \leftrightarrow 1 - p$) gives the equivariant class $\{\delta : \delta(x) = 1 - \delta(n-x),\ x = 0, \ldots, n\}$ by (6.4.1). (b) $\widehat{p}_B(x) = (x+\alpha)/(n+\alpha+\beta)$ satisfies $\widehat{p}_B(x) = 1 - \widehat{p}_B(n-x)$ iff $(x+\alpha) = (x+\beta)$, i.e. iff $\alpha = \beta$. The equivariant Bayes estimators are $\delta_\alpha(x) = (x+\alpha)/(n+2\alpha)$, $\alpha > 0$. (c) *"Smallest MSE" is read as minimax, since* $\mathrm{MSE}(\delta_\alpha)$ *depends on* p. $\mathrm{MSE}(\delta_\alpha, p) = \frac{np(1-p) + \alpha^2(1-2p)^2}{(n+2\alpha)^2}$, whose maximum over $p \in [0,1]$ is $\frac{\max(n/4, \alpha^2)}{(n+2\alpha)^2}$ (the numerator is $\alpha^2 + (n - 4\alpha^2)p(1-p)$, maximized at $p(1-p) = 1/4$ or 0). This is minimized over $\alpha > 0$ at $\alpha = \sqrt{n}/2 = \sqrt{n/4}$ (decreasing for $\alpha < \sqrt{n}/2$, increasing after), giving the constant (minimax) MSE $\frac{n/4}{(n+\sqrt{n})^2}$ — the estimator of Exercise 7.33.

Exercise 7.35

Central idea. d_P is the posterior mean under a flat prior; shifting all observations by c shifts it by c, and for the normal/uniform the integrals collapse to $\bar{X}$ and the midrange. **Solution.** (a) With $u = t - c$, $\int t \prod f(x_i + c - t)\,dt = \int (u + c) \prod f(x_i - u)\,du$, so $d_P(\mathbf{x} + c) = d_P(\mathbf{x}) + c$: location-equivariant (the "invariance" of Example 7.3.6). (b) $\prod \phi(x_i - t) \propto \exp(-\frac{n}{2}(t - \bar{x})^2)$; hence $d_P = \int t\, e^{-n(t-\bar{x})^2/2} dt \,/ \int e^{-n(t-\bar{x})^2/2} dt = \bar{x}$. (c) $\prod_i \mathbf{1}_{[-1/2, 1/2]}(x_i - t) = \mathbf{1}_{[x_{(n)} - 1/2,\ x_{(1)} + 1/2]}(t)$; thus d_P is the midpoint of $[x_{(n)} - \frac{1}{2}, x_{(1)} + \frac{1}{2}]$, i.e. $\frac{1}{2}(X_{(1)} + X_{(n)})$.

Exercise 7.36

Central idea. The substitution $u = ct$ pulls out c^r; each family then reduces to gamma-type integrals (or a truncation at $1/X_{(n)}$). **Solution.** (a) With $u = ct$: numerator $= c^{-(n+r)} \int u^{n+r-1} \prod f(ux_i)\, du$, denominator $= c^{-(n+2r)} \int u^{n+2r-1} \prod f(ux_i)\, du$; the ratio scales by c^r. (b) With $f = \phi$: $\prod \phi(tx_i) \propto e^{-t^2 \sum x_i^2/2}$. For $r = 2$, $\int_0^\infty t^{n+1} e^{-at^2/2} dt = 2^{n/2}\Gamma(\frac{n}{2}+1)/a^{(n+2)/2}$ ($a = \sum x_i^2$) and similarly for t^{n+3}; the ratio is $a/(n+2)$. Pitman estimator of σ^2: $\sum X_i^2/(n+2)$. (c) $f(x) = e^{-x}\mathbf{1}_{(x>0)}$: $\prod f(tx_i) = e^{-t\sum x_i}$ ($t > 0$). For $r = 1$: numerator $= n!/(\sum x_i)^{n+1}$, denominator $= (n+1)!/(\sum x_i)^{n+2}$; ratio $= (\sum X_i)/(n+1)$. (d) $f = \mathbf{1}_{[0,1]}$: $\prod f(tx_i) = \mathbf{1}_{[0,1/x_{(n)}]}(t)$. For $r = 1$: numerator $= (1/x_{(n)})^{n+1}/(n+1)$, denominator $= (1/x_{(n)})^{n+2}/(n+2)$; Pitman estimator of θ: $\frac{n+2}{n+1} X_{(n)}$.

Exercise 7.37

Central idea. $M = \max |X_i|$ is complete sufficient (differentiate $\int_0^\theta g(m) m^{n-1} dm \equiv 0$), so Lehmann–Scheffé applied to the unbiased $(n+1)M/n$ gives the UMVUE. **Solution.** Let $M = \max_i |X_i|$; $P_\theta(M \le m) = (m/\theta)^n$, $0 \le m \le \theta$, so $E_\theta M = n\theta/(n+1)$ and $\delta = \frac{n+1}{n} M$ is unbiased for θ. M is sufficient (the likelihood $\propto \theta^{-n}\mathbf{1}_{[m\le\theta]}$ factors through M) and complete: $E_\theta[g(M)] = 0\ \forall \theta > 0$ gives $\int_0^\theta g(m) m^{n-1} dm \equiv 0$, whence $g(\theta)\theta^{n-1} = 0$ a.e. by differentiation. By Lehmann–Scheffé, $\frac{n+1}{n} \max_i |X_i|$ is the best (UMVUE) unbiased estimator of θ.

Exercise 7.38

Central idea. The CRLB is attained iff the score is affine in $\delta - g(\theta)$; read off g from the score's linear structure. **Solution.** (a) Yes: $\frac{\partial}{\partial\theta} \log L = \frac{n}{\theta} + \sum \log X_i = -n(\delta - \frac{1}{\theta})$ with $\delta = -\frac{1}{n}\sum \log X_i$. Since $-\log X_1 \sim$ exponential(θ), $E\delta = 1/\theta =: g(\theta)$, $\mathrm{Var}(\delta) = \frac{1}{n\theta^2} = \frac{[g'(\theta)]^2}{nI(\theta)}$ (as $I(\theta) = 1/\theta^2$): the CRLB is attained for $g(\theta) = 1/\theta$. (b) Yes: $\frac{\partial}{\partial\theta} \log L = \frac{n}{\theta \log\theta} - \frac{n}{\theta - 1} + \frac{\sum x_i}{\theta} = \frac{n}{\theta}(\bar{x} - g(\theta))$ with $g(\theta) = \frac{\theta}{\theta-1} - \frac{1}{\log\theta}$. Indeed $E_\theta X_1 = \frac{\log\theta}{\theta-1}\int_0^1 x\theta^x dx = \frac{\theta}{\theta-1} - \frac{1}{\log\theta} = g(\theta)$, so $\bar{X}$ is unbiased for $g(\theta)$ and, the score being affine in $\bar{X} - g(\theta)$, attains the CRLB.

Exercise 7.39

Central idea. Differentiate $\int f(x \mid \theta)\, dx = 1$ twice and rewrite $\partial^2 f/\partial\theta^2$ via $\partial^2 \log f/\partial\theta^2$. **Solution.** Lemma 7.3.11 (the Bartlett identities): under regularity permitting differentiation under the integral, $0 = \frac{d}{d\theta}\int f = \int \frac{\partial f}{\partial\theta} = \int \frac{\partial \log f}{\partial\theta} f$, i.e. $E_\theta[\frac{\partial}{\partial\theta} \log f(X \mid \theta)] = 0$. Differentiating again, $0 = \int \frac{\partial^2 f}{\partial\theta^2}\, dx$; since $\frac{\partial^2}{\partial\theta^2} \log f = \frac{1}{f}\frac{\partial^2 f}{\partial\theta^2} - (\frac{\partial}{\partial\theta} \log f)^2$, taking expectations gives $E_\theta[\frac{\partial^2}{\partial\theta^2} \log f(X \mid \theta)] = -E_\theta[(\frac{\partial}{\partial\theta} \log f(X \mid \theta))^2]$.

Exercise 7.40

Central idea. For Bernoulli, $I(p) = 1/[p(1-p)]$ and $\mathrm{Var}(\bar{X}) = p(1-p)/n$ — exactly the CRLB. **Solution.** $\log f(x \mid p) = x \log p + (1-x)\log(1-p)$; $\frac{\partial^2}{\partial p^2} \log f = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}$, so $I(p) = \frac{1}{p(1-p)}$. The CRLB for unbiased estimators of p is $\frac{1}{nI(p)} = \frac{p(1-p)}{n} = \mathrm{Var}_p(\bar{X})$. Hence $\bar{X}$ attains the bound and is the best (UMVUE) unbiased estimator of p.

Exercise 7.41

Central idea. Minimize $\sigma^2 \sum a_i^2$ subject to $\sum a_i = 1$ — Lagrange gives equal weights. **Solution.** (a) $E[\sum a_i X_i] = \mu \sum a_i = \mu$ iff $\sum a_i = 1$. (b) $\mathrm{Var}(\sum a_i X_i) = \sigma^2 \sum a_i^2$; minimizing $\sum a_i^2$ under $\sum a_i = 1$ (Lagrange: $2a_i = \lambda$) gives $a_i \equiv 1/n$, i.e. $\bar{X}$, with variance σ^2/n — the BLUE.

Exercise 7.42

Central idea. Inverse-variance weighting: minimizing $\sum a_i^2 \sigma_i^2$ under $\sum a_i = 1$ puts weight proportional to precision. **Solution.** (a) Unbiasedness needs $\sum a_i = 1$; $\mathrm{Var}(\sum a_i W_i) = \sum a_i^2 \sigma_i^2$.

Lagrange: $2a_i\sigma_i^2 = \lambda$, so $a_i \propto 1/\sigma_i^2$; normalizing gives $a_i^* = (1/\sigma_i^2)/\sum_j(1/\sigma_j^2)$, i.e. W^*. (b) $\mathrm{Var}(W^*) = \sum(a_i^*)^2\sigma_i^2 = \sum(1/\sigma_i^2)/\left(\sum_j 1/\sigma_j^2\right)^2 = 1/\sum(1/\sigma_i^2)$.

Exercise 7.43

Central idea. Write the variance ratio as $\sum b_i q_i$; the constraints $\sum q_i = \sum q_i/b_i = 1$ force the worst case onto two extreme b's, and the resulting quadratic in $b_{\min}$ has zero discriminant. **Solution.** (a) Since $1/\sigma_i^2 = q_i^*/\mathrm{Var}(W^*)$, $\mathrm{Var}(W) = \sum q_i^2\sigma_i^2 = \mathrm{Var}(W^*)\sum b_i q_i$ with $b_i = q_i/q_i^*$; also $\sum q_i/b_i = \sum q_i^* = 1$. Maximizing $\sum b_i q_i$ under $\sum q_i = \sum q_i/b_i = 1$, $q_i \ge 0$: extreme points use at most two b's, say $b_i < b_j$, giving value $b_i + b_j - b_ib_j = 1 - (1-b_i)(1-b_j)$. Since $\sum b_i q_i^* = 1$ forces $b_{\min} \le 1 \le b_{\max}$, the maximum over $[b_{\min}, b_{\max}]^2$ is $b_{\min} + b_{\max} - b_{\min}b_{\max}$. With $r = b_{\max}/b_{\min} = (1+\lambda)/(1-\lambda)$, $\frac{1}{1-\lambda^2} = \frac{(r+1)^2}{4r}$, and $\frac{(r+1)^2}{4r} - (b_{\min} + rb_{\min} - rb_{\min}^2) = r\left(b_{\min} - \frac{r+1}{2r}\right)^2 \ge 0$. Hence $\mathrm{Var}(W)/\mathrm{Var}(W^*) \le 1/(1-\lambda^2)$. (b) For $q_i = 1/k$, $b_i = (1/k)/q_i^*$ so $b_{\max}/b_{\min} = \sigma_{\max}^2/\sigma_{\min}^2 =: r$. Thus $\mathrm{Var}(\bar W)/\mathrm{Var}(W^*) \le (r+1)^2/(4r)$: the ordinary mean is at most $(r+1)^2/(4r)$ times worse than optimal (equality only when the variances take two values in the right proportion); it is near-optimal when $\sigma_{\max}^2/\sigma_{\min}^2 \approx 1$ and can be arbitrarily bad when the ratio is large.

Exercise 7.44

Central idea. $\bar X$ is complete sufficient, so Lehmann–Scheffé blesses $\bar X^2 - 1/n$; Stein's identity gives $E\bar X^4 = \theta^4 + 6\theta^2/n + 3/n^2$, whence the variance exceeds the CRLB $4\theta^2/n$ by $2/n^2$. **Solution.** $\bar X$ is complete sufficient for θ and $E_\theta \bar X^2 = \theta^2 + 1/n$, so by Lehmann–Scheffé $\bar X^2 - 1/n$ is the UMVUE of θ^2. Let $W = \bar X \sim \mathrm{n}(\theta, 1/n)$. Stein's identity, $E_\theta[(W-\theta)h(W)] = \frac{1}{n}E_\theta[h'(W)]$: with $h(w) = w^2$, $E[W^3] = \theta^3 + 3\theta/n$; with $h(w) = w^3$, $E[W^4] = \theta E[W^3] + 3E[W^2]/n = \theta^4 + 6\theta^2/n + 3/n^2$. Hence $\mathrm{Var}(\bar X^2 - 1/n) = E[W^4] - (E[W^2])^2 = 4\theta^2/n + 2/n^2$. The CRLB for θ^2 is $[2\theta]^2/(n\cdot 1) = 4\theta^2/n$; the UMVUE's variance is larger by $2/n^2$.

Exercise 7.45

Central idea. $\mathrm{MSE}(aS^2)$ is quadratic in a with minimizer $a^* = \sigma^4/(\sigma^4 + \mathrm{Var}(S^2))$; under normality this is $(n-1)/(n+1)$, and heavier/lighter tails shift it down/up. **Solution.** (a) $E[aS^2 - \sigma^2]^2 = \mathrm{Var}(aS^2) + [E(aS^2) - \sigma^2]^2 = a^2\mathrm{Var}(S^2) + (a-1)^2\sigma^4$. (b) Standard (Exercise 5.8b): expanding $S^2 = \frac{1}{n-1}\sum(X_i - \bar X)^2$ and using independence gives $\mathrm{Var}(S^2) = \frac{1}{n}(\kappa - \frac{n-3}{n-1}\sigma^4)$. (c) For $Z \sim \mathrm{n}(0,1)$, $E[Z^4] = 3$ (Lemma 3.6.5), so $\kappa = 3\sigma^4$ and $\mathrm{Var}(S^2) = 2\sigma^4/(n-1)$. Minimizing $a^2\frac{2}{n-1} + (a-1)^2$ gives $a^* = \frac{n-1}{n+1}$. (d) *Erratum: as printed, the formula* $a = \frac{n-1}{(n+1)+\frac{(\kappa-1)(n-1)}{n}}$ *cannot be correct.* Minimizing the quadratic in (a) gives $a^* = \sigma^4/(\sigma^4 + \mathrm{Var}(S^2))$. With (b), and writing $K = \kappa/\sigma^4$ for the standardized kurtosis, this is

$$a^* = \frac{n-1}{(n+1) + \frac{(K-3)(n-1)}{n}} \qquad \left(K = \frac{\kappa}{\sigma^4}\right).$$

Proof of the erratum: (i) the printed $(\kappa - 1)$ mixes the raw fourth moment κ (units of X^4) with the dimensionless 1 — dimensionally inconsistent with (b)'s definition of κ; (ii) under normality ($K = 3$) the printed formula gives, e.g. for $n = 2$, $a = \frac{1}{3 + (3\sigma^4 - 1)/2} \ne \frac{1}{3}$, contradicting the $\frac{n-1}{n+1}$ of part (c); (iii) for a uniform distribution ($K = 1.8 < 3$) the printed formula can give $a < \frac{n-1}{n+1}$, contradicting (e)(ii) below. The intended formula has $(K-3) = (\kappa/\sigma^4 - 3)$ in place of $(\kappa - 1)$, and (e)'s "$\kappa > 3$"/"$\kappa < 3$" likewise refer to the standardized kurtosis $K = \kappa/\sigma^4$. (e) With $a^* = \frac{n-1}{(n+1)+(K-3)(n-1)/n}$: (i) if $K > 3$ the denominator exceeds $n+1$, so $a^* < \frac{n-1}{n+1}$; (ii) if $K < 3$ the denominator is smaller than $n+1$, so $a^* > \frac{n-1}{n+1}$, and $a^* < 1 \iff (3-K)(n-1) < 2n$, which holds since $K \ge 1$ (Jensen) gives $(3-K)(n-1) \le 2(n-1) < 2n$.

Exercise 7.46

Central idea. $E_\theta X_1 = 3\theta/2$ gives the MOM; the likelihood $L \propto \theta^{-3}$ on $[X_{(3)}/2, X_{(1)}]$ is maximized at the left endpoint; Rao–Blackwell improves only the non-sufficient MOM estimator. **Solution.** (a) $E_\theta X_1 = 3\theta/2$, so $\widetilde\theta = 2\bar X/3$. (b) $L(\theta \mid \mathbf{x}) = \theta^{-3}\mathbf{1}_{[x_{(3)}/2 \le \theta \le x_{(1)}]}$, strictly decreasing in θ on the feasible interval, so $\widehat\theta = X_{(3)}/2$.

Since $E_\theta X_{(3)} = 2\theta - \theta/4 = 7\theta/4$, $E_\theta\hat{\theta} = 7\theta/8$; thus $k = 8/7$ makes $(8/7)\hat{\theta}$ unbiased. (c) $(X_{(1)}, X_{(3)})$ is sufficient. The MOM $\tilde{\theta} = 2\bar{X}/3$ is not a function of it, so $E[\tilde{\theta} \mid X_{(1)}, X_{(3)}]$ is a strictly better (Rao–Blackwell) unbiased estimator. The MLE $\hat{\theta} = X_{(3)}/2$ is already a function of the sufficient statistic, so sufficiency cannot improve it. (d) $\bar{x} = 1.16$: MOM $\tilde{\theta} = 2(1.16)/3 \approx 0.773$; MLE $\hat{\theta} = 1.33/2 = 0.665$ (feasible since $x_{(1)} = 0.86 \geq 0.665$).

Exercise 7.47

Central idea. With $X_i \sim \mathrm{n}(\rho, \sigma^2)$ (σ^2 known), $\bar{X}$ is complete sufficient; unbias $\pi\bar{X}^2$ by subtracting $\pi\sigma^2/n$. **Solution.** Write $X_i = \rho + \varepsilon_i$, $\varepsilon_i \sim \mathrm{n}(0, \sigma^2)$ iid, σ^2 known; the area is $\tau(\rho) = \pi\rho^2$. Since $E_\rho\bar{X}^2 = \rho^2 + \sigma^2/n$, $\delta = \pi(\bar{X}^2 - \sigma^2/n)$ is unbiased for $\pi\rho^2$. $\bar{X}$ is complete sufficient for ρ, and δ is a function of $\bar{X}$, so by Lehmann–Scheffé δ is the best (UMVUE) unbiased estimator of the area.

Exercise 7.48

Central idea. The MLE is $\bar{X}$ (CRLB from 7.40); Rao–Blackwellizing $X_1X_2X_3X_4$ on the complete sufficient sum gives a hypergeometric probability. **Solution.** (a) The MLE is $\bar{X}$, and $\mathrm{Var}_p(\bar{X}) = p(1-p)/n = 1/[nI(p)]$ is the CRLB (Exercise 7.40). (b) $E_p[X_1X_2X_3X_4] = p^4$. $S = \sum_{i=1}^n X_i$ is complete sufficient; by Lehmann–Scheffé the UMVUE is $E[X_1X_2X_3X_4 \mid S]$. Given $S = s$, $(X_1, \ldots, X_n)$ is uniform over binary n-tuples with s ones, so $E[X_1X_2X_3X_4 \mid S = s] = P(X_1 = \cdots = X_4 = 1 \mid S = s) = \binom{s}{4}/\binom{n}{4}$. Hence the best unbiased estimator of p^4 is $\dfrac{S(S-1)(S-2)(S-3)}{n(n-1)(n-2)(n-3)}$.

Exercise 7.49

Central idea. Here exponential(λ) uses the textbook mean/scale convention: $f(x \mid \lambda) = \lambda^{-1}e^{-x/\lambda}$. The previous impossibility claim incorrectly treated λ as a rate.

Solution. (a) $Y = \min_i X_i$ has survival $P_\lambda(Y > y) = e^{-ny/\lambda}$, so $EY = \lambda/n$ and nY is unbiased for λ. (b) $\bar{X}$ is also unbiased, and for $n > 1$,

$$\mathrm{Var}(\bar{X}) = \lambda^2/n < \lambda^2 = \mathrm{Var}(nY).$$

It is the UMVUE because $\sum X_i$ is complete sufficient. In fact, symmetry of the exponential sample conditional on its sum gives $E(nY \mid \sum X_i) = \bar{X}$, so this is a Rao–Blackwell improvement. For $n = 1$ the estimators coincide. (c) $n = 12$, $Y = 50.1$, and $\sum x_i = 1497.9$. The estimates are $nY = 601.2$ hours and $\bar{X} = 124.825$ hours.

Exercise 7.50

Central idea. $\bar{X} \perp S$ with known variances; the optimal convex combination is inverse-variance weighting; $\bar{X} - cS$ is a nonzero unbiased estimator of zero. **Solution.** (a) $E_\theta[a\bar{X} + (1-a)cS] = a\theta + (1-a)\theta = \theta$. (b) $\mathrm{Var}(cS) = c^2\mathrm{Var}(S) = c^2\theta^2(1 - \frac{1}{c^2}) = \theta^2(c^2 - 1)$ (since $E_\theta S = \theta/c$ and $E_\theta S^2 = \theta^2$), and $\bar{X} \perp S$, so $\mathrm{Var}(a\bar{X} + (1-a)cS) = a^2\theta^2/n + (1-a)^2\theta^2(c^2-1)$. Minimizing in a: $\frac{a}{n} = (1-a)(c^2-1)$, i.e. $a^* = \dfrac{c^2-1}{1/n + c^2 - 1} = \dfrac{n(c^2-1)}{1 + n(c^2-1)}$. (c) $(\bar{X}, S^2)$ is sufficient (two-parameter exponential family). It is not complete: $\bar{X} - cS$ satisfies $E_\theta[\bar{X} - cS] = 0$ for all $\theta > 0$ but is not a.s. zero.

Exercise 7.51

Central idea. Dropping the unbiasedness constraint $a_1 + a_2 = 1$ strictly lowers the minimum MSE; truncating at 0 helps since $\theta > 0$; the family is $\frac{1}{\theta}g(x/\theta)$, i.e. scale. **Solution.** *Note:* the textbook cites Exercise 7.50 here. (a) $E_\theta(\theta - a_1\bar{X} - a_2cS)^2/\theta^2 = a_1^2/n + a_2^2v + (1 - a_1 - a_2)^2$ with $v = c^2 - 1 = \mathrm{Var}(cS)/\theta^2$

(using $\bar{X} \perp cS$; note $\mathrm{Var}(cS) = c^2\mathrm{Var}(S) = \theta^2(c^2-1)$ since $E_\theta S = \theta/c$). FOCs: $a_1/n = a_2 v = 1-a_1-a_2 =: d$, so $a_1^* = nd^*$, $a_2^* = d^*/v$ with $d^* = 1/(1+n+1/v)$; $T^* = a_1^*\bar{X} + a_2^* cS$. (b) Exercise 7.50(b) minimizes over the unbiased subclass $\{a_1 + a_2 = 1\} \subsetneq \mathcal{T}$; since $a_1^* + a_2^* = (n+1/v)/(n+1+1/v) < 1$, T^* is not in that subclass, and minimizing over the strictly larger $\mathcal{T}$ gives a strictly smaller minimum MSE. (c) For $\theta > 0$: if $T^* \geq 0$, $T^{*+} = T^*$; if $T^* < 0$, $(\theta - 0)^2 < (\theta - T^*)^2$. Hence $E_\theta(\theta - T^{*+})^2 < E_\theta(\theta - T^*)^2$. (d) Scale parameter: $f(x \mid \theta) = \frac{1}{\theta\sqrt{2\pi}} e^{-(x-\theta)^2/(2\theta^2)} = \frac{1}{\theta} g(x/\theta)$ with $g(u) = \phi(u-1)$, the scale-family form (not $f(x-\theta)$).

Exercise 7.52

Central idea. Conditionally on $S = \sum X_i$, the sample is multinomial — giving $E[S^2 \mid \bar{X}] = \bar{X}$ directly — and Lehmann–Scheffé makes this the UMVUE story in miniature. **Solution.** (a) $\sum X_i$ is complete sufficient (Poisson exponential family) and $\bar{X}$ is unbiased for λ; by Lehmann–Scheffé $\bar{X}$ is the UMVUE. (b) Given $\sum X_i = s$, $(X_1, \ldots, X_n) \sim \text{multinomial}(s; \frac{1}{n}, \ldots, \frac{1}{n})$, so $X_1 \mid s \sim \text{binomial}(s, 1/n)$ and $E[\sum X_i^2 \mid s] = n\big[\frac{s}{n}(1-\frac{1}{n}) + \frac{s^2}{n^2}\big] = s(1-\frac{1}{n}) + \frac{s^2}{n}$. Hence $E[S^2 \mid \bar{X}] = \frac{1}{n-1}(E[\sum X_i^2 \mid S] - \frac{S^2}{n}) = \frac{S}{n} = \bar{X}$. Then $\mathrm{Var}(S^2) = \mathrm{Var}(E[S^2 \mid \bar{X}]) + E[\mathrm{Var}(S^2 \mid \bar{X})] = \mathrm{Var}(\bar{X}) + E[\mathrm{Var}(S^2 \mid \bar{X})] > \mathrm{Var}(\bar{X})$, since S^2 is not a.s. constant given $\bar{X}$. (c) Yes: it is the Lehmann–Scheffé theorem in action. In general, if T is complete sufficient and W is unbiased for $\tau(\theta)$, then $E[W \mid T]$ is the (unique) best unbiased estimator; here $W = S^2$, $T = \bar{X}$, and the identity $E[S^2 \mid \bar{X}] = \bar{X}$ explicitly exhibits the UMVUE $\bar{X}$ as the Rao–Blackwellization of S^2.

Exercise 7.53

Central idea. $W + aU$ stays unbiased; choose a to kill the covariance term and strictly reduce variance at θ_0. **Solution.** Theorem 7.3.20: W is best unbiased iff it is uncorrelated with every unbiased estimator of zero. For fixed a, $W_a = W + aU$ is unbiased for $\tau(\theta)$, with $\mathrm{Var}_{\theta_0}(W_a) = \mathrm{Var}_{\theta_0}(W) + a^2\mathrm{Var}_{\theta_0}(U) + 2a\,\mathrm{Cov}_{\theta_0}(W, U)$. Minimizing in a gives $a^* = -\mathrm{Cov}_{\theta_0}(W,U)/\mathrm{Var}_{\theta_0}(U)$ (if $\mathrm{Var}_{\theta_0}(U) = 0$ then $U = 0$ a.s. and the covariance is 0), with $\mathrm{Var}_{\theta_0}(W_{a^*}) = \mathrm{Var}_{\theta_0}(W) - \mathrm{Cov}_{\theta_0}(W,U)^2/\mathrm{Var}_{\theta_0}(U) < \mathrm{Var}_{\theta_0}(W)$. Hence W is beaten at θ_0 and cannot be best unbiased.

Exercise 7.54

Central idea. With $A = \sum X_i \sim \text{gamma}(n, \theta)$, $B = \sum Y_i \sim \text{gamma}(n, 1/\theta)$, write $T = \theta\sqrt{G_2/G_1}$ and $U = (\theta A)(B/\theta)$ for iid $G_1, G_2 \sim \text{gamma}(n, 1)$; inverse-gamma moments give everything. **Solution.** In Exercise 6.37, $X_i \sim$ exponential(rate θ), $Y_i \sim$ exponential(rate $1/\theta$), independent, so $A := \sum X_i \sim \text{gamma}(n, \theta)$, $B := \sum Y_i \sim \text{gamma}(n, 1/\theta)$, independent. (a) $\log L = -\theta A - B/\theta$; $\partial/\partial\theta = -A + B/\theta^2 = 0$ gives $\hat{\theta} = \sqrt{B/A} = T$. With $G_1 = \theta A$, $G_2 = B/\theta$ iid gamma$(n, 1)$: $U = AB = G_1 G_2$ is free of θ (ancillary), and $T = \theta\sqrt{G_2/G_1}$. Since $E[G^{\pm 1/2}] = \Gamma(n \pm 1/2)/\Gamma(n)$, $E[T] = \theta\,\Gamma(n+1/2)\Gamma(n-1/2)/\Gamma(n)^2$; and $E[T^2] = \theta^2 E[G_2]E[1/G_1] = \theta^2 \cdot n \cdot \frac{1}{n-1} = \theta^2\Gamma(n+1)\Gamma(n-1)/\Gamma(n)^2$. (b) $E[1/A] = \theta/(n-1)$ (inverse-gamma mean), so $E[Z_1] = \theta$; $\mathrm{Var}(1/A) = \theta^2/[(n-1)^2(n-2)]$, so $\mathrm{Var}(Z_1) = \theta^2/(n-2)$. $E[Z_2] = \theta$, $\mathrm{Var}(Z_2) = \theta^2/n$. (c) $Z_1 \perp Z_2$; $\mathrm{Var}(aZ_1 + (1-a)Z_2) = \theta^2\big[\frac{a^2}{n-2} + \frac{(1-a)^2}{n}\big]$, minimized at $a^* = \frac{n-2}{2(n-1)}$ with variance $\frac{\theta^2}{2(n-1)}$. The bias-corrected MLE $T_c = T\,\Gamma(n)^2/[\Gamma(n+1/2)\Gamma(n-1/2)]$ is unbiased with $\mathrm{Var}(T_c) = \theta^2\big[\frac{n}{n-1}c^2 - 1\big]$, $c = \Gamma(n)^2/[\Gamma(n+1/2)\Gamma(n-1/2)] < 1$; at $n = 3$, $\mathrm{Var}(T_c) \approx 0.230\,\theta^2 < 0.25\,\theta^2 = \mathrm{Var}(a^*Z_1 + (1-a^*)Z_2)$, and T_c stays ahead asymptotically: $\mathrm{Var}(T_c) = \theta^2/(2n) + 3\theta^2/(8n^2) + o(1/n^2)$ versus $\theta^2/[2(n-1)] = \theta^2/(2n) + \theta^2/(2n^2) + o(1/n^2)$ (confirmed numerically at $n = 4, 5, 6, 10, 50, 200, 2000$ as well).

Exercise 7.55

Central idea. In each case the extreme order statistic is complete sufficient; Lehmann–Scheffé then blesses the (unique) unbiased function of it. **Solution.** (a) $X_{(n)}$ is complete sufficient; $E_\theta[X_{(n)}^r] = n\theta^r/(n+r)$, so $\frac{n+r}{n}X_{(n)}^r$ is the UMVUE of θ^r. (b) For integer $r \geq 1$, the explicit estimator is $X_{(1)}^r - (r/n)X_{(1)}^{r-1}$: if $M = \theta + Z$, $Z \sim \text{Exp(rate } n)$, integration by parts gives $E[M^r] - (r/n)E[M^{r-1}] = \theta^r$. Also, $X_{(1)}$ is complete sufficient; writing $X_{(1)} = \theta + Z$, $Z \sim$ exponential(n), $E_\theta[X_{(1)}^j]$ is a degree-j polynomial in θ with leading coefficient 1. Triangular inversion yields the unique degree-r polynomial p_r with $E_\theta[p_r(X_{(1)})] = \theta^r$

(e.g. $p_1(x) = x - 1/n$, $p_2(x) = x^2 - 2x/n$), which by Lehmann–Scheffé is the UMVUE. (c) Put $Y_i = e^{-X_i}$; then $Y_i \sim \text{uniform}(e^{-b}, \eta)$ iid with $\eta := e^{-\theta} > e^{-b}$, and $Y_{(n)} = e^{-X_{(1)}}$ is complete sufficient for η. For φ to satisfy $E_\eta[\varphi(Y_{(n)})] = (-\log\eta)^r = \theta^r$ for all $\eta > e^{-b}$,

$$\int_{e^{-b}}^{\eta} \varphi(y)(y - e^{-b})^{n-1}\, dy = \frac{(\eta - e^{-b})^n(-\log\eta)^r}{n} \qquad \forall\, \eta > e^{-b}.$$

Both sides vanish at $\eta = e^{-b}$; differentiating in η gives

$$\varphi(\eta) = (-\log\eta)^r - \frac{r(\eta - e^{-b})}{n\eta}(-\log\eta)^{r-1} = (-\log\eta)^{r-1}\left[(-\log\eta) - \frac{r(\eta - e^{-b})}{n\eta}\right].$$

Hence, for integer $r \geq 1$, the UMVUE of θ^r is the closed-form

$$\varphi(Y_{(n)}) = X_{(1)}^{r-1}\left[X_{(1)} - \frac{r}{n}\left(1 - e^{\,X_{(1)} - b}\right)\right],$$

a function of the complete sufficient $Y_{(n)}$, hence best unbiased by Lehmann–Scheffé. (Sanity check: as $b \to \infty$, $e^{X_{(1)}-b} \to 0$ and this tends to $X_{(1)}^r - \frac{r}{n}X_{(1)}^{r-1}$, reproducing the untruncated-exponential UMVUEs $p_1(x) = x - 1/n$, $p_2(x) = x^2 - 2x/n$ of part (b).)

Exercise 7.56

Central idea. Rao–Blackwell gives an unbiased function of T; completeness makes it the only one, hence best. **Solution.** $\phi(T)$ is unbiased by the tower property. If $\psi(T)$ is any other unbiased estimator based on T, then $E_\theta[\phi(T) - \psi(T)] = 0$ for all θ, so completeness of T forces $\phi(T) = \psi(T)$ a.s. Thus ϕ is the unique unbiased estimator that is a function of T; for any unbiased h', Rao–Blackwell gives $\text{Var}_\theta(h') \geq \text{Var}_\theta(E[h' \mid T]) = \text{Var}_\theta(\phi)$, so ϕ is best unbiased (the Lehmann–Scheffé theorem).

Exercise 7.57

Central idea. Rao–Blackwellize the indicator on the complete sufficient total $S = \sum_1^{n+1} X_i$; given $S = s$ the vector is uniform over weight-s binary strings. **Solution.** (a) $E_p[T] = P_p(\sum_1^n X_i > X_{n+1}) = h(p)$. (b) $S = \sum_{i=1}^{n+1} X_i$ is complete sufficient; the UMVUE is $\varphi(s) = E[T \mid S = s] = P(\sum_1^n X_i > X_{n+1} \mid S = s)$. Given $S = s$: with probability $\binom{n}{s}/\binom{n+1}{s} = (n+1-s)/(n+1)$, $X_{n+1} = 0$ (then $\sum_1^n X_i = s > 0 \iff s \geq 1$); with probability $s/(n+1)$, $X_{n+1} = 1$ (then $\sum_1^n X_i = s - 1 > 1 \iff s \geq 3$). Hence $\varphi(0) = 0$, $\varphi(s) = \frac{n+1-s}{n+1}$ for $s = 1, 2$, and $\varphi(s) = 1$ for $s \geq 3$ — the best unbiased estimator of $h(p)$.

Exercise 7.58

Central idea. $|X|$ is sufficient (f depends on x only through $|x|$); T wastes information by distinguishing 1 from -1, and Rao–Blackwell recovers $|X|$. **Solution.** (a) $L(\theta \mid x) = (\theta/2)^{|x|}(1-\theta)^{1-|x|}$: for $x = 0$, maximized at $\hat\theta = 0$; for $|x| = 1$, at $\hat\theta = 1$. So $\hat\theta = |X| = \mathbf{1}_{\{X \neq 0\}}$. (b) $E_\theta[T] = 2P_\theta(X = 1) = 2 \cdot \theta/2 = \theta$. (c) $|X|$ is sufficient and T is not a function of it. $E[T \mid |X|] = \mathbf{1}_{\{|X|=1\}} = |X|$ (given $|X| = 1$, $X = \pm 1$ with equal probability, so $E[T \mid |X| = 1] = \frac{1}{2} \cdot 2 = 1$). $\text{Var}_\theta(T) = 2\theta - \theta^2 > \theta(1-\theta) = \text{Var}_\theta(|X|)$ for $0 < \theta \leq 1$; thus $|X|$ (the MLE, in fact the UMVUE) is strictly better.

Exercise 7.59

Central idea. Normalize $(S^2)^{p/2}$ using its chi-square moment. When μ is unknown, the complete sufficient statistic is the **pair** $(\bar X, S^2)$, not S^2 alone.

Solution. For $n \geq 2$, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, so

$$E_{\mu,\sigma^2}[(S^2)^{p/2}] = \sigma^p \frac{2^{p/2}\Gamma((n-1+p)/2)}{(n-1)^{p/2}\Gamma((n-1)/2)}.$$

Thus

$$\delta = \frac{(n-1)^{p/2}\Gamma((n-1)/2)}{2^{p/2}\Gamma((n-1+p)/2)}(S^2)^{p/2}$$

is unbiased for σ^p and has finite variance for $p > 0$. The normal family is a full two-parameter exponential family with natural statistic $(\sum X_i, \sum X_i^2)$ and natural parameter space $\{(\eta_1, \eta_2) : \eta_1 \in \mathbb{R}, \eta_2 < 0\}$, which is open. Therefore the equivalent pair $(\bar{X}, S^2)$ is complete sufficient. Since δ is a function of this pair, Lehmann–Scheffé proves it is the UMVUE. The estimator is correct, but S^2 alone is not sufficient for the family with unknown mean.

Exercise 7.60

Central idea. Use completeness of $S = \sum X_i \sim \text{gamma}(k, \beta)$, $k = n\alpha$. Existence of the inverse moment requires $k > 1$, and finite variance requires $k > 2$.

Solution. With gamma scale β, $E(S^{-1}) = 1/[\beta(k-1)]$ for $k > 1$, giving the unique unbiased function of the complete sufficient statistic

$$\delta(S) = \frac{k-1}{S}.$$

For $k > 2$, $E(S^{-2}) = 1/[\beta^2(k-1)(k-2)]$, hence

$$\operatorname{Var}_\beta \delta = \frac{1}{\beta^2(k-2)}.$$

Lehmann–Scheffé gives the finite-variance UMVUE in this regime. For $1 < k \le 2$, δ is integrable and unbiased but has infinite variance; there is no finite-variance unbiased estimator, since Rao–Blackwellization would otherwise give a finite-variance unbiased function of S, contradicting completeness.

For $k \le 1$ there is no integrable unbiased estimator. Indeed, if such an estimator existed, Rao–Blackwellization would yield an integrable $d(S)$ with, on writing $t = 1/\beta$,

$$\int_0^\infty d(s) s^{k-1} e^{-ts}\, ds = \Gamma(k) t^{1-k} \quad (t > 0).$$

The left side tends to zero as $t \to \infty$ by dominated convergence, using absolute integrability at any fixed $t_0 > 0$. The right side is constant and nonzero for $k = 1$, and diverges for $k < 1$. This is impossible. An unqualified claim of a finite-variance best unbiased estimator therefore needs the condition $n\alpha > 2$.

Exercise 7.61

Central idea. The χ^2_ν density of $\nu S^2/\sigma^2$ contributes $-\frac{\nu}{2}(s^2/\sigma^2)$ from the exponential and $+\frac{\nu}{2}\log(s^2/\sigma^2)$ from the Jacobian/power — exactly Stein's loss up to affine transformation. **Solution.** For the stated $\nu > 0$, $f(s^2 \mid \sigma^2) = \frac{\nu}{\sigma^2} f_{\chi^2_\nu}(\nu s^2/\sigma^2)$, so

$$\log L = -\frac{\nu}{2}\frac{s^2}{\sigma^2} + \frac{\nu}{2}\log\frac{s^2}{\sigma^2} + K_3,$$

i.e. $K_1 = -\nu/2$, $K_2 = -\nu/2$, K_3 collecting terms free of σ^2. Hence $-\log L(\sigma^2 \mid s^2) = \frac{\nu}{2}[\frac{s^2}{\sigma^2} - \log\frac{s^2}{\sigma^2} - 1] + \text{const} = \frac{\nu}{2}L(\sigma^2, s^2) + \text{const}$, where $L(\sigma^2, a) = \frac{a}{\sigma^2} - 1 - \log\frac{a}{\sigma^2}$ is Stein's loss of Example 7.3.27 (with $a = s^2$): maximizing the normal likelihood in σ^2 is equivalent to minimizing Stein's loss.

Exercise 7.62

Central idea. $\delta^\pi = (1-\eta)\bar{X} + \eta\mu$ is of the form $a\bar{X} + b$; bias–variance plus $E_\pi[(\theta-\mu)^2] = \tau^2$ gives the Bayes risk. **Solution.** (a) $E_\theta[\delta] - \theta = (a-1)\theta + b$, $\operatorname{Var}_\theta(\delta) = a^2\sigma^2/n$; risk = variance + squared bias. (b) The posterior mean is $\delta^\pi = (1-\eta)\bar{X} + \eta\mu$ with $\eta = \frac{\sigma^2}{n\tau^2+\sigma^2}$ (since $\frac{\sigma^2/n}{\tau^2+\sigma^2/n} = \eta$); applying (a) with $a = 1-\eta$, $b = \eta\mu$ gives the stated risk. (c) $B(\pi, \delta^\pi) = E_\pi[R(\theta, \delta^\pi)] = (1-\eta)^2\frac{\sigma^2}{n} + \eta^2\tau^2 = \tau^2[\frac{n\tau^2\sigma^2}{(n\tau^2+\sigma^2)^2} + \frac{\sigma^4}{(n\tau^2+\sigma^2)^2}] = \tau^2\frac{\sigma^2}{n\tau^2+\sigma^2} = \tau^2\eta$.

Exercise 7.63

Central idea. Each risk is a parabola $(1-\eta)^2+\eta^2\mu^2$; the concentrated prior wins near the prior mean but loses in the tails, crossing at $|\mu|\approx 1.54$. **Solution.** Here $n=\sigma^2=1$. For $\pi=\mathrm{n}(0,1)$: $\eta=1/2$, $\delta^\pi=X/2$, $R(\mu,\delta^\pi)=\frac{1+\mu^2}{4}$. For $\pi=\mathrm{n}(0,10)$: $\eta=1/11$, $\delta^\pi=\frac{10}{11}X$, $R(\mu,\delta^\pi)=\frac{100+\mu^2}{121}$. Both graphs are upward parabolas with minima at $\mu=0$ (values 0.25 and ≈ 0.826); they cross where $\frac{1+\mu^2}{4}=\frac{100+\mu^2}{121}$, i.e. $|\mu|=\sqrt{279/117}\approx 1.544$. Comment: the concentrated $\mathrm{n}(0,1)$ prior shrinks harder toward 0, giving much smaller risk when μ is near 0 but larger risk when $|\mu|\gtrsim 1.54$; the diffuse $\mathrm{n}(0,10)$ prior is more robust to prior misspecification. Stronger prior concentration helps exactly when the prior is "right".

Exercise 7.64

Central idea. Independence plus product prior makes the posterior factor; the additive loss then separates into n independent minimizations. **Solution.** The posterior is $\pi(\theta\mid\mathbf{x})=\prod_{i=1}^n\pi_i(\theta_i\mid x_i)$ (independent likelihoods $\times$ product prior). The posterior expected loss of $\mathbf{a}=(a_1,\dots,a_n)$ is $E\big[\sum_i L(\theta_i,a_i)\mid\mathbf{x}\big]=\sum_i E[L(\theta_i,a_i)\mid x_i]$, minimized by minimizing each summand separately — i.e. $a_i=\delta_i^{\pi_i}(x_i)$. Hence the coordinate-wise rule is Bayes for the sum loss.

Exercise 7.65

Central idea. Minimizing the full posterior LINEX loss, including its linear term, gives $-\frac{1}{c}\log E[e^{-c\theta}\mid X]$; under a normal posterior the mgf yields the closed form $\bar X-c\sigma^2/(2n)$. **Solution.** (a) As a function of $d=a-\theta$, $L=e^{cd}-cd-1$: for $c=1$ it rises exponentially for $d>0$ (overestimation penalized heavily) and linearly for $d<0$; $c=0.5$ is milder; $c=0.2$ is nearly symmetric (approaching $\frac{c^2}{2}d^2$ as $c\to 0$). (b) $E[L(\theta,a)\mid X]=e^{ca}E[e^{-c\theta}\mid X]-ca+cE[\theta\mid X]-1$; $\frac{d}{da}=ce^{ca}E[e^{-c\theta}\mid X]-c=0$ gives $a=-\frac{1}{c}\log E[e^{-c\theta}\mid X]$ (a minimum by convexity in a). (c) With flat prior, $\theta\mid\bar X\sim\mathrm{n}(\bar X,\sigma^2/n)$; $E[e^{-c\theta}\mid\bar X]=\exp(-c\bar X+c^2\sigma^2/(2n))$ (normal mgf), so $\delta^B(\bar X)=\bar X-c\sigma^2/(2n)$. (d) Write $\delta^B-\theta=Z-c\sigma^2/(2n)$ with $Z:=\bar X-\theta\mid\bar X\sim\mathrm{n}(0,\sigma^2/n)$. Posterior LINEX loss of δ^B: $E[e^{cZ-c^2\sigma^2/(2n)}]-cE[Z-c\sigma^2/(2n)]-1=1+\frac{c^2\sigma^2}{2n}-1=\frac{c^2\sigma^2}{2n}$. For $\bar X$: $E[e^{cZ}]-1=e^{c^2\sigma^2/(2n)}-1>\frac{c^2\sigma^2}{2n}$ (since $e^t-1>t$), confirming δ^B is better under LINEX. (e) Under squared error: posterior expected loss of δ^B is $\frac{\sigma^2}{n}+\frac{c^2\sigma^4}{4n^2}$ (variance plus squared bias), versus $\frac{\sigma^2}{n}$ for $\bar X$ — under symmetric loss the unbiased $\bar X$ wins, as expected.

Exercise 7.66

Central idea. The leave-one-out means combine to $\sum_{i\neq j}X_iX_j/[n(n-1)]$ — a U-statistic that is exactly unbiased here, and by completeness (Lehmann–Scheffé) the UMVUE. **Solution.** (a) $E_\theta\bar X^2=\theta^2+\theta(1-\theta)/n\neq\theta^2$: biased (upward). (b) With $T_n=\bar X^2$, $T_n^{(i)}=\big(\frac{n\bar X-X_i}{n-1}\big)^2$. Since $\sum_i(n\bar X-X_i)^2=\sum_i X_i^2+n^2(n-2)\bar X^2$,

$$\mathrm{JK}(T_n)=n\bar X^2-\frac{\sum_i X_i^2+n^2(n-2)\bar X^2}{n(n-1)}=\frac{\sum_{i\neq j}X_iX_j}{n(n-1)}=\frac{Y(Y-1)}{n(n-1)},\quad Y=\sum X_i$$

(using $X_i^2=X_i$). (c) $E_\theta[Y(Y-1)]=n(n-1)\theta^2$, so $E_\theta[\mathrm{JK}(T_n)]=\theta^2$: unbiased — the jackknife removed the bias entirely. (d) Yes: $\mathrm{JK}(T_n)$ is a function of the complete sufficient statistic Y and is unbiased, hence by Lehmann–Scheffé it is the best (UMVUE) unbiased estimator of θ^2.

Exercise 7.67

Central idea. Split $|a_nW_n+b_n-\theta|\le|a_n||W_n-\theta|+|a_n-1||\theta|+|b_n|$ and let the deterministic terms vanish, leaving the consistency of W_n. **Solution.** Theorem 10.1.5: if W_n is consistent for θ, $a_n\to 1$, $b_n\to 0$, then $U_n=a_nW_n+b_n$ is consistent for θ. For $\varepsilon>0$,

$$|U_n-\theta|\le|a_n||W_n-\theta|+|a_n-1||\theta|+|b_n|.$$

Choose N so that for $n \geq N$: $|a_n| \leq 2$, $|a_n - 1||\theta| < \varepsilon/3$, $|b_n| < \varepsilon/3$. Then $P_\theta(|U_n - \theta| \geq \varepsilon) \leq P_\theta(2|W_n - \theta| \geq \varepsilon/3) = P_\theta(|W_n - \theta| \geq \varepsilon/6) \to 0$ by consistency of W_n. Hence U_n is consistent for θ.

Chapter 8

Exercise 8.1

Central idea. A large-sample z-test of $H_0 : \theta = 1/2$: the observed $z \approx 3.8$ gives a two-sided p-value of about 1.5×10^{-4}, far below any conventional level.

Solution. Let $X \sim \text{binomial}(1000, \theta)$ count heads and test $H_0 : \theta = 1/2$. Under H_0, $EX = 500$, $\text{sd}(X) = \sqrt{250} \approx 15.81$, so

$$z = \frac{560 - 500}{\sqrt{250}} \approx 3.79.$$

The (normal-approximated) two-sided p-value is $2\Phi(-3.79) \approx 0.00015$. No — it is not reasonable to assume fairness; the data reject H_0 at any conventional α (the deviation is nearly 4 standard errors).

Exercise 8.2

Central idea. A one-sided Poisson p-value: $P(X \leq 10 \mid \lambda = 15) \approx 0.12$, not significant at conventional levels.

Solution. Test $H_0 : \lambda \geq 15$ versus $H_1 : \lambda < 15$ with $X = 10$ observed. The p-value is the smallest α at which H_0 rejects, i.e.

$$p = \sup_{\lambda \geq 15} P_\lambda(X \leq 10) = P_{\lambda=15}(X \leq 10) = \sum_{k=0}^{10} e^{-15} \frac{15^k}{k!} \approx 0.119.$$

Since $p \approx 0.12 > 0.05$, the drop from 15 to 10 is within ordinary Poisson fluctuation; the claim of a drop is not justified at the usual significance levels.

Exercise 8.3

Central idea. The profile likelihood over $\theta \leq \theta_0$ is flat at the unconstrained MLE when $\bar{Y} \leq \theta_0$ and pinned at θ_0 otherwise, so the likelihood ratio is a decreasing function of $\sum Y_i$ on the rejection side.

Solution. With $S = \sum Y_i$, $L(\theta \mid \mathbf{y}) = \theta^S(1-\theta)^{m-S}$. The unconstrained MLE is $\hat{\theta} = S/m$. The numerator of $\lambda(\mathbf{y})$ maximizes over $\theta \leq \theta_0$: if $S/m \leq \theta_0$ the maximum is at $\hat{\theta}$ (so $\lambda = 1$); if $S/m > \theta_0$ the log-likelihood, concave in θ, is maximized at the boundary θ_0. Hence, for $S > m\theta_0$,

$$\lambda(\mathbf{y}) = \frac{\theta_0^S(1-\theta_0)^{m-S}}{(S/m)^S(1-S/m)^{m-S}},$$

and $\log\lambda = S\log\frac{\theta_0 m}{S} + (m-S)\log\frac{(1-\theta_0)m}{m-S}$, whose derivative in S is $\log\frac{\theta_0(1-S/m)}{(1-\theta_0)(S/m)} < 0$ for $S > m\theta_0$. So λ equals 1 for $S \leq m\theta_0$ and is strictly decreasing in S for $S > m\theta_0$; thus $\{\lambda(\mathbf{y}) < c\} = \{S > b\}$ for a suitable b.

Exercise 8.4

Central idea. For a discrete model the pmf value $f(\mathbf{x} \mid \theta)$ literally is $P_\theta(\mathbf{X} = \mathbf{x})$, so each supremum in λ is a maximum probability of the observed sample.

Solution. By Definition 8.2.1,

$$\lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta \mid \mathbf{x})}{\sup_{\theta \in \Theta} L(\theta \mid \mathbf{x})}, \qquad L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta).$$

If $f(\cdot \mid \theta)$ is a pmf, then $f(\mathbf{x} \mid \theta) = P_\theta(\mathbf{X} = \mathbf{x})$, the probability of the observed sample under θ. Hence $\sup_{\Theta_0} L$ is the largest probability assignable to the observed sample by any null parameter value, and $\sup_{\Theta} L$ is the largest probability assignable by any parameter value at all. The ratio $\lambda(\mathbf{x})$ therefore compares the best null explanation of the data against the best overall explanation.

Exercise 8.5

Central idea. The scale MLE is the sample minimum; profiling it out leaves a one-parameter problem in $T = \sum \log(X_i / \min X_i)$, and under H_0 the log-ratios are $n-1$ iid exponentials by the memoryless property, so $2T \sim \chi^2_{2(n-1)}$.

Solution. Write $m = \min_i X_i$. The likelihood is $L(\theta, v) = \theta^n v^{n\theta} (\prod x_i)^{-(\theta+1)} I(v \le m)$. (a) For fixed θ, L increases in v, so $\hat{v} = m$. Profiling: $\ell(\theta) = n \log\theta + n\theta \log m - (\theta+1) \sum \log x_i$; $\ell'(\theta) = n/\theta + n \log m - \sum \log x_i = 0$ gives

$$\hat{v} = \min_i X_i, \qquad \hat{\theta} = \frac{n}{\sum_{i=1}^n \log(X_i/\hat{v})} = \frac{n}{T}.$$

(b) Under H_0, $\hat{v}_0 = m$ as well, so with $T = \sum \log(x_i/m)$,

$$\log\lambda = -n \log\hat{\theta} + (\hat{\theta} - 1)T = n\big[1 - u + \log u\big], \quad u = T/n,$$

using $\hat{\theta} = n/T$. The function $k(u) = 1 - u + \log u \le 0$ equals 0 only at $u = 1$, strictly increasing on $(0,1)$ and strictly decreasing on $(1,\infty)$, with $k(u) \to -\infty$ at both ends. Hence $\{\log\lambda \le \log c\} = \{u \le c_1'\} \cup \{u \ge c_2'\}$, i.e. $\{T \le c_1\} \cup \{T \ge c_2\}$ with $0 < c_1 < c_2$ (and $T \ge 0$ always, $T = 0$ iff all X_i are equal). (c) Under H_0 ($\theta = 1$), $Y_i = \log(X_i/v)$ are iid exponential(1). Then $T = \sum_i (Y_i - \min_j Y_j)$: one term is 0 and, by the memoryless property, the other $n-1$ terms are iid exponential(1), independent of $\min_j Y_j$ (this is the hint's conditional argument — the conditional distribution of the ratios given the minimum is free of the minimum). Thus $T \sim \text{Gamma}(n-1, 1)$ and $2T \sim \chi^2_{2(n-1)}$: $2(n-1)$ degrees of freedom.

Exercise 8.6

Central idea. With the mean-parametrization, both MLEs are sample means; the likelihood ratio collapses to a function of the single fraction T, which under H_0 is the ratio of independent gammas with common scale — hence beta.

Solution. (Here exponential(θ) means mean θ: $f(x \mid \theta) = \theta^{-1} e^{-x/\theta}$.) Let $S_X = \sum X_i$, $S_Y = \sum Y_i$. $L(\theta, \mu) = \theta^{-n} \mu^{-m} \exp(-S_X/\theta - S_Y/\mu)$. (a) Unconstrained MLEs: $\hat{\theta} = S_X/n$, $\hat{\mu} = S_Y/m$. Under H_0 ($\theta = \mu$): $\hat{\theta}_0 = (S_X + S_Y)/(n+m)$. The exponential factors equal $e^{-(n+m)}$ in both, so

$$\lambda = \frac{\hat{\theta}_0^{-(n+m)}}{\hat{\theta}^{-n} \hat{\mu}^{-m}} = \frac{n^n m^m}{(n+m)^{n+m}} T^n (1-T)^m, \qquad T = \frac{S_X}{S_X + S_Y}.$$

(b) The bracketed constant is fixed, and $g(T) = T^n(1-T)^m$ on $(0,1)$ is maximized at $T = n/(n+m)$, strictly increasing before and strictly decreasing after. Hence $\{\lambda \le c\} = \{T \le c_1\} \cup \{T \ge c_2\}$: the LRT rejects for T far from $n/(n+m)$ in either direction. (c) Under H_0, $S_X \sim \text{Gamma}(n, \theta)$ and $S_Y \sim \text{Gamma}(m, \theta)$ independently (shape–scale form, common scale θ). For independent gammas with common scale, $S_X/(S_X + S_Y) \sim \text{beta}(n, m)$, independent of θ.

Exercise 8.7

Central idea. (a) Profiling out the scale λ leaves a monotone profile in the location θ, so the LRT rejects for large $\min X_i / \bar{X}$. (b) Profiling out β gives a closed-form profile likelihood in γ alone; the LRT has no closed-form critical region and is computed numerically.

Solution. (a) With $m = \min x_i$, $L(\theta, \lambda) = \lambda^{-n} \exp(-\sum (x_i - \theta)/\lambda)\, I(\theta \le m)$. For fixed $\theta \le m$, $\hat{\lambda}(\theta) = \bar{x} - \theta$, and the profile log-likelihood $\ell_p(\theta) = -n \log(\bar{x} - \theta) - n$ is strictly increasing in θ. Under H_0 the maximizer

is $\hat\theta_0 = \min(0, m)$; unconstrained it is $\hat\theta = m$. Thus

$$\lambda(\mathbf{x}) = \begin{cases} \left(\dfrac{\bar x - m}{\bar x}\right)^n, & m \ge 0, \\ 1, & m < 0, \end{cases}$$

so the LRT rejects iff $T(\mathbf{x}) = m/\bar x \ge c'$ (the $m < 0$ case is consistent since then $T < 0 < c'$). The cutoff c' is set from $\sup_{\theta \le 0} P_{\theta,\lambda}(T \ge c') = \alpha$; at $\theta = 0$, $T = \min E_i/\bar E$ for iid standard exponentials E_i, free of λ. (b) The Weibull pdf is $f(x \mid \gamma, \beta) = \frac{\gamma}{\beta}(x/\beta)^{\gamma-1}\exp(-(x/\beta)^\gamma)$, $x > 0$. $\ell(\gamma,\beta) = n\log\gamma - n\gamma\log\beta + (\gamma-1)\sum\log x_i - \sum(x_i/\beta)^\gamma$. For fixed γ, $\hat\beta(\gamma)^\gamma = \frac{1}{n}\sum x_i^\gamma$, giving the profile $\ell_p(\gamma) = n\log\gamma - n\log(\frac{1}{n}\sum x_i^\gamma) + (\gamma-1)\sum\log x_i - n$. Under H_0 ($\gamma = 1$), $\hat\beta = \bar x$ and $\ell_p(1) = -n\log\bar x - n$. Hence

$$\lambda(\mathbf{x}) = \frac{L(1, \bar x)}{\sup_{\gamma,\beta} L(\gamma,\beta)}, \qquad -2\log\lambda = 2\big[\sup_\gamma \ell_p(\gamma) - \ell_p(1)\big],$$

and the LRT rejects for large $-2\log\lambda$. The supremum over γ has no closed form (it solves $n/\gamma - n\log\hat\beta(\gamma) + \sum\log x_i - \sum(x_i/\hat\beta(\gamma))^\gamma\log(x_i/\hat\beta(\gamma)) = 0$), so in practice the statistic is maximized numerically; asymptotically $-2\log\lambda \approx \chi^2_1$ under H_0.

Exercise 8.8

Central idea. In both families the nuisance mean θ profiles out in closed form ($\hat\theta = \bar x$ under H_1), leaving an LRT that is a function of the sufficient statistics $(\bar x, \sum(X_i - \bar x)^2)$ with only the null-constrained $\hat\theta_0$ differing between the two parametrizations.

Solution. (a) $L(a,\theta) = (2\pi a\theta)^{-n/2}\exp(-\sum(x_i-\theta)^2/(2a\theta))$, $\theta > 0$, $a > 0$. For fixed θ, $\hat a(\theta) = \sum(x_i - \theta)^2/(n\theta)$, and the profile is $-\frac{n}{2}\log(\sum(x_i-\theta)^2/n) - \frac{n}{2}$ (up to constants), minimized at $\hat\theta = \bar x$. So $\sup_{H_1} L = (2\pi)^{-n/2}[\sum(x_i-\bar x)^2/n]^{-n/2}e^{-n/2}$. Under H_0 ($a = 1$): $\ell(\theta) = -\frac{n}{2}\log\theta - \sum(x_i-\theta)^2/(2\theta)$; $\ell'(\theta) = 0$ gives $\theta^2 + \theta - \frac{1}{n}\sum x_i^2 = 0$, i.e. $\hat\theta_0 = \frac{1}{2}(-1 + \sqrt{1 + 4\sum x_i^2/n})$. The LRT rejects for small

$$\lambda(\mathbf{x}) = \frac{(2\pi\hat\theta_0)^{-n/2}\exp(-\sum(x_i-\hat\theta_0)^2/(2\hat\theta_0))}{(2\pi)^{-n/2}[\sum(x_i-\bar x)^2/n]^{-n/2}e^{-n/2}},$$

a function of $(\bar x, \sum(x_i - \bar x)^2)$. (b) Now $L(a,\theta) = (2\pi a\theta^2)^{-n/2}\exp(-\sum(x_i-\theta)^2/(2a\theta^2))$. For fixed θ, $\hat a(\theta) = \sum(x_i-\theta)^2/(n\theta^2)$; profiling again gives $-\frac{n}{2}\log(\sum(x_i-\theta)^2/n) - \frac{n}{2}$, minimized at $\hat\theta = \bar x$, so $\sup_{H_1} L$ is the same expression as in (a). Under H_0 ($a = 1$): $\ell(\theta) = -n\log\theta - \sum(x_i-\theta)^2/(2\theta^2)$; $\ell'(\theta) = 0$ gives $-n\theta^2 + \sum(x_i-\theta)^2 + \theta\sum(x_i-\theta) = 0$, i.e. $\theta^2 + \bar x\,\theta - \frac{1}{n}\sum x_i^2 = 0$, so $\hat\theta_0 = \frac{1}{2}(-\bar x + \sqrt{\bar x^2 + 4\sum x_i^2/n})$. The LRT rejects for small $\lambda(\mathbf{x}) = L(1,\hat\theta_0)/\sup_{H_1} L$, again a function of $(\bar x, \sum(x_i-\bar x)^2)$.

Exercise 8.9

Central idea. The LRT statistic never exceeds 1, and writing out "$\lambda \le 1$" for these two parametrizations is exactly AM $\ge$ GM and GM $\ge$ HM.

Solution. (a) $L(\lambda) = (\prod\lambda_i)\exp(-\sum\lambda_i y_i)$. Under H_0 ($\lambda_i \equiv \lambda$): $\hat\lambda = 1/\bar y$, $\sup L = \bar y^{-n}e^{-n}$. Unconstrained: $\hat\lambda_i = 1/y_i$, $\sup L = (\prod y_i)^{-1}e^{-n}$. Hence

$$\lambda(\mathbf{y}) = \frac{(\bar Y)^{-n}}{(\prod_i Y_i)^{-1}} \le 1,$$

so $(\bar Y)^{-n} \le (\prod_i Y_i)^{-1}$, i.e. $\prod_i Y_i \le \bar Y^{\,n}$, the AM–GM inequality $\bar Y \ge (\prod_i Y_i)^{1/n}$. (b) With $X_i = 1/Y_i$, $f_{X_i}(x \mid \lambda_i) = \lambda_i e^{-\lambda_i/x}/x^2$. Under H_0: $\ell(\lambda) = n\log\lambda - \lambda\sum(1/x_i) - 2\sum\log x_i$, maximized at $\hat\lambda = n/\sum(1/x_i)$ (the

harmonic mean), giving $\sup L = [n/\sum(1/x_i)]^n e^{-n}(\prod x_i)^{-2}$. Unconstrained: $\hat\lambda_i = x_i$, $\sup L = (\prod x_i)^{-1}e^{-n}$. Thus

$$\lambda(\mathbf{x}) = \frac{[n/\sum_i(1/X_i)]^n}{\prod_i X_i} \le 1,$$

i.e. $[n/\sum_i(1/X_i)]^n \le \prod_i X_i$, the GM–HM inequality $(\prod_i X_i)^{1/n} \ge n/\sum_i(1/X_i)$.

Exercise 8.10

Central idea. The gamma posterior turns $P(\lambda \le \lambda_0 \mid \mathbf{x})$ into a chi-squared tail probability after the scale change $\text{gamma}(a,b) =_d \frac{b}{2}\chi^2_{2a}$.

Solution. With the C&B gamma(α,β) (mean $\alpha\beta$), the posterior is $\lambda \mid \mathbf{x} \sim \text{gamma}(\alpha + \sum x_i,\ \beta/(1+n\beta))$. (a)

$$P(H_0 \mid \mathbf{x}) = P(\lambda \le \lambda_0 \mid \mathbf{x}) = F_{\text{gamma}}\Big(\lambda_0;\ \alpha + \textstyle\sum x_i,\ \frac{\beta}{1+n\beta}\Big), \qquad P(H_1 \mid \mathbf{x}) = 1 - P(H_0 \mid \mathbf{x}),$$

where F_{gamma} is the cdf. The Bayes test (0–1 loss) decides H_1 iff $P(H_1 \mid \mathbf{x}) > 1/2$. (b) $\text{gamma}(5/2, 2) = \chi^2_5$, and the posterior is $\text{gamma}(5/2 + \sum x_i,\ 2/(1+2n))$. Since $\text{gamma}(a,b) =_d \frac{b}{2}\,\chi^2_{2a}$,

$$\lambda \mid \mathbf{x} \ =_d\ \frac{1}{1+2n}\,\chi^2_{5+2\sum x_i},$$

so $P(H_0 \mid \mathbf{x}) = P(\chi^2_{5+2\sum x_i} \le (1+2n)\lambda_0)$, with integer df $5 + 2\sum x_i$. Look up $(1+2n)\lambda_0$ in a $\chi^2_{5+2\sum x_i}$ table: decide H_1 iff the upper-tail probability exceeds $1/2$ (equivalently, iff $(1+2n)\lambda_0$ is below the median of that chi-squared distribution).

Exercise 8.11

Central idea. The inverted-gamma posterior makes $P(\sigma^2 \le 1 \mid s^2)$ a chi-squared tail probability that is strictly decreasing in s^2, so the Bayes rule is $\{s^2 < c(\alpha,\beta)\}$ — the same one-sided form as the LRT acceptance region $\{s^2 \le c'\}$, and the two free prior parameters can match any fixed c'.

Solution. In Exercise 7.23's IG(α,β), β is a scale: $1/\sigma^2 \sim$ gamma(shape α, rate $1/\beta$). The posterior is $\sigma^2 \mid s^2 \sim \text{IG}(\alpha^*,\beta^*)$ with $\alpha^* = \alpha + \frac{n-1}{2}$ and $1/\beta^* = 1/\beta + \frac{(n-1)s^2}{2}$, so $1/\sigma^2 \mid s^2 \sim \text{gamma}(\alpha^*, \text{rate } 1/\beta^*)$. (a) $P(\sigma^2 \le 1 \mid s^2) = P(\chi^2_{2\alpha^*} \ge 2/\beta^*)$, which is continuous and strictly decreasing in s^2 (through β^*), falling from $P(\chi^2_{2\alpha+(n-1)} > 2/\beta)$ at $s^2 = 0$ to 0. Hence $\{P(\sigma \le 1 \mid s^2) > 1/2\} = \{s^2 < c\}$ where $c = [\text{med}(\chi^2_{2\alpha+n-1}) - 2/\beta]/(n-1)$ (empty if $2/\beta \ge \text{med}(\chi^2_{2\alpha+n-1})$). (b) The LRT based on S^2: with $L(\sigma^2) \propto (\sigma^2)^{-(n-1)/2}\exp(-(n-1)s^2/(2\sigma^2))$, $\lambda(s^2) = 1$ for $s^2 \le 1$ and $= (s^2)^{(n-1)/2}e^{-(n-1)(s^2-1)/2}$, strictly decreasing, for $s^2 > 1$. So the LRT acceptance region is $\{s^2 \le c'\}$ for a fixed $c' \ge 1$ set by the size. Both regions are one-sided intervals in s^2; choosing α large enough that $\text{med}(\chi^2_{2\alpha+n-1}) > (n-1)c'$ and then $\beta = 2/[\text{med}(\chi^2_{2\alpha+n-1}) - (n-1)c'] > 0$ makes $c = c'$. Yes — such prior parameters exist.

Exercise 8.12

Central idea. Both power functions depend on the data only through the standardized effect $\sqrt{n}\,\mu/\sigma$; the one-sided power is $\Phi(\sqrt{n}\mu/\sigma - z_\alpha)$ and the two-sided power adds the reflected tail.

Solution. (a) The size-α LRT/UMP test rejects for $\bar X > z_\alpha\sigma/\sqrt{n}$. With $z_{.05} = 1.645$,

$$\gamma(\mu) = P_\mu\left(\bar X > z_\alpha\frac{\sigma}{\sqrt{n}}\right) = \Phi\left(\frac{\sqrt{n}\,\mu}{\sigma} - 1.645\right),$$

increasing from 0 to 1, equal to .05 at $\mu = 0$. (b) The two-sided size-α test rejects for $|\bar X| > z_{\alpha/2}\sigma/\sqrt{n}$. With $z_{.025} = 1.96$,

$$\gamma(\mu) = \Phi\left(\frac{\sqrt{n}\,\mu}{\sigma} - 1.96\right) + \Phi\left(-\frac{\sqrt{n}\,\mu}{\sigma} - 1.96\right),$$

symmetric in μ, equal to .05 at $\mu = 0$ and $\to 1$ as $|\mu| \to \infty$. In both cases larger n steepens the curve (power rises faster in $|\mu|/\sigma$).

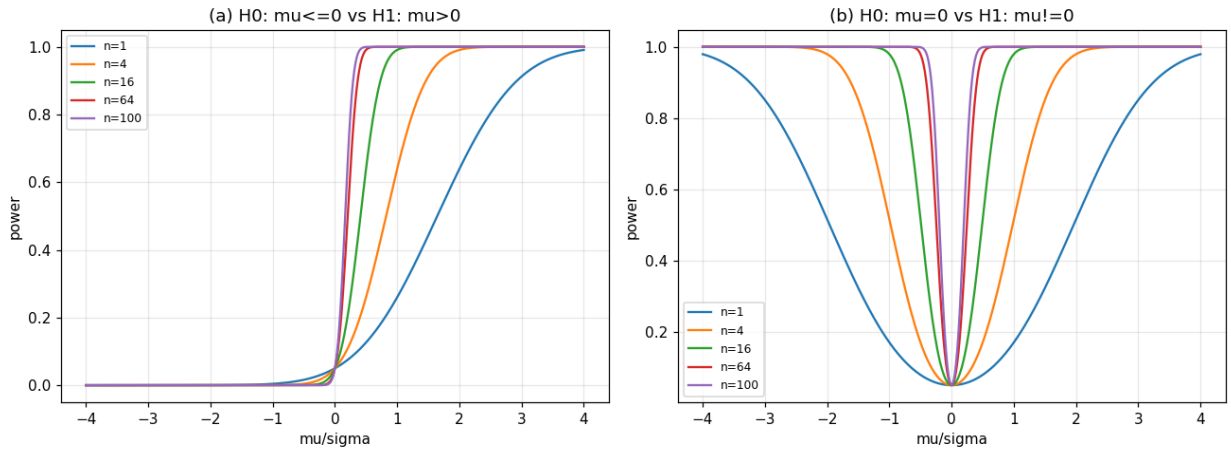


Figure 3: Power functions for 8.12

Exercise 8.13

Central idea. ϕ_1 has size .05; matching it forces $C = 2 - \sqrt{0.1} \approx 1.684$. The two closed-form power functions both start at .05 but cross at $\theta \approx .184$ — neither test dominates — so the "better" test for (d) is the UMP test of 8.33 ($n = 2$: reject iff $\max X_i \geq 1$ or $\min X_i \geq 1 - \sqrt{.05}$).

Solution. (a) ϕ_1 has size $P_0(X_1 > .95) = .05$. Under H_0, $S = X_1 + X_2$ has the triangular density $f(s) = s$ on $(0, 1)$, $2 - s$ on $(1, 2)$. For $C \in (1, 2)$, $P_0(S > C) = (2 - C)^2/2 = .05$ gives $C = 2 - \sqrt{0.1} \approx 1.6838$. (b) $\gamma_1(\theta) = P_\theta(X_1 > .95) = \min(1, \theta + .05)$ for $\theta \geq 0$ (i.e. $\theta + .05$ on $[0, .95]$, then 1). For ϕ_2, write $X_i = \theta + U_i$, $U_i \sim$ iid $U(0, 1)$; $\gamma_2(\theta) = P(U_1 + U_2 > C - 2\theta)$. With $t = C - 2\theta$:

$$\gamma_2(\theta) = \begin{cases} \frac{(2-t)^2}{2} = \frac{(2-C+2\theta)^2}{2}, & 0 \leq \theta \leq \frac{C-1}{2} \approx .342, \\ 1 - \frac{t^2}{2} = 1 - \frac{(C-2\theta)^2}{2}, & .342 \leq \theta \leq C/2 \approx .842, \\ 1, & \theta \geq .842. \end{cases}$$

Both start at .05 when $\theta = 0$; γ_1 initially lies above γ_2 (e.g. $\gamma_1(.05) = .1 > .0866 \approx \gamma_2(.05)$), but the curves cross at $\theta \approx .184$, after which γ_2 dominates: γ_2 reaches 1 at $\theta \approx .842$ while γ_1 reaches 1 only at $\theta = .95$.

(c) Disprove. "More powerful" requires $\gamma_2(\theta) \geq \gamma_1(\theta)$ for all $\theta > 0$, but at $\theta = .05$, $\gamma_1 = .1 > .0866 \approx \gamma_2$.

(d) The UMP test of 8.33 does the job. For $n = 2$, $\alpha = .05$ it rejects iff $Y_2 \geq 1$ or $Y_1 \geq k$ with $k = 1 - \sqrt{.05} \approx .776$ (size $P_0(Y_1 \geq k) = (1 - k)^2 = .05$, since $Y_2 < 1$ a.s. under H_0). Because it is UMP (proved in 8.33), its power $\gamma^*(\theta) = 1.05 - (1 - \theta)^2$ on $[0, k]$, $= 1$ for $\theta \geq k$, dominates γ_2 at every $\theta > 0$ — strictly on $(0, .842)$, e.g. $\gamma^*(.05) = .1475 > .0866 \approx \gamma_2(.05)$. (Note ϕ_1 itself does not work: γ_2 overtakes γ_1 past $\theta \approx .184$.)

Exercise 8.14

Central idea. Match the two normal-approximated error probabilities at a common cutoff: $.02n = 2z_{.01}\sqrt{.49 \cdot .51\, n}$, giving $n \approx 13{,}520$.

Solution. Let $S = \sum X_i$ and reject iff $S > c$. By the CLT, $S \approx n(.49, .49 \cdot .51)$ under H_0 and $\approx n(.51, .49 \cdot .51)$ under H_1. Setting both error probabilities to .01,

$$c = .49n + z_{.01}\sqrt{.2499n} = .51n - z_{.01}\sqrt{.2499n}, \quad z_{.01} \approx 2.326.$$

Hence $.02n = 2(2.326)\sqrt{.2499n}$, i.e. $\sqrt{n} = 2(2.326)\sqrt{.2499}/.02 \approx 116.27$, so $n \approx 13{,}520$ (round up). The cutoff is $c \approx .49n + 2.326\sqrt{.2499n} \approx 6760$: reject H_0 if more than about 6760 of the 13,520 trials are successes.

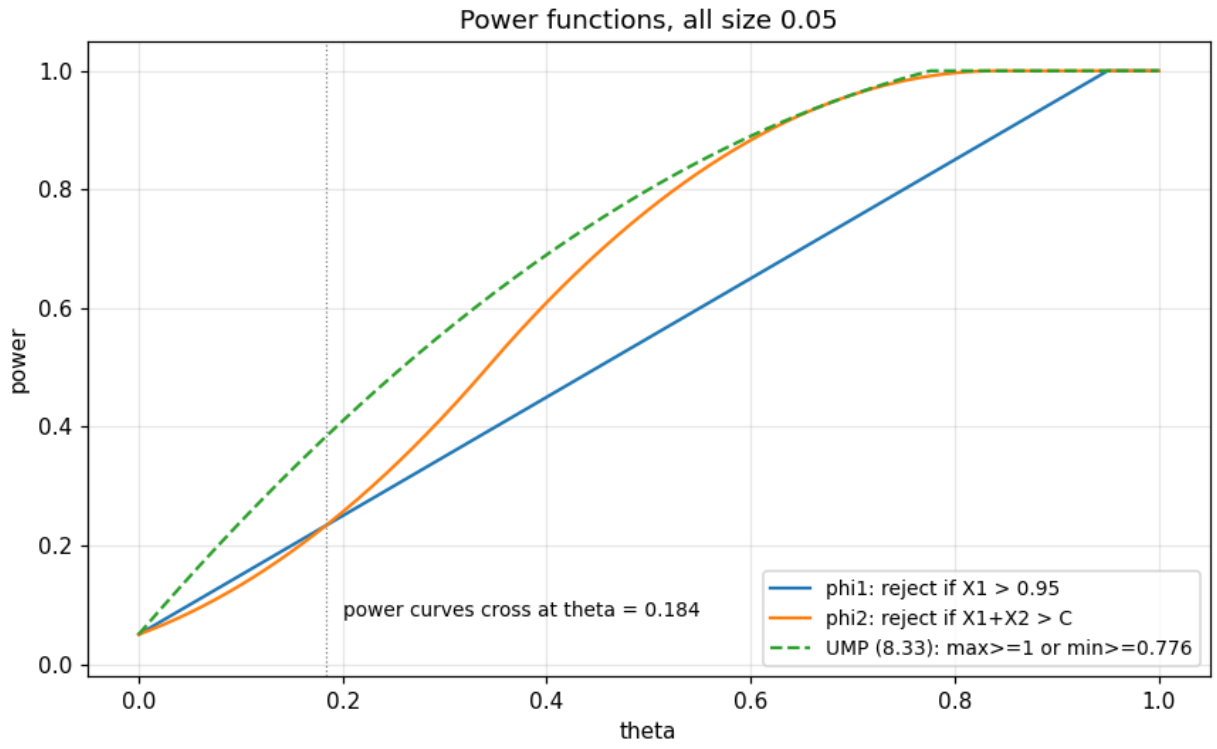


Figure 4: Power functions for 8.13

Exercise 8.15

Central idea. The Neyman–Pearson likelihood ratio is monotone increasing in $\sum X_i^2$ (because $\sigma_1 > \sigma_0$ flips the sign of the exponent), and $\sum X_i^2/\sigma_0^2 \sim \chi_n^2$ under H_0 pins down c.

Solution. The likelihood ratio is

$$\frac{L(\sigma_1 \mid \mathbf{x})}{L(\sigma_0 \mid \mathbf{x})} = \left(\frac{\sigma_0}{\sigma_1}\right)^n \exp\left[-\frac{\sum x_i^2}{2}\left(\frac{1}{\sigma_1^2} - \frac{1}{\sigma_0^2}\right)\right] = \left(\frac{\sigma_0}{\sigma_1}\right)^n \exp\left[\frac{\sum x_i^2}{2}\left(\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2}\right)\right].$$

Since $\sigma_0 < \sigma_1$, $\frac{1}{\sigma_0^2} - \frac{1}{\sigma_1^2} > 0$, so the ratio is a strictly increasing function of $T(\mathbf{x}) = \sum x_i^2$. By Neyman–Pearson, the MP test rejects iff the ratio exceeds k, equivalently iff $\sum X_i^2 > c$. Under H_0, $\sum X_i^2/\sigma_0^2 \sim \chi_n^2$, so size α requires $P(\chi_n^2 > c/\sigma_0^2) = \alpha$, i.e.

$$c = \sigma_0^2\, \chi_{n,\alpha}^2,$$

the upper-α point of χ_n^2.

Exercise 8.16

Central idea. A test that always rejects has rejection probability 1 under every parameter value — hence true size 1 and power 1; always accepting gives size 0 and power 0.

Solution. (a) $\phi \equiv 1$: size $= \sup_{\Theta_0} E_\theta \phi = 1$, power $\gamma(\theta) = 1$ for all $\theta \in \Theta_1$. True Type I error probability 1, Type II error probability 0. Choosing α after the fact to force rejection does not produce a level-α test — it produces a size-1 test masquerading as one. (b) $\phi \equiv 0$: size $= 0$, power $\equiv 0$. True Type I error probability 0, Type II error probability 1: H_0 is never rejected even when blatantly false.

Exercise 8.17

Central idea. $-\log X_i \sim$ exponential(μ), so this is the two-sample exponential problem in disguise (cf. 8.6); T is the beta-distributed fraction of the total "exponential mass" in the X sample.

Solution. The beta($\mu, 1$) pdf is $\mu x^{\mu-1}$ on $(0,1)$. Let $A = -\sum \log X_i > 0$, $B = -\sum \log Y_i > 0$. (a) $L(\mu,\theta) = \mu^n \theta^m \exp(-(\mu-1)\sum \log X_i - (\theta-1)\sum \log Y_i)$. Unconstrained MLEs: $\hat{\mu} = n/A$, $\hat{\theta} = m/B$. Under H_0 ($\mu = \theta = \phi$): $\hat{\phi} = (n+m)/(A+B)$. The $\exp(\sum \log X_i + \sum \log Y_i)$ factors cancel, giving

$$\lambda = \frac{(n+m)^{n+m}}{n^n m^m} \cdot \frac{A^n B^m}{(A+B)^{n+m}}.$$

(b) With $T = \frac{\sum \log X_i}{\sum \log X_i + \sum \log Y_i} = \frac{A}{A+B} \in (0,1)$,

$$\lambda = \frac{(n+m)^{n+m}}{n^n m^m}\, T^n (1-T)^m,$$

and $T^n(1-T)^m$ is maximized at $T = n/(n+m)$, strictly monotone on each side. Hence the LRT rejects for $T \le c_1$ or $T \ge c_2$. (c) Since $-\log X_i \sim$ exponential(μ) (mean $1/\mu$) and $-\log Y_i \sim$ exponential(θ), under H_0 ($\mu = \theta$) $A \sim$ Gamma(n, μ), $B \sim$ Gamma(m, μ) independently, so $T = A/(A+B) \sim$ beta(n, m), free of μ. A size-.10 test: reject if $T < b_{n,m,.05}$ or $T > b_{n,m,.95}$, the lower/upper 5% points of beta(n, m) (any $c_1 < c_2$ with $P(T < c_1) + P(T > c_2) = .10$ works; equal tails is standard).

Exercise 8.18

Central idea. Power is the two-tail normal probability $P(|Z+\delta| > c)$ with noncentrality $\delta = \sqrt{n}(\theta - \theta_0)/\sigma$; fixing $\alpha = .05$ gives $c = 1.96$, and $\beta = .25$ at one-σ displacement solves to $n = 7$.

Solution. (a) Under θ, $(\bar{X} - \theta_0)/(\sigma/\sqrt{n}) =_d Z + \delta$ with $Z \sim n(0,1)$ and $\delta = \sqrt{n}(\theta - \theta_0)/\sigma$. Hence

$$\gamma(\theta) = P_\theta(|Z+\delta| > c) = \big[1 - \Phi(c-\delta)\big] + \Phi(-c-\delta), \qquad \delta = \frac{\sqrt{n}(\theta - \theta_0)}{\sigma}.$$

(b) Type I error .05 gives $c = z_{.025} = 1.96$. The Type II error $\beta(\theta) = 1 - \gamma(\theta) = \Phi(c-\delta) - \Phi(-c-\delta)$ decreases as $|\theta - \theta_0|$ grows, so requiring $\beta \le .25$ for $|\theta - \theta_0| \ge \sigma$ amounts to $\beta(\theta_0 + \sigma) = .25$. With $\delta = \sqrt{n}$,

$$\Phi(1.96 - \sqrt{n}) - \Phi(-1.96 - \sqrt{n}) = .25.$$

The second term is negligible, so $1.96 - \sqrt{n} \approx \Phi^{-1}(.25) = -0.6745$, i.e. $\sqrt{n} \approx 2.63$, $n \approx 6.9$. Take $n = 7$ (with $n = 6$, $\beta \approx .31 > .25$; with $n = 7$, $\beta \approx .246$). Answer: $c = 1.96$, $n = 7$.

Exercise 8.19

Central idea. The Neyman–Pearson ratio $e^{y-\sqrt{y}}/(2\sqrt{y})$ is U-shaped in $\sqrt{y}$ (minimum 1/2 at $y = 1$), so the MP rejection region is two-sided, $\{Y < .0765\} \cup \{Y > 3.638\}$; for simple vs. simple this is automatically UMP.

Solution. $P_\theta(Y \le y) = P(X \le y^{1/\theta}) = 1 - e^{-y^{1/\theta}}$, so $f_Y(y \mid \theta) = \frac{1}{\theta} y^{1/\theta - 1} e^{-y^{1/\theta}}$, $y > 0$. Under H_0, $Y \sim$ exponential(1); under H_1, $f_Y(y \mid 2) = \frac{1}{2\sqrt{y}} e^{-\sqrt{y}}$. The likelihood ratio

$$r(y) = \frac{f_Y(y \mid 2)}{f_Y(y \mid 1)} = \frac{e^{\,y-\sqrt{y}}}{2\sqrt{y}}, \qquad \frac{d}{dt} \log r(t^2) = \frac{(2t+1)(t-1)}{t},\ t = \sqrt{y},$$

decreases on $0 < t < 1$ and increases on $t > 1$, with minimum $r(1) = 1/2$. By Neyman–Pearson the MP test rejects for $r(Y) > k$, i.e. $\{Y < a\} \cup \{Y > b\}$ with $r(a) = r(b) = k$. Size .10 under $Y \sim$ exponential(1): $(1 - e^{-a}) + e^{-b} = .10$ gives (numerically) $k \approx 1.479$, $a \approx 0.0765$, $b \approx 3.638$. Since H_0, H_1 are simple, this MP test is UMP. Type II error: under H_1, $P(Y \le y) = 1 - e^{-\sqrt{y}}$, so

$$P_{H_1}(\text{accept}) = P_{H_1}(a \le Y \le b) = (1 - e^{-\sqrt{b}}) - (1 - e^{-\sqrt{a}}) \approx 0.610.$$

Exercise 8.20

Central idea. The likelihood ratios $6, 5, 4, 3, 2, 1, 0.84$ order the outcomes; the four smallest-x outcomes carry exactly .04 of H_0 probability, so no randomization is needed.

Solution. $r(x) = f(x \mid H_1)/f(x \mid H_0)$: $6, 5, 4, 3, 2, 1, 0.79/.94 \approx 0.84$ for $x = 1, \dots, 7$. Neyman–Pearson: reject for the largest ratios. $P_{H_0}(X \in \{1, 2, 3, 4\}) = .04$ exactly, so the MP size-.04 test is

$$\phi(x) = 1 \text{ if } x \in \{1, 2, 3, 4\}, \qquad 0 \text{ otherwise,}$$

with no randomization needed (α is attainable). Type II error:

$$P_{H_1}(\text{accept}) = P_{H_1}(X \in \{5, 6, 7\}) = .02 + .01 + .79 = .82$$

(power .18).

Exercise 8.21

Central idea. The textbook proof is algebraic in ϕ, f_0, f_1 — replacing integrals by sums changes nothing, and attainability removes the boundary randomization.

Solution. Let f_0, f_1 be the pmfs under H_0, H_1 and let ϕ^* be the NP test: $\phi^*(x) = 1$ if $f_1(x) > kf_0(x)$, 0 if $f_1(x) < kf_0(x)$ (attainability means $P_0(f_1 = kf_0) = 0$ or the boundary can be assigned without randomization while keeping size α). Let ϕ be any test with $E_0\phi \le \alpha = E_0\phi^*$. Then

$$E_1\phi^* - E_1\phi = \sum_x (\phi^*(x) - \phi(x))(f_1(x) - kf_0(x)) + k(E_0\phi^* - E_0\phi).$$

In the sum, each summand is ≥ 0: where $f_1 > kf_0$, $\phi^* = 1 \ge \phi$; where $f_1 < kf_0$, $\phi^* = 0 \le \phi$; where equal, the factor vanishes. The last term is ≥ 0 since $k \ge 0$ and $E_0\phi^* - E_0\phi \ge 0$. Hence $E_1\phi^* \ge E_1\phi$: ϕ^* is most powerful. The argument is identical to the continuous case with $\sum$ in place of $\int$.

Exercise 8.22

Central idea. (a) The NP ratio $(3/2)^{10}(1/3)^S$ is decreasing in S, and $P_{1/2}(S \le 2) = 56/1024 \approx .0547$ is exactly attainable. (c) Simple vs. simple: Neyman–Pearson delivers a UMP test for every $\alpha \in [0, 1]$.

Solution. Let $S = \sum X_i \sim \text{binomial}(10, p)$. (a) $\frac{L(1/4)}{L(1/2)} = (\frac{1/4}{1/2})^S(\frac{3/4}{1/2})^{10-S} = (3/2)^{10}(1/3)^S$, strictly decreasing in S; reject iff $S \le c$. $P_{1/2}(S \le 1) = 11/1024 \approx .0107$, $P_{1/2}(S \le 2) = 56/1024 \approx .05469$: so $c = 2$ gives size exactly .0547. MP test: reject iff $S \le 2$. Power $= P_{1/4}(S \le 2) = \sum_{k=0}^{2} \binom{10}{k}(1/4)^k(3/4)^{10-k} \approx .0563 + .1877 + .2816 \approx .526$. (b) $\gamma(p) = P_p(S \ge 6) = \sum_{k=6}^{10} \binom{10}{k}p^k(1-p)^{10-k}$, strictly increasing in p (MLR). Size $= \sup_{p \le 1/2} \gamma(p) = \gamma(1/2) = 1 - \frac{638}{1024} = \frac{386}{1024} \approx .377$. The power curve rises from 0 (at $p = 0$) through .377 at $p = 1/2$ to 1 (at $p = 1$), steepest near $p = 1/2$. (c) All of them: for simple H_0 vs. simple H_1 the Neyman–Pearson test is UMP for every $\alpha \in [0, 1]$ (randomization on the boundary is used when α is not a binomial tail probability, as it is here).

Exercise 8.23

Central idea. The beta$(\theta, 1)$ family has MLR in X since $f(x \mid \theta_2)/f(x \mid \theta_1) = (\theta_2/\theta_1)x^{\theta_2 - \theta_1}$ is increasing in x; Karlin–Rubin then hands over the UMP test directly.

Solution. $f(x \mid \theta) = \theta x^{\theta-1}$, $0 < x < 1$; $P_\theta(X > t) = 1 - t^\theta$, increasing in θ. (a) Size $= \sup_{\theta \le 1} P_\theta(X > 1/2) = \sup_{\theta \le 1}(1 - (1/2)^\theta) = 1/2$ (attained at $\theta = 1$). Power: $\gamma(\theta) = 1 - (1/2)^\theta$ for $\theta > 1$, increasing and concave, from $1/2$ at $\theta = 1$ up to 1 as $\theta \to \infty$ (and $\gamma(\theta) \to 0$ as $\theta \to 0^+$). (b) $f(x \mid 2)/f(x \mid 1) = 2x$, strictly increasing in x; NP rejects iff $X > c$ with $P_1(X > c) = 1 - c = \alpha$, i.e. reject iff $X > 1 - \alpha$. (c) Yes. For $\theta_2 > \theta_1$, $f(x \mid \theta_2)/f(x \mid \theta_1) = (\theta_2/\theta_1)x^{\theta_2 - \theta_1}$ is increasing in x: MLR in X. By Karlin–Rubin, $\{X > 1 - \alpha\}$ is UMP size α for $H_0 : \theta \le 1$ vs. $H_1 : \theta > 1$ (size is $\sup_{\theta \le 1}(1 - (1 - \alpha)^\theta) = \alpha$ at $\theta = 1$).

Exercise 8.24

Central idea. With singletons the suprema vanish and the LRT is just the reciprocal likelihood ratio — the Neyman–Pearson test in disguise.

Solution. For $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$,

$$\lambda(\mathbf{x}) = \frac{L(\theta_0 \mid \mathbf{x})}{L(\theta_1 \mid \mathbf{x})} = \left(\frac{L(\theta_1 \mid \mathbf{x})}{L(\theta_0 \mid \mathbf{x})}\right)^{-1}.$$

The LRT rejects for $\lambda(\mathbf{x}) \le c$, i.e. for $L(\theta_1)/L(\theta_0) \ge 1/c =: k$ — exactly the Neyman–Pearson form (with randomization on $\{L(\theta_1) = kL(\theta_0)\}$ if needed to hit the size). Yes, equivalent: the LRT of simple vs. simple *is* the NP most powerful test.

Exercise 8.25

Central idea. In each case the log likelihood ratio is affine in the natural sufficient statistic with slope of the sign of $\theta_2 - \theta_1$.

Solution. It suffices to check the joint (or sufficient-statistic) ratio for $\theta_2 > \theta_1$. (a) For a sample, with $S = \sum X_i$: $\frac{L(\theta_2)}{L(\theta_1)} = \exp\left[\frac{\theta_2-\theta_1}{\sigma^2} S - \frac{n(\theta_2^2-\theta_1^2)}{2\sigma^2}\right]$, increasing in S (hence in $\bar{X}$). (b) With $S = \sum X_i$: $\frac{L(\theta_2)}{L(\theta_1)} = \left(\frac{\theta_2}{\theta_1}\right)^S e^{-n(\theta_2-\theta_1)}$, increasing in S since $\theta_2/\theta_1 > 1$. (c) $\frac{f(x|\theta_2)}{f(x|\theta_1)} = \left(\frac{\theta_2}{\theta_1}\right)^x \left(\frac{1-\theta_2}{1-\theta_1}\right)^{n-x}$; the log-ratio is $x \log \frac{\theta_2(1-\theta_1)}{\theta_1(1-\theta_2)} + \text{const}$, increasing in x because $\theta_2 > \theta_1$ implies $\frac{\theta_2}{1-\theta_2} > \frac{\theta_1}{1-\theta_1}$.

Exercise 8.26

Central idea. (a) An increasing likelihood ratio tilts odds $P_\theta(X > t)/P_\theta(X \le t)$ upward. (b) Stochastic order only constrains cdfs, so a likelihood ratio can wiggle non-monotonically while cdfs stay ordered — e.g. ratios $(.5, 2, 1)$ on $\{1, 2, 3\}$.

Solution. (a) Let $\theta_1 < \theta_2$ and $r(x) = f(x \mid \theta_2)/f(x \mid \theta_1)$, increasing. For any t (ignoring null-set technicalities),

$$P_{\theta_2}(X \le t) = E_{\theta_1}[r(X)\mathbf{1}_{X \le t}] \le r(t) P_{\theta_1}(X \le t),$$
$$P_{\theta_2}(X > t) = E_{\theta_1}[r(X)\mathbf{1}_{X > t}] \ge r(t) P_{\theta_1}(X > t),$$

since $r(X) \le r(t)$ on $\{X \le t\}$ and $\ge r(t)$ on $\{X > t\}$. Dividing: $\frac{P_{\theta_2}(X>t)}{P_{\theta_2}(X \le t)} \ge \frac{P_{\theta_1}(X>t)}{P_{\theta_1}(X \le t)}$, so $P_{\theta_2}(X > t) \ge P_{\theta_1}(X > t)$: $F_{\theta_2}(t) \le F_{\theta_1}(t)$ for all t, i.e. stochastically increasing. (b) On $\{1, 2, 3\}$ let $f(\cdot \mid \theta_1) = (.5, .25, .25)$ and $f(\cdot \mid \theta_2) = (.25, .5, .25)$. Then $F_{\theta_1} = (.5, .75, 1)$, $F_{\theta_2} = (.25, .75, 1)$: $F_{\theta_2} \le F_{\theta_1}$ everywhere, so the family is stochastically increasing. But $f(x \mid \theta_2)/f(x \mid \theta_1) = (.5, 2, 1)$ is not monotone in x: no MLR.

Exercise 8.27

Central idea. The ratio is $\frac{c(\theta_2)}{c(\theta_1)} \exp([w(\theta_2) - w(\theta_1)]t)$ — an exponential in t with positive slope exactly when w increases.

Solution. For $\theta_2 > \theta_1$,

$$\frac{g(t \mid \theta_2)}{g(t \mid \theta_1)} = \frac{c(\theta_2)}{c(\theta_1)} \exp([w(\theta_2) - w(\theta_1)]t),$$

and $w(\theta_2) - w(\theta_1) > 0$ since w is increasing; hence the ratio is strictly increasing in t: MLR in T. Examples: (i) $n(\theta, 1)$: $w(\theta) = \theta$; (ii) Poisson(θ) (for $T = \sum X_i$): $w(\theta) = \log\theta$; (iii) binomial(n, θ): $w(\theta) = \log \frac{\theta}{1-\theta}$ (equivalently exponential(θ) with $T = \sum X_i$: $w(\theta) = -1/\theta$).

Exercise 8.28

Central idea. The log-ratio's derivative is twice the difference of two logistic cdfs, hence positive; Karlin–Rubin then gives UMP one-sided tests $\{X > \log((1-\alpha)/\alpha)\}$, while two-sided UMP tests do not exist.

Solution. (a) For $\theta_2 > \theta_1$, put $d = \theta_2 - \theta_1 > 0$, $u = x - \theta_1$:

$$\frac{f(x \mid \theta_2)}{f(x \mid \theta_1)} = e^{-d}\left(\frac{1+e^u}{1+e^{u-d}}\right)^2, \qquad \frac{d}{du}\log(\cdot) = 2\left[\frac{e^u}{1+e^u} - \frac{e^{u-d}}{1+e^{u-d}}\right] > 0,$$

since the logistic cdf $F(u) = e^u/(1+e^u)$ is strictly increasing. So the ratio is strictly increasing in x: MLR. (b) By (a), $f(x \mid 1)/f(x \mid 0)$ is increasing in x; NP rejects iff $X > c$. The logistic(0) cdf is $F(x) = 1/(1+e^{-x})$, so $P_0(X > c) = 1 - F(c) = 1/(1+e^c) = \alpha$ gives $c = \log\frac{1-\alpha}{\alpha}$. For $\alpha = .2$: reject iff $X > \log 4 \approx 1.386$. Type II error $= P_1(X \le c) = F(c-1) = 1/(1+e^{1-c}) = 1/(1+e/4) \approx .595$. (c) MLR plus Karlin–Rubin: $\{X > \log((1-\alpha)/\alpha)\}$ is UMP size α for $H_0 : \theta \le 0$ vs. $H_1 : \theta > 0$ (size $= \sup_{\theta \le 0} P_\theta(X > c) = P_0(X > c) = \alpha$ by stochastic monotonicity). In general for the logistic location family: UMP tests exist for one-sided hypotheses (threshold tests in X), but no UMP test exists for two-sided $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \ne \theta_0$ — a UMP two-sided test would have to be simultaneously UMP against $\theta > \theta_0$ and $\theta < \theta_0$, which the one-sided forms contradict (same argument as for the normal location family).

Exercise 8.29

Central idea. The ratio $(1+x^2)/(1+(x-1)^2)$ takes the value 2 at both $x = 1$ and $x = 3$ with $\{r > 2\} = (1,3)$ — the NP interval test — but the interval slides with the alternative, so no one test is MP for every $\theta_1 > 0$.

Solution. $f(x \mid \theta) = \frac{1}{\pi(1+(x-\theta)^2)}$. (a) $\frac{f(x|1)}{f(x|0)} = \frac{1+x^2}{1+(x-1)^2}$: values $1/2, 2, 2.5, 2$ at $x = 0, 1, 2, 3$ and $\to 1$ as $|x| \to \infty$ — not monotone in x. No MLR. (b) NP rejects for $\frac{1+x^2}{1+(x-1)^2} > k$. At $k = 2$: $1 + x^2 > 2(1 + (x-1)^2) \iff -x^2 + 4x - 3 = -(x-1)(x-3) > 0 \iff 1 < x < 3$. So ϕ is the NP test with $k = 2$, hence MP of its size. With $F_\theta(x) = \frac{1}{2} + \frac{1}{\pi}\arctan(x-\theta)$: Type I $= P_0(1 < X < 3) = F_0(3) - F_0(1) = \frac{\arctan 3 - \arctan 1}{\pi} \approx .148$; Type II $= 1 - P_1(1 < X < 3) = 1 - \frac{\arctan 2 - \arctan 0}{\pi} \approx .648$. (c) Disprove. If ϕ^* were UMP size $\alpha \approx .148$ for $H_0 : \theta \le 0$ vs. $H_1 : \theta > 0$, it would be MP (a.e. unique) for $\theta = 0$ vs. each $\theta_1 > 0$. But the NP test for $\theta = 0$ vs. $\theta = 2$ rejects $\{\frac{1+x^2}{1+(x-2)^2} > k\}$ — e.g. at $k = 2$ the interval $(4 - \sqrt{7},\ 4 + \sqrt{7})$, centered at 4, not $(1,3)$ — a different test. Contradiction. In general: the Cauchy location family admits no UMP test for one-sided (hence no two-sided) hypotheses; without MLR, Karlin–Rubin is unavailable and the MP rejection regions genuinely depend on the alternative.

Exercise 8.30

Central idea. The ratio is monotone in x^2 but x^2 is U-shaped in x — folding via $|X|$ restores monotonicity on $(0,\infty)$, and symmetry makes the sign ancillary.

Solution. (a) For $\theta_2 > \theta_1$, $\frac{f(x|\theta_2)}{f(x|\theta_1)} = \frac{\theta_2}{\theta_1} \cdot \frac{\theta_1^2 + x^2}{\theta_2^2 + x^2}$, which as a function of x strictly decreases on $(-\infty, 0)$ and strictly increases on $(0,\infty)$ (it is increasing in x^2): not monotone in x. No MLR. (b) $f(x \mid \theta)$ depends on x only through $x^2 = |x|^2$, and by symmetry $P_\theta(X > 0 \mid |X| = t) = 1/2$ for all θ: the conditional distribution of X given $|X|$ is free of θ, so $|X|$ is sufficient (factorization with $h \equiv 1$ on the sign). $Y = |X|$ has pdf $f_Y(y \mid \theta) = \frac{2\theta}{\pi(\theta^2 + y^2)}$, $y > 0$, and

$$\frac{f_Y(y \mid \theta_2)}{f_Y(y \mid \theta_1)} = \frac{\theta_2}{\theta_1} \cdot \frac{\theta_1^2 + y^2}{\theta_2^2 + y^2}, \qquad \frac{d}{d(y^2)}\log(\cdot) = \frac{\theta_2^2 - \theta_1^2}{(\theta_1^2 + y^2)(\theta_2^2 + y^2)} > 0,$$

strictly increasing in y on $(0,\infty)$: the family of $|X|$ has MLR.

Exercise 8.31

Central idea. $T = \sum X_i \sim \text{Poisson}(n\lambda)$ is sufficient with MLR in T, so Karlin–Rubin gives the UMP test $\{T > c\}$. For (b), normal approximations at $\lambda = 1$ and $\lambda = 2$ give two equations in n and c.

Solution. (a) For $\lambda_2 > \lambda_1$,

$$\frac{f_T(t \mid \lambda_2)}{f_T(t \mid \lambda_1)} = e^{-n(\lambda_2-\lambda_1)} \left(\frac{\lambda_2}{\lambda_1}\right)^t,$$

strictly increasing in t: MLR in T. By Karlin–Rubin, the UMP level-α test rejects for large T: $\phi(t) = 1$ if $t > c$, 0 if $t < c$ (randomizing at $t = c$ if needed for exact size), where $P_{\lambda_0}(T > c) \le \alpha$ determines c. (b) By CLT, $T \approx n(n\lambda, n\lambda)$. Size: $c = n + z_{.95}\sqrt{n} = n + 1.645\sqrt{n}$. Power at $\lambda = 2$: $P(T > c \mid \lambda = 2) \approx 1 - \Phi\left(\frac{c-2n}{\sqrt{2n}}\right) = .9$, so $\frac{c-2n}{\sqrt{2n}} = -1.2816$. Hence $n + 1.645\sqrt{n} = 2n - 1.2816\sqrt{2n}$; with $t = \sqrt{n}$, $t = 1.645 + 1.2816\sqrt{2} \approx 3.457$, $n \approx 11.95$. Take $n = 12$ (then $c \approx 17.7$ and the power is $\approx .901$).

Exercise 8.32

Central idea. (a) MLR in $\bar{X}$ plus Karlin–Rubin. (b) A UMP test would have to be simultaneously most powerful against $\theta_1 > \theta_0$ and $\theta_2 < \theta_0$, but the Neyman–Pearson tests for these point in opposite directions — impossible since $\bar{X}$ has full support.

Solution. (a) The $n(\theta, 1)$ family has MLR in $\bar{X}$. By Karlin–Rubin the UMP level-α test of $\theta \ge \theta_0$ versus $\theta < \theta_0$ rejects for small $\bar{X}$: $\bar{X} < \theta_0 - z_\alpha/\sqrt{n}$ (size exactly α since $\bar{X}$ is continuous). (b) Suppose ϕ^* were UMP level α. Fix $\theta_1 > \theta_0$ and $\theta_2 < \theta_0$; then ϕ^* is MP level α for each point alternative. The likelihood ratio $f(\mathbf{x} \mid \theta_1)/f(\mathbf{x} \mid \theta_0)$ is strictly increasing in $\bar{X}$, so by Neyman–Pearson (a.e. uniqueness, and $P(\bar{X} = c) = 0$) $\phi^* = \mathbf{1}_{\{\bar{X} > c_1\}}$ a.s.; likewise $\phi^* = \mathbf{1}_{\{\bar{X} < c_2\}}$ a.s. Hence $\mathbf{1}_{\{\bar{X} > c_1\}} = \mathbf{1}_{\{\bar{X} < c_2\}}$ a.s. But under θ_0, $\bar{X} \sim n(\theta_0, 1/n)$ has support all of $\mathbb{R}$: if $c_1 < c_2$, they differ on $\{\bar{X} > c_2\}$ (positive probability); if $c_1 \ge c_2$, they differ on $\{\bar{X} < c_2\}$ (positive probability). Contradiction. Hence no UMP test exists.

Exercise 8.33

Central idea. Under H_0, $Y_n < 1$ a.s., so size comes entirely from Y_1. The likelihood ratio takes only the values 0, 1, ∞, and the test's power matches the Neyman–Pearson power at every $\theta_1 > 0$ (for $\theta_1 > k$ it is 1, trivially optimal).

Solution. (a) Under $\theta = 0$, $X_i \in (0, 1)$ so $Y_n < 1$ a.s. and $P_0(\text{reject}) = P_0(Y_1 \ge k) = (1-k)^n$. Setting $(1-k)^n = \alpha$ gives $k = 1 - \alpha^{1/n}$. (b) For $\theta \ge 1$, $Y_n \ge \theta \ge 1$ a.s., so $\gamma(\theta) = 1$. For $0 < \theta < 1$: $P_\theta(Y_n \ge 1) = 1 - (1-\theta)^n$. If $0 < \theta \le k$, then $\{Y_n < 1, Y_1 \ge k\} = \{k \le X_i < 1 \ \forall i\}$ has probability $(1-k)^n = \alpha$; if $\theta > k$, $Y_1 \ge \theta > k$ a.s. Hence, with γ the power function,

$$\gamma(\theta) = \begin{cases} 1 - (1-\theta)^n + \alpha, & 0 < \theta \le k, \\ 1, & \theta > k, \end{cases} \qquad \gamma(0) = \alpha,$$

continuous at $\theta = k$. (c) Fix $\theta_1 > 0$. Now $f(\mathbf{x} \mid \theta_1)/f(\mathbf{x} \mid 0)$ equals ∞ on $\{Y_1 > \theta_1, 1 \le Y_n < \theta_1 + 1\}$, 1 on $\{Y_1 > \theta_1, Y_n < 1\}$, and 0 elsewhere. By Neyman–Pearson the MP level-α test rejects $\{r = \infty\}$, accepts $\{r = 0\}$, randomizing on $\{r = 1\}$. If $0 < \theta_1 \le k$: $P_0(r = 1) = P_0(Y_1 > \theta_1) = (1-\theta_1)^n \ge \alpha$, and since $Y_1 > \theta_1$ a.s. under θ_1, the NP power is $P_{\theta_1}(Y_n \ge 1) + \alpha = 1 - (1-\theta_1)^n + \alpha = \gamma(\theta_1)$. If $\theta_1 > k$: our test has power $\gamma(\theta_1) = 1$, the maximum possible. Since our test has $E_0 = \alpha$ and any level-α test has $E_0 \le \alpha$, it is MP against every $\theta_1 > 0$: UMP. (d) For $\theta > 1 > k$, $\gamma(\theta) = 1 \ge .8$ for every n. E.g. $n = 1$, $k = 1 - .1 = .9$ gives a UMP level-.10 test with power 1 for all $\theta > 1$.

Exercise 8.34

Central idea. (a) $P_\theta(T > c) = 1 - F_0(c - \theta)$ shifts monotonically. (b) With $r(t) = g(t \mid \theta_2)/g(t \mid \theta_1)$ nondecreasing, compare $r(t)$ to $r(c)$ on $\{T > c\}$ and $\{T \le c\}$; one of the two comparisons always goes the right way.

Solution. (a) If θ is a location parameter, $T - \theta$ has a distribution F_0 free of θ, so $P_\theta(T > c) = 1 - F_0(c - \theta)$, nondecreasing in θ. (b) Let $\theta_1 < \theta_2$ (equality is trivial) and $r(t) = g(t \mid \theta_2)/g(t \mid \theta_1)$, nondecreasing by MLR. Then $P_{\theta_2}(T > c) = \int_c^\infty r(t) g(t \mid \theta_1)\, dt \ge r(c) P_{\theta_1}(T > c)$ and $P_{\theta_2}(T \le c) = \int_{-\infty}^c r(t) g(t \mid \theta_1)\, dt \le r(c) P_{\theta_1}(T \le$

c). If $r(c) \geq 1$ the first inequality gives the result; if $r(c) < 1$ the second gives $P_{\theta_2}(T \leq c) < P_{\theta_1}(T \leq c)$, i.e. the same conclusion.

Exercise 8.35

Central idea. (a) $E[T'] = E[X]\,E[(Y/\nu)^{-1/2}]$ by independence; the χ^2 negative moment is a gamma integral. (c) Write the pdf as a normal variance-mixture and apply Karlin's composition theorem: the normal location kernel is TP2, and the mixing kernel is TP2 because $\log w(e^x)$ is concave.

Solution. (a) $E[T'] = E[X]\,E[(Y/\nu)^{-1/2}] = \mu\sqrt{\nu}\,E[Y^{-1/2}]$. For $Y \sim \chi^2_\nu$, $E[Y^{-1/2}] = \frac{\Gamma((\nu-1)/2)}{\sqrt{2}\,\Gamma(\nu/2)}$ $(\nu > 1)$. Hence (with $\delta^2 = \mu^2$)

$$E[T'] = \mu\sqrt{\frac{\nu}{2}}\,\frac{\Gamma((\nu-1)/2)}{\Gamma(\nu/2)} \quad (\nu > 1),$$

which is 0 if $\mu = 0$. Also $E[T'^2] = E[X^2]E[\nu/Y] = (1+\mu^2)\frac{\nu}{\nu-2}$ $(\nu > 2)$, so

$$\mathrm{Var}(T') = \frac{\nu(1+\delta^2)}{\nu-2} - \delta^2\frac{\nu}{2}\left[\frac{\Gamma((\nu-1)/2)}{\Gamma(\nu/2)}\right]^2 \quad (\nu > 2).$$

(b) If $\delta = 0$, $(\delta t)^k$ is 1 for $k = 0$ and 0 for $k \geq 1$, so only $k = 0$ survives: $f_{T'}(t \mid 0) = \frac{\Gamma((\nu+1)/2)}{\Gamma(1/2)\Gamma(\nu/2)\sqrt{\nu}(1+t^2/\nu)^{(\nu+1)/2}}$, the central t_ν pdf. (c) Use the signed noncentrality $\delta = \mu$. If restricting to $\mu \geq 0$, this agrees with $\sqrt{\mu^2}$; for negative means, the sign cannot be discarded since the distribution is reflected. Conditional on $Y = y$, $T' \sim n(\delta/\sqrt{y/\nu}, 1/(y/\nu))$, so with $u = \sqrt{y/\nu}$,

$$f(t \mid \delta) = \int_0^\infty \phi(tu - \delta)\,w(u)\,du, \quad w(u) = 2\nu u^2 f_{\chi^2_\nu}(\nu u^2) = \frac{2(\nu/2)^{\nu/2}}{\Gamma(\nu/2)}u^\nu e^{-\nu u^2/2}.$$

For $t > 0$, $s = tu$ gives $f(t \mid \delta) = \int_0^\infty \phi(s-\delta)K_+(t,s)\,ds$ with $K_+(t,s) = \frac{1}{t}w(s/t)\mathbf{1}_{s>0}$. Now $\phi(s-\delta)$ is TP2 in (s,δ) (normal location MLR), and K_+ is TP2 in (t,s) on $(0,\infty)^2$: for $t_2 > t_1 > 0$, $s_2 > s_1 > 0$,

$$\frac{K_+(t_2,s_2)K_+(t_1,s_1)}{K_+(t_2,s_1)K_+(t_1,s_2)} = \frac{w(a)w(b)}{w(c)w(d)} \geq 1,$$

where $a = s_2/t_2$, $b = s_1/t_1$, $c = s_1/t_2$, $d = s_2/t_1$ satisfy $ab = cd$, $c < \min(a,b)$, $d > \max(a,b)$; with $h(x) = \log w(e^x) = C + \nu x - \frac{\nu}{2}e^{2x}$ and $h''(x) = -2\nu e^{2x} < 0$, concavity of h gives $h(\log a) + h(\log b) \geq h(\log c) + h(\log d)$. By Karlin's basic composition formula, f is TP2 in (t,δ) for $t > 0$. For $t < 0$, $s = tu$ gives $f(t \mid \delta) = \int_{-\infty}^0 \phi(s-\delta)K_-(t,s)\,ds$, $K_-(t,s) = \frac{1}{|t|}w(s/t)\mathbf{1}_{s<0}$, TP2 on $(-\infty,0)^2$ by the same concavity (now with $c > \max(a,b)$, $d < \min(a,b)$). Hence $R(t) = f(t \mid \delta_2)/f(t \mid \delta_1)$ $(\delta_2 > \delta_1)$ is increasing on each half-line; R is continuous, so for $t_1 < 0 < t_2$, $R(t_1) \leq R(0) \leq R(t_2)$. Thus MLR in δ on $\mathbb{R}$.

Exercise 8.36

Central idea. Choose a two-tail test matching both endpoint rejection probabilities; convexity proves that it uniformly improves the original test.

Solution. (a) Define

$$\psi(x) = 1\{x < 0.05 \text{ or } x > 0.55\}.$$

Under $\theta = 1$, its rejection probability is $0.05 + 0.45 = 0.5$. Under $\theta = 2$, it is $1 - 0.95^2 + 0.45^2 = 0.3$. More generally $E_\theta\psi = 1 - 0.95^\theta + 0.45^\theta$. To prove dominance, put $u = 1 - x$, $h(u) = \psi(1-u) - \phi(1-u)$. The supplied endpoint probabilities imply $\int_0^1 h(u)\,du = \int_0^1 uh(u)\,du = 0$. Moreover $h \geq 0$ outside $(0.45, 0.95)$ and $h \leq 0$ inside it, since $0 \leq \phi \leq 1$. Let $L_\theta(u)$ be the affine line joining $u^{\theta-1}$ at $u = 0.45$ and $u = 0.95$. For $\theta < 1$ or $\theta > 2$, $u^{\theta-1}$ is convex, so its difference from this chord is nonnegative outside the two cutpoints and nonpositive between them. Hence

$$E_\theta(\psi - \phi) = \theta\int_0^1 h(u)[u^{\theta-1} - L_\theta(u)]\,du \geq 0.$$

For $1 < \theta < 2$, the function is concave, so the inequality reverses: $E_\theta\psi \le E_\theta\phi$. At both endpoints equality holds. Thus the new test has no greater rejection probability at any null parameter and no smaller power at any alternative parameter; it uniformly improves risk under 0–1 testing loss. If ϕ differs from ψ on a set of positive measure, the inequalities are strict away from the endpoints. Both supplied numbers are essential. This resolves the earlier open doubt without imposing unbiasedness or a minimax interpretation.

(b) Since $f(x \mid 1) \equiv 1$ on $(0,1)$, $\gamma(\theta) = \int_0^1 \phi(x)\theta(1-x)^{\theta-1}dx$, and differentiating under the integral gives $\gamma'(1) = \int_0^1 \phi(x)[1+\ln(1-x)]dx$. We need $\int_0^1 \phi = .1$ and $\int_0^1 \phi(x)\ln(1-x)dx = -.1$. Take a two-sided nonrandomized test $\phi = \mathbf{1}_{[0,u]} + \mathbf{1}_{[1-s,1]}$ with $u+s = .1$. Then $\int \phi \ln(1-x)dx = -(1-u)\ln(1-u) - u + s\ln s - s$; with $u = .1 - s$ this equals $-.1$ iff $s\ln s = (.9+s)\ln(.9+s)$, solved by $s \approx .0196$, $u \approx .0804$. So $\phi(x) = \mathbf{1}_{[0,.0804]}(x) + \mathbf{1}_{[.9804,1]}(x)$ is two-sided, satisfies $E_1\phi = .1$, and $\gamma'(1) = .1 - .1 = 0$.

Exercise 8.37

Central idea. The (profile) likelihood ratio is 1 for $\bar X \le \theta_0$ and strictly decreasing in $\bar X$ for $\bar X > \theta_0$; MLR in $\bar X$ plus Karlin–Rubin gives UMP. With σ^2 unknown the same computation yields the one-sided t-test.

Solution. (a) $P_\theta(\bar X > \theta_0 + z_\alpha\sigma/\sqrt{n}) = 1 - \Phi(z_\alpha - \sqrt{n}(\theta-\theta_0)/\sigma) \le \alpha$ for $\theta \le \theta_0$, equality at θ_0: size α. LRT: $\lambda(\mathbf{x}) = \sup_{\theta\le\theta_0} L(\theta)/\sup_\theta L(\theta)$; L is maximized at $\bar X$ unconstrained and at $\min(\bar X, \theta_0)$ under H_0, so $\lambda = 1$ if $\bar X \le \theta_0$ and $\lambda = \exp(-n(\bar X - \theta_0)^2/2\sigma^2)$ if $\bar X > \theta_0$. Thus $\{\lambda < c\} = \{\bar X > \theta_0 + c'\}$: the LRT. (b) $n(\theta, \sigma^2)$ has MLR in $\bar X$; Karlin–Rubin says $\{\bar X > c\}$ is UMP level α (with $c = \theta_0 + z_\alpha\sigma/\sqrt{n}$). (c) At $\theta = \theta_0$, $\sqrt{n}(\bar X - \theta_0)/S \sim t_{n-1}$; for $\theta < \theta_0$ the rejection probability is $P(t_{n-1}(\delta) > t_{n-1,\alpha}) < \alpha$ ($\delta < 0$), so size α. LRT: maximizing first over σ^2 ($\hat\sigma^2_\theta = \frac{1}{n}\sum(X_i-\theta)^2$) then over $\theta \le \theta_0$ gives numerator $\frac{1}{n}\sum(X_i-\bar X)^2$ if $\bar X \le \theta_0$, $\frac{1}{n}\sum(X_i-\theta_0)^2$ if $\bar X > \theta_0$; denominator $\frac{1}{n}\sum(X_i-\bar X)^2$. Hence $\lambda = 1$ if $\bar X \le \theta_0$ and $\lambda = \left(\frac{\sum(X_i-\bar X)^2}{\sum(X_i-\theta_0)^2}\right)^{n/2}$ if $\bar X > \theta_0$. Since $\frac{\sum(X_i-\theta_0)^2}{\sum(X_i-\bar X)^2} = 1 + \frac{n(\bar X-\theta_0)^2}{(n-1)S^2}$ is strictly increasing in $\bar X$ for $\bar X > \theta_0$, $\{\lambda < c\} = \{\sqrt{n}(\bar X - \theta_0)/S > c'\}$: the one-sided t-test.

Exercise 8.38

Central idea. Under H_0, $\sqrt{n}(\bar X - \theta_0)/S \sim t_{n-1}$; the LRT reduces to a monotone function of $|\bar X - \theta_0|/S$.

Solution. (a) Under θ_0, $T = \sqrt{n}(\bar X - \theta_0)/S \sim t_{n-1}$, so $P_{\theta_0}(|T| > t_{n-1,\alpha/2}) = \alpha$: size α (point null). (b) $\lambda(\mathbf{x}) = \sup_{\sigma^2} L(\theta_0,\sigma^2)/\sup_{\theta,\sigma^2} L(\theta,\sigma^2) = \left(\frac{\sum(X_i-\bar X)^2}{\sum(X_i-\theta_0)^2}\right)^{n/2}$, since $\hat\sigma^2 = \frac{1}{n}\sum(X_i-\theta)^2$ in each case. Then $\{\lambda < c\} = \left\{\frac{\sum(X_i-\theta_0)^2}{\sum(X_i-\bar X)^2} > c'\right\} = \left\{\frac{n(\bar X-\theta_0)^2}{(n-1)S^2} > c''\right\} = \{|T| > c'''\}$: the two-sided t-test (with $c''' = t_{n-1,\alpha/2}$ for size α).

Exercise 8.39

Central idea. Differencing reduces the bivariate problem to a one-sample t problem for the W_i.

Solution. (a) Each (X_i, Y_i) is bivariate normal, so $W_i = X_i - Y_i$ is normal with $E[W_i] = \mu_X - \mu_Y =: \mu_W$ and $\mathrm{Var}(W_i) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y =: \sigma_W^2$; the pairs being iid, the W_i are iid. (b) $H_0 \iff \mu_W = 0$. From (a), $\bar W$ and S_W^2 are the usual one-sample mean and variance of iid normals, independent, so under $\mu_W = 0$, $T_W = \bar W/\sqrt{S_W^2/n} \sim t_{n-1}$: the paired-sample t test (reject for $|T_W| > t_{n-1,\alpha/2}$ at level α).

Exercise 8.40

Central idea. In (u, v) coordinates the likelihood factors into a conditional piece $g(v \mid u)$ whose maximum doesn't involve $\mu_X - \mu_Y$ (so it cancels) and a marginal piece $h(u)$ giving exactly the one-sample t LRT on the differences.

Solution. (a) Put $U = X - Y$, $V = X + Y$ (constant Jacobian). Then $f_{X,Y}(x,y) = g(v \mid u)h(u)$ with h the $n(\mu_X - \mu_Y, \sigma_W^2)$ marginal of U and g the normal conditional of $V \mid U$. The full likelihood is

$\prod_i g(v_i \mid u_i)\prod_i h(u_i)$. The parameters of g (conditional regression coefficients and variance) are unrestricted both under H_0 ($E[U]=0$) and under H_1, so $\sup \prod g$ is identical in numerator and denominator and cancels. Hence $\lambda = \sup_{E[U]=0,\sigma_W^2}\prod h(u_i) / \sup_{E[U],\sigma_W^2}\prod h(u_i)$, the ordinary one-sample normal LRT for $H_0 : E[U]=0$ versus $H_1 : E[U]\neq 0$ based on $U_i = X_i - Y_i$. (b) By 8.38(b), that LRT rejects for large $|T_W| = |\bar W|/\sqrt{S_W^2/n}$ — precisely the (two-sided) paired-sample t test of 8.39.

Exercise 8.41

Central idea. With common σ^2 the profile MLEs give λ as a decreasing function of $|T|$; under H_0, $\bar X - \bar Y$ is normal independent of the pooled χ^2.

Solution. (a) $\sup_{H_0} L$ uses $\hat\mu = \frac{n\bar X + m\bar Y}{n+m}$ (pooled mean) and $\hat\sigma_0^2 = \frac{1}{n+m}(\sum(X_i-\hat\mu)^2 + \sum(Y_i-\hat\mu)^2)$; $\sup L$ uses $\hat\mu_X = \bar X$, $\hat\mu_Y = \bar Y$, $\hat\sigma^2 = \frac{1}{n+m}(\sum(X_i-\bar X)^2 + \sum(Y_i-\bar Y)^2)$. So $\lambda = (\hat\sigma^2/\hat\sigma_0^2)^{(n+m)/2}$, and $\hat\sigma_0^2 = \hat\sigma^2 + \frac{nm}{(n+m)^2}(\bar X - \bar Y)^2$ (between-group sum of squares). Thus $\{\lambda < c\} = \left\{\frac{(\bar X-\bar Y)^2}{S_p^2(1/n+1/m)} > c'\right\} = \{|T| > c''\}$: reject for large $|T|$. (b) Under H_0, $\bar X - \bar Y \sim n(0, \sigma^2(1/n+1/m))$ independent of $(n+m-2)S_p^2/\sigma^2 \sim \chi^2_{n+m-2}$ (sums of independent χ^2's, independent of the means). Hence $T \sim t_{n+m-2}$; reject $|T| > t_{n+m-2,\alpha/2}$ for level α. (c) Core: $n=14$, $\bar x = 1249.86$, $s_X^2 = 591.4$; periphery: $m=9$, $\bar y = 1261.33$, $s_Y^2 = 176.0$. $S_p^2 = \frac{13(591.4)+8(176.0)}{21} = 433.1$. $T = \frac{1249.86-1261.33}{\sqrt{433.1\,(1/14+1/9)}} = -1.291$ on 21 df; two-sided $p = 2P(t_{21} > 1.291) \approx .211$. Fail to reject: no significant evidence the mean ages differ.

Exercise 8.42

Central idea. Satterthwaite: match the first two moments of $S_X^2/n + S_Y^2/m$ to $c\chi^2_\nu$; then T' is (normal numerator) over (root of an independent approximate-χ^2/df).

Solution. (a) $S_X^2/n + S_Y^2/m = \frac{\sigma_X^2}{n}\frac{\chi^2_{n-1}}{n-1} + \frac{\sigma_Y^2}{m}\frac{\chi^2_{m-1}}{m-1}$ (independent). Approximating by $c\chi^2_\nu$ and matching mean and variance: $c\nu = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}$, $2c^2\nu = \frac{2\sigma_X^4}{n^2(n-1)} + \frac{2\sigma_Y^4}{m^2(m-1)}$, giving $\nu = \frac{(\sigma_X^2/n+\sigma_Y^2/m)^2}{\sigma_X^4/(n^2(n-1))+\sigma_Y^4/(m^2(m-1))}$, $c = (\sigma_X^2/n + \sigma_Y^2/m)/\nu$. Dividing by $\sigma_X^2/n+\sigma_Y^2/m$ gives the χ^2_ν/ν approximation; substituting S_X^2, S_Y^2 for σ_X^2, σ_Y^2 gives $\hat\nu$. (b) Under H_0, $\bar X - \bar Y \sim n(0, \sigma_X^2/n + \sigma_Y^2/m)$ independent of (S_X^2, S_Y^2), so $T' = \frac{n(0,1)}{\sqrt{(S_X^2/n+S_Y^2/m)/(\sigma_X^2/n+\sigma_Y^2/m)}} \approx \frac{n(0,1)}{\sqrt{\chi^2_{\hat\nu}/\hat\nu}} \sim t_{\hat\nu}$. (c) $T' = \frac{-11.48}{\sqrt{591.4/14+176.0/9}} = \frac{-11.48}{\sqrt{61.80}} = -1.46$, $\hat\nu = \frac{(61.80)^2}{591.4^2/(14^2\cdot 13)+176.0^2/(9^2\cdot 8)} \approx 20.6$; two-sided $p \approx .159$. Same conclusion: fail to reject. (d) $F = S_X^2/S_Y^2 = 591.4/176.0 = 3.36$ on $(13, 8)$ df; the Example 5.4.1 two-sided $p \approx .092$. Not significant at .05: no strong evidence the variances differ (supporting the pooled test in 8.41).

Exercise 8.43

Central idea. *As printed this problem contains errata: with the usual unbiased s_X^2, s_Y^2 (as in 8.42), the factors $n_1(n_1-1)$ make the pooled quantity $n_1\chi^2_{n_1-1}+n_2\chi^2_{n_2-1}$ rather than $\chi^2_{n_1+n_2-2}$, so the displayed statistic is not Student t; the F statement and the maximizer are likewise incorrect as stated. The corrected versions are proved below.*

Solution. Let s_X^2, s_Y^2 be the usual unbiased sample variances. Then $\frac{(\bar X-\bar Y)-(\mu_1-\mu_2)}{\sigma\sqrt{1/n_1+\rho^2/n_2}} \sim n(0,1)$, independent of $V := \frac{(n_1-1)s_X^2+(n_2-1)s_Y^2/\rho^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$ (since $(n_1-1)s_X^2/\sigma^2 \sim \chi^2_{n_1-1}$ and $(n_2-1)s_Y^2/(\rho^2\sigma^2) \sim \chi^2_{n_2-1}$, independently). Hence the (corrected) statistic

$$T := \frac{(\bar X-\bar Y)-(\mu_1-\mu_2)}{\sqrt{\frac{1}{n_1}+\frac{\rho^2}{n_2}}\sqrt{\frac{(n_1-1)s_X^2+(n_2-1)s_Y^2/\rho^2}{n_1+n_2-2}}} \sim t_{n_1+n_2-2}.$$

(As printed, with $n_1(n_1-1)s_X^2$, the bracket over σ^2 equals $n_1\chi^2_{n_1-1}+n_2\chi^2_{n_2-1}$, not $\chi^2_{n_1+n_2-2}$, so the ratio is not Student t.) Moreover $\frac{s_Y^2/\rho^2}{s_X^2} = \frac{\chi^2_{n_2-1}/(n_2-1)}{\chi^2_{n_1-1}/(n_1-1)} \sim F_{n_2-1,n_1-1}$; as printed, $\frac{n_2 s_Y^2}{\rho^2 n_1 s_X^2} = \frac{n_2}{n_1}F_{n_2-1,n_1-1}$, not F_{n_1-1,n_2-1}.

Finally, for the corrected T, $|T|$ is maximized over $\rho^2 > 0$ by minimizing $(\frac{1}{n_1}+\frac{\rho^2}{n_2})((n_1-1)s_X^2+(n_2-1)s_Y^2/\rho^2)$; differentiating gives $(\rho^2)^2 = \frac{n_2(n_2-1)s_Y^2}{n_1(n_1-1)s_X^2}$, i.e. $\rho^2 = \sqrt{\frac{n_2(n_2-1)}{n_1(n_1-1)}}\,\frac{s_Y}{s_X}$ (For the originally printed denominator with the extra n_i factors, the optimizing variance ratio is instead $\frac{n_2\sqrt{n_2-1}s_Y}{n_1\sqrt{n_1-1}s_X}$. The printed proposal both reverses the dependence and uses variances where standard deviations are needed. Maximizing T itself requires attention to the sign of its numerator; the denominator minimization maximizes $|T|$.)

Exercise 8.44

Central idea. Test 3 is the two-sided z-test $|\bar{X}-\theta_0| > \sigma z_{\alpha/2}/\sqrt{n}$ (for $n(\theta,\sigma^2)$ with σ known); its power $\beta_3(\theta) = P(|Z+\delta| > z_{\alpha/2})$ is minimized at $\delta = 0$ by symmetry and the unimodality of ϕ.

Solution. Example 8.3.20's Test 3 rejects $H_0 : \theta = \theta_0$ iff $\bar{X} > \theta_0 + \sigma z_{\alpha/2}/\sqrt{n}$ or $\bar{X} < \theta_0 - \sigma z_{\alpha/2}/\sqrt{n}$. Under θ_0, $P(\text{reject}) = P(|Z| > z_{\alpha/2}) = \alpha$: level α (point null). The power is $\beta_3(\theta) = P_\theta(|Z+\delta| > z_{\alpha/2})$ with $\delta = \sqrt{n}(\theta-\theta_0)/\sigma$, $Z \sim n(0,1)$. β_3 is even in δ; for $\delta > 0$,

$$\frac{d}{d\delta}\beta_3 = \phi(z_{\alpha/2}-\delta) - \phi(z_{\alpha/2}+\delta) > 0,$$

since $(z_{\alpha/2}+\delta)^2 - (z_{\alpha/2}-\delta)^2 = 4z_{\alpha/2}\delta > 0$ and ϕ decreases in $|x|$. Hence β_3 is minimized at $\delta = 0$ with value α: $\beta_3(\theta) \geq \alpha$ for all $\theta \neq \theta_0$. Unbiased.

Exercise 8.45

Central idea. Each m-test is just the one-sided z-test on m observations; its power $1-\Phi(z_\alpha - \sqrt{m}(\theta-\theta_0)/\sigma)$ is increasing in both θ and m.

Solution. For fixed m, $\bar{X}_m \sim n(\theta,\sigma^2/m)$, so $P_\theta(\text{reject}) = 1-\Phi\left(z_\alpha - \frac{\sqrt{m}(\theta-\theta_0)}{\sigma}\right)$, increasing in θ and equal to α at $\theta = \theta_0$: size α. For $\theta > \theta_0$ it exceeds α: unbiased. The power curves ($\alpha = .05$, $\sigma = 1$, versus $\theta-\theta_0$) rise more steeply for larger m: the full-sample ($m = 4$) test is uniformly most powerful among the four, the $m = 1$ test weakest — all are unbiased size-α tests.

Exercise 8.46

Central idea. (a) Each one-sided piece has probability $< \alpha/2$ except at its own endpoint, so the supremum over $[\theta_1,\theta_2]$ is $< \alpha$. (b) $|\bar{X}-\bar{\theta}|/\sqrt{S^2/n} = |t_{n-1}(\delta)|$ is stochastically increasing in $|\delta|$, so the null rejection probability is maximized at the endpoints. (c) Conditioning on S reduces everything to normal CDFs; a calculus lemma gives monotonicity.

Solution. Write $c = t_{n-1,\alpha/2}$, $SE = \sqrt{S^2/n}$. (a) $P_\theta(\bar{X} > \theta_2 + c\cdot SE)$ is increasing in θ (conditional on S it is $1-\Phi(\cdot)$, increasing in θ), equal to $\alpha/2$ at $\theta = \theta_2$; similarly $P_\theta(\bar{X} < \theta_1 - c\cdot SE) \leq \alpha/2$ with equality only at θ_1. Hence for every $\theta \in [\theta_1,\theta_2]$, $P_\theta(\text{reject}) < \alpha/2+\alpha/2 = \alpha$ (at θ_2 the second term is $< \alpha/2$ by stochastic ordering; likewise at θ_1). So $\sup_{H_0} P_\theta < \alpha$: not size α (it is conservative). (b) $P_\theta(|\bar{X}-\bar{\theta}| > k\cdot SE) = P(|t_{n-1}(\delta)| > k)$ with $\delta = \sqrt{n}(\theta-\bar{\theta})/\sigma$. By the MLR, $|t_{n-1}(\delta)|$ is stochastically increasing in $|\delta|$, so on $[\theta_1,\theta_2]$ this is maximized at the endpoints $|\delta| = \sqrt{n}(\theta_2-\theta_1)/(2\sigma)$. Since $P(|t_{n-1}(\delta_0)| > k)$ decreases continuously from 1 to 0 in k, there is a unique k with $P(|t_{n-1}(\sqrt{n}(\theta_2-\theta_1)/(2\sigma))| > k) = \alpha$: size α. (c) For (b): if $\theta > \theta_2$, then $|\delta| > \sqrt{n}(\theta_2-\theta_1)/(2\sigma)$, so $P_\theta = P(|t_{n-1}(\delta)| > k) \geq \alpha$ (MLR); similarly for $\theta < \theta_1$. Unbiased. For (a): let $s_a = \sup_{H_0} P_\theta < \alpha$. Conditional on $S = s\sigma$, $P_\theta(\text{reject}) = E_s[h(a)]$ where $a = \sqrt{n}(\theta_2-\theta)/\sigma$, $k_0 = \sqrt{n}(\theta_2-\theta_1)/\sigma$, $h(a) = 1-\Phi(a/s+c)+\Phi((a-k_0)/s-c)$. Now $h'(a) = \frac{1}{s}[\phi((a-k_0)/s-c)-\phi(a/s+c)]$, and $|(a-k_0)/s-c|^2 - |a/s+c|^2 = (k_0/s+2c)(k_0-2a)/s$, so $h'(a) < 0$ for $a < k_0/2$. For $\theta > \theta_2$, $a < 0 < k_0/2$, so h is strictly decreasing in a, i.e. P_θ strictly increasing in θ; thus $P_\theta > P_{\theta_2}$ for $\theta > \theta_2$. On $(\bar{\theta},\theta_2)$, $a \in (0,k_0/2)$, $h'(a) < 0$, $da/d\theta < 0$, so P_θ is strictly increasing there; by symmetry about $\bar{\theta}$, $\max_{[\theta_1,\theta_2]} P_\theta = P_{\theta_1} = P_{\theta_2} = s_a$. Hence $P_\theta > s_a$ for $\theta > \theta_2$, and symmetrically for $\theta < \theta_1$: unbiased of size s_a.

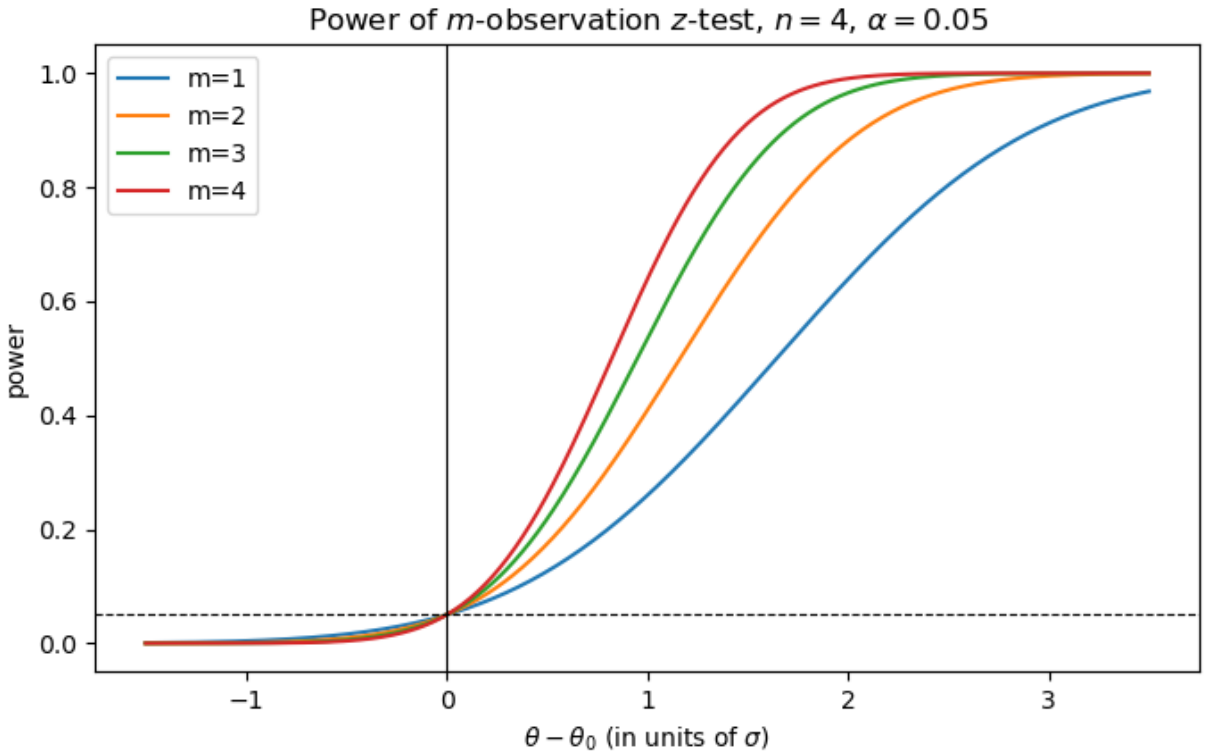


Figure 5: Power functions of the m-observation z-tests, $n = 4$, $\alpha = 0.05$.

Exercise 8.47

Central idea. Each one-sided problem is a two-sample t-test (as in 8.37(c)); the intersection–union test (TOST) rejects $H_0 = H_0^- \cup H_0^+$ iff both one-sided tests reject, hence level α, with the bound sharp as $\sigma \to 0$.

Solution. (a) With $\Delta = \mu_X - \mu_Y$, this is $H_0^- : \Delta \le -\delta$ vs $H_1^- : \Delta > -\delta$. Exactly as in 8.37(c) (two-sample version), the LRT rejects for large $\frac{\bar{X}-\bar{Y}+\delta}{\sqrt{S_p^2(1/n+1/m)}}$; at $\Delta = -\delta$ this is t_{n+m-2}, and for $\Delta < -\delta$ the rejection probability is $< \alpha$ (noncentral t with negative noncentrality): size α LRT. (b) By symmetry, the size α LRT rejects H_0^+ iff $T^+ = \frac{\bar{X}-\bar{Y}-\delta}{\sqrt{S_p^2(1/n+1/m)}} \le -t_{n+m-2,\alpha}$. (c) $H_0 = H_0^- \cup H_0^+$: use the IUT — reject H_0 iff both H_0^- and H_0^+ are rejected, i.e. iff $T^- \ge t_{n+m-2,\alpha}$ and $T^+ \le -t_{n+m-2,\alpha}$. For $\Delta \in H_0^-$, $P(\text{reject } H_0) \le P(\text{reject } H_0^-) \le \alpha$; similarly on H_0^+: level α. (The two one-sided tests procedure, TOST.) (d) At $\Delta = -\delta$, $T^- \sim t_{n+m-2}$ so $P(T^- \ge t_\alpha) = \alpha$, while $T^+ = T^- - \frac{2\delta}{\sqrt{S_p^2(1/n+1/m)}} \to -\infty$ a.s. as $\sigma \to 0$, so $P(T^+ \le -t_\alpha) \to 1$. Hence $P(\text{reject } H_0) = P(T^- \ge t_\alpha, T^+ \le -t_\alpha) \to \alpha$ as $\sigma \to 0$; the level-α bound is best possible, i.e. the test has size α.

Exercise 8.48

Central idea. Under $H_0 : p_1 = p_2 = p$, (S_1, S_2) has a joint binomial pmf in which the p-terms factor through $S = S_1 + S_2$ and cancel in the conditional distribution.

Solution. In Example 8.3.30, $S_1 \sim \text{binomial}(n_1, p_1)$, $S_2 \sim \text{binomial}(n_2, p_2)$ independent; under H_0, $p_1 = p_2 = p$. With $S = S_1 + S_2$,

$$P(S_1 = s_1 \mid S = s) = \frac{P(S_1 = s_1, S_2 = s - s_1)}{P(S = s)} = \frac{\binom{n_1}{s_1}\binom{n_2}{s-s_1}p^s(1-p)^{n_1+n_2-s}}{\binom{n_1+n_2}{s}p^s(1-p)^{n_1+n_2-s}} = \frac{\binom{n_1}{s_1}\binom{n_2}{s-s_1}}{\binom{n_1+n_2}{s}},$$

the hypergeometric pmf (population n_1+n_2 with n_1 "successes", s draws) — free of the nuisance parameter p.

Exercise 8.49

Central idea. Each p-value is $\sup_{H_0} P_\theta(\text{statistic} \geq \text{observed})$, attained at the boundary by MLR/stochastic ordering.

Solution. (a) $p = P_{\theta=1/2}(X \geq 7) = 2^{-10}\sum_{k=7}^{10}\binom{10}{k} = 176/1024 \approx .172$. (b) $p = P_{\lambda=1}(X \geq 3) = 1 - e^{-1}(1+1+1/2) = 1 - 2.5e^{-1} \approx .0803$. (c) $T = \sum X_i = 9 \sim \text{Poisson}(3\lambda)$; $p = P_{\lambda=1}(T \geq 9) = 1 - e^{-3}\sum_{k=0}^{8} 3^k/k! \approx .0038$.

Exercise 8.50

Central idea. The posterior is proportional to the normal likelihood times $e^{-|\theta|/a}$; as $a \to \infty$ the prior flattens to (improper) uniform and the posterior tends to $n(\bar{x}, \sigma^2/n)$, whose tail is 1 minus the one-sided p-value.

Solution. (a) By sufficiency,

$$P(\theta > K \mid \mathbf{x}, a) = \frac{\int_K^\infty \exp\left(-\frac{n}{2\sigma^2}(\bar{x}-\theta)^2 - \frac{|\theta|}{a}\right) d\theta}{\int_{-\infty}^\infty \exp\left(-\frac{n}{2\sigma^2}(\bar{x}-\theta)^2 - \frac{|\theta|}{a}\right) d\theta}.$$

(b) As $a \to \infty$, $e^{-|\theta|/a} \to 1$ pointwise and dominated convergence (dominated by the integrable normal kernel) gives

$$\lim_{a\to\infty} P(\theta > K \mid \mathbf{x}, a) = \frac{\int_K^\infty \exp(-\frac{n}{2\sigma^2}(\bar{x}-\theta)^2)d\theta}{\int_{-\infty}^\infty \exp(-\frac{n}{2\sigma^2}(\bar{x}-\theta)^2)d\theta} = 1 - \Phi\left(\frac{\sqrt{n}(K-\bar{x})}{\sigma}\right).$$

(c) The classical one-sided p-value (observed $\bar{x}$) is $p = \sup_{\theta\leq 0} P_\theta(\bar{X} \geq \bar{x}) = 1 - \Phi(\sqrt{n}\bar{x}/\sigma)$. With $K = 0$, the limiting posterior is $P(\theta > 0 \mid \mathbf{x}) \to \Phi(\sqrt{n}\bar{x}/\sigma) = 1 - p$: in the one-sided problem the (flat-prior) posterior probability of H_1 is exactly one minus the p-value — no conflict.

Exercise 8.51

Central idea. $p(\mathbf{x}) = \sup_{\Theta_0} P_\theta(W \geq W(\mathbf{x}))$; monotonicity of c_α in α turns "$W(\mathbf{x}) \geq c_\alpha$" into "$\alpha \geq p(\mathbf{x})$".

Solution. Let $p(\mathbf{x}) = \sup_{\theta\in\Theta_0} P_\theta(W(\mathbf{X}) \geq W(\mathbf{x}))$ (Definition 8.3.9) and $A = \{\alpha : W(\mathbf{x}) \geq c_\alpha\}$ (levels rejecting on $\mathbf{x}$). Since $\sup_{\Theta_0} P_\theta(W \geq c_\alpha) = \alpha$, c_α is nonincreasing in α. If $c_\alpha \leq W(\mathbf{x})$ then $\{W \geq W(\mathbf{x})\} \subseteq \{W \geq c_\alpha\}$, so $p(\mathbf{x}) \leq \alpha$; hence $p(\mathbf{x}) \leq \inf A$. If $c_\alpha > W(\mathbf{x})$ then $\{W \geq c_\alpha\} \subset \{W \geq W(\mathbf{x})\}$, so $\alpha \leq p(\mathbf{x})$; hence $\sup\{\alpha : c_\alpha > W(\mathbf{x})\} \leq p(\mathbf{x})$, and by monotonicity this sup equals $\inf A$. Thus $\inf A = p(\mathbf{x})$: the p-value is the smallest level at which H_0 rejects.

Exercise 8.52

Central idea. For $\theta \in \Theta_{j_0}$, $\max_j p_j \geq p_{j_0}$, so $\{p \leq \alpha\} \subseteq \{p_{j_0} \leq \alpha\}$; and $p(\mathbf{x}) \leq \alpha$ iff every $p_j(\mathbf{x}) \leq \alpha$.

Solution. (a) A p-value is valid if $P_\theta(p(\mathbf{X}) \leq \alpha) \leq \alpha$ for $\theta \in \Theta_0$. If $\theta \in \bigcup_j \Theta_j$, then $\theta \in \Theta_{j_0}$ for some j_0, and $p(\mathbf{X}) = \max_j p_j(\mathbf{X}) \geq p_{j_0}(\mathbf{X})$. Hence $\{p(\mathbf{X}) \leq \alpha\} \subseteq \{p_{j_0}(\mathbf{X}) \leq \alpha\}$, so $P_\theta(p(\mathbf{X}) \leq \alpha) \leq P_\theta(p_{j_0}(\mathbf{X}) \leq \alpha) \leq \alpha$ (validity of p_{j_0} on Θ_{j_0}). Thus p is valid for H_0. (b) Reject H_0 at level α iff $p(\mathbf{x}) \leq \alpha$ iff $\max_j p_j(\mathbf{x}) \leq \alpha$ iff $p_j(\mathbf{x}) \leq \alpha$ for every j iff each H_{0j} is rejected at level α: exactly the IUT for $H_0 = \bigcup_j \Theta_j$.

Exercise 8.53

Central idea. With $z = \sqrt{n}\bar{x}/\sigma$, the Bayes factor is $(1 + n\tau^2/\sigma^2)^{-1/2}\exp\left(\frac{z^2}{2}\frac{n\tau^2/\sigma^2}{1+n\tau^2/\sigma^2}\right)$; at fixed z the $(1+n)^{-1/2}$ penalty drives the posterior to 1 as $n \to \infty$ — Lindley's paradox.

Solution. (a) $P(\theta=0)+P(\theta\neq 0)=\frac{1}{2}+\frac{1}{2}\int n(0,\tau^2)=\frac{1}{2}+\frac{1}{2}=1$: proper. (b) With prior odds 1, $P(\theta=0\mid\mathbf{x})=(1+B)^{-1}$ where $B=\frac{\int f(\mathbf{x}|\theta)\,\frac{1}{2}n(\theta;0,\tau^2)d\theta}{\frac{1}{2}f(\mathbf{x}|0)}$. Since $\sum(x_i-\theta)^2-\sum x_i^2=n(\bar{x}-\theta)^2-n\bar{x}^2$, completing the square gives

$$B=\frac{1}{\sqrt{1+n\tau^2/\sigma^2}}\exp\left(\frac{z^2}{2}\,\frac{n\tau^2/\sigma^2}{1+n\tau^2/\sigma^2}\right),\qquad z=\frac{\sqrt{n}\,\bar{x}}{\sigma},$$

so $P(\theta=0\mid\mathbf{x})=\left[1+\frac{1}{\sqrt{1+n\tau^2/\sigma^2}}\exp\left(\frac{z^2}{2}\,\frac{n\tau^2/\sigma^2}{1+n\tau^2/\sigma^2}\right)\right]^{-1}$. (c) Two-sided z-test: $p=2(1-\Phi(|z|))=2\Phi(-|z|)$. (d) With $\sigma^2=\tau^2=1$: $P(\theta=0\mid\mathbf{x})=\left[1+\frac{1}{\sqrt{1+n}}e^{\frac{z^2}{2}\frac{n}{n+1}}\right]^{-1}$, $z=\sqrt{n}\bar{x}$. i. For $n=9$:

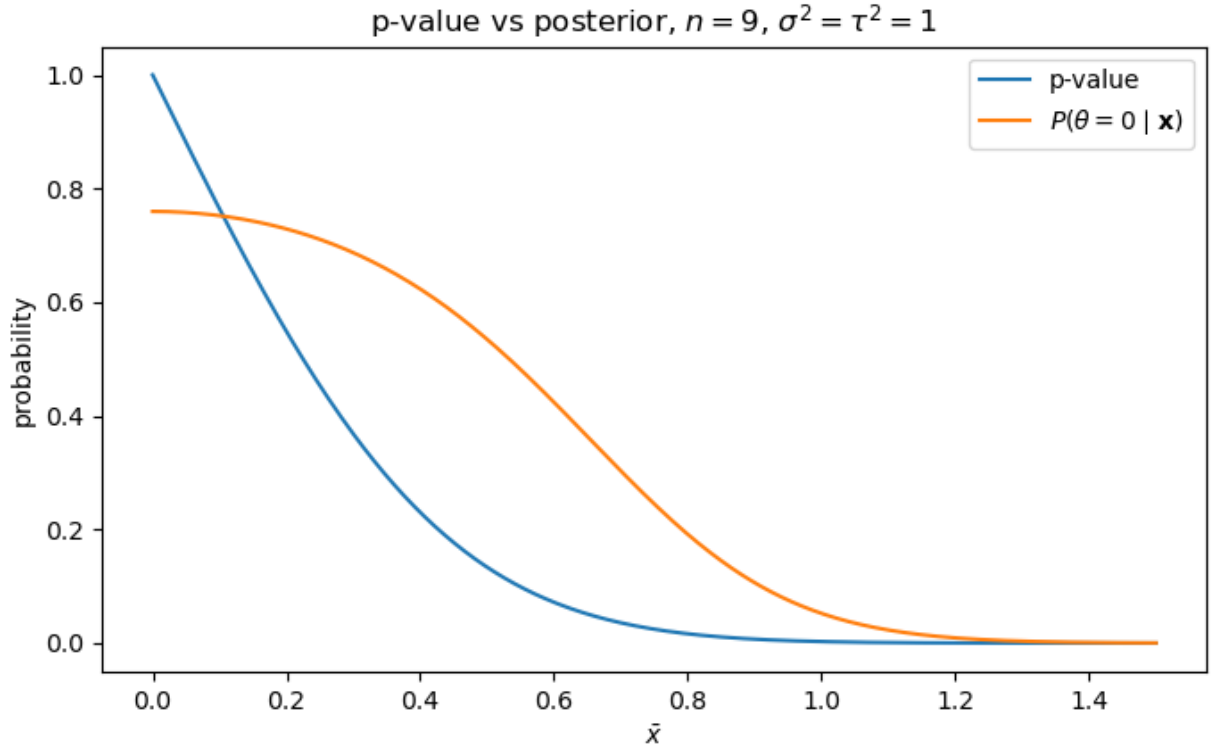


Figure 6: p-value versus posterior probability $P(\theta=0\mid\mathbf{x})$ as functions of $\bar{x}$, $n=9$.

the posterior lies above the p-value for moderately large $\bar{x}$ (e.g. at $\bar{x}=0.65$, $p\approx.05$ but posterior $\approx.35$). ii. At $\bar{x}=Z_{\alpha/2}/\sqrt{n}$ ($z=1.96$, $p\equiv.05$): $B=\frac{1}{\sqrt{1+n}}\exp\left(\frac{1.96^2}{2}\,\frac{n}{n+1}\right)\sim\frac{e^{1.92}}{\sqrt{n}}\to 0$, so $P(\theta=0\mid\mathbf{x})=(1+B)^{-1}\to 1$ as $n\to\infty$: with p-value fixed at .05, the posterior increasingly favors H_0 — Lindley's paradox.

Exercise 8.54

Central idea. The conjugate posterior is $\theta\mid\mathbf{x}\sim n(\mu_n,\sigma_n^2)$; with $z=\sqrt{n}\bar{x}/\sigma$ the posterior is $\Phi(-z/\sqrt{1+\sigma^2/(n\tau^2)})\geq\Phi(-z)=p$, with equality as $\tau^2\to\infty$.

Solution. (a) $\theta\mid\mathbf{x}\sim n(\mu_n,\sigma_n^2)$ with $\sigma_n^2=(n/\sigma^2+1/\tau^2)^{-1}$, $\mu_n=\frac{n\tau^2}{n\tau^2+\sigma^2}\bar{x}$. Hence $P(\theta\leq 0\mid\mathbf{x})=\Phi(-\mu_n/\sigma_n)=\Phi\left(-z/\sqrt{1+\sigma^2/(n\tau^2)}\right)$, $z=\sqrt{n}\bar{x}/\sigma$. (b) $p=\sup_{\theta\leq 0}P_\theta(\bar{X}\geq\bar{x})=P_0(\bar{X}\geq\bar{x})=1-\Phi(z)=\Phi(-z)$. (c) With $\sigma^2=\tau^2=1$: posterior $=\Phi(-z/\sqrt{1+1/n})$, $p=\Phi(-z)$. For $\bar{x}>0$ ($z>0$), $-z/\sqrt{1+1/n}>-z$, and Φ increasing, so posterior $>p$ always. (d) As $\tau^2\to\infty$, $\sqrt{1+\sigma^2/(n\tau^2)}\to 1$, so $P(\theta\leq 0\mid\mathbf{x})\to\Phi(-z)=p$: with a diffuse prior the one-sided posterior equals the p-value — no Lindley paradox here.

Exercise 8.55

Central idea. Under (8.3.13), $R(\theta) = c(\theta - \theta_0)^2\beta(\theta)$ on H_0 and $b(\theta_0 - \theta)(1 - \beta(\theta))$ on H_1, where $\beta(\theta) = \Phi(-z_\alpha - (\theta - \theta_0))$; larger α shifts risk from H_1 to H_0.

Solution. With $d = \theta - \theta_0$ and $\beta(\theta) = P_\theta(X < \theta_0 - z_\alpha) = \Phi(-z_\alpha - d)$: for $\theta \geq \theta_0$, $R(\theta) = c\,d^2\beta(\theta)$; for $\theta < \theta_0$, $R(\theta) = b(-d)(1 - \beta(\theta))$. (a) $b = c = 1$:

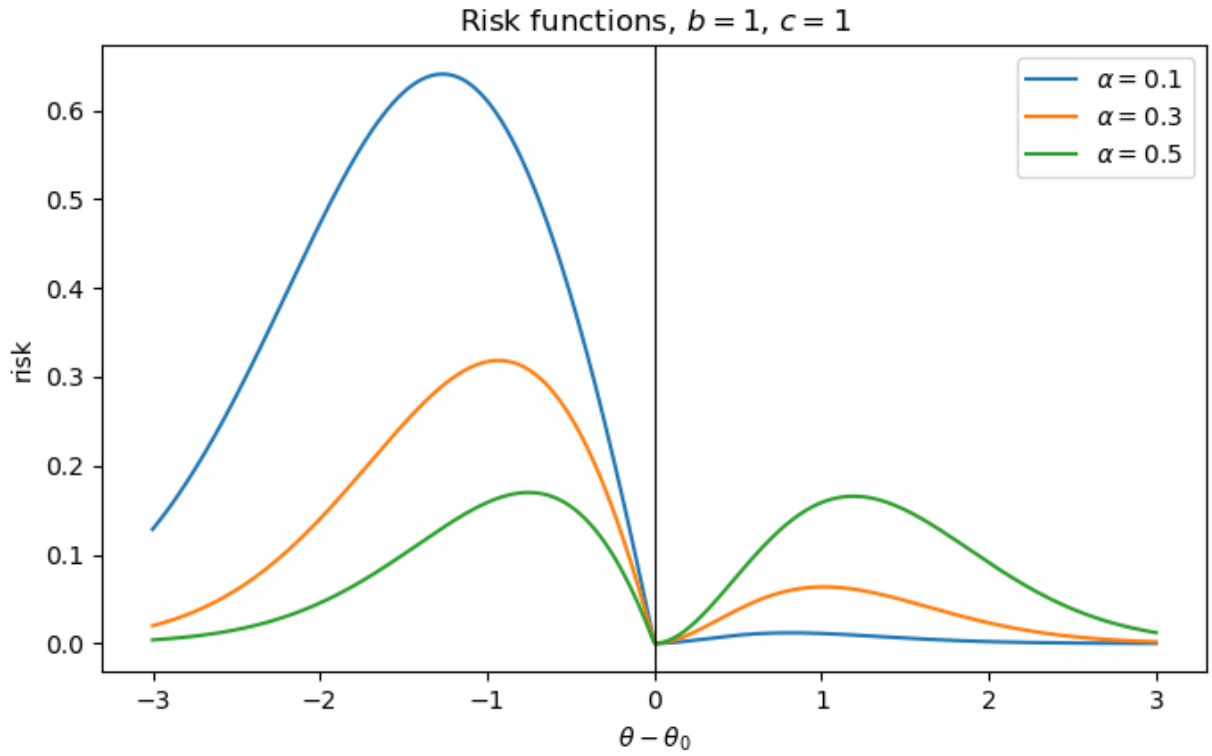


Figure 7: Risk functions for $\alpha = .1, .3, .5$ with $b = c = 1$.

(b) $b = 3$, $c = 1$:

In both, larger α (easier rejection) lowers risk on H_1 ($\theta < \theta_0$) at the cost of higher risk on H_0; with $b = 3$ the H_1 risks are tripled, favoring larger α. (c) The powers $\beta(\theta) = \Phi(-z_\alpha - d)$:

Risk is power scaled by the loss: on H_0, $R = c\,d^2\beta$ (grows quadratically in d times power); on H_1, $R = b|d|(1 - \beta)$ (grows linearly in $|d|$ times Type II probability). The crossing of the risk curves reflects the α tradeoff seen in the powers.

Exercise 8.56

Central idea. Test I rejects for small X — evidence *for* H_0 — so its risk is perverse on H_0; but on H_1 just above 1/3 it is Test I that has the smaller risk. Neither test dominates the other.

Solution. Under 0–1 loss, $R(p) = P_p(\text{reject})$ on H_0 ($p \leq 1/3$) and $1 - P_p(\text{reject})$ on H_1. $P_p(\text{I rejects}) = P_p(X \leq 1) = (1 - p)^4(1 + 4p)$; $P_p(\text{II rejects}) = P_p(X \geq 4) = p^4(5 - 4p)$.

On H_0, Test II is far better: at $p = 1/3$, Test I rejects with probability $(2/3)^4(7/3) \approx .46$ versus $\approx .045$ for Test II. But on H_1 just above 1/3, Test I has the *smaller* risk (at $p = .35$: .572 vs .946; at $p = .4$: .663 vs .913); the risks are equal at $p = 1/2$ (.8125 each), and only for large p does Test II win decisively (risk $\to 0$

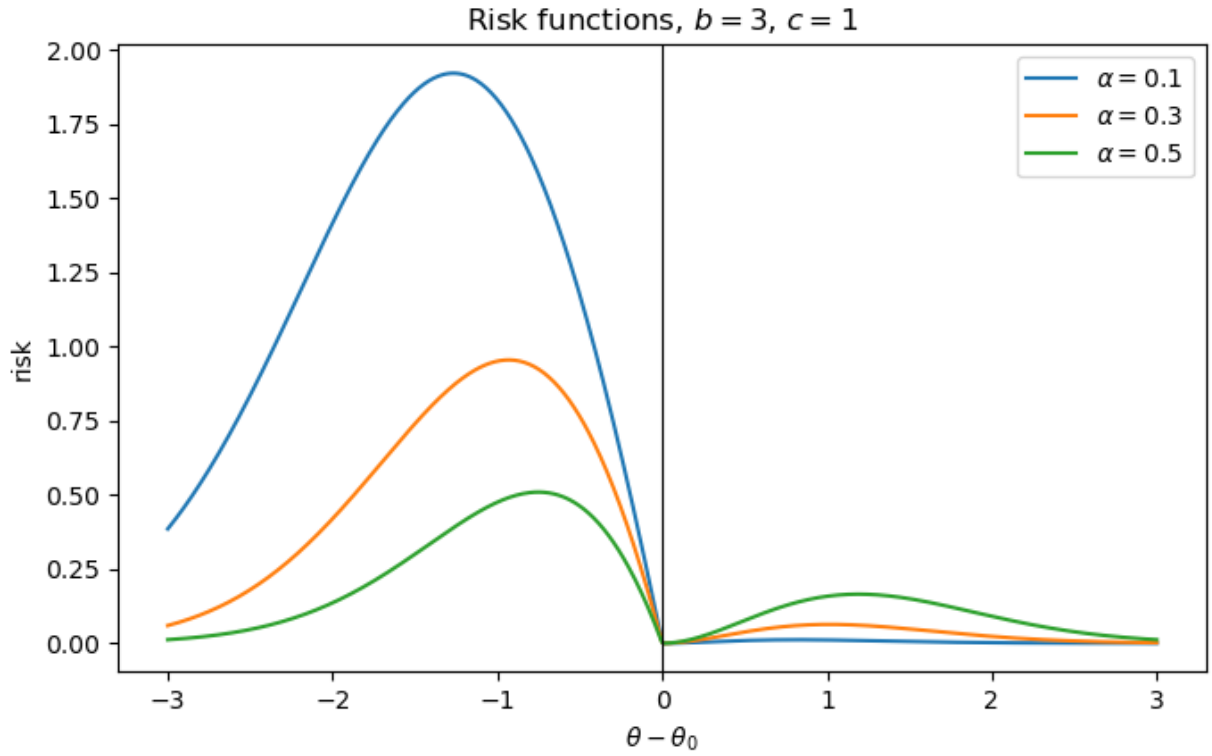


Figure 8: Risk functions for $\alpha = .1, .3, .5$ with $b = 3, c = 1$.

at $p = 1$ while Test I's stays near 1). So neither test uniformly dominates — Test I is not inadmissible, it is simply perverse on H_0 and competitive on H_1 near the boundary.

Exercise 8.57

Central idea. The interval test δ wastes rejection probability where it hurts (under H_0) and misses it where it helps (under H_1); the monotone test δ_c with $c \approx 1.10$ (matched to δ's risk at $\mu = 0$) dominates it, as MLR theory predicts.

Solution. Under 0–1 loss, $R(\mu, \delta) = \Phi(2 - \mu) - \Phi(1 - \mu)$ for $\mu \leq 0$ and $1 - \Phi(2 - \mu) + \Phi(1 - \mu)$ for $\mu > 0$; $R(\mu, \delta_c) = 1 - \Phi(c - \mu)$ for $\mu \leq 0$ and $\Phi(c - \mu)$ for $\mu > 0$. Matching risks at the boundary $\mu = 0$ ($1 - \Phi(c) = \Phi(2) - \Phi(1)$) gives $c = \Phi^{-1}(1 - \Phi(2) + \Phi(1)) \approx 1.10$. Then $R(\mu, \delta_c) \leq R(\mu, \delta)$ for all μ (verified numerically; equality only at $\mu = 0$):

δ_c shifts rejection mass from the interval $(1, 2)$ — which under H_0 ($\mu \leq 0$) only creates Type I error — out to (c, ∞), where it buys power under H_1; by the MLR of $n(\mu, 1)$ in X, the monotone test is uniformly better.

Exercise 8.58

Central idea. With $c_I = 8$, $c_{II} = 3$ in (8.3.12), $\max_\theta R(\theta, \alpha) = \max(8\alpha, 3(1-\alpha))$; the minimax α equalizes the two.

Solution. Example 8.3.31 tests $H_0 : \theta \geq \theta_0$ vs $H_1 : \theta < \theta_0$ with $X \sim n(\theta, 1)$ under loss (8.3.12) with $c_I = 8$ (Type I), $c_{II} = 3$ (Type II). For the test rejecting $\{X < \theta_0 - z_\alpha\}$: on H_0 ($\theta \geq \theta_0$), $R(\theta, \alpha) = 8P_\theta(X < \theta_0 - z_\alpha) = 8\Phi(-z_\alpha - (\theta - \theta_0))$, decreasing in θ, maximized at $\theta = \theta_0$ with value $8\Phi(-z_\alpha) = 8\alpha$; on H_1 ($\theta < \theta_0$), $R(\theta, \alpha) = 3P_\theta(X \geq \theta_0 - z_\alpha) = 3\Phi(z_\alpha + (\theta - \theta_0))$, increasing in θ, with supremum $3\Phi(z_\alpha) = 3(1-\alpha)$

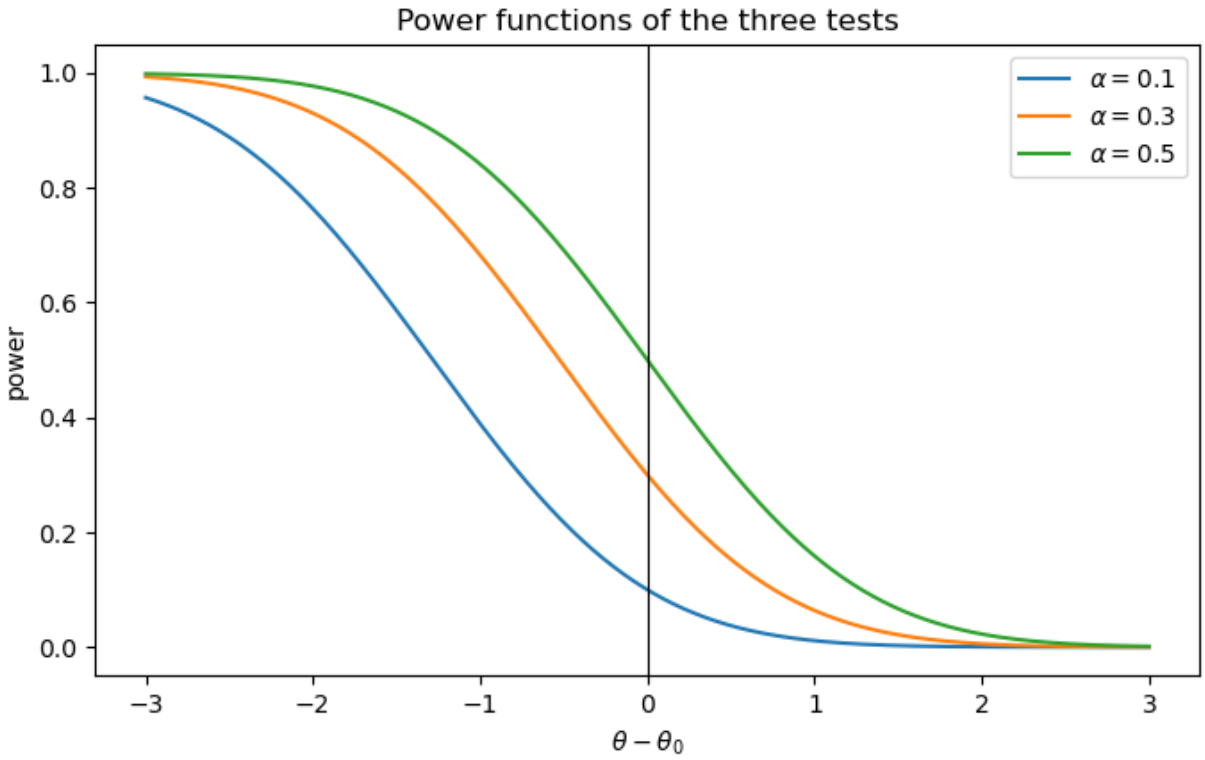


Figure 9: Power functions of the three tests.

as $\theta \uparrow \theta_0$. Hence $\max_\theta R(\theta, \alpha) = \max(8\alpha, 3(1-\alpha))$, minimized when $8\alpha = 3(1-\alpha)$, i.e. $\alpha = 3/11 \approx .273$ (minimax value $24/11 \approx 2.18$).

Chapter 9

Exercise 9.1

Central idea. Bonferroni alone gives only $\geq$; the ordering $L \leq U$ makes the two failure events disjoint, upgrading the bound to equality.

Solution. $P_\theta(L \leq \theta \leq U) = 1 - P_\theta(\{L(X) > \theta\} \cup \{U(X) < \theta\})$. Now $P_\theta(L(X) > \theta) = \alpha_1$ and $P_\theta(U(X) < \theta) = \alpha_2$. Since $L(x) \leq U(x)$ for every x, $\{L(X) > \theta\} \cap \{U(X) < \theta\} = \emptyset$ (it would require $U(X) < \theta < L(X) \leq U(X)$). Hence the union has probability exactly $\alpha_1 + \alpha_2$, and the coverage is $1 - \alpha_1 - \alpha_2$.

Exercise 9.2

Central idea. The interval is calibrated for the mean θ, not for a new observation: $X_{n+1} - \bar{X}$ has variance $1 + 1/n$, strictly larger than the $1/n$ the interval assumes.

Solution. $p = P_\theta(|X_{n+1} - \bar{X}| \leq 1.96/\sqrt{n})$. Since $X_{n+1} - \bar{X} \sim n(0, 1 + 1/n)$,

$$p = P\left(|Z| \leq \frac{1.96}{\sqrt{1 + 1/n}}\right) < P(|Z| \leq 1.96) = .95,$$

so p is less than .95.

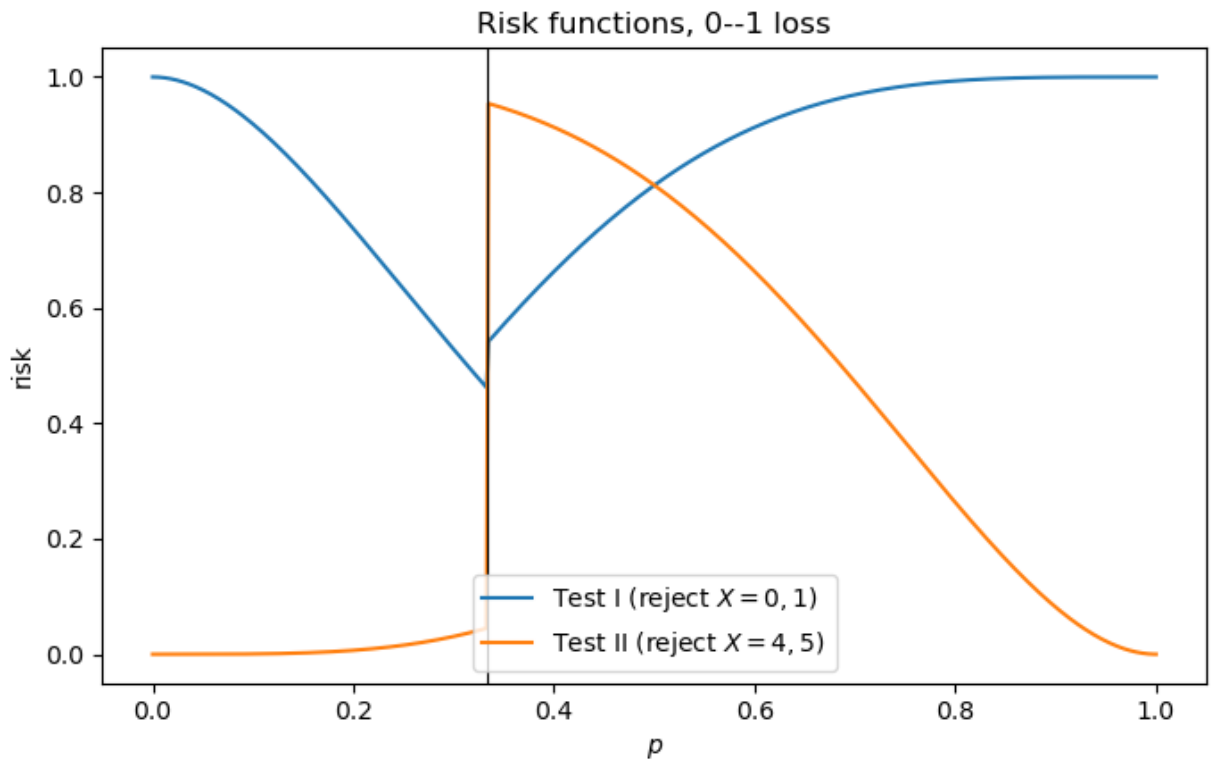


Figure 10: Risk functions of Test I and Test II under 0–1 loss.

Exercise 9.3

Central idea. $X_{(n)}$ is sufficient for β, and $(X_{(n)}/\beta)^{n\alpha_0} \sim \text{Uniform}(0,1)$ is an exact pivot.

Solution. (a) $P_\beta(X_{(n)} \le t) = (t/\beta)^{n\alpha_0}$ for $0 < t < \beta$, so $Q = (X_{(n)}/\beta)^{n\alpha_0} \sim U(0,1)$. Then

$$P_\beta(\beta \le U(X)) = .95 \iff P_\beta(Q \ge .05) = .95 \iff U(x) = \frac{x_{(n)}}{.05^{1/(n\alpha_0)}},$$

a 95% upper confidence limit for β. (b) From 7.10, $\hat\beta = x_{(n)} = 25.0$ and $\hat\alpha = n \big/ \sum_{i=1}^n \log(x_{(n)}/x_i) = 14/1.1116 = 12.59$ (data: $22.0, 23.9, 20.9, 23.8, 25.0, 24.0, 21.7, 23.8, 22.8, 23.1, 23.1, 23.5, 23.0, 23.0$). The 95% upper limit is $25.0/.05^{1/(14\times 12.59)} = 25.43$.

Exercise 9.4

Central idea. With known zero means, $\sum X_i^2/\sigma_X^2 \sim \chi^2_n$; the log-LRT is a U-shaped function $h(w)$ of $w = W/\lambda_0$ where $W = (\sum Y_j^2/m)/(\sum X_i^2/n)$, so the LRT rejects for w too small or too large.

Solution. (a) Unconstrained MLEs: $\hat\sigma_X^2 = \sum X_i^2/n$, $\hat\sigma_Y^2 = \sum Y_j^2/m$; constrained MLE $\tilde\sigma_X^2 = (\sum X_i^2 + \sum Y_j^2/\lambda_0)/(n+m)$. Writing $w = W/\lambda_0$,

$$-2\log\lambda = (n+m)\log\frac{n+mw}{n+m} - m\log w \equiv h(w),$$

with $h(1) = 0$, h decreasing on $(0,1)$ and increasing on $(1,\infty)$. Reject H_0 for $h(W/\lambda_0) > c_\alpha$. (b) Under H_0, $W/\lambda_0 = \frac{\sum Y_j^2/(m\sigma_Y^2)}{\sum X_i^2/(n\sigma_X^2)} \sim F_{m,n}$. The rejection region is $\{W/\lambda_0 < w_L\} \cup \{W/\lambda_0 > w_U\}$ with $h(w_L) = h(w_U) = c_\alpha$ and $P_{H_0}(\text{RR}) = \alpha$. (c) Inverting, $C = \{\lambda_0 : w_L \le W/\lambda_0 \le w_U\} = [W/w_U,\ W/w_L]$, a $1-\alpha$ interval for λ (equal-tail version: $[W/F_{m,n,\alpha/2},\ W/F_{m,n,1-\alpha/2}]$).

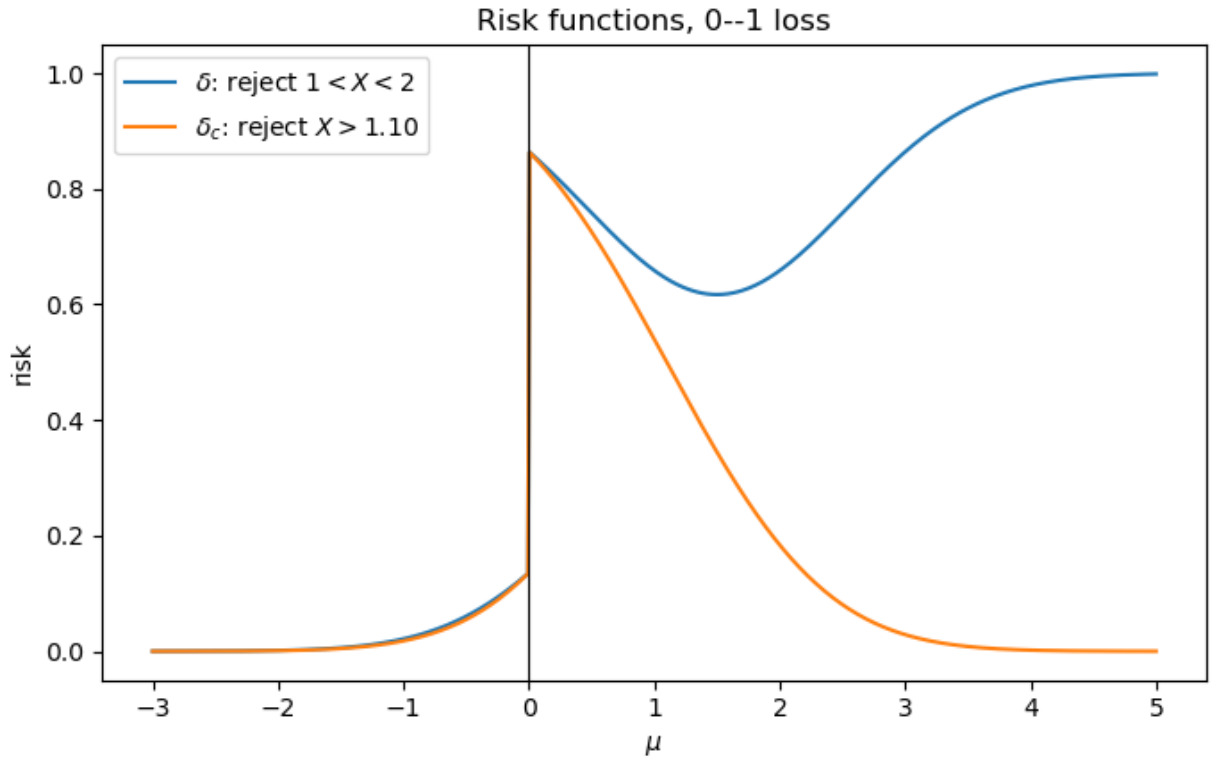


Figure 11: Risk functions of δ (reject $1 < X < 2$) and δ_c (reject $X > 1.10$).

Exercise 9.5

Central idea. Mirror Example 9.2.5: the UMP test rejects for small X, and monotonicity of the critical value in p_0 (MLR) makes the inverted set an interval $[0, U(x))$.

Solution. (a) By MLR and Karlin–Rubin, the UMP level-α test rejects for $X \le k(p_0)$; $A(p_0) = \{X > k(p_0)\}$ (with boundary randomization). $C(x) = \{p_0 : x \in A(p_0)\} = \{p_0 : k(p_0) < x\} = [0, U(x))$ since $k(p_0)$ is nondecreasing in p_0, with $U(x) = \sup\{p_0 : k(p_0) < x\}$. Then $P_p(p \in C(X)) = P_p(X \in A(p)) \ge 1 - \alpha$. (b) $U(x)$ solves $P_p(X \le x) = \sum_{y=0}^{x} \binom{n}{y} p^y (1-p)^{n-y} = \alpha$ for $x < n$, with $U(n) = 1$. Compare (9.2.8): the lower bound $L(x)$ instead solves $P_p(X \ge x) = \alpha$ (with $L(0) = 0$).

Exercise 9.6

Central idea. The LRT acceptance region $\{p_0 : L(p_0)/L(\hat{p}) \ge k\}$ is a level set of the likelihood — by definition a highest-density region — and its boundary is transcendental, unlike the quadratic Wilson interval.

Solution. (a) With $x = \sum x_i$, $\lambda = p_0^x (1-p_0)^{n-x} / [(x/n)^x (1 - x/n)^{n-x}]$ and

$$-2 \log \lambda = 2 \left[x \log \frac{x}{n p_0} + (n - x) \log \frac{n - x}{n(1 - p_0)} \right].$$

$C(x) = \{p_0 : -2 \log \lambda \le c\}$ (c from the exact null distribution, or $\chi^2_{1,\alpha}$ asymptotically). (b) $C(x) = \{p_0 : p_0^x (1-p_0)^{n-x} \ge k'\}$ is exactly the HDR of the likelihood shape $p_0 \mapsto p_0^x (1-p_0)^{n-x}$. It is not (10.4.4), the Wilson score interval solving the quadratic $|\hat{p} - p| / \sqrt{p(1-p)/n} = z_{\alpha/2}$; the LRT boundary equation above is transcendental in p_0, so the endpoints differ.

Exercise 9.7

Central idea. Profile out the nuisance parameter θ: the LRT compares the constrained MLE $\tilde{\theta}(a_0)$ with the unconstrained MLE, and the confidence set is the a_0-slice of the likelihood-ratio acceptance region.

Solution. (a) $\ell(\theta, a) = -\frac{n}{2}\log(a\theta) - \sum(x_i-\theta)^2/(2a\theta)$. For fixed a, the constrained MLE is the positive root of $n\theta^2 + na\theta - \sum x_i^2 = 0$,

$$\tilde{\theta}(a) = \frac{-a + \sqrt{a^2 + 4\overline{x^2}}}{2}, \qquad \overline{x^2} = \frac{1}{n}\sum x_i^2.$$

The $1-\alpha$ set is $C(x) = \{a_0 : L(\tilde{\theta}(a_0), a_0)/L(\hat{\theta}, \hat{a}) \geq k_\alpha\}$ (asymptotically $-2\log\text{ratio} \leq \chi^2_{1,\alpha}$). (b) $\ell(\theta, a) = -n\log\theta - \frac{n}{2}\log a - \sum(x_i-\theta)^2/(2a\theta^2)$; the constrained MLE is the positive root of $na\theta^2 + n\bar{x}\,\theta - \sum x_i^2 = 0$,

$$\tilde{\theta}(a) = \frac{-\bar{x} + \sqrt{\bar{x}^2 + 4a\,\overline{x^2}}}{2a},$$

and $C(x)$ has the same profile-LRT form as in (a).

Exercise 9.8

Central idea. Standardize by location and scale in different ways; any statistic computed from the standardized residuals $(X_i-\theta)/\sigma$ is free of (θ, σ).

Solution. With $\bar{X}$, S the sample mean and sd: $\frac{\bar{X}-\theta}{\sigma/\sqrt{n}}$, $\frac{\bar{X}-\theta}{S/\sqrt{n}}$, $\frac{(n-1)S^2}{\sigma^2}$, $\frac{X_{(1)}-\theta}{\sigma}$, $\frac{X_{(n)}-\bar{X}}{S}$ (others: $\max_i \frac{|X_i-\bar{X}|}{S}$, $\sum_i(\frac{X_i-\bar{X}}{S})^3$, $\frac{X_{(n)}-X_{(1)}}{\sigma}$, …). Each is a function of the iid standardized variables $(X_i-\theta)/\sigma \sim f$, hence pivotal.

Exercise 9.9

Central idea. Write $X_i = \mu + \sigma Z_i$ with Z_i iid $\sim f$; in each family the parameters cancel from the stated quantity.

Solution. In the location family $f(x-\mu)$ (σ known $= 1$), $\bar{X} - \mu = \bar{Z}$ is free of μ. In the scale family $\frac{1}{\sigma}f(x/\sigma)$ ($\mu = 0$), $\bar{X}/\sigma = \bar{Z}$ is free of σ. In the location–scale family, $\frac{\bar{X}-\mu}{S} = \frac{\bar{Z}}{S_Z}$ is free of both μ and σ.

Exercise 9.10

Central idea. This is the change-of-variables formula: the Jacobian $|\partial Q/\partial t|$ absorbs the θ-dependence, leaving the density of Q equal to g.

Solution. (a) (9.2.11) is $f(t \mid \theta) = g(Q(t,\theta))\left|\frac{\partial}{\partial t}Q(t,\theta)\right|$. By Theorem 2.1.5, for $q = Q(t,\theta)$ monotone in t, $f_Q(q \mid \theta) = f_T(t \mid \theta)/|\partial q/\partial t| = g(q)$ on the range of Q — independent of θ. Hence $Q(T,\theta)$ is pivotal. (b) $\left|\frac{\partial}{\partial t}F_\theta(t)\right| = f(t \mid \theta)$, so (9.2.11) holds with $g \equiv 1$. Then $Q(T,\theta) = F_\theta(T) \sim \text{Uniform}(0,1)$, a pivot.

Exercise 9.11

Central idea. $F_T(T \mid \theta_0) \sim \text{Uniform}(0,1)$ by the probability integral transform, so the acceptance probability is exactly the length of $[\alpha_1, 1-\alpha_2]$.

Solution. Since T is continuous, $U = F_T(T \mid \theta_0) \sim U(0,1)$, and $P_{\theta_0}(\alpha_1 \leq U \leq 1-\alpha_2) = 1-\alpha_2-\alpha_1 = 1-\alpha$: an α-level acceptance region. Inverting, $C(t) = \{\theta_0 : \alpha_1 \leq F_T(t \mid \theta_0) \leq 1-\alpha_2\}$ satisfies $P_\theta(\theta \in C(T)) = P_\theta(\alpha_1 \leq F_T(T \mid \theta) \leq 1-\alpha_2) = 1-\alpha$.

Exercise 9.12

Central idea. The normal pivot is exact for every $\theta > 0$. Inverting it requires considering all observed $\bar{x}$, including negative values, because a normal sample mean can be negative.

Solution. Let $z = z_{\alpha/2} > 0$ denote the upper-tail standard normal cutoff. Then

$$Q = \frac{\sqrt{n}(\bar{X} - \theta)}{\sqrt{\theta}} \sim N(0,1), \qquad C(\bar{x}) = \{\theta > 0 : n(\bar{x} - \theta)^2 \le z^2\theta\}.$$

This set has exact coverage $1 - \alpha$. Put $D = 4n\bar{x} + z^2$. When $D \ge 0$ the quadratic roots are

$$r_\pm = \bar{x} + \frac{z^2}{2n} \pm \frac{z}{2n}\sqrt{D}.$$

For $D > 0$, $C = [r_-, r_+] \cap (0, \infty)$; this includes negative $\bar{x}$ with $\bar{x} > -z^2/(4n)$. At $\bar{x} = 0$, it is $(0, z^2/n]$. At $D = 0$, it is the singleton $\{z^2/(4n)\}$. At $D < 0$, it is empty. Empty realizations do not invalidate the exact coverage identity, which follows directly from the pivot. The earlier positive-$\bar{x}$ formula was correct but omitted possible observations.

Exercise 9.13

Central idea. The definition is $Y = -1/\log X$, the reciprocal of $-\log X$. Consequently $F_Y(y) = e^{-\theta/y}$ for $y > 0$, and $\theta/Y = -\theta \log X \sim \text{Exp}(1)$ is a pivot.

Solution. (a) For $y > 0$, $P(Y \le y) = P(X \le e^{-1/y}) = e^{-\theta/y}$. Hence

$$P_\theta(\theta \in [Y/2, Y]) = P_\theta(\theta \le Y \le 2\theta) = e^{-1/2} - e^{-1} =: c_0 = 0.2386512185.$$

This is constant in θ, so it is also the confidence coefficient.

(b) With $Q = \theta/Y \sim \text{Exp}(1)$, any fixed $0 \le a < b$ satisfying $e^{-a} - e^{-b} = c_0$ gives the exact confidence set $[aY, bY] \cap (0, \infty)$. Equivalently, $X^\theta \sim U(0,1)$: endpoints $0 < A < B \le 1$ with $B - A = c_0$ give $[\log B/\log X, \log A/\log X] \cap (0, \infty)$.

(c) The original interval corresponds to $(a, b) = (1/2, 1)$. Among intervals obtained from **fixed endpoints of this pivot**, length is $(b - a)Y$. Since $b = -\log(e^{-a} - c_0)$,

$$\frac{d}{da}(b - a) = \frac{c_0}{e^{-a} - c_0} > 0.$$

Thus the shortest interval in this class takes $a = 0$ and $b = -\log(1 - c_0) = 0.2726637062$:

$$C(Y) = (0,\ -Y \log(1 - c_0)].$$

It has exactly the same confidence coefficient and is shorter for every $Y > 0$ than $[Y/2, Y]$. The decimal .239 is only a rounding of c_0; use c_0, not the rounded decimal, to preserve the exact coefficient. This comparison does not assert optimality among all possible confidence procedures.

Exercise 9.14

Central idea. Bonferroni for (a); for (b) the independence of $\bar{X}$ and S^2 sharpens the bound to the exact product $(1 - \alpha_1)(1 - \alpha_2)$.

Solution. (a) Take $\alpha_1 + \alpha_2 = \alpha$ (e.g. $\alpha/2$ each). With $k = t_{n-1,\alpha_1/2}$, $a = \chi^2_{n-1,1-\alpha_2/2}$, $b = \chi^2_{n-1,\alpha_2/2}$ (upper-tail cutoffs), each set is $1 - \alpha_i$, so $P(\text{both}) \ge 1 - \alpha_1 - \alpha_2 = 1 - \alpha$ by Bonferroni. (b) By independence of $\bar{X}$ and S^2,

$$P\left(|\bar{X} - \mu| \le \tfrac{k\sigma}{\sqrt{n}},\ a \le \tfrac{(n-1)S^2}{\sigma^2} \le b\right) = P(|Z| \le k)\, P(a \le \chi^2_{n-1} \le b) = (1 - \alpha_1)(1 - \alpha_2) \ge 1 - \alpha,$$

with $k = z_{\alpha_1/2}$ and a, b as in (a). (c) (a) is a rectangle — simpler, but conservative (Bonferroni) with wider t-based μ-slices. (b) has exact product coverage and narrower z-based μ-slices, but the μ-bounds involve the unknown σ, so the joint region is σ-dependent (trapezoidal), not a product of fixed intervals.

Exercise 9.15

Central idea. The sign of the quadratic coefficient is the significance of the denominator mean; the discriminant measures whether the data pin down the ratio at all.

Solution. The set is $\{A\theta^2 + B\theta + C \le 0\}$ with $A = \bar{x}^2 - \frac{t^2_{n-1,\alpha/2}}{n-1}S_x^2$, $B = -2(\bar{x}\bar{y} - \frac{t^2}{n-1}S_{xy})$, $C = \bar{y}^2 - \frac{t^2}{n-1}S_y^2$; roots $[-B \pm \sqrt{B^2 - 4AC}]/(2A)$. (a) $A > 0 \iff |\bar{x}|/(S_x/\sqrt{n-1}) > t_{n-1,\alpha/2}$ (μ_X significantly nonzero) and $B^2 > 4AC$: the set is the bounded interval $[\theta_1, \theta_2]$ — the ratio is well-identified. (b) $A < 0$ ($\bar{x}$ not significant) and $B^2 > 4AC$: $(-\infty, \theta_1] \cup [\theta_2, \infty)$ — the ratio could be arbitrarily large positive or negative; the confidence set is unbounded. (c) $B^2 < 4AC$: this forces $A < 0$ (the $A > 0$ subcase, which would give the empty set, cannot occur for a Fieller set — the A, B, C built from real data always satisfy $B^2 \ge 4AC$ when $A > 0$), so the set is the whole line. Tell the experimenter nothing can be ruled out: the data are consistent with any ratio value; the denominator mean is indistinguishable from zero, so the ratio is ill-determined and more/better data are needed.

Exercise 9.16

Central idea. One-sided hypotheses invert to one-sided bounds (the bound points away from the alternative); the two-sided hypothesis gives the usual z-interval.

Solution. (a) $A(\theta_0) = \{|\bar{x} - \theta_0| \le z_{\alpha/2}\sigma/\sqrt{n}\}$; $C = [\bar{x} - z_{\alpha/2}\sigma/\sqrt{n},\ \bar{x} + z_{\alpha/2}\sigma/\sqrt{n}]$. (b) UMP test rejects for small $\bar{x}$: $A(\theta_0) = \{\bar{x} \ge \theta_0 - z_\alpha\sigma/\sqrt{n}\}$; $C = (-\infty,\ \bar{x} + z_\alpha\sigma/\sqrt{n}]$. (c) UMP test rejects for large $\bar{x}$: $A(\theta_0) = \{\bar{x} \le \theta_0 + z_\alpha\sigma/\sqrt{n}\}$; $C = [\bar{x} - z_\alpha\sigma/\sqrt{n},\ \infty)$.

Exercise 9.17

Central idea. Extremes are the natural pivots for bounded-support families: $X_{(1)} - \theta$ for location, $X_{(n)}/\theta$ for scale.

Solution. (a) $Q = X_{(1)} - \theta$: $P(Q > q) = (1/2 - q)^n$, $-1/2 \le q \le 1/2$, a pivot. With $P(a \le Q \le b) = 1 - \alpha$, e.g. equal-tail $a = \frac{1}{2} - (1 - \alpha/2)^{1/n}$, $b = \frac{1}{2} - (\alpha/2)^{1/n}$: $[X_{(1)} - b,\ X_{(1)} - a]$. (b) $P(X_{(n)} \le t) = (t/\theta)^{2n}$, so $Q = X_{(n)}/\theta$ has cdf u^{2n} on $[0, 1]$, a pivot. Shortest $1 - \alpha$ interval (cf. 9.37): $P(\alpha^{1/(2n)} \le Q \le 1) = 1 - \alpha$ gives $[y,\ y/\alpha^{1/(2n)}]$.

Exercise 9.18

Central idea. Most-probable ordering can create disconnected Sterne confidence sets. The displayed modified acceptance regions invert to connected intervals at the intentionally low nominal coefficient 0.442; 44.2% is not undercoverage relative to that nominal level.

Solution. (a) $P_p(X = 0) = (1-p)^3$, $P_p(X = 1) = 3p(1-p)^2$, $P_p(X = 2) = 3p^2(1-p)$, $P_p(X = 3) = p^3$. $P_p(X = 1)$ is maximized at $p = 1/3$ (value 4/9); $P_p(X = 2)$ at $p = 2/3$ (value 4/9). (b) At $p = 1/3 + \epsilon$: $P(X = 0) = (2/3 - \epsilon)^3 = 8/27 - \frac{4}{3}\epsilon + O(\epsilon^2)$ while $P(X = 2) = 3(1/3 + \epsilon)^2(2/3 - \epsilon) = 6/27 + \epsilon + O(\epsilon^2)$. The difference is $2/27 - \frac{7}{3}\epsilon + O(\epsilon^2) > 0$ for small ϵ: just right of 1/3, the far-tail point 0 outranks the central point 2 under most-probable ordering. (c) Inverting, $C(x) = \{p : x \in A(p)\}$: $C(0) = [0, .305)$, $C(1) = (.238, .634)$, $C(2) = (.362, .762)$, $C(3) = (.695, 1]$ — all intervals. But $\inf_p P_p(p \in C(X)) = .442$: e.g. as $p \downarrow .305$ only $x = 1$ covers, giving $3(.305)(.695)^2 = .442$; at $p = .238$, $(.762)^3 = .442$. This modified construction achieves the intended approximately 44.2% coefficient and connected intervals. The printed three-decimal cutpoints are rounded: for example $3(0.305)(0.695)^2 = 0.441951375$ and $0.762^3 = 0.442450728$. Exact cutpoints solve the corresponding equations at 0.442; the defect being illustrated is disconnectedness of the original Sterne sets, not coverage below the stated nominal level.

Exercise 9.19

Central idea. Same argument as part (a) with the monotonicity reversed: when $F_T(t \mid \theta)$ increases in θ, small cdf values correspond to small θ.

Solution. Assume $F_T(t \mid \theta)$ is increasing in θ for each t, and put $\alpha_1 + \alpha_2 = \alpha$. By the probability integral transform, $\{t : \alpha_1 \le F_T(t \mid \theta_0) \le 1 - \alpha_2\}$ is a $1-\alpha$ acceptance region. Define $\theta_L(t)$, $\theta_U(t)$ by $F_T(t \mid \theta_U) = 1-\alpha_2$, $F_T(t \mid \theta_L) = \alpha_1$ (unique since $1-\alpha_2 > \alpha_1$ and F_T is continuous and strictly monotone in θ where it matters). Increasingness gives $F_T(t \mid \theta) < \alpha_1 \iff \theta < \theta_L(t)$ and $F_T(t \mid \theta) > 1-\alpha_2 \iff \theta > \theta_U(t)$, whence $\{\theta : \alpha_1 \le F_T(t \mid \theta) \le 1 - \alpha_2\} = [\theta_L(t), \theta_U(t)]$, a $1-\alpha$ interval.

Exercise 9.20

Central idea. (a) follows by applying Exercise 2.10 to $-T$. *Erratum:* the set in (b) as printed is empty — the inequalities must be reversed, after which Bonferroni on the stochastic-dominance bounds gives $1-\alpha$.

Solution. (a) $\bar{F}_T(T \mid \theta) = P(T \ge T \mid \theta) = P(-T \le -T \mid \theta) = F_{-T}(-T \mid \theta)$; Exercise 2.10 applied to $-T$ gives $P_\theta(\bar{F}_T(T \mid \theta) \le x) \le x$. (b) *Erratum — as stated this is false.* Take $Y \sim \text{Poisson}(10\lambda)$, $y = 6$, $\alpha_1 = \alpha_2 = .05$: $\{F_Y(6 \mid \lambda) \le .05\} = \{\lambda \ge 1.184\}$ (since $P(Y \le 6 \mid \lambda)$ decreases in λ) and $\{\bar{F}_Y(6 \mid \lambda) \le .05\} = \{\lambda \le .261\}$; the intersection is $\emptyset$, not a $1-\alpha$ set. The correct set reverses the inequalities: $C = \{\theta : F_T(T \mid \theta) \ge \alpha_1 \text{ and } \bar{F}_T(T \mid \theta) \ge \alpha_2\}$ (this matches the continuous case 9.2.12, where $\bar{F} = 1 - F$). Then

$$P_\theta(\theta \notin C) \le P_\theta(F_T(T \mid \theta) < \alpha_1) + P_\theta(\bar{F}_T(T \mid \theta) < \alpha_2) \le \alpha_1 + \alpha_2 = \alpha$$

by (a). In case (i) (F_T decreasing in θ), $\{F_T(t \mid \theta) \ge \alpha_1\} = (-\infty, \theta_U(t)]$ and $\{\bar{F}_T(t \mid \theta) \ge \alpha_2\} = [\theta_L(t), \infty)$ by (c), so $C = [\theta_L(T), \theta_U(T)]$. (c) $\bar{F}_T(t \mid \theta) = 1 - \lim_{s \uparrow t} F_T(s \mid \theta)$; each $F_T(s \mid \cdot)$ decreasing implies $\bar{F}_T(t \mid \cdot)$ nondecreasing. (d) Case (ii): F_T increasing in θ, so $\bar{F}_T$ decreasing. With $\bar{F}_T(t \mid \theta_U) = \alpha_1$, $F_T(t \mid \theta_L) = \alpha_2$: the corrected set $\{\theta : F_T(T \mid \theta) \ge \alpha_2,\ \bar{F}_T(T \mid \theta) \ge \alpha_1\} = [\theta_L(T), \theta_U(T)]$ is a $1-\alpha$ interval by the Bonferroni argument in (b).

Exercise 9.21

Central idea. The binomial–beta identity turns Clopper–Pearson tail equations into beta quantiles, and the beta–F relation turns those into F cutoffs.

Solution. $L(x)$ solves $P_L(X \ge x) = \alpha/2$, i.e. $P(Y \le L) = \alpha/2$ for $Y \sim \text{beta}(x, n-x+1)$. Since $\frac{Y}{1-Y} \cdot \frac{n-x+1}{x} \sim F_{2x,2(n-x+1)}$, $\frac{L}{1-L} \cdot \frac{n-x+1}{x} = F_{2x,2(n-x+1),1-\alpha/2} = 1/F_{2(n-x+1),2x,\alpha/2}$, giving

$$L(x) = \frac{1}{1 + \frac{n-x+1}{x} F_{2(n-x+1),2x,\alpha/2}}.$$

Numerical check: for $n = 10$, $x = 3$, $\alpha = .05$, direct solution of $P_p(X \ge 3) = .025$ gives $L = .066740$, matching the formula above (using $\frac{n-x}{x}$ in place of $\frac{n-x+1}{x}$ would give .075553 instead). The upper endpoint: $U(x)$ solves $P_U(X \le x) = \alpha/2$, i.e. $P(Z \ge U) = \alpha/2$ for $Z \sim \text{beta}(x+1, n-x)$, yielding the stated form (verified: .652453 both ways).

Exercise 9.22

Central idea. For the negative binomial (counting failures), tail probabilities are *decreasing* in p, so the Clopper–Pearson equations are reversed relative to the binomial: L solves $P_L(X \le x) = \alpha/2$ and U solves $P_U(X \ge x) = \alpha/2$.

Solution. With X = failures before r successes: $P_p(X \le x) = P_p(\text{Bin}(x+r, p) \ge r) = P(\text{beta}(r, x+1) \le p)$ and $P_p(X \ge x) = P_p(\text{Bin}(x+r-1, p) \le r-1) = P(\text{beta}(r, x) \ge p)$. Since $P_p(X \le x)$ increases in p, L solves $P_L(X \le x) = \alpha/2$, i.e. L is the $\alpha/2$ quantile of $\text{beta}(r, x+1)$: $\frac{L}{1-L} \cdot \frac{x+1}{r} = F_{2r,2(x+1),1-\alpha/2} = 1/F_{2(x+1),2r,\alpha/2}$, giving the stated lower bound. Since $P_p(X \ge x)$ decreases in p, U solves $P_U(X \ge x) = \alpha/2$, i.e. U is the $1-\alpha/2$ quantile of $\text{beta}(r, x)$: $\frac{U}{1-U} \cdot \frac{x}{r} = F_{2r,2x,\alpha/2}$, giving the stated upper bound. (If $x = 0$, $L = 0$.) Numerically ($r = 3, x = 7, \alpha = .1$): direct solution gives $[.087264, .549642]$, matching the formulas.

Exercise 9.23

Central idea. The LRT statistic $2[Y\log(Y/(n\lambda_0)) - (Y - n\lambda_0)]$ is χ^2_1 asymptotically; inverting it gives an interval nearly identical to the exact χ^2-quantile interval of 9.2.15.

Solution. (a) With $Y \sim \text{Poisson}(n\lambda)$, $\lambda(Y;\lambda_0) = (Y/(n\lambda_0))^Y e^{-(Y-n\lambda_0)}$ and $-2\log\lambda = 2[Y\log(Y/(n\lambda_0)) - (Y-n\lambda_0)]$. The $1-\alpha$ LRT interval is $\{\lambda_0 : -2\log\lambda \le \chi^2_{1,\alpha}\}$, versus 9.2.15's exact $[\chi^2_{2Y,1-\alpha/2}/(2n),\ \chi^2_{2(Y+1),\alpha/2}/(2n)]$; both are centered near Y/n and agree closely. (b) $n = 9$, $\sum x = 558$, $\bar{x} = 62$. LRT 90% interval: $[57.78,\ 66.42]$; Example 9.2.15 interval: $[57.75,\ 66.49]$ — essentially identical.

Exercise 9.24

Central idea. The endpoints $L(x), U(x)$ are increasing in x, so $\{x : L(x) \le \lambda \le U(x)\}$ is a contiguous block of integers — and it changes only when λ crosses an endpoint.

Solution. The first display is just $P_\lambda(\lambda \in [L(X), U(X)]) = \sum_x P_\lambda(X = x)$ over x with $L(x) \le \lambda \le U(x)$. Since $L(x) = \chi^2_{2x,1-\alpha/2}/2$ and $U(x) = \chi^2_{2(x+1),\alpha/2}/2$ are strictly increasing in x, $\{x : L(x) \le \lambda\} = \{x \ge x_l(\lambda)\}$ and $\{x : U(x) \ge \lambda\} = \{x \le x_u(\lambda)\}$; intersecting gives the contiguous sum. As λ increases, the block $[x_l(\lambda), x_u(\lambda)]$ changes only when λ passes some $L(x)$ or $U(x)$ — hence the coverage is a step function with jumps exactly at the interval endpoints.

Exercise 9.25

Central idea. The pivot $n(Y-\mu) \sim \text{Exp}(1)$ has a strictly decreasing density, so the shortest pivotal interval hugs 0 — and the LRT reproduces exactly this interval.

Solution. Pivotal/shortest: $Q = n(Y-\mu) \sim \text{Exp}(1)$; by 9.41 the shortest $1-\alpha$ set for Q is $[0, -\log\alpha]$, giving $\mu \in [Y + (\log\alpha)/n,\ Y]$, length $-(\log\alpha)/n$. LRT: $\lambda = e^{-n(Y-\mu_0)} I(\mu_0 \le Y)$; $-2\log\lambda = 2n(Y-\mu_0) \sim \chi^2_2$ exactly, so the $1-\alpha$ LRT interval is $[Y - \chi^2_{2,\alpha}/(2n),\ Y] = [Y + (\log\alpha)/n,\ Y]$ — identical to the shortest pivotal interval (since $\chi^2_{2,\alpha} = -2\log\alpha$). Example 9.2.13's equal-tail cdf interval is $[Y + \log(\alpha/2)/n,\ Y + \log(1-\alpha/2)/n]$, strictly longer (e.g. $\alpha = .1$: $2.94/n$ vs $2.30/n$).

Exercise 9.26

Central idea. The gamma prior (scale convention, as in the text) is conjugate for the beta$(\theta, 1)$ shape parameter: the posterior is gamma$(n + r,\ [1/\lambda - \sum \log x_i]^{-1})$.

Solution. $L(\theta \mid x) = \theta^n (\prod x_i)^{\theta-1}$; with $\pi(\theta) \propto \theta^{r-1} e^{-\theta/\lambda}$,

$$\pi(\theta \mid x) \propto \theta^{n+r-1} \exp(-\theta(1/\lambda - \textstyle\sum \log x_i)),$$

i.e. $\theta \mid x \sim \text{gamma}(n + r,\ [1/\lambda - \sum_{i=1}^n \log x_i]^{-1})$ (shape–scale). A $1-\alpha$ credible set is e.g. the equal-tail interval $[G^{-1}(\alpha/2),\ G^{-1}(1-\alpha/2)]$ for this gamma law.

Exercise 9.27

Central idea. The inverted gamma is conjugate in both cases; the posterior is unimodal, so the HPD region is $\{\pi(\cdot \mid x) \ge k\}$ — an interval with equal posterior density at the endpoints.

Solution. (a) (exponential(λ) = mean λ.) $L(\lambda) \propto \lambda^{-n} e^{-\sum x_i/\lambda}$; posterior $\pi(\lambda \mid x) \propto \lambda^{-(a+n)-1} \exp(-(1/b + \sum x_i)/\lambda)$, i.e. $IG(a+n,\ (1/b + \sum x_i)^{-1})$. The HPD set $\{\lambda : \pi(\lambda \mid x) \ge k\}$ is the interval $[\lambda_L, \lambda_U]$ solving $\pi(\lambda_L \mid x) = \pi(\lambda_U \mid x)$ and $\int_{\lambda_L}^{\lambda_U} \pi(\lambda \mid x)\, d\lambda = 1 - \alpha$ (numerically). (b) $L(\sigma^2) \propto (\sigma^2)^{-(n-1)/2} \exp(-(n-1)s^2/(2\sigma^2))$; posterior $IG(a + (n-1)/2,\ [1/b + (n-1)s^2/2]^{-1})$. Same HPD construction as (a). (c) As $a \to 0$, $b \to \infty$, $\pi(\sigma^2) \to (\sigma^2)^{-1}$ (Jeffreys) and the posterior tends to $IG((n-1)/2,\ 2/((n-1)s^2))$; the limiting HPD set is $\{\sigma^2 : \pi(\sigma^2 \mid x) \ge k\}$ for this limiting posterior, i.e. the interval $[\sigma^2_L, \sigma^2_U]$ with equal posterior density at the endpoints and posterior probability $1 - \alpha$.

Exercise 9.28

Central idea. This is the normal–inverse-gamma conjugate family; the marginal posterior of θ is a t with $2a+n$ df — never $n-1$ — so the classical t_{n-1} set cannot be recovered exactly.

Solution. (a) $n(\theta \mid \mu, \sigma^2/\kappa_0)$ with $\kappa_0 = 1/\tau^2$ times $IG(a,b)$ is the normal–inverse-gamma family, conjugate for (θ, σ^2): the posterior is $N\text{-}IG(\mu_n, \kappa_n, a_n, b_n)$ with $\kappa_n = \kappa_0 + n$, $\mu_n = (\kappa_0\mu + n\bar{x})/\kappa_n$, $a_n = a + n/2$, $1/b_n = 1/b + \frac{1}{2}[(n-1)s^2 + \frac{\kappa_0 n}{\kappa_n}(\bar{x}-\mu)^2]$. (b) Marginally, $\frac{\theta-\mu_n}{\sqrt{1/(\kappa_n a_n b_n)}} \sim t_{2a_n}$; a $1-\alpha$ credible set is $\mu_n \pm t_{2a_n,\alpha/2}\sqrt{1/(\kappa_n a_n b_n)}$. (c) No. The Bayes set is always $t_{2a_n} = t_{2a+n}$-based with $2a+n > n$, while the classical set is t_{n-1}-based. In the diffuse limit $\tau^2 \to \infty$, $a \to 0$, $b \to \infty$: $\mu_n \to \bar{x}$, $\kappa_n \to n$, $a_n \to n/2$, $1/b_n \to (n-1)s^2/2$, giving $\bar{x} \pm t_{n,\alpha/2}\frac{s\sqrt{n-1}}{n}$ — close to but not equal to $\bar{x} \pm t_{n-1,\alpha/2}s/\sqrt{n}$. (The residual $(\sigma^2)^{-1/2}$ factor in the $n(\mu, \tau^2\sigma^2)$ prior keeps the limiting prior at $(\sigma^2)^{-3/2}$, not Jeffreys' $(\sigma^2)^{-1}$.)

Exercise 9.29

Central idea. The posterior is beta$(a+x,\ b+n-x)$; the beta–F transformation puts it in Clopper–Pearson form with prior pseudo-counts.

Solution. (a) With $x = \sum x_i$, $p \mid x \sim \text{beta}(a+x,\ b+n-x)$; e.g. equal-tail $[B^{-1}(\alpha/2),\ B^{-1}(1-\alpha/2)]$. (b) Via $\frac{Y}{1-Y} \cdot \frac{v}{u} \sim F_{2u,2v}$ for $Y \sim \text{beta}(u,v)$:

$$\frac{1}{1+\frac{b+n-x}{a+x}F_{2(b+n-x),2(a+x),1-\alpha/2}} \le p \le \frac{\frac{a+x}{b+n-x}F_{2(a+x),2(b+n-x),\alpha/2}}{1+\frac{a+x}{b+n-x}F_{2(a+x),2(b+n-x),\alpha/2}}.$$

This is algebraically the Clopper–Pearson F-form with the data counts augmented by prior pseudo-counts (a successes, b failures): it shrinks the interval toward $a/(a+b)$ relative to 9.21, with the difference vanishing as n grows.

Exercise 9.30

Central idea. The posterior is gamma$(a+\sum x_i,\ \text{rate } n+1/b)$, so T is exactly χ^2; the confidence interval's endpoints, rescaled into T-units, drift to $+\infty$ relative to T's mean because of the factor $(nb+1)/(nb) > 1$.

Solution. (a) $\lambda \mid x \sim \text{gamma}(a+\sum x_i,\ [n+1/b]^{-1})$ (shape–scale, Example 9.2.16), so $2(n+1/b)\lambda = \frac{2(nb+1)}{b}\lambda \sim \chi^2_{2(a+\sum x_i)}$. (b) $M(t) = E\exp(t(\chi^2_\nu - \nu)/\sqrt{2\nu}) = (1-2t/\sqrt{2\nu})^{-\nu/2}e^{-t\nu/\sqrt{2\nu}}$; $\log M(t) = -\frac{\nu}{2}\log(1-2t/\sqrt{2\nu}) - t\sqrt{\nu/2} = t^2/2 + o(1) \to t^2/2$. (c) (9.2.21) is $P(\frac{1}{2n}\chi^2_{2\sum x,1-\alpha/2} \le \lambda \le \frac{1}{2n}\chi^2_{2(\sum x+1),\alpha/2})$ under the posterior. In T-units: $P(c\,\chi^2_{2\sum x,1-\alpha/2} \le T \le c\,\chi^2_{2(\sum x+1),\alpha/2})$ with $c = \frac{nb+1}{nb} = 1 + \frac{1}{nb} > 1$. Standardizing, the lower cutoff is

$$\frac{c\,\chi^2_{2\sum x,1-\alpha/2} - 2(a+\sum x)}{\sqrt{4(a+\sum x)}} = \frac{\sqrt{\sum x}}{nb} + O(1) \to \infty,$$

since $\chi^2_{2\sum x,1-\alpha/2} = 2\sum x + O(\sqrt{\sum x})$. Hence the credible probability $\to 0$. (The argument uses only $c > 1$, so it holds for every $b > 0$ — including $b = 1/n$, contrary to the text's "unless $b = 1/n$" aside, which appears to be an overstatement; numerically the probability decays to 0 for $b = 1/n$ as well.)

Exercise 9.31

Central idea. Condition on Y to get moments and mgf; the credible set's endpoints, rescaled, sit a factor $nb/(nb+1) < 1$ below where χ^2_{2Y} concentrates, so the standardized upper limit diverges to $-\infty$.

Solution. (a) $E(\chi^2_{2Y}) = E[E(\chi^2_{2Y} \mid Y)] = E(2Y) = 2\lambda$; $\text{Var} = E(4Y) + \text{Var}(2Y) = 4\lambda + 4\lambda = 8\lambda$. $M(t) = E[(1-2t)^{-Y}] = \exp(\lambda((1-2t)^{-1} - 1)) = \exp(-\lambda + \lambda/(1-2t))$. For the standardized variable, $\log M(t/\sqrt{8\lambda}) - t \cdot 2\lambda/\sqrt{8\lambda} = \lambda(2s + 4s^2 + o(s^2)) - t\sqrt{\lambda/2}$ with $s = t/\sqrt{8\lambda}$, $= t^2/2 + o(1) \to t^2/2$. (b) (9.2.22): $P(\frac{nb}{nb+1}\chi^2_{2(Y+a),1-\alpha/2} \le \chi^2_{2Y} \le \frac{nb}{nb+1}\chi^2_{2(Y+a),\alpha/2})$, $Y \sim \text{Poisson}(n\lambda)$. Standardizing χ^2_{2Y}, the upper

limit becomes

$$\frac{\frac{nb}{nb+1}\chi^2_{2(Y+a),\alpha/2} - 2\lambda}{\sqrt{8\lambda}} = \frac{-\frac{2\lambda}{nb+1} + O_p(\sqrt{\lambda})}{\sqrt{8\lambda}} \to -\infty,$$

since $\chi^2_{2(Y+a),\alpha/2} = 2(Y+a) + O_p(\sqrt{Y})$ and $Y/(n\lambda) \xrightarrow{p} 1$. Hence the coverage probability $\to P(Z \le -\infty) = 0$.

Exercise 9.32

Central idea. With prior $\theta \sim n(\mu, \tau^2)$, write $\gamma = \sigma^2/(n\tau^2)$; both the coverage of the Bayes set and the credibility of the classical set reduce to normal probabilities over Z-intervals whose centers drift to $\pm\infty$ in the unfavorable regimes.

Solution. From Example 9.2.18, $\delta^B(\bar{x}) = w\mu + (1-w)\bar{x}$, $\mathrm{Var}(\theta \mid \bar{x}) = \sigma^2\tau^2/(\sigma^2 + n\tau^2)$, $w = \sigma^2/(\sigma^2 + n\tau^2)$; put $\gamma = \sigma^2/(n\tau^2) = w/(1-w)$. (a) $\theta - \delta^B(\bar{X}) = w(\theta - \mu) - (1-w)\frac{\sigma}{\sqrt{n}}Z$. Dividing $|\theta - \delta^B| \le z_{\alpha/2}\sqrt{\mathrm{Var}}$ by $(1-w)\sigma/\sqrt{n}$ and using $\sqrt{\mathrm{Var}}/[(1-w)\sigma/\sqrt{n}] = \sqrt{1+\gamma}$ gives the stated Z-interval. (b) With $\tau = \sigma/\sqrt{n}$, $\gamma = 1$; the interval is $[-\sqrt{2}\, z_{\alpha/2} + (\theta-\mu)/(\sigma/\sqrt{n}),\ \sqrt{2}\, z_{\alpha/2} + (\theta-\mu)/(\sigma/\sqrt{n})]$. For fixed $\theta \ne \mu$, as $\sigma^2/n \to 0$ both endpoints $\to +\infty$ or both $\to -\infty$, so the probability $\to 0$: not a $1-\alpha$ confidence set. (c) If $\theta = \mu$, the probability is $P(|Z| \le \sqrt{1+\gamma}\, z_{\alpha/2}) = 2\Phi(\sqrt{1+\gamma}\, z_{\alpha/2}) - 1$, increasing in $\gamma \in (0, \infty)$ from $1-\alpha$ (as $\gamma \to 0$) to 1 (as $\gamma \to \infty$): bounded away from zero, min $1-\alpha$, max 1. (d) Under the posterior, $(\theta - \delta^B(\bar{x}))/\sqrt{\mathrm{Var}} \sim n(0,1)$; writing $\theta - \bar{x} = (\theta - \delta^B) - w(\bar{x} - \mu)$ and dividing by $\sqrt{\mathrm{Var}}$ gives the stated form (using $w/\sqrt{\mathrm{Var}} = \gamma/(\sqrt{1+\gamma}\,\sigma/\sqrt{n})$). As $|\bar{x}| \to \infty$ the shift $\to \pm\infty$, so the probability $\to 0$: not bounded away from zero, hence the classical set is not a $1-\alpha$ credible set.

Exercise 9.33

Central idea. For $\mu \ne 0$ only one side of $C_a(x)$ binds, reducing coverage to a one-sided normal tail $\Phi(1.645) = .95$; under the flat prior the posterior is $n(x, 1)$ and the credible probability interpolates between .90 (symmetric case) and .95.

Solution. (a) For $\mu > 0$: $\min\{0, X-a\} \le 0 < \mu$ always, so $\mu \in C_a \iff \mu \le \max\{0, X+a\} \iff X \ge \mu - a$: probability $\Phi(a) = .95$. For $\mu < 0$: $\max\{0, X+a\} \ge 0 > \mu$ always, so $\mu \in C_a \iff \min\{0, X-a\} \le \mu \iff X \le \mu + a$: probability $\Phi(a) = .95$. For $\mu = 0$: $0 \in C_a(x)$ always, coverage 1. (b) $\mu \mid x \sim n(x, 1)$. For $|x| \le a$: $C_a(x) = [x-a, x+a]$, credible probability $P(|Z| \le a) = 2\Phi(1.645) - 1 = .90$. For $x > a$: $C_a(x) = [0, x+a]$, probability $\Phi(a) - \Phi(-x) = .95 - \Phi(-x)$, increasing from .90 to .95 as $x \to \infty$; symmetrically for $x < -a$.

Exercise 9.34

Central idea. Known σ: deterministic length gives $n \ge (8z_{.025})^2$; unknown σ: the random length condition becomes a chi-squared probability.

Solution. (a) Length $= 2z_{.025}\sigma/\sqrt{n} \le \sigma/4 \iff n \ge (8 \times 1.96)^2 = 245.86$, so $n = 246$. (b) Length $= 2t_{n-1,.025}s/\sqrt{n} \le \sigma/4 \iff (n-1)s^2/\sigma^2 \le (n-1)n/(64t^2_{n-1,.025})$. Since $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, need $P(\chi^2_{n-1} \le (n-1)n/(64t^2_{n-1,.025})) \ge .90$, giving $n = 276$ ($n = 275$ gives $.8956 < .90$).

Exercise 9.35

Central idea. $E[\text{length}] = 2z_{\alpha/2}\sigma/\sqrt{n}$ versus $2t_{n-1,\alpha/2}E[S]/\sqrt{n}$; the t quantile exceeds z by more than $E[S]$ falls short of σ.

Solution. (a) $E[L] = 2z_{\alpha/2}\sigma/\sqrt{n}$ (nonrandom). (b) $E[L] = 2t_{n-1,\alpha/2}E[S]/\sqrt{n}$ with $E[S] = \sigma\sqrt{2/(n-1)}\,\Gamma(n/2)/\Gamma((n-1)/2)$. The ratio (b)/(a) is $\frac{t_{n-1,\alpha/2}}{z_{\alpha/2}} \cdot \frac{E[S]}{\sigma} > 1$, decreasing to 1 as $n \to \infty$ (e.g. $5.17, 1.95, 1.33, 1.12, 1.03$ for $n = 2, 3, 5, 10, 30$ at $\alpha = .05$): not knowing σ^2 always costs expected length.

Exercise 9.36

Central idea. $T-\theta \sim \text{Exp}(\text{rate } n(n+1)/2)$ is supported on $[0,\infty)$, so the shortest interval $[T+a, T+b]$ pushes b to 0 and puts all the probability at the left.

Solution. Joint pdf $= \exp(\theta\sum i - \sum x_i)\prod_i I_{[i\theta,\infty)}(x_i) = e^{-\sum x_i} e^{\theta n(n+1)/2} I_{[\theta,\infty)}(T)$; by factorization T is sufficient. $P(T > t) = \prod_i e^{i\theta - it} = e^{-(t-\theta)n(n+1)/2}$ $(t > \theta)$, so $W = T-\theta \sim \text{Exp}(m)$, $m = n(n+1)/2$. $P(\theta \in [T+a, T+b]) = P(-b \le W \le -a) = P(0 \le W \le -a)$ (needs $a \le 0 \le b$) $= 1 - e^{ma}$; length $b-a$ is minimized at $b = 0$, $a = (\log\alpha)/m$. Shortest: $[T + (\log\alpha)/m,\ T]$, length $-\log(\alpha)/m$.

Exercise 9.37

Central idea. $P(Y/\theta \le u) = u^n$ is free of θ; among $[Y/b, Y/a]$ with $b^n - a^n = 1-\alpha$, the length $Y(1/a - 1/b)$ decreases in a, forcing $a^n = \alpha$, $b = 1$.

Solution. $P_\theta(Y/\theta \le u) = P_\theta(Y \le u\theta) = u^n$, $0 \le u \le 1$: pivotal. A pivotal interval is $[Y/b, Y/a]$ with $P(a \le Q \le b) = b^n - a^n = 1-\alpha$, length $Y(1/a - 1/b)$. With $b = (a^n + 1 - \alpha)^{1/n}$, $\frac{d}{da}(1/a - 1/b) = -1/a^2 + a^{n-1}/b^{n+1} < 0$ (since $a < b$), so length decreases in a; max feasible $a = \alpha^{1/n}$ $(b = 1)$. Shortest: $[Y,\ Y/\alpha^{1/n}]$.

Exercise 9.38

Central idea. For fixed length, sliding the interval trades density at the endpoints; unimodality makes the total probability single-peaked in the slide position, maximized at equal endpoint densities.

Solution. (a) FTC: $\frac{d}{da}[F(a+c) - F(a)] = f(a+c) - f(a)$. (b) Let m be the mode. For $a + c < m$, f increasing gives $f(a+c) - f(a) > 0$; for $a > m$, < 0. On $(m-c, m)$, $h(a) = f(a+c) - f(a)$ has $h'(a) = f'(a+c) - f'(a) < 0$ (since $a + c > m > a$), with $h(m-c) > 0 > h(m)$: exactly one root, the unique maximizer. (c) If $[u, v]$, $v - u < c^*$, had $\int_u^v f = 1-\alpha$, then $\int_u^{u+c^*} f \ge 1-\alpha$, but (b) gives $\int_a^{a+c^*} f \le \int_{a^*}^{a^*+c^*} f = 1-\alpha$ with equality only at a^*; hence $\int_u^{u+c^*} f = 1-\alpha$, $u = a^*$, and $\int_u^v f < 1-\alpha$, contradiction. So $[a^*, a^* + c^*]$ is shortest.

Exercise 9.39

Central idea. Shortest $\Rightarrow f(a) = f(b)$ (9.38); symmetry about the mode then forces a, b equidistant from the mode, i.e. equal tails.

Solution. By 9.38/Theorem 9.3.2 the shortest interval satisfies $f(a) = f(b)$. Let m be the mode (= center of symmetry). $a \ge m$ is impossible (then f strictly decreasing gives $f(a) > f(b)$); similarly $b \le m$. So $a < m < b$, and $f(a) = f(2m-a) = f(b)$ with f strictly decreasing on $[m, \infty)$ gives $b = 2m - a$. Hence $P(X \le a) = P(X \ge 2m - a) = P(X \ge b)$; the two tails are equal, each $\alpha/2$.

Exercise 9.40

Central idea. Equal endpoint densities plus symmetry $\Rightarrow$ symmetric interval; apply to the likelihood (in θ) and to the posterior.

Solution. From 9.39, $b = 2m - a$: the optimal interval is $[m-k, m+k]$ with $k = m - a$. (a) If $L(\theta \mid x)$ is symmetric about the MLE $\hat\theta$, the likelihood region (9.2.7) $\{\theta : L(\theta \mid x) \ge kL(\hat\theta \mid x)\} = \{\theta : L(\theta \mid x) \ge k'\}$ is symmetric about $\hat\theta$ (provided the cutoff k' is on the likelihood scale, not parameter-distorted). (b) If $\pi(\theta \mid x)$ is symmetric about its mean (= mode), the HPD region $\{\theta : \pi(\theta \mid x) \ge k\}$ from Corollary 9.3.10 is symmetric about the posterior mean by the same argument.

Exercise 9.41

Central idea. For fixed length, probability mass is maximized by hugging 0 where the density is highest; hence any $[a, b]$, $a > 0$, can be slid left and shrunk.

Solution. (a) For fixed length c, $\int_0^c f \geq \int_a^{a+c} f$ for all $a \geq 0$ since f is decreasing. Given $[a,b]$ with $\int_a^b f = 1-\alpha$ and $c = b-a$: $\int_0^c f \geq 1-\alpha$, so shrinking to $c' \leq c$ with $\int_0^{c'} f = 1-\alpha$ gives a shorter valid interval. Hence $[0, b^*]$, $\int_0^{b^*} f = 1-\alpha$, is shortest. (b) In 9.2.13 the pivot $Q = n(Y-\mu) \sim \text{Exp}(1)$ has strictly decreasing density e^{-q} on $[0,\infty)$. By (a), the shortest $1-\alpha$ set for Q is $[0, -\log\alpha]$, i.e. $0 \leq n(Y-\mu) \leq -\log\alpha$: $[Y + (\log\alpha)/n,\ Y]$.

Exercise 9.42

Central idea. The β-interval has length $\propto 1/a - 1/b$, not $b-a$; differentiating the coverage constraint gives $db/da = f(a)/f(b)$, and the length FOC becomes $f(a)a^2 = f(b)b^2$.

Solution. (a) Minimize $\phi(a) = 1/a - 1/b(a)$ subject to $\int_a^{b(a)} f_Y = 1-\alpha$. Differentiating the constraint: $-f(a) + f(b)b'(a) = 0$, so $b'(a) = f(a)/f(b)$. $\phi'(a) = -1/a^2 + b'(a)/b^2 = -1/a^2 + f(a)/(f(b)b^2) = 0 \iff f(a)a^2 = f(b)b^2$. (b) $Y = X/\beta \sim \text{gamma}(k,1)$ is a pivot; $\{\beta : x/b \leq \beta \leq x/a\}$ has length $x(1/a - 1/b)$. By (a), the shortest $1-\alpha$ interval uses a, b with $\int_a^b f_Y = 1-\alpha$ and $f_Y(b)b^2 = f_Y(a)a^2$.

Exercise 9.43

Central idea. Neyman–Pearson logic: include t iff its "length cost" $g(t)$ is small relative to its "probability value" $f(t)$.

Solution. (a) For any C' with $\int_{C'} f \geq 1-\alpha = \int_C f$:

$$\int_{C'} g - \int_C g = \int_{C'\setminus C} g - \int_{C\setminus C'} g \geq \lambda \int_{C'\setminus C} f - \lambda \int_{C\setminus C'} f = \lambda\left(\int_{C'} f - \int_C f\right) \geq 0,$$

using $g \geq \lambda f$ off C and $g < \lambda f$ on C. So $C = \{g < \lambda f\}$ minimizes length. (b) 9.37: length $Y(1/a - 1/b) = Y\int_a^b t^{-2}dt$, so $g(t) = t^{-2}$, $f(t) = nt^{n-1}$ on $[0,1]$. $C = \{t^{-2} < \lambda n t^{n-1}\} = \{t > c\}$: $C = (c,1]$, $1 - c^n = 1-\alpha$, $c = \alpha^{1/n}$ — the interval $[Y, Y/\alpha^{1/n}]$. 9.42: length $x\int_a^b t^{-2}dt$, $g(t) = t^{-2}$, $f = \text{gamma}(k,1)$ pdf; $C = \{t : t^{-2} < \lambda f(t)\}$ is an interval $[a,b]$ (unimodality) with $1/a^2 = \lambda f(a)$, $1/b^2 = \lambda f(b) \iff f(a)a^2 = f(b)b^2$, as in 9.42(a).

Exercise 9.44

Central idea. Inverting a UMP test's acceptance region yields a UMA confidence set; MLR gives the test direction (large $Y = \sum X_i$ for Poisson; small x for logistic with $H_1 : \theta < \theta_0$).

Solution. (a) $Y = \sum X_i \sim \text{Poisson}(n\lambda)$ has MLR; UMP level-α test rejects for large Y. $A(\lambda_0) = \{Y \leq c(\lambda_0)\}$ (with randomization); since $c(\lambda_0)$ increases in λ_0, $C(y) = \{\lambda_0 : y \in A(\lambda_0)\} = [\underline{\lambda}(y),\ \infty)$ where $\underline{\lambda}(y)$ solves $P_{\underline{\lambda}}(Y \geq y) = \alpha$. Via $P_\lambda(Y \geq y) = P(\chi^2_{2y} \leq 2n\lambda)$: $C(y) = [\chi^2_{2y,1-\alpha}/(2n),\ \infty)$, UMA (a lower confidence bound, as befits a one-sided test against $\lambda > \lambda_0$). (b) The logistic location family is log-concave, hence MLR; the UMP test of $H_0 : \theta = \theta_0$ vs $H_1 : \theta < \theta_0$ rejects for small x: $A(\theta_0) = \{x > \theta_0 + c_0\}$ with $P_0(X \leq c_0) = \alpha$. Inverting, $C(x) = \{\theta_0 : \theta_0 < x - c_0\} = (-\infty,\ x - c_0]$, UMA. For standard logistic $F(t) = 1/(1+e^{-t})$, $c_0 = \log(\alpha/(1-\alpha))$, so $U(x) = x + \log((1-\alpha)/\alpha)$.

Exercise 9.45

Central idea. *Erratum:* as printed, C^* has coverage α, not $1-\alpha$ — the cutoff must be $\chi^2_{2n,1-\alpha}$ (and the "$\chi^2_{240,.7} = 251.046$" in (d) is really $\chi^2_{240,.3}$). With the correction, Madansky's minimum-based interval beats the UMA interval in expected length.

Solution. (exponential(λ) = mean λ, pdf $\lambda^{-1}e^{-x/\lambda}$.) (a) MLR in $-\sum X_i$: UMP size-α test rejects for small $\sum X_i$, i.e. reject iff $2\sum X_i/\lambda_0 < \chi^2_{2n,1-\alpha}$ (lower-α quantile). (b) *Erratum.* Inverting $A(\lambda_0) = \{2\sum X_i/\lambda_0 \geq \chi^2_{2n,1-\alpha}\}$ gives the UMA interval $\{\lambda : 0 < \lambda \leq 2\sum X_i/\chi^2_{2n,1-\alpha}\}$ — with $\chi^2_{2n,1-\alpha}$, not $\chi^2_{2n,\alpha}$ as printed. Proof:

as printed, $P_\lambda(\lambda \le 2\sum X_i/\chi^2_{2n,\alpha}) = P(\chi^2_{2n} \ge \chi^2_{2n,\alpha}) = \alpha \ne 1-\alpha$. (c) $E[\text{Length}] = 2n\lambda/\chi^2_{2n,1-\alpha}$. (d) With $1-\alpha = .3$: the correct cutoff is $\chi^2_{240,.3} = 251.046$ (the text mislabels it $\chi^2_{240,.7}$; the true upper-.7 cutoff is $\approx$ 228.05). $E[\text{Length}(C^*)] = 240\lambda/251.046 = .956\lambda$. For C^M: $X_{(1)} \sim \text{Exp}(\text{mean } \lambda/n)$, $E[\text{Length}] = \frac{\lambda/n}{-\log .99} = \frac{\lambda}{120(0.0100503)} = .829\lambda < .956\lambda$. Coverage is actually $0.99^{120} = 0.2993803913$, slightly below 0.3. Thus this rounded interval is only approximately a 30% interval. For exact 30% coverage, use $C^M = \{0 < \lambda \le X_{(1)}/d\}$ with $d = -\log(0.3)/120 = 0.01003310670$. Its expected length is $\lambda/[-\log(0.3)] = 0.8305835451\lambda$, still shorter than the UMA interval.

Exercise 9.46

Central idea. Inversion swaps the roles: $P_\theta(\theta' \in C(X)) = P_\theta(X \in A(\theta'))$, so unbiasedness of the test ($\le 1-\alpha$ acceptance of false values) is exactly unbiasedness of the set ($\le 1-\alpha$ false coverage).

Solution. $P_\theta(\theta \in C(X)) = P_\theta(X \in A(\theta)) \ge 1-\alpha$: a $1-\alpha$ set. For $\theta' \ne \theta$: $P_\theta(\theta' \in C(X)) = P_\theta(X \in A(\theta')) \le 1-\alpha$ since $A(\theta')$ is unbiased level α. Hence C is unbiased.

Exercise 9.47

Central idea. For one-sided bounds unbiasedness is one-sided: an upper bound need only cover false values *above* θ with probability $\le 1-\alpha$ — which follows from the unbiasedness of the one-sided UMP test it inverts.

Solution. For the upper bound $C = (-\infty, \bar X + z_\alpha\sigma/\sqrt{n}]$, the relevant false values are $\theta' > \theta$: $P_\theta(\theta' \in C) = P_\theta(\bar X \ge \theta' - z_\alpha\sigma/\sqrt{n}) = 1 - \Phi((\theta'-\theta)/(\sigma/\sqrt{n}) - z_\alpha) \le 1 - \Phi(-z_\alpha) = 1-\alpha$. Similarly the lower bound $[\bar X - z_\alpha\sigma/\sqrt{n}, \infty)$ satisfies $P_\theta(\theta' \in C) = \Phi((\theta'-\theta)/(\sigma/\sqrt{n}) + z_\alpha) \le 1-\alpha$ for $\theta' < \theta$. (Equivalently, each inverts a UMP — hence unbiased — one-sided test.)

Exercise 9.48

Central idea. The LRT statistic is monotone in $t = \sqrt{n}(\bar x - \theta_0)/s$; one- and two-sided inversions give the one- and two-sided t-intervals, and both LRTs are unbiased tests.

Solution. (a) The LRT of $H_0 : \theta = \theta_0$ vs $H_1 : \theta < \theta_0$ rejects for small $t = \sqrt{n}(\bar x - \theta_0)/s$: $-2\log\lambda = n\log(1 + t^2/(n-1))$, reject iff $t \le -t_{n-1,\alpha}$. $A(\theta_0) = \{\theta_0 \le \bar x + t_{n-1,\alpha}s/\sqrt{n}\}$; inversion is exact. (b) For $H_1 : \theta \ne \theta_0$, reject iff $|t| > t_{n-1,\alpha/2}$; $A(\theta_0) = \{|\bar x - \theta_0| \le t_{n-1,\alpha/2}s/\sqrt{n}\}$, inverting gives (9.2.14). (c) The one-sided LRT is UMP, hence unbiased; inverting gives the unbiased bound as in 9.47. The two-sided t-test is unbiased (Example 8.3.20/Exercise 8.38); by 9.46 the inverted interval is unbiased.

Exercise 9.49

Central idea. Conditioning on the observed σ, test (a) is sensible; test (b) gains unconditional power by always rejecting when $\sigma = 10$ — but its inverted "confidence set" is empty whenever $\sigma = 10$ is observed.

Solution. (a) Conditionally on σ, $P_{\theta_0}(X > \theta_0 + z_\alpha\sigma \mid \sigma) = \alpha$, so unconditionally level α. $A(\theta_0) = \{X \le \theta_0 + z_\alpha\sigma\}$; $C(x,\sigma) = \{\theta_0 : \theta_0 \ge x - z_\alpha\sigma\} = [x - z_\alpha\sigma, \infty)$, a $1-\alpha$ lower bound using the *observed* σ. **Audit note.** The exercise asks to "show that" test (b) is more powerful, but the power comparison is invalid: because $(\alpha-p)/(1-p) < \alpha$, the cutoff $z_{(\alpha-p)/(1-p)}$ is *larger* than z_α, not smaller. Numerically ($\alpha = .1$, $p = .05$), test (a) has the higher power for small shifts ($\delta = .25$: .149 vs .131; $\delta = 1$: .376 vs .304; $\delta = 2$: .733 vs .666) while test (b) wins only for large shifts ($\delta = 3$: .917 vs .920; $\delta = 5$: .961 vs 1.000) — the powers cross, so neither test uniformly dominates. The level calculation and the empty-set inversion below are correct.

(b) Level: $p \cdot 1 + (1-p)P_{\theta_0}(X > \theta_0 + z_{(\alpha-p)/(1-p)} \mid \sigma = 1) = p + (\alpha - p) = \alpha$. Inverting: $A(\theta_0) = \{\sigma = 1,\ X \le \theta_0 + z_{(\alpha-p)/(1-p)}\}$, so $C = [x - z_{(\alpha-p)/(1-p)}, \infty)$ if $\sigma = 1$ and $C = \emptyset$ if $\sigma = 10$ — empty with probability $p > 0$, yet $P_\theta(\theta \in C) = (1-p)(1 - \frac{\alpha-p}{1-p}) = 1-\alpha$.

Exercise 9.50

Central idea. A bijection of the parameter space transports coverage and false-coverage probabilities unchanged, hence preserves UMA; injectivity is exactly what is needed.

Solution. (a) $C^*(x) = \{1/\theta : a(x) \le \theta \le b(x)\}$: if D^* is any $1-\alpha$ set for $\eta = 1/\theta$, then $D = \{1/\eta : \eta \in D^*\}$ is $1-\alpha$ for θ, and $P_\eta(\eta' \in D^*) = P_\theta(\theta' \in D) \ge P_\theta(\theta' \in C) = P_\eta(\eta' \in C^*)$ for $\eta' \ne \eta$ by UMA of C. If $a(x) < 0 < b(x)$, $\theta \mapsto 1/\theta$ maps $[a,b]$ to $(-\infty, 1/a] \cup [1/b, \infty)$ — neither an interval nor bounded. (b) h strictly increasing is a bijection: the same transport argument gives UMA of $\{h(\theta) : h(a(X)) \le h(\theta) \le h(b(X))\}$. The condition can be relaxed to h strictly monotone (increasing or decreasing); without injectivity UMA need not be preserved (e.g. $h(\theta) = \theta^2$ folds $\pm\theta$ together, scrambling false-coverage probabilities).

Exercise 9.51

Central idea. $\bar{X} - \theta$ is pivotal in a location family, so the coverage event depends on θ only through $\bar{X} - \theta$.

Solution. $P_\theta(\theta \in C) = P_\theta(\bar{x} - k_1 \le \theta \le \bar{x} + k_2) = P_\theta(-k_2 \le \bar{X} - \theta \le k_1)$. Since $\bar{X} - \theta$ has pdf $f_{\bar{X}}(\cdot)$ free of θ, this equals $\int_{-k_2}^{k_1} f_{\bar{X}}(t)\,dt$, independent of θ.

Exercise 9.52

Central idea. All four are "minimize something subject to $\int_a^b f_{n-1} = 1-\alpha$"; the Lagrange conditions collapse to $f_{n+k}(a) = f_{n+k}(b)$ for $k = 2, 3, 1$ because $t^m f_{n-1}(t) \propto f_{n-1+2m}(t)$.

Solution. Write $W = (n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$. (a) $-2\log\lambda = W - n\log W + n\log n - n \equiv g(W)$; $\{g(W) \le c\} = [a,b]$ with $g(a) = g(b) \iff a - n\log a = b - n\log b \iff a^{n/2}e^{-a/2} = b^{n/2}e^{-b/2} \iff f_{n+2}(a) = f_{n+2}(b)$. (b) Minimize $(n-1)s^2(1/a - 1/b)$ s.t. $\int_a^b f_{n-1} = 1-\alpha$: Lagrange gives $1/a^2 = \lambda f_{n-1}(a)$, $1/b^2 = \lambda f_{n-1}(b)$, i.e. $a^2 f_{n-1}(a) = b^2 f_{n-1}(b) \iff f_{n+3}(a) = f_{n+3}(b)$. (c) False coverage $h(t) = P(a/t \le \chi^2_{n-1} \le b/t)$ $(t = \sigma^2/\sigma_0^2)$; unbiasedness $\Rightarrow h'(1) = 0 \iff a f_{n-1}(a) = b f_{n-1}(b) \iff f_{n+1}(a) = f_{n+1}(b)$. Minimizing the endpoint ratio b/a s.t. coverage gives the same FOC. (d) By definition of equal tails. (e) $n = 3$ (χ^2_2), $\alpha = .1$ (lengths $\times (n-1)s^2$): (a) LRT: $a = .2065$, $b = 12.5214$, length 4.76; (b) minimum-length: $a = .2104$, $b = 18.0077$, length 4.70; (c) shortest-unbiased: $a = .1676$, $b = 7.8643$, length 5.84; (d) equal-tail: $a = .1026$, $b = 5.9915$, length 9.58. So the order is minimum-length < LRT < shortest-unbiased < equal-tail; the equal-tail interval is the longest of the four.

Exercise 9.53

Central idea. The risk trades deterministic length $2bc\sigma$ against coverage $P(|Z| \le c)$; the FOC $2b\sigma = 2\phi(c)$ either has no positive solution (point estimator wins) or recovers the z-interval.

Solution. (a) $R = bE[\text{Length}] - P_\mu(\mu \in C(X)) = 2bc\sigma - P(|X-\mu|/\sigma \le c) = b(2c\sigma) - P(-c \le Z \le c)$. (b) $\frac{d}{dc}P(-c \le Z \le c) = 2\phi(c) = \frac{2}{\sqrt{2\pi}}e^{-c^2/2}$; increasing in $c \ge 0$ since ϕ decreases. (c) At $c = 0$ the derivative is $2b\sigma - 2/\sqrt{2\pi} > 0$ and increasing, so R is strictly increasing: minimum at $c = 0$. (d) $2b\sigma = \frac{2}{\sqrt{2\pi}}e^{-c^2/2} \iff c^2 = -2\log(b\sigma\sqrt{2\pi})$. With b such that $c = z_{\alpha/2}$, $C(x) = [x \pm z_{\alpha/2}\sigma]$ is the usual $1-\alpha$ interval.

Exercise 9.54

Central idea. $(X-\mu)/S \sim t_v$ replaces Z; the FOC $f_{t_v}(c) = bM$ is the t analogue of $2b\sigma = 2\phi(c)$, and $t_v \to$ normal as $v \to \infty$.

Solution. (a) $R = \frac{b}{\sigma}2cE[S] - P_\mu(|X-\mu| \le cS) = 2bcM - P(|T| \le c)$ with $T = \frac{(X-\mu)/\sigma}{S/\sigma} \sim t_v$; $P(|T| \le c) = 2P(T \le c) - 1$. (b) $\frac{dR}{dc} = 2bM - 2f_{t_v}(c) = 0$. With $f_{t_v}(c) = \frac{\Gamma((v+1)/2)}{\sqrt{v\pi}\,\Gamma(v/2)}(1 + c^2/v)^{-(v+1)/2}$ and $M = \sqrt{2/v}\,\Gamma((v+1)/2)/\Gamma(v/2)$, $f_{t_v}(c) = bM$ simplifies to $b = \frac{1}{\sqrt{2\pi}}(\frac{v}{v+c^2})^{(v+1)/2}$. (c) As $v \to \infty$: $(\frac{v}{v+c^2})^{(v+1)/2} = (1 + c^2/v)^{-(v+1)/2} \to e^{-c^2/2}$, $M \to 1$, $T \xrightarrow{d} Z$. The FOC becomes $b = \frac{1}{\sqrt{2\pi}}e^{-c^2/2}$, i.e. $2b = \frac{2}{\sqrt{2\pi}}e^{-c^2/2}$ — exactly 9.53's FOC with $b\sigma$ replaced by b, as the loss here scales length by b/σ instead of b.

Exercise 9.55

Central idea. $R(C) - R(C') = E[I(s \geq K)(2bcs - P(|X - \mu| \leq cs \mid s))] \geq (2bcK - 1)P(s \geq K)$: when s is large the length penalty dominates any possible coverage gain, so deleting the interval there *reduces* risk — the loss is at fault, not the t-interval.

Solution. $R(C) - R(C') = b \cdot 2cE[SI(S \geq K)] - P(|X - \mu| \leq cS,\ S \geq K) = E[I(S \geq K)(2bcS - P(|X - \mu| \leq cS \mid S))]$. Since $P(|X - \mu| \leq cS \mid S) \leq 1$ and $S \geq K$ on the event,

$$R(C) - R(C') \geq (2bcK - 1)P(S \geq K) \geq 0,$$

with strict inequality for every (μ, σ) (as $P(S \geq K) > 0$) whenever $K > 1/(2bc)$, e.g. $K = 1/(bc)$. Thus the "do exactly the wrong thing" rule C' dominates C — an indictment of the loss function, which over-penalizes length when s is large.

Exercise 9.56

Central idea. Minimize posterior expected loss pointwise: include θ iff its posterior density exceeds the length price b.

Solution. Posterior expected loss: $E[L(\theta, C) \mid x] = b \operatorname{Len}(C) - P(\theta \in C \mid x) = \int_C [b - \pi(\theta \mid x)]\, d\theta$. This is minimized by including exactly those θ with $b - \pi(\theta \mid x) < 0$: $C^\pi = \{\theta : \pi(\theta \mid x) \geq b\}$ (the Neyman–Pearson argument).

Exercise 9.57

Central idea. Prediction adds the future observation variance. Tolerance requires a separate population-content parameter p, not $p = 1 - \alpha$.

Solution. (a) A future observation is independent of the sample, and $X_{n+1} - \bar{X} \sim N(0, \sigma^2(1 + 1/n))$. Thus the prediction interval is $\bar{X} \pm z_{\alpha/2}\sigma\sqrt{1 + 1/n}$, with upper-tail quantile notation. (b) The displayed statement cannot hold for arbitrary p because its width does not depend on p. A valid conservative replacement is

$$\bar{X} \pm \sigma \left[z_{(1-p)/2} + \frac{z_{\alpha/2}}{\sqrt{n}} \right].$$

The fixed interval $[\mu - \sigma z_{(1-p)/2}, \mu + \sigma z_{(1-p)/2}]$ contains population probability p. On the event $|\bar{X} - \mu| \leq \sigma z_{\alpha/2}/\sqrt{n}$, which has probability $1 - \alpha$, the random interval contains that fixed interval. Hence with confidence at least $1 - \alpha$ it contains at least proportion p of the population. The earlier answer solved only the special case $p = 1 - \alpha$. The first-printing errata print $z_{p/2}$; with the statement's content convention p, the required cutoff is $z_{(1-p)/2}$. (c) For unknown variance, $(X_{n+1} - \bar{X})/(S\sqrt{1 + 1/n}) \sim t_{n-1}$, giving $\bar{X} \pm t_{n-1,\alpha/2} S\sqrt{1 + 1/n}$.

Exercise 9.58

Central idea. Only the signs $P(X_i \lessgtr m) = 1/2$ matter (binomial); prediction uses symmetry of ranks among $n + 1$ iid variables; tolerance uses $U_i = F(X_i) \sim U(0, 1)$.

Solution. (a) $P(m \leq X_{(n)}) = 1 - P(\text{all } X_i < m) = 1 - (1/2)^n$; similarly $P(m \geq X_{(1)}) = 1 - (1/2)^n$; $P(X_{(1)} \leq m \leq X_{(n)}) = 1 - 2(1/2)^n$ (equalities for continuous F). (b) Among $n+1$ iid (continuous) variables all rank orders are equally likely: $P(X_{n+1} \leq X_{(n)}) = n/(n+1)$, $P(X_{(1)} \leq X_{n+1} \leq X_{(n)}) = (n-1)/(n+1)$. (c) With continuous F, $U_i = F(X_i)$ iid $U(0, 1)$. One-sided: $P(F(X_{(n)}) \geq p) = 1 - P(\text{all } U_i < p) = 1 - p^n$. Two-sided: $P(U_{(n)} - U_{(1)} \geq p) = 1 - \iint_{v-u<p} n(n-1)(v-u)^{n-2} du dv = 1 - [np^{n-1} - (n-1)p^n] = 1 - p^n - n(1-p)p^{n-1}$.

Chapter 10

Exercise 10.1

Central idea. The mean is linear in θ: $E_\theta X = \theta/3$, so $3\bar{X}$ is unbiased for θ and consistent by the WLLN.

Solution. $E_\theta X_1 = \int_{-1}^{1} x \cdot \frac{1}{2}(1+\theta x)\,dx = \frac{1}{2}\left(0 + \theta \cdot \frac{2}{3}\right) = \theta/3$. Hence $\tilde{\theta} = 3\bar{X}$ satisfies $E_\theta \tilde{\theta} = \theta$. Moreover $E_\theta X_1^2 = \int_{-1}^{1} x^2 \cdot \frac{1}{2}(1+\theta x)\,dx = 1/3 < \infty$, so $\mathrm{Var}_\theta(X_1) < \infty$ and the WLLN gives $\bar{X} \xrightarrow{p} \theta/3$. By continuity, $3\bar{X} \xrightarrow{p} \theta$; i.e., $3\bar{X}$ is consistent for θ (and unbiased).

Exercise 10.2

Central idea. Split $|a_n W_n + b_n - \theta| \le |a_n||W_n - \theta| + |a_n - 1||\theta| + |b_n|$; the deterministic pieces vanish, leaving only $|W_n - \theta|$.

Solution. For any $\eta > 0$,

$$|U_n - \theta| = |a_n(W_n - \theta) + (a_n - 1)\theta + b_n| \le |a_n||W_n - \theta| + |a_n - 1||\theta| + |b_n|.$$

Since $a_n \to 1$, $|a_n| \le 2$ eventually; choose N so that for $n \ge N$, $|a_n - 1||\theta| < \eta$ and $|b_n| < \eta$. Then for $n \ge N$,

$$\{|U_n - \theta| > 4\eta\} \subseteq \{2|W_n - \theta| > 2\eta\} = \{|W_n - \theta| > \eta\},$$

so $P(|U_n - \theta| > 4\eta) \le P(|W_n - \theta| > \eta) \to 0$ by consistency of W_n. Since $\eta > 0$ is arbitrary, $U_n \xrightarrow{p} \theta$.

Exercise 10.3

Central idea. The log-likelihood derivative factors as $n(W - \theta - \theta^2)/(2\theta^2)$; only the positive root is admissible, and the approximate variance is $1/(nI(\theta))$.

Solution. (a) With $S(\theta) = \sum(x_i - \theta)^2 = nW - 2n\theta\bar{x} + n\theta^2$,

$$l(\theta) = -\tfrac{n}{2}\log\theta - \frac{S(\theta)}{2\theta} + c, \qquad \frac{dl}{d\theta} = -\frac{n}{2\theta} + \frac{n(W - \theta^2)}{2\theta^2} = \frac{n\,(W - \theta - \theta^2)}{2\theta^2}.$$

Setting this to zero gives $\theta^2 + \theta - W = 0$, with roots $\frac{-1 \pm \sqrt{1+4W}}{2}$. Since $W > 0$ a.s. and $\theta > 0$, the MLE is the positive root

$$\hat{\theta} = \frac{-1 + \sqrt{1 + 4W}}{2}$$

(the negative root is < 0; $l \to -\infty$ as $\theta \downarrow 0$ or $\theta \to \infty$, so this critical point is the global maximizer). (b) From $l'(\theta) = n(W - \theta - \theta^2)/(2\theta^2)$,

$$l''(\theta) = \frac{n}{2} \cdot \frac{(-\theta - 2\theta^2)\theta - 2(W - \theta - \theta^2)\theta}{\theta^4} \cdot \theta = \frac{n}{2} \cdot \frac{-\theta - 2\theta^2 - 2(W - \theta - \theta^2)}{\theta^3}.$$

Since $E_\theta W = \theta + \theta^2$, $E_\theta l''(\theta) = \frac{n}{2} \cdot \frac{-\theta - 2\theta^2}{\theta^3} = -\frac{n(1+2\theta)}{2\theta^2}$, so $I(\theta) = \frac{1+2\theta}{2\theta^2}$ and

$$\mathrm{Var}(\hat{\theta}) \approx \frac{1}{nI(\hat{\theta})} = \frac{2\hat{\theta}^2}{n(2\hat{\theta} + 1)}.$$

(The delta method applied to W, using $\hat{\theta} = g(W)$ with $g(w) = (-1 + \sqrt{1+4w})/2$, gives the same value.)

Exercise 10.4

Central idea. Each is a smooth function of sample means, so the delta method applies; for (a) the cross terms cancel beautifully, leaving the familiar OLS variance $\sigma^2/(nEX^2)$.

Solution. Write each estimator as $g(\bar{U}, \bar{V})$ and use $\mathrm{Var} \approx \nabla g^T \Sigma \nabla g / n$ with Σ the one-observation covariance. (a) $T_a = \overline{XY}/\overline{X^2}$, $E[XY] = \beta(\mu^2+\tau^2)$, $E[X^2] = \mu^2+\tau^2$: approximate mean β. With $U = XY$, $V = X^2$: $\mathrm{Var}(V) = 2\tau^2(2\mu^2+\tau^2)$, $\mathrm{Var}(U) = \beta^2\mathrm{Var}(V) + (\mu^2+\tau^2)\sigma^2$, $\mathrm{Cov}(U,V) = \beta\,\mathrm{Var}(V)$ (using independence of X, ε). At $(\beta(\mu^2+\tau^2), \mu^2+\tau^2)$, $\nabla g = (1, -\beta)/(\mu^2+\tau^2)$, and

$$\nabla g^T \Sigma \nabla g = \frac{\mathrm{Var}(U) - 2\beta\,\mathrm{Cov}(U,V) + \beta^2\mathrm{Var}(V)}{(\mu^2+\tau^2)^2} = \frac{(\mu^2+\tau^2)\sigma^2}{(\mu^2+\tau^2)^2},$$

so $\mathrm{Var}(T_a) \approx \dfrac{\sigma^2}{n(\mu^2+\tau^2)}$. (b) $T_b = \bar{Y}/\bar{X}$: approximate mean $\beta\mu/\mu = \beta$. With $U = Y$, $V = X$: $\mathrm{Var}(U) = \beta^2\tau^2 + \sigma^2$, $\mathrm{Var}(V) = \tau^2$, $\mathrm{Cov}(U,V) = \beta\tau^2$; $\nabla g = (1, -\beta)/\mu$ gives $\nabla g^T \Sigma \nabla g = \sigma^2/\mu^2$, so $\mathrm{Var}(T_b) \approx \dfrac{\sigma^2}{n\mu^2}$. (c) $T_c = \frac{1}{n}\sum Y_i/X_i$ is a sample mean. Formally $E[Y_i/X_i] = \beta + E[\varepsilon_i/X_i] = \beta$ by independence ($E\varepsilon_i = 0$), and the ratio-formula $E[Y/X] \approx \beta$ (second-order terms cancel), $\mathrm{Var}(Y/X) \approx \sigma^2/\mu^2$, so $\mathrm{Var}(T_c) \approx \dfrac{\sigma^2}{n\mu^2}$. *Caveat:* this is formal — for normal X, $E|1/X| = \infty$, so Y/X has Cauchy-like tails, infinite variance, and the CLT does not strictly apply; the delta-method value should be read as a large-sample heuristic. The ranking is nonetheless clear: (a), the OLS estimator, has the smallest approximate variance.

Exercise 10.5

Central idea. $\bar{X}_n$ has a density bounded away from zero near 0, so $E[1/\bar{X}_n^2] = \infty$; truncating away from 0 bounds T_n, and for $\delta < |\mu|$ the bad region has vanishing probability since $\bar{X}_n \xrightarrow{p} \mu$.

Solution. Here X_i are iid $n(\mu, \sigma^2)$, so $\bar{X}_n$ has a continuous density f_n with $f_n(0) > 0$. (a) $E[T_n^2] = nE[1/\bar{X}_n^2] \geq n \min_{|x|\leq\delta} f_n(x) \int_{-\delta}^{\delta} x^{-2}dx = \infty$ for any $\delta > 0$. Hence $\mathrm{Var}(T_n) = \infty$. (b) Deleting $(-\delta, \delta)$ forces $|\bar{X}_n| \geq \delta$ a.s., so $T_n^2 = n/\bar{X}_n^2 \leq n/\delta^2$ is bounded; a bounded variable has finite variance. (c) For fixed $\delta < |\mu|$, $\{|\bar{X}_n| < \delta\} \subseteq \{|\bar{X}_n - \mu| > |\mu| - \delta\}$, and $P(|\bar{X}_n - \theta| > |\mu| - \delta) \to 0$ since $\bar{X}_n \xrightarrow{p} \mu$. Thus $P(\bar{X}_n \in (-\delta, \delta)) \to 0$: the region driving the infinite variance has vanishing probability. (For $\delta \geq |\mu|$ this fails — the claim is about a small neighborhood of 0.)

Exercise 10.6

Central idea. Condition on the mixture indicator and use total variance; the cdf is a mixture whose contaminating piece vanishes because $1 - p_n \to 0$ — convergence in distribution with exploding variance.

Solution. Write $Y_n \mid W_n = 1 \sim n(0,1)$, $Y_n \mid W_n = 0 \sim n(0, \sigma_n^2)$ with $P(W_n = 1) = p_n$. (a) $EY_n = E[E[Y_n \mid W_n]] = 0$. By total variance, $\mathrm{Var}(Y_n) = E[\mathrm{Var}(Y_n \mid W_n)] + \mathrm{Var}(E[Y_n \mid W_n]) = p_n \cdot 1 + (1-p_n)\sigma_n^2 + 0$. (b) $P(Y_n < a) = p_n P(Z < a) + (1-p_n)P(Z < a/\sigma_n)$. Since $a/\sigma_n \to 0$, $P(Z < a/\sigma_n) \to 1/2$, and $(1-p_n) \to 0$, the second term vanishes while $p_n P(Z < a) \to P(Z < a)$. Hence $Y_n \xrightarrow{d} n(0,1)$ even though $\mathrm{Var}(Y_n) = p_n + (1-p_n)\sigma_n^2 \to \infty$: the asymptotic variance (1) and the limiting variance (∞) disagree.

Exercise 10.7

Central idea. Consistency is the continuous mapping theorem (Theorem 10.1.6); efficiency is the delta method, whose asymptotic variance is exactly the Cramér–Rao bound for $\tau(\theta)$.

Solution. Consistency: $\hat{\theta} \xrightarrow{p} \theta$ and τ continuous imply $\tau(\hat{\theta}) \xrightarrow{p} \tau(\theta)$ (this is Theorem 10.1.6). For efficiency,

the proof of Theorem 10.1.12 gives $\sqrt{n}(\hat{\theta}-\theta) \xrightarrow{d} n(0, 1/I(\theta))$. By the delta method (Theorem 5.5.24),

$$\sqrt{n}(\tau(\hat{\theta})-\tau(\theta)) \xrightarrow{d} n\left(0, \frac{[\tau'(\theta)]^2}{I(\theta)}\right),$$

and $[\tau'(\theta)]^2/I(\theta)$ is precisely the Cramér–Rao lower bound for estimators of $\tau(\theta)$ (Theorem 7.3.9 applied to τ). Hence $\tau(\hat{\theta})$ attains the bound: it is asymptotically efficient. *Remark:* the efficiency step uses the delta method, which needs τ differentiable at θ (with $\tau'(\theta) \neq 0$ for the stated normal limit); continuity alone gives only the consistency part.

Exercise 10.8

Central idea. The score is a sum of iid mean-zero terms (regularity lets the derivative pass under the integral); the CLT and WLLN do the rest.

Solution. (a) $l'(\theta_0 \mid X) = \sum_i \frac{f'(X_i|\theta_0)}{f(X_i|\theta_0)} = \sum_i W_i$, so $\frac{1}{\sqrt{n}} l'(\theta_0 \mid X) = \sqrt{n} \cdot \frac{1}{n}\sum_i W_i$. Now $E_{\theta_0} W_1 = \int \frac{f'(x|\theta_0)}{f(x|\theta_0)} f(x \mid \theta_0)\, dx = \frac{d}{d\theta}\int f(x \mid \theta)\, dx\big|_{\theta_0} = 0$ by regularity, and $\mathrm{Var}(W_1) = E_{\theta_0} W_1^2 = I(\theta_0)$. By the CLT, $\frac{1}{\sqrt{n}}\sum_i W_i \xrightarrow{d} n[0, I(\theta_0)]$. (b) Differentiating $l'(\theta \mid X) = \sum_i f'(X_i \mid \theta)/f(X_i \mid \theta)$,

$$l''(\theta_0 \mid X) = \sum_i \left[\frac{f''(X_i \mid \theta_0)}{f(X_i \mid \theta_0)} - \left(\frac{f'(X_i \mid \theta_0)}{f(X_i \mid \theta_0)}\right)^2\right] = \sum_i \frac{f''(X_i \mid \theta_0)}{f(X_i \mid \theta_0)} - \sum_i W_i^2,$$

so $-\frac{1}{n} l''(\theta_0 \mid X) = \frac{1}{n}\sum_i W_i^2 - \frac{1}{n}\sum_i f''(X_i \mid \theta_0)/f(X_i \mid \theta_0)$. Moreover $E_{\theta_0} W_1^2 = I(\theta_0)$ and $E_{\theta_0}[f''(X_1 \mid \theta_0)/f(X_1 \mid \theta_0)] = \frac{d^2}{d\theta^2}\int f(x \mid \theta)\, dx\big|_{\theta_0} = 0$. By the WLLN the two averages converge in probability to $I(\theta_0)$ and 0, so $-\frac{1}{n} l''(\theta_0 \mid X) \xrightarrow{p} I(\theta_0)$.

Exercise 10.9

Central idea. $T = \sum X_i$ is complete sufficient; Rao–Blackwellize the naive unbiased indicators $I(X_1 = 0)$, $I(X_1 = 1)$ onto T. Both UMVUEs turn out asymptotically efficient (ARE = 1).

Solution. $T = \sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda)$ is complete sufficient. (a) $E[I(X_1 = 0)] = e^{-\lambda}$. The UMVUE is $E[I(X_1 = 0) \mid T = t] = P(X_1 = 0 \mid T = t) = \frac{e^{-\lambda} e^{-(n-1)\lambda}((n-1)\lambda)^t/t!}{e^{-n\lambda}(n\lambda)^t/t!} = \left(\frac{n-1}{n}\right)^t$. So $\delta_1(T) = \left(\frac{n-1}{n}\right)^T$. (b) $E[I(X_1 = 1)] = \lambda e^{-\lambda}$. Similarly $P(X_1 = 1 \mid T = t) = \frac{\lambda e^{-\lambda} e^{-(n-1)\lambda}((n-1)\lambda)^{t-1}/(t-1)!}{e^{-n\lambda}(n\lambda)^t/t!} = \frac{t}{n}\left(\frac{n-1}{n}\right)^{t-1}$ (and 0 for $t = 0$). So $\delta_2(T) = \frac{T}{n}\left(\frac{n-1}{n}\right)^{T-1}$. (c) The MLEs are $e^{-\bar{X}}$ and $\bar{X}e^{-\bar{X}}$. By the delta method their asymptotic variances are the Cramér–Rao bounds: for $\tau_1(\lambda) = e^{-\lambda}$, $[\tau_1']^2/(nI(\lambda)) = e^{-2\lambda}/(n/\lambda) = \lambda e^{-2\lambda}/n$; for $\tau_2(\lambda) = \lambda e^{-\lambda}$, $\lambda e^{-2\lambda}(1-\lambda)^2/n$. Now $\delta_1(T) = \exp(T\log(1-1/n)) = e^{-\bar{X}}(1 + o_p(1))$ and $\delta_2(T) = \bar{X}e^{-\bar{X}}(1 + o_p(1))$, so both UMVUEs have the same asymptotic variances as the MLEs: ARE = 1 in both cases (with ARE as in Definition 10.1.16). Preference: the UMVUEs — exactly unbiased, asymptotically efficient, and simple closed forms; the MLE is asymptotically equivalent but biased in finite samples. (d) $n = 15$, $T = 104$, $\bar{x} = 6.933$. For $e^{-\lambda}$: UMVUE $(14/15)^{104} = 0.000765$; MLE $e^{-6.933} = 0.000975$. For $\lambda e^{-\lambda}$: UMVUE $(104/15)(14/15)^{103} = 0.005685$; MLE $6.933\, e^{-6.933} = 0.006758$.

Exercise 10.10

Central idea. $\tau(p) = p(1-p)$ has $\tau'(p) = 1 - 2p$; away from 1/2 the delta method gives efficiency, but at $p = 1/2$ the linear term vanishes and the second-order (chi-squared) term takes over at rate n.

Solution. (a) For $p \neq 1/2$, $\tau'(p) = 1 - 2p \neq 0$, so by the delta method $\sqrt{n}(\hat{p}(1-\hat{p}) - p(1-p)) \xrightarrow{d} n(0, (1-2p)^2 p(1-p))$. The Cramér–Rao bound for $\tau(p)$ is $[\tau'(p)]^2/(nI(p)) = (1-2p)^2 p(1-p)/n$; the MLE attains it, hence is asymptotically efficient. (b) At $p = 1/2$, $\tau'(1/2) = 0$, $\tau''(1/2) = -2$. Theorem 5.5.26

with $\sigma^2 = p(1-p) = 1/4$ gives $n(\hat{p}(1-\hat{p}) - 1/4) \xrightarrow{d} \frac{(1/4)(-2)}{2}\chi_1^2 = -\frac{1}{4}\chi_1^2$. (c) With $T \sim \text{Binomial}(n, p)$ and $U = \hat{p}(1-\hat{p}) = T/n - T^2/n^2$, binomial moments give the exact

$$\text{Var}[\hat{p}(1-\hat{p})] = \frac{p(1-p)(n-1)[n(1-2p)^2 + 6p(1-p) - 1]}{n^3},$$

in particular $\frac{n-1}{8n^3}$ at $p = 1/2$. Yes — the reason is now clear: the first-order delta-method variance $(1-2p)^2 p(1-p)/n$ is the $O(1/n)$ term, which vanishes at $p = 1/2$; the exact formula shows the surviving terms are $O(1/n^2)$, a second-order effect the linear approximation cannot see.

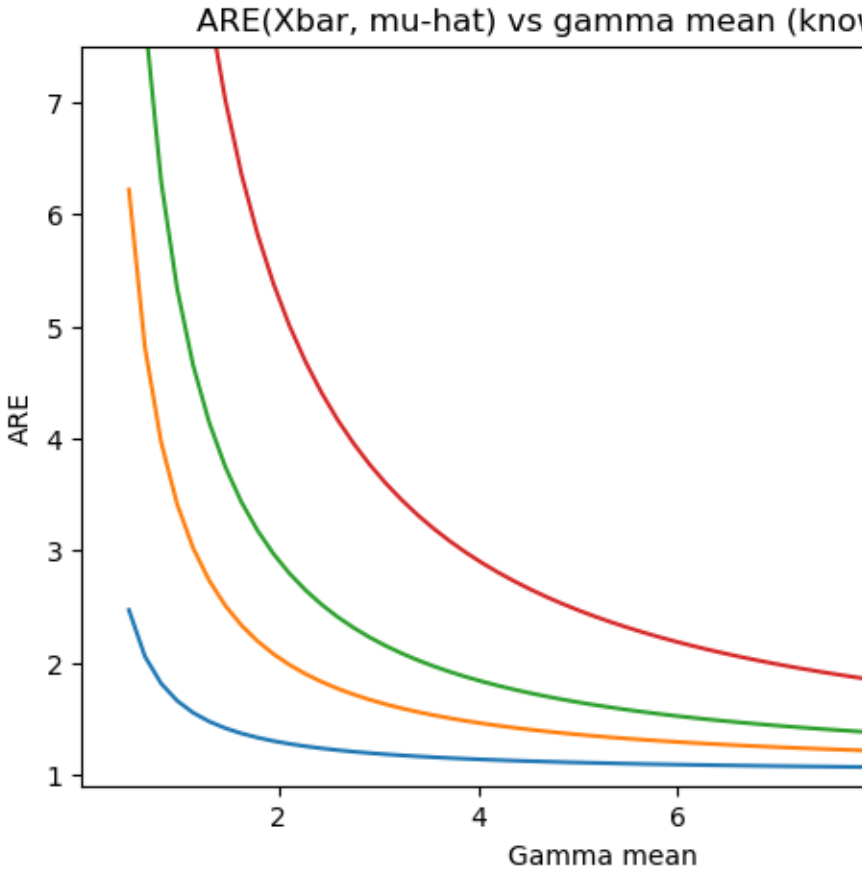
Exercise 10.11

Central idea. For the gamma mean with known scale, $I_1(\mu) = \psi'(\mu/\beta)/\beta^2$ gives $\text{ARE}(\bar{X}, \hat{\mu}) = \alpha\psi'(\alpha)$ with $\alpha = \mu/\beta$; with β unknown the MLE of μ collapses to $\bar{X}$.

Solution. With $f(x \mid \mu, \beta) = \Gamma(\mu/\beta)^{-1}\beta^{-\mu/\beta}x^{\mu/\beta-1}e^{-x/\beta}$, $\partial^2 \log f/\partial\mu^2 = -\psi'(\alpha)/\beta^2$ ($\alpha = \mu/\beta$), so $I_1(\mu) = \psi'(\alpha)/\beta^2$. Since $\text{Var}(\bar{X}) = \beta\mu/n$,

$$\text{ARE}(\bar{X}, \hat{\mu}) = \frac{\beta\mu}{1/I_1(\mu)} = \frac{\mu}{\beta}\psi'\left(\frac{\mu}{\beta}\right) = \alpha\psi'(\alpha).$$

(a) Plotting $\alpha\psi'(\alpha)$ against μ for $\beta \in \{1, 3, 5, 10\}$ reproduces Figure 10.1.1 (higher β = higher curve; all curves $\downarrow 1$ as $\mu \to \infty$). E.g. for $\beta = 5$: $\mu = 1, 2, \ldots, 10$ gives $5.25348, 2.91014, 2.18173, 1.83958, 1.64493, 1.52085, 1.43550, 1.37349, 1.326$



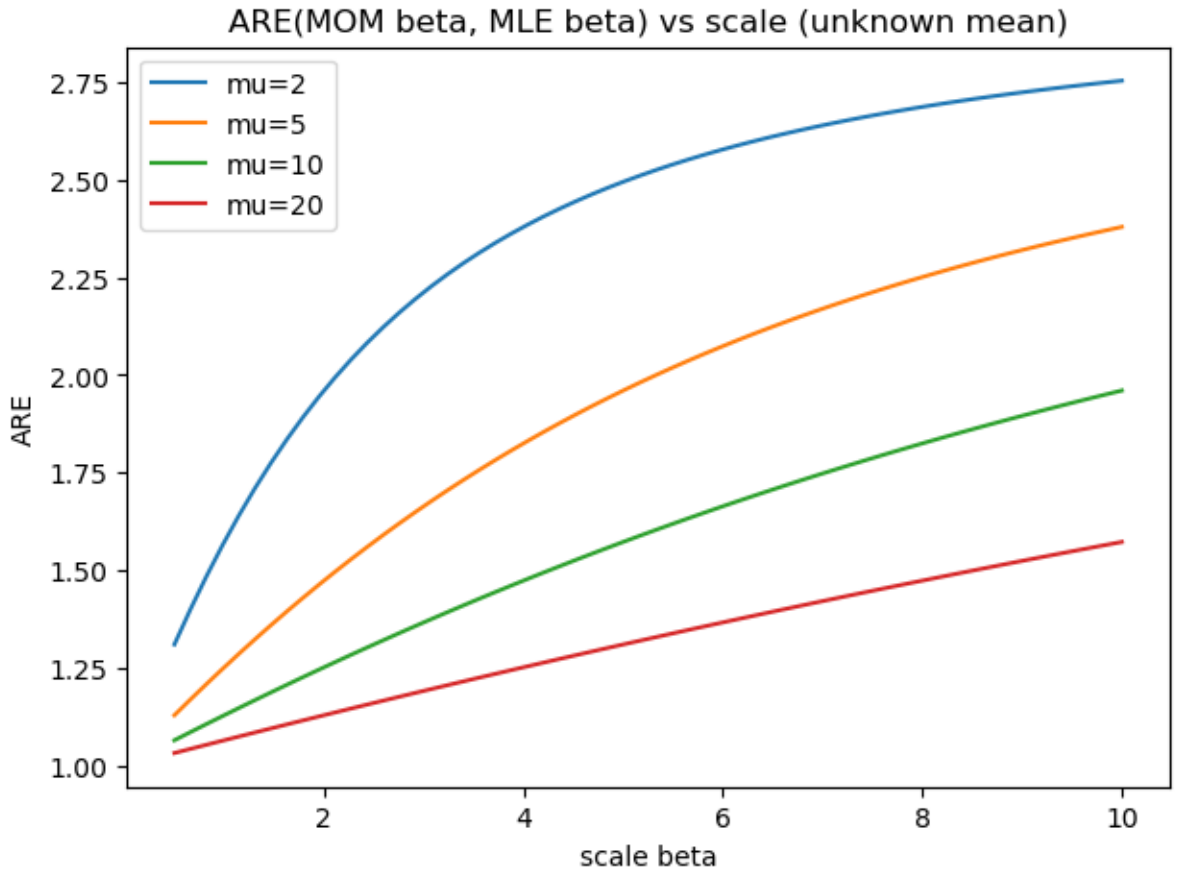
matching Example 12.6.7's Mathematica output.

(b) *As stated this is incorrect.* With β unknown, the score equation $\partial l/\partial\beta = -n\alpha/\beta + \sum x_i/\beta^2 = 0$ gives $\hat{\alpha}\hat{\beta} = \bar{X}$ at the joint MLE, i.e. $\hat{\mu}_{\text{MLE}} = \bar{X}$ exactly — the MOM and ML estimators of μ coincide, so $\text{ARE}(\bar{X}, \hat{\mu}) = 1$ trivially, NOT $\alpha\psi'(\alpha)$. The "sameness" claim fails: knowing β genuinely helps (the (μ, β) information matrix is not block-diagonal), so the curve in (a) is specific to known β. (Book

erratum.) (c) *As printed this is incorrect.* With μ known, the MOM estimator is $\tilde{\beta} = (m_2 - \mu^2)/\mu$ (from $E[X^2] = \mu^2 + \mu\beta$), while the MLE solves the digamma equation $\psi(\mu/\hat{\beta}) + \log\hat{\beta} = 1 + \overline{\log x} - \bar{x}/\mu$ — a function of $(\bar{x}, \overline{\log x})$, not of m_2. They are different statistics and different estimators: e.g. on $x = (2.1, 4.5, 6.3, 8.8, 3.2, 7.7, 5.5, 9.1, 1.2, 6.6)$ with $\mu = 6$ known, $\tilde{\beta} = 0.156$ but $\hat{\beta} = 1.867$. This is a book erratum. (d) With μ unknown, $\tilde{\beta} = (m_2 - \bar{X}^2)/\bar{X}$ (MOM; replacing its variance numerator by unbiased S^2 is only asymptotically equivalent) vs. $\hat{\beta} = \bar{X}/\hat{\alpha}$ (MLE). Delta method on (m_1, m_2) gives asymptotic variance of $\tilde{\beta}$ as $\beta^2(2\alpha + 3)/(\alpha n)$; from the Fisher information, $\hat{\beta}$ has asymptotic variance $\beta^2\psi'(\alpha)/[(\alpha\psi'(\alpha) - 1)n]$. Hence (book convention)

$$\mathrm{ARE}(\tilde{\beta}, \hat{\beta}) = \frac{(2\alpha + 3)(\alpha\psi'(\alpha) - 1)}{\alpha\psi'(\alpha)},$$

a function of $\alpha = \mu/\beta$ only, decreasing to 1 as $\alpha \to \infty$ (e.g. $\alpha = 1$: 1.96; $\alpha = 5$: 1.25; $\alpha = 10$: 1.13).



Exercise 10.12

Central idea. d_n equals $\bar{X}$ (resp. $a\bar{X}$) with probability tending to 1 when $\theta \neq 0$ (resp. $\theta = 0$), because the $n^{-1/4}$ threshold separates the two regimes.

Solution. $d_n = \bar{X}$ if $|\bar{X}| \geq n^{-1/4}$, $a\bar{X}$ otherwise, with X_i iid $n(\theta, 1)$. If $\theta \neq 0$: $P(d_n \neq \bar{X}) = P(|\bar{X}| < n^{-1/4}) \leq P(|\bar{X} - \theta| > |\theta|/2) \to 0$ (eventually $n^{-1/4} < |\theta|/2$). Hence $\sqrt{n}(d_n - \theta) = \sqrt{n}(\bar{X} - \theta) + o_p(1) \xrightarrow{d} n(0, 1)$. If $\theta = 0$: $P(d_n \neq a\bar{X}) = P(|\bar{X}| \geq n^{-1/4}) = P(|\sqrt{n}\bar{X}| \geq n^{1/4}) \to 0$ since $\sqrt{n}\bar{X} = O_p(1)$. Hence $\sqrt{n}(d_n - 0) = a\sqrt{n}\bar{X} \xrightarrow{d} n(0, a^2)$. So $\sqrt{n}(d_n - \theta) \xrightarrow{d} n[0, v(\theta)]$ with $v(\theta) = 1$ $(\theta \neq 0)$, a^2 $(\theta = 0)$; for $a < 1$ this beats the Cramér–Rao bound at the single point $\theta = 0$ — superefficiency on a Lebesgue-null set.

Exercise 10.13

Central idea. Averages over all n^n resamples inherit the empirical distribution's moments by symmetry; the only subtlety is the n^n vs. $n^n - 1$ divisor.

Solution. (a)(b) $\bar{x} = (2+4+9+12)/4 = 6.75$. Each resample mean $\bar{x}_i^*$ averages n iid draws from the empirical distribution, so $E^*[\bar{x}^*] = \bar{x}$; averaging over all $4^4 = 256$ resamples gives $\bar{\bar{x}}^* = 6.75$. For the variance, $\mathrm{Var}^*(\bar{X}^*) = \hat{\sigma}^2/n$ with $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2 = 62.75/4 = 15.6875$, i.e. 3.921875; with the sample-variance divisor $n^n - 1 = 255$ this becomes $\frac{256}{255} \times 3.921875 = 3.937 \approx 3.94$. (c) The population version of the bootstrap variance is $\hat{\sigma}^2/n = [\frac{1}{n}\sum(x_i - \bar{x})^2]/n$, exactly the usual estimate of $\mathrm{Var}(\bar{X})$ when σ^2 is estimated with divisor n (the MLE) rather than $n-1$. (d) List the 35 distinct multisets of size 4 from $\{2,4,9,12\}$, each weighted by its multinomial multiplicity $\frac{4!}{m_1!m_2!m_3!m_4!}$ (e.g. $\{2,2,2,4\}$ has weight 4 and mean 2.5; $\{2,4,9,12\}$ has weight 24 and mean 6.75). Then $\bar{\bar{x}}^* = \frac{1}{256}\sum w_j \bar{x}_j^* = 6.75$ and $\mathrm{Var}^* = \frac{1}{255}\sum w_j(\bar{x}_j^* - \bar{\bar{x}}^*)^2 = 3.94$, using weights w_j summing to 256. (e) For general $X_1, \ldots, X_n$: $E^*[X_1^*] = \bar{X}$, so $E^*[\bar{X}^*] = \bar{X}$ (part (b)); $\mathrm{Var}^*(\bar{X}^*) = \frac{1}{n}\mathrm{Var}^*(X_1^*) = \frac{1}{n} \cdot \frac{1}{n}\sum(X_i - \bar{X})^2$ (part (c)).

Exercise 10.14

Central idea. The nonparametric bootstrap resamples the data (model-free but limited to the empirical distribution); the parametric bootstrap simulates from the fitted model (smooth, but inherits model risk).

Solution. (a) For $-1.81, 0.63, 2.22, 2.41, 2.95, 4.16, 4.24, 4.53, 5.09$ ($S^2 = 4.82$), resampling $n = 9$ values with replacement ($B = 20000$) and computing S^{2*} each time gives $\mathrm{Var}_B^*(S^2) \approx 4.46$, close to the book's parametric bootstrap 4.33 (and below the normal-theory 5.81). The nonparametric version assumes nothing about normality — an advantage when the model is doubtful — but is jerkier for small n. (b) For the anchovy counts $19, 32, \ldots, 23$ ($n = 13$, $\bar{x} = 18.69$, $S^2 = 44.90$), the nonparametric bootstrap distribution of S^{2*} has mean ≈ 41.4 and sd ≈ 15.7, versus the Poisson parametric bootstrap's mean 18.86, sd 7.68 (Figure 5.6.2). The data are strongly overdispersed ($S^2 \gg \bar{x}$), which the nonparametric bootstrap honestly reflects while the Poisson parametric bootstrap suppresses — a clear win for the nonparametric approach when the model is wrong; its disadvantage is coarseness from only $n = 13$ distinct values. (c) MLEs: $\hat{\mu} = \bar{x} = 2.303$; solving $\psi(\alpha) - \log\alpha = \overline{\log x} - \log\bar{x}$ gives $\hat{\alpha} = 1.215$, $\hat{\beta} = \bar{x}/\hat{\alpha} = 1.896$, so $\widehat{\mathrm{Var}} = \hat{\alpha}\hat{\beta}^2 = 4.367$. A parametric bootstrap ($B = 4000$ from gamma$(\hat{\alpha}, \hat{\beta})$, refitting each time) gives $\mathrm{SE}(\hat{\mu}) \approx 0.83$ and $\mathrm{SE}(\widehat{\mathrm{Var}}) \approx 4.35$ — the variance estimate is very noisy, as expected with $n = 6$.

Exercise 10.15

Central idea. For fixed data, bootstrap resamples are iid draws from the bootstrap distribution: sample variances converge to the population variance (SLLN), and averages of iid unbiased copies converge to their mean (WLLN).

Solution. (a) Conditional on the data, $\hat{\theta}_1^*, \ldots, \hat{\theta}_B^*$ are iid from the bootstrap distribution of $\hat{\theta}^*$. By the SLLN, $\frac{1}{B-1}\sum_{i=1}^{B}(\hat{\theta}_i^* - \bar{\hat{\theta}}^*)^2 \xrightarrow{a.s.} E^*[(\hat{\theta}^* - E^*\hat{\theta}^*)^2] = \mathrm{Var}^*(\hat{\theta})$. (b) For fixed B, each $\mathrm{Var}_{B_i}^*(\hat{\theta})$ is the sample variance of B iid bootstrap draws, hence the $\mathrm{Var}_{B_i}^*$ are iid across i with $E[\mathrm{Var}_{B_i}^*(\hat{\theta})] = \mathrm{Var}^*(\hat{\theta})$ (the $B-1$ divisor makes it unbiased). By the WLLN, $\frac{1}{m}\sum_{i=1}^{m}\mathrm{Var}_{B_i}^*(\hat{\theta}) \xrightarrow{p} \mathrm{Var}^*(\hat{\theta})$.

Exercise 10.16

Central idea. With $g(p) = p(1-p)$, $g''(p) = -2$, the second-order delta method gives $\mathrm{Var}(g(\hat{p})) \approx 2\sigma^4/n^2 = 2[\mathrm{Var}(\hat{p})]^2$; near $p = 1/2$ the first-order term vanishes, which is exactly why the ordinary delta method collapses.

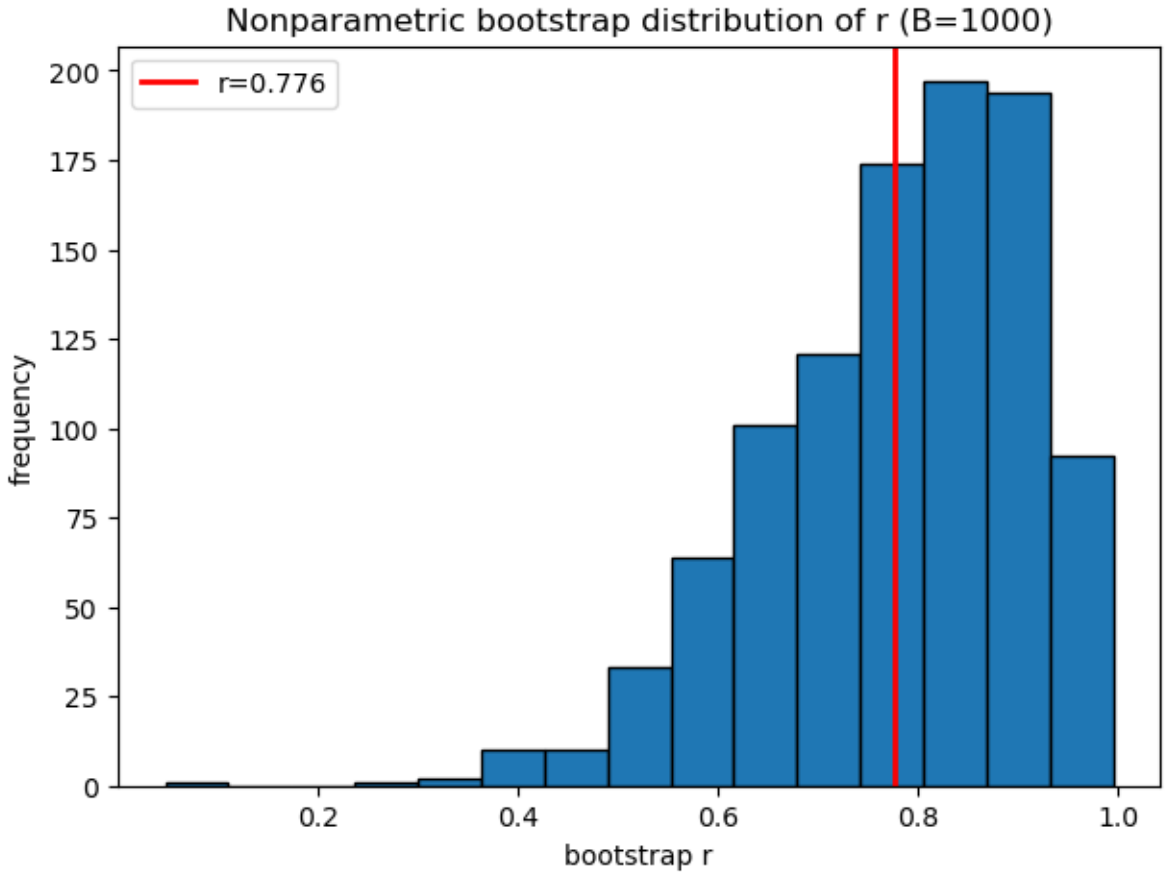
Solution. Theorem 5.5.26: $n(g(\hat{p}) - g(p)) \xrightarrow{d} \frac{\sigma^2 g''(p)}{2}\chi_1^2$ with $\sigma^2 = p(1-p)$. Hence $\mathrm{Var}(g(\hat{p})) \approx \frac{1}{n^2}\left(\frac{\sigma^2 g''(p)}{2}\right)^2 \cdot 2$. With $g''(p) = -2$: $= \frac{2\sigma^4}{n^2} = 2[\mathrm{Var}(\hat{p})]^2$. (a) *As printed this is incorrect.* At $\hat{p} = 11/24$, $n = 24$:

$2[\hat{p}(1-\hat{p})/n]^2 = 2(0.0103449)^2 = 0.000214 \approx .00021$, not .00007. (The printed .00007 is the *first-order* delta-method value $(1-2\hat{p})^2\mathrm{Var}(\hat{p}) = 7.18 \times 10^{-5}$.) This is a book erratum. (b) *As printed this is incorrect.* The exact variance, from the binomial moments of $T \sim \mathrm{Binomial}(24, 11/24)$ applied to $\hat{p}(1-\hat{p}) = T/n - T^2/n^2$, is 0.000271 (confirmed by direct summation and simulation), not .00529. This is a book erratum. (c) The (first-order) Delta Method is bad because $g'(p) = 1 - 2p \approx 0$ near $p = 1/2$ ($\hat{p} = 11/24 = 0.458$): the linear term it relies on nearly vanishes, so it reports .00007 while missing the dominant quadratic fluctuation (true value .00027, a 4× underestimate). The second-order Delta Method does better (.00021, within $\sim 20\%$) since it captures the quadratic term. For the nonparametric bootstrap of Bernoulli data, $n\hat{p}^* \mid \hat{p} \sim \mathrm{Binomial}(n, \hat{p})$ exactly. Therefore its conditional variance is the exact binomial variance formula with p replaced by $\hat{p}$; at $\hat{p} = 11/24$ it is 0.0002710672072. This establishes the numerical claim directly. At fixed $p \neq 1/2$, $2[\mathrm{Var}(\hat{p})]^2$ is the quadratic contribution alone, not the full second-order normal approximation, which also includes the linear term and equals 0.0002858458887 here. At $p = 1/2$ the quadratic term is the leading term. Do not infer moment convergence from convergence in distribution alone or claim automatic second-order bootstrap accuracy at a stationary point.

Exercise 10.17

Central idea. Three standard errors for r: nonparametric bootstrap (model-free), parametric bootstrap (bivariate normal plug-in), and the delta method $(1-r^2)/\sqrt{n}$ — the first-order approximation is optimistic.

Solution. (a) The 2024 CRC reprint prints (5.58, 2.81) (likely a typo for the canonical (558, 2.81) in Efron 1982; see addendum). With the printed value, $r = 0.4927656216$. (b) Resampling the 15 pairs with replacement ($B = 1000$): $\mathrm{SE}^*(r) \approx 0.20$; the histogram is left-skewed with mass piling near 1 (correlation is bounded above).



(c) Bivariate normal MLEs: $\hat{\mu}_X = 563.4$, $\hat{\mu}_Y = 3.095$, $\hat{\sigma}_X = 154.1$, $\hat{\sigma}_Y = 0.235$, $\hat{\rho} = 0.493$. Simulating 1000 samples of 15 pairs: $\mathrm{SE}^* \approx 0.19$. (d) $\mathrm{SE}(r) \approx (1-r^2)/\sqrt{15} = (1 - 0.243)/3.873 = 0.196$. This is smaller than both bootstrap SEs (0.19–0.20): the first-order delta method is optimistic for $n = 15$, missing the

skewness the bootstrap captures. Approximate pdf: $r \dot{\sim} n(0.493, 0.196^2)$. (e) $z = \frac{1}{2}\log\frac{1+r}{1-r} \dot{\sim} n(\zeta, \frac{1}{n-3})$ with $\zeta = \frac{1}{2}\log\frac{1+\rho}{1-\rho}$; here $z = 0.540$, SE $= 1/\sqrt{12} = 0.289$. The approximate pdf of r follows by transformation: $f_R(r) = \phi((z(r) - 0.540)/0.289) \cdot \frac{1}{0.289(1-r^2)}$, $z(r) = \frac{1}{2}\log\frac{1+r}{1-r}$ — shifted left relative to (d)'s symmetric normal, better reflecting the skewness.

Exercise 10.18

Central idea. Total variance on the mixture indicator: average of the component variances plus the variance of the component means.

Solution. Let $Y_i \in \{0, 1\}$ indicate contamination. $E[\mathrm{Var}(X_i \mid Y_i)] = (1-\delta)\sigma^2 + \delta\tau^2$ and $E[X_i \mid Y_i]$ equals θ or μ, so $\mathrm{Var}(E[X_i \mid Y_i]) = (\theta-\mu)^2\delta(1-\delta)$. Hence $\mathrm{Var}(X_i) = (1-\delta)\sigma^2 + \delta\tau^2 + \delta(1-\delta)(\theta-\mu)^2$, and $\mathrm{Var}(\bar{X}) = \mathrm{Var}(X_1)/n$ gives the formula. For Cauchy contamination $\tau^2 = \infty$, so $\mathrm{Var}(\bar{X}) = \infty$ for any $\delta > 0$.

Exercise 10.19

Central idea. $\mathrm{Var}(\bar{X})$ sums all pairwise covariances; persistent correlation leaves an $O(1)$ term, while geometrically decaying correlation contributes only $O(1/n)$.

Solution. (a) $\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\left[n\sigma^2 + n(n-1)\rho\sigma^2\right] = \frac{\sigma^2}{n} + \frac{n-1}{n}\rho\sigma^2 \to \rho\sigma^2 \neq 0$. (b) *As printed the formula has a sign error (verified against direct summation): the "+" before* $\frac{1-\rho^n}{1-\rho}$ *should be "−".* Indeed $\sum_{i<j}\rho^{|i-j|} = \sum_{k=1}^{n-1}(n-k)\rho^k = \frac{\rho\left[(n-1)-n\rho+\rho^n\right]}{(1-\rho)^2} = \frac{\rho}{1-\rho}\left(n - \frac{1-\rho^n}{1-\rho}\right)$ (check $n = 2$: both sides equal ρ; the printed "+" version gives $\rho(3-2\rho-\rho^2)/(1-\rho)^2 \neq \rho$). The correct formula is

$$\mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} + \frac{2\sigma^2}{n^2}\frac{\rho}{1-\rho}\left(n - \frac{1-\rho^n}{1-\rho}\right),$$

which is $O(1/n) \to 0$ since the bracket is $O(n)$. (c) Iterating, $X_i = \rho^{i-1}X_1 + \sum_{j=1}^{i-1}\rho^{i-1-j}\delta_j$. In stationarity $\mathrm{Var}(X_1) = \rho^2\mathrm{Var}(X_1)+1$, i.e. $\mathrm{Var}(X_1) = 1/(1-\rho^2) = \sigma^2$, and the δ_j are independent of X_1, so $\mathrm{Cov}(X_1, X_i) = \rho^{i-1}\sigma^2$, giving $\mathrm{Corr}(X_1, X_i) = \rho^{i-1}$.

Exercise 10.20

Central idea. The breakdown value is the largest contamination fraction the estimator survives: one bad point breaks the mean, half the sample breaks the median, and αn bad points break the α-trimmed mean.

Solution. (a) Driving a single observation ($1/n$ of the sample) to ∞ sends $\bar{X}_n \to \infty$; hence no positive contamination fraction is safe and $b = 0$. (b) If fewer than half the observations are driven to ∞, the median remains among the uncontaminated values (bounded); if more than half are driven to ∞, the median follows them to ∞. Hence $b = 1/2$. (c) Deleting the αn extremes: driving at most αn observations to ∞ leaves $\bar{X}_n^\alpha$ computed from uncontaminated values (bounded), while driving more than αn to ∞ contaminates the retained middle block, sending $\bar{X}_n^\alpha \to \infty$. Thus $b = \alpha$, and $0 < \alpha < 1/2$ gives $0 < b < 1/2$.

Exercise 10.21

Central idea. S^2 inherits the mean's fragility (one outlier inflates it without bound); the MAD inherits the median's: it needs a majority of bad points to break.

Solution. (a) Send $X_1 \to \infty$: then $\bar{X} \to \infty$ and $(X_1 - \bar{X})^2 \to \infty$, so $S^2 \to \infty$ from a single bad observation; $b = 0$. (b) If fewer than half the X_i are contaminated, m stays bounded and a majority of the $|X_i - m|$ stay bounded, so their median stays bounded. If more than half are driven to ∞, then $m \to \infty$ and a majority of $|X_i - m| \to \infty$, so MAD$\to \infty$. Hence $b = 1/2$.

Exercise 10.22

Central idea. For odd n, $\{M_n \le t\} = \{\sum Y_i \ge (n+1)/2\}$ turns the median into a binomial tail; the threshold is a difference quotient converging to $-2af(\mu)$.

Solution. (a) With $Y_i = I(X_i \le \mu + a/\sqrt{n}) \sim \text{Bernoulli}(p_n)$, $p_n = F(\mu + a/\sqrt{n})$, and n odd, $\{M_n \le \mu + a/\sqrt{n}\} = \{\sum_i Y_i \ge (n+1)/2\}$. Standardizing gives the displayed identity. (b) $p_n = F(\mu + a/\sqrt{n}) \to F(\mu) = 1/2$ by continuity. Write the threshold as

$$\frac{\frac{(n+1)/2 - np_n}{\sqrt{n}}}{\sqrt{p_n(1-p_n)}}, \qquad \frac{(n+1)/2 - np_n}{\sqrt{n}} = -\sqrt{n}\left[F\left(\mu + \frac{a}{\sqrt{n}}\right) - F(\mu)\right] + \frac{1}{2\sqrt{n}} \to -aF'(\mu) = -af(\mu),$$

the limit form of the derivative. Since $\sqrt{p_n(1-p_n)} \to 1/2$, the threshold $\to -2af(\mu)$. (c) $P(Z \ge -2af(\mu)) = P(Z \le 2af(\mu)) = \Phi(2af(\mu)) = \Phi\big(\frac{a}{1/[2f(\mu)]}\big)$, which is exactly the cdf of $n\big(0, \frac{1}{[2f(\mu)]^2}\big)$ evaluated at a. Convergence of cdfs at all a gives $\sqrt{n}(M_n - \mu) \xrightarrow{d} n(0, 1/[2f(\mu)]^2)$. (Because p_n varies with n, the Y_i form a triangular array, so the ordinary CLT does not apply directly — a Lindeberg-type argument is needed for full rigor.)

Exercise 10.23

Central idea. $\text{ARE}(M_n, \bar{X}) = \text{Var}(\bar{X})/\text{Var}(M_n) = 4\sigma^2 f(0)^2$ for symmetric location families; heavy tails inflate σ^2 but not $1/f(0)^2$, favoring the median.

Solution. For symmetric f with median 0, $\text{Var}(M_n) \approx 1/[4nf(0)^2]$, so $\text{ARE}(M_n, \bar{X}) = 4\sigma^2 f(0)^2$. (a) Normal: $4\sigma^2/(2\pi\sigma^2) = 2/\pi = 0.637 \approx .64$. Logistic ($f(0) = 1/4$, $\sigma^2 = \pi^2/3$): $4(\pi^2/3)(1/16) = \pi^2/12 = 0.823 \approx .82$. Double exponential ($f(0) = 1/2$, $\sigma^2 = 2$): $4 \cdot 2 \cdot 1/4 = 2$. (b) Under $X = \sigma Z$, $\text{Var}(\bar{X})$ scales by σ^2 while $f_X(0) = f_Z(0)/\sigma$, so $4\sigma_X^2 f_X(0)^2 = 4\sigma^2\sigma_Z^2 \cdot f_Z(0)^2/\sigma^2$ — the σ cancels. (c) For t_ν, $f(0) = \Gamma(\frac{\nu+1}{2})/(\sqrt{\nu\pi}\,\Gamma(\frac{\nu}{2}))$, $\sigma^2 = \nu/(\nu-2)$:

ν	3	5	10	25	50	∞
ARE	1.62	0.96	0.76	0.68	0.66	0.64

Heavier tails (small ν) strongly favor the median (ARE > 1 at $\nu = 3$); as $\nu \to \infty$ the ARE decreases to the normal value $2/\pi$. (d) Here $\sigma^2 = (1-\delta) + \delta\sigma^2$ and $f(0) = [(1-\delta) + \delta/\sigma]/\sqrt{2\pi}$, so ARE $= \frac{2}{\pi}[(1-\delta) + \delta/\sigma]^2[(1-\delta) + \delta\sigma^2]$:

$\delta \backslash \sigma$	1	3	5	10
0.01	0.64	0.68	0.78	1.24
0.05	0.64	0.83	1.29	3.46
0.10	0.64	1.00	1.83	5.75
0.20	0.64	1.24	2.61	8.90

More/wilder contamination (larger δ, σ) dramatically favors the median — the mean's variance explodes while the median's barely moves.

Exercise 10.24

Central idea. It is Slutsky's theorem: a standardized sum converging in distribution, divided by an average converging in probability.

Solution. From the Taylor expansion, $\sqrt{n}(\hat{\theta}_M - \theta_0) = \dfrac{-\frac{1}{\sqrt{n}}\sum_{i=1}^n \psi(X_i - \theta_0)}{\frac{1}{n}\sum_{i=1}^n \psi'(X_i - \theta_0)}$. By (10.2.4) the numerator $\xrightarrow{d} n(0, E_{\theta_0}[\psi(X-\theta_0)]^2)$ (CLT, using $E_{\theta_0}\psi(X-\theta_0) = 0$), and by (10.2.5) the denominator $\xrightarrow{p} E_{\theta_0}\psi'(X-\theta_0)$ (WLLN). Slutsky's theorem gives

$$\sqrt{n}(\hat{\theta}_M - \theta_0) \xrightarrow{d} n\left(0, \frac{E_{\theta_0}[\psi(X-\theta_0)]^2}{[E_{\theta_0}\psi'(X-\theta_0)]^2}\right),$$

which is (10.2.6).

Exercise 10.25

Central idea. Symmetric ρ has odd derivative ψ; odd $\times$ even integrates to zero, so the true θ is the zero of $E_\theta\psi(X-\theta)$ that (10.2.6) centers on.

Solution. From $\rho(-u) = \rho(u)$, differentiating gives $-\psi(-u) = \psi(u)$, i.e. ψ is odd. With $u = x - \theta$ and f symmetric ($f(-u) = f(u)$),

$$\int \psi(x-\theta)f(x-\theta)\,dx = \int \psi(u)f(u)\,du = 0,$$

since $\psi(u)f(u)$ is odd. Thus $\theta_0 = \theta$ (the true value) satisfies $E_{\theta_0}\psi(X-\theta_0) = 0$ (uniquely, as ψ is increasing), and (10.2.6) gives $\sqrt{n}(\hat{\theta}_M - \theta) \xrightarrow{d} n(0, E_\theta\psi^2/(E_\theta\psi')^2)$: asymptotically normal with mean θ.

Exercise 10.26

Central idea. For Huber scores, average derivatives estimate the central probability; the expected score can be differentiated by moving the distribution while holding its argument fixed.

Solution. (a) $\psi'(u) = 1\{|u| < k\}$ almost everywhere, so $A = E\psi'(\epsilon) = P(|\epsilon| < k)$. For symmetric errors,

$$B = E\psi(\epsilon)^2 = \int_{-k}^{k} u^2 f(u)\,du + 2k^2\int_k^\infty f(u)\,du.$$

The asymptotic variance of $\sqrt{n}(\hat{\theta}_M - \theta)$ is B/A^2; the approximate variance of $\hat{\theta}_M$ itself is $B/(nA^2)$. (b) The inspected textbook already gives the correct moving-distribution derivative. Write $X = \theta_0 + \epsilon$ under P_{θ_0}, and hold the argument parameter θ fixed:

$$\frac{d}{d\theta_0}E_{\theta_0}\psi(X-\theta)\bigg|_{\theta_0=\theta} = \frac{d}{d\theta_0}E\psi(\epsilon+\theta_0-\theta)\bigg|_{\theta_0=\theta} = E\psi'(\epsilon).$$

For Huber ψ, dominated convergence applies since ψ is Lipschitz and the continuous error law assigns zero probability to its two kinks. Differentiating $E_\theta\psi(X-\theta_0)$ instead gives $-A$; moving both parameters gives zero.

Exercise 10.27

Central idea. The contaminated cdf $F_\delta = (1-\delta)F + \delta\Delta_x$ shifts the median; implicit differentiation of $F(a_\delta) = 1/[2(1-\delta)]$ gives the influence $\pm 1/[2f(m)]$ — bounded, unlike the mean's.

Solution. (a) $T(F_\delta) = (1-\delta)\mu + \delta x$, so $IF(\bar{X}, x) = \lim_{\delta\to 0}[T(F_\delta) - T(F)]/\delta = x - \mu$. (b) $F_\delta = (1-\delta)F + \delta\Delta_x$ gives $P_{F_\delta}(X \le a) = (1-\delta)F(a) + \delta\mathbf{1}\{x \le a\}$, i.e. the two cases shown. The median solves $(1-\delta)F(m) + \delta\mathbf{1}\{x \le m\} = 1/2$: if $x > m$, $F(m) = 1/[2(1-\delta)]$; if $x \le m$, $F(m) = (1/2 - \delta)/(1-\delta)$, giving the two cases (each consistent on its stated region). (c) Put $a_\delta = F^{-1}(1/[2(1-\delta)])$, so $F(a_\delta) = 1/[2(1-\delta)]$. Differentiating in

δ: $f(a_\delta)a'_\delta = 1/[2(1-\delta)^2]$, and at $\delta = 0$, $a_0 = m = F^{-1}(1/2)$, so $a'_0 = 1/[2f(m)]$ — the displayed limit. For $x < m$, with $b_\delta = F^{-1}((1/2-\delta)/(1-\delta))$: $f(b_\delta)b'_\delta = \frac{d}{d\delta}\frac{1/2-\delta}{1-\delta} = \frac{-1/2}{(1-\delta)^2}$, so $b'_0 = -1/[2f(m)]$. Hence

$$IF(M,x) = \begin{cases} \frac{1}{2f(m)} & x > m, \\ -\frac{1}{2f(m)} & x < m, \end{cases}$$

bounded in x, in contrast to $IF(\bar{X}, x) = x - \mu$.

Exercise 10.28

Central idea. Check the joins at $x = \pm k$: ρ matches ($k^2/2 = k^2 - k^2/2$) and ρ' matches ($\pm k$ from both sides).

Solution. $\rho(x) = x^2/2$ for $|x| \le k$, $k|x| - k^2/2$ for $|x| \ge k$. At $x = k$: left value $k^2/2$, right value $k^2 - k^2/2 = k^2/2$; similarly at $-k$. So ρ is continuous. $\rho'(x) = x$ for $|x| < k$ and $k\,\mathrm{sgn}(x)$ for $|x| > k$: at $x = k$ both one-sided limits are k; at $x = -k$ both are $-k$. So ρ' is continuous (it is ψ in (10.2.7)).

Exercise 10.29

Central idea. (10.2.9) is Cauchy–Schwarz, with equality iff $\psi \propto l'$; for a location family the efficient choice is $\psi(u) = -\frac{d}{du}\log f(u)$.

Solution. (a) (10.2.9), $\mathrm{ARE}(\hat{\theta}_M, \hat{\theta}) = \frac{[E_\theta(\psi(X-\theta)l'(\theta|X))]^2}{E_\theta\psi^2\, E_\theta(l')^2} \le 1$, is Cauchy–Schwarz for the inner product $(g,h) \mapsto E_\theta[gh]$, with equality iff ψ is a.s. proportional to l'. If $\psi = cl'$, the numerator is $c^2[E(l')^2]^2$ and the denominator $c^2E(l')^2 \cdot E(l')^2$: ratio 1. (b) For location families $l'(\theta \mid x) = -\frac{f'(x-\theta)}{f(x-\theta)}$, so efficiency needs $\psi(u) \propto -\frac{d}{du}\log f(u)$. (i) $-\frac{d}{dx}(-x^2/2) = x = \psi$. (iii) $-\frac{d}{dx}[-\log(1+x^2)] = 2x/(1+x^2) = \psi$. (iv) $-\frac{d}{dx}\log f = x$ ($|x| < c$), $c\,\mathrm{sgn}(x)$ ($|x| > c$) $= \max\{-c, \min(c,x)\}$. (ii) *As printed this is incorrect:* $-\frac{d}{dx}\log f = -\frac{d}{dx}[-x - 2\log(1+e^{-x})] = \tanh(x/2)$, not $\tanh(x)$. The efficient ψ for the stated logistic pdf is $\tanh(x/2)$; $\tanh(x)$ is not proportional to the score, so it is not asymptotically efficient. (Book erratum.)

Exercise 10.30

Central idea. For bounded odd scores with nonzero opposite limits, the displayed formula gives $\eta = 1$, $b^* = 1/2$; the normal (mean) ψ is unbounded, hence $b^* = 0$. AREs come from (10.2.9) by numerical integration.

Solution. (a) (i) Normal: $\psi(x) = x$ unbounded — the formula does not apply; $b^* = 0$ (the mean). (ii) Logistic $\psi = \tanh$: $\psi(\pm\infty) = \pm 1$, $\eta = \min\{1,1\} = 1$, $b^* = 1/2$. (iii) Cauchy $\psi = 2x/(1+x^2)$ tends to zero at both infinities, so the displayed expression for η is $0/0$ and cannot be applied. A finite-M ratio does not repair that theorem; breakdown must be proved for the specified estimator and rule for selecting a root or objective minimum. (iv) Least informative: $\psi(\pm\infty) = \pm c$, $\eta = 1$, $b^* = 1/2$. Use the corrected logistic score $\tanh(x/2)$ from 10.29. The score choices are efficient for their respective models; logistic and least-informative scores satisfy the stated nonzero-limit breakdown calculation. The Cauchy score is bounded, but its breakdown conclusion needs a separate argument and an estimator-selection rule. (b) (i) Huber ψ: bounded with $\psi(\pm\infty) = \pm k$: $\eta = 1$, $b^* = 1/2$. (ii) Tukey and (iii) Andrews scores vanish identically sufficiently far out, so both give an undefined $0/0$ in the displayed formula. Their estimating equations may have multiple roots. Boundedness and oddness alone do not justify a universal breakdown value of $1/2$; specify scale estimation and root/global-minimum selection before making that claim. (c) From (10.2.9) (larger = better, max 1), with the following fixed tuning constants (the printed unsquared Tukey score is not the usual squared biweight), using a scale-one Laplace error law (Huber $k = 1.345$, Tukey $c = 4.685$, Andrews $c = 1.339$):

	Huber	Tukey	Andrews
Normal	0.95	0.98	0.95
Double exp.	0.70	0.65	0.70

The normal efficiencies are approximately 0.95000, 0.98358, and 0.95004. These efficiency calculations do not establish breakdown for the redescending scores; for double-exponential data (where the MLE is the median) they keep 65–70% efficiency — the robustness–efficiency compromise.

Exercise 10.31

Central idea. Under H_0 the pooled variance estimate gives a standard normal pivot whose square is χ^2_1; the Pearson T^* collapses algebraically to the same T, while T^{**} uses the unpooled (Wald) variance.

Solution. (a) Under H_0 ($p_1 = p_2 = p$), $\hat{p}_1 - \hat{p}_2$ is asymptotically $n(0, p(1-p)(1/n_1 + 1/n_2))$ by independence and the CLT, and $\hat{p} \xrightarrow{p} p$; Slutsky gives $\frac{\hat{p}_1 - \hat{p}_2}{\sqrt{(1/n_1+1/n_2)\hat{p}(1-\hat{p})}} \xrightarrow{d} n(0,1)$, whose square is $T \xrightarrow{d} \chi^2_1$. (b) $S_1 - \frac{n_1 S}{n} = \frac{n_2 S_1 - n_1 S_2}{n} = \frac{n_1 n_2}{n}(\hat{p}_1 - \hat{p}_2)$, and similarly for the other cells. Summing $\frac{(\text{obs}-\text{exp})^2}{\text{exp}}$:

$$T^* = \left[\frac{n_1 n_2}{n}(\hat{p}_1 - \hat{p}_2)\right]^2 \cdot n\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\left(\frac{1}{S} + \frac{1}{F}\right) = \frac{(\hat{p}_1 - \hat{p}_2)^2}{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\hat{p}(1-\hat{p})} = T,$$

using $S/n = \hat{p}$, $F/n = 1 - \hat{p}$. Hence $T^* \xrightarrow{d} \chi^2_1$. (c) Same CLT, but the denominator $\xrightarrow{p} \sqrt{p(1-p)(1/n_1 + 1/n_2)}$ without pooling: $T^{**} \xrightarrow{d} n(0,1)$, $(T^{**})^2 \xrightarrow{d} \chi^2_1$. $(T^{**})^2 \neq T^*$ because the denominators differ (unpooled $\hat{p}_1(1-\hat{p}_1)/n_1 + \hat{p}_2(1-\hat{p}_2)/n_2$ vs. pooled $(1/n_1 + 1/n_2)\hat{p}(1-\hat{p})$). (d) T ($= T^*$) is the score test: preferable under H_0 (pooled variance is more stable; locally most powerful). T^{**} is the Wald test: preferable when a signed/directional test or a confidence interval for $p_1 - p_2$ is wanted, or when the H_0 variance assumption is doubted. (e) $\hat{p}_1 = 34/40 = 0.85$, $\hat{p}_2 = 19/35 = 0.543$, $\hat{p} = 53/75 = 0.707$: $T = \frac{(0.307)^2}{(1/40+1/35)(0.707)(0.293)} = 8.50$. Since $\chi^2_{1,0.95} = 3.84$, reject H_0 (p-value 0.0036): carbolic acid is significantly associated with lower mortality.

Exercise 10.32

Central idea. Under $p_1 = p_2$ the constrained MLEs are $(X_1 + X_2)/(2m)$, and the Pearson sum collapses to $(X_1 - X_2)^2/(X_1 + X_2)$; the conditional version $p_1/(p_1 + p_3) = p_2/(p_2 + p_4)$ is genuinely different.

Solution. (a) Under $H_0 : p_1 = p_2$, maximizing the multinomial likelihood gives $\hat{p}_1 = \hat{p}_2 = (X_1 + X_2)/(2m)$, so the expected counts are $(X_1 + X_2)/2$ for cells 1, 2. Then

$$\sum_{i=1,2} \frac{(X_i - (X_1 + X_2)/2)^2}{(X_1 + X_2)/2} = 2 \cdot \frac{[(X_1 - X_2)/2]^2}{(X_1 + X_2)/2} = \frac{(X_1 - X_2)^2}{X_1 + X_2},$$

(the remaining cells contribute nothing under the constraint as stated). (b) The second hypothesis is $P(\text{change} \mid \text{initially agree}) = P(\text{change} \mid \text{initially disagree})$, i.e. $\frac{p_1}{p_1+p_3} = \frac{p_2}{p_2+p_4}$. This differs from $p_1 = p_2$: e.g. $(p_1, p_2, p_3, p_4) = (0.1, 0.1, 0.7, 0.1)$ satisfies $p_1 = p_2$ but gives conditionals $0.1/0.8 = 0.125 \neq 0.5 = 0.1/0.2$. It is a test of independence in the 2×2 table, testable by the χ^2 statistics of Exercise 10.31.

Exercise 10.33

Central idea. The log-likelihood Taylor expansion multiplies the squared estimation error by observed information.

Solution. Correction to the displayed statement: use $(\theta_0 - \hat{\theta})\sqrt{-\ell_n''(\hat{\theta})}$, not division by that square root. Let $J_n(\hat{\theta}) = -\ell_n''(\hat{\theta})$ be total observed information and $I_1(\theta_0)$ information per observation. Under the

usual interior, identifiability, consistency, smoothness and information-convergence conditions,

$$\sqrt{n}(\hat{\theta}-\theta_0) \Rightarrow N(0, I_1(\theta_0)^{-1}), \qquad J_n(\hat{\theta})/n \to_p I_1(\theta_0).$$

Thus

$$\sqrt{J_n(\hat{\theta})}(\hat{\theta}-\theta_0) = \sqrt{J_n(\hat{\theta})/n}\,\sqrt{n}(\hat{\theta}-\theta_0) \Rightarrow N(0,1).$$

Expanding about the likelihood maximizer, whose score is zero, gives

$$-2\log\Lambda_n = 2[\ell_n(\hat{\theta}) - \ell_n(\theta_0)] = J_n(\hat{\theta})(\hat{\theta}-\theta_0)^2 + o_p(1) \Rightarrow \chi^2_1.$$

The original division is $O_p(n^{-1})$, so it converges to zero. The multiplication correction is essential.

Exercise 10.34

Central idea. The Bernoulli LRT is a scaled Kullback–Leibler divergence $2n[\hat{p}\log(\hat{p}/p_0) + (1-\hat{p})\log((1-\hat{p})/(1-p_0))]$; simulation confirms the χ^2_1 limit.

Solution. (a) $\lambda(x) = \frac{p_0^S(1-p_0)^{n-S}}{\hat{p}^S(1-\hat{p})^{n-S}}$ with $S = \sum x_i$, $\hat{p} = S/n$. Hence

$$-2\log\lambda(x) = 2\left[S\log\frac{\hat{p}}{p_0} + (n-S)\log\frac{1-\hat{p}}{1-p_0}\right] = 2n\left[\hat{p}\log\frac{\hat{p}}{p_0} + (1-\hat{p})\log\frac{1-\hat{p}}{1-p_0}\right].$$

(b) Simulating $M = 20000$ samples with $n = 20$, $p_0 = 0.5$: simulated mean 1.03 (vs. 1 for χ^2_1), $P(-2\log\lambda > 3.84) = 0.043$ (vs. nominal 0.05), and simulated 0.90/0.95/0.99 quantiles $(3.29, 3.29, 7.71)$ vs. χ^2_1 quantiles $(2.71, 3.84, 6.64)$ — reasonable agreement, slightly conservative in the tail, as expected for $n = 20$.

Exercise 10.35

Central idea. Wald $= (\hat{\theta}-\theta_0)/\mathrm{se}(\hat{\theta})$: for μ the MLE is $\bar{X}$ with se $\sigma/\sqrt{n}$; for σ (with μ known) the MLE is $\hat{\sigma} = \sqrt{\frac{1}{n}\sum(X_i-\mu)^2}$ with se $\hat{\sigma}/\sqrt{2n}$ from $I(\sigma) = 2n/\sigma^2$.

Solution. (a) The MLE of μ is $\hat{\mu} = \bar{X}$ with $\mathrm{se}(\hat{\mu}) = \sigma/\sqrt{n}$ (σ known), so the Wald statistic is $(\bar{X}-\mu_0)/(\sigma/\sqrt{n}) = \sqrt{n}(\bar{X}-\mu_0)/\sigma$. (b) With μ known, the MLE of σ is $\hat{\sigma} = \sqrt{\frac{1}{n}\sum_{i=1}^n (X_i-\mu)^2}$. From $l(\sigma) = -n\log\sigma - \sum(X_i-\mu)^2/(2\sigma^2)$, $I(\sigma) = -E[l'']/n = 2n/\sigma^2$, so $\mathrm{se}(\hat{\sigma}) = 1/\sqrt{I(\hat{\sigma})} = \hat{\sigma}/\sqrt{2n}$. The Wald statistic is

$$W = \frac{\hat{\sigma}-\sigma_0}{\hat{\sigma}/\sqrt{2n}} = \sqrt{2n}\,\frac{\hat{\sigma}-\sigma_0}{\hat{\sigma}} \;\xrightarrow{d}\; n(0,1) \text{ under } H_0.$$

Exercise 10.36

Central idea. $\hat{\beta} = \bar{X}/\alpha$; $I(\beta) = n\alpha/\beta^2$, so the Wald statistic is $\sqrt{n\alpha}(\hat{\beta}-\beta_0)/\hat{\beta}$, with S substituted for the model-based se in (c).

Solution. (a) $l(\beta) = -n\alpha\log\beta - \sum x_i/\beta + c$; $l'(\beta) = -n\alpha/\beta + \sum x_i/\beta^2 = 0$ gives $\hat{\beta} = \bar{X}/\alpha$. (b) $l''(\beta) = n\alpha/\beta^2 - 2\sum x_i/\beta^3$, so $I(\beta) = -E[l'']/n = \alpha/\beta^2$ and $\mathrm{se}(\hat{\beta}) = \hat{\beta}/\sqrt{n\alpha}$. The Wald statistic (MLE in both numerator and denominator) is

$$W = \frac{\hat{\beta}-\beta_0}{\hat{\beta}/\sqrt{n\alpha}} = \sqrt{n\alpha}\,\frac{\hat{\beta}-\beta_0}{\hat{\beta}}.$$

(c) $\mathrm{Var}(\hat{\beta}) = \beta^2/(n\alpha) = \mathrm{Var}(X)/(n\alpha^2)$, estimated by $S^2/(n\alpha^2)$; hence se $= S/(\alpha\sqrt{n})$ and

$$W = \frac{\hat{\beta}-\beta_0}{S/(\alpha\sqrt{n})} = \alpha\sqrt{n}\,\frac{\hat{\beta}-\beta_0}{S}.$$

Exercise 10.37

Central idea. Score $= l'(\theta_0)/\sqrt{I(\theta_0)}$: for μ this is the usual z-statistic; for σ it is a centered, scaled sum of squared deviations.

Solution. (a) $l(\mu) = -\sum(X_i-\mu)^2/(2\sigma^2) + c$, $l'(\mu) = n(\bar{X}-\mu)/\sigma^2$, $I(\mu) = n/\sigma^2$. At μ_0: $S = l'(\mu_0)/\sqrt{I(\mu_0)} = \frac{n(\bar{X}-\mu_0)/\sigma^2}{\sqrt{n}/\sigma} = \sqrt{n}(\bar{X}-\mu_0)/\sigma$. (b) With μ known, $l(\sigma) = -n\log\sigma - \sum(X_i-\mu)^2/(2\sigma^2)$, $l'(\sigma) = -n/\sigma + \sum(X_i-\mu)^2/\sigma^3$, and $I(\sigma_0) = 2n/\sigma_0^2$ (from 10.35). The score statistic is

$$S = \frac{l'(\sigma_0)}{\sqrt{I(\sigma_0)}} = \frac{-n/\sigma_0 + \sum(X_i-\mu)^2/\sigma_0^3}{\sqrt{2n}/\sigma_0} = \frac{\sum_{i=1}^n (X_i-\mu)^2 - n\sigma_0^2}{\sigma_0^2\sqrt{2n}} \xrightarrow{d} n(0,1) \text{ under } H_0.$$

Exercise 10.38

Central idea. $l'(\beta) = -n\alpha/\beta + \sum X_i/\beta^2$ and $I(\beta) = n\alpha/\beta^2$ give $S = \sqrt{n}(\bar{X}-\alpha\beta_0)/(\beta_0\sqrt{\alpha})$.

Solution. $l'(\beta) = -n\alpha/\beta + \sum_{i=1}^n X_i/\beta^2$, $I(\beta_0) = n\alpha/\beta_0^2$ (from 10.36). Hence

$$S = \frac{l'(\beta_0)}{\sqrt{I(\beta_0)}} = \frac{-n\alpha/\beta_0 + \sum X_i/\beta_0^2}{\sqrt{n\alpha}/\beta_0} = \frac{\sqrt{n}\,(\bar{X}-\alpha\beta_0)}{\beta_0\sqrt{\alpha}} \xrightarrow{d} n(0,1) \text{ under } H_0.$$

Exercise 10.39

Central idea. Simulate size and power ($n = 15$, nominal $\alpha = 0.1$, Huber $k = 1.5$): the (10.3.6) plug-in and the bootstrap variance both match the book's Z_{GW}; the median test is competitive only for heavy tails.

Solution. *Note:* (10.3.8) as printed has denominator $[1 - \frac{1}{n}\sum I(|x_i - \hat{\theta}_M| < k)]^2$, which estimates $[P(|X| \geq k)]^2$; to match (10.3.7) it should be $[\frac{1}{n}\sum I(|x_i - \hat{\theta}_M| \leq k)]^2$ (used below). Rejection rates of $|Z| > z_{0.05}$ (600 simulations, $B = 200$ bootstrap):

	Normal		Double exp.		t_5	
	θ_0 (size)	$\theta_0 + .5\sigma$ (power)	size	power	size	power
(10.3.6) plug-in	0.123	0.653	0.127	0.712	0.098	0.690
(10.3.8) corrected	0.123	0.653	0.127	0.712	0.098	0.690
bootstrap var	0.122	0.672	0.122	0.708	0.097	0.678
median (boot SE)	0.108	0.480	0.082	0.655	0.085	0.557

(a) The (10.3.6) variance $\frac{\frac{1}{n}\sum\psi^2}{(\frac{1}{n}\sum\psi')^2}$ performs essentially identically to (the corrected) (10.3.8) — both are consistent plug-ins for (10.2.6); neither dominates, and (10.3.6) is the more direct plug-in. (b) The bootstrap variance is fully automatic (no analytic work) and matches the plug-ins on both size and power — desirable as a general-purpose alternative. (c) The median-based test has the most conservative size but notably less power for normal data (0.48 vs. 0.65); for double-exponential data it is nearly competitive (0.66 vs. 0.71), as expected since the median is the MLE there. All tests share the book's finding: z-cutoffs are slightly too small (sizes 0.10–0.13 above nominal 0.10).

Exercise 10.40

Central idea. For the Poisson, the score is $n(\bar{X}-\lambda)/\lambda$ and the expected information is n/λ; their ratio is exactly $(\bar{X}-\lambda)/\sqrt{\lambda/n}$ — i.e. this statistic *is* Wilks' Q in (10.4.3).

Solution. $\log L(\lambda \mid x) = n\bar{x}\log\lambda - n\lambda - \sum \log(x_i!)$. Then $\frac{\partial}{\partial\lambda}\log L = \frac{n\bar{x}}{\lambda} - n = \frac{n(\bar{x}-\lambda)}{\lambda}$ and $\frac{\partial^2}{\partial\lambda^2}\log L = -\frac{n\bar{x}}{\lambda^2}$, so $-E_\lambda[\frac{\partial^2}{\partial\lambda^2}\log L] = \frac{n\lambda}{\lambda^2} = \frac{n}{\lambda}$. Therefore

$$\frac{\frac{\partial}{\partial\lambda}\log L(\lambda \mid \mathbf{X})}{\sqrt{-E_\lambda\left(\frac{\partial^2}{\partial\lambda^2}\log L(\lambda \mid \mathbf{X})\right)}} = \frac{n(\bar{X}-\lambda)/\lambda}{\sqrt{n/\lambda}} = \frac{\bar{X}-\lambda}{\sqrt{\lambda/n}},$$

which is Wilks' statistic Q of (10.4.3) — the score interval, asymptotically optimal (shortest) by Wilks (1938).

Exercise 10.41

Central idea. Wilks' Q for p is $\sqrt{n}\,[r(1-p) - p\bar{X}]/\sqrt{r(1-p)}$; for the mean $\mu = r(1-p)/p$ the pivot $(\bar{X}-\mu)/\sqrt{\mu(\mu+r)/(nr)}$ inverts to a quadratic, with $\pm 1/(2n)$ continuity correction.

Solution. (a) $\log L(p \mid x) = nr\log p + \sum x_i \log(1-p) + c$ (r known). $\frac{\partial}{\partial p}\log L = \frac{nr}{p} - \frac{n\bar{x}}{1-p} = \frac{n[r(1-p)-p\bar{x}]}{p(1-p)}$, $-E_p[\frac{\partial^2}{\partial p^2}\log L] = \frac{nr}{p^2(1-p)}$. Hence Wilks' statistic is

$$Q = \frac{\sqrt{n}\,[r(1-p) - p\bar{X}]}{\sqrt{r(1-p)}} \,\dot{\sim}\, n(0,1),$$

and $\{p : |Q| \le z_{\alpha/2}\}$ is the interval — i.e. $n[r - p(r+\bar{X})]^2 \le z_{\alpha/2}^2 r(1-p)$, a quadratic in p whose roots are the endpoints. (b) $E\bar{X} = \mu = r(1-p)/p$, $\mathrm{Var}(\bar{X}) = r(1-p)/(np^2) = \mu(\mu+r)/(nr)$, so $\frac{\bar{X}-\mu}{\sqrt{\mu(\mu+r)/(nr)}} \xrightarrow{d} n(0,1)$. Inverting $|\bar{X}-\mu|/\sqrt{\mu(\mu+r)/(nr)} \le z_{\alpha/2}$ gives the quadratic $\mu^2(1 - \frac{z^2}{nr}) - \mu(2\bar{X} + \frac{z^2}{n}) + \bar{X}^2 \le 0$; its roots are the endpoints. Continuity correction ($\bar{X}$ takes values k/n): the upper endpoint is the larger root with $\bar{X}$ replaced by $\bar{X} + 1/(2n)$, the lower endpoint the smaller root with $\bar{X} - 1/(2n)$ (as in Example 10.4.6). (c) Aphid data $155, 104, 66, 50, 36, 40, 30, 35, 42$ ($n = 9$, $\bar{x} = 62$, $S^2 = 1735.75$): MOM $\hat{r} = \bar{x}^2/(S^2 - \bar{x}) = 2.30$. The 90% NB interval for the mean is $(45.2, 97.8)$ (with continuity correction $(45.2, 97.9)$). The Poisson-based 90% intervals are $(57.7, 66.3)$ (Wald), $(57.8, 66.5)$ (score), $(57.8, 66.4)$ (LRT) — much narrower, because the Poisson model forces Var = mean while the data are wildly overdispersed ($S^2 = 1736 \gg 62$); the NB interval honestly reflects the extra-Poisson variation.

Exercise 10.42

Central idea. $-2\log(L(\theta \mid x)/L(\hat{\theta} \mid x)) \le \chi^2_{1,\alpha}$ is the same inequality as $L(\theta \mid x)/L(\hat{\theta} \mid x) \ge e^{-\chi^2_{1,\alpha}/2}$.

Solution. (10.4.5) is $\{\theta : -2\log\frac{L(\theta|x)}{L(\hat{\theta}|x)} \le \chi^2_{1,\alpha}\}$; (9.2.7) is $\{\theta : \frac{L(\theta|x)}{L(\hat{\theta}|x)} \ge c\}$. For fixed α, $-2\log\frac{L(\theta|x)}{L(\hat{\theta}|x)} \le \chi^2_{1,\alpha} \iff \frac{L(\theta|x)}{L(\hat{\theta}|x)} \ge e^{-\chi^2_{1,\alpha}/2}$, so with $c = e^{-\chi^2_{1,\alpha}/2}$ the two sets coincide.

Exercise 10.43

Central idea. At the boundary the score equation degenerates, so graft on the exact Clopper–Pearson endpoint $(1-p)^n = \alpha/2$; truncating to $[0,1]$ only removes impossible values.

Solution. (a) When $y = 0$ ($\hat{p} = 0$) the Wald/score formulas degenerate. Inverting the exact test (Section 9.2.3): the set of p with $P_p(Y = 0) = (1-p)^n \ge \alpha/2$ is $p \le 1 - (\alpha/2)^{1/n}$ — the exact one-sided upper limit. Similarly $y = n$ gives $P_p(Y = n) = p^n \ge \alpha/2$, i.e. $p \ge (\alpha/2)^{1/n}$. These are the Clopper–Pearson boundary endpoints, the sensible exact replacements. (b) $p \in [0,1]$ by definition; any interval extending outside $[0,1]$ contains impossible values. Intersecting with $[0,1]$ strictly shortens such intervals and cannot reduce coverage (no true p lies outside $[0,1]$) — an improvement on both length and coverage, complementing (a)'s boundary fix.

Exercise 10.44

Central idea. The "plus four" interval $\tilde{p} \pm z_{\alpha/2}\sqrt{\tilde{p}(1-\tilde{p})/(n+4)}$ mimics the score interval's shrinkage; exact coverage/length calculations show it tracks the score interval closely and dominates Wald.

Solution. Exact coverage and expected length (95% nominal):

$n = 20$, p	0.1	0.3	0.5	0.7	0.9
coverage: score	0.957	0.975	0.959	0.975	0.957
coverage: Agresti–Coull	0.957	0.975	0.959	0.975	0.957
coverage: Wald	0.876	0.947	0.959	0.947	0.876
length: score	0.263	0.364	0.393	0.364	0.263
length: AC	0.292	0.370	0.393	0.370	0.292

(At $n = 40$ the pattern is the same: AC coverage 0.944–0.962 vs. score 0.943–0.962, Wald dipping to 0.914.) Yes — the Agresti–Coull interval is a reasonable alternative: its coverage is virtually identical to the score interval's (never below nominal here, unlike Wald's dip to 0.876), its length is within a few percent of the score length, and the formula is no harder than Wald's.

Exercise 10.45

Central idea. Shift $\hat{p}$ by $\pm 1/(2n)$ before solving the score quadratic: each endpoint moves outward, widening the interval; the widening lifts the coverage dips above $1-\alpha$.

Solution. The corrected endpoints solve $\frac{|(\hat{p}\pm 1/(2n))-p|}{\sqrt{p(1-p)/n}} \le z_{\alpha/2}$: the upper endpoint is the larger root of $(\hat{p} + \frac{1}{2n} - p)^2 = z_{\alpha/2}^2 \frac{p(1-p)}{n}$, i.e.

$$p: \quad \left(1 + \frac{z_{\alpha/2}^2}{n}\right) p^2 - \left(2\hat{p} + \frac{1}{n} + \frac{z_{\alpha/2}^2}{n}\right) p + \left(\hat{p} + \frac{1}{2n}\right)^2 \le 0 \quad \text{(larger root)},$$

and the lower endpoint is the smaller root of the same quadratic with $\hat{p} - \frac{1}{2n}$. Wider: at the uncorrected upper root p_U, $|(\hat{p} + \frac{1}{2n}) - p_U| = z_{\alpha/2}\sqrt{p_U(1-p_U)/n} + \frac{1}{2n} > z_{\alpha/2}\sqrt{p_U(1-p_U)/n}$, so p_U lies strictly inside the corrected interval — the corrected upper (lower) endpoint is larger (smaller). E.g. $n = 20$, $y = 10$: uncorrected $(0.299, 0.701)$, corrected $(0.279, 0.722)$. Coverage: over a grid of $p \in [0.02, 0.98]$ ($n = 20$, 95% nominal), the uncorrected score interval's minimum coverage is 0.927 (dips below 0.95), while the corrected interval's minimum is 0.959 — uniformly higher, maintaining $\ge 1-\alpha$ throughout.

Exercise 10.46

Central idea. Simulate coverage ($n = 15$, nominal 0.9): median-with-bootstrap-SE intervals hold their level across distributions; Huber-with-bootstrap-variance intervals undercover slightly, like the book's plug-in versions.

Solution. Intervals: (a) $M_n \pm z_{0.05}\,\mathrm{SE}^*_{\mathrm{boot}}(M_n)$; (b) $\hat{\theta}_M \pm z_{0.05}\sqrt{V^*_{\mathrm{boot}}/n}$ ($B = 200$, 1500 simulations):

Underlying pdf	(a) median + bootstrap SE	(b) Huber + bootstrap var
Normal	0.891	0.875
t_5	0.893	0.871
Logistic	0.897	0.880
Double exponential	0.923	0.869

(a) The median intervals are robust: coverage stays near 0.90 for all four distributions (best for the double exponential, 0.923, where the median is the MLE). (b) The Huber bootstrap-variance intervals

undercover slightly (0.87–0.88) across the board — the bootstrap variance is automatic and avoids density estimation, but inherits the z-cutoff optimism seen in Table 10.4.8. Neither is perfect; the median interval is the more robust of the two.

Exercise 10.47

Central idea. For small p, $pX_i \dot{\sim} \text{Gamma}(r,1)$ so $2p\sum X_i \dot{\sim} \chi^2_{2nr}$; shortest intervals put equal density (not equal tail) at the endpoints.

Solution. (a) By Exercise 2.38, as $p \to 0$, $pX_i \xrightarrow{d} \text{Gamma}(r,1)$; hence $p\sum_{i=1}^n X_i \xrightarrow{d} \text{Gamma}(nr,1)$ and $2p\sum X_i \xrightarrow{d} \chi^2_{2nr}$. Therefore $P\big(\chi^2_{2nr,1-\alpha/2} \le 2p\sum X_i \le \chi^2_{2nr,\alpha/2}\big) \approx 1-\alpha$; solving for p gives the displayed interval. (b) For $1-\alpha$ coverage with upper-γ and upper-$(1-\alpha+\gamma)$ quantiles, the interval is $[\chi^2_{2nr,1-\alpha+\gamma}/(2\sum x),\ \chi^2_{2nr,\gamma}/(2\sum x)]$ with length $\propto \chi^2_{2nr,\gamma} - \chi^2_{2nr,1-\alpha+\gamma}$. Minimizing in $\gamma \in [0,\alpha]$: $\frac{d}{d\gamma}\chi^2_{2nr,\gamma} = 1/f(\chi^2_{2nr,\gamma})$ (quantile derivative), so the first-order condition is $f(\chi^2_{2nr,\gamma}) = f(\chi^2_{2nr,1-\alpha+\gamma})$ — equal χ^2_{2nr} density at the two endpoints (not equal tails). Solve numerically; e.g. $nr=10$, $\alpha=0.05$: $\gamma^* = 0.0372$ (vs. equal-tail 0.025), length $\propto 24.02$ vs. 24.58.

Exercise 10.48

Central idea. $\hat\theta = \bar Y/\bar X$; Example 5.5.27's delta-method variance plus the CLT gives a Wald interval with plug-in moments.

Solution. Estimate θ by $\hat\theta = \bar Y/\bar X$. From Example 5.5.27 (delta method for a ratio),

$$\text{Var}(\hat\theta) \approx \frac{1}{n}\left[\frac{\sigma_Y^2}{\mu_X^2} + \frac{\mu_Y^2\sigma_X^2}{\mu_X^4} - 2\frac{\mu_Y\rho\sigma_X\sigma_Y}{\mu_X^3}\right] = \frac{\sigma_Y^2 + \theta^2\sigma_X^2 - 2\theta\rho\sigma_X\sigma_Y}{n\mu_X^2},$$

and by the multivariate CLT/delta method, $(\hat\theta-\theta)/\sqrt{\text{Var}(\hat\theta)} \xrightarrow{d} n(0,1)$. Substituting sample moments gives the approximate $1-\alpha$ interval

$$\hat\theta \pm z_{\alpha/2}\sqrt{\frac{S_Y^2 + \hat\theta^2 S_X^2 - 2\hat\theta\, r S_X S_Y}{n\bar X^2}},$$

with S_X^2, S_Y^2 the sample variances and r the sample correlation. (This is the Wald/delta-method analogue of Fieller's exact set; unlike Fieller's it is always a bounded interval, at the cost of the normal approximation.)

Chapter 11

Exercise 11.1

Central idea. The Delta Method: a one-term Taylor expansion linearizes g, so the variance scales by the squared derivative; choosing g' proportional to $1/\sqrt{v}$ cancels the θ-dependence.

Solution. (a) Expand g about θ: $g(Y) \approx g(\theta) + g'(\theta)(Y-\theta)$. Then $\text{Var}(g(Y)) \approx [g'(\theta)]^2\text{Var}(Y) = [g'(\theta)]^2 v(\theta)$. (b) By the Fundamental Theorem of Calculus, $\frac{d}{dy}g^*(y) = 1/\sqrt{v(y)}$, so $\frac{d}{d\theta}g^*(\theta) = 1/\sqrt{v(\theta)}$. Plugging into (a): $\text{Var}(g^*(Y)) \approx [1/\sqrt{v(\theta)}]^2 v(\theta) = 1$, a constant independent of θ.

Exercise 11.2

Central idea. Each is (up to an irrelevant constant factor) the integral $\int dy/\sqrt{v(y)}$ for the stated mean–variance relationship.

Solution. (a) Poisson: $v(\theta)=\theta$, so $\int dy/\sqrt{y} = 2\sqrt{y} \propto \sqrt{y}$. (b) For the binomial count, $\theta = np$ and $v(\theta) = \theta(1-\theta/n)$. Then $\int \frac{dy}{\sqrt{y(1-y/n)}} = \sqrt{n}\int \frac{dy}{\sqrt{y(n-y)}}$. With $y = n\sin^2 t$, $dy = 2n\sin t\cos t\,dt$ and $\sqrt{y(n-y)} = n\sin t\cos t$, giving $2\sqrt{n}\,t = 2\sqrt{n}\sin^{-1}(\sqrt{y/n}) \propto \sin^{-1}(\sqrt{y/n})$. (c) $\int dy/\sqrt{Ky^2} = \frac{1}{\sqrt{K}}\log y \propto \log y$.

Exercise 11.3

Central idea. L'Hôpital's rule gives continuity at $\lambda = 0$; inverting the Exercise 11.1(b) construction ($g' = 1/\sqrt{v}$) recovers the variance function each power stabilizes.

Solution. (a) For $\lambda \neq 0$, $(y^\lambda - 1)/\lambda$ is continuous in λ. At $\lambda = 0$, l'Hôpital gives $\lim_{\lambda\to 0}(y^\lambda - 1)/\lambda = \lim_{\lambda\to 0} y^\lambda \log y = \log y$, which is the defined value — hence continuous. (b) For $\lambda \neq 0$, $\frac{d}{dy} g^*_\lambda(y) = y^{\lambda-1}$; for $\lambda = 0$, $\frac{d}{dy}\log y = 1/y = y^{-1}$, the same formula. By 11.1(b), stabilization requires $[g'(\theta)]^2 v(\theta)$ constant, i.e. $v(\theta) \propto \theta^{2(1-\lambda)}$ (take $v(\theta) = \theta^{2(1-\lambda)}$). Check: $\lambda = 0$ gives $v(\theta) = \theta^2$, agreeing with 11.2(c).

Exercise 11.4

Central idea. Exercise 10.17(d) gives $\sqrt{n}(r-\rho) \to n(0, (1-\rho^2)^2)$; the Delta Method with $g(r) = \operatorname{artanh}(r)$ has $g'(\rho) = 1/(1-\rho^2)$, exactly canceling the asymptotic variance. The $1/(n-3)$ refinement is a first-order expansion match.

Solution. (a) Let $g(r) = \frac{1}{2}\log\frac{1+r}{1-r} = \operatorname{artanh}(r)$, so $g'(r) = 1/(1-r^2)$. From 10.17(d), $\sqrt{n}(r-\rho) \xrightarrow{d} n(0, (1-\rho^2)^2)$. By the Delta Method, $\sqrt{n}(g(r)-g(\rho)) \xrightarrow{d} n(0, (1-\rho^2)^2 \cdot [1/(1-\rho^2)]^2) = n(0,1)$, i.e. $g(r)-g(\rho) \approx n(0, 1/n)$. (b) The mean $\rho/(2(n-1)) \approx 0$ for small ρ. For the variance: $\frac{1}{n-3} - \frac{1}{n-1} = \frac{2}{(n-3)(n-1)} \approx \frac{2}{(n-1)^2}$ for moderate n, while $\frac{4-\rho^2}{2(n-1)^2} \approx \frac{2}{(n-1)^2}$ for small ρ. Hence the variance $\approx 1/(n-1) + 2/(n-1)^2 \approx 1/(n-3)$.

Exercise 11.5

Central idea. Only the sums $\mu + \tau_i$ enter the means EY_{ij}; shifting μ up and every τ_i down by the same constant leaves the distribution untouched.

Solution. Under (11.2.2), $Y_{ij} \sim n(\mu + \tau_i, \sigma^2)$ independently (with the usual normal-error assumptions), so the distribution depends on $(\mu, \tau_1, \ldots, \tau_k)$ only through the k sums $\mu + \tau_i$. For any $c \neq 0$, $(\mu + c, \tau_1 - c, \ldots, \tau_k - c)$ is a distinct parameter vector with $(\mu + c) + (\tau_i - c) = \mu + \tau_i$ for every i, hence the identical distribution for all Y_{ij}. E.g. $(\mu, \tau_1, \ldots, \tau_k)$ and $(\mu + 1, \tau_1 - 1, \ldots, \tau_k - 1)$.

Exercise 11.6

Central idea. The normal likelihood factors through the cell means and within-cell sums of squares; independence across cells plus Lemma 5.3.3 ($\bar{X} \perp S^2$ for normal samples) gives (b); with σ^2 known the standardized cell means are exactly the canonical $n(\eta_i, 1)$ model.

Solution. (a) With $Y_{ij} \sim n(\theta_i, \sigma^2)$ independent, the likelihood is $\prod_i (2\pi\sigma^2)^{-n_i/2} \exp(-\sum_j (y_{ij} - \theta_i)^2/(2\sigma^2))$. Since $\sum_j (y_{ij} - \theta_i)^2 = \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 + n_i(\bar{y}_{i\cdot} - \theta_i)^2 = (n_i - 1)s_i^2 + n_i(\bar{y}_{i\cdot} - \theta_i)^2$, the likelihood depends on the data only through the $\bar{y}_{i\cdot}$ and s_i^2 (equivalently S_p^2), so these are sufficient by the Factorization Theorem. (b) The k samples are independent across i. By Lemma 5.3.3, for a normal sample $\bar{Y}_{i\cdot} \perp S_i^2$; hence each S_i^2 is independent of all $\bar{Y}_{j\cdot}$, and so is S_p^2, a function of the S_i^2 only. (c) With σ^2 known, the $\bar{Y}_{i\cdot} \sim n(\theta_i, \sigma^2/n_i)$ are sufficient for the θ_i. Standardizing, $X_i = \sqrt{n_i}\,\bar{Y}_{i\cdot}/\sigma \sim n(\eta_i, 1)$ independently with $\eta_i = \sqrt{n_i}\,\theta_i/\sigma$ — precisely the canonical model of Miscellanea 11.5.6 (independent unit-variance normals). Inference on θ_i is equivalent to inference on η_i.

Exercise 11.7

Central idea. A weighted version of the Lemma 5.3.2 sum-of-squares identity splits off one independent χ^2_1 piece at a time by induction.

Solution. $\bar{U}_i = \bar{Y}_{i\cdot} - \theta_i \sim n(0, \sigma^2/n_i)$ independently, and $(\bar{Y}_{i\cdot} - \bar{\bar{Y}}) - (\theta_i - \bar{\theta}) = \bar{U}_i - \bar{\bar{U}}$. Let $N_m = \sum_{i=1}^m n_i$,

$W_m = \sum_{i=1}^m n_i \bar{U}_i / N_m$. The weighted identity

$$\sum_{i=1}^{m} n_i (\bar{U}_i - W_m)^2 = \sum_{i=1}^{m-1} n_i (\bar{U}_i - W_{m-1})^2 + \frac{N_{m-1} n_m}{N_m} (\bar{U}_m - W_{m-1})^2$$

holds by algebra (write $\bar{U}_i - W_m = (\bar{U}_i - W_{m-1}) + (W_{m-1} - W_m)$; the cross term vanishes since $\sum_{i<m} n_i(\bar{U}_i - W_{m-1}) = 0$). Now $\bar{U}_m - W_{m-1} \sim n(0, \sigma^2/n_m + \sigma^2/N_{m-1}) = n(0, \sigma^2 N_m/(n_m N_{m-1}))$, independent of $\bar{U}_1, \ldots, \bar{U}_{m-1}$, so $\frac{N_{m-1} n_m}{N_m}(\bar{U}_m - W_{m-1})^2/\sigma^2 \sim \chi^2_1$ independently of the first sum. Induction from $m = 2$ (where the identity gives $\frac{n_1 n_2}{n_1+n_2}(\bar{U}_1 - \bar{U}_2)^2/\sigma^2 \sim \chi^2_1$) yields $\sum_{i=1}^k n_i(\bar{U}_i - \bar{\bar{U}})^2/\sigma^2 \sim \chi^2_{k-1}$.

Exercise 11.8

Central idea. A linear combination of independent normals is normal; means and variances add with squared coefficients.

Solution. The $\bar{Y}_{i\cdot}$ are independent with $\bar{Y}_{i\cdot} \sim n(\theta_i, \sigma^2/n_i)$. By Corollary 4.6.10, $\sum a_i \bar{Y}_{i\cdot}$ is normal with mean $\sum a_i E\bar{Y}_{i\cdot} = \sum a_i \theta_i$ and variance $\sum a_i^2 \mathrm{Var}(\bar{Y}_{i\cdot}) = \sigma^2 \sum a_i^2/n_i$.

Exercise 11.9

Central idea. Center the (11.2.7) pivot at δ: $(\sum a_i \bar{Y}_{i\cdot} - \sum a_i \theta_i)/(S_p \sqrt{\sum a_i^2/n_i}) \sim t_{N-k}$ is a pivot, independent of the unknown $\sum a_i \theta_i$.

Solution. By 11.8, $\sum a_i \bar{Y}_{i\cdot} \sim n(\sum a_i \theta_i, \sigma^2 \sum a_i^2/n_i)$, and by 11.6(b) it is independent of S_p^2 with $(N-k)S_p^2/\sigma^2 \sim \chi^2_{N-k}$. Hence

$$T = \frac{\sum_{i=1}^k a_i \bar{Y}_{i\cdot} - \sum_{i=1}^k a_i \theta_i}{S_p \sqrt{\sum_{i=1}^k a_i^2/n_i}} \sim t_{N-k}.$$

(a) Reject H_0 if $|T_0| > t_{N-k,\alpha/2}$ where $T_0 = (\sum a_i \bar{Y}_{i\cdot} - \delta)/(S_p \sqrt{\sum a_i^2/n_i})$. (b) Reject H_0 if $T_0 > t_{N-k,\alpha}$.

Exercise 11.10

Central idea. The $\mathbf{a}_i$ form a Helmert hierarchy peeling off one treatment at a time, so sequential tests pin down the ordering; orthogonality is $\sum a_i b_i/n_i = 0$ (Exercise 11.11). The fertilizer contrasts each isolate one nutrient (K, Mg dose, Mg level, Zn) against a shared reference, a scientifically natural set.

Solution. (a) With equal $n_i = n$: $\mathbf{a}_1$ compares the control θ_1 to the average of the four new treatments; $\mathbf{a}_2$ compares θ_2 to the average of $\theta_3, \theta_4, \theta_5$; $\mathbf{a}_3$ compares θ_3 to the average of θ_4, θ_5; $\mathbf{a}_4$ compares θ_4 to θ_5. Testing them in order hierarchically determines the ranking: e.g. a significant $\mathbf{a}_1$ test says the control differs from the new-treatment average (with the sign saying which is larger); then $\mathbf{a}_2$ locates θ_2 relative to the remaining three; $\mathbf{a}_3$ locates θ_3 relative to θ_4, θ_5; $\mathbf{a}_4$ orders θ_4 vs θ_5. Together the four tests can establish a complete ordering such as $\theta_1 < \bar{\theta}_{\text{new}}$, $\theta_2 > \text{avg}(\theta_3, \theta_4, \theta_5)$, etc. (b) By 11.11, $\mathrm{Cov}(\sum a_i \bar{Y}_{i\cdot}, \sum b_i \bar{Y}_{i\cdot}) = \sigma^2 \sum a_i b_i/n_i = (\sigma^2/n)\,\mathbf{a}\cdot\mathbf{b}$ with equal n_i. Directly: $\mathbf{a}_1\cdot\mathbf{a}_2 = -\frac{1}{4} + 3(\frac{1}{4}\cdot\frac{1}{3}) = 0$; $\mathbf{a}_1\cdot\mathbf{a}_3 = -\frac{1}{4} + 2(\frac{1}{4}\cdot\frac{1}{2}) = 0$; $\mathbf{a}_1\cdot\mathbf{a}_4 = -\frac{1}{4} + \frac{1}{4} = 0$; $\mathbf{a}_2\cdot\mathbf{a}_3 = -\frac{1}{3} + 2(\frac{1}{3}\cdot\frac{1}{2}) = 0$; $\mathbf{a}_2\cdot\mathbf{a}_4 = -\frac{1}{3} + \frac{1}{3} = 0$; $\mathbf{a}_3\cdot\mathbf{a}_4 = -\frac{1}{2} + \frac{1}{2} = 0$. All dot products are 0, so all pairs are uncorrelated. (c) With equal n, $\mathbf{a}_1\cdot\mathbf{a}_2 = (1)(-1) = -1 \neq 0$, so they are not orthogonal (similarly $\mathbf{a}_2\cdot\mathbf{a}_4 = 1 \neq 0$). Treatments (Mg, K, Zn): 1=(0, 0, 0), 2=(0, 200, 0), 3=(50, 200, 0), 4=(200, 200, 0), 5=(0, 200, 200). $\mathbf{a}_1$: treatment 2 vs 1 = potassium effect; $\mathbf{a}_2$: avg(3,4) vs 2 = magnesium effect at fixed K; $\mathbf{a}_3$: 3 vs 4 = magnesium dose effect; $\mathbf{a}_4$: 5 vs 2 = zinc effect. Each contrast isolates exactly one nutrient question using treatment 2 as the common reference — scientifically sensible despite non-orthogonality.

Exercise 11.11

Central idea. Independence across treatment groups kills all cross terms; only $\mathrm{Var}(\bar{Y}_{i\cdot}) = \sigma^2/n_i$ survives.

Solution. $\mathrm{Cov}(\sum_i a_i \bar{Y}_{i\cdot}, \sum_j b_j \bar{Y}_{j\cdot}) = \sum_{i,j} a_i b_j \mathrm{Cov}(\bar{Y}_{i\cdot}, \bar{Y}_{j\cdot})$. By independence across i, $\mathrm{Cov}(\bar{Y}_{i\cdot}, \bar{Y}_{j\cdot}) = 0$ for $i \neq j$ and $= \sigma^2/n_i$ for $i = j$. Hence the covariance is $\sigma^2 \sum_i a_i b_i/n_i$, which is 0 iff $\sum a_i b_i/n_i = 0$.

Exercise 11.12

Central idea. Each pairwise t^2 is an $F_{1,N-k}$ statistic; Exercise 5.8a rewrites the sum of all squared pairwise differences of the k cell means as $2k$ times the treatment sum of squares.

Solution. (a) $\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot}$ is the contrast with $a_i = 1$, $a_{i'} = -1$. By 11.8–11.9, $(\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot})/(S_p\sqrt{2/n}) \sim t_{N-k}$ under H_0, so its square is the test statistic. (b) $\sum_{i,i'} t_{ii'}^2 = \frac{n}{2S_p^2}\sum_{i,i'}(\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot})^2$ (diagonal terms are 0). Apply Exercise 5.8a to the k numbers $\bar{Y}_{1\cdot}, \ldots, \bar{Y}_{k\cdot}$: $\frac{1}{k-1}\sum_i(\bar{Y}_{i\cdot} - \bar{\bar{Y}})^2 = \frac{1}{2k(k-1)}\sum_{i,i'}(\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot})^2$. With equal n, $\mathrm{SSB} = n\sum_i(\bar{Y}_{i\cdot} - \bar{\bar{Y}})^2$, so $\sum_{i,i'}(\bar{Y}_{i\cdot} - \bar{Y}_{i'\cdot})^2 = 2k \cdot \mathrm{SSB}/n$. Thus $\frac{1}{k(k-1)}\sum_{i,i'} t_{ii'}^2 = \frac{1}{k(k-1)} \cdot \frac{n}{2S_p^2} \cdot \frac{2k\,\mathrm{SSB}}{n} = \frac{\mathrm{SSB}/(k-1)}{S_p^2} = F$.

Exercise 11.13

Central idea. The LRT statistic reduces to $(1 + \mathrm{SSB}/\mathrm{SSW})^{-N/2}$, monotone decreasing in the F ratio.

Solution. Unrestricted MLEs: $\hat{\theta}_i = \bar{Y}_{i\cdot}$, $\hat{\sigma}^2 = \sum_{i,j}(Y_{ij} - \bar{Y}_{i\cdot})^2/N = \mathrm{SSW}/N$. Under H_0 ($\theta_i \equiv \theta$): $\hat{\theta} = \bar{\bar{Y}}$, $\hat{\sigma}_0^2 = \sum_{i,j}(Y_{ij} - \bar{\bar{Y}})^2/N = \mathrm{SST}/N = (\mathrm{SSB} + \mathrm{SSW})/N$. The likelihood ratio is

$$\lambda = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}^2}\right)^{-N/2} = \left(\frac{\mathrm{SSB} + \mathrm{SSW}}{\mathrm{SSW}}\right)^{-N/2} = \left(1 + \frac{\mathrm{SSB}}{\mathrm{SSW}}\right)^{-N/2},$$

which is strictly decreasing in SSB/SSW, hence strictly decreasing in $F = \frac{\mathrm{SSB}/(k-1)}{\mathrm{SSW}/(N-k)}$. So $\{\lambda < c\} = \{F > c'\}$: the LRT rejects for large F, the F test of (11.2.14).

Exercise 11.14

Central idea. Without the $\sum a_i = 0$ restriction, Cauchy–Schwarz gives the unconstrained maximum $\sum c_i v_i^2$; with $v_i = \bar{Y}_{i\cdot} - \theta_i$, $c_i = n_i$ this is χ_k^2, gaining one degree of freedom over the contrast case.

Solution. First, by Cauchy–Schwarz, $(\sum a_i v_i)^2 = (\sum \frac{a_i}{\sqrt{c_i}} \cdot v_i\sqrt{c_i})^2 \le (\sum a_i^2/c_i)(\sum c_i v_i^2)$, with equality at $a_i \propto c_i v_i$. Hence $\max_{\mathbf{a}}(\sum a_i v_i)^2/(\sum a_i^2/c_i) = \sum c_i v_i^2$. Now let $T_{\mathbf{a}} = \sum a_i(\bar{Y}_{i\cdot} - \theta_i)/\sqrt{S_p^2 \sum a_i^2/n_i}$. Adapting Theorem 11.2.10: the simultaneous intervals hold iff $\max_{\mathbf{a}} T_{\mathbf{a}}^2 \le \mathbf{M}^2$. With $v_i = \bar{Y}_{i\cdot} - \theta_i$, $c_i = n_i$, the identity gives $\max_{\mathbf{a}} T_{\mathbf{a}}^2 = \sum n_i(\bar{Y}_{i\cdot} - \theta_i)^2/S_p^2$. Since $\sqrt{n_i}(\bar{Y}_{i\cdot} - \theta_i)/\sigma$ are iid $n(0,1)$, the numerator/$\sigma^2 \sim \chi_k^2$, independent of S_p^2 (by 11.6(b)), so $\max_{\mathbf{a}} T_{\mathbf{a}}^2 \sim kF_{k,N-k}$. Taking $\mathbf{M}^2 = kF_{k,N-k,\alpha}$ gives simultaneous coverage $1-\alpha$ for all linear combinations.

Exercise 11.15

Central idea. $t_{\nu,\alpha/2}^2 = F_{1,\nu,\alpha}$, and the quantiles of $kF_{k,\nu}$ increase with k; Scheffé pays for simultaneity over all contrasts with a larger critical value.

Solution. (a) Since $t_\nu^2 \overset{d}{=} F_{1,\nu}$, $t_{\nu,\alpha/2}^2 = F_{1,\nu,\alpha} = 1 \cdot F_{1,\nu,\alpha}$. The distributions $kF_{k,\nu}$ are stochastically increasing in k, so their upper-α quantiles increase in k: for $k - 1 \ge 1$, $(k-1)F_{k-1,\nu,\alpha} \ge 1 \cdot F_{1,\nu,\alpha} = t_{\nu,\alpha/2}^2$. Taking square roots gives the inequality. (b) The Scheffé interval for a contrast $\sum a_i\theta_i$ has half-width $\sqrt{(k-1)F_{k-1,N-k,\alpha}}\, S_p\sqrt{\sum a_i^2/n_i}$, while the single-contrast t interval (from (11.2.7)) has half-width $t_{N-k,\alpha/2}\, S_p\sqrt{\sum a_i^2/n_i}$. By (a) with $\nu = N-k$, the Scheffé multiplier is $\ge$ the t multiplier, so Scheffé intervals are always at least as wide. (c) The Scheffé test rejects $H_0 : \sum a_i\theta_i = 0$ when $|T| > \sqrt{(k-1)F_{k-1,N-k,\alpha}}$, the t test when $|T| > t_{N-k,\alpha/2}$, for the same statistic T. Since the Scheffé cutoff is larger, whenever Scheffé rejects the t test also rejects — the Scheffé rejection region is smaller, so its power is $\le$ the t test's power at every alternative.

Exercise 11.16

Central idea. Equality of all means is the intersection of all pairwise equalities; the union–intersection principle then rejects the global null when any pairwise t test rejects.

Solution. (a) $\theta_1 = \cdots = \theta_k$ holds iff every pair is equal, i.e. iff $\theta_i - \theta_j = 0$ for all i, j; negating, some pair differs iff not all are equal. (b) $H_0 : \theta \in \bigcap_{i,j} \Theta_{ij}$ and $H_1 : \theta \in \bigcup_{i,j} \Theta_{ij}^c$. This is exactly the union–intersection form $H_0 = \bigcap_\gamma \Theta_{0\gamma}$ vs $H_1 = \bigcup_\gamma \Theta_{1\gamma}$. The UIT rejects H_0 iff at least one component test rejects: for each pair (i, j), run the two-sample t test of $\theta_i = \theta_j$, $|\bar{Y}_{i\cdot} - \bar{Y}_{j\cdot}|/(S_p\sqrt{1/n_i + 1/n_j}) > t_{N-k,\alpha/2}$, and reject the ANOVA H_0 if any pair rejects. Multiplicity is handled as in Miscellanea 11.5.2 (e.g. Bonferroni or Tukey adjustment) to control the experimentwise error.

Exercise 11.17

Central idea. Protection works because a false difference assertion requires the F gate to open first, and the F test has level α under the global null; without the gate, $\binom{k}{2}$ chances at error α each inflate the familywise rate.

Solution. (a) Under the global null (all means equal), a false assertion of difference can only occur if the pairwise t tests are performed, which requires the F test to reject. Hence

$$P(\text{false assertion} \mid H_0) \le P(F \text{ rejects} \mid H_0) = \alpha,$$

regardless of k. (b) Without the F gate, under H_0 each of the $m = \binom{k}{2}$ pairwise t tests falsely rejects with probability α, so $P(\text{at least one false} \mid H_0) = P(\bigcup_{pairs}\{\text{reject}\})$, which can be as large as $\min(1, m\alpha)$ — exceeding α for $k \ge 3$ (e.g. approaching $1-(1-\alpha)^m$ for nearly independent tests). Each single comparison still has $P(\text{false difference}) = \alpha$: the comparisonwise rate. (c) Fish toxin data ($n = (4, 6, 5, 4)$, $N = 19$, $k = 4$): $\bar{y} = (23.0, 32.83, 21.0, 13.0)$, $S_p^2 = 12.72$, $F = 26.09$ on $(3, 15)$ df — significant at $\alpha = 0.05$ (indeed $p \approx 4 \times 10^{-6}$), so proceed. With $t_{15,0.025} = 2.13$:

Pair	$\lvert t\rvert$	Significant?
Toxin 1 – Toxin 2	4.27	yes
Toxin 1 – Toxin 3	0.84	no
Toxin 1 – Control	3.96	yes
Toxin 2 – Toxin 3	5.48	yes
Toxin 2 – Control	8.61	yes
Toxin 3 – Control	3.34	yes

Conclusions: Toxin 2 produces the most liver deterioration (mean 32.8), the control the least (13.0); Toxins 1 and 3 are indistinguishable (23.0 vs 21.0); every other pair differs significantly.

Exercise 11.18

Central idea. Conditioning on the event that suggested the cutoff ($Y > b$) can only inflate the rejection probability, since $\{Y > a\} \subset \{Y > b\}$.

Solution. (a) Since $a > b$, $\{Y > a\} \subset \{Y > b\}$, so $P(Y > a \mid Y > b) = P(Y > a)/P(Y > b)$. With $P(Y > b) < 1$, dividing by it strictly increases: $P(Y > a \mid Y > b) > P(Y > a)$. (b) Suppose the data suggest testing at a cutoff a slightly below the observed statistic — the hypothesis is entertained only because the data already exceeded some threshold $b < a$ (the "suggestion" event). The nominal size $P(Y > a)$ is computed unconditionally, but the actual rejection probability of this data-suggested procedure is $P(Y > a \mid Y > b) > P(Y > a)$: data snooping inflates the true Type I error above the advertised level.

Exercise 11.19

Central idea. Write $S_i = X_1 + \cdots + X_i$ and $T_i = S_i/S_{i+1}$; the T_i are independent Beta variables, and $Y_i = (1-T_i)/T_i$ turns them into independent beta-prime (scaled F) ratios, all independent of the total $Y_n = S_n$.

Solution. (a) Put $S_i = \sum_{j=1}^{i} X_j$ and $\Lambda_i = \sum_{j=1}^{i} \lambda_j$, so $S_i \sim \text{gamma}(\Lambda_i, 1)$. For $i = 1, \ldots, n-1$ let $T_i = S_i/S_{i+1}$. Since S_i and $X_{i+1} = S_{i+1} - S_i$ are independent gammas, $T_i \sim \text{Beta}(\Lambda_i, \lambda_{i+1})$ independent of S_{i+1}; working backward from $i = n-1$, the T_i are mutually independent and independent of $S_n = Y_n \sim \text{gamma}(\Lambda_n, 1)$. Now $Y_i = X_{i+1}/S_i = (S_{i+1} - S_i)/S_i = (1-T_i)/T_i$. If $T \sim \text{Beta}(a, b)$ then $(1-T)/T$ is beta-prime(b, a), so the Y_i are mutually independent with $Y_i \sim$ beta-prime$(\lambda_{i+1}, \Lambda_i)$, i.e. $\frac{\Lambda_i}{\lambda_{i+1}} Y_i \sim F_{2\lambda_{i+1}, 2\Lambda_i}$, and all are independent of $Y_n \sim \text{gamma}(\Lambda_n, 1)$. (b) These are independent sequential F ratios: each new observation's contribution X_{i+1}, scaled by the accumulated sum S_i, is independent of everything before it. In ANOVA the same mechanism produces the independent F statistics of sequential (Type I) sums of squares — each factor's extra sum of squares, scaled by what came before, tested independently of the earlier terms.

Exercise 11.20

Central idea. Under H_0 the between-treatment sum of squares is $(k-1)\sigma^2$ in expectation (one degree lost estimating $\bar{\bar{Y}}$); writing $F = (\text{MSB}/\sigma^2)/(\text{MSW}/\sigma^2)$ as a ratio of independent scaled chi-squares is exactly the Example 5.3.5 construction.

Solution. (a) Under H_0, $\bar{Y}_{i\cdot} \sim n(\theta, \sigma^2/n_i)$ independently. $\sum n_i(\bar{Y}_{i\cdot} - \bar{\bar{Y}})^2 = \sum n_i(\bar{Y}_{i\cdot} - \theta)^2 - N(\bar{\bar{Y}} - \theta)^2$, so its expectation is $\sum n_i(\sigma^2/n_i) - N(\sigma^2/N) = k\sigma^2 - \sigma^2 = (k-1)\sigma^2$. Dividing by $k-1$ is unbiased for σ^2. (b) Example 5.3.5 derives the F distribution as the ratio of two independent scaled chi-squares, cf. (5.3.8). Under H_0, $\text{MSB}/\sigma^2 = [\sum n_i(\bar{Y}_{i\cdot} - \bar{\bar{Y}})^2/\sigma^2]/(k-1) \sim \chi^2_{k-1}/(k-1)$ (by 11.20(a)/11.7) and $\text{MSW}/\sigma^2 = S_p^2/\sigma^2 \sim \chi^2_{N-k}/(N-k)$, independent by 11.6(b). Hence $F = (\text{MSB}/\sigma^2)/(\text{MSW}/\sigma^2) \sim F_{k-1,N-k}$ under H_0 — the ANOVA F test by the same variance-ratio method.

Exercise 11.21

Central idea. SST = SSB + SSW via the cross term vanishing; the diet means differ enormously ($F \approx 110$); the F statistic is the average of the six ordered pairwise t^2 values.

Solution. (a) Cell means: Low 3.508, Medium 9.274, High 24.926; grand mean 12.569. SSB = 1228.27, SSW = 66.74, SST = 1295.01.

Source	df	SS	MS	F
Diet (between)	2	1228.27	614.14	110.43
Error (within)	12	66.74	5.561	
Total	14	1295.01		

The F of 110.43 is overwhelming evidence that protein level affects weight gain. (b) Write $y_{ij} - \bar{\bar{y}} = (y_{ij} - \bar{y}_{i\cdot}) + (\bar{y}_{i\cdot} - \bar{\bar{y}})$. Squaring and summing, the cross term is $2\sum_i(\bar{y}_{i\cdot} - \bar{\bar{y}})\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$ since $\sum_j(y_{ij} - \bar{y}_{i\cdot}) = 0$ for each i. Hence $\sum_{i,j}(y_{ij} - \bar{\bar{y}})^2 = \sum_i n_i(\bar{y}_{i\cdot} - \bar{\bar{y}})^2 + \sum_{i,j}(y_{ij} - \bar{y}_{i\cdot})^2$. (c) With $n = 5$, $S_p^2 = 5.561$: $t_{12}^2 = 14.95$ (Low–Med), $t_{13}^2 = 206.21$ (Low–High), $t_{23}^2 = 110.13$ (Med–High). Summing over all 3×3 ordered pairs (diagonal terms 0, each pair counted twice): $\frac{1}{3\cdot 2} \cdot 2(14.95 + 206.21 + 110.13) = 110.43 = F$.

Exercise 11.22

Central idea. Write $\bar{Y}_{i\cdot} - \bar{\bar{Y}} = (\theta_i - \bar{\theta}) + (\text{mean-zero noise})$; the noise contributes $(k-1)\sigma^2$ (Exercise 11.20a) and the signal contributes $\sum n_i(\theta_i - \bar{\theta})^2$.

Solution. MSW $= S_p^2$, and $E[\text{MSW}] = \sigma^2$ since each S_i^2 is unbiased. For MSB, write $\bar{Y}_{i\cdot} = \theta_i + e_i$, $\bar{\bar{Y}} = \bar{\theta} + \bar{e}$ with $\bar{\theta} = \sum n_i\theta_i/N$. Then SSB $= \sum n_i[(\theta_i - \bar{\theta}) + (e_i - \bar{e})]^2 = \sum n_i(\theta_i - \bar{\theta})^2 + 2\sum n_i(\theta_i - \bar{\theta})(e_i - \bar{e}) + \sum n_i(e_i - \bar{e})^2$. The middle term has mean 0; the last has mean $(k-1)\sigma^2$ by 11.20(a). Hence

$$E[\text{MSB}] = \sigma^2 + \frac{\sum_{i=1}^k n_i(\theta_i - \bar{\theta})^2}{k-1}, \qquad E[\text{MSW}] = \sigma^2.$$

Under H_0 both equal σ^2, which is why $F = \text{MSB}/\text{MSW}$ detects treatment effects.

Exercise 11.23

Central idea. Averaging over blocks leaves $EY_{ij} = \mu + \tau_i$; the $\sum a_i = 0$ condition annihilates the shared block component, leaving only σ^2/r per unit weight — which equals $\frac{1}{r}(\sigma^2 + \sigma_B^2)(1-\rho)$ since $\rho = \sigma_B^2/(\sigma^2 + \sigma_B^2)$.

Solution. (a) $Y_{ij} = \mu + \tau_i + B_j + \varepsilon_{ij}$ with $B_j \sim n(0, \sigma_B^2) \perp \varepsilon_{ij} \sim n(0, \sigma^2)$. $EY_{ij} = \mu + \tau_i$; $\operatorname{Var} Y_{ij} = \operatorname{Var}(B_j) + \operatorname{Var}(\varepsilon_{ij}) = \sigma_B^2 + \sigma^2$. (b) $\operatorname{Cov}(Y_{ij}, Y_{i'j'}) = \sigma_B^2$ if $j = j'$, $i \neq i'$ (shared block), else 0. So $\operatorname{Cov}(\bar{Y}_{i\cdot}, \bar{Y}_{i'\cdot}) = \sigma_B^2/r$ for $i \neq i'$ and $\operatorname{Var}(\bar{Y}_{i\cdot}) = (\sigma_B^2 + \sigma^2)/r$. Then

$$\operatorname{Var}\left(\sum a_i \bar{Y}_{i\cdot}\right) = \frac{\sigma^2 + \sigma_B^2}{r}\sum a_i^2 + \frac{\sigma_B^2}{r}\sum_{i \neq i'} a_i a_{i'} = \frac{1}{r}\left[\sigma^2 \sum a_i^2 + \sigma_B^2\left(\sum a_i\right)^2\right] = \frac{\sigma^2}{r}\sum a_i^2,$$

using $\sum a_i = 0$. Since $\rho = \sigma_B^2/(\sigma^2 + \sigma_B^2)$, $(\sigma^2 + \sigma_B^2)(1-\rho) = \sigma^2$, giving the stated form.

Exercise 11.24

Central idea. Marginally $X_i \sim n(0, \tau^2 + 1)$; $E[(p-2)/S] = 1/(\tau^2+1)$ because $E[1/\chi_p^2] = 1/(p-2)$ — so the Stein shrinkage factor unbiasedly estimates the Bayes shrinkage factor $\tau^2/(\tau^2+1)$, but the reciprocal-chi-square moment explodes for $p < 3$.

Solution. (a) $X_i = \theta_i + \varepsilon_i$ with $\theta_i \sim n(0, \tau^2) \perp \varepsilon_i \sim n(0,1)$; marginally $X_i \sim n(0, \tau^2+1)$, iid. Hence $S/(\tau^2+1) = \sum X_i^2/(\tau^2+1) \sim \chi_p^2$. (b) The Bayes estimator under the $n(0,\tau^2)$ prior is $E[\theta_i \mid X_i] = [\tau^2/(\tau^2+1)]X_i$. Now

$$E\left[\frac{p-2}{S}\right] = \frac{p-2}{\tau^2+1}E\left[\frac{1}{S/(\tau^2+1)}\right] = \frac{p-2}{\tau^2+1}\cdot\frac{1}{p-2} = \frac{1}{\tau^2+1},$$

using $E[1/Y] = 1/(p-2)$ for $Y \sim \chi_p^2$, $p > 2$ (since $\chi_p^2 = \text{gamma}(p/2, 2)$ in the text's scale convention and $E[Y^{-1}] = \frac{1}{2}\Gamma(p/2-1)/\Gamma(p/2) = 1/(p-2)$). Thus $E[1-(p-2)/S] = \tau^2/(\tau^2+1)$: the Stein factor $(1-(p-2)/S)$ is an unbiased (empirical Bayes) estimate of the Bayes shrinkage factor. (c) For $Y \sim \chi_2^2 = \text{Exp}(1/2)$, $E[1/Y] = \int_0^\infty \frac{1}{y}\frac{1}{2}e^{-y/2}dy = \infty$ (diverges like $\int_0 dy/y$ at 0); for $p = 1$ the density $\propto y^{-1/2}$ near 0 gives $\int_0 y^{-3/2}dy = \infty$. So $E[(p-2)/S]$ is meaningless for $p < 3$.

Exercise 11.25

Central idea. Setting the gradient of the quadratic $\text{RSS}(c,d) = \sum(y_i - c - dx_i)^2$ to zero gives the normal equations; the Hessian $2\begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$ is positive definite ($nS_{xx} > 0$), so the critical point is the global minimum.

Solution. (a) $\frac{\partial \text{RSS}}{\partial c} = -2\sum_i(y_i - c - dx_i) = 0$ gives $nc + d\sum x_i = \sum y_i$; $\frac{\partial \text{RSS}}{\partial d} = -2\sum_i x_i(y_i - c - dx_i) = 0$ gives $c\sum x_i + d\sum x_i^2 = \sum x_i y_i$. (b) With $a = \bar{y} - b\bar{x}$, $b = S_{xy}/S_{xx}$: $na + b\sum x_i = n\bar{y} - bn\bar{x} + bn\bar{x} = \sum y_i$; and $a\sum x_i + b\sum x_i^2 = n\bar{x}\bar{y} - bn\bar{x}^2 + b\sum x_i^2 = n\bar{x}\bar{y} + bS_{xx} = n\bar{x}\bar{y} + S_{xy} = \sum x_i y_i$ (since $S_{xy} = \sum x_i y_i - n\bar{x}\bar{y}$). (c) The Hessian is $2\begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$ with leading minor $2n > 0$ and determinant $4(n\sum x_i^2 - (\sum x_i)^2) = 4nS_{xx} > 0$ (unless all x_i equal), hence positive definite: (a, b) is the unique global minimizer.

Exercise 11.26

Central idea. S_{xx} is convex in each coordinate separately, so at the maximum every x_i sits at an endpoint; among endpoint allocations, $j(n-j)(f-e)^2/n$ is maximized at $j = n/2$.

Solution. Fix all coordinates but x_k: $S_{xx} = \sum x_i^2 - (\sum x_i)^2/n$ is quadratic in x_k with positive leading coefficient $(1 - 1/n) > 0$, hence convex in x_k — its maximum over $[e, f]$ is at an endpoint. So at any maximizer, each $x_i \in \{e, f\}$. If j of them equal e (and $n - j$ equal f), then $\bar{x} = (je + (n-j)f)/n$ and

$$S_{xx} = j(e - \bar{x})^2 + (n-j)(f - \bar{x})^2 = \frac{j(n-j)}{n}(f-e)^2,$$

maximized over j at $j = n/2$ (n even). Hence half at e, half at f maximizes S_{xx} (and minimizes $\operatorname{Var} b = \sigma^2/S_{xx}$).

Exercise 11.27

Central idea. Minimize $\sum c_i^2$ subject to the two unbiasedness constraints $\sum c_i = 1$, $\sum c_i x_i = 0$ (Lagrange); the solution is $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$, the least squares intercept.

Solution. Seek $\sum c_i Y_i$ with $E\sum c_i Y_i = \alpha \sum c_i + \beta \sum c_i x_i \equiv \alpha$, i.e. $\sum c_i = 1$, $\sum c_i x_i = 0$, minimizing $\operatorname{Var} = \sigma^2 \sum c_i^2$. Lagrange: $2c_i = \lambda_1 + \lambda_2 x_i$, so $c_i = a + bx_i$. The constraints give $na + b\sum x_i = 1$ and $a\sum x_i + b\sum x_i^2 = 0$, i.e. $a + b\bar{x} = 1/n$ and, substituting, $bS_{xx} = -\bar{x}$. Hence $b = -\bar{x}/S_{xx}$, $a = 1/n + \bar{x}^2/S_{xx}$, and

$$c_i = \frac{1}{n} - \frac{\bar{x}(x_i - \bar{x})}{S_{xx}}.$$

The BLUE is $\sum c_i Y_i = \bar{Y} - \bar{x}\sum_i \frac{(x_i - \bar{x})}{S_{xx}} Y_i = \bar{Y} - \hat{\beta}\bar{x} = \hat{\alpha}$, the least squares intercept (the Hessian/constrained-quadratic argument confirms the minimum).

Exercise 11.28

Central idea. Profile the likelihood: for fixed σ^2 the likelihood is maximized at the least squares $(\hat{\alpha}, \hat{\beta})$; optimizing the remaining $-\frac{n}{2}\log\sigma^2 - \mathrm{RSS}/(2\sigma^2)$ gives RSS/n.

Solution. The log likelihood is $-\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_i (y_i - \alpha - \beta x_i)^2$. For fixed σ^2, maximizing over (α, β) minimizes $\sum_i (y_i - \alpha - \beta x_i)^2$, attained at the least squares estimators $(\hat{\alpha}, \hat{\beta})$ with value RSS. Maximizing $-\frac{n}{2}\log\sigma^2 - \mathrm{RSS}/(2\sigma^2)$ over $\sigma^2 > 0$: derivative $-\frac{n}{2\sigma^2} + \frac{\mathrm{RSS}}{2\sigma^4} = 0$ gives $\hat{\sigma}^2 = \mathrm{RSS}/n = \frac{1}{n}\sum_i (y_i - \hat{\alpha} - \hat{\beta}x_i)^2$ (second derivative negative there).

Exercise 11.29

Central idea. Bilinearity of covariance gives the six-term decomposition in (b); Lemma 11.3.2 reduces each covariance to a coefficient (c_i or d_i) times σ^2 — and the text's + sign in the first covariance is a typo for −, without which (11.3.28) does not follow.

Solution. (a) $E\hat{\alpha} = \alpha$, $E\hat{\beta} = \beta$, so $E\hat{\varepsilon}_i = (\alpha + \beta x_i) - \alpha - \beta x_i = 0$. (b) Write $\hat{\varepsilon}_i = Y_i + (-\hat{\alpha}) + (-x_i\hat{\beta})$ and expand by bilinearity:

$$\operatorname{Var}\hat{\varepsilon}_i = \operatorname{Var} Y_i + \operatorname{Var}\hat{\alpha} + x_i^2 \operatorname{Var}\hat{\beta} + 2\operatorname{Cov}(Y_i, -\hat{\alpha}) + 2\operatorname{Cov}(Y_i, -x_i\hat{\beta}) + 2\operatorname{Cov}(-\hat{\alpha}, -x_i\hat{\beta}),$$

which is the stated formula, since $\operatorname{Cov}(-\hat{\alpha}, -x_i\hat{\beta}) = x_i \operatorname{Cov}(\hat{\alpha}, \hat{\beta})$. (c) By Lemma 11.3.2, $\operatorname{Cov}(Y_i, \hat{\alpha}) = c_i\sigma^2$ and $\operatorname{Cov}(Y_i, \hat{\beta}) = d_i\sigma^2$, with $c_i = \frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}}$ and $d_i = \frac{x_i - \bar{x}}{S_{xx}}$ from 11.30(a). *Erratum:* the text prints the first covariance with a + sign, $\sigma^2(\frac{1}{n} + \frac{(x_i - \bar{x})\bar{x}}{S_{xx}})$ — this is wrong; the correct sign is −. Proof: (i) Lemma 11.3.2 gives $\operatorname{Cov}(Y_i, \hat{\alpha}) = c_i\sigma^2$, and 11.30(a) — the very next exercise, with the same estimators — prints $c_i = \frac{1}{n} - \frac{(x_i - \bar{x})\bar{x}}{S_{xx}}$, so the text contradicts itself; (ii) only the minus sign reproduces (11.3.28): with the + sign the computation below misses by $-4\bar{x}(x_i - \bar{x})/S_{xx}$ (checked numerically — the + version even

produces negative "variances"). Using the corrected covariances, with $\operatorname{Var} Y_i = \sigma^2$, $\operatorname{Var}\hat{\alpha} = \sigma^2 \sum_j x_j^2/(nS_{xx})$, $\operatorname{Var}\hat{\beta} = \sigma^2/S_{xx}$ and $\operatorname{Cov}(\hat{\alpha},\hat{\beta}) = -\sigma^2\bar{x}/S_{xx}$ (all from 11.30), part (b) gives

$$\frac{\operatorname{Var}\hat{\varepsilon}_i}{\sigma^2} = 1 - \frac{2}{n} + \frac{1}{S_{xx}}\left[\frac{1}{n}\sum_{j=1}^{n} x_j^2 + x_i^2 + 2\bar{x}(x_i - \bar{x}) - 2x_i(x_i - \bar{x}) - 2x_i\bar{x}\right].$$

Since $2\bar{x}(x_i-\bar{x}) - 2x_i(x_i-\bar{x}) = -2(x_i-\bar{x})^2$, the bracket equals $\frac{1}{n}\sum_j x_j^2 + x_i^2 - 2(x_i-\bar{x})^2 - 2x_i\bar{x}$, and $1 - 2/n = (n-2)/n$: exactly (11.3.28). Equivalently, after simplifying with $n^{-1}\sum_j x_j^2 = S_{xx}/n + \bar{x}^2$, the familiar result is $\operatorname{Var}(\hat{\varepsilon}_i) = \sigma^2[1 - 1/n - (x_i-\bar{x})^2/S_{xx}]$.

Exercise 11.30

Central idea. Direct coefficient matching from $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$; means, variances and covariances are inner products of the coefficient vectors, simplified with $\sum(x_i - \bar{x}) = 0$.

Solution. (a) $\hat{\beta} = S_{xY}/S_{xx} = \sum_i d_i Y_i$ with $d_i = (x_i - \bar{x})/S_{xx}$. $\hat{\alpha} = \bar{Y} - \bar{x}\hat{\beta} = \sum_i (\frac{1}{n} - \bar{x}d_i)Y_i = \sum_i c_i Y_i$ since $\frac{1}{n} - \bar{x}(x_i - \bar{x})/S_{xx} = c_i$. (b) $E\hat{\alpha} = \sum_i c_i(\alpha + \beta x_i) = \alpha\sum_i c_i + \beta\sum_i c_i x_i = \alpha$, because $\sum_i c_i = 1$ and $\sum_i c_i x_i = \bar{x} - \bar{x}\sum_i d_i x_i = \bar{x} - \bar{x} = 0$. $\operatorname{Var}\hat{\alpha} = \sigma^2\sum c_i^2$; $\sum c_i^2 = \sum[\frac{1}{n^2} - \frac{2\bar{x}(x_i-\bar{x})}{nS_{xx}} + \frac{\bar{x}^2(x_i-\bar{x})^2}{S_{xx}^2}] = \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} = \frac{S_{xx}+n\bar{x}^2}{nS_{xx}} = \frac{\sum x_i^2}{nS_{xx}}$ (using $\sum(x_i-\bar{x}) = 0$, $\sum(x_i-\bar{x})^2 = S_{xx}$, $\sum x_i^2 = S_{xx} + n\bar{x}^2$). Hence $\operatorname{Var}\hat{\alpha} = \sigma^2\sum x_i^2/(nS_{xx})$. (c) $\operatorname{Cov}(\hat{\alpha},\hat{\beta}) = \sigma^2\sum_i c_i d_i = \sigma^2\sum_i(\frac{1}{n} - \bar{x}d_i)d_i = \sigma^2(\frac{1}{n}\sum_i d_i - \bar{x}\sum_i d_i^2) = -\sigma^2\bar{x}/S_{xx}$, using $\sum_i d_i = \sum_i(x_i-\bar{x})/S_{xx} = 0$ and $\sum_i d_i^2 = 1/S_{xx}$.

Exercise 11.31

Central idea. $\operatorname{Cov}(\hat{\varepsilon}_i,\hat{\alpha}) = \sigma^2\sum_j e_j c_j$ and $\sum_j e_j c_j = c_i - \sum c_j^2 - x_i\sum c_j d_j = 0$ (and similarly for $\hat{\beta}$) by the four coefficient identities; normality upgrades uncorrelated to independent.

Solution. For fixed i, (11.3.30) gives $e_j = \delta_{ij} - c_j - d_j x_i$, so $\hat{\varepsilon}_i = \sum_j e_j Y_j$. By Lemma 11.3.2, $\operatorname{Cov}(\hat{\varepsilon}_i,\hat{\alpha}) = \sigma^2\sum_j e_j c_j$ and $\operatorname{Cov}(\hat{\varepsilon}_i,\hat{\beta}) = \sigma^2\sum_j e_j d_j$. Now $\sum_j e_j c_j = c_i - \sum_j c_j^2 - x_i\sum_j c_j d_j = [\frac{1}{n} - \frac{\bar{x}(x_i-\bar{x})}{S_{xx}}] - [\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}] - x_i[-\frac{\bar{x}}{S_{xx}}] = 0$, and $\sum_j e_j d_j = d_i - \sum_j c_j d_j - x_i\sum_j d_j^2 = \frac{x_i-\bar{x}}{S_{xx}} + \frac{\bar{x}}{S_{xx}} - \frac{x_i}{S_{xx}} = 0$, using $\sum c_j^2 = 1/n + \bar{x}^2/S_{xx}$, $\sum d_j^2 = 1/S_{xx}$, $\sum c_j d_j = -\bar{x}/S_{xx}$. So both covariances are 0; $(\hat{\varepsilon}_i,\hat{\alpha},\hat{\beta})$ are jointly normal (linear in the Y_j), hence independent.

Exercise 11.32

Central idea. Centering the predictor ($z_i = x_i - \bar{x}$, so $\bar{z} = 0$) orthogonalizes the design: the slope is unchanged, the intercept collapses to $\bar{Y}$, and the two estimators become uncorrelated.

Solution. (a) In the reparameterized model the predictor is $z_i = x_i - \bar{x}$. The least squares (hence MLE, under normality) slope is $\hat{\beta}' = \sum_i z_i(Y_i - \bar{Y})/\sum_i z_i^2$. Since $\sum_i z_i^2 = \sum_i(x_i-\bar{x})^2 = S_{xx}$ and $\sum_i z_i(Y_i-\bar{Y}) = \sum_i(x_i-\bar{x})(Y_i-\bar{Y}) = S_{xY}$, $\hat{\beta}' = S_{xY}/S_{xx} = \hat{\beta}$. (b) $\hat{\alpha}' = \bar{Y} - \hat{\beta}'\bar{z} = \bar{Y} - \hat{\beta}'\cdot 0 = \bar{Y}$. In general $\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x} \neq \bar{Y}$ (they differ unless $\bar{x} = 0$ or $\hat{\beta} = 0$), so $\hat{\alpha}' \neq \hat{\alpha}$. Since $EY_i = \alpha + \beta x_i = (\alpha + \beta\bar{x}) + \beta(x_i - \bar{x})$, the reparameterized intercept is $\alpha' = \alpha + \beta\bar{x}$, and $\hat{\alpha}' = \bar{Y} \sim n(\alpha', \sigma^2/n) = n(\alpha + \beta\bar{x}, \sigma^2/n)$. (c) $\hat{\alpha}' = \sum_i \frac{1}{n}Y_i$ and $\hat{\beta}' = \sum_i d_i Y_i$ with $d_i = (x_i - \bar{x})/S_{xx}$. $\operatorname{Cov}(\hat{\alpha}',\hat{\beta}') = \sigma^2\sum_i \frac{1}{n}d_i = \frac{\sigma^2}{nS_{xx}}\sum_i(x_i - \bar{x}) = 0$. Both are linear in the normal Y_is, hence jointly normal; uncorrelated $\Rightarrow$ independent.

Exercise 11.33

Central idea. The population regression slope is $\beta = \rho\sigma_Y/\sigma_X$, so $\beta = 0 \iff \rho = 0$; the usual t statistic for the slope rewrites as a function of r alone.

Solution. (a) From (11.3.25), the conditional distribution of Y given $X = x$ in the bivariate normal model is $n(\alpha + \beta x, \sigma^2)$ with $\beta = \rho\,\sigma_Y/\sigma_X$. Since $\sigma_X, \sigma_Y > 0$, $\beta = 0 \iff \rho = 0$. (b) $\hat\beta = S_{xy}/S_{xx}$ and $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, so $\hat\beta = r\sqrt{S_{yy}/S_{xx}}$. With $S^2 = \text{SSE}/(n-2)$ and $\text{SSE} = S_{yy} - S_{xy}^2/S_{xx} = S_{yy}(1-r^2)$:

$$\frac{\hat\beta}{S/\sqrt{S_{xx}}} = \frac{r\sqrt{S_{yy}}}{\sqrt{S_{yy}(1-r^2)/(n-2)}} = \sqrt{n-2}\,\frac{r}{\sqrt{1-r^2}}.$$

(c) Under $H_0 : \rho = 0$ (equivalently $H_0 : \beta = 0$), the statistic $T = r\sqrt{n-2}/\sqrt{1-r^2}$ is exactly the t statistic for $H_0 : \beta = 0$, so $T \sim t_{n-2}$. Reject $H_0 : \rho = 0$ at level α if $|T| > t_{n-2,\alpha/2}$; computing $|T| = \sqrt{n-2}\cdot\sqrt{r^2/(1-r^2)}$ needs only r^2 and n.

Exercise 11.34

Central idea. $n = 8$: $S_{xx} = 70$, $S_{xy} = 65$, $S_{yy} = 67.5$ give $F = 50.70$; the cross term in $(y_i - \bar y) = (y_i - \hat y_i) + (\hat y_i - \bar y)$ vanishes by the normal equations; $r^2 = S_{xy}^2/(S_{xx}S_{yy})$ is both squared correlations.

Solution. (a) $\bar x = 34$, $\bar y = 66.75$, $S_{xx} = 70$, $S_{xy} = 65$, $S_{yy} = 67.5$. $\hat\beta = 65/70 = 0.9286$, $\hat\alpha = 66.75 - 0.9286(34) = 35.179$. $\text{SSR} = S_{xy}^2/S_{xx} = 60.36$, $\text{SSE} = S_{yy} - \text{SSR} = 7.14$.

Source	df	SS	MS	F
Regression	1	60.36	60.36	50.70
Residual	6	7.14	1.19	
Total	7	67.50		

Height at age two is a strong predictor of adult height ($r^2 = 0.894$). (b) Write $y_i - \bar y = (y_i - \hat y_i) + (\hat y_i - \bar y)$. Squaring and summing, the cross term is $2\sum_i (y_i - \hat y_i)(\hat y_i - \bar y) = 2\hat\beta\sum_i (y_i - \hat y_i)(x_i - \bar x) = 0$, because $\sum_i (y_i - \hat y_i) = 0$ and $\sum_i x_i(y_i - \hat y_i) = 0$ are the normal equations (and $\hat y_i - \bar y = \hat\beta(x_i - \bar x)$). Hence $\sum_i (y_i - \bar y)^2 = \sum_i (\hat y_i - \bar y)^2 + \sum_i (y_i - \hat y_i)^2$, i.e. (11.3.36): $\text{SST} = \text{SSR} + \text{SSE}$. (c) $\hat y_i - \bar y = \hat\beta(x_i - \bar x)$, so $\sum_i (\hat y_i - \bar y)^2 = \hat\beta^2 S_{xx} = (S_{xy}/S_{xx})^2 S_{xx} = S_{xy}^2/S_{xx}$. (d) $r^2 = \text{SSR}/\text{SST} = S_{xy}^2/(S_{xx}S_{yy}) = \left[S_{xy}/\sqrt{S_{xx}S_{yy}}\right]^2$, the squared sample correlation of the (y_i, x_i) pairs. Also, $\sum_i (y_i - \bar y)(\hat y_i - \bar y) = \hat\beta S_{xy} = S_{xy}^2/S_{xx} = \text{SSR}$ and $\sum_i (\hat y_i - \bar y)^2 = \text{SSR}$ by (c), so the squared sample correlation of the $(y_i, \hat y_i)$ pairs is $\text{SSR}^2/(S_{yy}\cdot\text{SSR}) = \text{SSR}/S_{yy} = r^2$.

Exercise 11.35

Central idea. With $z_i = x_i^2$ this is regression through the origin; least squares, MLE, and UMVUE coincide at $\sum z_i Y_i/\sum z_i^2$, attaining the Cramér–Rao bound.

Solution. Put $z_i = x_i^2$; then $Y_i = \theta z_i + \varepsilon_i$. (a) Minimize $\sum_i (y_i - \theta z_i)^2$: $-2\sum_i z_i(y_i - \theta z_i) = 0$ gives $\hat\theta_{LS} = \dfrac{\sum_i z_i y_i}{\sum_i z_i^2} = \dfrac{\sum_i x_i^2 y_i}{\sum_i x_i^4}$. (b) The log likelihood is $-\frac{1}{2\sigma^2}\sum_i (y_i - \theta z_i)^2$ plus a constant; maximizing over θ is the same minimization, so $\hat\theta_{MLE} = \dfrac{\sum_i x_i^2 Y_i}{\sum_i x_i^4}$. (c) $\hat\theta = \sum_i z_i Y_i/\sum_i z_i^2$ is unbiased ($E\hat\theta = \theta\sum_i z_i^2/\sum_i z_i^2 = \theta$) with $\operatorname{Var}\hat\theta = \sigma^2/\sum_i z_i^2$. The score gives Fisher information $I(\theta) = \sum_i z_i^2/\sigma^2$, so the Cramér–Rao bound is $\sigma^2/\sum_i z_i^2 = \operatorname{Var}\hat\theta$: $\hat\theta$ attains the bound, hence is the best (uniformly minimum variance) unbiased estimator. Equivalently, $T = \sum_i z_i Y_i$ is complete sufficient for θ, and $\hat\theta = T/\sum_i z_i^2$ is unbiased — UMVUE by Lehmann–Scheffé.

Exercise 11.36

Central idea. Tower property: conditional unbiasedness $E[\hat\beta \mid \mathbf{X}] = \beta$ survives averaging over $\mathbf{X}$; but $\mathrm{Cov}(\hat\alpha, \hat\beta \mid \mathbf{X}) = -\sigma^2 \bar X / S_{xx}$ does not average to the fixed-X value — for normal X, $\bar X \perp S_{xx}$ and $E[1/S_{xx}] = 1/[\sigma_X^2 (n-3)]$.

Solution. (a) Conditionally on $\mathbf{X} = \mathbf{x}$, $\hat\alpha$ and $\hat\beta$ are the usual least squares estimators, hence unbiased: $E[\hat\alpha \mid \mathbf{X}] = \alpha$ and $E[\hat\beta \mid \mathbf{X}] = \beta$ for every $\mathbf{x}$ with $S_{xx} > 0$. By the tower property, $E\hat\alpha = E[E[\hat\alpha \mid \mathbf{X}]] = \alpha$ and $E\hat\beta = E[E[\hat\beta \mid \mathbf{X}]] = \beta$ — no assumption on the distribution of $\mathbf{X}$ is needed. (b) $\mathrm{Cov}(\hat\alpha, \hat\beta) = E[\mathrm{Cov}(\hat\alpha, \hat\beta \mid \mathbf{X})] + \mathrm{Cov}(E[\hat\alpha \mid \mathbf{X}], E[\hat\beta \mid \mathbf{X}])$. The second term is $\mathrm{Cov}(\alpha, \beta) = 0$; conditionally, $\mathrm{Cov}(\hat\alpha, \hat\beta \mid \mathbf{X}) = -\sigma^2 \bar X / S_{xx}$ (11.30(c)). For iid normal X_i, $\bar X$ is independent of S_{xx}, so $E[\bar X / S_{xx}] = E\bar X \cdot E[1/S_{xx}] = \mu_X \cdot \frac{1}{\sigma_X^2 (n-3)}$, using $S_{xx}/\sigma_X^2 \sim \chi^2_{n-1}$ and $E[1/\chi^2_{n-1}] = 1/(n-3)$ for $n > 3$. Hence the unconditional covariance is $-\dfrac{\sigma^2 \mu_X}{\sigma_X^2 \, (n-3)}$.

Exercise 11.37

Central idea. Regression through the origin: all information about β is in $T = \sum x_i Y_i$; $\mathrm{Var}\,\hat\beta = \sigma^2 / \sum x_i^2$ is minimized by pushing every x_i to the endpoint of $[A, B]$ with the largest $|x|$; the MPE is the MLE by symmetry and unimodality.

Solution. (a) $\ell(\beta) = -\frac{1}{2\sigma^2} \sum_i (y_i - \beta x_i)^2$ plus a constant; setting $\ell'(\beta) = \frac{1}{\sigma^2} \sum_i x_i (y_i - \beta x_i) = 0$ gives $\hat\beta = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2}$. $E\hat\beta = \beta \sum_i x_i^2 / \sum_i x_i^2 = \beta$; $\mathrm{Var}\,\hat\beta = \sigma^2 \sum_i x_i^2 / (\sum_i x_i^2)^2 = \sigma^2 / \sum_i x_i^2$. (b) $\ell''(\beta) = -\sum_i x_i^2 / \sigma^2$, so $I(\beta) = E[-\ell''(\beta)] = \sum_i x_i^2 / \sigma^2$, and the Cramér–Rao Lower Bound is $\sigma^2 / \sum_i x_i^2$. (c) $\hat\beta$ is unbiased and attains the CRLB, hence is the best (uniformly minimum variance) unbiased estimator. (d) $\mathrm{Var}\,\hat\beta = \sigma^2 / \sum_i x_i^2$ is minimized by maximizing $\sum_i x_i^2$ over $[A, B]^n$. Since $x_i^2 \le \max(A^2, B^2)$ for each i, place every x_i at the endpoint with the larger absolute value — all at A if $|A| > |B|$, all at B if $|B| > |A|$ (if $|A| = |B|$, any allocation attains $\sum_i x_i^2 = n \max(A^2, B^2)$). (e) $f(\mathbf{y} \mid \beta) \propto \exp\big(-\frac{\sum_i x_i^2}{2\sigma^2} (\beta - \hat\beta)^2\big)$: as a function of β it is symmetric and unimodal about the MLE $\hat\beta$. For a window of fixed length $2r$, $\int_{D-r}^{D+r} g(\beta)\, d\beta$ with g symmetric and unimodal about $\hat\beta$ is maximized by centering the window on the mode, $D = \hat\beta$. Hence the maximum probability estimator is $\hat\beta = \sum_i x_i Y_i / \sum_i x_i^2$, the MLE.

Exercise 11.38

Central idea. Least squares minimizes $\sum (y_i - \theta x_i)^2$ ignoring the Poisson mean–variance link; the likelihood exploits it, giving the complete sufficient statistic $T = \sum Y_i$ and the UMVUE $\sum Y_i / \sum x_i$ attaining the CRLB.

Solution. (a) Minimize $\sum_i (y_i - \theta x_i)^2$: $-2 \sum_i x_i (y_i - \theta x_i) = 0$ gives $\tilde\theta = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2}$. $E\tilde\theta = \theta \sum_i x_i^2 / \sum_i x_i^2 = \theta$, so the bias is 0. $\mathrm{Var}\,\tilde\theta = \sum_i x_i^2 \mathrm{Var}(Y_i) / (\sum_i x_i^2)^2 = \sum_i x_i^2 (\theta x_i) / (\sum_i x_i^2)^2 = \theta \sum_i x_i^3 / (\sum_i x_i^2)^2$, using $\mathrm{Var}(Y_i) = \theta x_i$ and independence. (b) $\ell(\theta) = \sum_i [y_i \log(\theta x_i) - \theta x_i - \log y_i!] = (\sum_i y_i) \log\theta - \theta \sum_i x_i +$ const; $\ell'(\theta) = \sum_i y_i / \theta - \sum_i x_i = 0$ gives $\hat\theta = \dfrac{\sum_i Y_i}{\sum_i x_i}$. $E\hat\theta = \theta \sum_i x_i / \sum_i x_i = \theta$ (unbiased); $\mathrm{Var}\,\hat\theta = \sum_i \theta x_i / (\sum_i x_i)^2 = \theta / \sum_i x_i$. (c) $T = \sum_i Y_i \sim \mathrm{Poisson}(\theta \sum_i x_i)$ is complete sufficient for θ; $\hat\theta = T / \sum_i x_i$ is unbiased, hence the UMVUE by Lehmann–Scheffé. Moreover $I(\theta) = E[-\ell''(\theta)] = E[\sum_i Y_i / \theta^2] = \sum_i x_i / \theta$, so the CRLB is $\theta / \sum_i x_i = \mathrm{Var}\,\hat\theta$: the best unbiased estimator attains the bound. (The LSE is also unbiased but less efficient: $\theta \sum_i x_i^3 / (\sum_i x_i^2)^2 \ge \theta / \sum_i x_i$ by Cauchy–Schwarz.)

Exercise 11.39

Central idea. By (11.3.44) (Exercise 11.40(b)), simultaneous coverage over all x_0 is the event that a χ^2_2 / χ^2_{n-2} ratio — i.e. an $F_{2,n-2}$ variable — falls below $2F_{2,n-2,\alpha}$.

Solution. The intervals (11.3.42) are, for every x_0,

$$\hat{\alpha}+\hat{\beta}x_0 \pm \sqrt{2F_{2,n-2,\alpha}}\cdot S\sqrt{\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}}.$$

All x_0 are covered simultaneously iff $\max_{x_0}\frac{(\hat{\alpha}+\hat{\beta}x_0-\alpha-\beta x_0)^2}{S^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)} \le 2F_{2,n-2,\alpha}$. By (11.3.44), verified in 11.40(b), the maximum equals $\dfrac{n(\bar{Y}-\mu_{\bar{Y}})^2+S_{xx}(\hat{\beta}-\beta)^2}{S^2}$, where $\mu_{\bar{Y}}=\alpha+\beta\bar{x}$. Now $n(\bar{Y}-\mu_{\bar{Y}})^2/\sigma^2 \sim \chi^2_1$ and $S_{xx}(\hat{\beta}-\beta)^2/\sigma^2 \sim \chi^2_1$ independently ($\bar{Y} \perp \hat{\beta}$), so the numerator/$\sigma^2 \sim \chi^2_2$, independent of SSE/$\sigma^2 \sim \chi^2_{n-2}$. Hence

$$\frac{n(\bar{Y}-\mu_{\bar{Y}})^2+S_{xx}(\hat{\beta}-\beta)^2}{S^2} = 2\cdot\frac{\chi^2_2/2}{\chi^2_{n-2}/(n-2)} \sim 2F_{2,n-2},$$

and $P\big(2F_{2,n-2}\le 2F_{2,n-2,\alpha}\big)=1-\alpha$: the simultaneous coverage probability is exactly $1-\alpha$.

Exercise 11.40

Central idea. The identity $\frac{a^2}{c}+\frac{b^2}{d}-\frac{(a+bt)^2}{c+dt^2}=\frac{(adt-bc)^2}{cd\,(c+dt^2)}\ge 0$ proves (a); applied with $c=S^2/n$ it gives the (11.3.44) maximum $\big[n(\bar{Y}-\mu_{\bar{Y}})^2+S_{xx}(\hat{\beta}-\beta)^2\big]/S^2$, and with $c=S^2(1+1/n)$ it gives the prediction-band maximum — the $n/(n+1)$ factor comes from the extra $+1$ in the prediction variance.

Solution. (a) For $c,d>0$,

$$\frac{a^2}{c}+\frac{b^2}{d}-\frac{(a+bt)^2}{c+dt^2}=\frac{(a^2d+b^2c)(c+dt^2)-cd(a+bt)^2}{cd\,(c+dt^2)}=\frac{(adt-bc)^2}{cd\,(c+dt^2)}\ge 0,$$

with equality at $t=bc/(ad)$ (if $a=0$ the ratio $\uparrow b^2/d$ as $|t|\to\infty$; if $b=0$ the maximum a^2/c is attained at $t=0$). Hence the supremum over t is $a^2/c+b^2/d$. (This is the Cauchy–Schwarz inequality in disguise.) (b) Write $(\hat{\alpha}+\hat{\beta}x_0)-(\alpha+\beta x_0)=(\bar{Y}-\mu_{\bar{Y}})+(\hat{\beta}-\beta)(x_0-\bar{x})$, using $\hat{\alpha}=\bar{Y}-\hat{\beta}\bar{x}$ and $\mu_{\bar{Y}}=\alpha+\beta\bar{x}$. Apply (a) with $t=x_0-\bar{x}$, $a=\bar{Y}-\mu_{\bar{Y}}$, $b=\hat{\beta}-\beta$, $c=S^2/n$, $d=S^2/S_{xx}$:

$$\sup_{x_0}\frac{\big((\hat{\alpha}+\hat{\beta}x_0)-(\alpha+\beta x_0)\big)^2}{S^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}=\frac{(\bar{Y}-\mu_{\bar{Y}})^2}{S^2/n}+\frac{(\hat{\beta}-\beta)^2}{S^2/S_{xx}}=\frac{n(\bar{Y}-\mu_{\bar{Y}})^2+S_{xx}(\hat{\beta}-\beta)^2}{S^2},$$

which is (11.3.44) — the gap in the proof of Theorem 11.3.6. (c) For a new observation the prediction variance carries an extra $+1$: apply (a) with the same a,b,t but $c=S^2(1+\frac{1}{n})=S^2\frac{n+1}{n}$, $d=S^2/S_{xx}$:

$$\sup_{t}\frac{\big((\bar{Y}-\mu_{\bar{Y}})+(\hat{\beta}-\beta)t\big)^2}{S^2\left[1+\frac{1}{n}+\frac{t^2}{S_{xx}}\right]}=\frac{(\bar{Y}-\mu_{\bar{Y}})^2}{S^2(n+1)/n}+\frac{(\hat{\beta}-\beta)^2}{S^2/S_{xx}}=\frac{\frac{n}{n+1}(\bar{Y}-\mu_{\bar{Y}})^2+S_{xx}(\hat{\beta}-\beta)^2}{S^2}.$$

Note: this is where the $n/(n+1)$ factor genuinely belongs — it comes from the $1+1/n$ in the prediction-interval denominator, not from (11.3.44). (d) Write $U=n(\bar{Y}-\mu_{\bar{Y}})^2/\sigma^2\sim\chi^2_1$, $V=S_{xx}(\hat{\beta}-\beta)^2/\sigma^2\sim\chi^2_1$, $W=\text{SSE}/\sigma^2\sim\chi^2_{n-2}$, mutually independent. The statistic in (c) is $M=\dfrac{U/(n+1)+V}{W/(n-2)}$. Moment-match the numerator $N=U/(n+1)+V$: $E[N]=1+\frac{1}{n+1}=\frac{n+2}{n+1}$, $\text{Var}(N)=2+\frac{2}{(n+1)^2}$. Approximating $N\doteq g\chi^2_h$ as in Example 7.2.3 (Patnaik), with $g=\text{Var}(N)/(2E[N])$ and $h=2E[N]^2/\text{Var}(N)$:

$$g=\frac{(n+1)^2+1}{(n+1)(n+2)},\qquad h=\frac{(n+2)^2}{(n+1)^2+1},$$

so $M\doteq gh\cdot F_{h,n-2}$ — a constant times an F variable, whose $1-\alpha$ quantile calibrates the displayed mean-error band. Interpreting it as a simultaneous prediction band requires defining and including the future error process.

Exercise 11.41

Central idea. With x_{24} observed and y_{24} missing, the E-step imputes $E[Y_{24} \mid x_{24}]$ and $E[Y_{24}^2 \mid x_{24}]$ from the conditional normal; the M-step is the complete-data MLE (sample moments) with imputed sufficient statistics. In (d) the roles reverse.

Solution. Let (x_i, y_i), $i = 1, \ldots, 23$ be complete, and write $\theta = (\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$. (a) Observed data: $\prod_{i=1}^{23} f(x_i, y_i; \theta) \cdot f_X(x_{24}; \theta)$, so

$$\ell_{obs}(\theta) = \sum_{i=1}^{23} \log f(x_i, y_i; \theta) + \log f_X(x_{24}; \theta),$$

f the bivariate normal density, f_X its $n(\mu_X, \sigma_X^2)$ marginal. The expected complete-data log likelihood: $Q(\theta \mid \theta^{(t)}) = \sum_{i=1}^{23} \log f(x_i, y_i; \theta) + E_{\theta^{(t)}}[\log f(x_{24}, Y_{24}; \theta) \mid x_{24}]$. (b) E-step: compute, under $\theta^{(t)}$, with $Y_{24} \mid x_{24} \sim n(\mu_Y^{(t)} + \rho^{(t)} \frac{\sigma_Y^{(t)}}{\sigma_X^{(t)}}(x_{24} - \mu_X^{(t)}),\ \sigma_Y^{2(t)}(1 - \rho^{2(t)}))$:

$$\hat{y}_{24} = E[Y_{24} \mid x_{24}], \qquad \widehat{y_{24}^2} = \mathrm{Var}(Y_{24} \mid x_{24}) + \hat{y}_{24}^2.$$

M-step: with $S_x = \sum_1^{24} x_i$, $S_{xx} = \sum_1^{24} x_i^2$ (known), $S_y = \sum_1^{23} y_i + \hat{y}_{24}$, $S_{yy} = \sum_1^{23} y_i^2 + \widehat{y_{24}^2}$, $S_{xy} = \sum_1^{23} x_i y_i + x_{24}\hat{y}_{24}$:

$$\mu_X = S_x/24,\ \mu_Y = S_y/24,\ \sigma_X^2 = S_{xx}/24 - \mu_X^2,\ \sigma_Y^2 = S_{yy}/24 - \mu_Y^2,\ \rho = \frac{S_{xy}/24 - \mu_X\mu_Y}{\sigma_X\sigma_Y}.$$

(c) Under Example 12.4.2's model — the regression of CO_2 (y) on O_2 (x) — the 24th animal has $x_{24} = 16.3$ observed but y_{24} missing, so it contributes no information about the regression: the MLEs are the 23-animal least squares, $y = 18.675 - 0.894x$ with $\hat{\sigma}^2 = .0503$ (the book's note in (d) confirms only (d) needs the bivariate-normal formulation). As a bivariate-normal alternative, EM from the 23 complete observations (MLEs: $\hat{\mu}_X = 18.33$, $\hat{\mu}_Y = 2.29$, $\hat{\sigma}_X^2 = 0.83$, $\hat{\sigma}_Y^2 = 0.72$, $\hat{\rho} = -0.964$) converges to

μ_X	μ_Y	σ_X^2	σ_Y^2	ρ
18.24	2.37	0.962	0.819	−0.969

with $E[Y_{24} \mid x_{24} = 16.3] = 4.10$ at the MLE (the missing CO_2 is imputed high, consistent with the strong negative correlation). Convergence was verified from two different starting values to the same fixed point; at the MLE, $\ell_{obs} = -31.85$. (d) Now $y_{24} = 4.2$ is observed and x_{24} missing. E-step: $X_{24} \mid y_{24} \sim n(\mu_X^{(t)} + \rho^{(t)} \frac{\sigma_X^{(t)}}{\sigma_Y^{(t)}}(y_{24} - \mu_Y^{(t)}),\ \sigma_X^{2(t)}(1 - \rho^{2(t)}))$; impute $\hat{x}_{24} = E[X_{24} \mid y_{24}]$, $\widehat{x_{24}^2} = \mathrm{Var}(X_{24} \mid y_{24}) + \hat{x}_{24}^2$. M-step: sample moments with $S_x = \sum_1^{23} x_i + \hat{x}_{24}$, $S_{xx} = \sum_1^{23} x_i^2 + \widehat{x_{24}^2}$, $S_{xy} = \sum_1^{23} x_i y_i + \hat{x}_{24} y_{24}$ (S_y, S_{yy} known). EM converges to

μ_X	μ_Y	σ_X^2	σ_Y^2	ρ
18.24	2.37	0.957	0.831	−0.969

with $E[X_{24} \mid y_{24} = 4.2] = 16.34$ at the MLE — close to the 16.3 supposed in (a)–(c), as expected.

Chapter 12

Exercise 12.1

Central idea. The closest point on the line to (x', y') is the one for which the displacement vector is orthogonal to the line's direction — equivalently, the point that makes the Pythagorean decomposition of squared distances hold for every other point on the line.

Solution. Parametrize points on $y = a + bx$ as $(u, a + bu)$. The squared distance from (x', y') is $D(u) = (x'-u)^2+(y'-a-bu)^2$. By Pythagoras, $(u, a+bu)$ is the closest point iff the displacement $(x'-u,\ y'-a-bu)$ is orthogonal to the line's direction $(1, b)$ for every competing point, i.e. $(x'-u, y'-a-bu)\cdot(1, b) = 0$. This gives $(x'-u)+b(y'-a-bu) = 0$, whence $u(1+b^2) = x'+b(y'-a)$, i.e.

$$\hat{x}' = \frac{by'+x'-ab}{1+b^2}, \qquad \hat{y}' = a + b\hat{x}' = a + \frac{b}{1+b^2}(by'+x'-ab),$$

which is (12.2.7). (Direct minimization of $D(u)$ gives the same first-order condition.)

Exercise 12.2

Central idea. Setting $f'(b) = 0$ clears denominators to a quadratic in b; the range of f is found by asking for which constants c the equation $f(b) = c$ has a real root, and the smaller root c_- is attained exactly at the "+" critical point.

Solution. Assume $S_{xy} \neq 0$ (otherwise f is minimized trivially at $b = 0$). Then

$$f'(b) = 0 \iff \frac{(-2S_{xy}+2bS_{xx})(1+b^2) - 2b(S_{yy}-2bS_{xy}+b^2S_{xx})}{(1+b^2)^2} = 0,$$

and the numerator simplifies to $-2S_{xy}+2b(S_{xx}-S_{yy})+2b^2S_{xy} = 0$, i.e.

$$b^2S_{xy} + b(S_{xx}-S_{yy}) - S_{xy} = 0,$$

whose roots are the stated values. For the range, $f(b) = c$ is $b^2(S_{xx}-c)-2bS_{xy}+(S_{yy}-c) = 0$, which has a real root iff its discriminant is nonnegative: $S_{xy}^2-(S_{xx}-c)(S_{yy}-c) \geq 0$, i.e. $c^2-c(S_{xx}+S_{yy})+S_{xx}S_{yy}-S_{xy}^2 \leq 0$. Hence f ranges over $[c_-, c_+]$ with $c_\pm = \frac{1}{2}[(S_{xx}+S_{yy}) \pm \sqrt{(S_{xx}-S_{yy})^2+4S_{xy}^2}]$. The minimum c_- is attained where the quadratic in b has a double root, $b = S_{xy}/(S_{xx}-c_-) = 2S_{xy}/(S_{xx}-S_{yy}+\sqrt{D})$ with $D = (S_{xx}-S_{yy})^2+4S_{xy}^2$; rationalizing gives $b = (\sqrt{D}-(S_{xx}-S_{yy}))/(2S_{xy})$, the "+" root.

Exercise 12.3

Central idea. Each nuisance-coordinate minimization is a convex quadratic. The squared weighted distance fails the triangle inequality; its square root is a metric.

Solution. (a) For $\lambda > 0$, $g_i(\xi) = (x_i-\xi)^2+\lambda(y_i-\alpha-\beta\xi)^2$ has second derivative $2(1+\lambda\beta^2) > 0$. Its unique minimum solves $\xi(1+\lambda\beta^2) = x_i+\lambda\beta(y_i-\alpha)$, giving the displayed ξ_i^*. (b) Extend the two-point expression to $D_\lambda((x,y),(u,v)) = (x-u)^2+\lambda(y-v)^2$. It satisfies zero self-distance, positivity for distinct points, and symmetry, but is not a metric: for $A = (0,0)$, $C = (1,0)$, $B = (2,0)$ and $\lambda = 1$, $D(A,B) = 4 > 1+1$. The correct metric is $d_\lambda(A,B) = \sqrt{(x-u)^2+\lambda(y-v)^2}$, the Euclidean norm after scaling the second coordinate by $\sqrt{\lambda}$. The expression in the exercise is its restriction when the second point lies on the fitted line. The earlier claim that $D(A,A) = \lambda(y-\alpha-\beta x)^2$ was false: that expression compares A to the different projected point $(x, \alpha+\beta x)$, not to itself.

Exercise 12.4

Central idea. The MLE solves the first-order condition $\lambda S_{xy}b^2+(S_{xx}-\lambda S_{yy})b-S_{xy} = 0$; implicit differentiation of this identity gives the sign of $d\hat{\beta}/d\lambda$, and the limits interpolate between the two ordinary regressions.

Solution. $\hat{\beta}(\lambda)$ is the "+" root of $\lambda S_{xy}b^2+(S_{xx}-\lambda S_{yy})b-S_{xy} = 0$ (the first-order condition for the concentrated likelihood (12.2.14)).

(a) Write $\sqrt{(S_{xx}-\lambda S_{yy})^2+4\lambda S_{xy}^2} = S_{xx}-\lambda S_{yy}+2\lambda S_{xy}^2/S_{xx}+o(\lambda)$ (Taylor, assuming $S_{xx} > 0$). The numerator becomes $2\lambda S_{xy}^2/S_{xx}+o(\lambda)$; dividing by $2\lambda S_{xy}$ gives S_{xy}/S_{xx}.

(b) Divide numerator and denominator by λ and let $\lambda \to \infty$: the numerator tends to $S_{yy} + \sqrt{S_{yy}^2} = 2S_{yy}$ and the denominator to $2S_{xy}$, giving S_{yy}/S_{xy}.

(c) Differentiating the first-order identity in λ: with $D = (S_{xx} - \lambda S_{yy})^2 + 4\lambda S_{xy}^2$,

$$\frac{d\hat{\beta}}{d\lambda}\sqrt{D} = \hat{\beta}\,(S_{yy} - S_{xy}\hat{\beta}).$$

If $S_{xy} > 0$ then $\hat{\beta} > 0$ (numerator $-(S_{xx}-\lambda S_{yy})+\sqrt{D} > 0$), and $\hat{\beta} \le S_{yy}/S_{xy}$: indeed $\hat{\beta} = 2S_{xy}/(S_{xx} - \lambda S_{yy} + \sqrt{D}) \le S_{yy}/S_{xy}$ is equivalent, after clearing denominators, to $(2S_{xy}^2 - S_{yy}S_{xx} + \lambda S_{yy}^2)^2 \le S_{yy}^2 D$, whose difference from the right side is $4S_{xy}^2(S_{xx}S_{yy} - S_{xy}^2) \ge 0$ by Cauchy–Schwarz. Hence $d\hat{\beta}/d\lambda \ge 0$. If $S_{xy} < 0$ then $\hat{\beta} < 0$, and $\hat{\beta} \ge S_{yy}/S_{xy}$ follows from $\sqrt{D} \le S_{xx} + \lambda S_{yy} \iff 4\lambda S_{xy}^2 \le 4\lambda S_{xx}S_{yy}$, again Cauchy–Schwarz; then $S_{xy}\hat{\beta} \le S_{yy}$ gives $S_{yy} - S_{xy}\hat{\beta} \ge 0$, so $d\hat{\beta}/d\lambda \le 0$.

(d) For $S_{xy} > 0$, (a)–(c) give $S_{xy}/S_{xx} = \hat{\beta}(0+) \le \hat{\beta}(1) \le \hat{\beta}(\infty) = S_{yy}/S_{xy}$; for $S_{xy} < 0$, $\hat{\beta}$ decreases from S_{xy}/S_{xx} to S_{yy}/S_{xy}, so $\hat{\beta}(1)$ again lies between them.

(e) With $\lambda = 1$ (orthogonal least squares): $S_{xx} = 9.095$, $S_{yy} = 1358.25$, $S_{xy} = 96.15$. OLS of y on x: $S_{xy}/S_{xx} = 10.57$; reciprocal of OLS of x on y: $S_{yy}/S_{xy} = 14.13$; EIV MLE: $\hat{\beta}(1) = 14.10$. Indeed $10.57 < 14.10 < 14.13$, the MLE lying between the two OLS slopes, close to the x-on-y reciprocal because $S_{yy} \gg S_{xx}$.

Exercise 12.5

Central idea. After concentrating out $\alpha, \beta, \xi_1, \ldots, \xi_n$, the profile likelihood (12.2.17) is a one-parameter scaled-χ^2-type likelihood in σ_δ^2.

Solution. From (12.2.17), with $A = \frac{\lambda}{1+\lambda\hat{\beta}^2}\sum_{i=1}^n (y_i - (\hat{\alpha} + \hat{\beta}x_i))^2$,

$$\log L = \frac{n}{2}\log\lambda - n\log(2\pi) - n\log\sigma_\delta^2 - \frac{A}{2\sigma_\delta^2}.$$

Differentiating in σ_δ^2: $-n/\sigma_\delta^2 + A/(2(\sigma_\delta^2)^2) = 0$, so $\hat{\sigma}_\delta^2 = A/(2n)$, i.e.

$$\hat{\sigma}_\delta^2 = \frac{1}{2n}\frac{\lambda}{1+\lambda\hat{\beta}^2}\sum_{i=1}^n \left(y_i - (\hat{\alpha} + \hat{\beta}x_i)\right)^2,$$

which is (12.2.18). It is a maximum since the second derivative is negative there.

Exercise 12.6

Central idea. (X_i, Y_i) is an affine (linear-plus-constant) transformation of the independent normals $(\xi_i, \delta_i, \epsilon_i)$, hence jointly normal; only the means, variances, and covariance need computing.

Solution. Write $X_i = \xi_i + \delta_i$, $Y_i = \alpha + \beta\xi_i + \epsilon_i$ with $\xi_i \sim n(\xi, \sigma_\xi^2)$, $\delta_i \sim n(0, \sigma_\delta^2)$, $\epsilon_i \sim n(0, \sigma_\epsilon^2)$ independent. As a linear function of independent normals, (X_i, Y_i) is bivariate normal with $EX_i = \xi$, $EY_i = \alpha + \beta\xi$, $\operatorname{Var} X_i = \sigma_\delta^2 + \sigma_\xi^2$, $\operatorname{Var} Y_i = \sigma_\epsilon^2 + \beta^2\sigma_\xi^2$, and $\operatorname{Cov}(X_i, Y_i) = \operatorname{Cov}(\xi_i + \delta_i, \alpha + \beta\xi_i + \epsilon_i) = \beta\sigma_\xi^2$, which is (12.2.19).

Exercise 12.7

Central idea. (a) The exponent is quadratic in ξ_i; completing the square reduces the integral to a normal normalizing constant times the minimized exponent, which collapses to the squared residual $(y_i - (\alpha + \beta x_i))^2$. (b) An improper prior produces a formal expression, not a genuine conditional distribution — and in a genuine structural model $E[Y \mid X] \neq \alpha + \beta X$.

Solution. (a) Write the exponent as $-\frac{1}{2}[A\xi_i^2 - 2B\xi_i + C]$ with $A = 1/\sigma_\delta^2 + \beta^2/\sigma_\epsilon^2$, $B = x_i/\sigma_\delta^2 + \beta(y_i - \alpha)/\sigma_\epsilon^2$, $C = x_i^2/\sigma_\delta^2 + (y_i - \alpha)^2/\sigma_\epsilon^2$. Completing the square, the integral equals $\frac{1}{2\pi\sigma_\delta\sigma_\epsilon}\sqrt{2\pi/A}\,\exp(B^2/(2A) - C/2)$. Since $\sqrt{2\pi/A} = \sqrt{2\pi}\,\sigma_\delta\sigma_\epsilon/\sqrt{\beta^2\sigma_\delta^2 + \sigma_\epsilon^2}$, the prefactor is $\frac{1}{\sqrt{2\pi}\sqrt{\beta^2\sigma_\delta^2+\sigma_\epsilon^2}}$. Moreover $C - B^2/A = \min_{\xi_i}\,[(x_i - \xi_i)^2/\sigma_\delta^2 + (y_i - \alpha - \beta\xi_i)^2/\sigma_\epsilon^2] = (y_i - (\alpha + \beta x_i))^2/(\beta^2\sigma_\delta^2 + \sigma_\epsilon^2)$ (the profile calculation of Section 12.2.3 with $\lambda = \sigma_\delta^2/\sigma_\epsilon^2$), giving the stated exponential.

(b) The calculation is purely formal: uniform$(-\infty, \infty)$ is not a probability distribution, so the integrated "marginal" is not a legitimate marginal and certainly not the conditional distribution of Y_i given X_i. In a genuine structural model with a proper ξ_i distribution, (X_i, Y_i) is bivariate normal (12.2.19) and $E[Y_i \mid X_i] = \alpha + \beta\xi + \frac{\beta\sigma_\xi^2}{\sigma_\delta^2+\sigma_\xi^2}(X_i - \xi)$ — the regression of Y on X is attenuated (slope $\beta\sigma_\xi^2/(\sigma_\delta^2+\sigma_\xi^2) \neq \beta$), not $\alpha + \beta X_i$. The formal integral also treats the random X_i as a fixed design point. So nothing here justifies OLS; the resemblance to a normal linear-regression pdf is an artifact of the improper prior.

Exercise 12.8

Central idea. The bivariate normal marginal (12.2.19) has five parameters but the structural model has six — one degree of freedom is unidentified; exhibit it by trading β against the variance components while holding the five moments fixed.

Solution. (a) The marginal is determined by the five quantities ξ, $\alpha + \beta\xi$, $\sigma_\delta^2 + \sigma_\xi^2$, $\sigma_\epsilon^2 + \beta^2\sigma_\xi^2$, $\beta\sigma_\xi^2$. Take $\theta_1 = (\alpha, \beta, \xi, \sigma_\delta^2, \sigma_\epsilon^2, \sigma_\xi^2) = (0, 1, 0, 2, 15, 4)$: means $(0, 0)$, $\operatorname{Var} X = 6$, $\operatorname{Var} Y = 19$, $\operatorname{Cov} = 4$. Take $\theta_2 = (0, 2, 0, 4, 11, 2)$: means $(0, 0 + 2\cdot 0) = (0, 0)$, $\operatorname{Var} X = 4 + 2 = 6$, $\operatorname{Var} Y = 11 + 4\cdot 2 = 19$, $\operatorname{Cov} = 2\cdot 2 = 4$. Both give the identical bivariate normal, but $\theta_1 \neq \theta_2$.

(b) The system (12.2.20) has five equations in six unknowns. E.g., with $\bar{x}, \bar{y}$ arbitrary and $S_{xx}/n = 6$, $S_{yy}/n = 19$, $S_{xy}/n = 4$: one solution is $\hat{\beta} = 1$, $\hat{\alpha} = \bar{y} - \bar{x}$, $\hat{\sigma}_\xi^2 = 4$, $\hat{\sigma}_\delta^2 = 2$, $\hat{\sigma}_\epsilon^2 = 15$; another is $\hat{\beta} = 2$, $\hat{\alpha} = \bar{y} - 2\bar{x}$, $\hat{\sigma}_\xi^2 = 2$, $\hat{\sigma}_\delta^2 = 4$, $\hat{\sigma}_\epsilon^2 = 11$. Both satisfy all five equations, so the MLE is not unique.

Exercise 12.9

Central idea. Nonnegativity of the three variance estimates forces $\hat{\beta}$ and S_{xy} to share a sign and squeezes $|\hat{\beta}|$ between the two OLS slopes.

Solution. (a) From (12.2.20), $\hat{\sigma}_\xi^2 = S_{xy}/(n\hat{\beta})$, so $\hat{\sigma}_\delta^2 = S_{xx}/n - \hat{\sigma}_\xi^2 = (1/n)(S_{xx} - S_{xy}/\hat{\beta}) \geq 0$ iff $S_{xx} \geq S_{xy}/\hat{\beta}$, and $\hat{\sigma}_\epsilon^2 = S_{yy}/n - \hat{\beta}^2\hat{\sigma}_\xi^2 = (1/n)(S_{yy} - \hat{\beta}S_{xy}) \geq 0$ iff $S_{yy} \geq \hat{\beta}S_{xy}$.

(b) Since $\hat{\sigma}_\xi^2 = S_{xy}/(n\hat{\beta}) \geq 0$, S_{xy} and $\hat{\beta}$ have the same sign, so $S_{xy}/\hat{\beta} = |S_{xy}|/|\hat{\beta}|$ and $\hat{\beta}S_{xy} = |\hat{\beta}||S_{xy}|$. Then $S_{xx} \geq |S_{xy}|/|\hat{\beta}|$ gives $|\hat{\beta}| \geq |S_{xy}|/S_{xx}$, and $S_{yy} \geq |\hat{\beta}||S_{xy}|$ gives $|\hat{\beta}| \leq S_{yy}/|S_{xy}|$.

Exercise 12.10

Central idea. Imposing $\sigma_\delta^2 = \lambda\sigma_\epsilon^2$ on the moment equations collapses them to the same quadratic for $\hat{\beta}$ as in the functional case; the variance identities then follow by substituting $\hat{\beta}$'s first-order condition.

Solution. (a) From (12.2.20): $\hat{\xi} = \bar{x}$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$, $\hat{\sigma}_\xi^2 = S_{xy}/(n\hat{\beta})$. Also $\hat{\sigma}_\epsilon^2 = (S_{yy} - \hat{\beta}S_{xy})/n$ and, with $\hat{\sigma}_\delta^2 = \lambda\hat{\sigma}_\epsilon^2$, $S_{xx}/n = \lambda(S_{yy} - \hat{\beta}S_{xy})/n + S_{xy}/(n\hat{\beta})$, i.e. $S_{xx} = \lambda S_{yy} - \lambda\hat{\beta}S_{xy} + S_{xy}/\hat{\beta}$. Multiplying by $\hat{\beta}$: $\lambda S_{xy}\hat{\beta}^2 + (S_{xx} - \lambda S_{yy})\hat{\beta} - S_{xy} = 0$, the same quadratic as the functional case, so $\hat{\beta}$ is (12.2.16) and $\hat{\sigma}_\delta^2 = \lambda(S_{yy} - \hat{\beta}S_{xy})/n$, $\hat{\sigma}_\xi^2 = S_{xy}/(n\hat{\beta})$, as in (12.2.21).

(b) With $\lambda = 1$ and the sums from Exercise 12.4: $\hat{\beta} = 14.10$, $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 12.075 - 14.10(1.1005) = -3.45$, $\hat{\sigma}_\epsilon^2 = (S_{yy} - \hat{\beta}S_{xy})/n = 0.285$, $\hat{\sigma}_\delta^2 = 0.285$, $\hat{\sigma}_\xi^2 = S_{xy}/(n\hat{\beta}) = 0.852$.

(c) Since $\hat\alpha = \bar y - \hat\beta\bar x$, $\sum_i (y_i - (\hat\alpha + \hat\beta x_i))^2 = S_{yy} - 2\hat\beta S_{xy} + \hat\beta^2 S_{xx} \equiv Q$. The claim is $(S_{xx} - S_{xy}/\hat\beta)(1 + \lambda\hat\beta^2) = \lambda Q$, i.e. $S_{xx} + \lambda\hat\beta^2 S_{xx} - S_{xy}/\hat\beta - \lambda\hat\beta S_{xy} - \lambda S_{yy} + 2\lambda\hat\beta S_{xy} - \lambda\hat\beta^2 S_{xx} = 0$, which simplifies to $[\lambda S_{xy}\hat\beta^2 + (S_{xx} - \lambda S_{yy})\hat\beta - S_{xy}]/\hat\beta = 0$, true by the quadratic in (a). Numerically both sides equal 2.2774, and since $\hat\sigma_\delta^2$(structural) $= (S_{xx} - S_{xy}/\hat\beta)/n$ while $\hat\sigma_\delta^2$(functional) $= \frac{\lambda}{2n(1+\lambda\hat\beta^2)} Q$, this is exactly $\hat\sigma_\delta^2$(structural) $= 2\hat\sigma_\delta^2$(functional).

(d) From (12.2.21), $\hat\sigma_\delta^2 = (1/n)(S_{xx} - S_{xy}/\hat\beta)$ and also $\hat\sigma_\delta^2 = \lambda\hat\sigma_\epsilon^2 = (\lambda/n)(S_{yy} - \hat\beta S_{xy})$; equating gives $S_{xx} - S_{xy}/\hat\beta = \lambda(S_{yy} - \hat\beta S_{xy})$.

Exercise 12.11

Central idea. Bilinearity of covariance plus the structural moments makes the two linear combinations uncorrelated; for bivariate normals, uncorrelated means independent, so the ordinary sample-correlation t-statistic is pivotal for every β.

Solution. (a) Expand by bilinearity: $\operatorname{Cov}(aY + bX, cY + dX) = ac \operatorname{Var} Y + ad \operatorname{Cov}(Y, X) + bc \operatorname{Cov}(X, Y) + bd \operatorname{Var} X$.

(b) With $a = \beta\lambda$, $b = 1$, $c = 1$, $d = -\beta$: $\operatorname{Cov} = \beta\lambda \operatorname{Var} Y_i + (1 - \lambda\beta^2)\operatorname{Cov}(X_i, Y_i) - \beta \operatorname{Var} X_i$. Using (12.2.19), $\operatorname{Var} Y_i = \sigma_\epsilon^2 + \beta^2\sigma_\xi^2$, $\operatorname{Cov}(X_i, Y_i) = \beta\sigma_\xi^2$, $\operatorname{Var} X_i = \lambda\sigma_\epsilon^2 + \sigma_\xi^2$:

$$\beta\lambda(\sigma_\epsilon^2 + \beta^2\sigma_\xi^2) + (1 - \lambda\beta^2)\beta\sigma_\xi^2 - \beta(\lambda\sigma_\epsilon^2 + \sigma_\xi^2) = 0,$$

all terms cancelling.

(c) $(\beta\lambda Y_i + X_i,\ Y_i - \beta X_i)$ are linear combinations of independent normals, hence bivariate normal; zero correlation implies independence. For iid pairs with zero true correlation, the sample correlation $r_\lambda(\beta)$ satisfies $\sqrt{n-2}\, r_\lambda(\beta)/\sqrt{1 - r_\lambda^2(\beta)} \sim t_{n-2}$ (Exercise 11.33), for every β — in particular at the true β. Hence

$$P\big(\beta \in C_\lambda(\hat\beta)\big) = P\left(\frac{(n-2)r_\lambda^2(\beta)}{1 - r_\lambda^2(\beta)} \le F_{1,n-2,\alpha}\right) = 1 - \alpha,$$

constant in β (here $t_{n-2}^2 \sim F_{1,n-2}$).

Exercise 12.12

Central idea. The score equation is a quadratic whose roots multiply to $-1/\lambda$; the map $\beta \mapsto -1/(\lambda\beta)$ swaps the two linear combinations defining r_λ up to scalars of opposite sign, flipping the correlation's sign but not its square.

Solution. (a) Maximizing (12.2.14) in β (with $\hat\alpha = \bar y - \beta\bar x$) minimizes $g(b) = (S_{yy} - 2bS_{xy} + b^2 S_{xx})/(1 + \lambda b^2)$; $g'(b) = 0$ iff $\lambda S_{xy} b^2 + (S_{xx} - \lambda S_{yy})b - S_{xy} = 0$. The product of its two roots is $-S_{xy}/(\lambda S_{xy}) = -1/\lambda$. The "+" root is the MLE $\hat\beta$; the other is $(-1/\lambda)/\hat\beta = -1/(\lambda\hat\beta)$.

(b) $r_\lambda(\beta)$ is the sample correlation of $A = \beta\lambda Y + X$ and $B = Y - \beta X$. Under $\beta' = -1/(\lambda\beta)$ (for $\beta \ne 0$): $\beta'\lambda Y + X = -(Y - \beta X)/\beta$ and $Y - \beta' X = (\beta\lambda Y + X)/(\lambda\beta)$, i.e. the new pair is $(c_1 B, c_2 A)$ with $c_1 c_2 = -1/(\lambda\beta^2) < 0$. Correlation is invariant under positive rescaling and flips sign under negative rescaling, so $r_\lambda(-1/(\lambda\beta)) = -r_\lambda(\beta)$.

(c) $C_\lambda(\hat\beta)$ is defined by $(n-2)r_\lambda^2(\beta)/(1 - r_\lambda^2(\beta)) \le F_{1,n-2,\alpha}$, which depends on β only through $r_\lambda^2(\beta) = r_\lambda^2(-1/(\lambda\beta))$ by (b).

Exercise 12.13

Central idea. The numerator of $r_\lambda(\beta)$ is (up to sign) the score quadratic, so it factors through its roots $\hat\beta$ and $-1/(\lambda\hat\beta)$; the denominator-minus-numerator collapses to $(1 + \lambda\beta^2)^2(S_{xx}S_{yy} - S_{xy}^2)$, and the score quadratic evaluated at $\hat\beta$ converts the remaining constants into $\hat\sigma_{\hat\beta}^2$.

Solution. (a) Squaring (12.2.22), $(\beta-\hat\beta)^2 \le t^2_{n-2,\alpha/2}\hat\sigma^2_{\hat\beta}/(n-2)$, and $t^2_{n-2,\alpha/2} = F_{1,n-2,\alpha}$.

(b) The numerator of $r_\lambda(\beta)$ is $\beta\lambda S_{yy} + (1-\beta^2\lambda)S_{xy} - \beta S_{xx} = -(\lambda S_{xy}\beta^2 + (S_{xx}-\lambda S_{yy})\beta - S_{xy}) = -\lambda S_{xy}(\beta-\hat\beta)(\beta+1/(\lambda\hat\beta))$ by Exercise 12.12(a). Writing $A = S_{xx}$, $B = S_{xy}$, $C = S_{yy}$, the denominator minus the squared numerator is

$$\begin{aligned}&(\lambda^2\beta^2 C + 2\lambda\beta B + A)(C - 2\beta B + \beta^2 A) - (\beta\lambda C + (1-\beta^2\lambda)B - \beta A)^2\\ &\qquad = (AC - B^2)(1+\lambda\beta^2)^2,\end{aligned}$$

the β^3 and β terms cancelling identically. Hence the stated identity for $r^2_\lambda/(1-r^2_\lambda)$ holds. Now

$$\frac{(n-2)r^2_\lambda}{1-r^2_\lambda} \le F \iff (n-2)\frac{\lambda^2 S^2_{xy}(\beta-\hat\beta)^2(\beta+1/(\lambda\hat\beta))^2}{(1+\lambda\beta^2)^2(S_{xx}S_{yy}-S^2_{xy})} \le F.$$

Since $\beta + 1/(\lambda\hat\beta) = (1+\lambda\hat\beta\beta)/(\lambda\hat\beta)$, this equals $\frac{(\beta-\hat\beta)^2}{\hat\sigma^2_{\hat\beta}/(n-2)}\frac{(1+\lambda\hat\beta\beta)^2}{(1+\lambda\beta^2)^2} \le F$ provided $\hat\sigma^2_{\hat\beta} = \hat\beta^2(S_{xx}S_{yy} - S^2_{xy})/S^2_{xy}$. Indeed, with $\hat\sigma^2_{\hat\beta} = (1+\lambda\hat\beta^2)^2(S_{xx}S_{yy}-S^2_{xy})/((S_{xx}-\lambda S_{yy})^2 + 4\lambda S^2_{xy})$, the requirement is $(1+\lambda\hat\beta^2)^2 S^2_{xy} = \hat\beta^2[(S_{xx}-\lambda S_{yy})^2 + 4\lambda S^2_{xy}]$, which follows from the score equation: $(S_{xx}-\lambda S_{yy})\hat\beta = S_{xy}(1-\lambda\hat\beta^2)$, so $\hat\beta^2(S_{xx}-\lambda S_{yy})^2 + 4\lambda S^2_{xy}\hat\beta^2 = S^2_{xy}(1-\lambda\hat\beta^2)^2 + 4\lambda S^2_{xy}\hat\beta^2 = S^2_{xy}(1+\lambda\hat\beta^2)^2$. Finally $(1+\lambda\hat\beta\beta)^2/(1+\lambda\beta^2)^2 \to_p 1$ since $\hat\beta \to_p \beta$, so C_G is the Creasy–Williams set with that factor replaced by its probability limit.

Exercise 12.14

Central idea. Each curve is an S-shape through $(-\alpha/\beta,\, 1/2) = (-1,\, 1/2)$ with slope $\beta/4$ there; larger $\alpha = \beta$ steepens the transition.

Solution. With $\pi(x) = e^{\alpha+\beta x}/(1+e^{\alpha+\beta x})$, all three curves pass through $(-1, 1/2)$ and increase from 0 to 1, getting steeper as $\alpha = \beta$ grows (slope at $x = -1$ is $\beta\pi(1-\pi) = \beta/4$):

Exercise 12.15

Central idea. Everything follows from $\pi(x) = e^{\alpha+\beta x}/(1+e^{\alpha+\beta x})$ and the chain rule; the logistic cdf satisfies $F' = F(1-F)$, which collapses the GLM score equations.

Solution. (a) $\pi(-\alpha/\beta) = e^0/(1+e^0) = 1/2$. (b) $\pi(-\alpha/\beta + c) = e^{\beta c}/(1+e^{\beta c})$ and $\pi(-\alpha/\beta - c) = e^{-\beta c}/(1+e^{-\beta c}) = 1/(1+e^{\beta c})$; they sum to 1. (c) $d\pi/dx = \beta e^{\alpha+\beta x}/(1+e^{\alpha+\beta x})^2 = \beta\pi(x)(1-\pi(x))$. (d) From (12.3.1), $\log\frac{\pi(x+1)}{1-\pi(x+1)} - \log\frac{\pi(x)}{1-\pi(x)} = \beta$; exponentiating gives (12.3.4). (e) Rearranged, this is $\frac{\pi(x+1)}{1-\pi(x+1)} = e^\beta\frac{\pi(x)}{1-\pi(x)}$. (f) With $F_i = F(\alpha+\beta x_i)$, $f_i = F'(\alpha+\beta x_i)$: $\frac{\partial}{\partial\alpha}\log(1-F_i) = -f_i/(1-F_i) = -F_i f_i/(F_i(1-F_i))$, and $\frac{\partial}{\partial\alpha}\log\frac{F_i}{1-F_i} = \frac{f_i}{F_i} + \frac{f_i}{1-F_i} = \frac{f_i}{F_i(1-F_i)}$, which is (12.3.6); then $\frac{\partial\log L}{\partial\alpha} = \sum_i (y_i - F_i)\frac{f_i}{F_i(1-F_i)}$, i.e. (12.3.7). Since $\partial F_i/\partial\beta = x_i f_i$, the same calculation with an extra x_i gives (12.3.8). (g) For $F(w) = e^w/(1+e^w)$, $f(w) = e^w/(1+e^w)^2 = F(w)(1-F(w))$, so $f_i/(F_i(1-F_i)) = 1$ and the score equations simplify to $\sum_i(y_i - F_i) = 0$, $\sum_i x_i(y_i - F_i) = 0$.

Exercise 12.16

Central idea. This is complete separation: the unconstrained binomial MLEs sit on the boundary $\pi(0) = 1$, which the logistic curve can only approach asymptotically.

Solution. (a) Unrestricted, the likelihood is $\pi(0)^{10}(1-\pi(0))^0 \cdot \pi(1)^5(1-\pi(1))^5$, maximized at the binomial MLEs $\hat\pi(0) = 10/10 = 1$, $\hat\pi(1) = 5/10 = 0.5$, with maximum value $(1/2)^{10}$.

(b) Under (12.3.2), $\pi(0) = e^\alpha/(1+e^\alpha) < 1$ for every finite α, so the value $(1/2)^{10}$ is unattainable at finite (α,β). But with $\alpha = -\beta \to \infty$: $\pi(0) = e^\alpha/(1+e^\alpha) \to 1$ and $\pi(1) = e^{\alpha+\beta}/(1+e^{\alpha+\beta}) = 1/2$ exactly, so the likelihood tends to $(1/2)^{10}$, the unrestricted maximum. Hence no finite maximizer exists — the MLE is "at infinity."

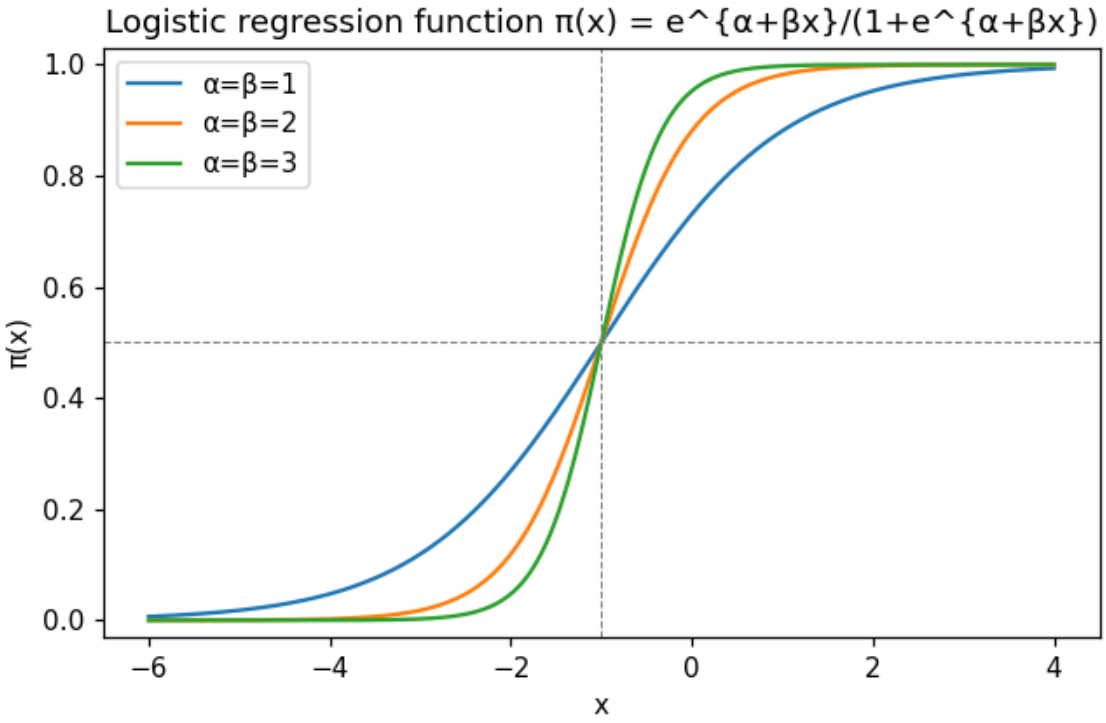


Figure 12: Logistic curves for = =1,2,3

Exercise 12.17

Central idea. Same Bernoulli likelihood with Φ in place of the logistic cdf; the score equations are nonlinear and solved by Newton–Raphson (the log-likelihood is strictly concave). The probit and logistic links are nearly indistinguishable on data.

Solution. (a) $L(\alpha, \beta \mid \mathbf{y}) = \prod_{i=1}^{n} \Phi(\alpha + \beta x_i)^{y_i}(1 - \Phi(\alpha + \beta x_i))^{1-y_i}$. With $\phi = \Phi'$,

$$\frac{\partial \log L}{\partial \alpha} = \sum_{i=1}^{n} (y_i - \Phi_i) \frac{\phi_i}{\Phi_i(1 - \Phi_i)}, \qquad \frac{\partial \log L}{\partial \beta} = \sum_{i=1}^{n} x_i (y_i - \Phi_i) \frac{\phi_i}{\Phi_i(1 - \Phi_i)},$$

where $\Phi_i = \Phi(\alpha + \beta x_i)$. Setting these to zero gives nonlinear equations solved numerically (Newton–Raphson/IRLS); strict concavity guarantees a unique solution when one exists.

(b) For the Challenger data (Table 12.3.1), Newton–Raphson gives $\hat{\alpha} = 9.20$, $\hat{\beta} = -0.141$ (compare logistic $\hat{\alpha} = 15.81$, $\hat{\beta} = -0.243$). The fits are virtually identical: e.g. fitted failure probabilities at 53/63/70/81°F are logistic $(.949, .623, .231, .020)$ vs probit $(.958, .626, .253, .013)$, and both give $P(\text{fail} \mid 31°\text{F}) \approx 1$. The probit coefficients are roughly the logistic ones divided by 1.6–1.8 (the usual scaling between the links); the logistic link has slightly heavier tails, but the data cannot distinguish the two models.

Exercise 12.18

Central idea. Grouped binomial likelihood with a quadratic logit; the LRT for $\gamma = 0$ is $2(\ell_{\text{quad}} - \ell_{\text{lin}}) \sim \chi^2_1$. The survival proportions rise then fall, which a linear logit cannot capture.

Solution. (a) With n_j birds and y_j^* survivors at age x_j, $\pi_j = e^{\alpha+\beta x_j+\gamma x_j^2}/(1+e^{\alpha+\beta x_j+\gamma x_j^2})$,

$$L(\alpha,\beta,\gamma) = \prod_j \pi_j^{y_j^*}(1-\pi_j)^{n_j-y_j^*}.$$

Since $f_j/(F_j(1-F_j)) = 1$ for the logistic link, the score equations are $\sum_j (y_j^* - n_j\pi_j)(1, x_j, x_j^2)' = 0$, solved by Newton–Raphson (strictly concave log-likelihood).

(b) Let $(\tilde\alpha, \tilde\beta)$ be the MLEs under $\gamma = 0$ and $(\hat\alpha, \hat\beta, \hat\gamma)$ the full MLEs. The LRT statistic $-2\log\lambda = 2[\log L(\hat\alpha,\hat\beta,\hat\gamma) - \log L(\tilde\alpha,\tilde\beta,0)]$ is approximately χ^2_1 under H_0; reject H_0 if it exceeds $\chi^2_{1,\alpha}$.

(c) Linear fit: $\tilde\alpha = 0.480$, $\tilde\beta = 0.0074$, $\log L = -457.57$. Quadratic fit: $\hat\alpha = -0.845$, $\hat\beta = 0.837$, $\hat\gamma = -0.103$, $\log L = -445.89$ (Newton–Raphson, gradients $< 10^{-12}$). The quadratic captures the hump: fitted survival rises to ≈ 0.70 near age 4 then declines, matching the observed proportions $(.455, .597, .714, .669, .667, .609, .519, .300, .250)$.

(d) $-2\log\lambda = 2(-445.89 + 457.57) = 23.36 \gg \chi^2_{1,.05} = 3.84$ ($p \approx 10^{-6}$): reject H_0. The quadratic model is decisively better — survival first increases then decreases with age, which the linear logit (with its near-zero slope) cannot represent.

Exercise 12.19

Central idea. The log-likelihood is linear in (α,β) with coefficients $(\sum Y_j^*, \sum Y_j^* x_j^*)$ — factorization is immediate; the Hessian involves only the fitted probabilities, so observed and expected information coincide.

Solution. (a) $L(\alpha,\beta) = \prod_j F_j^{Y_j^*}(1-F_j)^{n_j-Y_j^*}$ with $F_j = e^{\alpha+\beta x_j^*}/(1+e^{\alpha+\beta x_j^*})$. Then

$$\log L = \alpha\sum_j Y_j^* + \beta\sum_j Y_j^* x_j^* - \sum_j n_j \log(1+e^{\alpha+\beta x_j^*}),$$

so by the Factorization Theorem, $(\sum_j Y_j^*, \sum_j Y_j^* x_j^*)$ is sufficient for (α,β).

(b) From the score $\partial\log L/\partial\alpha = \sum_j (Y_j^* - n_j F_j)$: $\partial^2\log L/\partial\alpha^2 = -\sum_j n_j F_j(1-F_j)$, $\partial^2\log L/\partial\alpha\partial\beta = -\sum_j x_j^* n_j F_j(1-F_j)$, $\partial^2\log L/\partial\beta^2 = -\sum_j (x_j^*)^2 n_j F_j(1-F_j)$, using $dF_j/d(\alpha+\beta x_j^*) = F_j(1-F_j)$. None of these involve Y_j^*, so the observed information equals the expected information, giving (12.3.10).

Exercise 12.20

Central idea. With $\beta = 0$ the model collapses to iid Bernoulli trials; the boundary cases 0 or n successes push the logit to $\pm\infty$ but the likelihood ratio only needs likelihood values, which stay finite.

Solution. (a) With $\beta = 0$, $\pi(x) \equiv \pi = e^\alpha/(1+e^\alpha)$ and $L(\pi) = \pi^{\sum y_i}(1-\pi)^{n-\sum y_i}$, the ordinary Bernoulli likelihood, maximized at $\hat\pi = \sum y_i/n$ for $0 < \sum y_i < n$.

(b) Inverting the logit, $\hat\alpha_0 = \log(\hat\pi/(1-\hat\pi)) = \log\left((\sum y_i)/(n-\sum y_i)\right)$.

(c) If $\sum y_i = 0$, $\hat\pi = 0$ and $\log(\hat\pi/(1-\hat\pi)) = -\infty$; if $\sum y_i = n$, it is $+\infty$ — no finite $\hat\alpha_0$ exists. But the LRT statistic $-2\log\lambda = 2[\log L(\hat\alpha,\hat\beta) - \log L(\hat\alpha_0, 0)]$ needs only $\log L(\hat\alpha_0, 0) = \sum y_i \log\hat\pi + (n - \sum y_i)\log(1-\hat\pi) = 0$ (with $0\log 0 = 0$), which is well defined (equal to 0), so the test statistic is finite and well defined.

Exercise 12.21

Central idea. One application of the delta method to $g(\pi) = \log(\pi/(1-\pi))$: $g'(\pi) = 1/[\pi(1-\pi)]$ squares against $\operatorname{Var}\hat\pi = \pi(1-\pi)/n$.

Solution. $g(\pi) = \log(\pi/(1-\pi))$ has $g'(\pi) = 1/\pi + 1/(1-\pi) = 1/[\pi(1-\pi)]$. By the delta method,

$$\operatorname{Var} W \approx [g'(\pi)]^2 \operatorname{Var}\hat\pi = \frac{1}{\pi^2(1-\pi)^2}\cdot\frac{\pi(1-\pi)}{n} = \frac{1}{n\pi(1-\pi)},$$

and plugging in the MLE $\hat{\pi}$ gives the variance estimate $1/[n\hat{\pi}(1-\hat{\pi})]$.

Exercise 12.22

Central idea. Write the intercept as a linear combination $a = \sum c_i Y_i$; under variance contamination its variance inflates by the same factor $(1-\delta)\sigma^2 + \delta\tau^2$ as the slope's — small perturbations hurt it only mildly.

Solution. With $d_i = (x_i - \bar{x})/\sum_j (x_j - \bar{x})^2$, $b = \sum d_i Y_i$ and $a = \bar{y} - b\bar{x} = \sum_i c_i Y_i$ where $c_i = 1/n - \bar{x} d_i$. Under $\mathrm{Var}(\epsilon_i) = (1-\delta)\sigma^2 + \delta\tau^2$ for all i,

$$\mathrm{Var}(a) = \sum_i c_i^2 [(1-\delta)\sigma^2 + \delta\tau^2] = [(1-\delta)\sigma^2 + \delta\tau^2] \sum_i c_i^2.$$

Since $\sum_i d_i = 0$ and $\sum_i d_i^2 = 1/\sum_j (x_j - \bar{x})^2$,

$$\sum_i c_i^2 = \frac{1}{n} - 2\bar{x}\frac{1}{n}\sum_i d_i + \bar{x}^2 \sum_i d_i^2 = \frac{1}{n} + \frac{\bar{x}^2}{\sum_j (x_j - \bar{x})^2}.$$

Hence $\mathrm{Var}(a) = [(1-\delta)\sigma^2 + \delta\tau^2](1/n + \bar{x}^2/\sum_j (x_j - \bar{x})^2)$: exactly the uncontaminated variance with σ^2 replaced by the mixture variance. The variance inflation factor is $1 + \delta(\tau^2/\sigma^2 - 1)$. For fixed finite τ^2 it tends to one as $\delta \to 0$, but it is unbounded over contaminant scales even at arbitrarily small fixed $\delta > 0$. This calculation does not establish robustness to unrestricted contamination.

Exercise 12.23

Central idea. $E\epsilon_i = \delta\mu$ is constant across i; the slope estimator differences the x_i's out ($\sum d_i = 0$), while the intercept absorbs the common bias.

Solution. Here $E\epsilon_i = (1-\delta)\cdot 0 + \delta \cdot \mu = \delta\mu$ for every i.

(a) $b = \sum_i d_i Y_i$ with $d_i = (x_i - \bar{x})/\sum_j (x_j - \bar{x})^2$, so $Eb = \sum_i d_i(\alpha + \beta x_i + \delta\mu) = \beta + \delta\mu \sum_i d_i = \beta$ since $\sum_i d_i = 0$ and $\sum_i d_i x_i = 1$.

(b) $Ea = E\bar{y} - \bar{x}Eb = \alpha + \beta\bar{x} + \delta\mu - \beta\bar{x} = \alpha + \delta\mu$. Writing $\epsilon_i' = \epsilon_i - \delta\mu$ (so $E\epsilon_i' = 0$), the model is $Y_i = (\alpha + \delta\mu) + \beta x_i + \epsilon_i'$ — the bias is absorbed into the intercept and β estimation is unaffected.

Exercise 12.24

Central idea. $|y_i - \beta x_i| = |x_i| \cdot |t_i - \beta|$, so the LAD problem is a *weighted median* of the ratios $t_i = y_i/x_i$ with weights $|x_i|$; the k^* conditions are exactly the weighted-median balance conditions.

Solution. Terms with $x_i = 0$ are constant in β and can be dropped; assume $x_i \neq 0$. Then

$$\sum_i |y_i - \beta x_i| = \sum_i |x_i|\,|t_i - \beta| \equiv L(\beta), \qquad t_i = y_i/x_i,$$

a convex piecewise-linear function. Its subgradient at β is $\sum_i |x_{(i)}| s_i$ with $s_i = +1$ if $t_{(i)} < \beta$, -1 if $t_{(i)} > \beta$, and $s_i \in [-1, 1]$ if $t_{(i)} = \beta$. At $\beta = t_{(k^*+1)}$ this equals

$$\sum_{i \le k^*} |x_{(i)}| - \sum_{i > k^*+1} |x_{(i)}| + |x_{(k^*+1)}| u, \quad u \in [-1, 1].$$

With $W = \sum_i |x_{(i)}|$ and $W_L(k) = \sum_{i \le k} |x_{(i)}|$, the two conditions on k^* are $W_L(k^*) \le W/2$ and $W_L(k^*+1) > W/2$; choosing $u = (W - 2W_L(k^*))/|x_{(k^*+1)}| - 1 \in [-1, 1]$ makes the subgradient contain 0. Hence $t_{(k^*+1)}$ minimizes L — it is the weighted median of the t_i with weights $|x_i|$.

Exercise 12.25

Central idea. With two observations sharing an x, the LAD objective splits into $|y_1-v|+|y_2-v|$ (minimized by any v between y_1 and y_2) plus the third residual (minimized at zero by passing through (x_3, y_3)) — a whole interval of minimizers.

Solution. (a) For a line $y = a+bx$ put $v = a+bx_1$. The LAD objective is $|y_1-v|+|y_2-v|+|y_3-(a+bx_3)| \geq |y_1 - y_2| + 0 = |y_1 - y_2|$, with equality iff v lies between y_1 and y_2 and the line passes through (x_3, y_3). Any line not through (x_3, y_3) has objective $> |y_1-y_2|$. Hence the LAD lines are exactly the lines through (x_3, y_3) whose value at x_1 lies between y_1 and y_2 — not unique.

(b) Least squares: $\bar{x} = 130$, $\bar{y} = 14.733$, $S_{xx} = 54$, $S_{xy} = 28.5$, so $\hat{b} = 0.528$, $\hat{a} = \bar{y} - \hat{b}\bar{x} = -53.88$. LAD: lines through $(136, 17.9)$ with $y(127) = 17.9-9b \in [11.9, 14.4]$, i.e. slope $b \in [3.5/9,\ 6/9] = [0.389, 0.667]$ and intercept $a = 17.9 - 136b \in [-72.77, -34.99]$. The LS line ($b = 0.528$) lies inside the LAD range.

Exercise 12.26

Central idea. Bounded design values bound the average squared design, but do not ensure that it converges. For the plug-in score, use a deterministic neighborhood rather than treating an estimated threshold as independent of the data.

Solution. (a)(i) If $x_i = 1$, $q_n = n^{-1}\sum_i x_i^2 = 1$. (ii) If $|x_i| \leq 1$, $0 \leq q_n \leq 1$, but boundedness alone does not imply convergence or a positive limit. For example, successive blocks of zeros and ones, each longer than all previous blocks by a factor tending to infinity, have subsequences with $q_n \to 0$ and $q_n \to 1$. Retain $q_n \to q > 0$ as an additional assumption for the asymptotic-normality result. (b) Suppose $\hat{\beta}_L \to_p \beta$, errors are independent with median zero and continuous at zero, and $|x_i| \leq 1$. Put $S_i(t) = x_i \operatorname{sign}(Y_i - tx_i)$. The average at $t = \beta$ has mean zero and variance at most $1/n$, hence converges to zero in probability. For any fixed $\delta > 0$, on $|\hat{\beta}_L - \beta| \leq \delta$,

$$\frac{1}{n}\sum_i |S_i(\hat{\beta}_L) - S_i(\beta)| \leq \frac{2}{n}\sum_i 1\{|\epsilon_i| \leq \delta\}.$$

For $\eta > 0$, Markov's inequality gives a bound $P(|\hat{\beta}_L - \beta| > \delta) + 2P(|\epsilon| \leq \delta)/\eta$ for the probability that the discrepancy exceeds η. First let $n \to \infty$, then $\delta \downarrow 0$. This proves the assertion without a false independence assumption about $\hat{\beta}_L$ and ϵ_i.

Exercise 12.27

Central idea. Use the true-parameter weighted sign score for the CLT and differentiate its expectation. Convex LAD objective expansion replaces differentiation of the discontinuous empirical sign score.

Solution. Write $Y_i = \beta x_i + \epsilon_i$, $S_i(t) = x_i \operatorname{sign}(Y_i - tx_i)$, and assume independent identically distributed errors with continuous density at zero, median zero, $f(0) > 0$, bounded fixed design, and $q_n = n^{-1}\sum x_i^2 \to q > 0$. (a) At the true parameter,

$$Z_n = \frac{1}{\sqrt{n}}\sum_i S_i(\beta) = \frac{1}{\sqrt{n}}\sum_i x_i \operatorname{sign}(\epsilon_i) \Rightarrow N(0, q).$$

Each summand has mean zero and variance x_i^2; bounded design implies Lindeberg's condition. The printed plug-in assertion is false: for $x_i = 1$ and odd n, $\hat{\beta}_L$ is the sample median and the plug-in sign sum is zero almost surely. (b) $E_\beta S_i(t) = x_i[1 - 2F((t-\beta)x_i)]$, so

$$\frac{1}{n}\sum_i \frac{d}{dt} E_\beta S_i(t)\bigg|_{t=\beta} = -2f(0)q_n,$$

with a minus sign. To obtain the limiting law rigorously, for fixed h expand the LAD objective $Q_n(h) = \sum_i(|\epsilon_i - hx_i/\sqrt{n}| - |\epsilon_i|)$ using

$$|e - v| - |e| = -v\,\mathrm{sign}(e) + 2\int_0^v [1\{e \le s\} - 1\{e \le 0\}]\,ds.$$

Its linear term is $-hZ_n$. Continuity of f at zero makes the expectation of the integral terms $f(0)q_nh^2 + o(1)$; independence and bounded design make their centered sum $o_p(1)$. Convexity upgrades this to convergence on compact h sets and yields the argmin expansion

$$\sqrt{n}(\hat{\beta}_L - \beta) = \frac{Z_n}{2f(0)q_n} + o_p(1) \Rightarrow N\left(0, \frac{1}{4f(0)^2 q}\right).$$

This justifies the final law. The empirical sign score has derivative zero between its jumps, so an ordinary sample-score Taylor derivative is not a valid denominator.

Exercise 12.28

Central idea. Closed-form normal equation for the no-intercept model; the numerator is a CLT average and the denominator converges to σ_x^2 (with unit error variance as in Example 12.4.3).

Solution. Minimizing $\sum_i(y_i - \beta x_i)^2$ gives $-2\sum_i x_i(y_i - \beta x_i) = 0$, i.e. $\hat{\beta} = \sum_i x_i y_i / \sum_i x_i^2$. Then

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{\frac{1}{\sqrt{n}}\sum_i x_i\epsilon_i}{\frac{1}{n}\sum_i x_i^2}.$$

The numerator $\to_d n(0, \sigma_\epsilon^2\sigma_x^2)$ by the CLT ($E[x_i\epsilon_i] = 0$, Var $= \sigma_\epsilon^2 x_i^2$, x_i bounded) and the denominator $\to_p \sigma_x^2$; with $\sigma_\epsilon^2 = 1$ as in Example 12.4.3, Slutsky's theorem gives $\sqrt{n}(\hat{\beta} - \beta) \to_d n(0, 1/\sigma_x^2)$.

Exercise 12.29

Central idea. The median is the LAD location estimator; Taylor-expand the expected objective, with the empirical fluctuation handled by the convex expansion in 12.27.

Solution. Let m be the population median, assume f is continuous and positive at m, and write $\epsilon_i = Y_i - m$. In the corrected 12.27, take $x_i = 1$ and $q_n = 1$. The true-parameter sign sum $Z_n = n^{-1/2}\sum_i \mathrm{sign}(\epsilon_i)$ converges to $N(0, 1)$. The exact absolute-loss identity there gives

$$\sum_i(|\epsilon_i - h/\sqrt{n}| - |\epsilon_i|) = -hZ_n + f(m)h^2 + o_p(1),$$

uniformly on compact h sets. Its minimizing argument therefore satisfies $\sqrt{n}(M_n - m) = Z_n/[2f(m)] + o_p(1)$, for any chosen sample median, and

$$\sqrt{n}(M_n - m) \Rightarrow N\left(0, \frac{1}{4f(m)^2}\right).$$

The expected sign score has derivative $-2f(m)$; the empirical sign score has jumps and derivative zero between them. Substituting the latter into an ordinary differentiable Taylor formula is invalid.

Exercise 12.30

Central idea. Parametric bootstrap: treat the fitted line and $\hat{\sigma}^2$ as truth, simulate new data from the assumed error law, refit, and use the spread of refits as the standard error. (Note: "Double Exponential" and "Laplace" are two names for the same distribution.)

Solution. (a) From Example 12.4.2, the LAD fit to Table 12.4.1 is $\tilde{\alpha} = 18.59$, $\tilde{\beta} = -0.89$ (use the M-estimate fit analogously for the M-estimator standard error). (b) Compute residuals $\hat{e}_i = y_i - (\tilde{\alpha} + \tilde{\beta}x_i)$ and

$\hat{\sigma}^2 = \frac{1}{n-2}\sum_i \hat{e}_i^2$ ($n = 23$). (c) For $b = 1, \ldots, B$: draw $e_i^{*(b)} \overset{iid}{\sim} n(0, \hat{\sigma}^2)$, form $y_i^{*(b)} = \tilde{\alpha} + \tilde{\beta}x_i + e_i^{*(b)}$, and refit by LAD (resp. M-estimation) to get $(\tilde{\alpha}^{*(b)}, \tilde{\beta}^{*(b)})$. (d) With B large (e.g. $B = 1000$), the bootstrap standard errors are $\text{se}(\tilde{\alpha}) = \text{sd}\{\tilde{\alpha}^{*(b)}\}$, $\text{se}(\tilde{\beta}) = \text{sd}\{\tilde{\beta}^{*(b)}\}$. (e) The Double Exponential distribution *is* the Laplace distribution (pdf $\frac{1}{2b}e^{-|e|/b}$); matching variances to $\hat{\sigma}^2$ requires scale $b = \hat{\sigma}/\sqrt{2}$. Repeat (c)–(d) drawing $e_i^{*(b)}$ from this Laplace law instead of normal. At fixed design and the same stipulated error variance $\hat{\sigma}^2$, least-squares coefficient covariance is exactly $\hat{\sigma}^2(X^TX)^{-1}$ under both error laws; heavier tails do not make those standard errors larger. For LAD, the asymptotic covariance is $(X^TX)^{-1}/[4f_\epsilon(0)^2]$: its scalar factor is $\pi\hat{\sigma}^2/2$ under normal errors and $\hat{\sigma}^2/2$ under a variance-matched Laplace law. Thus Laplace LAD standard errors are asymptotically $1/\sqrt{\pi}$ times the normal LAD standard errors. M-estimator comparisons depend on the chosen score and tuning. Numerical bootstrap results still require actually fitting the data and simulating the requested refits; the algorithm alone does not complete those numerical parts.

Exercise 12.31

Central idea. Nonparametric bootstrap: resample the (centered) fitted residuals with replacement instead of assuming a parametric error law; everything else mirrors Exercise 12.30.

Solution. (a) As in Exercise 12.30(a): LAD fit $\tilde{\alpha} = 18.59$, $\tilde{\beta} = -0.89$ (resp. the M-estimate fit), with fitted residuals $\hat{e}_i = y_i - (\tilde{\alpha} + \tilde{\beta}x_i)$, $i = 1, \ldots, 23$. (b) Center the residuals, $\hat{e}_i^c = \hat{e}_i - \bar{\hat{e}}$ (they need not have mean zero for LAD/M fits). For $b = 1, \ldots, B$: draw $e_1^{*(b)}, \ldots, e_n^{*(b)}$ with replacement from $\{\hat{e}_i^c\}$, form $y_i^{*(b)} = \tilde{\alpha} + \tilde{\beta}x_i + e_i^{*(b)}$, and refit by LAD (resp. M-estimation) to obtain $(\tilde{\alpha}^{*(b)}, \tilde{\beta}^{*(b)})$. (c) With B large (e.g. $B = 1000$), report $\text{se}(\tilde{\alpha}) = \text{sd}\{\tilde{\alpha}^{*(b)}\}$ and $\text{se}(\tilde{\beta}) = \text{sd}\{\tilde{\beta}^{*(b)}\}$. Unlike the parametric bootstrap, no error distribution is assumed — the empirical residual distribution drives the resampling — so the resulting standard errors are valid under whatever error law generated the data, at the cost of discreteness when n is small.